

MAGNETOM

Startup System

System Start-up

Avanto

Document Version

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The Startup documentation is for Magnetom Avanto and Magnetom Avanto Dot. Magnetom Avanto Dot has different MSUP and Alarm box.

Magnetom Avanto Dot Ex-factory (Serial number 60xxx) has the MSUP K2306 (Aera, Skyra)

Magnetom Avanto Dot Upgrade (Serial number 62xxx) has the MSUP K2256 (Avanto a Tim).

1.1 Prerequisites

- Room lighting must be functioning in every room of the MR site.
- The system components have been positioned and installed according installation instructions.
- The RF integrity of the RF room was measured after installing the RF room and its components.



Only trained personnel, who have successfully completed the Avanto course, are allowed to perform the start-up. The warranty is invalid, if problems occur in the system due to work performed by untrained personnel or third parties.

1.2 Service Tools

The Service Tools required for starting up the MR system include:

- Service tool kit (complete)
 - RF-cable 2 m BNC (part-no.: 8733479, included in the kit) is required for the tune-up procedure.
- Set of standard tools
- ESD service kit
- Digital multimeter
- Protective conductor measurement device
- Helium syphon (dewar side)

1.2.1 Magnet system

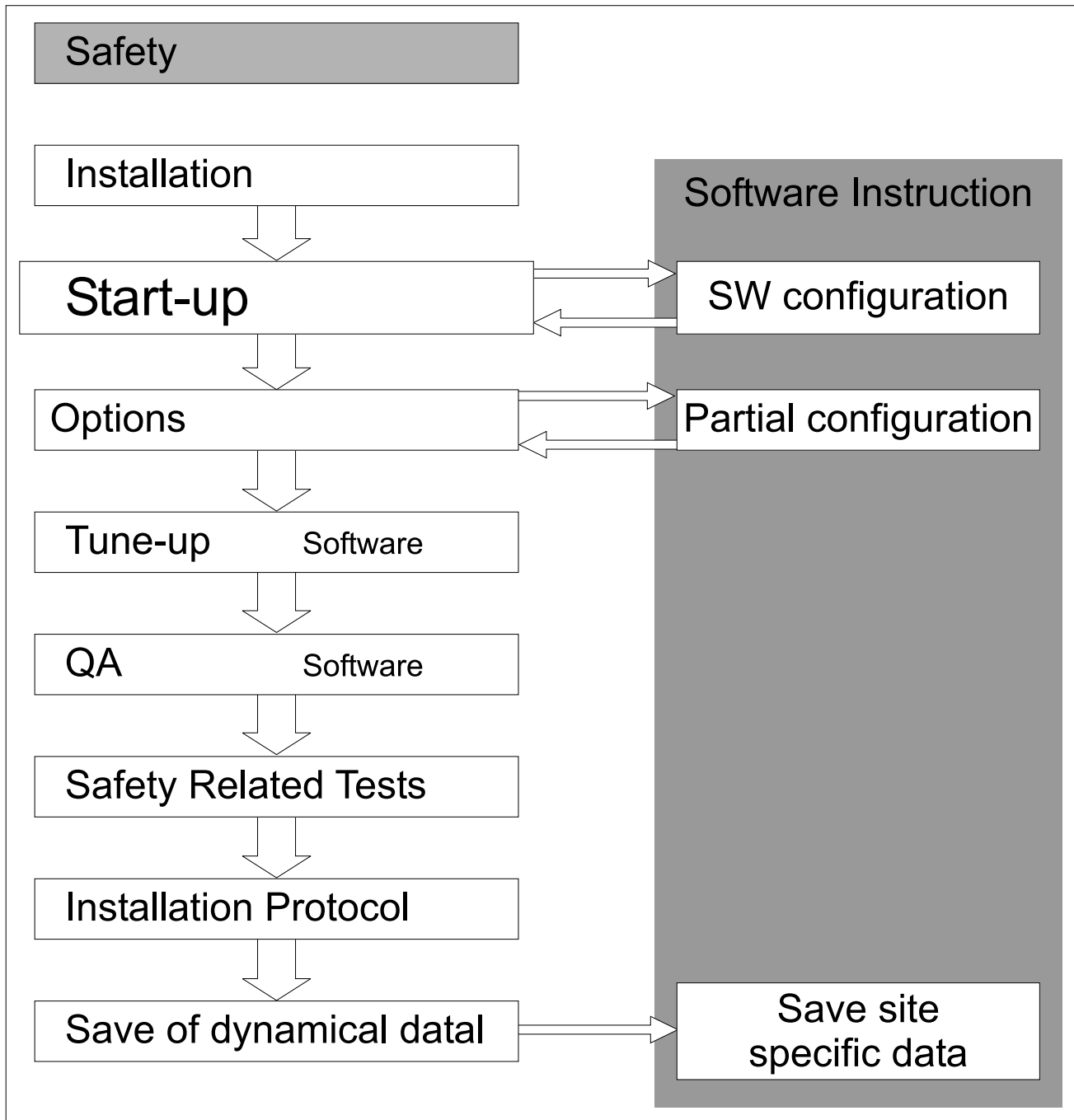
- Array shim device
- Heat gun
- Magnet power supply
- Protective clothing (face shield, gloves, safety shoes, etc.)
- Helium syphon (magnet side)
- MSUP Display Unit (only for Avanto, Espree, Avanto Dot Upgrade)

1.3 Overall Flowchart



System start-up should be performed in the order of the described steps. Complete each step before you start working on the next one.

Fig. 1: Startup flowchart



1.4 Safety

For your own safety and for the safety of third parties, as well as to protect the MR system and parts, observe the safety information in the register "General Safety Notes" and "MR-Specific Safety Information" of "Technical Documents" when performing work on the system.

Read the special safety information regarding:

- general safety for service personnel
- attaching or checking the "Strong Magnetic Field" warning sign
- attaching or checking the "Cardiac Pacemaker" warning sign
- bringing ferromagnetic materials into the magnet area
- safety information for working when the magnet is under field
- hazards involved in cryogen-filling
- working with heavy loads
- protective conductor test prior to switch-on and after repair work
- reduction of the magnetic field to 20 mT when quenching the magnet

Using the service platform

- Before using the service platform, perform the relevant safety checks.
- Install the platform as described in the Installation Instructions.
 - » Refer to (→ Safety Information / M6-020.812.01)

2.1 Safety



For safety information, refer to “MR-specific safety information”.

2.2 Checking the line voltage

- Lock all emergency shutdown buttons.
Prior to switch-on several adaptations to the line voltage have to be made (refer to the Installation manual, register xx, chapter xx).
- Ask your project manager about the on-site line voltage or perform your own measurement.

The on-site line voltage is:	VAC
The on-site line frequency is:	Hz



WARNING

High Voltage

Adhere to the warning. Otherwise death or serious injury may occur.

- » Verify that no line voltage is present at the ACC, GPA, SEP and LHe compressor.
- » Switch off the on-site power supply (ACC: F100, F102) and secure it with a lock to prevent anyone from inadvertently switching it on.

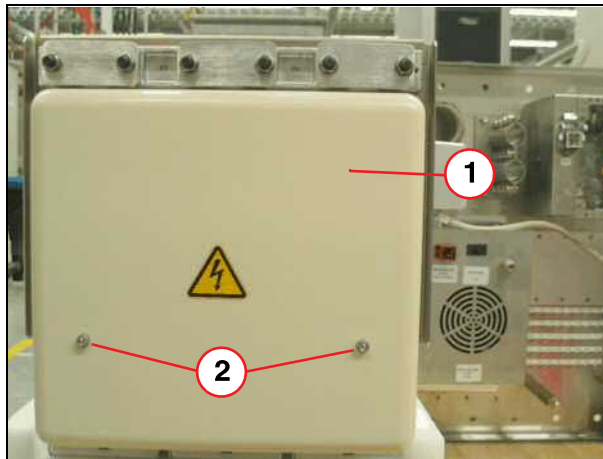


For detailed information, refer to the "Installation Manual, magnet cooling".

2.3 Checking the gradient connections

2.3.1 Gradient cables outside the RF room

Fig. 2: Protective cover, gradient filter (technical room)



- (1) Protective cover
- (2) Screws

- Dismounting the protective cover (→ 1/ Fig. 2 Page 15) of the gradient filter at the filter plate.

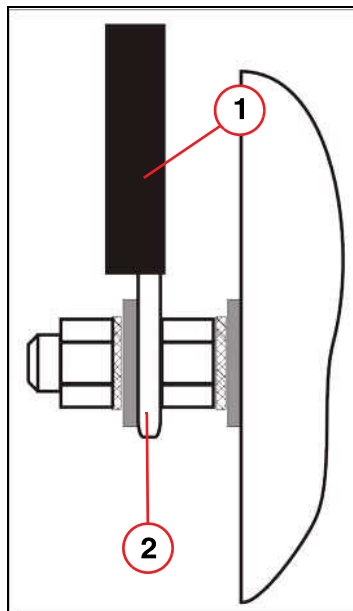


The cable shoes may point forward or backward. However, they cannot be turned by 180°.

Fig. 3: Filter plate 1



Fig. 4: Filter plate 1/1

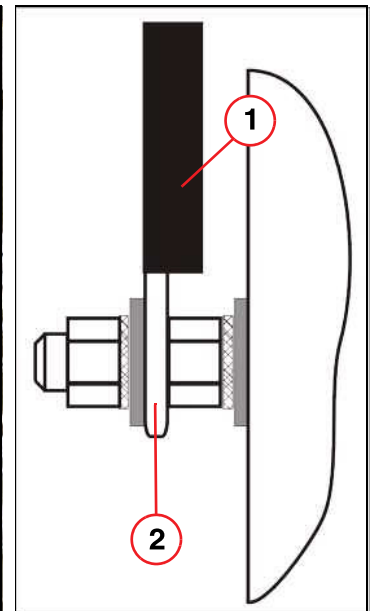


- (1) Gradient cable
- (2) Cable lug

Fig. 5: Filter plate 2

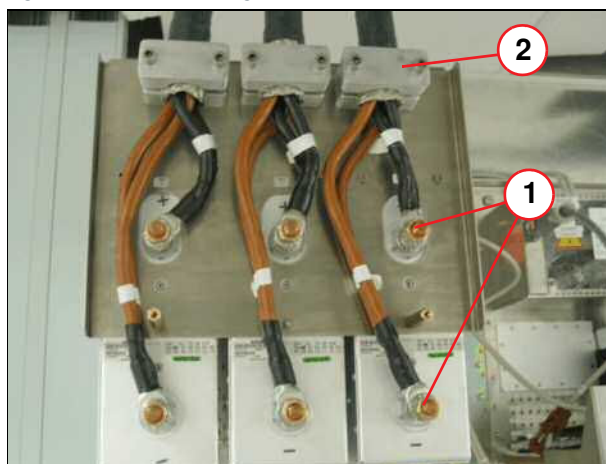


Fig. 6: Filter plate 2/1



- (1) Gradient cable
- (2) Cable lug

Fig. 7: Gradient filter gradient cable



- (1) Gradient cable
(2) Clamp screws

- Do a polarity check of the gradient cables (X, Y, and Z) to the gradient filters.
- Check the torque of the cable connections (→ 1/Fig. 7 Page 16) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 22 ± 1 Nm
- Check whether the cable clamps (→ 2/Fig. 7 Page 16) are tightened.
- Attach the protective cover.



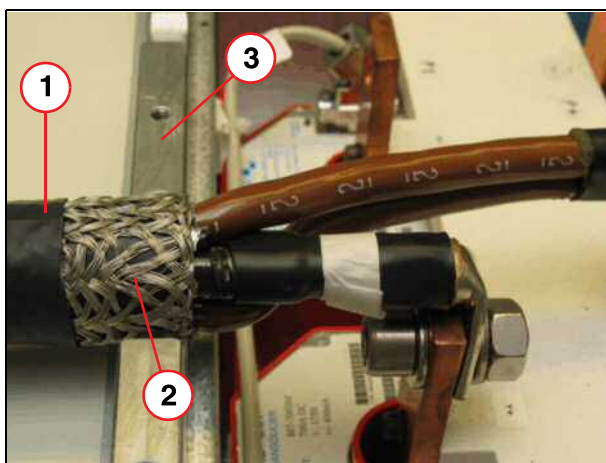
The cable clamps (→ 2/Fig. 7 Page 16) must clamp the shielding only. Ensure that the insulation is not clamped.

2.3.1.1 Check the connection at the GPA with a 1.5m cable set



Ensure that the gradient cables are connected as shown in (→ Fig. 8 Page 16). The shield must be clamped between the rails.

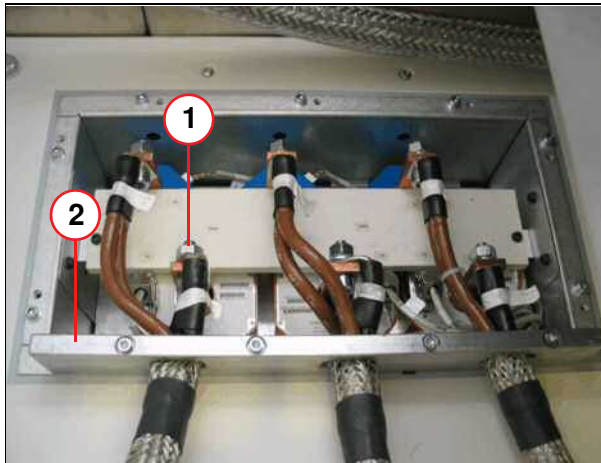
Fig. 8: GPA gradient cable



- (1) Gradient cable
(2) Shielding
(3) Rail

- Remove the protective cover.

Fig. 9: GPA gradient cable 1



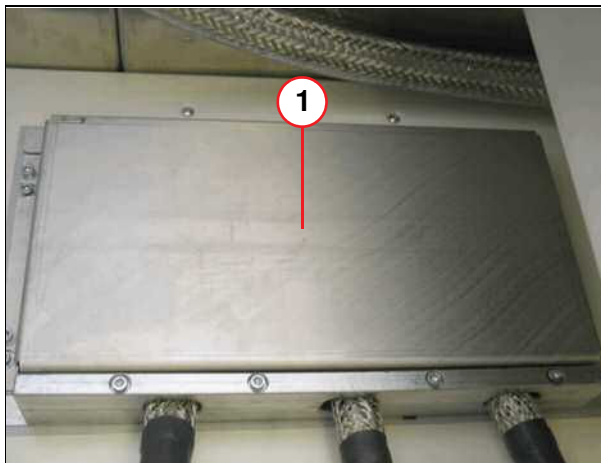
- (1) Gradient cable
(2) Rails

- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/Fig. 9 Page 17) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 40 ± 4 Nm.
- Check whether the cable clamp rail (→ 2/Fig. 9 Page 17) is tightened.



The cable clamp rail (→ 2/Fig. 9 Page 17) must clamp the shielding only. Ensure that the insulation is not clamped.

Fig. 10: GPA gradient cover 1



- (1) Gradient cover

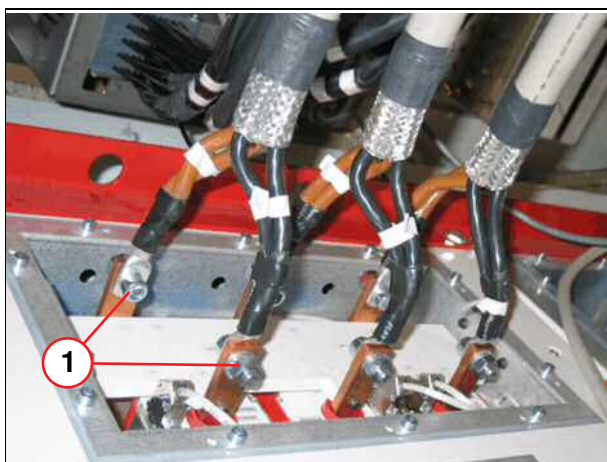
- Install the protective cover (→ 1/Fig. 10 Page 17).

2.3.1.2 Check the connection at the GPA with a cable set that is longer than 1.5m



The cable shoes may point forward or backward. However, they cannot be turned by 180° .

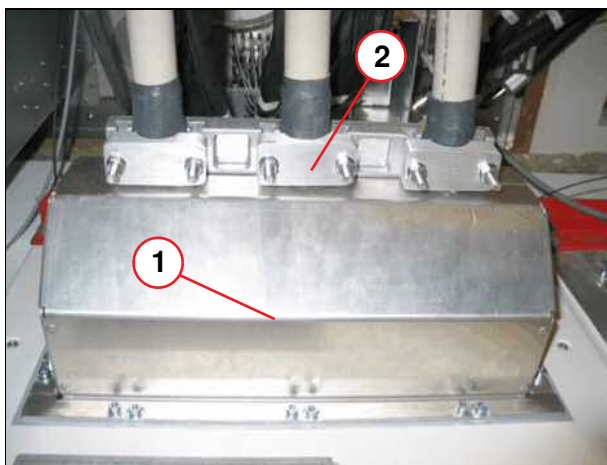
Fig. 11: GPA_gradient cable



(1) Gradient cable

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/ Fig. 11 Page 18) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 40 ± 4 Nm.

Fig. 12: GPA gradient cover



(1) Gradient cover
(2) Clamps

- Check whether the cable clamps (→ 2/ Fig. 12 Page 18) are tightened.
- Attach the protective cover (→ 1/ Fig. 12 Page 18).



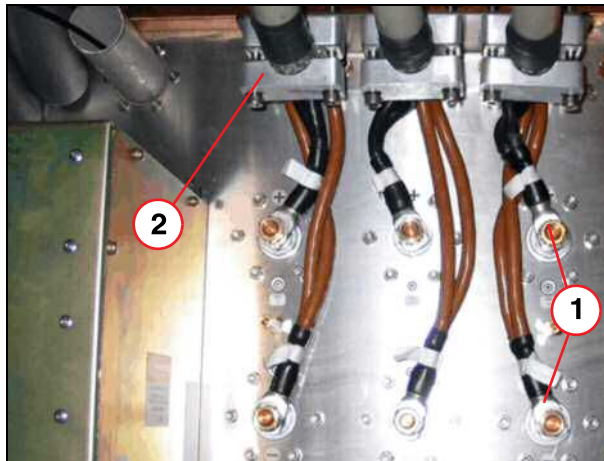
The cable clamps (→ 2/ Fig. 12 Page 18) must clamp the shielding only. Ensure that the insulation is not clamped.

2.3.2 Gradient cable inside the RF room



The cable shoes may point forward or backward. However, they cannot be turned by 180° .

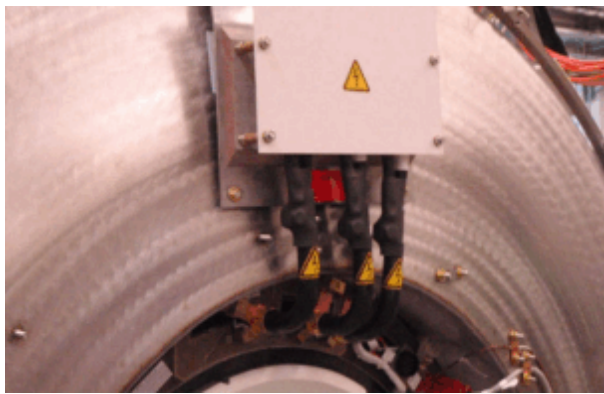
Fig. 13: Gradient filter, gradient cable UR



- (1) Gradient cable
(2) Clamps

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/Fig. 13 Page 19) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 22 ± 1 Nm.
- Check whether the cable clamps (→ 2/Fig. 13 Page 19) are tightened.
- Install the protective cover.

Fig. 14: Cover installed

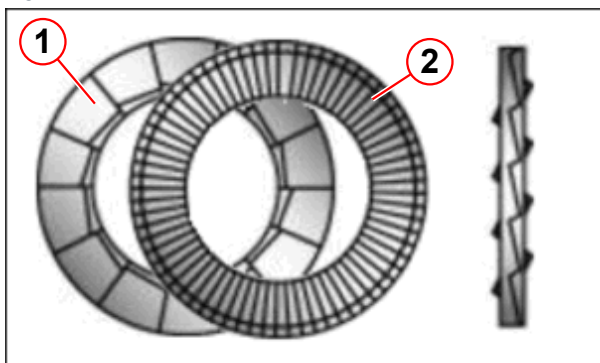


- Remove the protective cover at the connection plate.



The gradient cable cannot lie against the base frame of the cover nor secured to it.

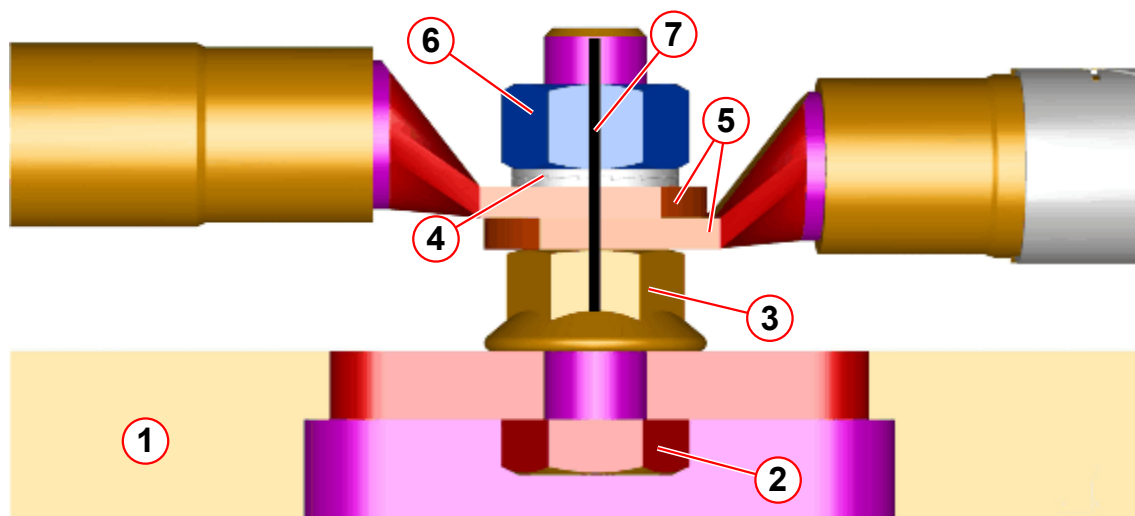
Fig. 15: NORD-LOCK washer



- (1) Cam side
(2) Ridged side

A Nordlock washer (consisting of two parts) has to be used between the cable shoe and the nut. Ensure that the washers are located correctly to one another, refer to (→ Fig. 15 Page 19).

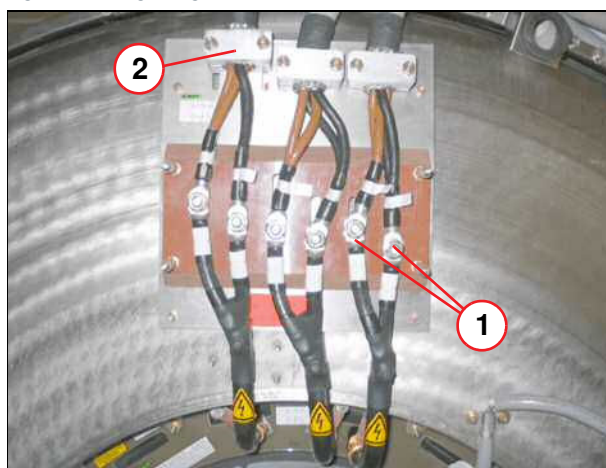
Fig. 16: Gradient connection plate at the magnet



- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19)
- (7) Mark with water proof pen

- Check the correct configuration of the connection according (→ Fig. 16 Page 20)

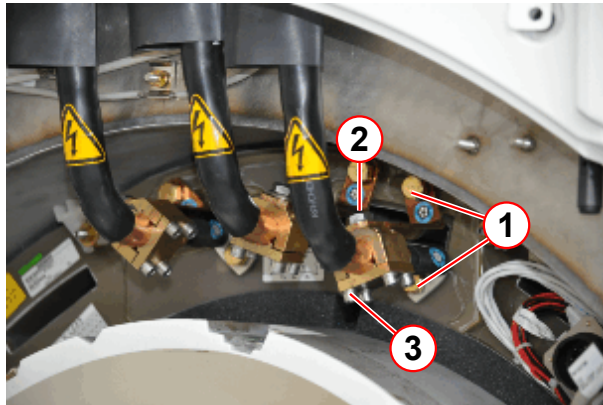
Fig. 17: Magnet gradient cable



- (1) Gradient cable
- (2) Clamp

- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/ Fig. 17 Page 20) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 40 ± 4 Nm.
- Check if the mark with the water proof pen is present (→ 7/ Fig. 16 Page 20)
- If no mark is present, mark the connection like in (→ 7/ Fig. 16 Page 20)
- Check whether the cable clamps (→ 2/ Fig. 17 Page 20) are tightened.
- Install the protective cover.

Fig. 18: Coaxial cable kit, installed



- (1) Cable connection at the gradient coil (M10 screws)
- (2) Tube connection (M10 screws)
- (3) Tube connection (M6 screws)

- Visually check the correct configuration of the gradient coil connection accordingly. (→ Fig. 18 Page 21)
- Check the torque of the cable connections on the gradient coil (→ 1/ Fig. 18 Page 21).
 - » There are 6 bolts that need to be checked.
 - » Apply a torque of 30 ± 3 Nm.
- Check the torque of the tube connection (→ 2/ Fig. 18 Page 21)
 - » There are 3 bolts that need to be checked
 - » Apply a torque of 30 ± 3 Nm.
- Check the torque of the tube connection. (→ 3/ Fig. 18 Page 21)
 - » There are 12 M6 screws that need to be checked.
 - » Apply a torque of 6.2 ± 1 Nm.



The cables or the cable lugs cannot contact the mounting material at any one location.



To avoid a short-circuit, check the cable insulation for damage. Make sure that the cables do not touch the conductive material at any one location.

2.4 Emergency shutdown buttons

For safety reasons, all emergency shutdown buttons must be tested prior to starting up the MR system.



If a UPS system is installed, the UPS system and the MR system must be shut off by pushing either one of the EPO buttons.



Refer to local codes/guidelines for the use of EPO buttons.

EPO	Function	
	o.k.	not o.k.
1		
2		
3		
4		
5		

2.5 Checking the water connection at the ACC



Check the correct water connection on top of the ACC, on top of the SEP, or at the IFP. Ensure that the connection to “WATER IN” and “WATER OUT” is correct. Incorrect water flow to the RFPA will damage the RFPA.

2.6 Switching on the system

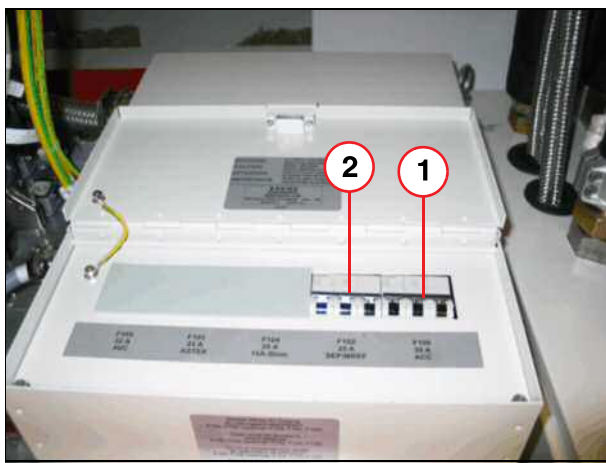
2.6.1 Circuit breakers

2.6.1.1 Circuit breakers ACC

Switch on the following circuit breakers in the ACC (top):

- F100 50A ACC
- F102 25A SEP/MREF

Fig. 19: ACC_F100_F102



(1) F100

(2) F102

Switch on the following circuit breakers in the ACC:

- F11 RFPA 1
- F29 Fan-Box / ACS-Fans
- F22 RF Room
- F24 RFIS
- F21 E-Shim (optional)
- F23 AMC/RFSU
- F1 UPS
- F2 MSUP-PS / ACS-Contr
- F30 Imagecalc
- F31 MRC
- F3 36V
- F4 24V

Fig. 20: ACC_fuses

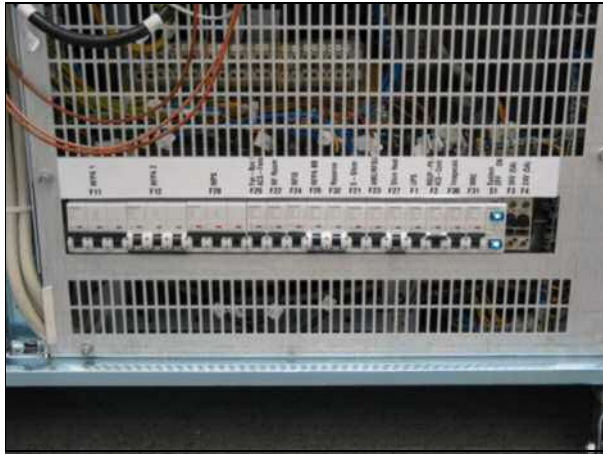
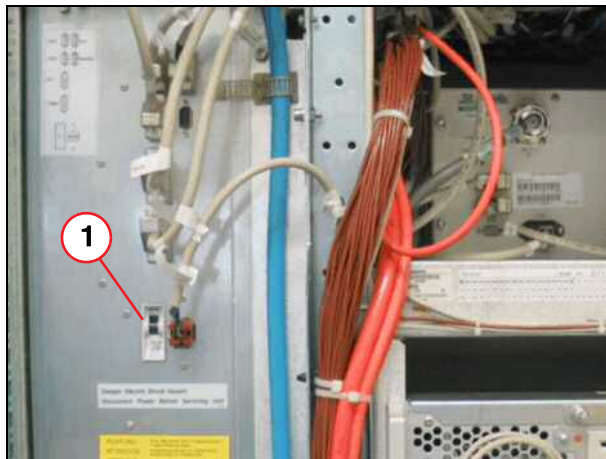


Fig. 21: Fuse F1 in the ACC



(1) F1

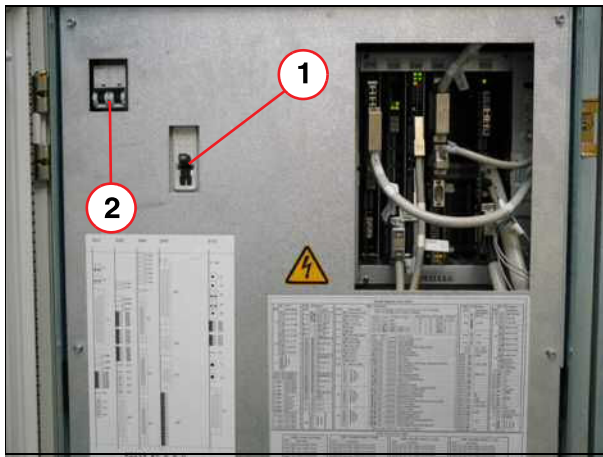
- Check whether the circuit breaker F1 (1) is switched to the "ON" position.

2.6.1.2 Circuit breakers GPA

Switch on the following circuit breakers in the GPA:

- F1
- F2

Fig. 22: GPA_fuses



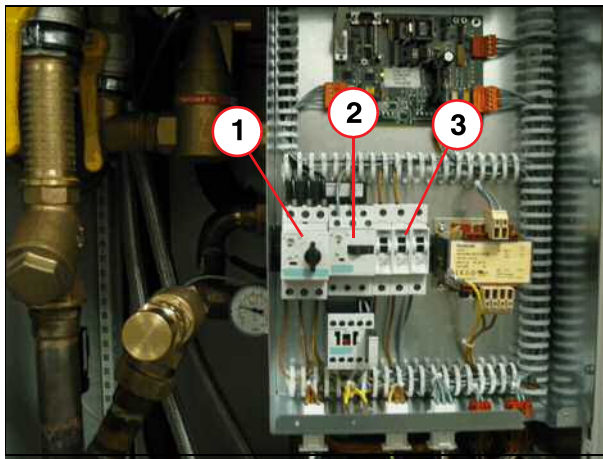
- (1) F1
- (2) F2

2.6.1.3 Circuit breakers SEP (Optional)

Switch on the following circuit breakers in the SEP:

- F1
- F3
- F4, F5, F6

Fig. 23: SEP_fuses



- (1) F1
- (2) F3
- (3) F4, F5, F6

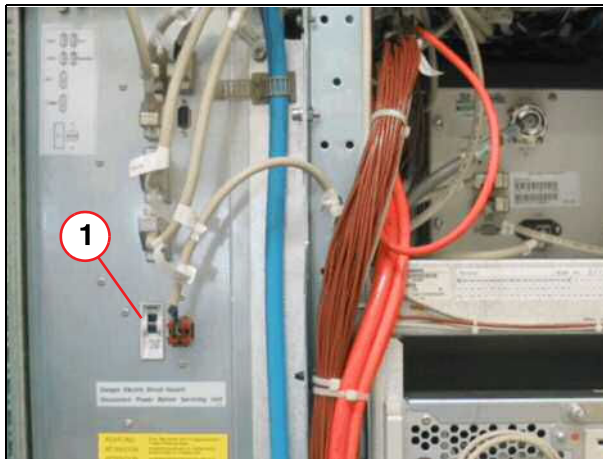
2.6.1.4 Switching on the helium compressor

i

The Avanto system can use the CSW-71 compressor or the F70 compressor
The start-up procedure for the F70 compressor is different than that for the CSW-71 compressor.
It is important to check the static helium gas pressure of the F70 compressor.

Switch-on procedure for the F70 helium compressor

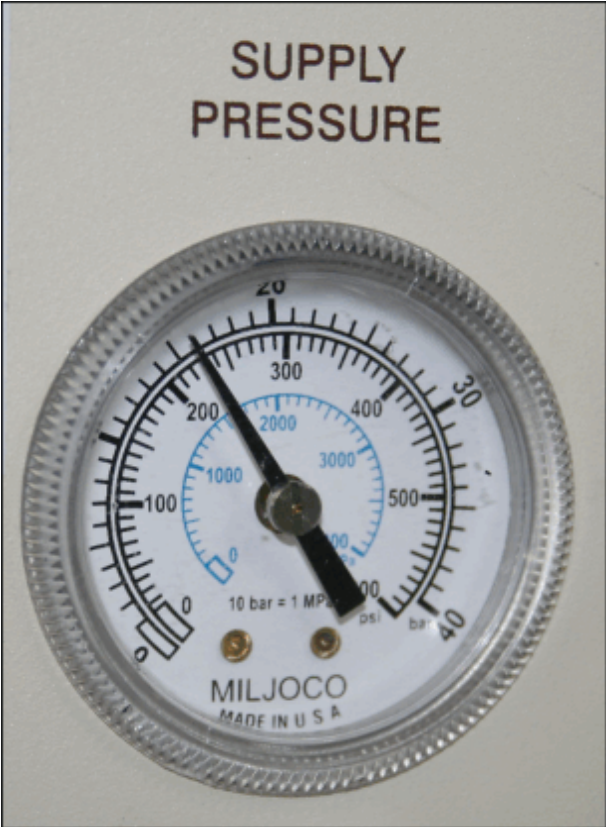
Fig. 24: Fuse F1 in the ACC



(1) F1

- Verify that **circuit breaker F1 (1)** is switched on.

Fig. 25: F70 Static pressure reading



- Check the **helium pressure** at the compressor. (→ Tab. 1 Page 28)

Equalization (static) pressure at 20⁰ C (68⁰ F) for 12 to 20m long gas lines:

Tab. 1 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

- i

If the pressure shown on the helium pressure manometer does not match the pressure shown in the table above, refill the helium compressor with helium (grade 99.999%).
- i

After the compressor has been running for 24 hours, check the static and dynamic pressures again.
If necessary, adjust the pressure.
Perform a final check prior to system handover.

- Check to ensure that the configuration selector switch is set correctly

i

Configured in the factory for configuration 1.

Before the compressor’s **main power switch** is switched on, the **configuration selector switch** has to be set to configuration 2.

The **ON** and **OFF** buttons are **disabled** in configuration mode 2, as the compressor is controlled via the pressure within the magnet.

Tab. 2 Configuration mode switch

Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10 KGM shield cooling	Enabled
Mode 2	DOWN	4 KGM recondensing	Disabled

Fig. 26: 10K APD Cold head



- Configuration mode 1**
- The switch position is **UP**.
 - 10k shield cooler system for Symphony and Trio system.

Fig. 27: 4K Sumitomo Cold head

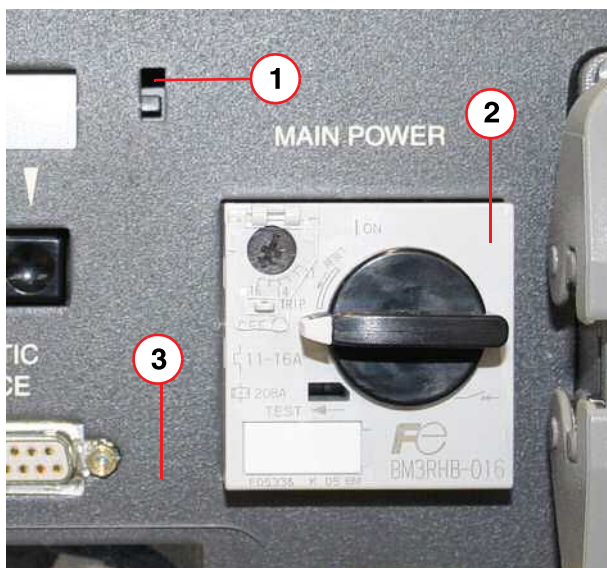
**Configuration mode 2**

- The switch position is **DOWN**.
- 4k recondensing system for Avanto and Espreo.



Do not change the switch position after the main power has been switched on.

Fig. 28: F70 Configuration switch



- (1) Configuration selector switch
- (2) Main power switch
- (3) Front panel

- Use a flat blade screwdriver to move the switch up or down.
 - Switch **up** = **configuration 1**
 - Switch **down** = **configuration 2**

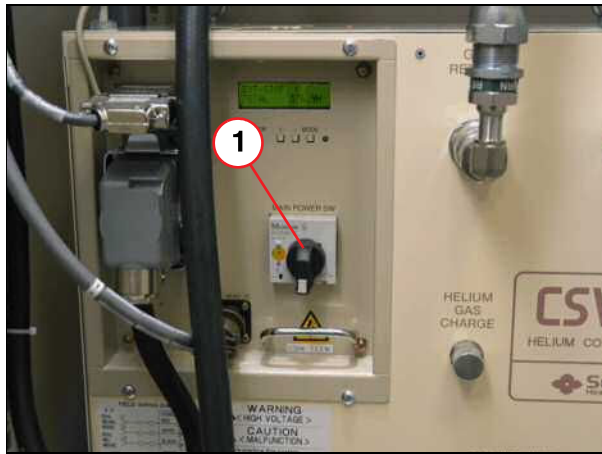
- Turn ON the mains switch to start the compressor

Switch-on procedure for the CSW-71 helium compressor

Switch on the following circuit breakers in the MREF:

- MAIN POWER SW

Fig. 29: Compressor_switch



(1) MAIN POWER SW

2.6.2 Alarm box (Mat. number 10096765)

Fig. 30: Alarm box



The alarm box indicates the system status via LEDs:

- LINE POWER
- HELIUM LEVEL
- MAG STOP
- PRESSURE
- COMPRESSOR
- BATTERY LOW
- EIS
- COMMS FAULT

The alarm box contains the following switch functions:

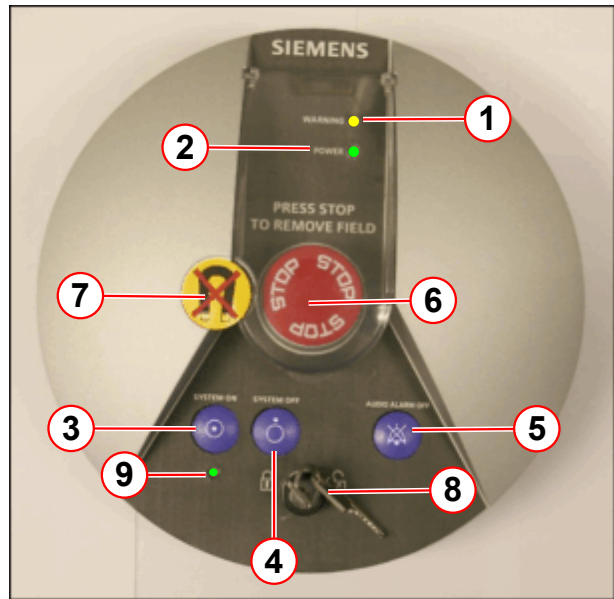
Previous status	Push button... to change status to...		
	Host Standby	System ON	System OFF
MRC OFF System OFF	MRC ON System remains OFF	MRC ON System ON	no action
MRC ON System OFF	no action	MRC remains ON System ON	MRC OFF System OFF
MRC ON System ON	MRC remains ON System OFF	no action	MRC OFF System OFF

Figure showing alarm box

2.6.3 Alarm box (Mat. number 10523563)

The alarm box indicates the system status via LEDs:

Fig. 31: Alarm box - LEDs and buttons



- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

After starting the compressor unit, the **alarm box** will indicate:-

- **LINE POWER (1):**
 - » LED lights up green:

Check the alarm box status and LEDs. If the red LED is illuminated, troubleshoot.

2.6.3.1 LEDs Status

Tab. 3 OR98/99 alarm box LEDs

LED	Description
LINE VOLTAGE	Power is supplied to the alarm box
SYSTEM WARNING	Liquid helium level below warning thresholds Quench/switch heater error or ERDU button/battery error Magnet pressure too high or low Compressor unit stopped (pressure switch activated) ERDU battery voltage < 22 V EIS drive error

2.7 Warning signs



Prior to starting up the magnet system make sure that the warning labels are posted! The warning labels are available in different languages. Pick the correct language and the correct field strength ($\rightarrow$ 1/Fig. 32 Page 34). The following pictures are examples.

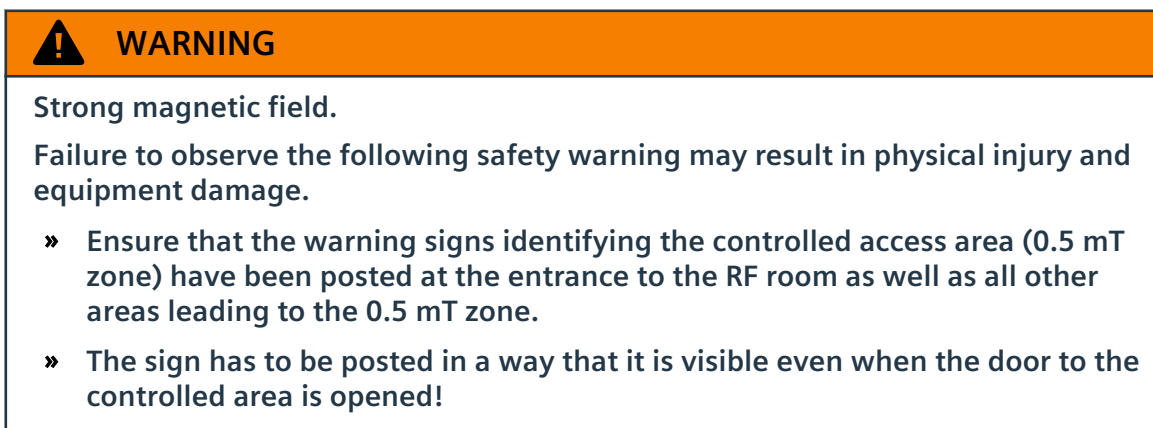


Fig. 32: Warning sign



(1) Field strength

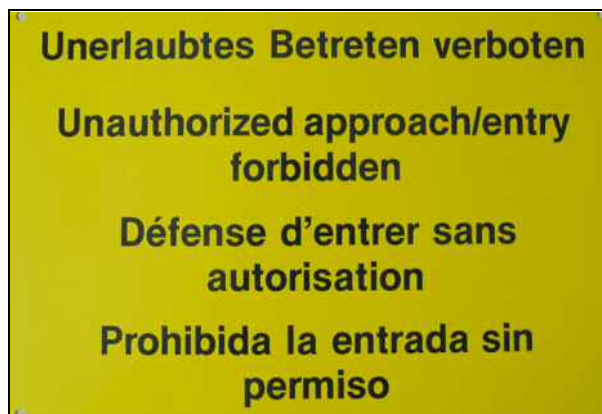
Fig. 33: Warning sign "Pacemaker"



Fig. 34: Warning sign "Filling-up"



Fig. 35: Warning sign "Unauthorized"



2.8 T-class (optional)

If the MR system includes the **T-class** and **removable table** options, the T-class accent strips must be attached to the left and right side of the patient table.

Fig. 36: T-class (example Avanto)



(1) T-class

Fig. 37: T-class (example Symphony, A Tim System)



(1) T-class



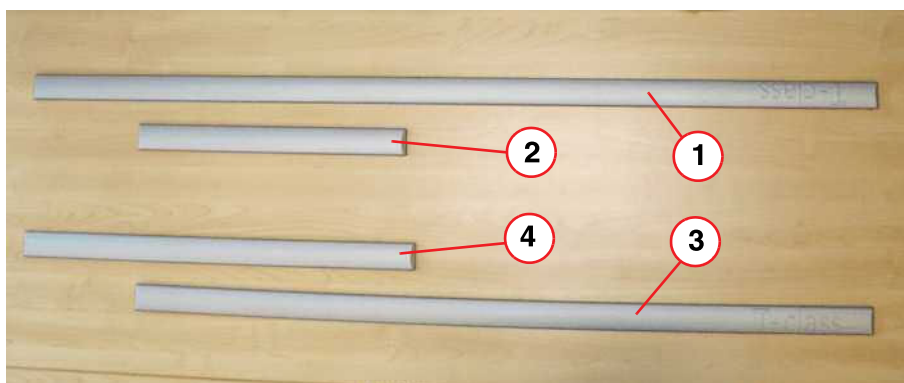
Determine the side to be used for docking the trolley together with the customer.



If the optional breast biopsy device is available, the accent strips are not used.

The T-class accent strips are stored at the system by the service provider (e.g. on the patient table).

Fig. 38: Accent strips for the removable table



- (1) Accent strip for the right side of the patient table with docking at the left (length: 1275 mm)
- (2) Accent strip for the left side of the patient table with docking at the left (length: 415 mm)
- (3) Accent strip for the left side of the patient table with docking at the right (length: 1075 mm)
- (4) Accent strip for the right side of the patient table with docking at the right (length: 613 mm)

- Remove the caps at the left and right side of the patient table (support frame) to prepare for gluing the accent strips.
- Clean the surface of the patient table to prepare for gluing the accent strips.

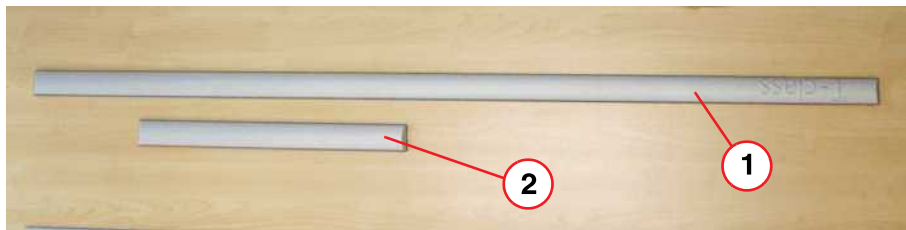
- Move the tabletop into the magnet.

2.8.1 Removable table

There are two different types of accent strips for the removable table, depending on the docking position of the trolley (left or right).

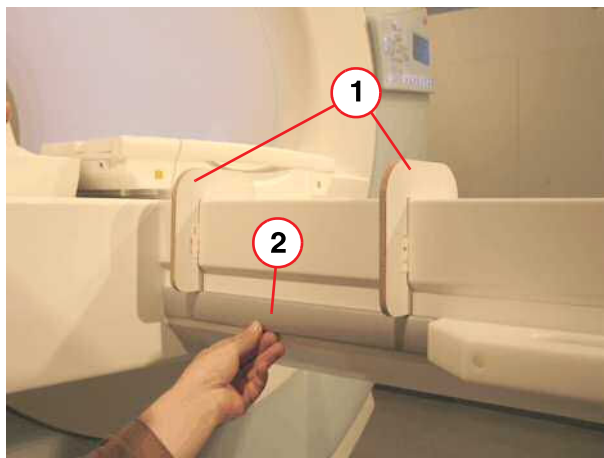
2.8.1.1 Docking position at the left side of the patient table

Fig. 39: Accent strips for the removable table with docking at the left side



- (1) Accent strip for the right side of the patient table with docking at the left side (length: 1275 mm)
- (2) Accent strip for the left side of the patient table with docking at the left (length: 415 mm)

Fig. 40: Affix the T-class accent strips (docking on left side)

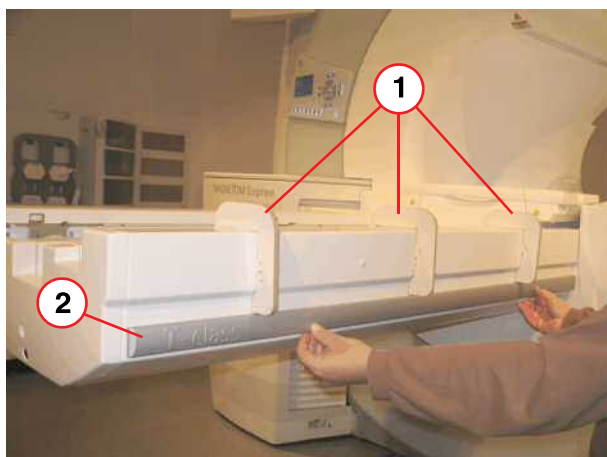


- (1) Adjustment tool
- (2) T-class accent strip (length: 415 mm)

Left side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 415 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

Fig. 41: Affix the T-class accent strips (right side)



- (1) Adjustment tool
(2) T-class accent strip (length: 1275 mm)

Right side of the patient table:

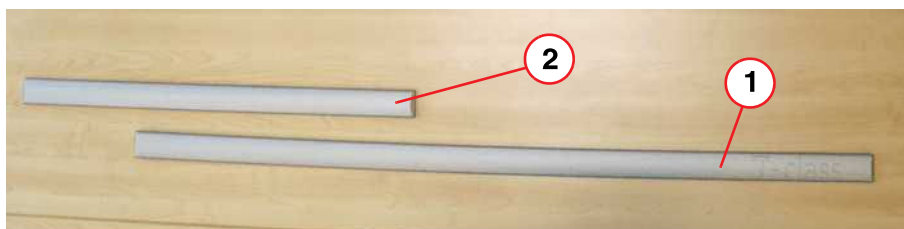
- Affix the adjustment tools (1) at the patient table (support frame).
 - Remove the protective foil from the T-class accent strips (length: 1275 mm).
 - Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
 - Remove the adjustment tools.
- If the mechanical end stop was removed, install it again and deactivate the emergency release of the patient table.



The accent strips for the docking position at the right side of the patient table are not required.

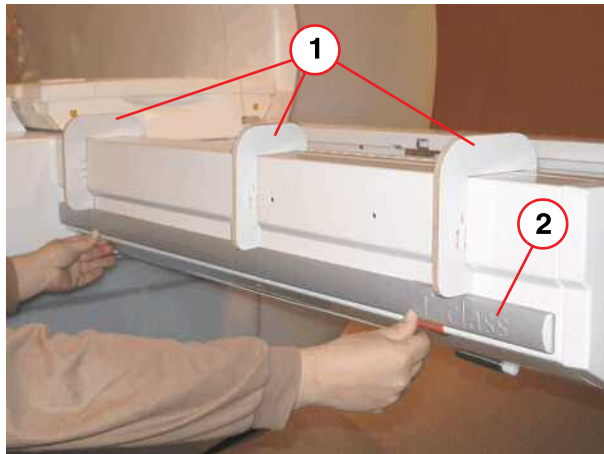
2.8.1.2 Docking position at the right side of the patient table

Fig. 42: Accent strips for the removable table with docking at the right side



- (1) Accent strip for the left side of the patient table with docking at the right (length: 1075 mm)
(2) Accent strip for the right side of the patient table with docking at the right (length: 613 mm)

Fig. 43: Affix the T-class accent strips (left side)

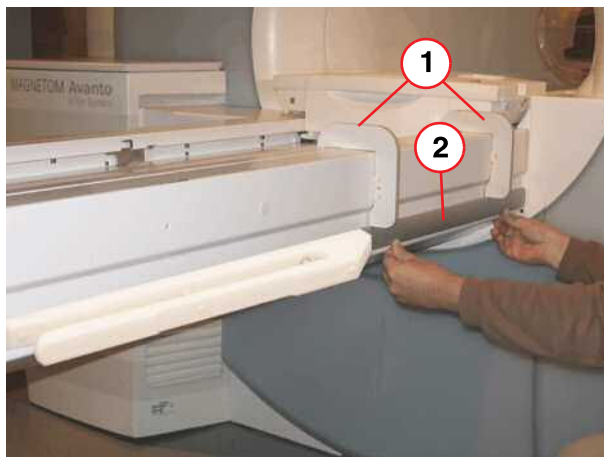


- (1) Adjustment tool
(2) T-class accent strip (length: 1075 mm)

Left side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 1075 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

Fig. 44: Affix the T-class accent strips (docking on right side)



- (1) Adjustment tool
(2) T-class accent strip (length: 613 mm)

Right side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 1275 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

- If the mechanical end stop was removed, install it again and deactivate the emergency release of the patient table.



The accent strips for the docking position at the left side of the patient table are not required.

3.1 General



Software configuration information is not published as hardcopy documentation. For more information, refer to the Online Help.

Perform the software configuration of the system (Options/Service/Local Service/Configuration). Most of the software configuration steps performed are completely automatic. Follow the instructions on-screen. The pop-up windows will prompt you to perform the required actions. Refer to the Online Help for detailed information.

- Check the MeasSettings in the Service Software/Configuration/Menu.

3.1.1 Remote Diagnostics



It is important to install and configure the Rdiag, to get support from the HSC in case you need it during installation.

For detailed information regarding rdiag configuration, refer to "Installation Software" chapter 7.

3.1.1.1 Introduction

Remote Diagnostics allows you to establish a connection from a remote location in order to provide remote service to the system, on-line technical assistance, automated reporting of system quality assurance data (QDB, EDB), magnet monitoring data, and system utilization data (customer option).

Only router connections are supported for new installations equipped with the syngo MR system.

3.1.1.2 Supported connections

- ISDN router (router provided by SIEMENS)
- Customer ISDN router
- Direct connection via Internet Security EndPoint (VPN tunneling) Authentication: SHA1 or MD5 supported encryption: DES or 3DES
- Router with analog modem interface via modem (router provided by SIEMENS)
- Customer router with analog modem interface via modem

3.1.1.3 Responsibility

The project manager is responsible for planning a remote connection in cooperation with the customer.

3.1.1.4 Support

Only IE 6.0 has been tested and released for remote service as service browser on syngo MR systems running NUMARIS MR 2004V. The "SIEMENS Remote Services TM" will only support router connections for NEWLY delivered systems. See <http://cs.med.siemens.de/ForService>, "Remote Services" for the most recent information regarding the router configuration. The project manager is responsible for planning a router connection to the Siemens Remote Diagnostic server. In case you have questions regarding your preconfigured router from Siemens, please contact srs_final@med.siemens.de or your responsible remote diagnostic technician.

3.2 TestTools

Mandatory testing of the TestTools is required at the end of the installation.

Instead of running all tests at one sitting, it is recommended to perform them individually over a period of time (e.g., during wait times, lunch break, etc.).

To get an idea of how much time is needed for the tests, please refer to the following table. Please plan to perform the tests accordingly:

Test time in hh:mm:ss	functional	communication/status	interactive	complete
AMC	00:25:50	00:01:19	00:05:00	00:31:09
RF	00:54:21	00:00:45	03:00:08	03:55:14
RCCS-int			02:30:00 ¹	
GPA	00:24:54	00:00:20		00:25:14
IMAGER	00:05:50			00:05:50 ²
HOST	00:00:05		00:10:00	00:10:05
All comm. tests		00:02:19		

1. In case the system is equipped with eight receive channels only the time required for the RCCS-int test is approx. 45 minutes.
2. In case the system is equipped with eight receive channels only the time required for the IMAGER test is approx. 4 minutes.

Normally it is not necessary to perform the cable-tests in the RF-section. Only in case of problems while trouble-shooting they should be used to isolate the defective component.

The following tests can be performed prior to Tune-Up/QA as they do not require a tuned system:

- AMC complete
- Imager complete
- Host complete
- Communication tests complete



Before handing over system, please ensure that all tests of the TestTools have been performed and are within specification.



When the patient table was removed for moving the system into the site it is highly recommended to run the RCCS-int tests prior to shimming of the magnet.



The RCCS-int test requires no interaction, only a pop-up has to be confirmed at the start of the test (not at the start of the test session!).

4.1 Setting the correct water pressure



Do not use any connection to a drinking water system if it becomes necessary to refill water.

4.1.1 System with IFP / Chiller KCC215 or KSC215

Check the water pressure supply and water pressure return via SeSo

- Select the menu item "Magnet & Cooling".
- Select "Cooling Status" in the "Action" menu.
- Read out the water pressure, supply.
- Read out the water pressure, return.
- Calculate the mathematical difference between the water pressure supply and water pressure return.
 - » The difference must range between 4.0 bar to 4.5 bar.

Fig. 45: Cooling status

Magnet & Cooling			
Event Log Reports Home Help			
Action			
Cooling Status			
Cabinet	Air Temperature Supply		26.1°C
	Air Temperature Return		29.6°C
ACS	Water Temperature GC		23.2°C
	Water Temperature Cabinet		OK
	Water Pressure Supply		5.7bar
	Water Pressure Return		1.2bar
	Water Flow GC		OK
	Water Flow Cabinet		OK
	Water Flow Total		OK
	Water Flow Mref		OK
Ready			
Update			

- » Refill if water pressure return shows < 0.5 bar



Water pressure return should be in the range of 1.0 - 1.2 bar depending on the position of the chiller.

- Adjust to the correct pressure difference with the TA valve of the chiller

Fig. 46: Flow control valve (TA valve)



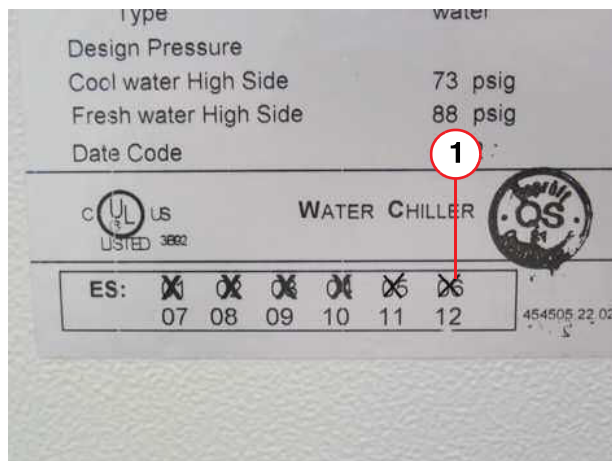
4.1.2 System with SEP

The bypass valve of the SEP has to be set before calibrating the flow sensors.

Product affected: SEP

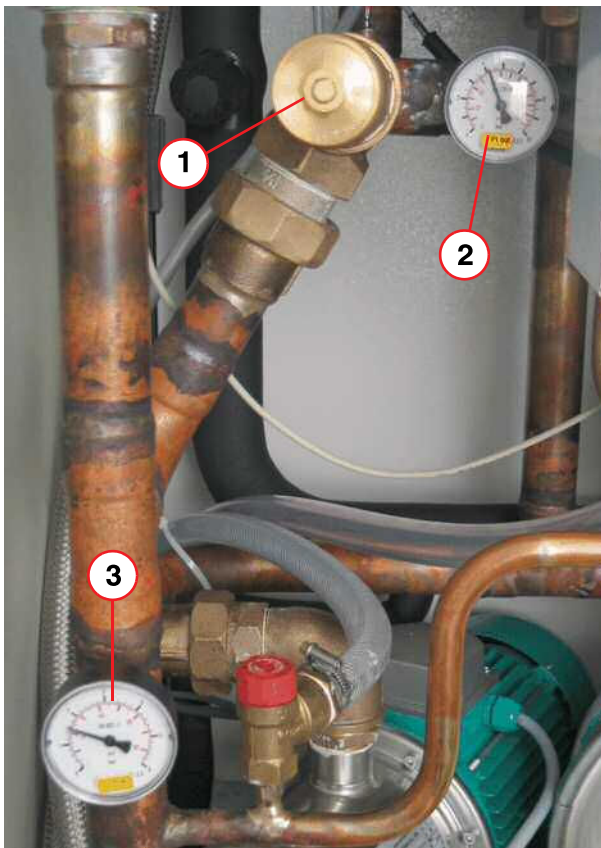
- » SEP with revision level ES 06

Fig. 47: Revision level of SEP



(1) ES 06

Fig. 48: SEP



- (1) Bypass valve
- (2) High pressure Plout
- (3) Low pressure Plin

Prerequisites:

- System is filled with water to the required pressure.
- Water pump of the SEP is switched on.

Setting the bypass valve:

Fig. 49: Bypass valve



- (1) Bypass valve
- (2) High pressure Plout

- Make sure that the pressures Pl_{out} and Pl_{in} are within specification.
- Calculate the mathematical difference between high pressure (Pl_{out}) and low pressure (Pl_{in})
 - » This difference (Pl_{out} minus Pl_{in}) must be at 50 Hz: 3.0 ± 0.3 bar
 - » This difference (Pl_{out} minus Pl_{in}) must be at 60 Hz: 3.8 ± 0.3 bar

In case you need to correct:

- Remove the cup of the bypass valve.
- Adjust the bypass valve ($\rightarrow$ 1/ Fig. 49 Page 47); make sure that the pressure difference is within specifications

4.2 Magnet & Cooling

The water flow sensors have to be calibrated using the service software ("SeSo").

Prerequisites



Make sure that the water pressure is within specification.

- Valves in the water circuits are open: at the compressor, the ACS and RF filter plate.
- Water pump of the SEP or the Chiller is on.



CAUTION

High water pressure and incorrect water flow through the RFPA.

Incorrect water flow through the RFPA will damage the RFPA.

- » Check the water pressure supply and return under Magnet & Cooling / Status.
- » If the water pressure return is higher than the water pressure supply: switch off the water pump and check water connections and water flow.
- » Check the event log for error message "Return pressure too high".

4.2.1 Avanto Dot and Verio Dot system

4.2.1.1 Flow sensor calibration and MSUP start-up (Avanto Dot, Verio Dot)



Ensure that the water flow through the GC, GPA, and RFPA is correct.
Check that the valves are open.

- Access the service menu in the SeSo at the MRC
- Select the menu item "Magnet & Cooling".

Cooling Setup

- Select "Cooling Setup" in the "Action" menu.

Fig. 50: Cooling Setup

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup**
 - Magnet Setup

Magnet & Cooling - Cooling Setup

Cabinet

Air Return Temperature Max. 38.0 °C

ACS

Water Supply Pressure Max. 6.0 bar

Water Return Pressure Min. 0.5 bar

Max. 2.8 bar

Water Supply Temperature Max. 250.0 °C

Water Flow GC Min. 20.0 l/min

Water Flow Cabinet Min. 50.0 l/min

Water Flow Total Min. 100.0 l/min

Water Flow Mref Min. 6.0 l/min

SEP

Secondary Water Set Point 20.0 °C

Secondary Water Temperature Max. 25.0 °C

Min. 15.0 °C

Flow Sensors

Sensor	Raw Data 1	Flow Value 1	Raw Data 2	Flow Value 2
GC	1106.0	100.0	1065.0	250.0
Supply	1062.0	300.0	979.0	700.0
Return	930.0	600.0	898.0	1200.0
Mref	959.0	50.0	934.0	100.0

Calibrates Sensor GC

Calibrates Sensor Supply (GPA, RFPA)

Calibrates Sensor Return (GC, GPA, RFPA, ACS)

Calibrates Sensor Mref

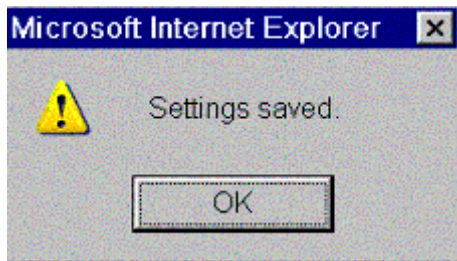
☐ Save values as default

Ready.

- Select "Sensor GC" and click "Calibrate" to calibrate the sensor of the gradient coil water circuit.

- After successful calibration, the following pop-up confirmation appears:

Fig. 51: Settings saved



- Click "OK"
- Continue with "Sensor Supply"
This is the sensor for the water supply to the GPA and RFPA.
- Continue with "Sensor Return"
This is the sensor for the total return water flow from the gradient coil, GPA, RFPA and ACS.
- Continue with "Sensor Mref"
This is the sensor for the water flow to the compressor of the refrigerator system.

4.2.1.2 Setting the helium level warning (Avanto Dot, Verio Dot)



The helium level warning has to be set to 50% if the system has a router connection to the Siemens Remote Diagnostic server with the proactive Event Monitoring.

If the system has **not** a router connection to the Siemens Remote Diagnostic server the helium level warning has to be set as described below.

- Select the menu item "Magnet & Cooling".

- Select "Magnet Setup" in the "Action" menu.

Fig. 52: Magnet Setup - Dot Upgrade

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup
 - Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.1	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.1	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Warning	Min.	55	%
Pressure Heater Resistance		20.5	Ohm
Shield Temperature Warning 50K Link	Max.	74.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	0.0	Ohm
	Nitrogen Temperature	0.0	Ohm
	Helium Temperature	0.0	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1054	
	Revision Number	0	
Magnet	Serial Number	21176154	
	Revision Number	0	
Magnet Current		466.64	A

MSUP Data

Downloads MSUP data from MSUP to host

Uploads MSUP data from host to MSUP

MSUP Logfiles

Downloads all MSUP logfiles to the host

☐ Save values as default

Ready.

- Set the helium warning 3% lower than the actual value.
 - » A helium loss of 3% will generate a warning.

4.2.1.3 Checking the parameters (Avanto Dot, Verio Dot)

- Select the menu item "Magnet & Cooling".
- Select "Magnet Setup" in the "Action" menu.
- Check and compare the magnet-specific values with the values in the SMT document.
- Check "Time Setting"; the default setting is 02:00.
- Select "Cooling Setup".

- Check the set point of the water temperature; the default setting is 20°C.

4.2.1.4 Checking the current Magnet & Cooling values (Avanto Dot, Verio Dot)

- Select "Magnet Status" in the "Action menu."
 - Perform a supervisory test.

Fig. 53: Magnet Status - Avanto Dot

Magnet & Cooling - Magnet Status

Magnet Values

Time	He / %	50K L / K	50K B / K	CHT / K	MP / psiA	PHAP / W	CP / V	Status
2012-01-03 02:00:20	93.3	43.6	49.1	42.4	15.31	0.72	1.026	OK
2012-01-02 02:00:19	93.3	43.6	49.1	42.4	15.30	0.72	1.026	OK
2012-01-01 02:00:19	93.3	43.6	49.1	42.4	15.30	0.23	1.037	OK
2011-12-31 02:00:20	93.0	43.6	49.1	42.4	15.30	0.56	1.020	OK
2011-12-30 08:21:57	93.0	43.6	48.5	42.4	15.31	0.72	1.026	OK

Starts the supervisory test

Helium Level

Helium Level 93.3 %

Starts the helium measurement

Real Time Values

Quench Heater 1	21.8 Ohm
Quench Heater 2	21.9 Ohm
Switch Heater	125.6 Ohm
EROU Button 1	OK
EROU Button 2	OK
EROU Button 3	OK
EIS	Off
Battery Voltage (21.50 V ... 25.00 V)	23.75 V
Magnet Pressure (MP)	15.31 psiA
Pressure Heater Average Power (PHAP)	0.69 W
Pressure Heater Status	Off
50K Link	43.6 K
50K Bore	49.1 K
Cold Head Temperature (CHT)	42.4 K
Compressor Pressure (CP)	1.026 V

Ready:

- Select "Cooling Status" in the "Action" menu.

Fig. 54: Cooling Status - Avanto Dot, Verio Dot

The screenshot shows the 'Magnet & Cooling - Cooling Status' page in the Siemens Med Service Software. The left sidebar contains a navigation menu with 'HOME' and 'Magnet & Cooling' sections. Under 'Magnet & Cooling', there is an 'Actions' menu with 'Cooling Status' selected. The main content area displays the following data:

Cabinet

Air Supply Temperature	26.7 °C
Air Return Temperature	27.4 °C

ACS

Water Temperature GC	20.0 °C
Water Temperature Cabinet	OK
Water Supply Pressure	3.90 bar
Water Return Pressure	0.90 bar
Water Flow GC	OK
Water Flow Cabinet	OK
Water Flow Total	OK
Water Flow Mref	OK

SEP

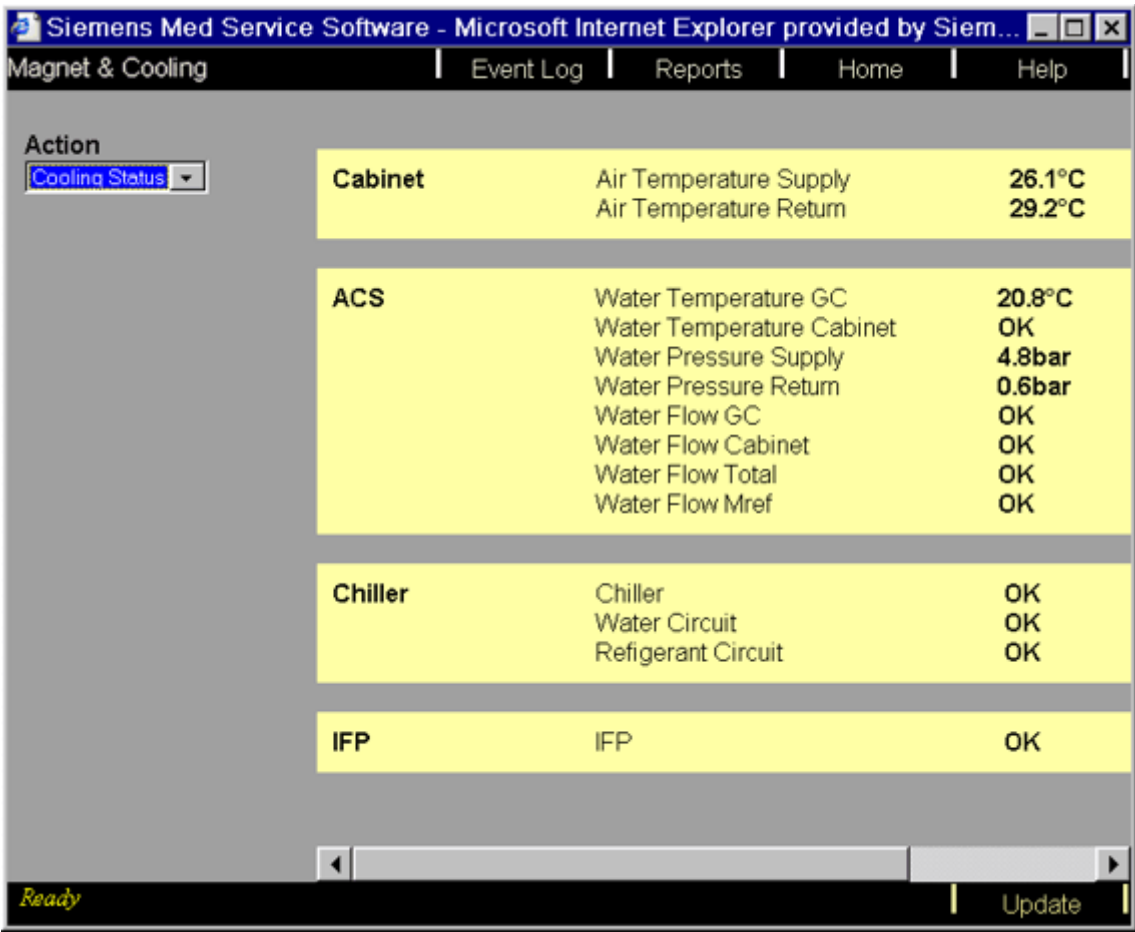
Pump	OK
Pump Overload	OK
Service Switch	OK
Primary Water Temperature	7.9 °C
Secondary Water Temperature	19.9 °C
Secondary Water Set Point	20.0 °C

Coil Flow Sensors

Sensor	Raw Data1	Flow Value 1	Raw Data 2	Flow Value 2
GC	1106.0	100.0	1065.0	250.0
Supply	1062.0	300.0	979.0	700.0
Return	930.0	600.0	898.0	1200.0
Mref	959.0	50.0	934.0	100.0

At the bottom of the page, there is a status bar with 'Ready.' on the left and an 'Update' button on the right.

Fig. 55: Cooling Status IFP



4.2.1.5 Downloading the MSUP data (Avanto Dot, Verio Dot)

Magnet Setup

- Select "Magnet Setup" in the "Action" menu.

Fig. 56: Magnet Setup - Dot Upgrade

The screenshot shows the 'Magnet & Cooling - Magnet Setup' window. The sidebar on the left has a 'Magnet & Cooling' section with a 'Magnet Setup' link highlighted. The main area contains several sections:

- Magnet**: A table of settings for Helium Level Probe 1 and 2, Helium Level Warning, Pressure Heater Resistance, Shield Temperature Warning 50K Link and Bore, CCR Calibration (Room, Nitrogen, Helium Temperature), Time Setting, MSUP (Serial and Revision Numbers), Magnet (Serial and Revision Numbers), and Magnet Current (466.64 A).
- MSUP Data**: A section with 'Download' and 'Upload' buttons. A red circle with the number '1' points to the 'Download' button.
- MSUP Logfiles**: A section with a 'Download' button. A red circle with the number '2' points to the 'Download' button.
- Save values as default**: A checkbox at the bottom.

The bottom status bar shows 'Ready.' and buttons for 'Save', 'Default', and 'Update'.

(1) Download of MSUP data

(2) Download of MSUP files

- Click "Download" (→ 1/ Fig. 56 Page 55) to save the magnet-specific values to the HOST.

4.2.2 Avanto and Espree system

4.2.2.1 Flow sensor calibration and MSUP start-up (Avanto, Espree with MSUP K2256 or K2306)



Ensure that the water flow through the GC, GPA, and RFPA is correct.
Check that the valves are open.

- Access the service menu in the SeSo at the MRC
- Select the menu item "Magnet & Cooling".
- Select "Initialization" in the "Action" menu.

Fig. 57: MSUP K2256 - Initialization



Fig. 58: Initialization Avanto, Espree IFP

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Action: Initialization

Magnet

Helium Level	Probe 1	min(0%)	85.2	Ohm	max(100%)	5.0	Ohm
	Probe 2	min(0%)	85.2	Ohm	max(100%)	5.0	Ohm
Pressure Heater	Resistance		19.2	Ohm			
Time Setting		Daily at	02:00	o'Clock			
Magnet Current							463.50 A

Calibration of Flow Sensors

	Raw Data 1	Flow Value 1	Raw Data 2	Flow Value 2
	l(10 min)		l(10 min)	
☒ Sensor GC	1039	100	1012	250
☐ Sensor Supply (GPA, RFPA)	1229	300	1184	700
☐ Sensor Return (GC, GPA, RFPA, ACS)	1051	600	997	1200
☐ Sensor Mref	1091	50	1066	100

Ready | Save Settings | Default | Upload | Download | Calibrate

Fig. 59: MSUP K2306 - Initialization

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HOME

Magnet & Cooling

- Actions
- Cooling Status
- Magnet Status
- Initialization
- Alarm Settings

Magnet & Cooling - Initialization

Magnet

Helium Level Probe 1	Min. (0%)	84.6	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☐ Probe 1 ☐ Probe 2 ☒ Both Probes	
CCR Calibration	Room Temperature	982	Ohm
	Nitrogen Temperature	1260	Ohm
	Helium Temperature	2750	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	10:20	O'Clock
Magnet Current			509.88 A

MSUP Data

Downloads the magnet setup/calibration data to the host [Download]

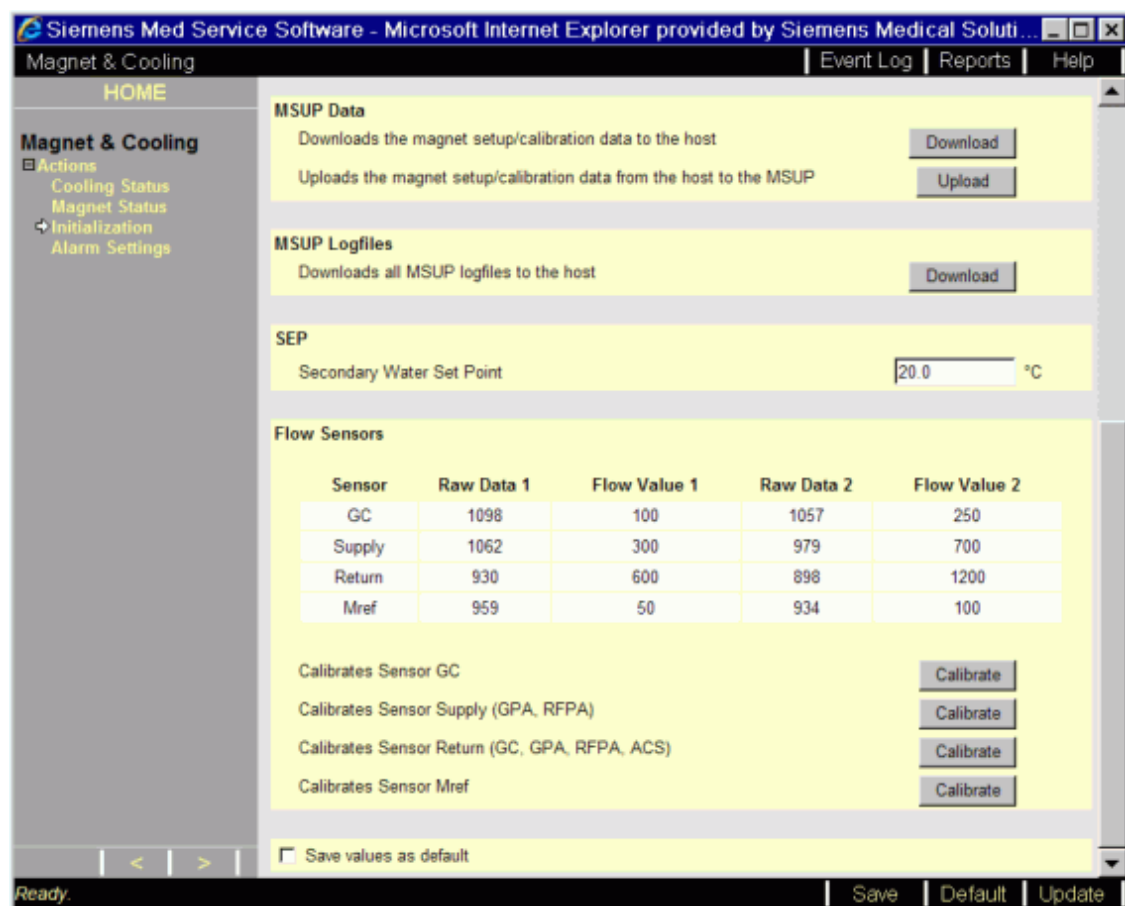
Uploads the magnet setup/calibration data from the host to the MSUP [Upload]

MSUP Logfiles

Downloads all MSUP logfiles to the host [Download]

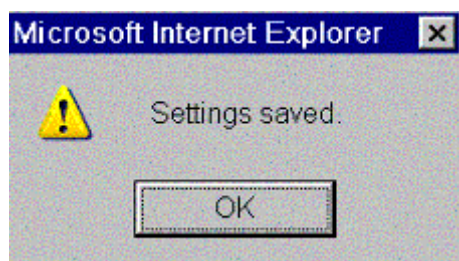
Ready | Save | Default | Update

Fig. 60: MSUP K2306 - Initialization



- Select "Sensor GC" and click "Calibrate" to calibrate the sensor of the gradient coil water circuit.
- After successful calibration, the following pop-up confirmation appears:

Fig. 61: Settings saved



- Click "OK"
- Continue with "Sensor Supply"
This is the sensor for the water supply to the GPA and RFPA.
- Continue with "Sensor Return"
This is the sensor for the total return water flow from the gradient coil, GPA, RFPA and ACS.

- Continue with "Sensor Mref"

This is the sensor for the water flow to the compressor of the refrigerator system.

4.2.2.2 Setting the helium level warning (Avanto, Espree with MSUP K2256 or K2306)



The helium level warning has to be set to 50% if the system has a router connection to the Siemens Remote Diagnostic server with the proactive Event Monitoring.

If the system has **not** a router connection to the Siemens Remote Diagnostic server the helium level warning has to be set as described below.

- Select the menu item "Magnet & Cooling".
- Select "Alarm Settings" in the "Action" menu.

Fig. 62: K2256 Magnet & Cooling - Alarm Settings

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Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Initialization
 - Alarm Settings

Magnet & Cooling - Alarm Settings

Parameter	Unit	Value
Helium Level	Min.	60 %
Shield Temperature Warning 50K Link	Max.	74 K
Shield Temperature Warning 50K Bore	Max.	83 K
Air Return Temperature	Max.	38.0 °C
Water Supply Pressure	Max.	6.0 bar
Water Return Pressure	Max.	2.8 bar
	Min.	0.2 bar
Water Supply Temperature	Max.	25.0 °C
Water Flow GC	Min.	20 l/min
Water Flow Cabinet	Min.	50 l/min
Water Flow Total	Min.	100 l/min
Water Flow Mref	Min.	6 l/min
Secondary Water Temperature	Max.	25.0 °C
	Min.	15.0 °C

☐ Save values as default

Ready. | Save | Default | Update

Fig. 63: MSUP K2306 - Alarm Settings

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Magnet & Cooling | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
- Cooling Status
- Magnet Status
- Initialization
- Alarm Settings**

Magnet & Cooling - Alarm Settings

Magnet		
Helium Level Warning	Min.	45 %
Shield Temperature Warning 50K Link	Max.	71 K
Shield Temperature Warning 50K Bore	Max.	85 K

Cabinet		
Air Return Temperature	Max.	38.0 °C

ACS		
Water Supply Pressure	Max.	6.0 bar
Water Return Pressure	Max.	2.8 bar
	Min.	0.5 bar
Water Supply Temperature	Max.	25.0 °C
Water Flow GC	Min.	20 l/min
Water Flow Cabinet	Min.	50 l/min
Water Flow Total	Min.	100 l/min
Water Flow Mref	Min.	6 l/min

SEP		
Secondary Water Temperature	Max.	25.0 °C
	Min.	15.0 °C

☐ Save values as default

Ready. | Save | Default | Update

- Set the helium warning 3% lower than the actual value.
 - » A helium loss of 3% will generate a warning.

4.2.2.3 Checking the parameters (Avanto, Espree with MSUP K2256 or K2306)

- Select the menu item "Magnet & Cooling".
- Select "Initialization" in the "Action" menu.
- Check and compare the magnet-specific values with the values in the SMT document.
- Check "Time Setting"; the default setting is 02:00.
- Check the set point of the water temperature; the default setting is 20 °C.

4.2.2.4 Checking the current Magnet & Cooling values (Avanto, Espree with MSUP K2256 or K2306)

- Select "Magnet Status" in the "Action" menu.
By selecting "Magnet Status", a helium measurement will be performed.

Fig. 64: MSUP K2256 - Magnet Status

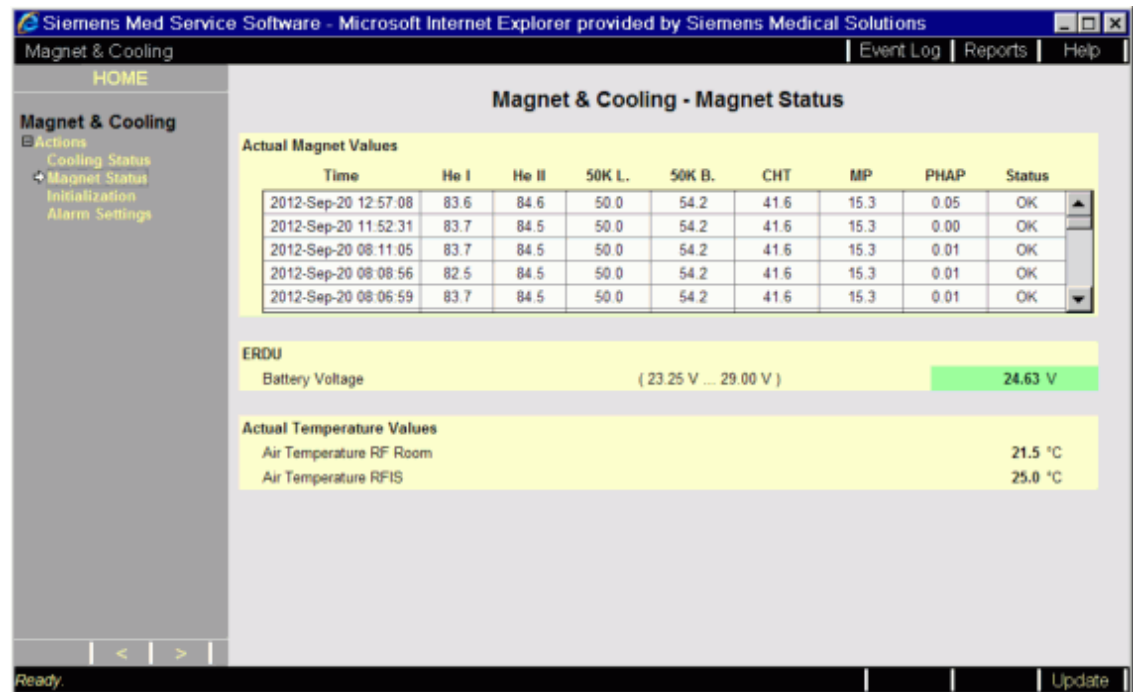
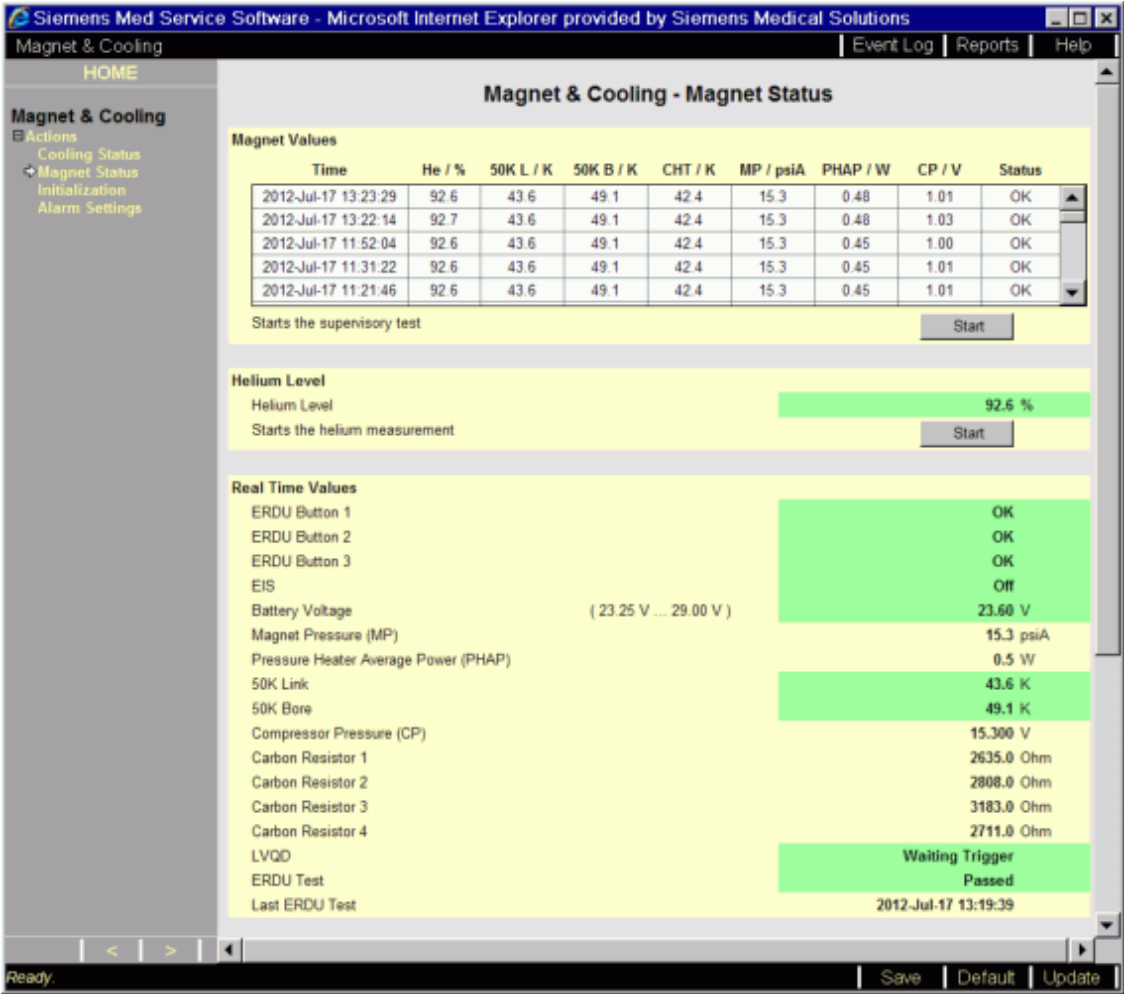


Fig. 65: MSUP K2306 - Magnet Status



- Select "Cooling Status" in the "Action" menu.

Fig. 66: Cooling Status SEP

Magnet & Cooling

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Action

Cooling Status

Cabinet	Air Temperature Supply	122.2°C
	Air Temperature Return	25.5°C

ACS	Water Temperature GC	22.6°C
	Water Temperature Cabinet	OK
	Water Pressure Supply	4.0bar
	Water Pressure Return	0.9bar
	Water Flow GC	OK
	Water Flow Cabinet	OK
	Water Flow Total	OK
	Water Flow Mref	OK

SEP	Pump	OK
	Pump Overload	OK
	Service Switch	OK
	Water Temperature (primary)	11.6 °C
	Water Temperature (secondary)	19.9 °C
	Water (secondary) set point	20.0 °C

Ready | Update

Fig. 67: Cooling Status IFP

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Magnet & Cooling

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Action

Cooling Status

Cabinet	Air Temperature Supply	26.1°C
	Air Temperature Return	29.2°C
ACS	Water Temperature GC	20.8°C
	Water Temperature Cabinet	OK
	Water Pressure Supply	4.8bar
	Water Pressure Return	0.6bar
	Water Flow GC	OK
	Water Flow Cabinet	OK
	Water Flow Total	OK
	Water Flow Mref	OK
Chiller	Chiller	OK
	Water Circuit	OK
	Refrigerant Circuit	OK
IFP	IFP	OK

Ready

Update

4.2.2.5 Downloading the MSUP data (Avanto, Espree with MSUP K2256 or K2306)

- Select "Initialization" in the "Action" menu.
- Click "Download" to save the magnet setup/calibration data to the HOST.

4.2.3 Setting the warning level (only for systems with IFP)

The warning level of "Water Pressure Return" depends on the installation position of the chiller.

Installation position of the Chiller

1. Chiller is above the MR System

» Warning level =

0.1 bar (over pressure) + 0.01 bar/m (flow resistance) + 0.1 bar/m (difference of level)

e.g. Chiller is 10 m above the MR system, length of piping is 20 m

Warning level = 0.1 bar + 0.01 bar/m * 20 m + 0.1 bar/m * 10 m = 1.3 bar

2. Chiller and MR system are on the same level, but the pipe routing is above the MR system

» Warning level =

0.1 bar (over pressure) + 0.01 bar/m (flow resistance) + 0.1 bar/m (difference of level)

e.g. pipe routing is 5 m above the MR system, length of piping is 20 m

Warning level = 0.1 bar + 0.01 bar/m * 20 m + 0.1 bar/m * 5 m = 0.8 bar

3. MR system is higher than the Chiller

» Warning level = 0.4 bar

4. MR system & Chiller & pipe routing are on the same level

» Warning level = 0.4 bar

4.2.3.1 Changing the warning level

1. Open SESO Magnet & Cooling
2. Enter new value of "Water Pressure Return"
3. Press "Save Settings" button.

4.3 Serial number set-up



Required only on Tim upgrade systems.

Serial numbers of the magnet and the MSUP have to be set-up in the MSUP.

Using a Laptop or the HyperTerminal of the MR system to do the set-up.

- Connect the HOST via the RS232 cable to the MSUP X5. RS232 cable of the MPS can be used.
- Use the HyperTerminal with the following settings:

Baud rate	9600
Data bits	8
Parity	No
Stop bit	1
Flow control	No
Terminal emulation	VT100

To enter the magnet and the MSUP serial / revision number via the HyperTerminal, the following commands should be performed:

- Type **"Run"** or **"R"** to enter the process display

Fig. 68: HyperTerminal

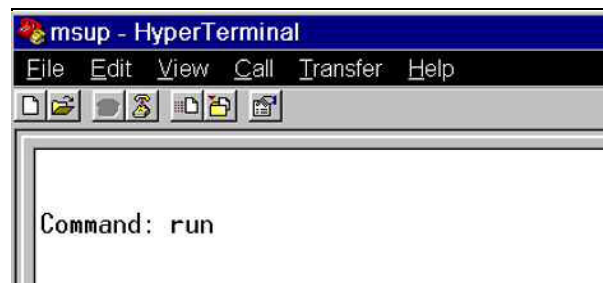
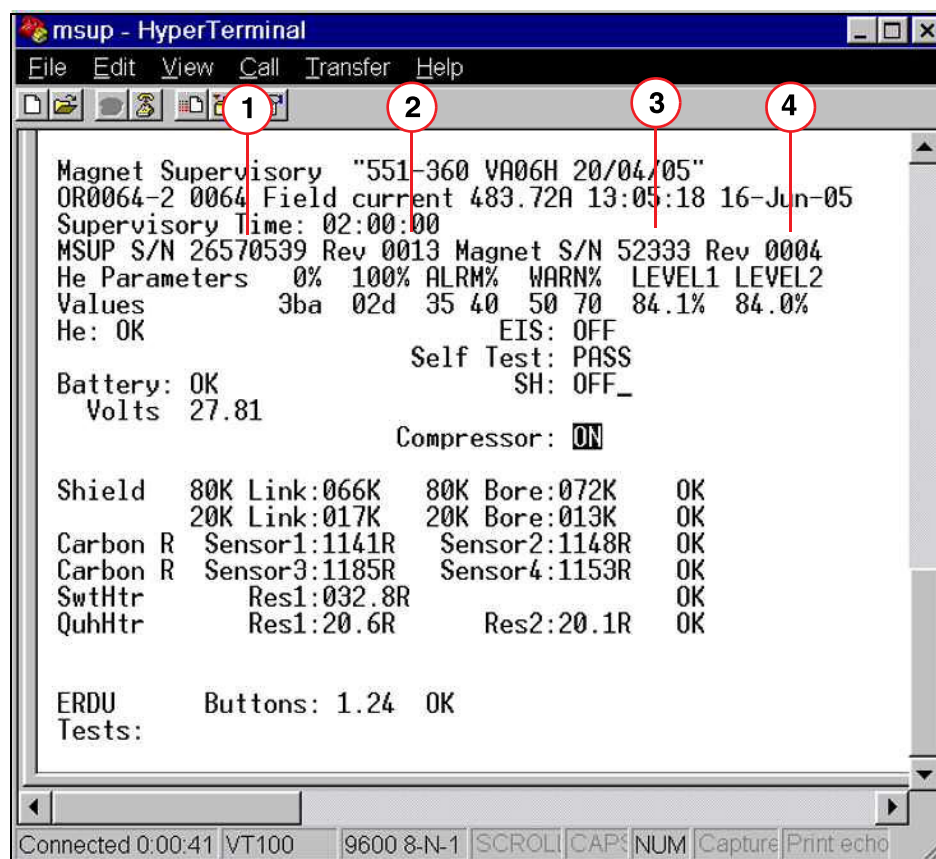


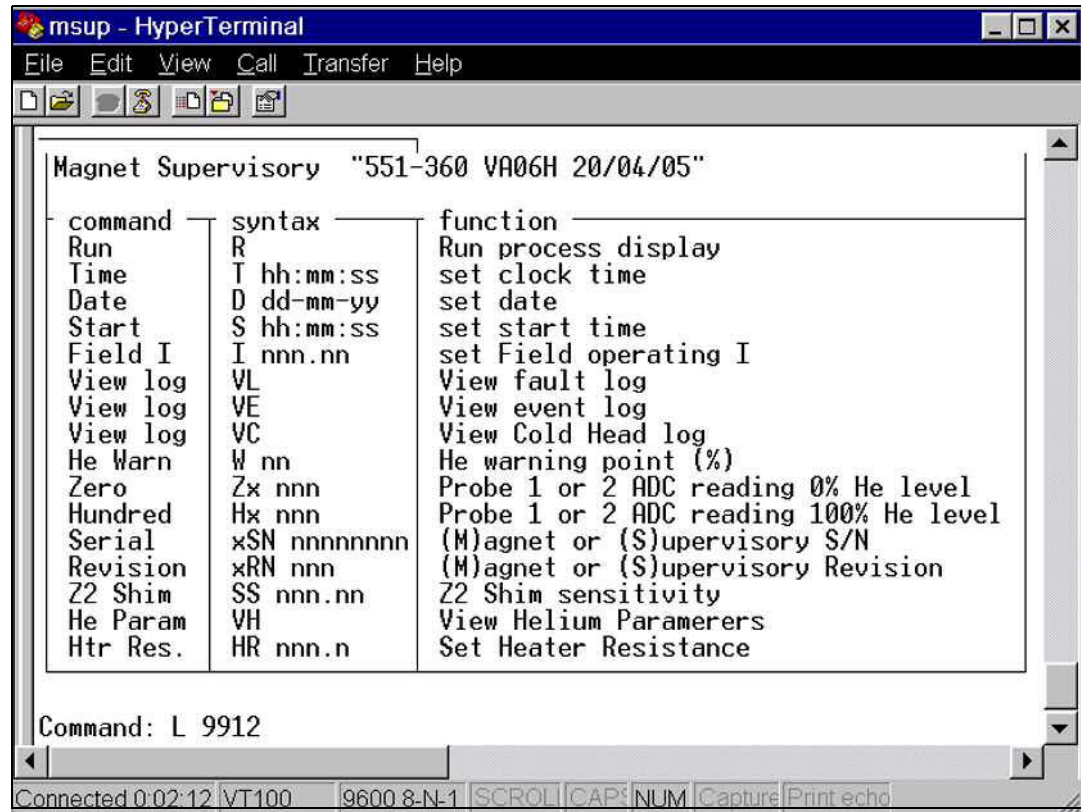
Fig. 69: Magnet Supervisory Run Process Display



- (1) MSUP Serial Number
- (2) MSUP Revision Number
- (3) Magnet Serial Number
- (4) Magnet Revision Number

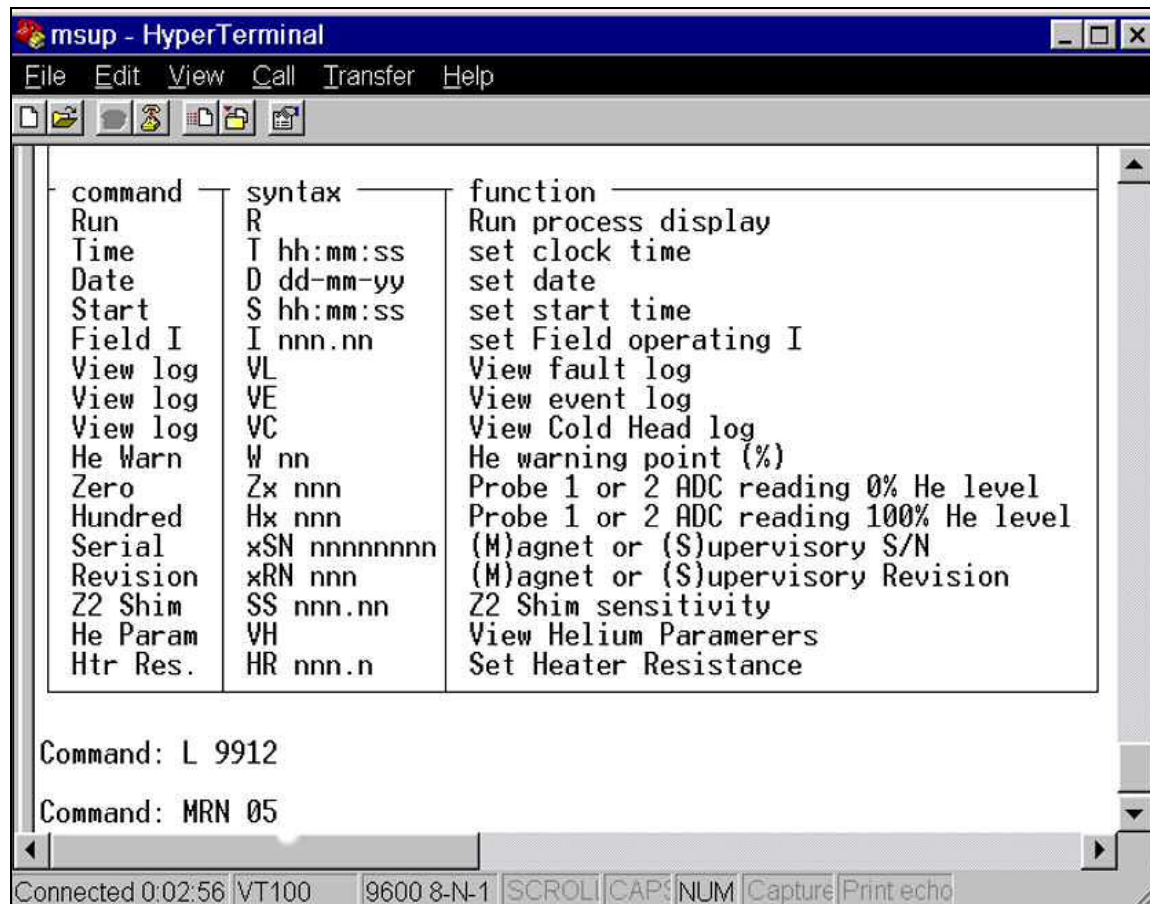
- Switch back to the command display by pressing "ESC"

Fig. 70: Magnet Supervisory Command Menu



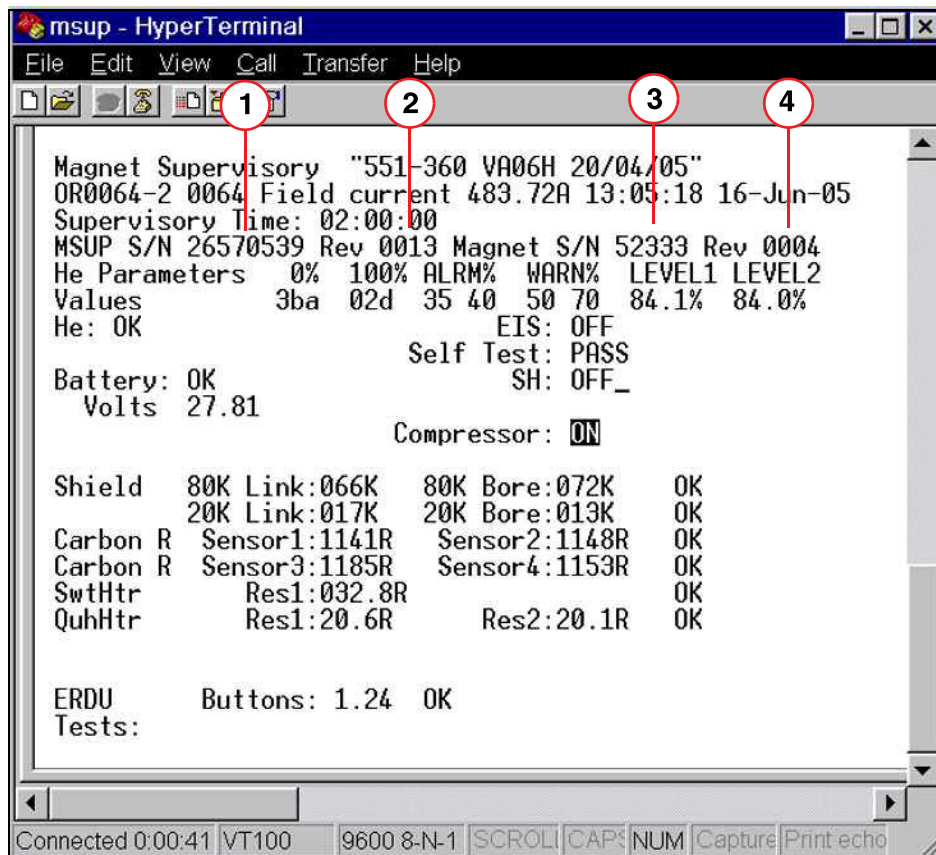
- Enter "L 9912" to unlock the access for changing the serial/revision numbers.

Fig. 71: Magnet Supervisory Command Menu



- Enter the magnet/MSUP serial/revision number
 - » Magnet Serial Number -> Type "MSN xxxxx" Return (e.g. 54321)
 - » Magnet Revision Number -> Type "MRN xx" Return (e.g. 13)
 - » MSUP Serial Number -> Type "SSN xxxxx" Return (e.g. 54321)
 - » MSUP Revision Number -> Type "SRN xx" Return (e.g.14)
- Check the serial/revision number
 - » Type "Run" or "R"

Fig. 72: Magnet Supervisory Run Process Display



- (1) MSUP Serial Number
- (2) MSUP Revision Number
- (3) Magnet Serial Number
- (4) Magnet Revision Number

- Disconnect the connection.



Filling procedure is described for MSUP K2306 and MSUP K2256.

K2306 is for Avanto Dot Ex-factory, Verio Dot Ex-factory, Aera, Skyra.

K2256 is for Avanto Dot upgrade and Verio Dot upgrade.

5.1 Avanto Dot Ex-factory - MSUP K2306 (10523587)

5.1.1 Filling the static system

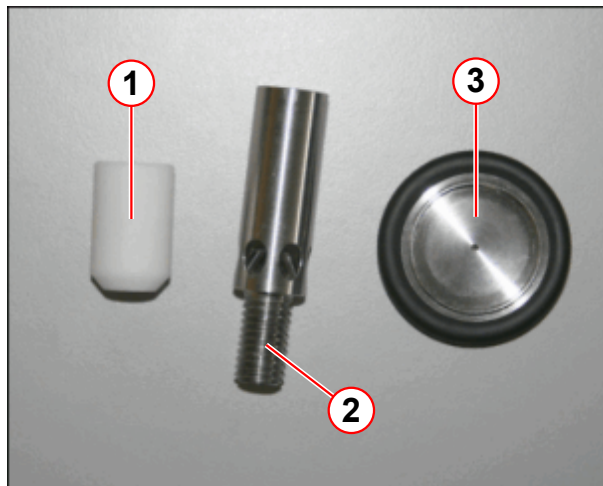
5.1.1.1 General

Static systems

The magnet design does not include a permanently fitted syphon. A short magnet service syphon must be fitted when filling with liquid helium. The service syphon is **tooling** only and must be **removed immediately** after filling has been completed.

This procedure can be performed on magnets **at field** and magnets **at zero field**.

Fig. 73: Upgrade kit 10129438 - De-ice tooling



- (1) PTFE tip
- (2) De-ice helium fill diffuser
- (3) NW25 flow restrictor

The Magnet service syphon kit is now upgraded to include **de-ice tooling**.

The de-ice tooling is used for **helium top fill**, and **de-icing of the syphon port**.

- Use for **helium top filling**:
 - PTFE tip
 - 6 hole metal de-ice helium fill diffuser
- Use for **de-icing of the syphon port**:
 - PTFE tip
 - 6 hole metal de-ice helium fill diffuser
 - NW25 flow resistor

The new 6 hole diffuser is **more efficient**, and must be used **in place of** the 60 mm PTFE diffuser.



Only Siemens trained and accredited CSEs are allowed to perform this work.



During filling, do not exceed 95%.



To recover a potentially warm magnet, check the magnet temperatures in the Bottom filling procedure (→ Bottom filling the magnet / Page 114). Contact Magnet HSC for advice.



It is recommended that fills are performed during service visits and before ramping. The minimum level for filling is 40%. Wait at least 2 hours after filling before ramping the magnet.



If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

Venting system



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

» **Always ensure the cold head bypass valve is CLOSED during helium filling.**

This document describes the filling procedure for the **OR105** magnet.

Fig. 74: Mk2 OR105 Venting System



5.1.1.2 Safety

 WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the magnet room.
- » Never transport the GHe steel cylinder into the magnet room.



Ensure that the Quench Line has been correctly installed and safe before performing Installation and Startup procedures.

 WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, helium gas may be vented to cool the syphon.
- » Ensure that the turret venting is connected to quench pipe.
- » Oxygen monitors must be activated at all times.

 CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

- » Protective clothing, gloves and eye protection must be worn.
- » Filling must be performed by trained personnel only.

 CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.

**CAUTION**

Risk of quench during filling

If the magnet quenches during helium transfer

- » Stop the helium transfer if possible and safe to do so.
- » Follow Post Quench procedures

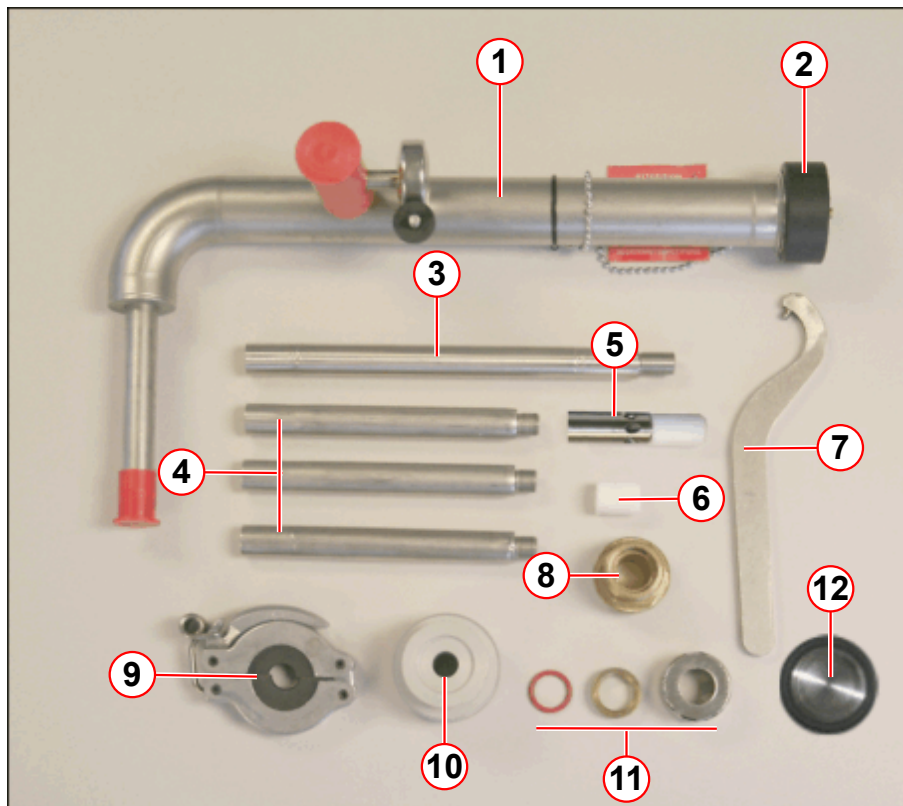
NOTICE

Mishandling of the dewar.

- » The dewar-relevant safety regulations of the manufacturer must be observed.

Equipment required

Fig. 75: Magnet service syphon kit



- (1) Service syphon assembly
- (2) Syphon end cap
- (3) 200 mm extension leg (1 off)
- (4) 140 mm extension legs (3 off)
- (5) 6 hole helium fill diffuser tip (top fill)
- (6) 22mm PTFE seal (bottom fill)
- (7) Syphon C-spanner/wrench
- (8) M28 brass syphon adaptor bush (OR105, OR122)
- (9) Syphon leg support clamp (low ceiling)
- (10) Nitrogen fill adaptor and O-ring seal
- (11) Syphon O-ring seal, brass syphon flange, syphon ring nut
- (12) NW25 flow restrictor

- Magnet service syphon kit (part no **10126651 (Revision 01)**). (→ Fig. 75 Page 75)
- Dewar syphon.
- Dewar neck adaptor, clamps, seals, and hose.
- Heat gun.
- Paper towelling.
- Helium gas supply (at least 99.9% pure).
- Pressure regulator 200 bar/ 0 - 1.5 bar.
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes.
- Replacement copper seal for syphon port on the turret.
- Service platform.

- Gas bottle wrench.
- A room or personal O₂ Monitor.

5.1.1.3 Site preparation

- Ensure warning signs are correctly positioned.
- Ensure there is a clear path to exit the imaging suite.

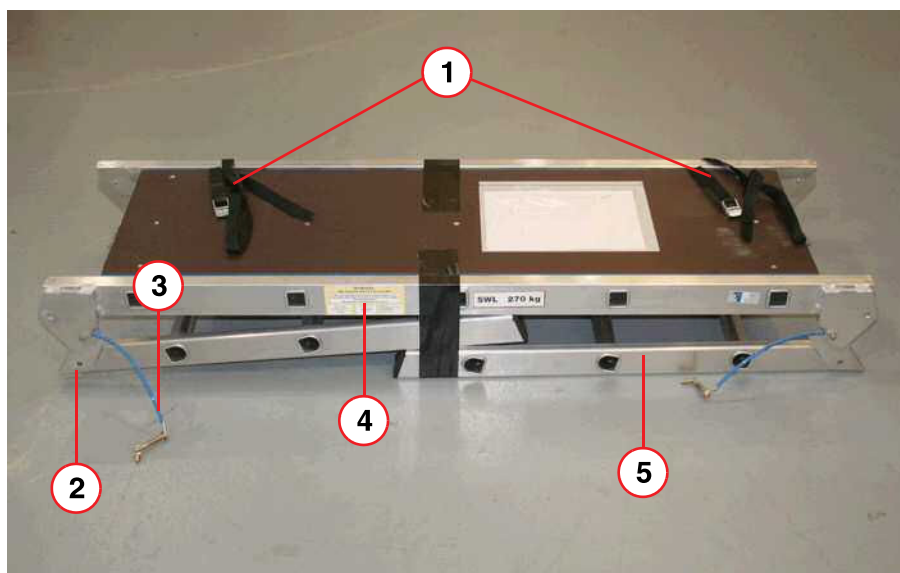
5.1.1.4 System preparation

Certain operations must be performed without delay to ensure efficient transfer:-

- Ensure that there is **open and clear access** to the magnet for **helium dewar(s)**.
- Ensure that there is **open and clear access** to the **syphon**.
- **Position** the **Service platform** next to the magnet. .
 - The platform is secured to the magnet, using **2 long straps** which are passed **through the magnet bore**.
- During installation, if the **MRC (MRAWP) Host** is not available, take a helium sample using the **MSUP**. (→ Helium level check at the MSUP / Page 95)
- **Remove** the appropriate **covers**.

Service platform installation

Fig. 76: Service platform - as delivered



- (1) Safety straps (2 off)
- (2) Hinged locking pin location (each end)
- (3) Retaining wire and hinged locking pin (each end)
- (4) Warning label
- (5) Step legs (each end)

Service platform safety

Before using the service platform read the following safety cautions and perform the required safety checks.



CAUTION

Risk of personal injury

Failure to observe the following precautions may result in personal injury.

- » Always check the platform and hinge locking pins for damage.
- » Always check the hinge locking pins are secured.
- » Always secure the platform to the magnet frame or with the two long straps through the magnet bore.
- » Always ascend and descend (Walk UP or DOWN) the steps facing the platform.
- » Where possible use the frame or covers for additional hand support.
- » Never climb onto the platform from the side.
- » Never jump from the steps or top of the platform.

Fig. 77: Correct method - Up to platform



Fig. 78: Incorrect method - Down from platform



**CAUTION****Risk of personal injury**

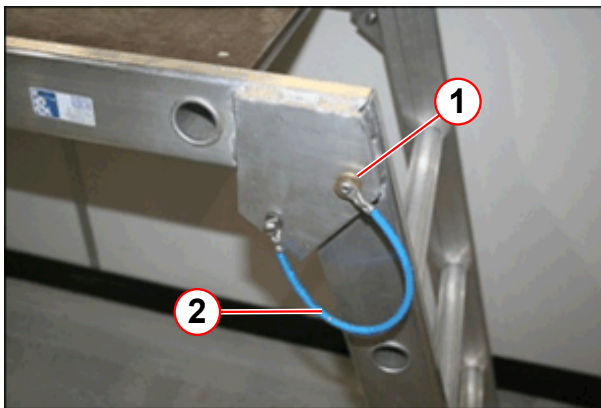
Carrying heavy components or tooling up or down the platform steps can lead to injury.

- » DO NOT carry heavy components up or down the platform steps.
- » Place heavy objects (i.e. cold head) onto the platform first.
- » Only lift the component when standing safely on the platform.

Platform assembly

- Remove any packaging.
- Unfold both step legs.

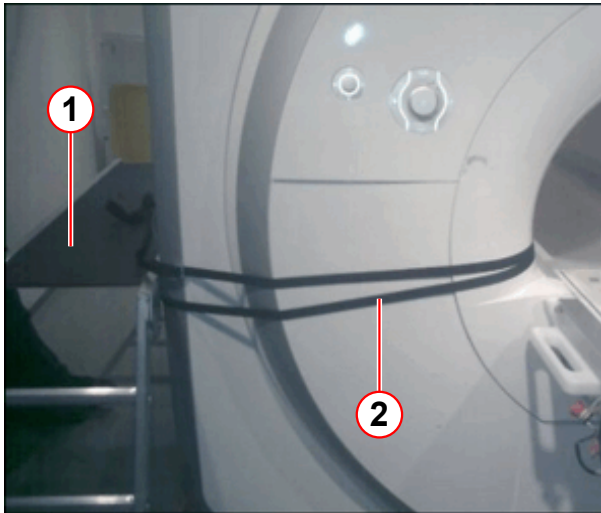
Fig. 79: Service platform - lock legs in position



- (1) Hinged locking pin
- (2) Retaining wire

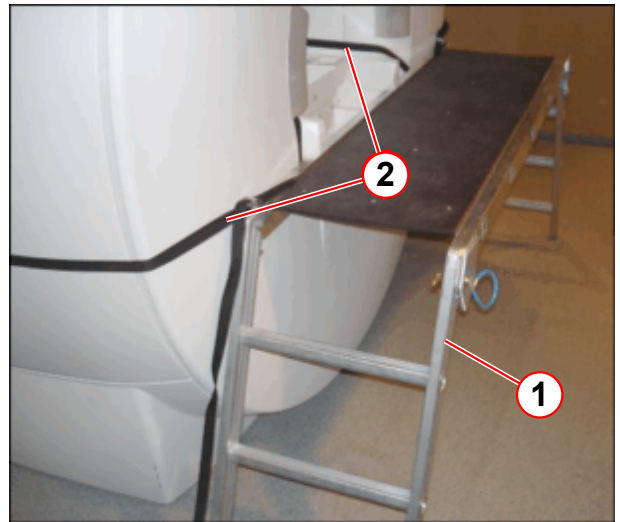
- Align the hole at the top of the leg with the **pin location hole** (pos 1).
- Push the **hinged locking pin** through the hole (pos 1) and fold back the hinge to secure.
- Repeat with the other leg.

Fig. 80: Securing the service platform (example)



- (1) Service platform
- (2) Long straps (fitted through the magnet bore)

Fig. 81: Securing the service platform (example)



- (1) Service platform
- (2) Long straps (fitted through the magnet bore)



Do not exceed the maximum load - the combined weight of 2 people must not exceed 270 kg.



The fiber optic connections in the top of RFIS are very fragile, so exercise due caution.

Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

The procedure covers all dewars that are externally pressurized with helium gas.



Refer to the dewar manufacturer's documentation when using side fill and/or self-pressurizing dewars.



Refer to the dewar manufacturer's safety guidelines and check the dewar valve functions before commencing the LHe fill.

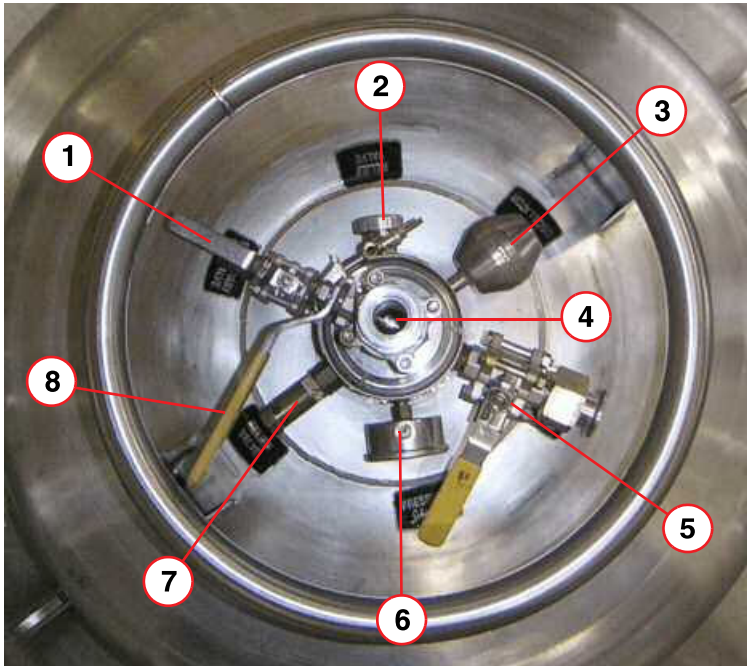
- If the dewar is **highly pressurized**, (greater than 0.3bar (4.4psi)), release the pressure by slowly opening the **primary ball valve** (→ 5/ Fig. 82 Page 80) to minimize the flash losses of helium.
 - This must be done in a **well ventilated area - NOT in the imaging suite.**



Check for the cause of the high pressure, i.e. was the low pressure relief valve closed? Inform the supplier if the cause cannot be found.

- Fit the **dewar syphon adaptor** (→ 3/Fig. 84 Page 82) to the top of the dewar.

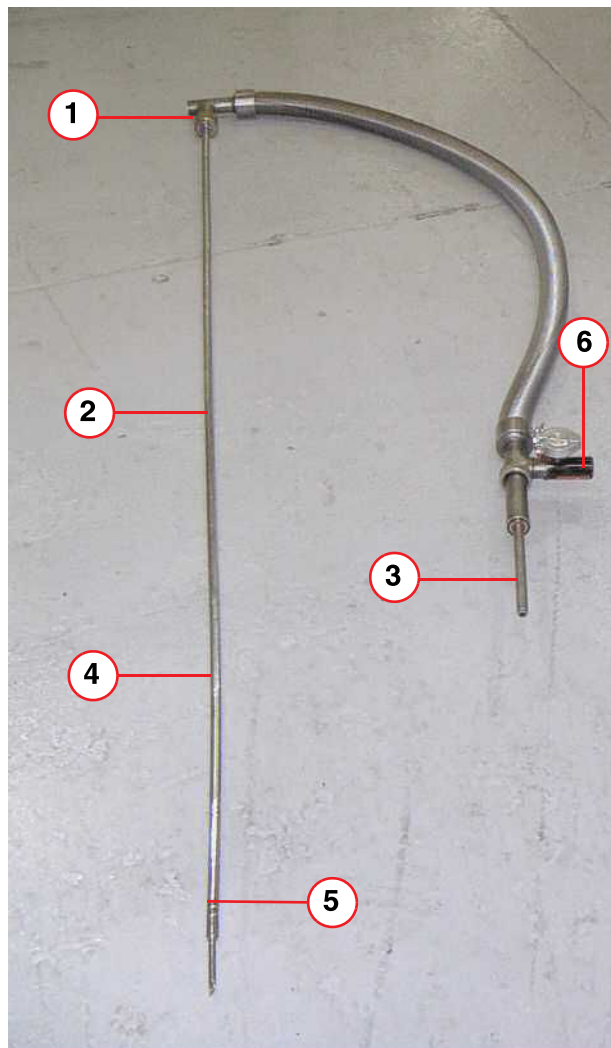
Fig. 82: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
 - (2) Relief valve
 - (3) Anti-oscillator
 - (4) Syphon entry port (shown closed)
 - (5) Primary ball valve (shown closed)
 - (6) Pressure gauge
 - (7) Relief valve
 - (8) Syphon entry valve
- **Close the primary ball valve** (→ 5/Fig. 82 Page 80) and **low pressure relief valve**. (→ 1/Fig. 82 Page 80)

Dewar syphon preparation

Fig. 83: Dewar syphon



- (1) Syphon flow valve
- (2) Syphon leg
- (3) Bayonet connection (male)
- (4) Extension leg
- (5) Sludge inhibitor
- (6) Syphon vacuum valve

- Fit the extension leg (s) and sludge inhibitor, to the dewar **syphon leg**.

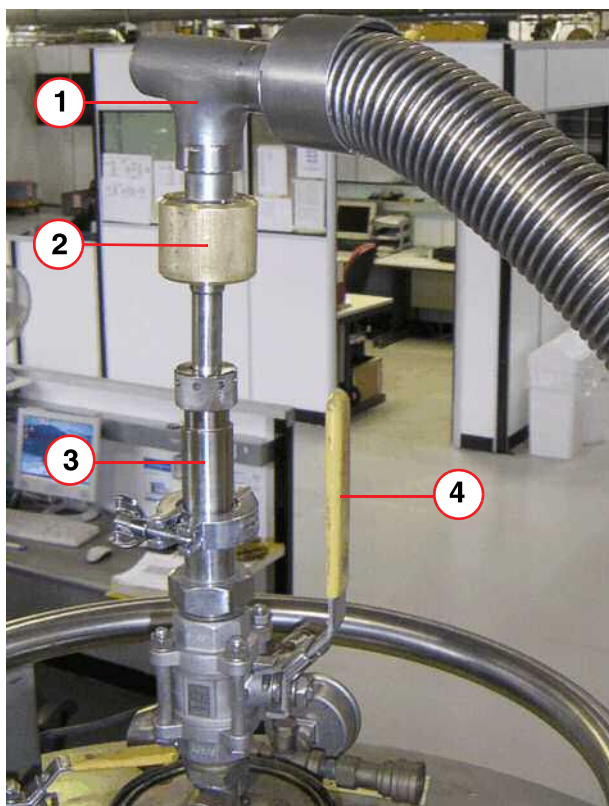


Do not open the syphon vacuum valve (6).

- Fit the **helium regulator** to the helium **gas bottle**.

Precool the dewar syphon

Fig. 84: Dewar syphon fitment



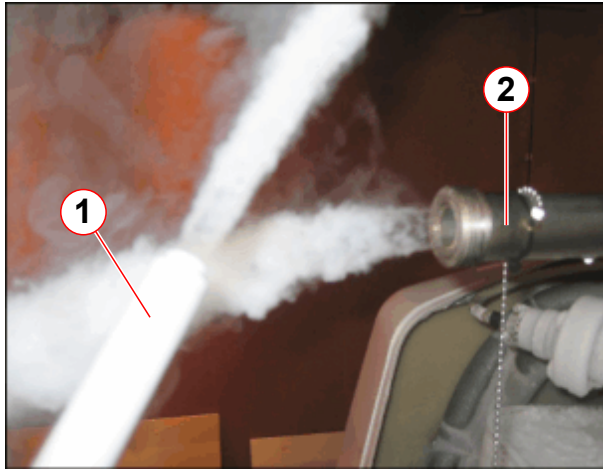
- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- Open the **syphon flow valve** (pos 2) two turns.
- Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.
(Do not fully tighten the ring nut at this stage).
- Open the **syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.

- Push the syphon **fully down** into the dewar, (very **slowly** to minimize pressure build-up and flash losses), until the **sludge inhibitor** touches the bottom.

- Fully tighten the **ring nut and seal**.

Fig. 85: Pre-cooling the syphons

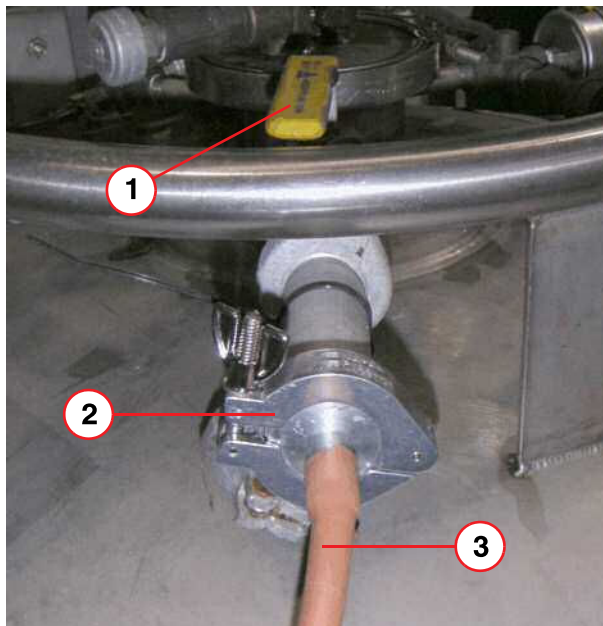


(1) Dewar syphon
(2) Service syphon

- Allow the syphon to cool until a dense white "cascading plume" appears.
- **Do not** allow high pressure to build in the dewar, only enough to cool the syphon.

- When the syphon is cold, close the **syphon flow valve** (→ 2/ Fig. 84 Page 82) and move the dewar so that the syphon can be connected without excessive bending.

Fig. 86: Dewar gas connection



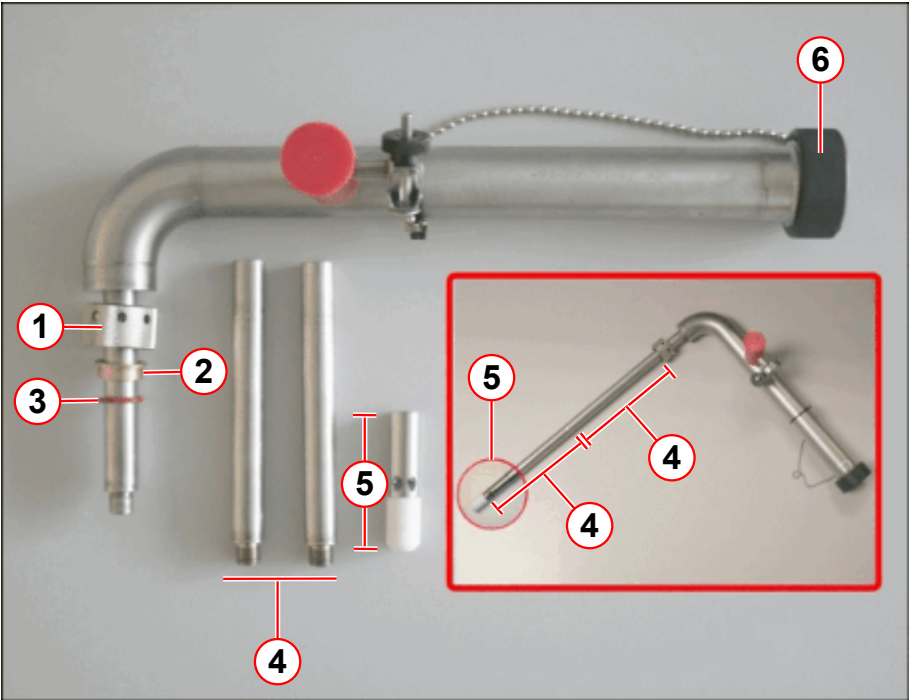
(1) Primary ball valve
(2) NW25 clamp and seal
(3) Gas supply line

- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the magnet if the magnet is at field.
- **Purge** the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an **NW25** clamp and seal (pos 2).

- **Open the primary ball valve** (→ 5/ Fig. 82 Page 80) and pressurize the dewar to **0.27bar (4 psi (G))**.

5.1.1.5 Assemble the service syphon

Fig. 87: Magnet service syphon assembly - top filling



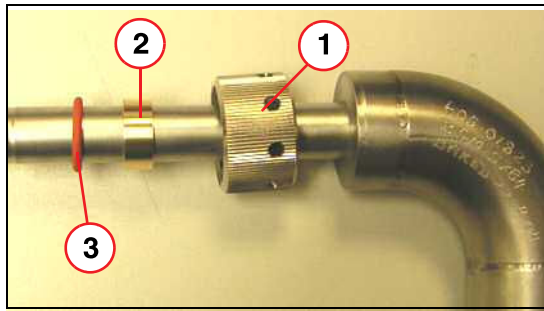
- (1) Ring nut
- (2) Brass flange
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) De-ice and helium fill diffuser and PTFE tip
- (6) Syphon end cap

Tab. 4 OR105 TOP filling syphon extension legs

Magnet type	140 mm leg	60 mm De-ice & helium fill diffuser & PTFE tip
Avanto (OR105)	2	1
The syphon diffuser should optimally be 50-80 mm above the syphon cone for top filling . If damaged, the PTFE diffuser tip (→ 5/Fig. 87 Page 84) must be replaced.		

- Check the **ceiling height**. (→ Low ceiling height / Page 106)
 - Assemble the **syphon** (only if it can be fitted), or
 - Assemble the **diffuser**, bottom **extension** and support **clamp**.

Fig. 88: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer, and the O-ring to the **service syphon leg**.

- Connect the **extension legs** and diffuser:
 - Screw the (2 off) **140 mm legs** together.
 - Connect the leg to the **syphon**.
 - Fit the 60 mm **diffuser** and PTFE tip.

5.1.1.6 Fit the magnet service syphon



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure the cold head bypass valve is CLOSED during helium filling.



CAUTION

Pressurized helium vessel.

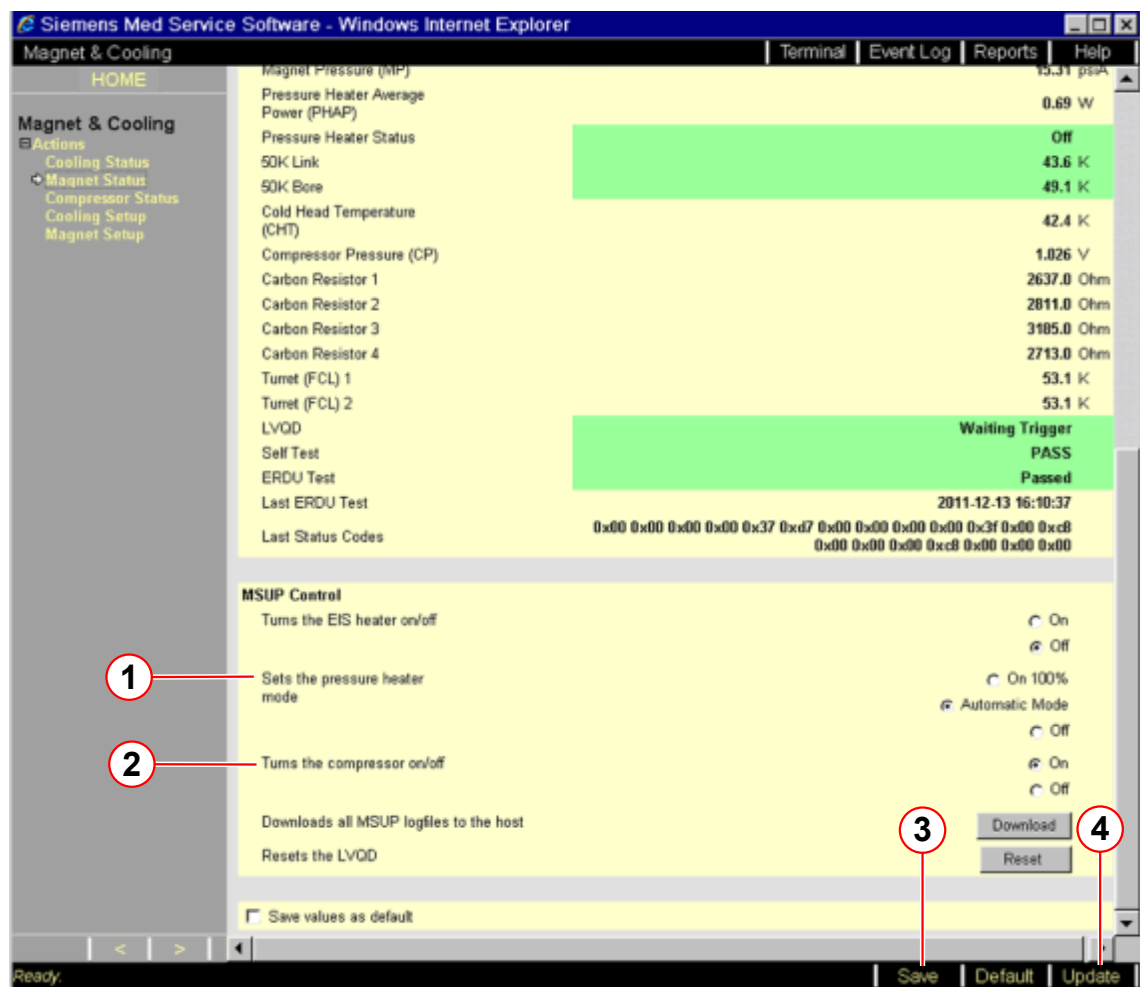
Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the fittings without first depressurizing the helium vessel.

Deactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **OFF** button (→ 1/Fig. 89 Page 86).
- Select **SAVE** (→ 3/Fig. 89 Page 86).

Fig. 89: Service screen Mag&Cooling - Magnet Status - MSUP Control



- (1) Sets the pressure heater mode
- (2) Sets the compressor on/off
- (3) Save
- (4) Update



The pressure heater will reactivate within 60 min.

Depressurize the helium vessel



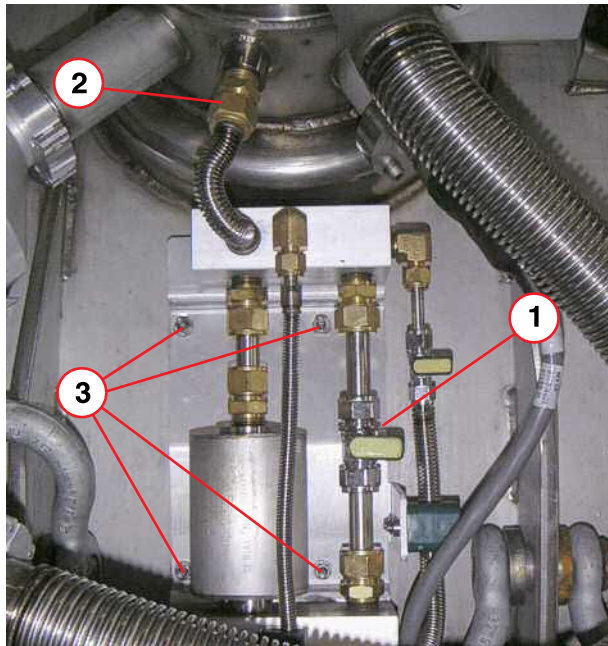
CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

- » To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.

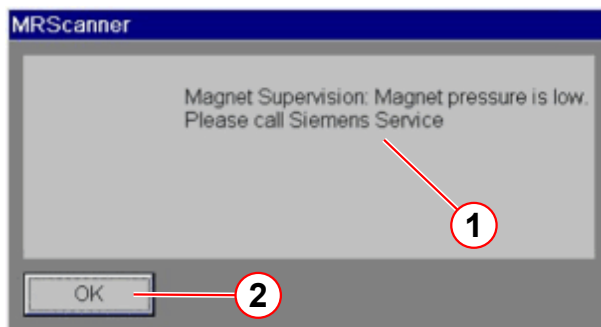
Fig. 90: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

- **Fully open the magnet bypass valve to depressurize the helium vessel.**
- Wait until **zero psi** is displayed on the pressure gauge.
- The **pressure switch** will activate.
 - » the **cold head** will turn **off**.
- **Close** the magnet bypass valve to prevent **air ingress**.
- **Warm up** the vent valve assembly.
 - » **Take care** not to damage any components when using a **heat gun**.

Fig. 91: MSUP - Magnet pressure is low

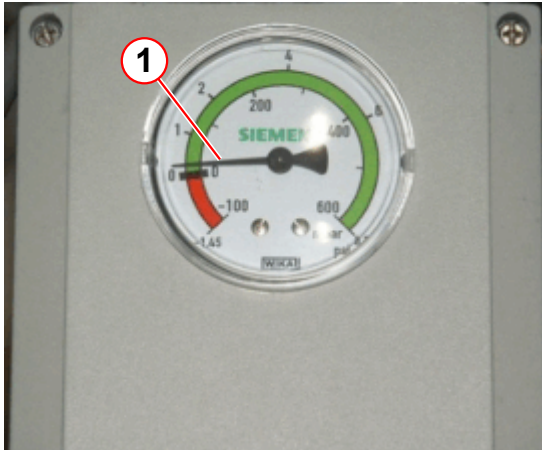


- (1) Pop-up message
- (2) Click OK to clear

On the **MR Scanner**:

- A pop-up **alarm message** is displayed.
 - » **Magnet pressure is low.**
 - » Click **OK** to clear the alarm.
 - The **alarm box** will sound when **pressure falls to alarm limits**.
 - » The yellow **WARNING LED** lights on the **alarm box**.
 - » The alarm **buzzer** is **activated**
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- Check the **pressure** is positive when the **compressor** switches **OFF**.

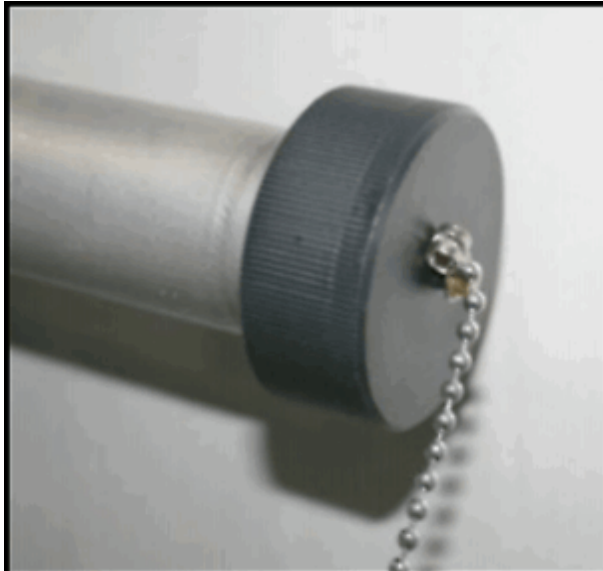
Fig. 92: WIKKA Pressure switch - with a small positive pressure



- Check the helium vessel has a **small positive pressure**.

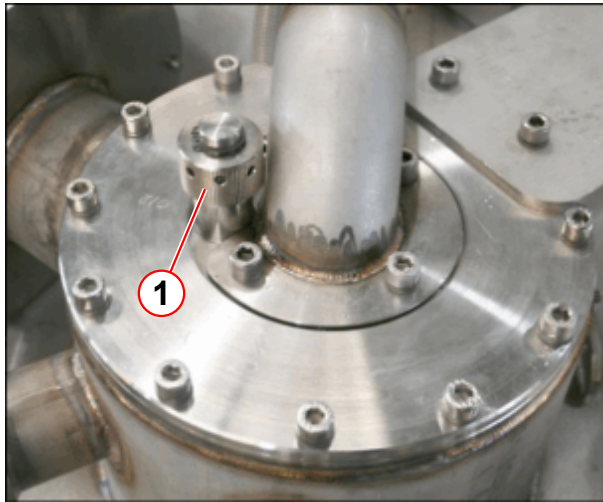
- Warm and dry the **syphon blanking plug** using a heat gun to ease removal.

Fig. 93: Syphon end cap



- Remove the syphon **end cap**.

Fig. 94: Fit/remove the syphon blanking plug



- **Remove** the **syphon blanking plug** using the syphon 'C' spanner.



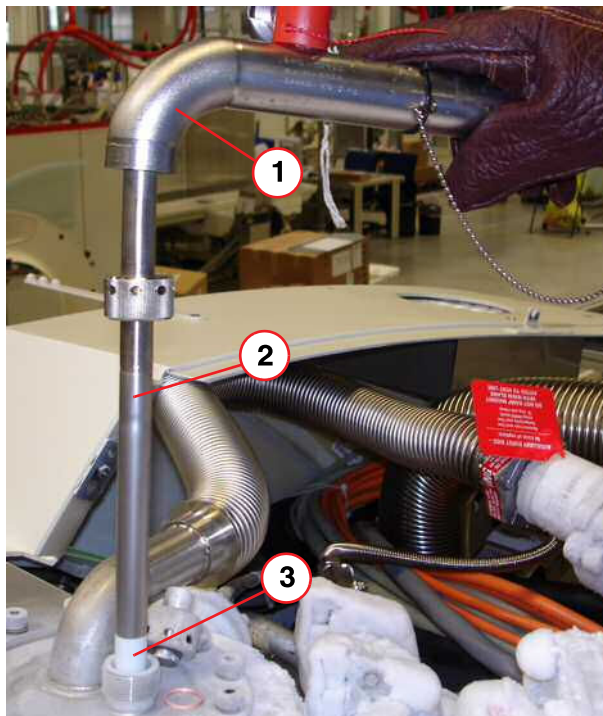
CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

Fig. 95: Helium syphon



- **Insert** the service syphon into the port without delay.
- Slowly lower the service syphon fully into the magnet.
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon C-spanner to form a **gas tight seal**.

- (1) Service Syphon
- (2) Two Extension Legs
- (3) PTFE Syphon Diffuser



There is a risk of quenching the magnet if the syphon is inserted too quickly.

- **Re-fit** the syphon **end cap**. (→ Fig. 93 Page 88)



If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

5.1.1.7 Precool the service syphon



The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use. (→ Dewar preparation / Page 79)).



If the temperature of the magnet is above 4.2K, a dense cascading plume will not form on cooling the service syphon.



CAUTION

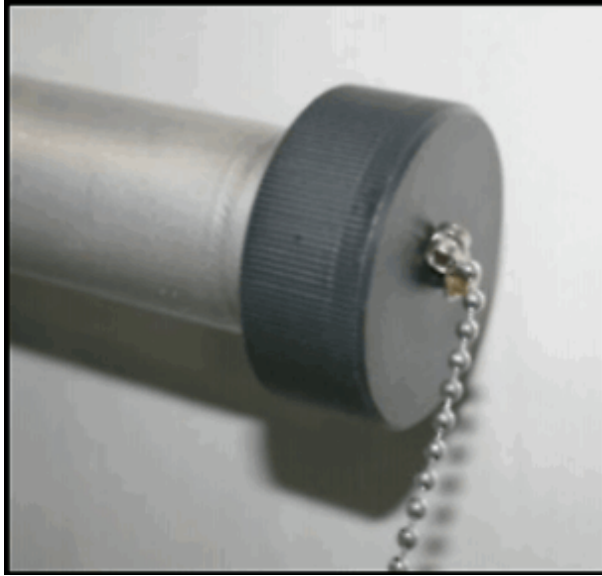
Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Cool down the **service syphon** as follows:-

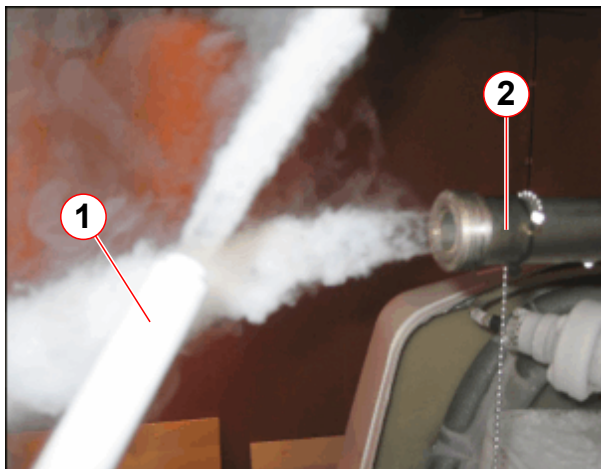
Fig. 96: Syphon end cap



- Ensure the **syphon end cap** is fitted and the **bypass valve** is closed.

- **At the MRC, reactivate** the **pressure heater** to achieve a positive pressure of about **0.3 psi (G)** in the magnet. (→ Reactivate the pressure heater / Page 98)
- **Deactivate** the pressure heater (→ Deactivate the pressure heater / Page 85).

Fig. 97: Pre-cooling the syphons



- (1) Dewar syphon
(2) Service syphon

- **Remove** the **syphon end cap**.
- Allow the syphon to vent until **fully cold**.
 - a "cascading plume" of helium gas appears.



CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

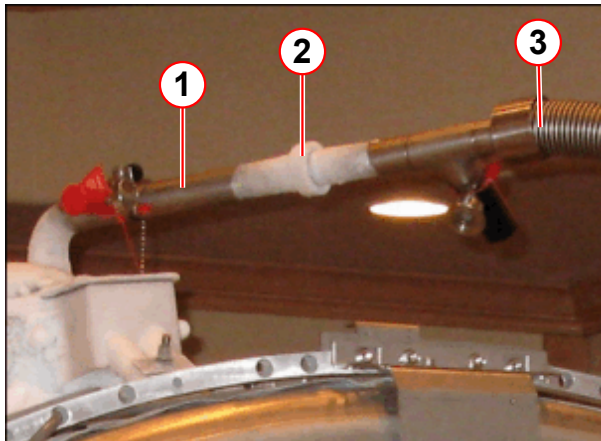
5.1.1.8 Liquid helium transfer - top fill



Before helium transfer, check the (→ Helium pressures at altitude / Page 107)

- Check the **dewar** gas supply pressure is set at **0.27bar (4psi (G))**.
- As the **service syphon** nears the point of being fully cold:
 - open the **dewar syphon flow valve** half way (2 turns). (→ Fig. 99 Page 93)
 - **precool the syphon** again until a dense white "cascading plume" appears.

Fig. 98: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, connect them together and secure the lock nut.
- **OPEN** the magnet bypass valve. This will:
 - » Improve efficiency.
 - » Improve the filling speed.



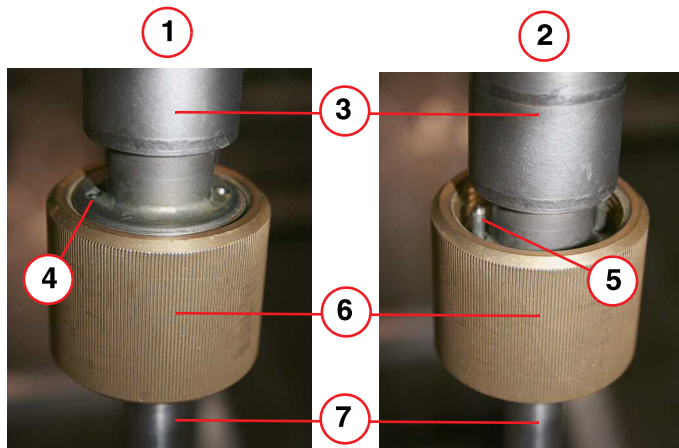
CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Fig. 99: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

Fig. 100: Helium filling



- Fully opening the **dewar syphon flow valve**.
 - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body (→ 4/Fig. 99 Page 93).
 - » From fully closed to fully open, is **4 turns**.
- Ensure the transfer dewar pressure is **4 psi (A)** at normal altitude. (→ Helium pressures at altitude / Page 107).

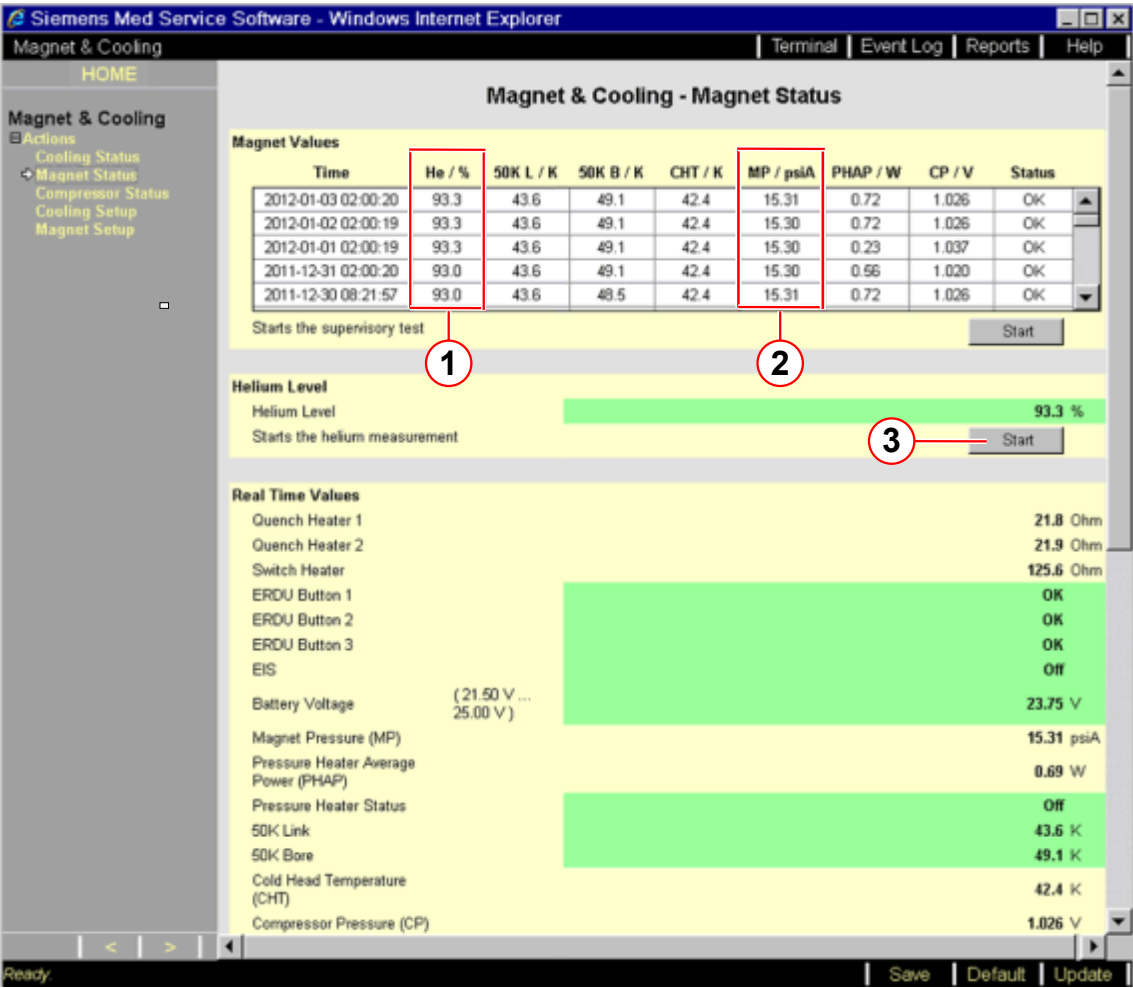
5.1.1.9 Perform a helium level check

Helium level check at the MRC

- Check the **liquid helium level** by using **Service Software** on **MRC (MRAWP) Host** or visiting the **MR Quality Home Page** if the system has SRS enabled.
 - Check the **liquid helium level** using the MRC.
 - Access the **Service Software** at the MRC.
 - Select the menu item **Magnet & Cooling - Magnet Status**.

- Press the **Start** button (→ 3/ Fig. 101 Page 94) to initiate the **helium measurement**. (→ 1/ Fig. 101 Page 94)

Fig. 101: Checking liquid helium and magnet pressure levels

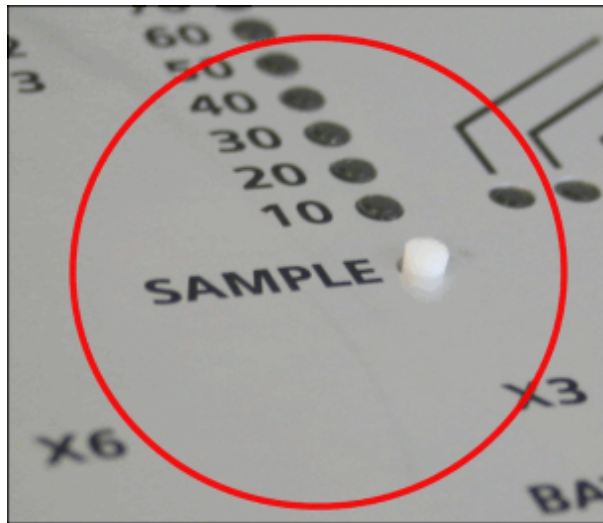


- (1) Liquid helium level
- (2) Magnet pressure
- (3) Start button

i Data is only accurate from the last refresh of the page, or from when the page was opened last.

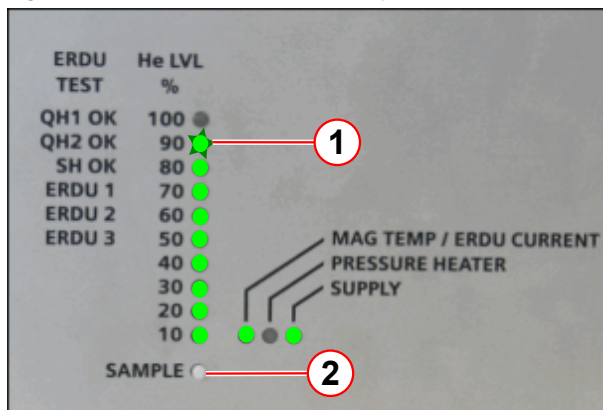
Helium level check at the MSUP

Fig. 102: MSUP 2306 - depress SAMPLE to activate



- Locate the SAMPLE switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.

Fig. 103: MSUP - Helium level sample



- (1) Sample switch
(2) Helium level

- The green LEDs illuminate and display the actual liquid helium level percentage.
- If the 90% LED is flashing, STOP the transfer. Refer to (→ Tab. 5 Page 95).

Tab. 5 MSUP Helium display

MSUP LEDs	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue to fill as required.
90% ON	As displayed	Stop transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATELY!
*100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED.



* The LEDs return to ON after approximately 30 seconds.



To avoid overfilling, the helium transfer must be stopped as soon as the 90% LED illuminates.

- Repeat the measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - » If not, stop the transfer and investigate the reason for poor transfer efficiency.



Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.



Maintain a pressure of about 0.27 bar (4 psi (G)) at the helium dewar. Too much or too little pressure will result in inefficiency.

At 4 psi (G), the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- » **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

5.1.1.10 Removing the service syphon

- **Depressurize** the helium vessel. (→ Depressurize the helium vessel / Page 86)

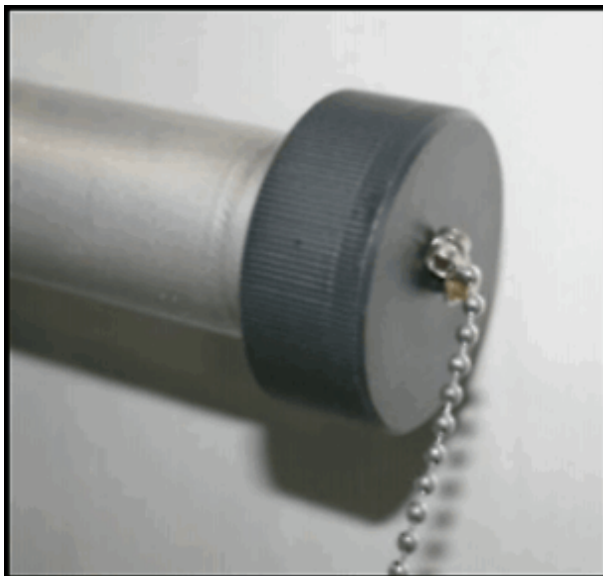
Fig. 104: Warm up the syphon joint with heat gun.



- **Warm** the connecting **syphon joint** to room temperature (using a **heat gun**) to remove ice.

- **Warm** the syphon **entry valve** to room temperature and dry.

Fig. 105: Syphon end cap



- **Warm** the syphon **end cap** ready for fitment onto the syphon.

- **Disconnect** the syphons.



CAUTION

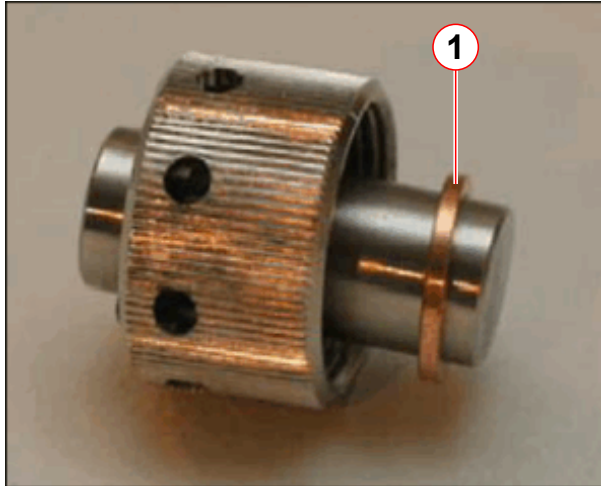
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

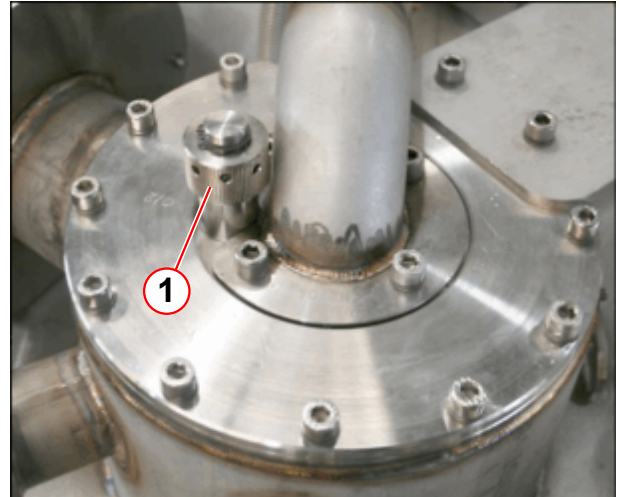
- **Close off** the end of the service syphon with the **syphon end cap**.
(→ Fig. 105 Page 97)
- **Immediately remove** the service **syphon** from the magnet.

Fig. 106: Syphon blanking plug



(1) Copper seal

Fig. 107: Fit/remove the syphon blanking plug



- **Immediately** fit the **syphon blanking plug** using a new copper seal.
(→ Fig. 106 Page 98)
- **Warm up the threads** on the plug and **carefully** tighten with the syphon 'C' spanner.
 - There might be ice on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. **Re-tighten** the plug.

⚠ CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.

- **Pressurize** the helium vessel to **15.3 psi (A)**.

Reactivate the pressure heater

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- At **MSUP Control - Sets the pressure heater mode** select the **On 100%** button.
(→ 1/Fig. 89 Page 86)
- Select **SAVE**. (→ Fig. 89 Page 86)

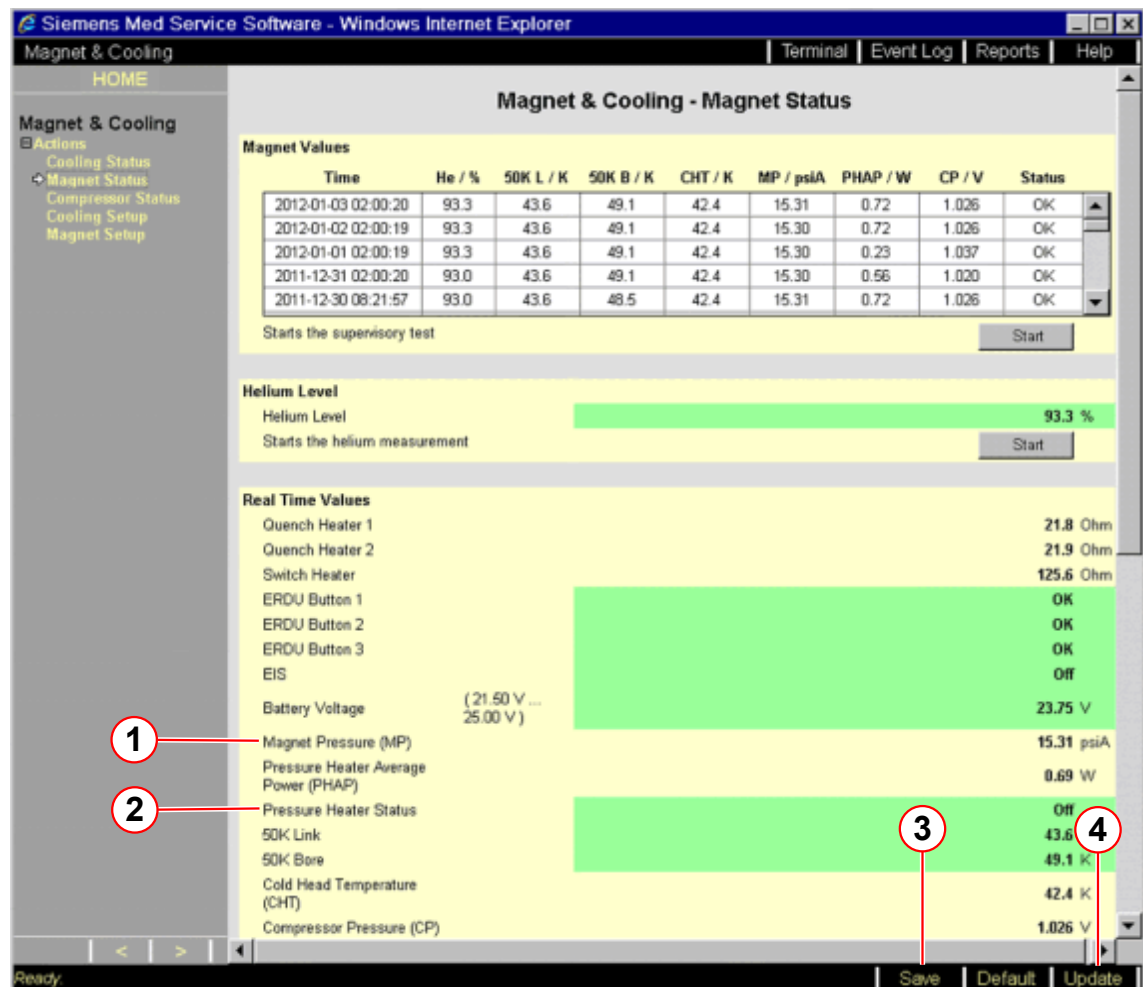
- **At Real-Time Values** - Monitor the **magnet pressure**.
 - Select **Update** to refresh data until pressure reaches **≥ 15.3 psi (A)**.

Data is only accurate from the last refresh of the page, or from when the page was opened last.

During repressurization, the cold head will switch on automatically.

If the cold head was turned off at MRC, select (→ Fig. 89 Page 86)

Fig. 108: Check magnet pressure



- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

- At **MSUP Control** - Sets the pressure heater mode select the **Automatic mode** button. (→ 1/ Fig. 89 Page 86)
- Select **SAVE**.

Leak check

- After helium fill has been completed, **leak check** the **turret**.



Wait at least 2 hours after completion of a helium fill before attempting to ramp the magnet.

- **Move the dewar** out of the imaging suite into a **well ventilated area**.
- Complete **depressurizing** the dewar by opening the **primary ball valve**. (→ 5/Fig. 82 Page 80)
- When at the gas pressure is at **zero**, **close** the **primary ball valve** on the dewar.
- **Remove** the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar. (→ 4/Fig. 82 Page 80)
- **Open** the **low-pressure dewar relief valve**. (→ 1/Fig. 82 Page 80)
- **Warm** both **syphons** with the heat gun **until dry**, and store safely.

5.1.1.11 Changing supply dewars during transfer

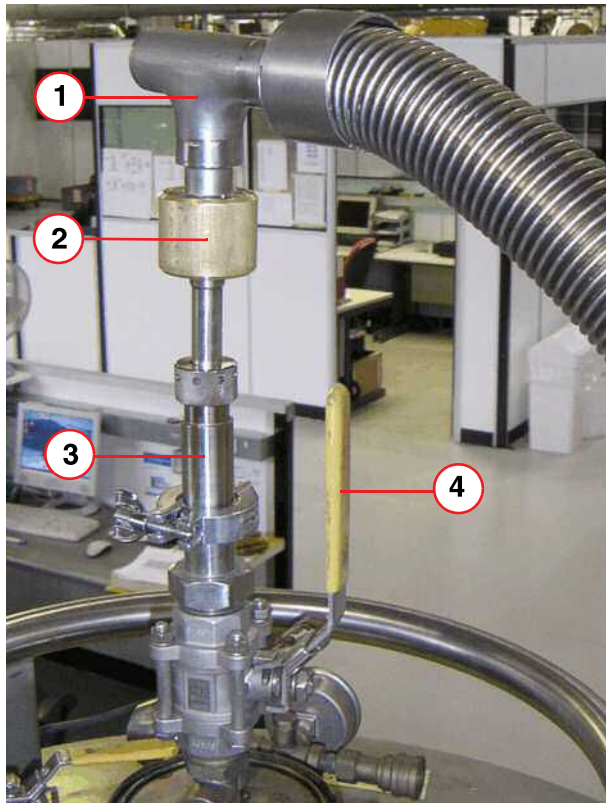


Whistling of the syphon (a high pitched sound), or a noticeable drop in dewar pressure are indications of an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, ready to receive the syphon.

Fig. 109: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- When the dewar empties, stop the transfer **immediately** by closing the dewar syphon flow valve (pos 2).



CAUTION

Risk of quench.

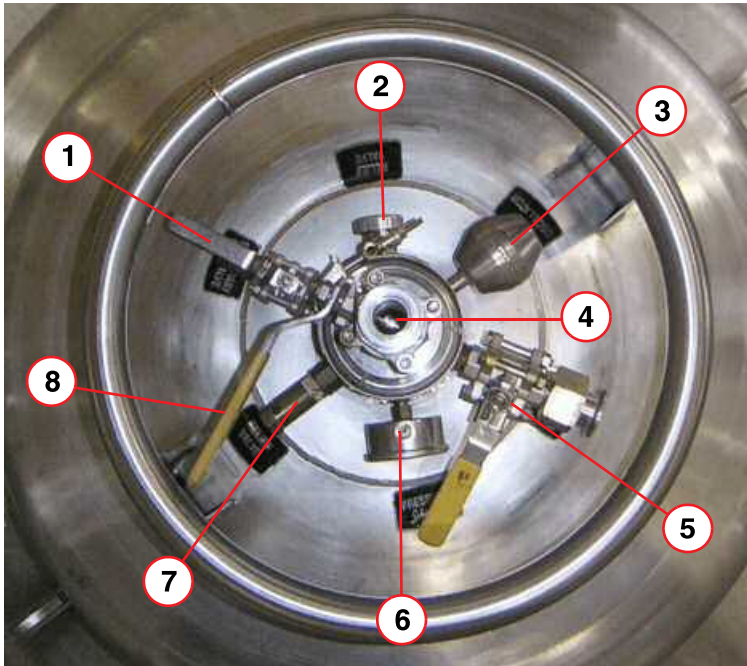
Failure to close the valve immediately, when the magnet is at field, will result in a quench.

» Close the valve immediately.

- Turn **off** the dewar gas supply:
 - remove the **gas connection**.
 - vent the dewar down to atmospheric pressure.
 - **Ensure the area is well ventilated.**

- Close the **primary dewar ball valve**. (→ 5/ Fig. 110 Page 102)

Fig. 110: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- Using a heat gun, **warm the syphon connecting joint** to room temperature.
- **Disconnect** the syphons.

CAUTION

Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting.
- » Wear protective gloves and safety glasses.

- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **service syphon end cap**. (→ Fig. 105 Page 97)

**CAUTION**

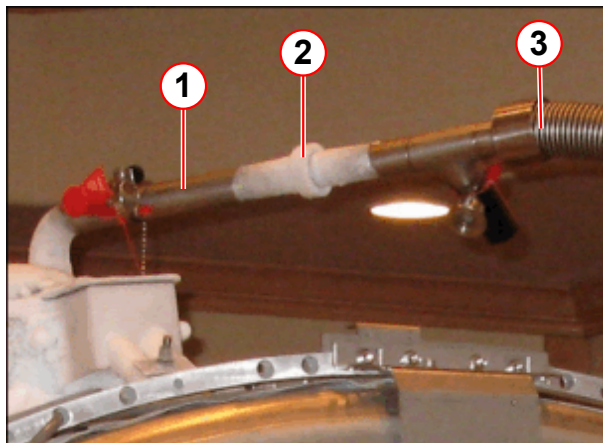
Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» Ensure the syphons are capped and plugged immediately and securely.

- Move the empty dewar close to the full dewar.
- Vent both storage dewars down to **atmospheric pressure**.
 - » **This must be done in a well ventilated area - NOT in the imaging suite.**
- **Open the low pressure relief valve** on the **empty** dewar. (→ 1/Fig. 110 Page 102)
- **Remove the dewar syphon and adaptor** from the empty dewar and **immediately** insert the syphon leg into the **full** dewar.
- **Close the syphon entry valve** (→ 4/Fig. 110 Page 102), of the **empty** dewar. and **open the low pressure relief valve**. (→ 1/Fig. 110 Page 102)
- **Close the low pressure relief valve** of the **full** dewar.
- **Purge** the gas supply line.
- **Connect** to the full dewar and **open** the dewar **primary ball valve**.
- Set the **gas pressure** to 0.27bar (4psi (G)).
- **Precool** the **dewar syphon**.
- Move the dewar next to the magnet system.
- **Precool** the **service syphon**.

Fig. 111: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, **connect** them together and **secure** the lock nut.

- **Continue** with the transfer.

5.1.1.12 Stopping the helium transfer

Fig. 112: Dewar manometer



- As soon as the helium level reads **95%**, or the dewar is empty, **stop the transfer** by closing the flow valve at the dewar syphon. (→ Helium level check at the MRC / Page 93) **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** (→ 6/Fig. 99 Page 93)
- Set the dewar gas pressure regulator to **zero**.
- Close the dewar **primary ball valve.** (→ 1/Fig. 86 Page 83)
- **Remove** the **gas line** from the **dewar**.
- Refer to (→ Changing supply dewars during transfer / Page 100) if the magnet is not full.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

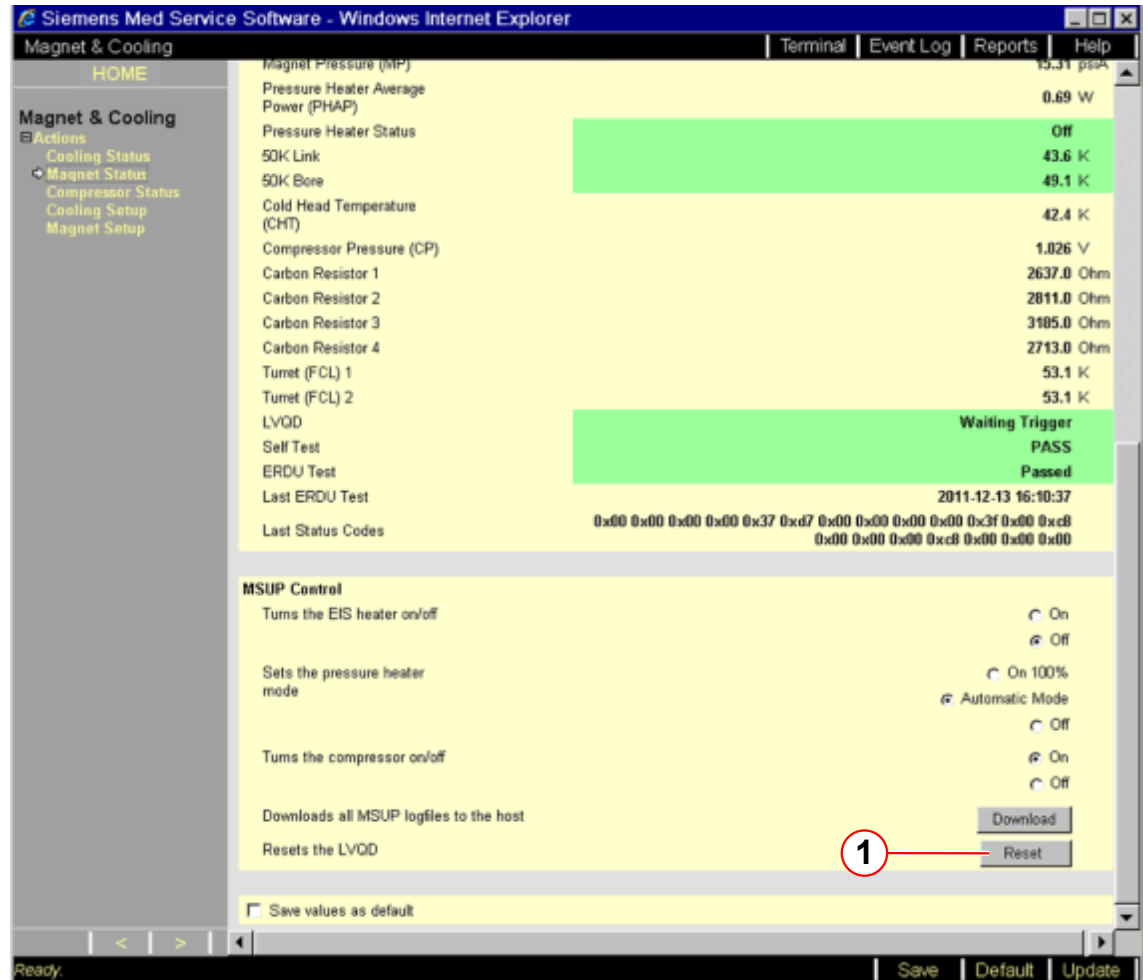
- » If the magnet is still at field, stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure drops.

5.1.1.13 Final work steps

After transfer, check:-

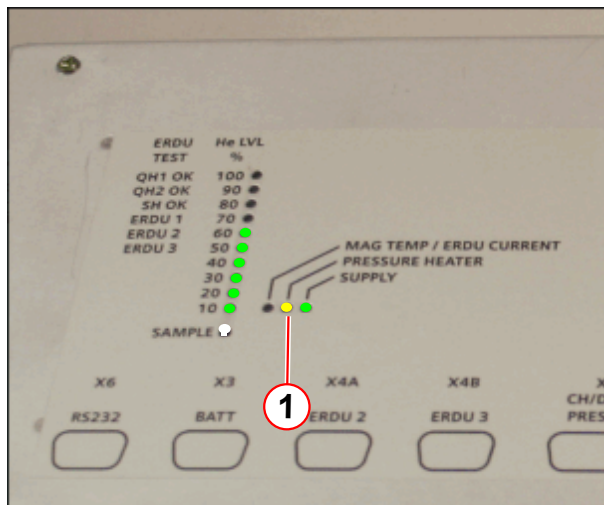
- Press the **LVQD reset button** to ensure the LVQD shows waiting trigger.

Fig. 113: LVQD reset button



- The **magnet bypass valve** is closed.
- The **pressure heater** is operating correctly.

Fig. 114: MSUP - Pressure heater LED



- The pressure heater **LED** on the **MSUP** is illuminated, or switching on and off.

(1) Pressure heater LED - yellow

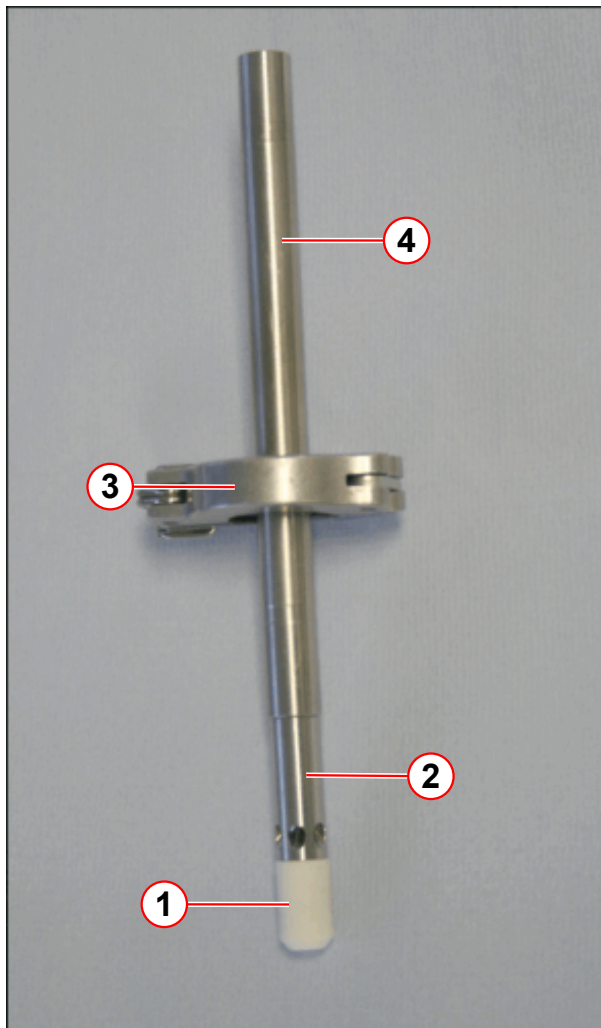
- **No alarm** signals are evident at the **MSUP** or the **alarm box**.
- The **compressor** is switched **on** and **running**.
- The **magnet pressure** is stable at **15.3 psi (A)**.
- The **turret** and **venting** system are **warm** and **dry**.
- The filling **port** and **venting** system have **no leaks**.
- The 4K **cold head flange** has **no leaks**.
- The used **dewars** are correctly **closed**, with their low pressure relief valves **open**.

5.1.1.14 Low ceiling height

Where magnets have been installed into rooms with a **low ceiling height**, it can be difficult to fit the assembled service syphon. To avoid dropping the extension leg(s) down into the syphon port, the **syphon leg support clamp** must be used (→ 3/ Fig. 115 Page 107).

- Check the **ceiling height**:
 - Assemble the syphon (only if it can be fitted), or
 - Assemble the diffuser, bottom extension and support clamp

Fig. 115: Syphon configuration for low ceiling height



- (1) PTFE extension tip (top fill)
- (2) Helium fill diffuser tip (top fill)
- (3) Syphon leg support clamp
- (4) Extension leg

Refer to Magnet Service syphon kit

- Remove the **syphon end cap**.
- Ensure the **ring nut, brass spacer and O-ring** are fitted correctly.
- Fit the **diffuser** to the extension leg.
- Fit **support clamp** to the extension leg.
- **Warm** the syphon **blanking plug** to ease removal.
- **Remove** the syphon **blanking plug** using the C-spanner.
- **Lower** the assembly into the **neck**.
- **Screw** on the second **extension leg**.
- **Move** the syphon **clamp** up the extension leg and lower extensions into the entry port, to allow room for the syphon body.
- **Screw** the **diffuser leg** to the syphon **body**.
- **Remove** the syphon **clamp** and slowly **lower the syphon fully** into the syphon entry.
- **Screw** the **syphon port ring** nut into place and **tighten** to make a gas-tight **seal**.
- **Re-fit** the syphon **end cap**.



Use the support clamp for removing the syphon.

5.1.1.15 Helium pressures at altitude

Helium transfer with magnet bypass valve closed

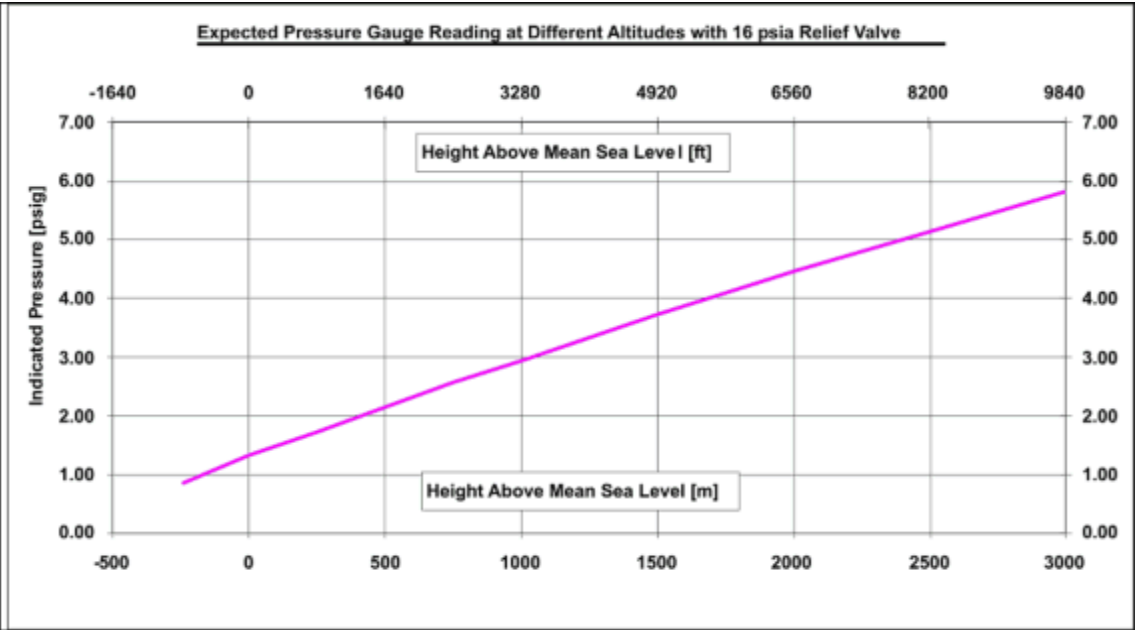
As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi. To calculate this:

- Determine the **magnet's altitude**
- Determine the **magnet gauge pressure** from the graphs below.

OR105 magnet - 16 psi (A) valve

Fig. 116: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 6 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- Add 4 psi to this number to calculate the **required transfer pressure**
 - » **Example of 16 psi (A) vent valve assembly**
Altitude (1000 m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)

Maximum dewar pressure is 10.5 psi (G).

5.1.1.16 Verifying the electronic low pressure switch

This procedure may be carried out in conjunction with any other process involving de-pressurization of the helium vessel, i.e. fitting the service syphon, cold head replacement. Pressure control of the helium vessel is managed by the **MSUP**, via the **Pressure heater control unit**. The low pressure switch must be checked **whenever the magnet is de-pressurized**.

When pressure in the helium vessel drops to **within 0.2 psi (G)** above atmospheric pressure, the **electrical pressure switch** will open and turn **off** the compressor.



Atmospheric pressure may differ, depending on the altitude and local weather conditions.

As the helium vessel is **pressurized**, the switch closes and the **compressor** will **re-start**.



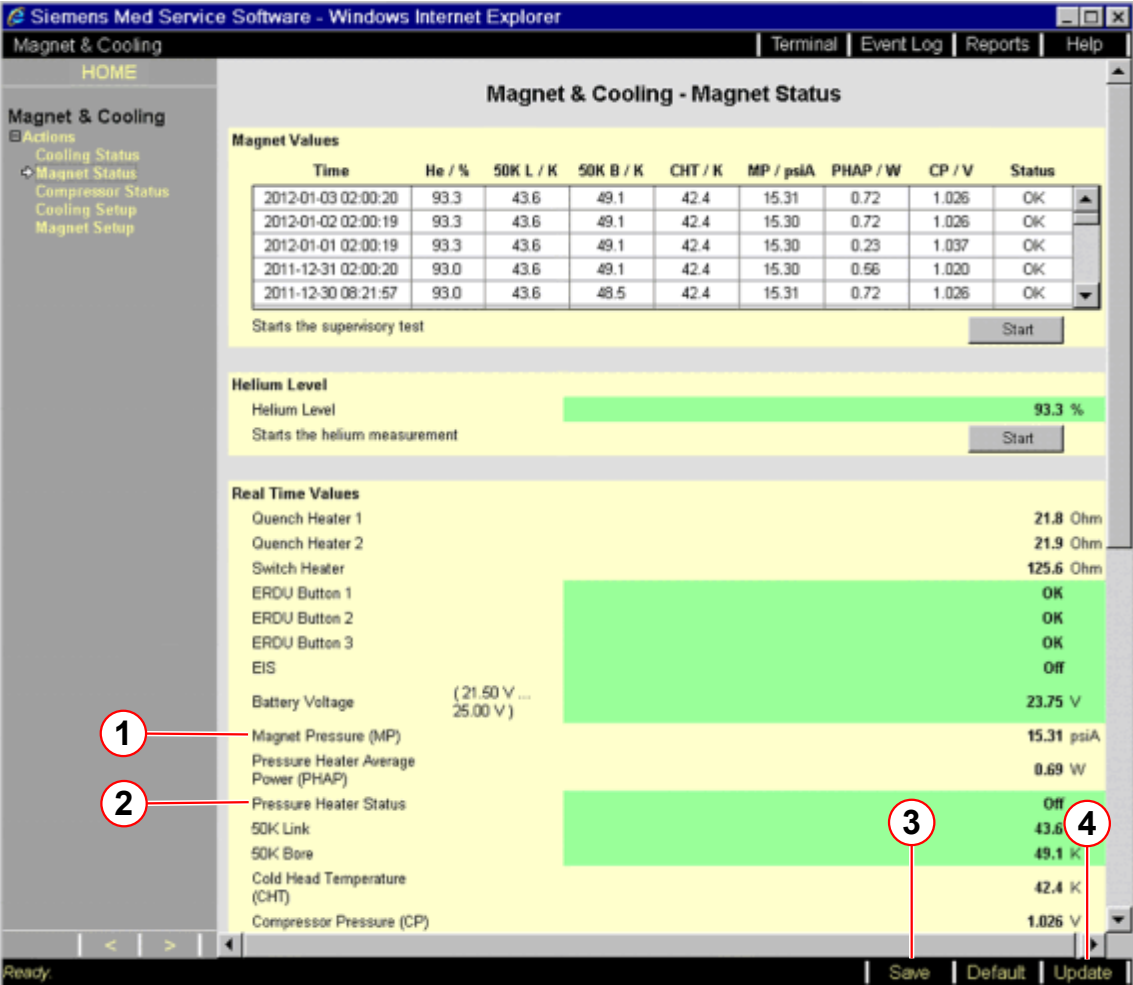
The pressure switch is factory calibrated to operate at 0.2 psi (G) above atmospheric pressure.



Perform this check after helium filling when the magnet syphon is removed.
The filling port is closed and the refrigerator switched on.

- Access the **Service Software** at the **MRC (MRAWP) Host**.
- Select the menu item **Magnet & Cooling - Magnet Status**.
- Check **magnet pressure** is normal **15.3 psi (A)**.

Fig. 117: Check magnet pressure

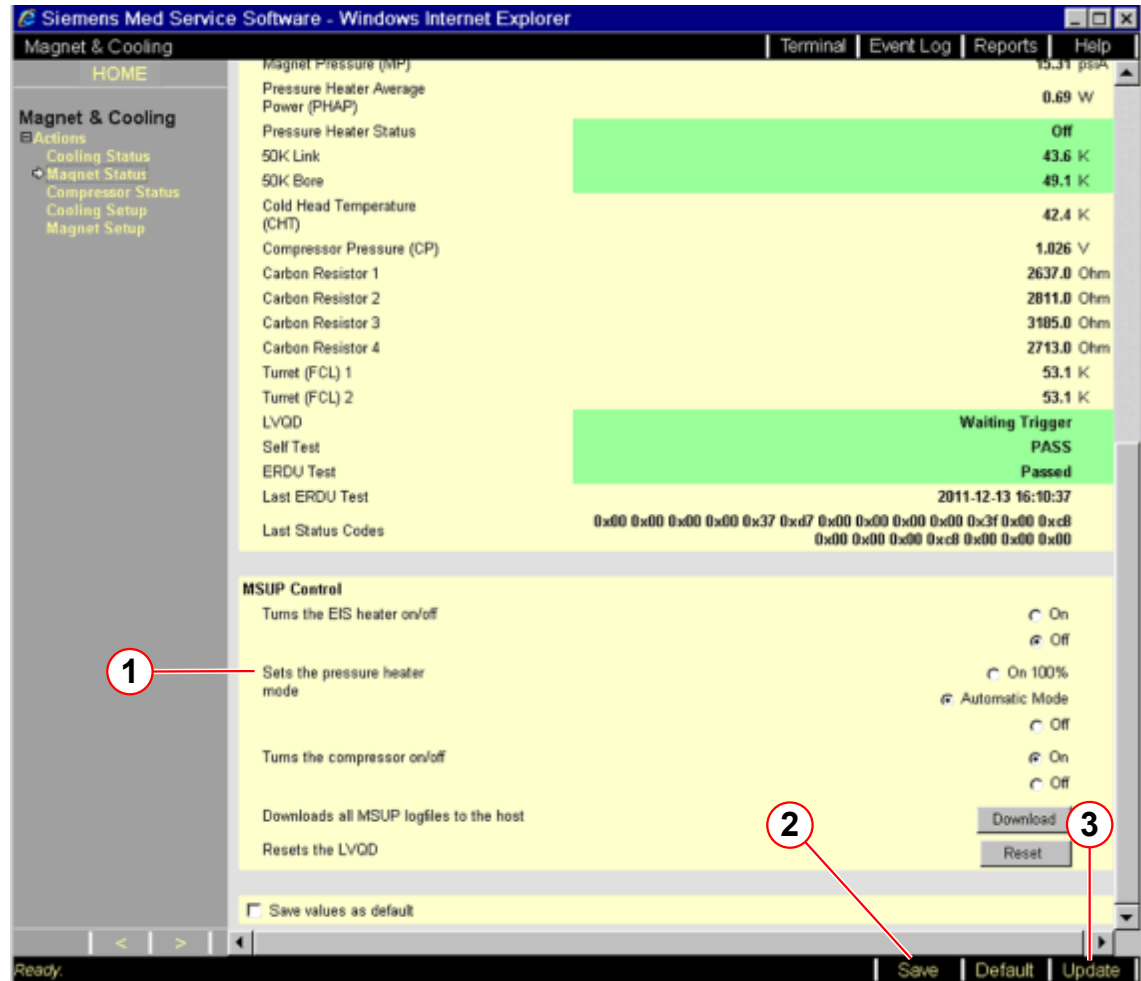


- (1) Magnet Pressure (MP)
 - (2) Pressure Heater Status
 - (3) Save
 - (4) Update
- Check the **compressor** is switched on and running.

Deactivate the pressure heater

- At **MSUP Control** - Sets the pressure heater mode select the **OFF** button.
(→ 1/ Fig. 118 Page 111)

Fig. 118: Service screen Mag&Cooling - Magnet Status - MSUP Control

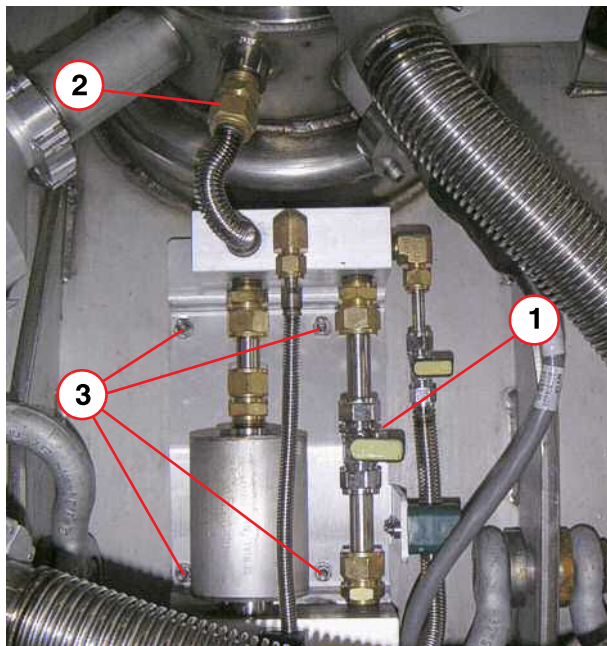


- (1) Sets the pressure heater mode
- (2) Save
- (3) Update

Depressurize the helium vessel

- Depressurize** to within **0.2 psi (A)** of atmospheric pressure.

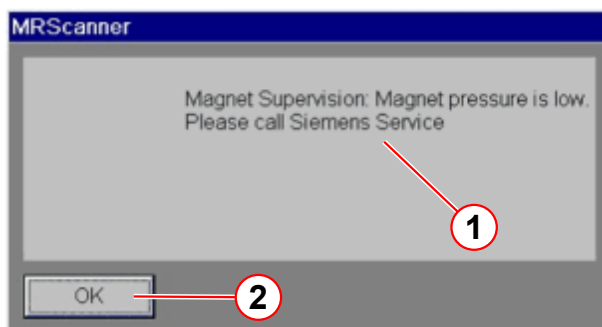
Fig. 119: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

- Fully open the **magnet bypass valve** to **depressurize** the helium vessel.
- The **pressure switch** will activate when system pressure is within **0.2 psi (A)** of atmospheric pressure.
 - » this turns the **cold head OFF**.
- **Close** the magnet **bypass valve** to prevent **air ingress**.
- **Warm up** the vent **valve assembly**.
 - » **Take care** not to damage any components when using a **heat gun**.

Fig. 120: MSUP - Magnet pressure is low



- (1) Pop-up message
- (2) Click OK to clear

On the **MR Scanner**:

- A pop-up **alarm message** is displayed.
 - » **Magnet pressure is low.**
 - » Click **OK** to clear the alarm.
- The **alarm box** will sound when **pressure falls to alarm limits**.
 - » The yellow **WARNING LED** lights on the alarm box.
 - » The alarm **buzzer** is **activated**
 - » Press **AUDIO ALARM OFF** to silence the alarm.

- Check the **pressure is positive** when the **compressor** switches **OFF**.
- **Repressurize** the helium vessel.¹

Reactivate the pressure heater

- At **MSUP Control** - **Sets the pressure heater mode** select the **On 100%** button.
(→ 1/ Fig. 89 Page 86)

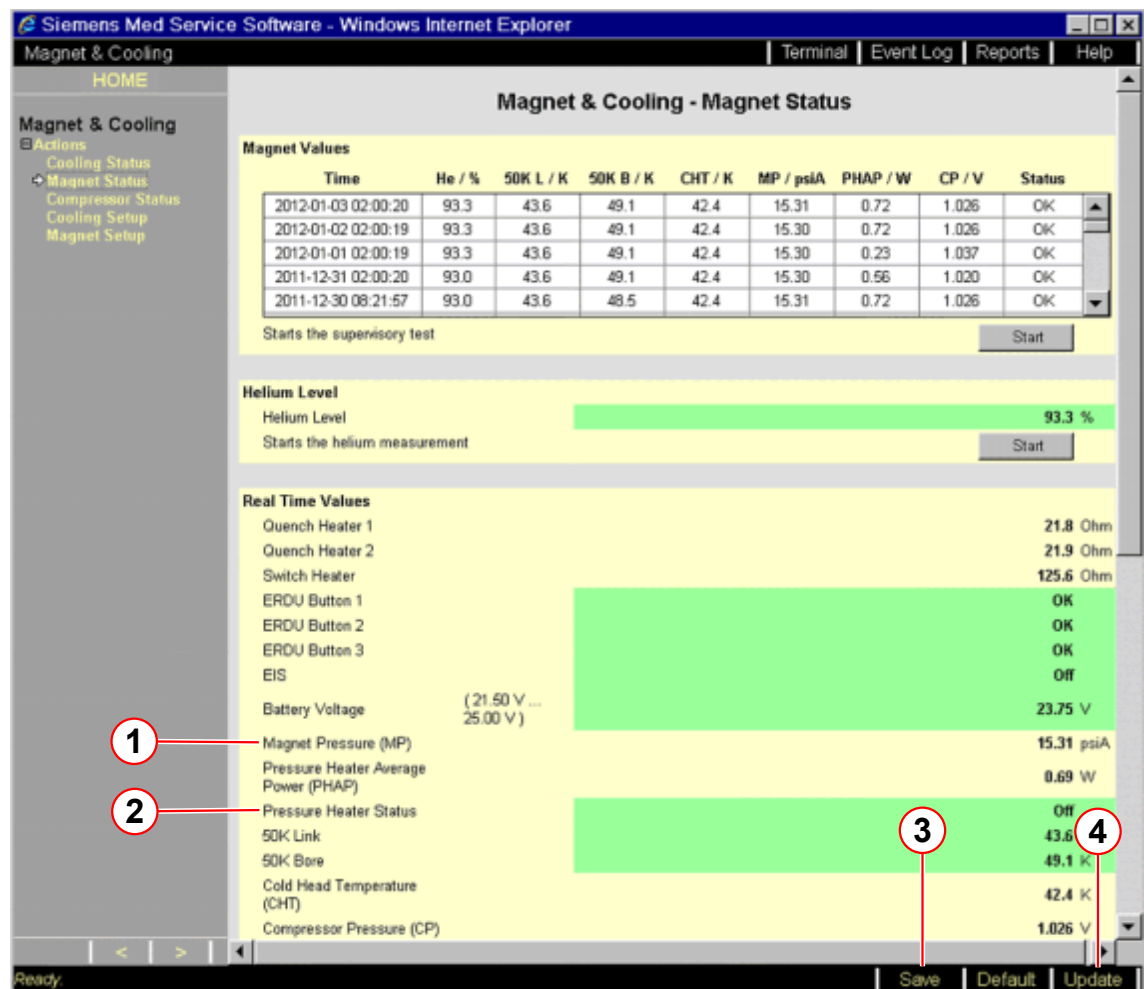
1. If the verification is being carried out with another process (i.e. fitting the Magnet Service Syphon (except for Mobile systems), or replacing the Cold head), follow the **re-pressurization** part of this procedure when those processes are complete.

- Click **UPDATE**. (→ 4/ Fig. 89 Page 86)
- Check that the **compressor** activates **automatically** as the pressure rises above **0.2 psi (A)** atmospheric.
- **At Real Time Values** - Monitor the **magnet pressure**.
 - Select **Update** to refresh data until pressure reaches ≥ 15.4 psi (A).



Data is only accurate from the last refresh of the page, or from when the page was last opened.

Fig. 121: Check magnet pressure



- (1) Magnet Pressure (MP)
- (2) Pressure Heater Status
- (3) Save
- (4) Update

- **At MSUP Control** - Sets the pressure heater mode select the **Automatic mode** button (→ 1/ Fig. 89 Page 86)
- Click **UPDATE**. (→ 4/ Fig. 89 Page 86)
- Cancel any **alarms**.



Check that the bypass valve is locked securely in the closed position.



Contact the Magnet HSC if the test fails.



As the cold head recondenses the helium gas, the system pressure is returned to 15.3 psi (A).

5.1.2 Bottom filling the magnet



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

» Always ensure the cold head bypass valve is **CLOSED** during helium filling.



The bottom fill procedure must **ONLY** be performed when the magnet is at **ZERO FIELD**.

Bottom filling of the magnet will be required if:-

- The liquid helium level -
 - Falls to **zero** percent,
 - **and**
 - The temperatures measured from the ceramic carbon resistors (**CCR**) and coil diodes, indicates temperatures **no warmer than 110K**.

The magnet must be filled through the **magnet syphon cone** to allow filling from the bottom of the helium vessel.



Inform Magnet HSC immediately if the helium level is zero. Magnet coil temperatures must be confirmed before continuing with the helium fill.

Possible causes for this include:-

- Post magnet quench, (ramping or from persistence).
- Faulty helium level probes.
- Alarm box buttons still depressed.
- Pressure heater is left on.
- Long transportation and storage times.

- Cold head or compressor failure.

5.1.2.1 Checking magnet temperatures

Fig. 122: OR105 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).		
This system requires 0 Z2 rings and 2.92 kg of iron to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms	Calibration values	
Liquid nitrogen temperature (77K)	1309 Ohms		
Liquid helium temperature (4.2K)	2994 Ohms		
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	Signed <i>AN Other</i>		
V Turner			
	Date 23/12/2011		
	Page 1 of 1		
	25-AD EPS 000 01		

To recover a magnet with a **bottom helium fill**, the following must be checked:-

Access the Service software at the **MRC (MRAWP) Host** to measure:

- **Helium level** values.
- **CCR** values.

Tab. 7 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380



Carbon 4/CCR4 is for measuring helium gas temperatures during ramping and is not applicable to filling.

5.1.2.2 Bottom fill procedure

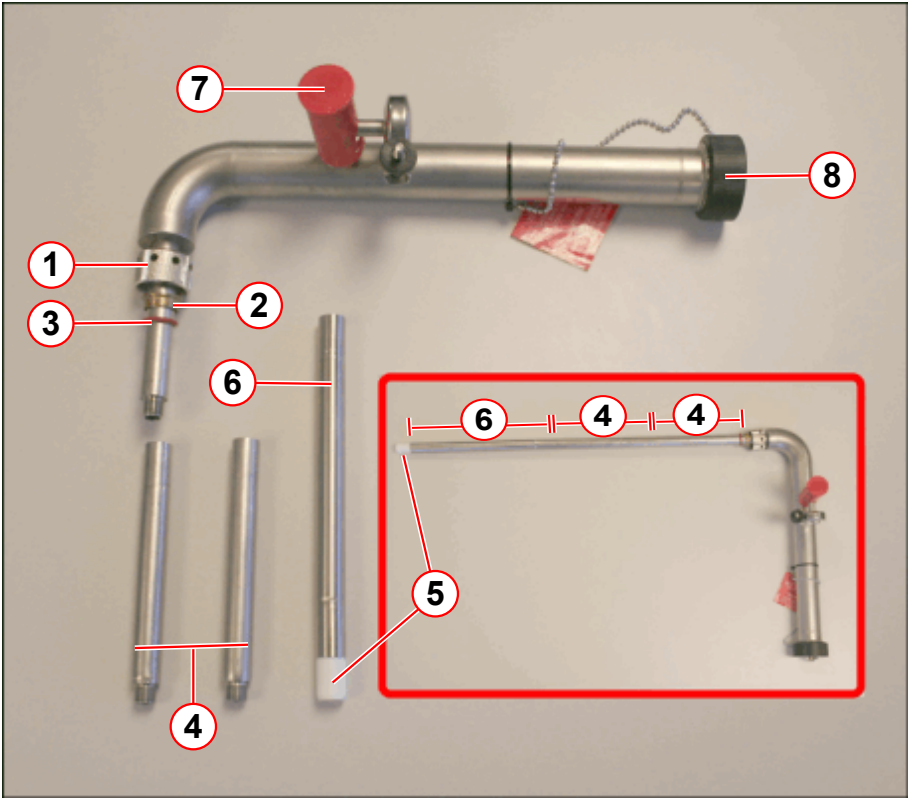


The bottom fill procedure must ONLY be performed when the magnet is at ZERO FIELD.

- Prepare the **dewar** and **dewar syphon**. (→ Dewar preparation / Page 79), (→ Dewar syphon preparation / Page 81)
- Check the **ceiling height**. (→ Low ceiling height / Page 106)
 - Assemble the **syphon** (only if it can be fitted), or
 - Assemble the **diffuser**, bottom **extension** and support **clamp**.

Assemble the service syphon

Fig. 123: Magnet service syphon assembly - bottom filling

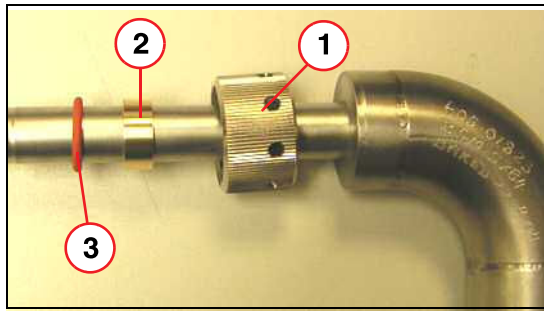


- (1) Ring nut
- (2) Brass spacer
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) 22 mm PTFE seal (bottom fill)
- (6) 200 mm syphon leg (1 off)
- (7) Vacuum valve
- (8) Syphon end cap

Tab. 8 OR105 BOTTOM filling syphon extension legs

Magnet type	140 mm leg	200 mm leg	22 mm PTFE seal
OR105	2	1	1

Fig. 124: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer and the O-ring to the **service syphon leg**.

Refer to table above (→ Tab. 4 Page 84)

- Connect the **extension legs** and the 22 mm **PTFE seal**:
 - Screw the 2 x **140 mm legs** together, and connect to the **syphon**.
 - Fit the 1 x **200 mm leg**.
 - Fit the **22 mm PTFE seal** to the 200mm leg.

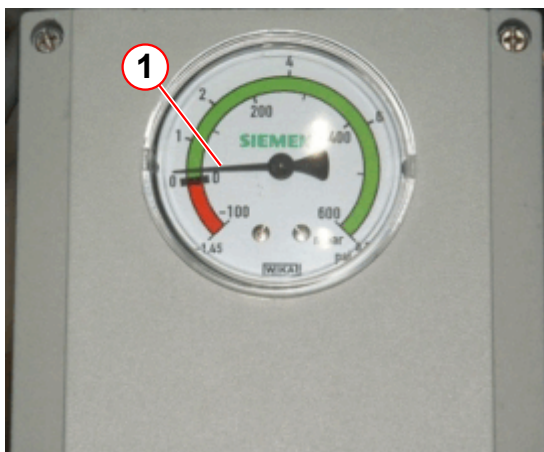


The minimum length required to reach the syphon cone is:
OR105 - 485 mm (+/- 5 mm)

Depressurize the helium vessel

- **Deactivate** the **pressure heater**. (→ Deactivate the pressure heater / Page 85)
- **Depressurize** the helium vessel (→ Depressurize the helium vessel / Page 111)
- Note the **pressure** (using the **MRC**, or **Laptop**) when the **compressor** switches **OFF**.

Fig. 125: WIKKA Pressure switch - with a small positive pressure



- **Check** the helium vessel has a **small positive pressure**.

- **Warm** and dry the **syphon blanking plug** using a heat gun to ease removal.
- **Dry** the syphon port.

- Remove the syphon **end cap**. (→ Fig. 105 Page 97)
- Remove the syphon **blanking plug**.

⚠ CAUTION

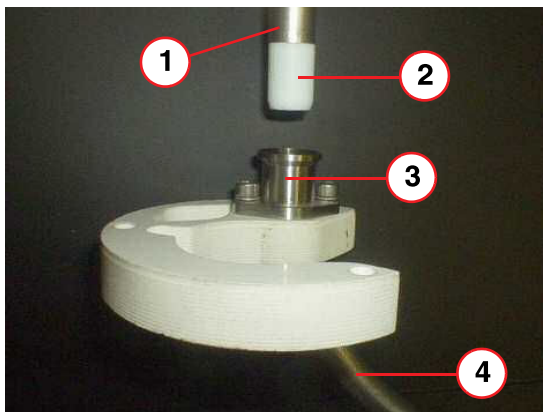
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

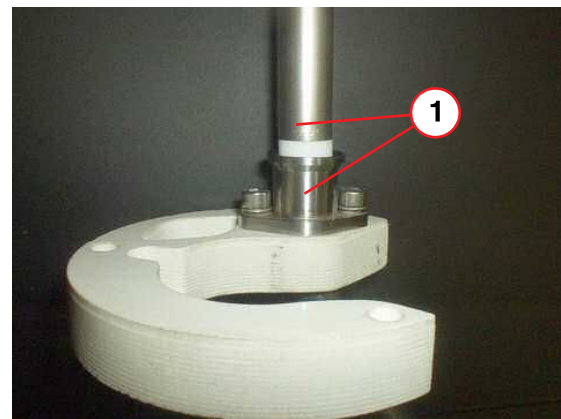
- Slowly lower the **service syphon** fully into the neck.
- The **syphon leg** must locate fully into the **syphon cone** - make sure it is **fully engaged**.

Fig. 126: Bottom fill syphon cone (pictorial view)



- (1) Syphon leg
- (2) PTFE Seal (Bottom Fill)
- (3) Syphon cone
- (4) Syphon tube

Fig. 127: Bottom fill - syphon leg fully engaged (pictorial view)



- (1) Syphon leg - fully engaged



Helium efficiency will be reduced if the syphon leg is not fully engaged in the syphon cone.

- Tighten the **syphon port ring nut** (→ 1/ Fig. 124 Page 119) by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the **syphon end cap**.

5.1.2.3 Liquid helium transfer - bottom fill



Before helium transfer, check (→ Helium pressures at altitude / Page 107)



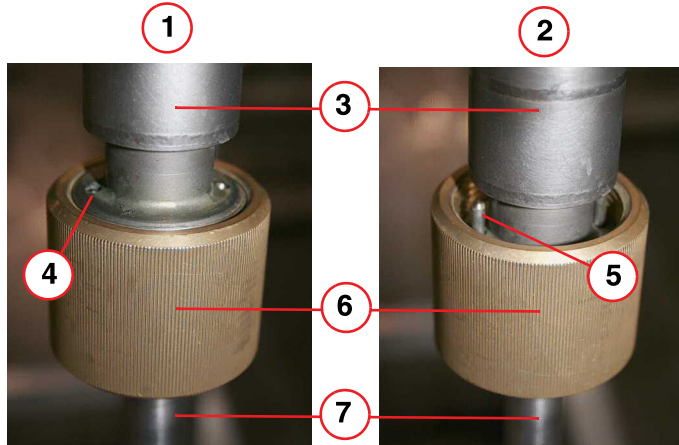
The helium dewar must be positioned close to the magnet, with the dewar syphon fitted.



For bottom filling a magnet at zero, there is no need to pre-cool both syphons.

- Check the dewar gas supply pressure is set at **0.27bar (4 psi (G))**.
- Open the dewar syphon flow valve half way (2 turns) to **purge** the syphon of air.

Fig. 128: Dewar syphon flow valves - open and closed positions

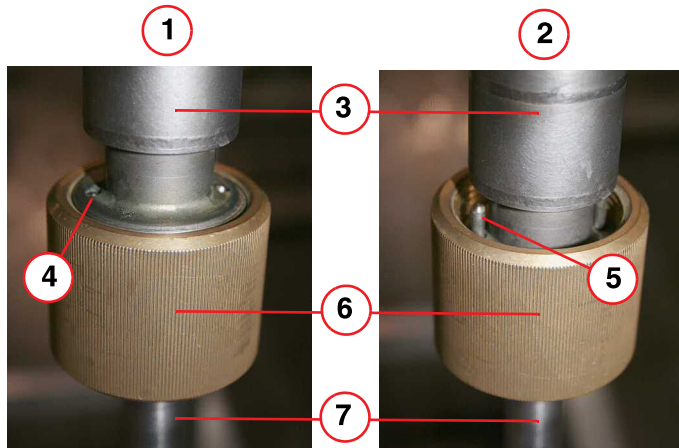


- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

Fig. 129: Helium filling



Fig. 130: Dewar syphon flow valves - open and closed positions

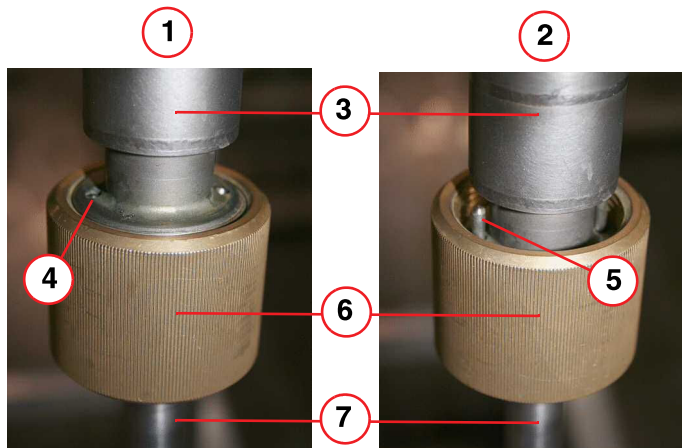


- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

Fig. 131: Helium filling



Fig. 132: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

Fig. 133: Helium filling



- **Close** the dewar syphon **flow valve**.
- **Remove** the service syphon end cap and **couple the two syphons together** and secure the **locknut**.
- **Open** the dewar syphon **flow valve**.
- Maximize the transfer by fully opening the **dewar syphon flow valve**.
 - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (→ 4/ Fig. 132 Page 122)
 - » From **fully closed** to **fully open**, is **4 turns**.
- Ensure the transfer dewar pressure is **4 psi (A)** at normal altitude. (→ Helium pressures at altitude / Page 107).

Monitor the CCRs

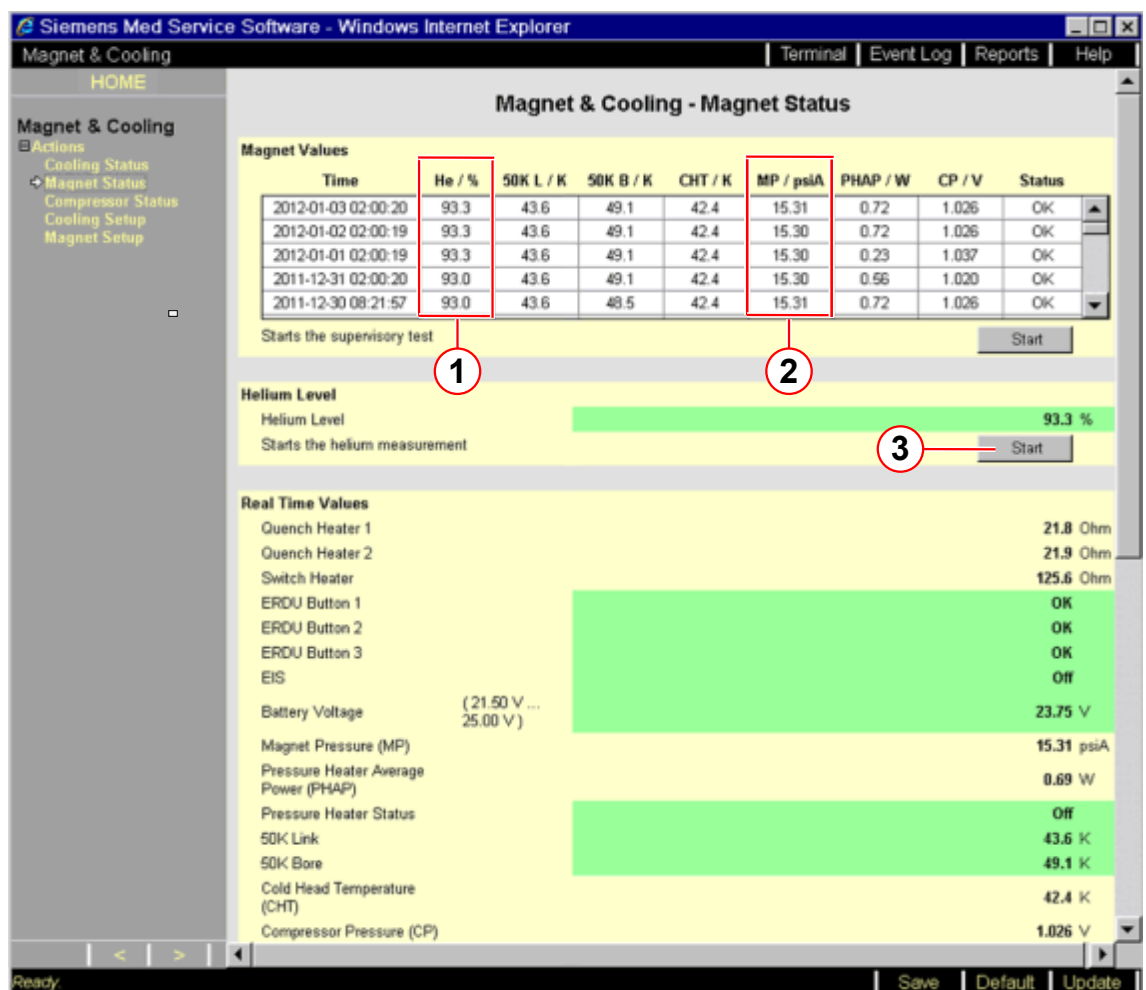
- Monitor the **CCRs** every **15 minutes**, to measure the **temperature drop**.
 - » Refer to the **acceptance documentation** (Installation Protocol) for required values.
 - » If the CCRs **do not** get cooler, **contact Magnet HSC**.
- When the required value is reached, **monitor** the **helium level** at the **Host** or **MSUP**.

5.1.2.4 Perform a helium level check

Helium level check at the MRC

- Check the **liquid helium level** by using **Service Software** on **MRC (MRAWP) Host** or visiting the **MR Quality Home Page** if the system has SRS enabled.
 - Check the **liquid helium level** using the MRC.
 - Access the **Service Software** at the MRC.
 - Select the menu item **Magnet & Cooling - Magnet Status**.
 - Press the **Start** button (→ 3/ Fig. 134 Page 123) to initiate the **helium measurement**. (→ 1/ Fig. 134 Page 123)

Fig. 134: Checking liquid helium and magnet pressure levels



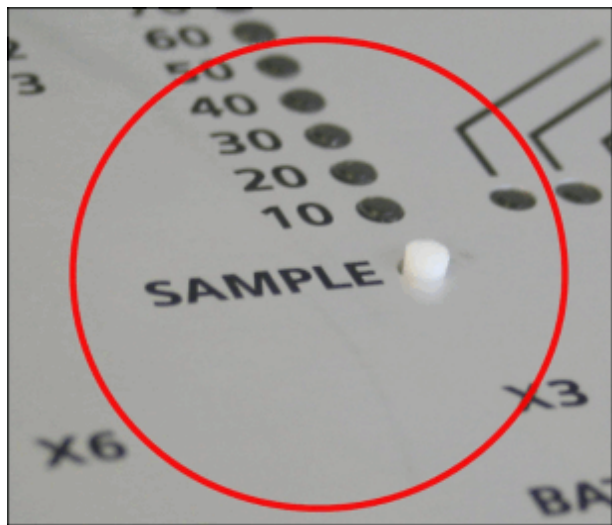
- (1) Liquid helium level
- (2) Magnet pressure
- (3) Start button



Data is only accurate from the last refresh of the page, or from when the page was opened last.

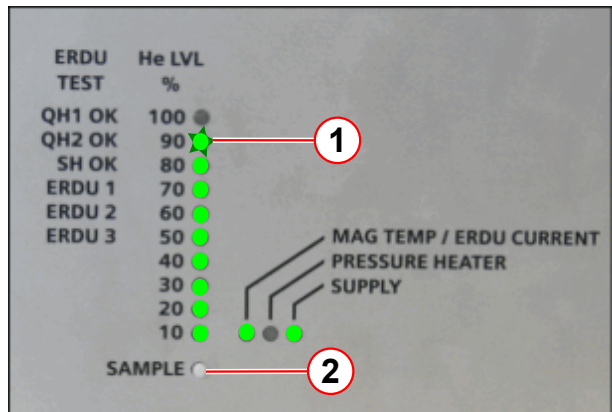
Helium level check at the MSUP

Fig. 135: MSUP 2306 - depress SAMPLE to activate



- Locate the SAMPLE switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.

Fig. 136: MSUP - Helium level sample



- (1) Sample switch
- (2) Helium level

- The green LEDs illuminate and display the actual liquid helium level percentage.
- If the 90% LED is flashing, STOP the transfer. Refer to (→ Tab. 9 Page 124).

Tab. 9 MSUP Helium display

MSUP LEDs	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue to fill as required.
90% ON	As displayed	Stop transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATELY!
*100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED.



* The LEDs return to ON after approximately 30 seconds.



To avoid overfilling, the helium transfer must be stopped as soon as the 90% LED illuminates.

- Repeat the measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - » If not, stop the transfer and investigate the reason for poor transfer efficiency.



Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.



Maintain a pressure of about 0.27 bar (4 psi (G)) at the helium dewar.
Too much or too little pressure will result in inefficiency.

At 4 psi (G), the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- » **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

5.1.2.5 Stopping the helium transfer

Fig. 137: Dewar manometer



- As soon as the level reads **95%**, or the dewar is empty, stop the transfer by closing the **flow valve** at the dewar syphon. (→ Helium level check at the MRC / Page 123) **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** (→ 6/Fig. 132 Page 122)
- Set the **dewar** gas pressure regulator to **zero**.
- Close the dewar **primary ball valve**. (→ 1/Fig. 86 Page 83)
- **Remove** the **gas line** from the **dewar**.
- Refer to (→ Changing supply dewars during transfer / Page 100) if the magnet is not full.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- » Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.

5.1.2.6 Removing the service syphon

- **Deactivate** the pressure heater. (→ Deactivate the pressure heater / Page 85)
- **Depressurize** the helium vessel. (→ Depressurize the helium vessel / Page 119)

! CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.

- Warm the **syphon joint** to room temperature (using a heat gun).
- Warm the syphon **entry valve** to room temperature and **dry**.
- Warm the syphon **end cap** ready for fitting onto the syphon.
- **Disconnect** the syphons.



CAUTION

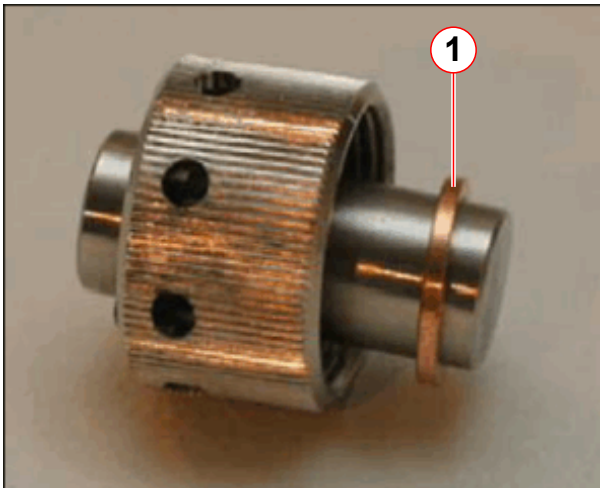
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

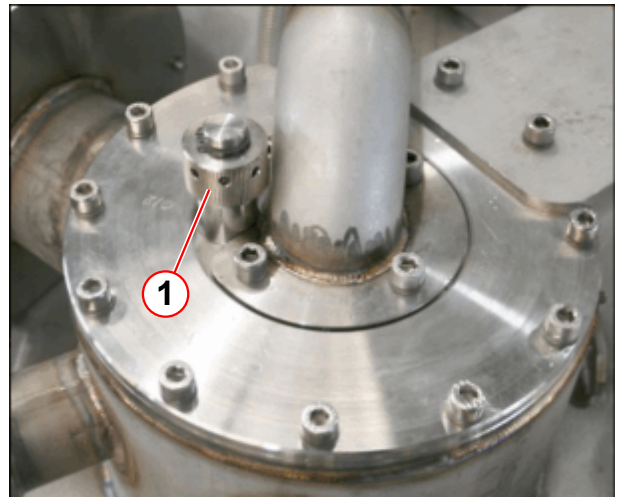
- **Immediately** close off the **end** of the service syphon with the **syphon end cap**.
- **Immediately remove** the service **syphon** from the magnet.

Fig. 138: Syphon blanking plug



(1) Copper seal

Fig. 139: Fit/remove the syphon blanking plug



- **Immediately** fit the **syphon blanking plug** using a new copper seal (→ Fig. 138 Page 127).
- **Warm up the threads** on the plug and **carefully** tighten with the syphon C-spanner.
 - There might be **ice** on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. **Re-tighten** the plug.
- **Warm up** the magnet bypass **valve assembly** and check the **bypass valve** is **closed**.

**CAUTION**

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.

- Pressurize the helium vessel to 15.3 psi (A).
 - **Reactivate** the pressure heater.(→ Reactivate the pressure heater / Page 112)
 - Check that all **pressure alarms** are deactivated.
 - Check the **compressor** is **running**.
- Warm the **extension legs** using a heat gun.
- Remove the **bottom fill extension leg** and **PTFE seal**.
- Warm the **turret and venting valves** to room temperature with a heat gun.
 - **remove** any **surplus water** with paper towelling.
- **Move the dewar** out of the imaging suite into a **well ventilated area**.
- Complete **depressurizing** the dewar by opening the **primary ball valve**.
(→ 5/Fig. 110 Page 102)
- When at the gas pressure is at **zero**, **close** the **primary ball valve** on the dewar.
- **Remove** the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar.
(→ 4/Fig. 110 Page 102)
- **Open** the **low-pressure dewar relief valve**. (→ 1/Fig. 110 Page 102)
- Warm both **syphons** with the heat gun **until dry**, and store safely.
- **Leak check** the turret and venting.



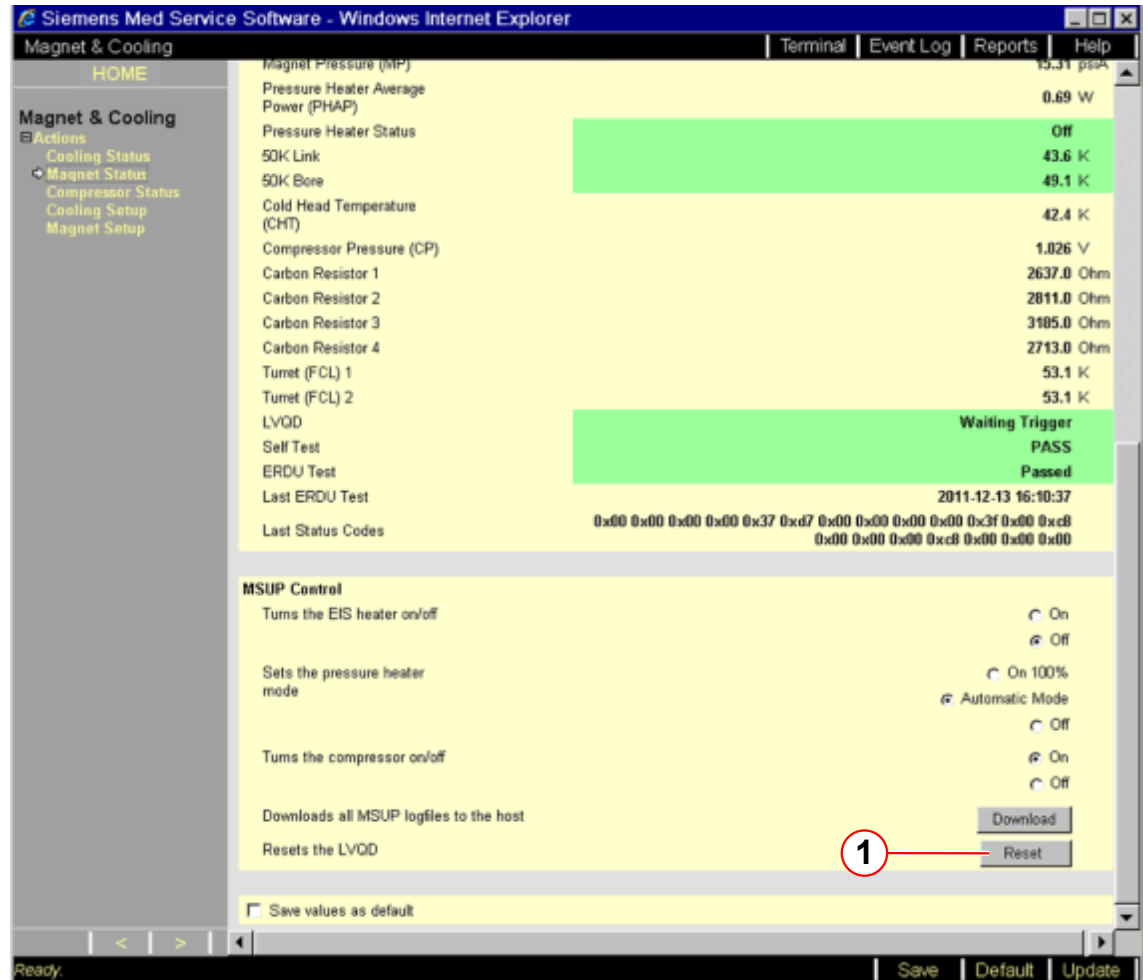
After a bottom fill, it is recommended that the first magnet energization is performed the next day. This will allow the magnet coils to thermally stabilize.

5.1.2.7 Final work steps

After transfer, check:-

- Press the **LVQD reset button** to ensure the functionality of the LVQD circuit.

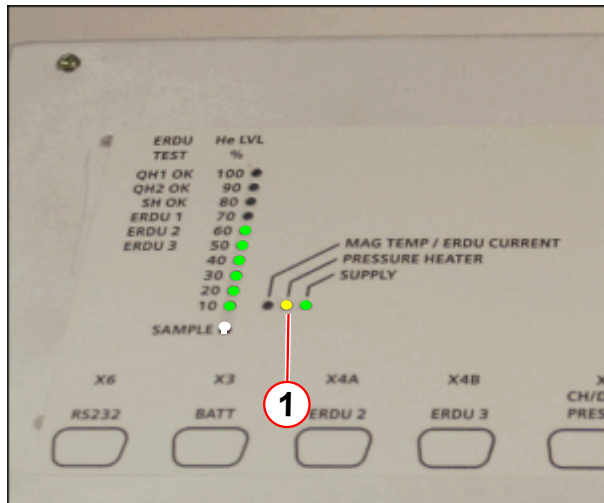
Fig. 140: LVQD reset button



- The magnet bypass valve is closed.

- The **pressure heater** is operating correctly.

Fig. 141: MSUP - Pressure heater LED



- The pressure heater **LED** on the **MSUP** is illuminated, or switching on and off.

(1) Pressure heater LED - yellow

- **No alarm** signals are evident at the **MSUP** or the **alarm box**.
- The **compressor** is switched **on** and **running**.
- The **magnet pressure** is stable at **15.3 psi (A)**.
- The **turret** and **venting** system are **warm** and **dry**.
- The filling **port** and **venting** system have **no leaks**.
- The 4K **cold head flange** has **no leaks**.
- The used **dewars** are correctly **closed**, with their low pressure relief valves **open**.

5.2 Avanto, Avanto Dot Upgrade - MSUP K2256 (10118361, 10142046, 3861296 or 7758209)

5.2.1 Filling the static system

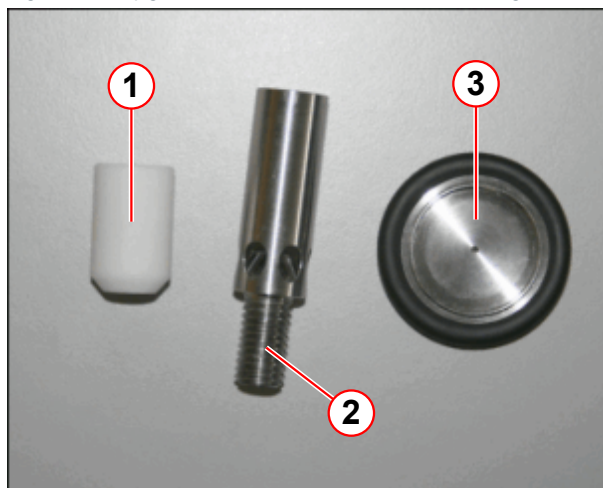
5.2.1.1 General

Static Systems

The magnet design does not include a permanently fitted syphon. A short magnet service syphon must be fitted when filling with liquid helium. The service syphon is tooling only, and **must be removed** immediately after filling has been completed.

This procedure can be performed on magnets at field and magnets at zero field.

Fig. 142: Upgrade kit 10129438 - De-ice tooling



- (1) PTFE tip
- (2) De-ice helium fill diffuser
- (3) NW25 flow restrictor

The Magnet service syphon kit is now upgraded to include **de-ice tooling**. The de-ice tooling is used for **helium top fill**, and **de-icing of the syphon port**.

- Use for **helium top filling**:
 - PTFE tip
 - 6 hole metal de-ice helium fill diffuser
- Use for **de-icing of the syphon port**:
 - PTFE tip
 - 6 hole metal de-ice helium fill diffuser
 - NW25 flow resistor

The new 6 hole diffuser **is more efficient**, and must be used **in place of** the 60 mm PTFE diffuser.



Only Siemens trained and accredited CSEs are allowed to perform this work.



During filling, do not exceed 95%.



To recover a potentially warm magnet, refer to magnet temperatures in the Bottom filling procedure (→ Bottom filling the magnet / Page 176) Contact Magnet HSC for advice.



If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

Venting system



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

» Always ensure the cold head bypass valve is **CLOSED** during helium filling.

The document describes the procedure for magnets with **Mk1** (→ Fig. 143 Page 132), (→ Fig. 144 Page 133) and **Mk2** (→ Fig. 145 Page 133) venting systems.

Fig. 143: Mk1 OR105 Venting System



Fig. 144: OR122 Mk1 Venting System

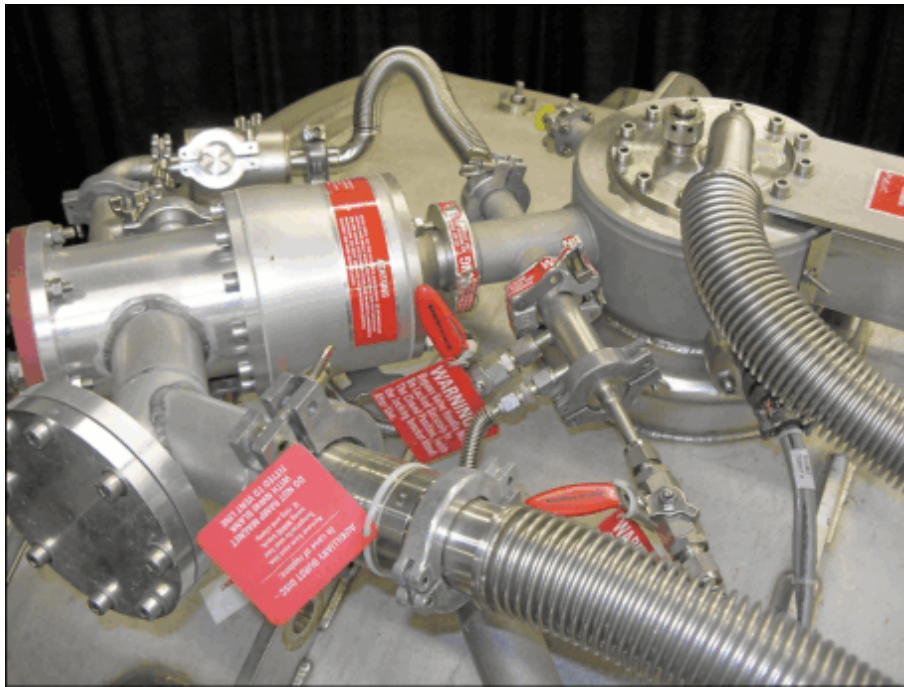


Fig. 145: Mk2 OR105 Venting System



5.2.1.2 Safety

 CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

- » Protective clothing, gloves and eye protection must be worn.
- » Filling must be performed by trained personnel only.

 WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the magnet room.
- » Never transport the GHe steel cylinder into the magnet room.



Ensure that the Quench Line has been correctly installed and safe before performing Installation and Startup procedures.

 WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, helium gas may be vented to cool the syphon.
- » Ensure that the turret venting is connected to quench pipe.
- » Oxygen monitors must be activated at all times.

 CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.

NOTICE

Mishandling of the dewar.

- » The dewar-relevant safety regulations of the manufacturer must be observed.



Monitor the filling procedure continuously by checking transfer pressure and the fill level.

Identify hazardous areas with warning signs.

**CAUTION**

Risk of quench during filling

If the magnet quenches during helium transfer

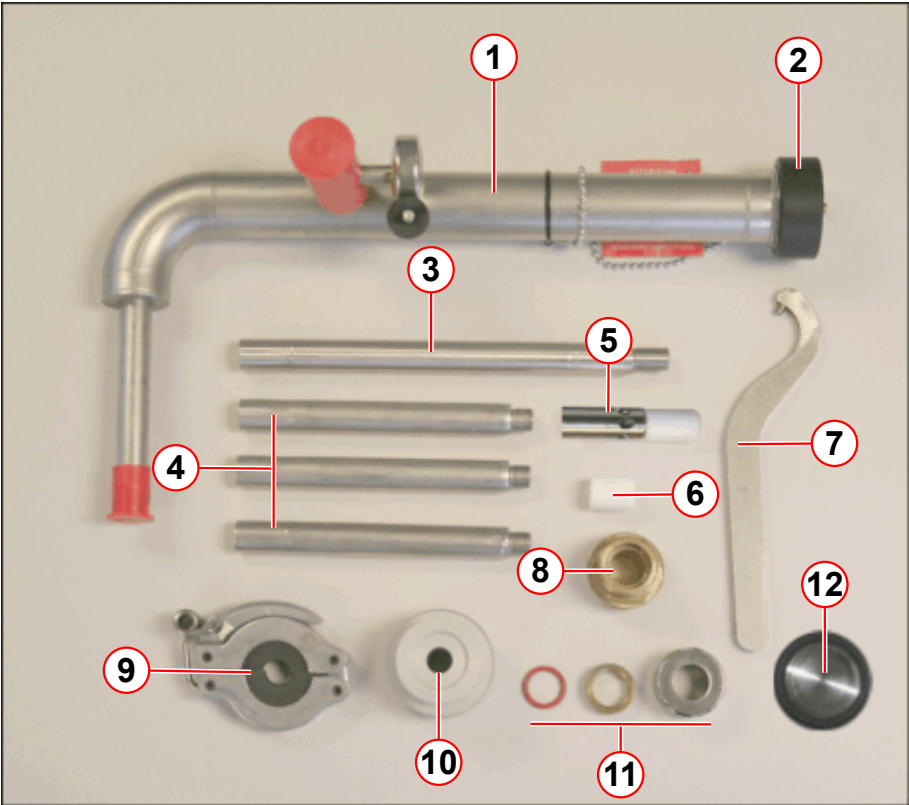
- » Stop the helium transfer if possible and safe to do so.
- » Follow Post Quench procedures



If the F3 fuse breaker is turned off during the filling procedure, there will be no power to the MSUP, and therefore no real time measurements can be taken.

5.2.1.3 Equipment required

Fig. 146: Magnet service syphon kit



- (1) Service syphon assembly
- (2) Syphon end cap
- (3) 200 mm extension leg (1 off)
- (4) 140 mm extension legs (3 off)
- (5) 6 hole helium fill diffuser tip (top fill)
- (6) 22mm PTFE seal (bottom fill)
- (7) Syphon C-spanner/wrench
- (8) M28 brass syphon adaptor bush (OR105, OR122)
- (9) Syphon leg support clamp (low ceiling)
- (10) Nitrogen fill adaptor and O-ring seal
- (11) Syphon O-ring seal, brass syphon flange, syphon ring nut
- (12) NW25 flow restricter

Tab. 10 Service syphon kit and upgrades

Part number	Revision	Comment
08396363	00	Service syphon kit
10126691	00	Upgrade kit (200 mm extension leg added).
10129438	00	Upgrade kit (de-ice tooling)
10126651	01	Upgrade Service syphon kit (includes upgrade kits)

- Magnet service syphon kit (part no **10126651 Revision 01**)

- Dewar syphon
- Dewar neck adaptor, clamps, seals, and hose
- Pressure service tool (part no **03861221**)
- MSUP-Display Unit
- Heat gun
- Paper towelling
- Helium gas supply (at least 99.9% pure)
- Pressure regulator 200 bar/ 0 - 1.5 bar
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes
- Replacement copper seal for syphon port on the turret
- Service platform
- Gas bottle wrench
- A room or personal O₂ Monitor

Mk2 venting only

- Syphon port adaptor (part no **10113551**)
- 30mm non-magnetic syphon spanner

5.2.1.4 Site preparation

- Ensure warning signs are correctly positioned.
- Ensure there is a clear path to exit the imaging suite.

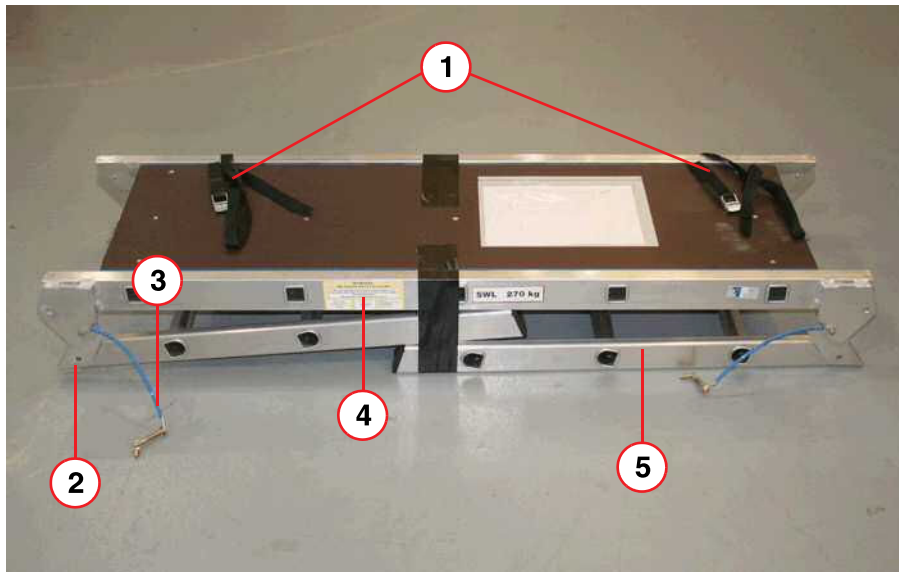
5.2.1.5 System preparation

Certain operations must be performed without delay to ensure efficient transfer:-

- Ensure that there is **open and clear access** to the magnet for **helium dewar(s)**.
- Ensure that there is **open and clear access** to the **syphon**.
- **Position** the **Service platform** next to the magnet. The platform must be secured to the magnet frame using the straps attached to the platform.
- Fit the MSUP-Display Unit to the MSUP to enable continuous monitoring of the pressure and LHe levels during this procedure. Ensure the MSUP is clearly visible.

Service platform installation

Fig. 147: Service platform - as delivered



- (1) Safety straps (2 off)
- (2) Hinged locking pin location (each end)
- (3) Retaining wire and hinged locking pin (each end)
- (4) Warning label
- (5) Step legs (each end)

Service platform safety

Before using the service platform **read the following safety cautions** and perform the **required safety checks**.



CAUTION

Risk of personal injury

Failure to observe the following precautions may result in personal injury.

- » Always check the platform and hinge locking pins for damage.
- » Always check the hinge locking pins are secured.
- » Always secure the platform to the magnet frame or with the two long straps through the magnet bore.
- » Always ascend and descend (Walk UP or DOWN) the steps facing the platform.
- » Where possible use the frame or covers for additional hand support.
- » Never climb onto the platform from the side.
- » Never jump from the steps or top of the platform.

Fig. 148: Correct method - Up to platform



Fig. 149: Incorrect method - Down from platform

**⚠ CAUTION****Risk of personal injury**

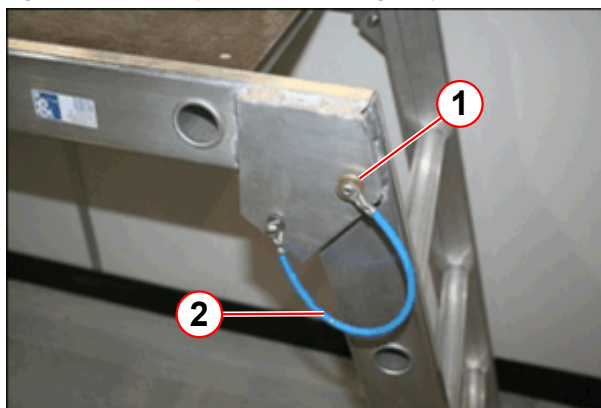
Carrying heavy components or tooling up or down the platform steps can lead to injury.

- » DO NOT carry heavy components up or down the platform steps.
- » Place heavy objects (i.e. cold head) onto the platform first.
- » Only lift the component when standing safely on the platform.

Platform assembly

- Remove any packaging.
- Unfold both step legs.

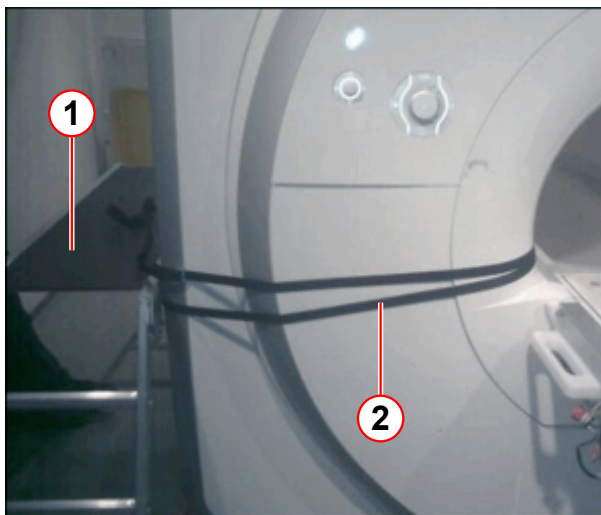
Fig. 150: Service platform - lock legs in position



- (1) Hinged locking pin
(2) Retaining wire

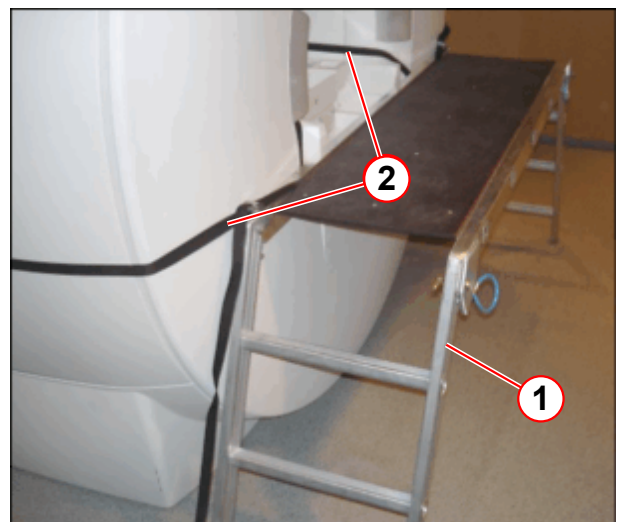
- Align the hole at the top of the leg with the **pin location hole** (pos 1).
- Push the **hinged locking pin** through the hole (pos 1) and fold back the hinge to secure.
- Repeat with the other leg.

Fig. 151: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)

Fig. 152: Securing the service platform (example)



- (1) Service platform
(2) Long straps (fitted through the magnet bore)



Do not exceed the maximum load - the combined weight of 2 people must not exceed 270 kg.



The fiber optic connections in the top of RFIS are very fragile, so exercise due caution.

Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

It covers all dewars that are externally pressurized with helium gas.



Refer to the dewar manufacturer's documentation when using side fill and/or self-pressurizing dewars.



Refer to the dewar manufacturer's safety guidelines and check the dewar valve functions before commencing the LHe fill.

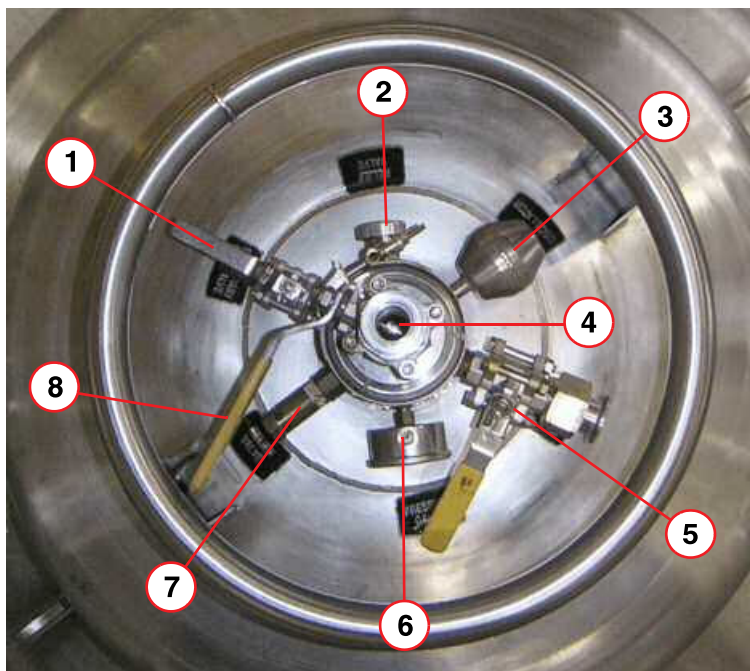
- If the dewar is **highly pressurized**, (greater than 0.3bar (4.4psi G)), release the pressure by slowly opening the **primary ball valve** (→ 5/Fig. 153 Page 141) to minimize the flash losses of helium.
 - This must be done in a **well ventilated area - NOT in the imaging suite.**



Check for the cause of the high pressure, i.e. was the low pressure relief valve closed? Inform the supplier if the cause cannot be found.

- Fit the **dewar syphon adaptor** (→ 3/Fig. 155 Page 143) to the top of the dewar.

Fig. 153: Dewar Ports (typical)

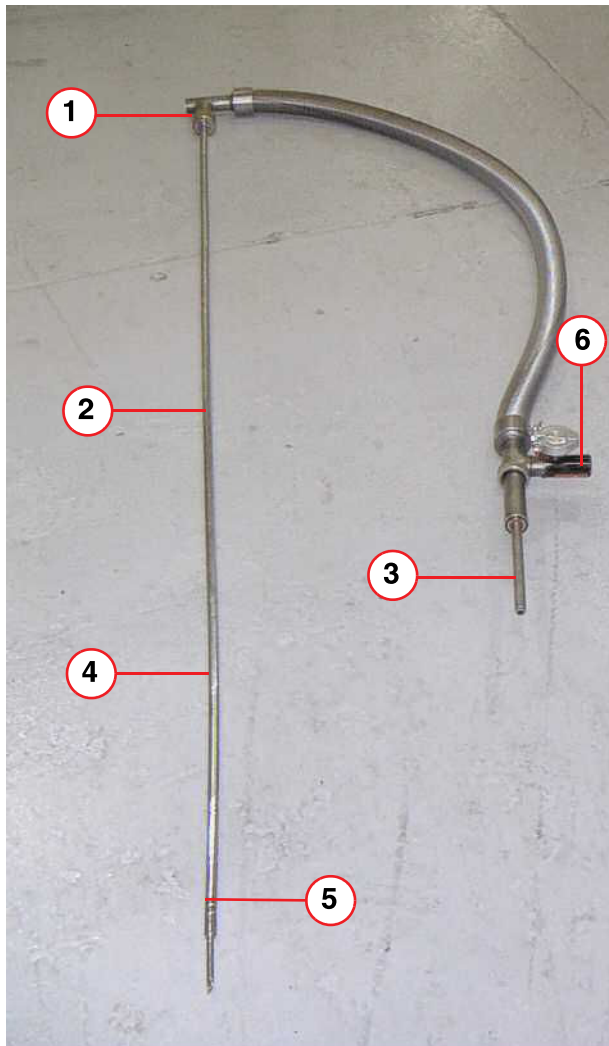


- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- **Close the primary ball valve** (→ 5/Fig. 153 Page 141) and **low pressure relief valve.** (→ 1/Fig. 153 Page 141)

Dewar syphon preparation

Fig. 154: Dewar syphon



- Fit the extension leg (s) and sludge inhibitor, to the dewar syphon leg.

- (1) Syphon flow valve
- (2) Syphon leg
- (3) Bayonet connection (male)
- (4) Extension leg
- (5) Sludge inhibitor
- (6) Syphon vacuum valve

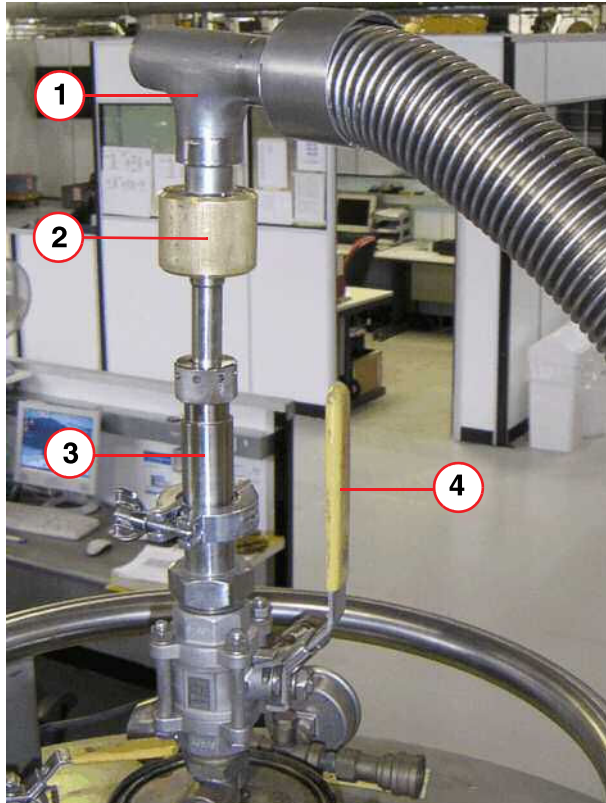


Do not open the syphon vacuum valve (6).

- Fit the **helium regulator** to the helium gas bottle.

Pre-cool the dewar syphon

Fig. 155: Dewar syphon fitment

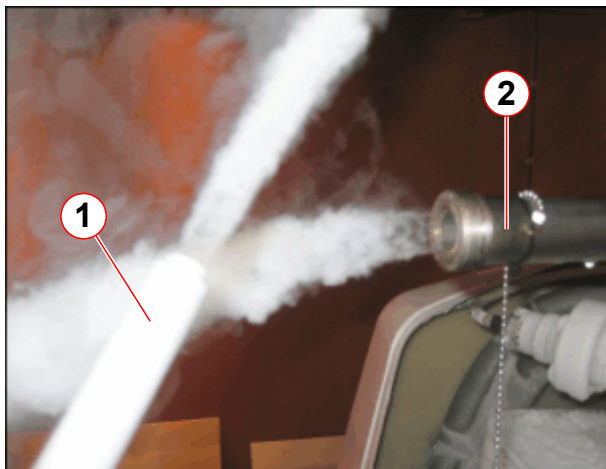


- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- Open the **syphon flow valve** (pos 2) two turns.
 - Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.
(Do not fully tighten the ring nut at this stage).
 - Open the **syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.
-
- Push the syphon **fully down** into the dewar, (very slowly to minimize pressure build-up and flash losses), until the sludge inhibitor touches the bottom.

- Fully tighten the **ring nut and seal**.

Fig. 156: Pre-cooling the syphons

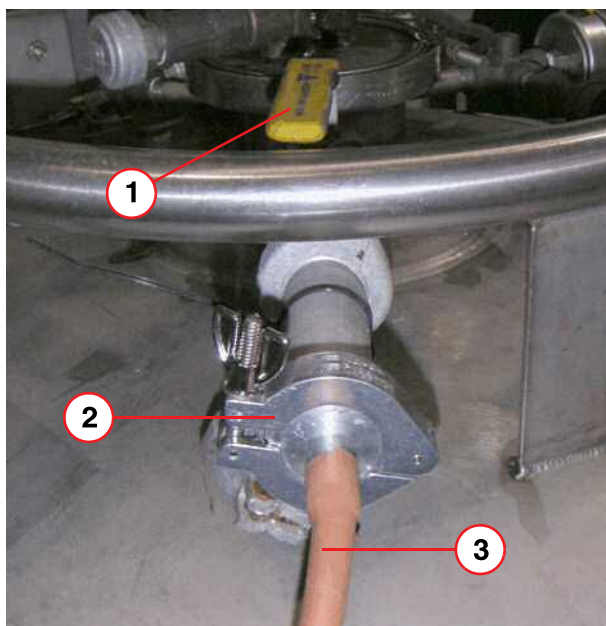


(1) Dewar syphon
(2) Service syphon

- Allow the syphon to cool until a dense white "cascading plume" appears.
- **Do not** allow high pressure to build in the dewar, only enough to cool the syphon.

- When the syphon is cold, close the **syphon flow valve** (→ 2/ Fig. 155 Page 143) and move the dewar so that the syphon can be connected without excessive bending.

Fig. 157: Dewar gas connection



(1) Primary ball valve
(2) NW25 clamp and seal
(3) Gas supply line

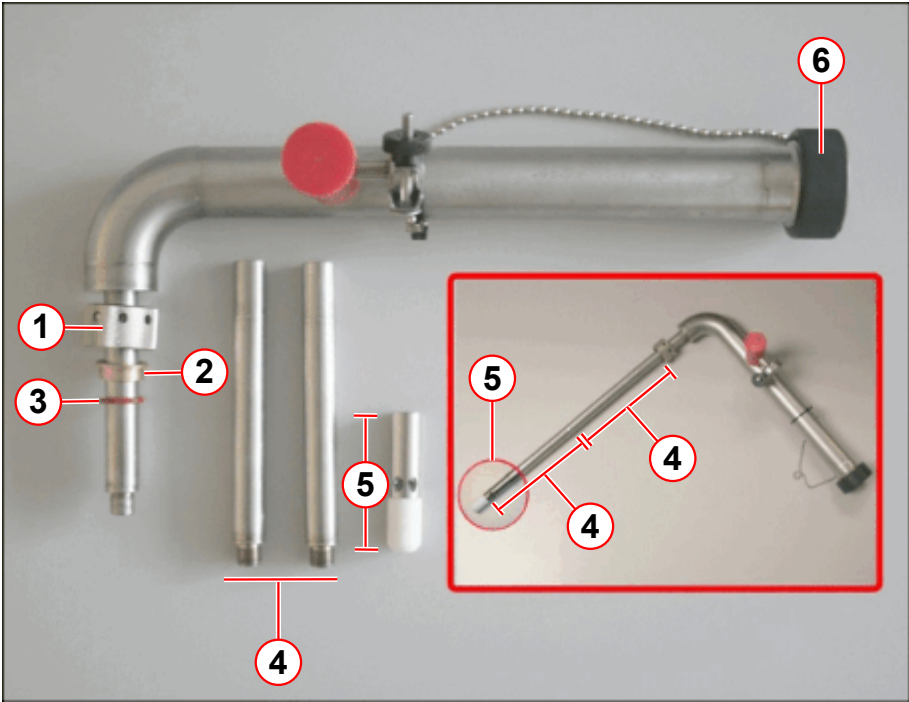
- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the magnet trailer end, if the magnet is at field.
- Purge the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an NW25 clamp and seal (pos 2).

- **Open the primary ball valve** (→ 5/ Fig. 153 Page 141) and pressurize the dewar to **0.27bar (4 psi (G))**.

Assemble the Service syphon

Prepare the syphon first, before depressurizing the helium vessel.

Fig. 158: Magnet service syphon assembly - top filling

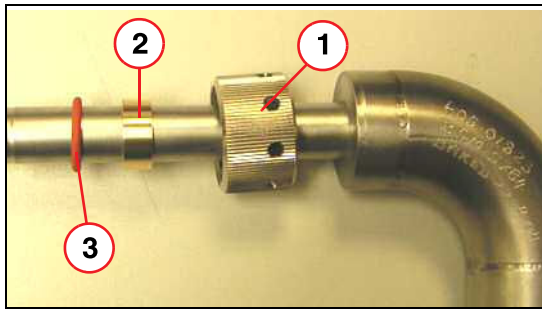


- (1) Ring nut
- (2) Brass flange
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) De-ice and helium fill diffuser and PTFE tip
- (6) Syphon end cap

Tab. 11 OR105 / OR122 TOP filling syphon extension legs

Magnet type	130* mm leg	60 mm De-ice & helium fill diffuser & PTFE tip
OR105 Avanto	2	1
OR122 Espree		
The syphon diffuser should optimally be 50-80mm above the syphon cone for top filling .		
Note: *Effective extension leg length		
If damaged, the PTFE diffuser tip must be replaced.		

Fig. 159: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer, and the O-ring to the **service syphon leg**.

Refer to table above (→ Tab. 11 Page 145)

- Connect the **extension legs** and diffuser:
 - Screw the (2 off) **140 mm legs** together.
 - Connect the leg to the **syphon**.
 - Fit the **60 mm diffuser** and **PTFE tip**.

5.2.1.6 Magnet with Mk1 Venting System

CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure the cold head bypass valve is CLOSED during helium filling.

Fit the magnet service syphon

CAUTION

Pressurized helium vessel.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the fittings without first depressurizing the helium vessel.

De-pressurize the helium vessel:-

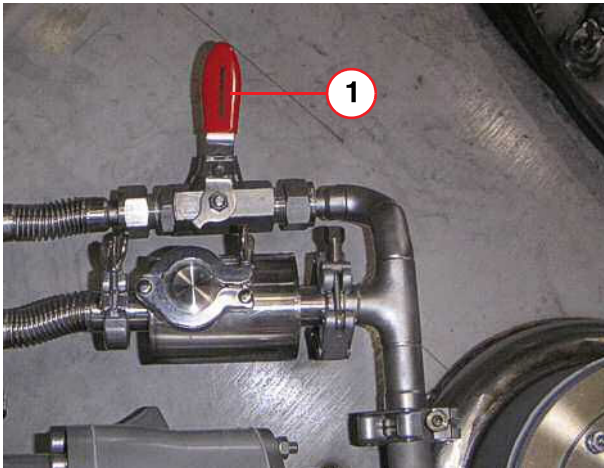
- De-activate the **pressure heater**.

i

If the pressure service tool is not available, disable the pressure heater by removing the F3 from the ACC cabinet. This also switches off the compressor.

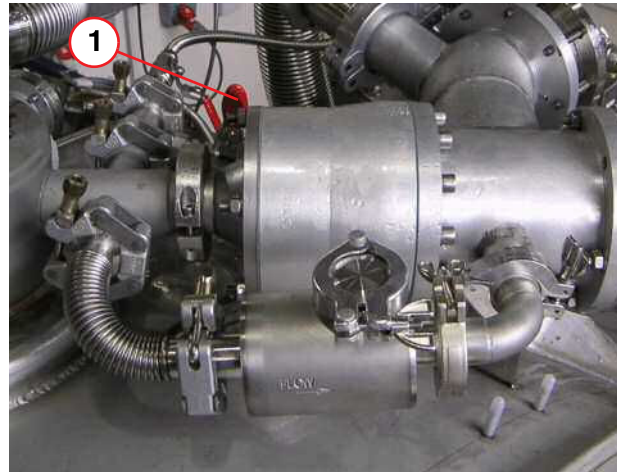
It may not be possible to measure, or monitor helium level during the transfer. This may result in helium wastage by overfilling the magnet.

Fig. 160: OR105 Mk1 Venting bypass valve



(1) Bypass valve (shown closed)

Fig. 161: OR122 Mk1 Venting bypass valve



(1) Bypass valve (shown closed)

- **Open** the **magnet bypass valve** slowly to depressurize the helium vessel.
- Note the **gauge pressure** when the **compressor** switches **off**.

Fig. 162: Mk1 Pressure manometer



(1) Scale of helium pressure

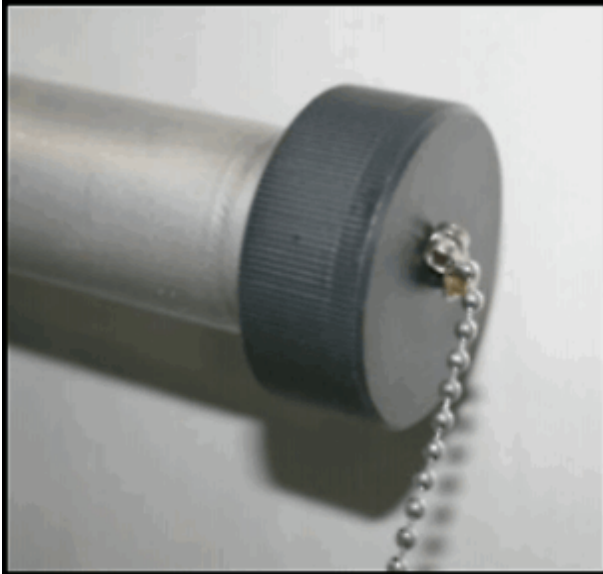
- **Close** the magnet bypass valve.
- Check the helium vessel has a **small positive pressure**.

(Note: This picture shows normal operating pressure).

- Warm and dry the **syphon blanking plug** using a heat gun to ease removal.

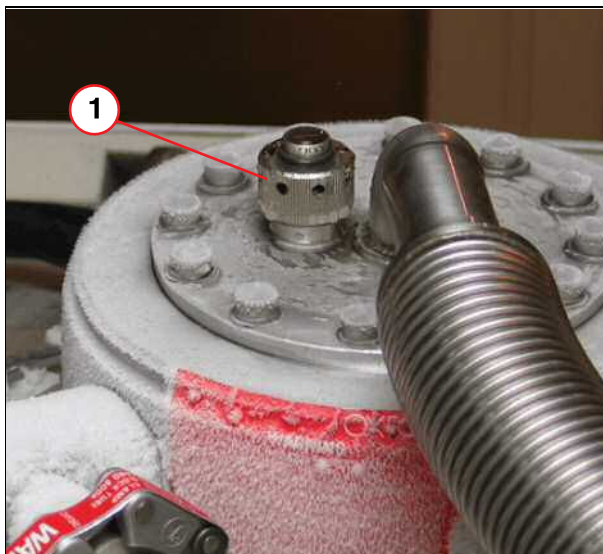
- Dry the syphon port.

Fig. 163: Syphon end cap



- Remove the syphon end cap.

Fig. 164: Mk1 Syphon entry port



- Remove the syphon blanking plug using the syphon C-spanner.

(1) Syphon blanking plug



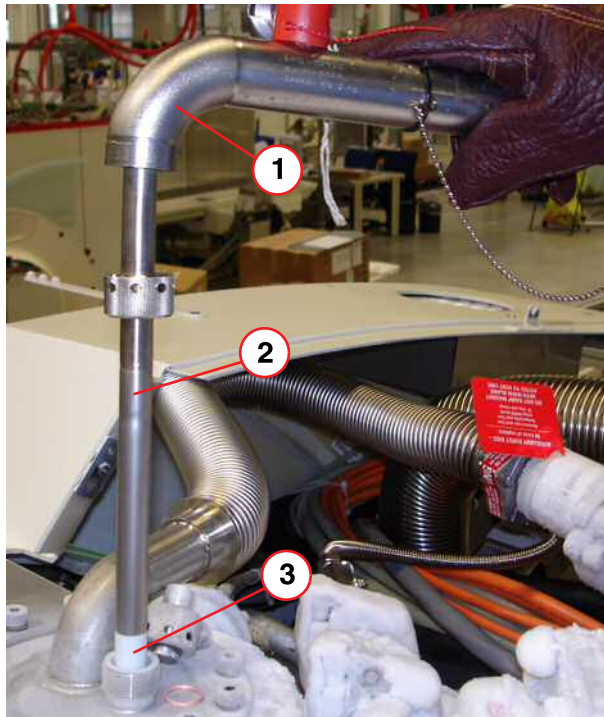
CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

Fig. 165: Helium syphon



- (1) Service Syphon
 (2) Two Extension Legs
 (3) PTFE Syphon Diffuser

- **Insert** the service syphon into the port without delay.
- Slowly lower the service syphon fully into the magnet.
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon C-spanner to form a **gas tight seal**.



There is a risk of quenching the magnet if the syphon is inserted too quickly.

- **Re-fit** the syphon **end cap**.



If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

Pre-cool the service syphon



The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use, refer to (→ Dewar preparation / Page 140)).

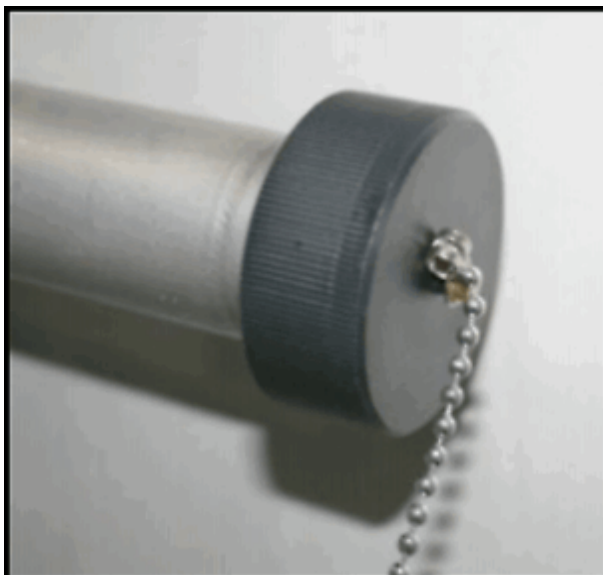
**CAUTION****Risk of quench**

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Cool down the **service syphon** as follows:-

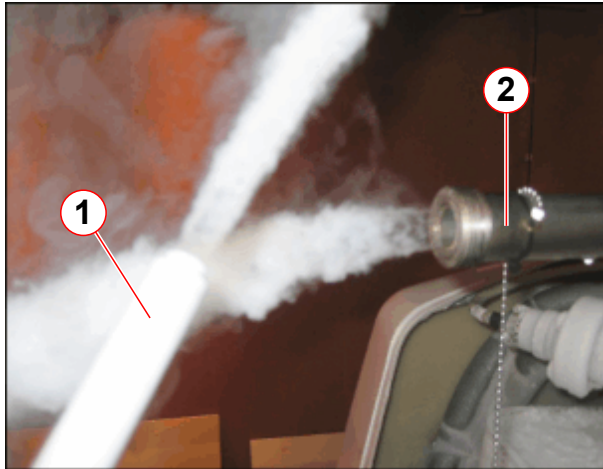
Fig. 166: Syphon end cap



- Ensure the **syphon end cap** is fitted and the **bypass valve** is **closed**.
- **Activate** the **pressure heater** to achieve a positive pressure of about **0.3 psi (A)** in the magnet.

- **De-activate** the pressure heater.

Fig. 167: Pre-cooling the syphons



(1) Dewar syphon
(2) Service syphon

- **Remove the syphon end cap.**
- Allow the syphon to vent until **fully cold**.
 - a "cascading plume" of helium gas appears.



CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

Liquid helium transfer - top fill



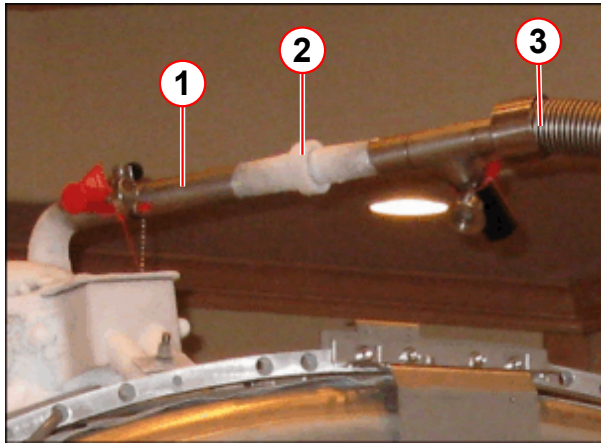
Before helium transfer, check the (→ Helium pressures at altitude / Page 107)

- Check the dewar gas supply pressure is set at **0.27bar (4 psi (G))**.
- As the **service syphon** nears the point of being fully cold:
 - open the **dewar syphon flow valve** half way (2 turns). (→ Fig. 169 Page 153)
 - **pre-cool the dewar syphon** again until a dense white "cascading plume" appears.



For helium filling magnets at ZERO FIELD, it is not necessary to pre-cool the service and dewar syphons.

Fig. 168: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, connect them together and secure the lock nut.
- OPEN the magnet bypass valve. This will:
 - » Improve efficiency.
 - » Improve the filling speed.

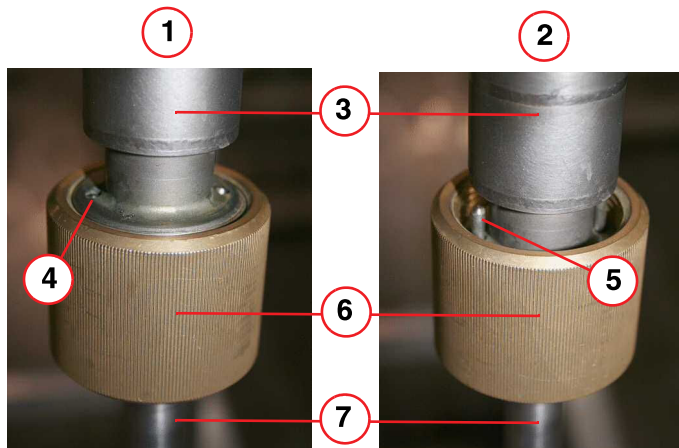
CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Fig. 169: Dewar syphon flow valves - open and closed positions



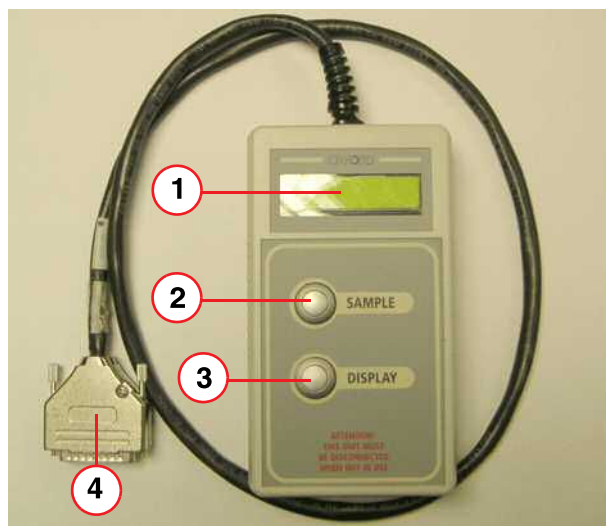
- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

Fig. 170: Helium filling



- Fully open the **dewar syphon flow valve**.
 - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (→ 4/ Fig. 169 Page 153)
 - » From fully **closed** to **fully open**, is **4 turns**.

Fig. 171: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

- Take an initial **helium level measurement** from the MSUP-Display Unit.
- Repeat the level measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - If not, stop the transfer and investigate the reason for poor transfer efficiency.
- **NOTE:** If F3 has been switched off, measurements can be taken, but only on battery for 2 minutes.



Maintain a pressure of about 0.27 bar (4 psi (G)) at the helium dewar.
Too much or too little pressure will result in inefficiency.

At **4 psi (G)**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**



CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Stopping the Helium Transfer

Fig. 172: Dewar manometer



- As soon as the level reads 95%, or the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** (→ 6/ Fig. 169 Page 153)
- Set the **dewar** gas pressure regulator to **zero**.
- Close the dewar **primary ball valve**. (→ 1/ Fig. 157 Page 144)
- Refer to (→ Changing supply dewars during transfer / Page 168) if the magnet is not full.



Do not fill above 95%.

Overfilling the magnet will cause the helium level to show S/C (short circuit).

If S/C is displayed, allow helium to boil off until a correct helium reading is confirmed. If this takes longer than 10 minutes contact Magnet HSC for advice.

The magnet may be ramped at a higher helium level as long as S/C is not displayed.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- » **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

Removing the service syphon

- **Open the magnet bypass valve** very slowly to depressurize the helium vessel to **0.1 psi (G) (7 mbar)** above atmospheric pressure.
- **CLOSE the magnet bypass valve.**
 - If the magnet bypass valve was open during the filling, leave it open until the magnet pressure has dropped to **0.1 psi (G) (7 mbar)**.
- Warm the **syphon joint** to room temperature (using a heat gun), to remove ice.
- Warm the **syphon entry valve** to room temperature and dry.
- Warm the **syphon end cap** ready for fitment onto the syphon.
- **De-couple the syphons.**
 - ensure the magnet bypass valve is closed **BEFORE** disconnecting the syphons.



CAUTION

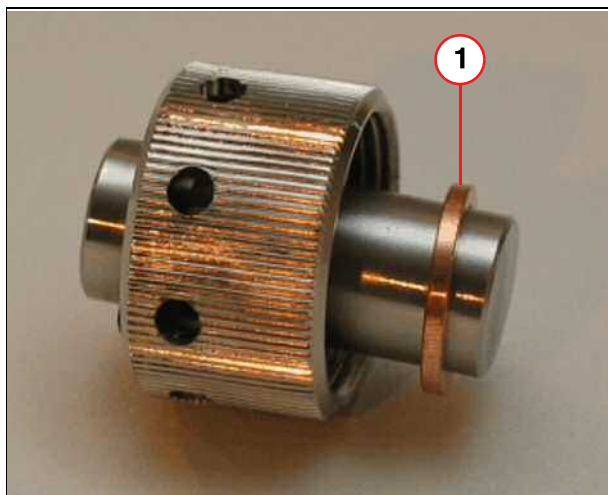
Risk of cold burns.

Cold gas will exhaust from the syphons when they are disconnected.

- » **Stand clear of the syphon joint when disconnecting.**
- » **Wear protective gloves and safety glasses.**

- **Close off** the end of the service syphon with the **syphon end cap**.
- **Immediately** remove the service **syphon** from the magnet.

Fig. 173: Mk1 syphon blanking plug



(1) Copper seal

Fig. 174: Mk1 Syphon entry port



(1) Syphon 'C' spanner

- Fit the **syphon blanking plug** using a new copper seal.
- **Warm** up the threads on the plug and **carefully** tighten with the syphon C-spanner.
 - There might be ice on the threads - **do not use undue force**. Damaged threads will result in a leak.
- **Warm** the **top plate** and **syphon plug** using a heat gun. Re-tighten the plug.
- **Warm** up the magnet bypass **valve assembly** and **check** the bypass valve is **closed**.

⚠ CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- **Re-activate** the pressure heater.
- Scroll to '**Pressure heater**' (Off/On,%) on the MSUP-Display Unit (#12).
 - Press the **SAMPLE** button to toggle on **100% duty cycle**.
 - When the pressure is at **15.3 psi (A)**, or above, press the **SAMPLE** button to toggle back to **20% duty cycle**.



Failure to do this may result in exceeding the 2 watt limit in the MSUP and disabling the pressure heater. An MSUP reset will be required to re-enable the heater.

- Check that the **pressure heater** activates with a 20% duty cycle. The yellow LED on the MSUP should also illuminate at the same duty cycle.
- Check that the **compressor** activates automatically as the pressure rises above atmospheric. Note the pressure at which it starts.

- **Warm the turret and venting valves** to room temperature with a heat gun.
 - remove any surplus water with paper towelling.



Wait at least 2 hours after completion of a helium fill before attempting to ramp the magnet.

- **Move the dewar** out of the imaging suite into a well ventilated area.
- **Remove** the gas line from the dewar.(→ 3/Fig. 157 Page 144)
- Complete depressurizing the dewar by opening the **primary ball valve**. (→ 5/Fig. 153 Page 141)
- When the gas pressure is at **zero**, remove the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar. (→ 4/Fig. 153 Page 141)
- **Open** the **low-pressure dewar relief valve**. (→ 1/Fig. 153 Page 141)
- **Close** the **primary dewar ball valve**. (→ 5/Fig. 153 Page 141)
- **Warm both syphons** with the heat gun until dry, and store safely.

5.2.1.7 Magnet With Mk2 Venting System



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure the cold head bypass valve is **CLOSED** during helium filling.

Fit the magnet service syphon



CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- » **Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.**

De-pressurize the helium vessel:-

- De-activate the **pressure heater**.



If the pressure service tool is not available, disable the pressure heater by removing the F3 from the ACC cabinet. This also switches off the compressor.

It may not be possible to measure, or monitor helium level during the transfer. This may result in helium wastage by overfilling the magnet.

Fig. 175: Mk2 Venting bypass valve



(1) Bypass valve (show open)

- Check the helium vessel has a **small positive pressure**.

Fig. 176: Mk2 Pressure Manometer

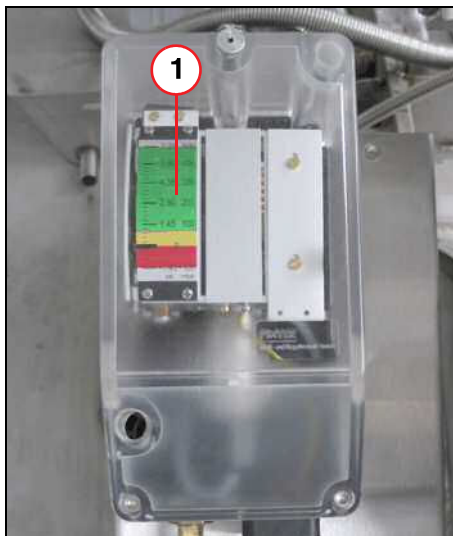


Fig. 177: WIKKA Pressure switch - with a small positive pressure

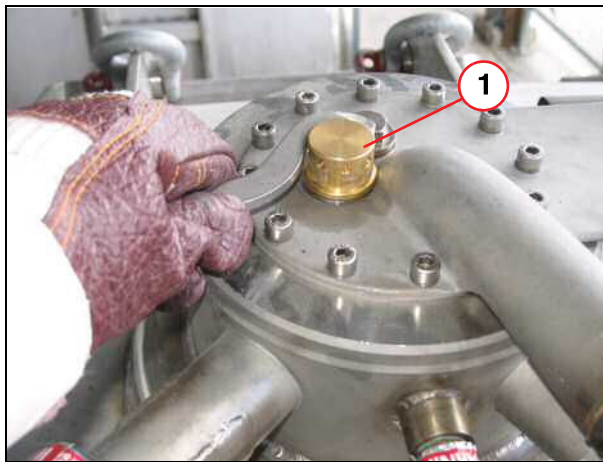


Fig. 178: Mk2 Venting system



- Warm the **blanking plug** using a heat gun to ease removal.
- Dry the syphon port.
- Remove the syphon **end cap**.

Fig. 179: Mk2 Syphon port port



- Remove the **syphon blanking plug** using the syphon C-spanner.

(1) Syphon blanking plug



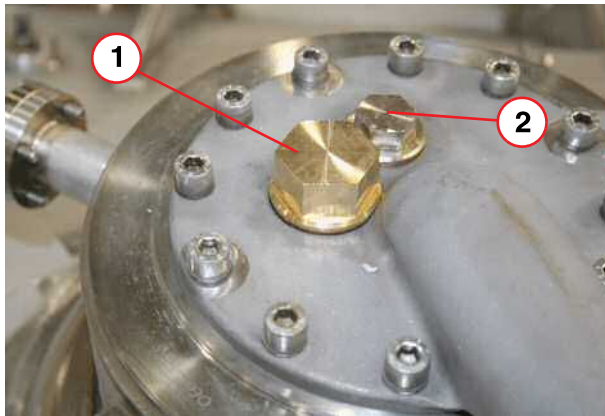
CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

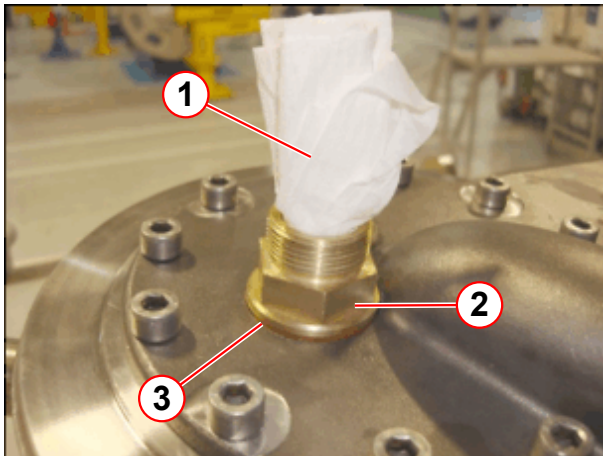
Fig. 180: Mk2 Syphon port entry



- (1) Syphon blanking plug
- (2) Sight glass

- On later systems the blanking plug has **hexagonal** flats and a non-magnetic 30mm spanner is needed.

Fig. 181: Mk2 - Fill adaptor plugged with paper towelling



- (1) Compacted paper towelling
- (2) Fill adaptor
- (3) Copper seal

- To prevent **air ingress**, plug the brass adaptor with **paper towelling** prior to fitment.

- Fit the **brass syphon fill adaptor** and a new **copper seal**.

Fig. 182: Mk2 Syphon fitment



- **Remove** the plug of **paper towel-ling** from brass adaptor port. Ensure the port is **clean** and **dry**.
- Guide the syphon into the **syphon fill adaptor**.



There is a risk of quenching the magnet if the syphon is inserted too quickly.

Fig. 183: Mk2 Syphon fitment



- **Slowly** lower the service syphon fully into the neck.
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon C-spanner to form an air-tight seal.
- Re-fit the **syphon end cap**.

Pre-cool the service syphon



The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use, refer to (→ Dewar preparation / Page 140)).

**CAUTION****Risk of quench**

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Cool down the **service syphon** as follows:-

- Ensure the **syphon end cap** is fitted, and the bypass valve is **closed**.
- Activate the **pressure heater** to achieve a positive pressure of about **0.3 psi (A)** in the magnet.
- **De-activate** the pressure heater.

Fig. 184: Syphon venting (cascading plume)



- Remove the service **syphon end cap** and allow the syphon to vent until it is fully cold and a dense white "cascading plume" appears.

**CAUTION****Risk of cold burns.**

The "cascading plume" of gas is very cold.

- » Wear protective gloves and safety glasses.

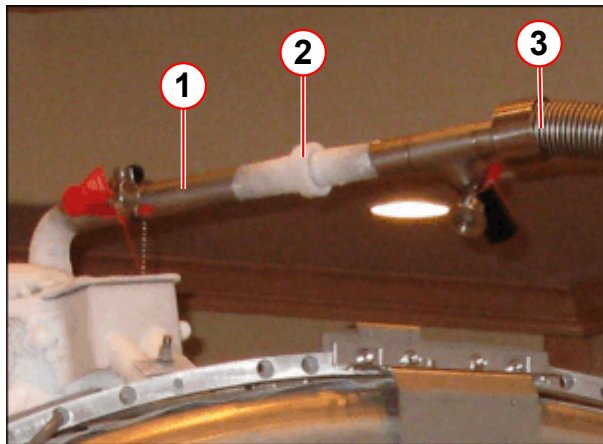
Liquid helium transfer - top fill



Before helium transfer, check the (→ Helium pressures at altitude / Page 107)

- Check the dewar gas supply pressure is set at **0.27bar (4 psi (G))**.
- As the **service syphon** nears the point of being fully cold, open the **dewar syphon flow valve** half way (2 turns) (→ Fig. 186 Page 164), and **pre-cool the dewar syphon** again until a dense white "cascading plume" appears

Fig. 185: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, connect them together and secure the lock nut.



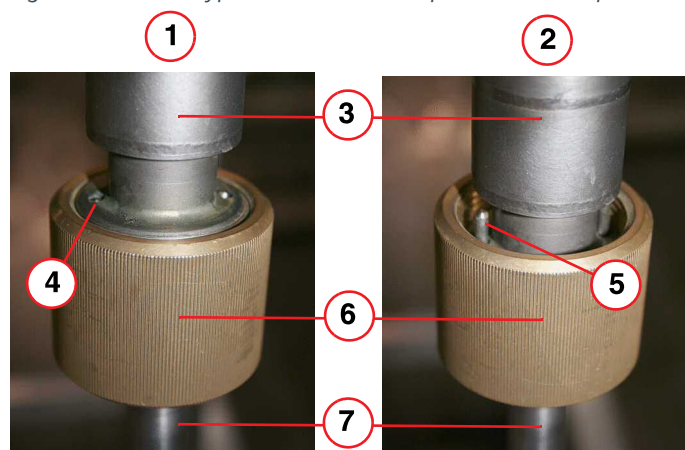
CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Fig. 186: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

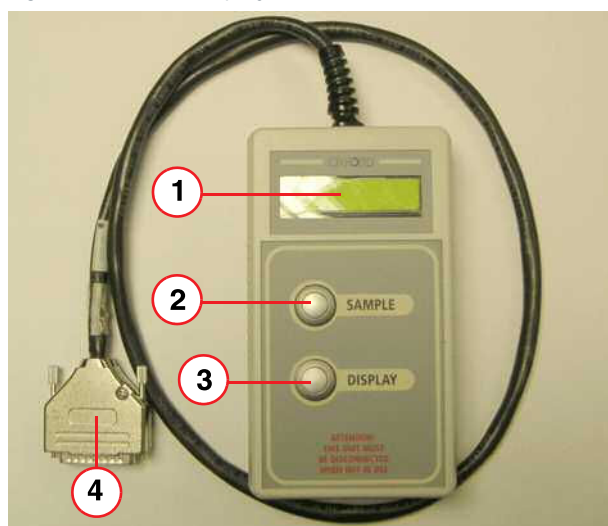
Fig. 187: Helium filling



- Fully open the **dewar syphon flow valve**.

- » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (→ 4/ Fig. 186 Page 164)
- » From fully closed to fully open, is **4 turns**.

Fig. 188: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

- Take an initial helium level measurement from the MSUP-Display Unit .
- Repeat the level measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - If not, stop the transfer and investigate the reason for poor transfer efficiency.
- **NOTE: If F3 has been switched off, measurements can be taken, but only on battery for 2 minutes.**



Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.



Maintain a pressure of about 0.27bar (4psig) at the helium dewar.
Too much or too little pressure will result in inefficiency.

At **4 psi (G)**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**



CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Stopping the helium transfer

Fig. 189: Dewar manometer



- As soon as the level reads 95%, or the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** (→ 6/ Fig. 186 Page 164)
- Set the dewar gas pressure regulator to **zero**.
- Close the dewar **primary ball valve.** (→ 1/ Fig. 157 Page 144)
- Refer to (→ Changing supply dewars during transfer / Page 168) if the magnet is not full.



Do not fill above 95%.

Overfilling the magnet will cause the helium level to show S/C (short circuit).

If S/C is displayed, allow helium to boil off until a correct helium reading is confirmed. If this takes longer than 10 minutes contact Magnet HSC for advice.

The magnet may be ramped at a higher helium level as long as S/C is not displayed.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- » **Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.**

Removing the Service Syphon

- **Open the magnet bypass valve** very slowly to depressurize the helium vessel to **0.1 psi (G) (7 mbar) above atmospheric pressure.**
- **CLOSE the magnet bypass valve.**
 - If the magnet bypass valve was open during the filling, leave it open until the magnet pressure has dropped to **0.1 psi (G) (7 mbar).**
- **Warm the syphon joint** to room temperature (using a heat gun).
- **Warm the syphon entry port** to room temperature and dry.
- **Warm the syphon end cap** ready for fitment onto the syphon.
- **De-couple the syphons.**
 - ensure the magnet bypass valve is closed **BEFORE** disconnecting the syphons.



CAUTION

Risk of cold burns.

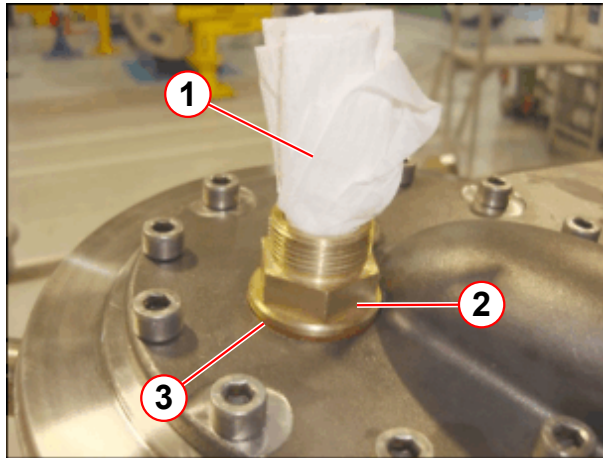
Cold gas will exhaust from the syphons when they are disconnected.

- » **Stand clear of the syphon joint when disconnecting.**
- » **Wear protective gloves and safety glasses.**

- **Close off the end of the service syphon with the syphon end cap.**

- **Immediately** remove the service syphon from the magnet.

Fig. 190: Mk2 - Fill adaptor plugged with paper towelling



- (1) Compacted paper towelling
- (2) Fill adaptor
- (3) Copper seal

- To prevent **air ingress**, temporarily block the brass **adaptor port** using a plug of **paper towelling**.

Fig. 191: Mk2 Blanking plug fitment



- Remove the brass **syphon fill adaptor** from the turret top plate and **immediately** fit the **syphon blanking plug** using a **new** copper seal.
- **Warm up the threads** on the plug and **carefully** tighten them with the 30mm non-magnetic spanner.
 - There may be ice on the threads- **do not use undue force**. Damaged threads will result in a leak.

- Warm the **top plate** and **syphon plug** using a heat gun. Re-tighten the plug.
- Warm up the magnet bypass **valve assembly** and **check** the bypass valve is **closed**.



CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.

- **Re-activate** the pressure heater.
- Scroll to '**Pressure heater**' (Off/On,%) on the MSUP-Display Unit (#12).
 - Press the **SAMPLE** button to toggle on **100% duty cycle**.

- When the pressure is at **15.3 psi (A)**, or above, press the **SAMPLE** button to toggle back to **20% duty cycle**.



Failure to do this may result in exceeding the 2 watt limit in the MSUP and disabling the pressure heater. An MSUP reset will be required to re-enable the heater.

- Check that the **pressure heater** activates with a 20% duty cycle. The yellow LED on the MSUP should also illuminate at the same duty cycle.
- Check that the **compressor** activates automatically as the pressure rises above atmospheric. Note the pressure at which it starts.
- **Warm** the **turret** and **venting valves** to room temperature with a heat gun.
 - remove any surplus water with paper towelling.



Wait at least 2 hours after completion of a helium fill before attempting to ramp the magnet.

- **Move the dewar** out of the imaging suite into a well ventilated area.
- Remove the **gas line** from the dewar.(→ 3/Fig. 157 Page 144)
- Complete depressurizing the dewar by opening the **primary ball valve**. (→ 5/Fig. 193 Page 170)
- When at the gas pressure is at **zero**, **close** the **primary ball valve** on the dewar.
- Remove the dewar syphon.
- **Immediately** close the **syphon entry valve** on top of the dewar. (→ 4/Fig. 193 Page 170)
- **Open** the **low-pressure dewar relief valve**. (→ 1/Fig. 193 Page 170)
- **Warm** both syphons with the heat gun until dry, and store safely.

Changing supply dewars during transfer

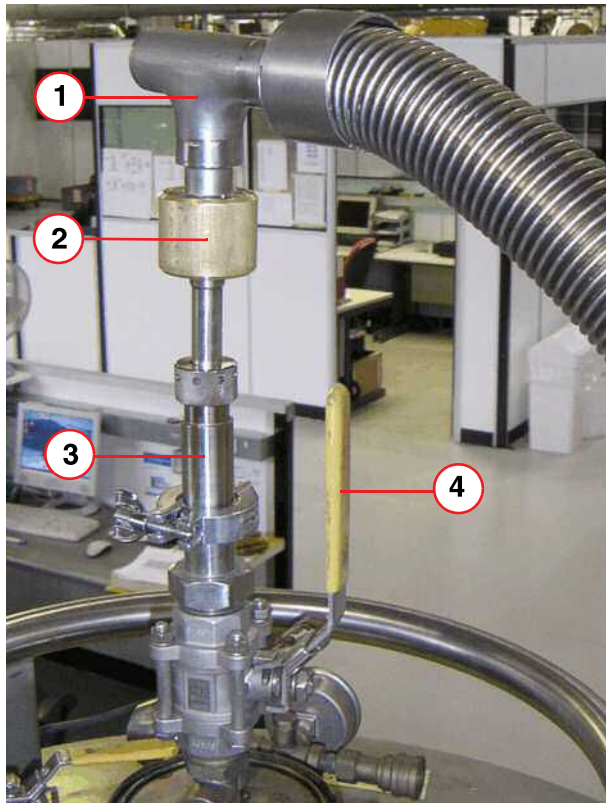


Whistling of the syphon (a high pitched sound), or a noticeable drop in dewar pressure are indications of an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, ready to receive the syphon.

Fig. 192: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- When the dewar empties, stop the transfer **immediately** by closing the **dewar syphon flow valve** (pos 2).



CAUTION

Risk of quench

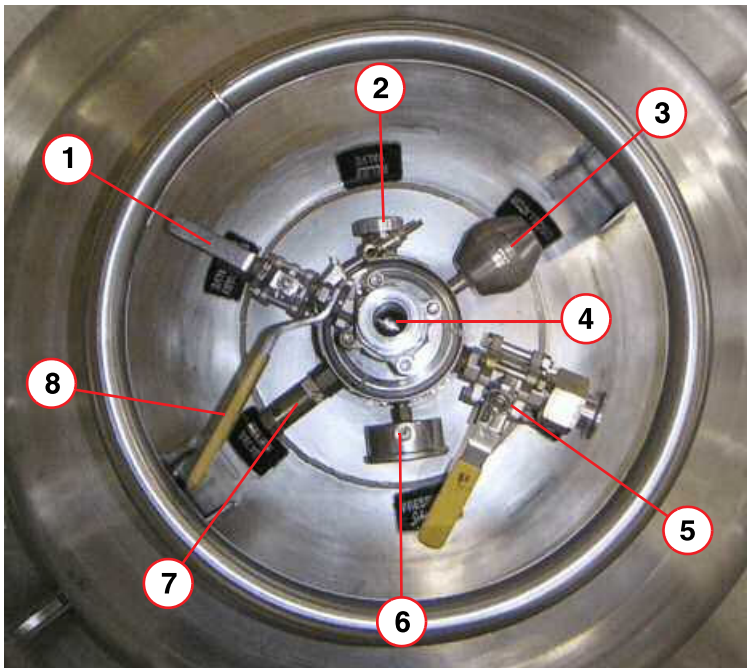
Failure to close the valve immediately, when the magnet is at field, will result in a quench.

- » Close the valve immediately.

- Turn **off** the dewar gas supply:
 - remove the **gas connection**.
 - vent the dewar down to atmospheric pressure.
 - **Ensure the area is well ventilated.**

- Close the primary dewar ball valve. (→ 5/ Fig. 193 Page 170)

Fig. 193: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
 - (2) Relief valve
 - (3) Anti-oscillator
 - (4) Syphon entry port (shown closed)
 - (5) Primary ball valve (shown closed)
 - (6) Pressure gauge
 - (7) Relief valve
 - (8) Syphon entry valve
- Using a heat gun, **warm the syphon connecting joint** to room temperature.
 - **Disconnect** the syphons.

CAUTION

Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting.
- » Wear protective gloves and safety glasses.

- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **service syphon end cap**.

**CAUTION**

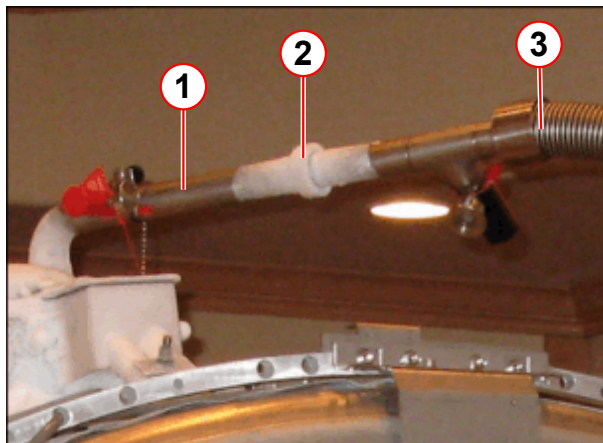
Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Ensure the syphons are capped and plugged immediately and securely.

- Move the empty dewar close to the full dewar.
- Vent both storage dewars down to **atmospheric pressure**.
 - » This must be done in a well ventilated area - NOT in the imaging suite.
- Open the **low pressure relief valve** on the **empty** dewar. (→ 1/ Fig. 193 Page 170)
- Remove the dewar syphon and adaptor from the empty dewar and **immediately** Insert the syphon leg into the full dewar.
- Close the **syphon entry valve** (→ 4/ Fig. 193 Page 170), of the **empty** dewar. and **open the low pressure relief valve**. (→ 1/ Fig. 193 Page 170)
- Close the **low pressure relief valve** of the **full** dewar.
- Purge the gas supply line.
- Connect to the full dewar and **open the dewar primary ball valve**.
- Set the **gas pressure** to 0.27bar (**4 psi (G)**).
- **Pre-cool the dewar syphon**. (→ Pre-cool the dewar syphon / Page 143)
- **Move the dewar** next to the magnet system.
- **Pre-cool the service syphon**. (→ Pre-cool the service syphon / Page 149)

Fig. 194: Dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, connect them together and secure the lock nut.

- Continue with the transfer.

5.2.1.8 Final work steps

After transfer, check:-

- The **magnet bypass valve** is **closed** and **locked**.

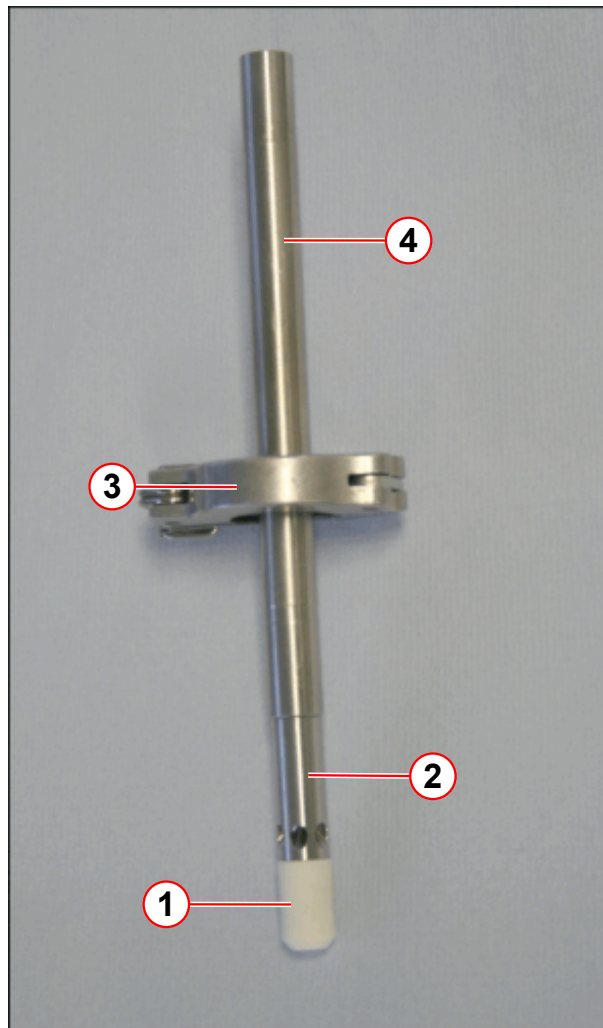
- The **pressure heater** is operating correctly.
- The pressure heater **LED** on the **MSUP** is illuminated.
- **No alarm signals** are evident on the MSUP-Display Unit, or the **Alarm box**.
- The **compressor** is switched **on** and running.
- **Manually** operate the **pressure heater** to repressurize the magnet.
 - Clear average power by manually turning off pressure heater at **15.3 psi (A)**.
- The **magnet pressure** is stable at **15.3 psi (A)**.
- The **turret** and **venting** system are **warm and dry**.
- The MSUP-Display Unit is disconnected.
- The filling port and venting system have no leaks. (syphon to **Venting Leak Check procedure**).
- The **4K cold head** flange has no leaks.
- The used **dewars** are correctly **closed**, with their low pressure relief valves **open**.

5.2.1.9 Low ceiling height

Where magnets have been installed into rooms with a **low ceiling height**, it can be difficult to fit the assembled service syphon. To avoid dropping the extension leg(s) down into the syphon port, the **syphon leg support clamp** must be used (→ 3/Fig. 195 Page 173).

- Check the **ceiling height**:
 - Assemble the syphon (only if it can be fitted), or
 - Assemble the diffuser, bottom extension and support clamp

Fig. 195: Syphon configuration for low ceiling height



- (1) PTFE extension tip (top fill)
- (2) Helium fill diffuser tip (top fill)
- (3) Syphon leg support clamp
- (4) Extension leg

Refer to Magnet Service syphon kit

- Remove the **syphon end cap**.
- Ensure the **ring nut, brass spacer and O-ring** are fitted correctly.
- Fit the **diffuser** to the extension leg.
- Fit **support clamp** to the extension leg.
- **Warm** the syphon **blanking plug** to ease removal.
- **Remove** the syphon **blanking plug** using the C-spanner.
- **Lower** the assembly into the **neck**.
- **Screw** on the second **extension leg**.
- **Move** the syphon **clamp** up the extension leg and lower extensions into the entry port, to allow room for the syphon body.
- **Screw** the **diffuser leg** to the syphon **body**.
- **Remove** the syphon **clamp** and slowly **lower the syphon fully** into the syphon entry.
- **Screw** the **syphon port ring** nut into place and **tighten** to make a gas-tight **seal**.
- **Re-fit** the syphon **end cap**.



Use the support clamp for removing the syphon.

5.2.1.10 Helium pressures at altitude

Helium transfer with magnet bypass valve closed

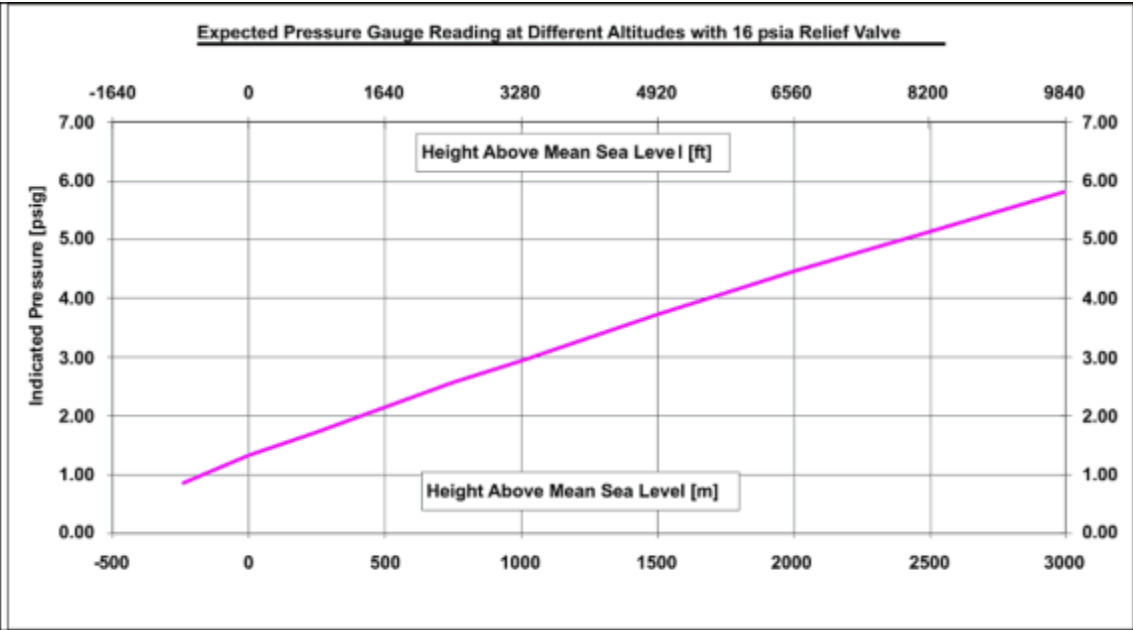
As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi. To calculate this:

- Determine the **magnet's altitude**
- Determine the **magnet gauge pressure** from the graph (→ Fig. 196 Page 174)

16 psi (A) valve

Fig. 196: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 12 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- Add 4 psi to this number to calculate the **required transfer pressure**
 - » **Example of 16 psi (A) vent valve assembly**
Altitude (1000m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)

i

Maximum dewar pressure is 10.5 psi (G).

5.2.1.11 Verifying the electronic low pressure switch

This procedure may be carried out in conjunction with any other process involving de-pressurization of the helium vessel, i.e. fitting the service syphon, cold head replacement.

i

DO NOT Switch off F3 (fuse) in the ACC - this will shut down the compressor and invalidate the Low Pressure Switch Test.

Pressure control of the helium vessel is managed by the **MSUP**, via the **Pressure Heater Control Unit**. The low pressure switch must be checked **whenever the magnet is de-pressurized**.

When pressure in the helium vessel drops to within 0.2 psi (G) above atmospheric pressure, the **electrical pressure switch** will open and turn **off** the compressor.



Atmospheric pressure may differ, depending on the altitude and local weather conditions.

As the helium vessel is pressurized, the switch closes and the compressor will re-start.



The pressure switch is factory calibrated to operate at 0.2 psi (G) above atmospheric pressure.



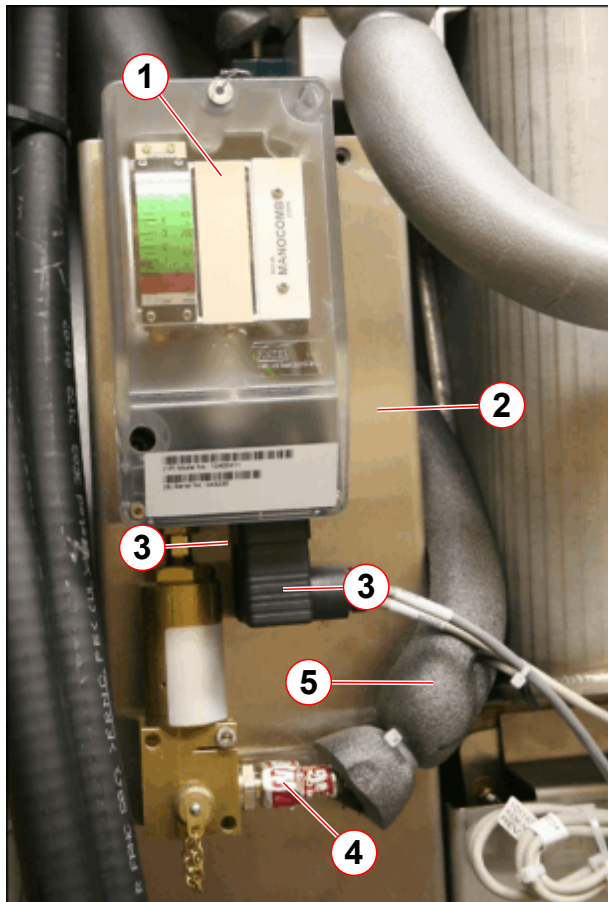
Perform this check after helium filling when the magnet syphon is *removed.

The filling port is closed and the refrigerator switched on.

Note: *Mobile systems have a fixed syphon.

Mk2 Manocomb Pressure switch

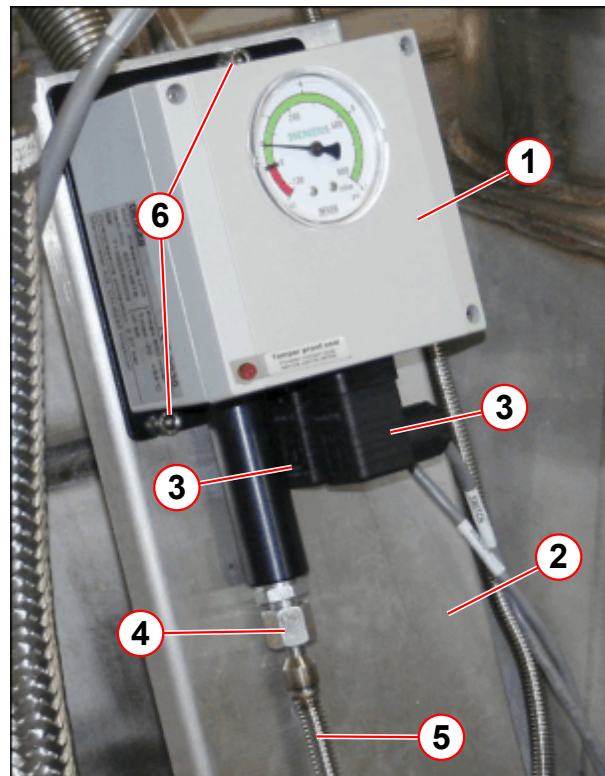
Fig. 197: Pressure switch assembly



- (1) Pressure switch assembly
- (2) Mounting plate
- (3) Magnet pressure sensor cables (front and rear)
- (4) Brass compression nut
- (5) Pressure line

WIKA Pressure switch

Fig. 198: WIKA Pressure switch assembly



- (1) Pressure switch assembly
- (2) Mounting plate
- (3) Magnet pressure sensor cables (front and rear)
- (4) Compression nut
- (5) Pressure line
- (6) M6 x 20mm cap head bolts (2 off))

- Check the **compressor** is switched **on** and running.
- Check **magnet pressure** is normal (15.3 psi (A)) (using the hand held display).

5.2.2 Bottom filling the magnet

**CAUTION**

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » Always ensure the cold head bypass valve is **CLOSED** during helium filling.



The bottom fill procedure must **ONLY** be performed when the magnet is at ZERO FIELD.

Bottom filling of the magnet will be required if:-

- The liquid helium level -
 - Falls to **zero** percent,
 - **and**
 - The temperatures measured from the carbon resistors and coil diodes, indicates temperatures **no warmer than** liquid nitrogen temperature. Refer to (→ Tab. 13 Page 180), (→ Fig. 199 Page 178), (→ Fig. 200 Page 179)

The magnet must be filled through the **magnet syphon cone** to allow filling from the bottom of the helium vessel.



Inform Magnet HSC immediately if the helium level is zero. Magnet coil temperatures must be confirmed before continuing with the helium fill.

Possible causes for this include:-

- Post magnet quench, (ramping or from persistence).
- Faulty helium level probes.
- Alarm box buttons still depressed.
- Pressure heater is left on.
- Long transportation and storage times.
- Cold head or compressor failure.

5.2.2.1 Checking Magnet Temperatures

Fig. 199: OR105/122 Magnet Acceptance Document (CR)

SIEMENS		Magnet Acceptance Documentation Installation Data		Siemens Magnet Technology Quality Regulations	
Cryostat Number	21199812	SMT Reference	21201632 02.09		
System Description 1.494T Actively Shielded Superconducting Twin Turret Magnet System OR105					
Measured Magnet Centre (Offset)		-1 (S) mm			
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet					
Operating Parameters					
Description	Specification value	Measured value			
Operating Current @ 1.494T	< 600 AMPS	509.95	AMPS		
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp		
Carbon Resistor Values					
POSITION	Room Temperature	Helium Temperature			
Top	162.4	1140			
One	134.4	903			
Two	132.3	994			
TDC	140.4	943			
Helium Level Probe Resistance (When using a 250 mA supply)					
		Probe 1	Probe 2		
@ 0 Resistance	Ohms	85.36	85.36		
0 % Calibration		774	774		
100 % Resistance	Ohms	0.22	0.22		
100 % Calibration		2	2		
Diode Table No		-1			
Pressure Heater Res	Ohms	19			
Final Temperature Sensor Recon Values					
50kK	50kK Lin	Coldhead	Recon Margin		
48	45	42	K 467 Millivolts		
Checked by	K Suraina		Signature		
Date	02/02/2009				

Fig. 200: OR105/122 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
MR Magnet Technology Quality Regulations			
Cryostat Number	21243234	SMT Reference	21245901 06.10
System Description		OR105	
1.494T Actively Shielded Superconducting Twin Turret Magnet System			
Measured Magnet Centre (Offset)		-2 (S) mm	
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet			
Operating Parameters			
Description	Specification value	Measured value	
Operating Current @ 1.494T	> 600 AMPS	510.05	AMPS
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp
Carbon Resistor Values			
POSITION	Room Temperature	Nitrogen(77K)	Helium Temperature
TDC d - e	878		2567
Top a - e	926		2938
Mid b - e	985		2715
Bottom c - e	877		2613
TDC Calibration Values:	850.4	1122.9	2731.7
Helium Level Probe Resistance (When using a 250 mA supply)			
		Probe 1	Probe 2
@ 0 Resistance	Ohms	85.36	85.36
0 % Calibration		774	774
100 % Resistance	Ohms	0.22	0.22
100 % Calibration		2	2
Diode Table No		-1	
Pressure Heater Res	Ohms	19	
Final Temperature Sensor Recon Values			
50kK	50kK Lin	Coldhead	Recon Margin
53	47	44	530 mW
Checked by		Signature	
K Suraina			
Date			
02/06/2010			
Page 1 of 1		25-AD EPS 000 01	

To recover a magnet with a **bottom helium fill**, the following must be checked:-

- Use the MSUP-Display Unit to measure:-
 - **Helium level** values

- Carbon resistor/ CCR values

Tab. 13 Expected Carbon Resistor Values

41 way plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e (Bottom)	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500
Carbon Four d - e (Top)	120 - 160	130 - 180	700 - 1500
All Ceramic Carbon Resistors (CCRs)	850 - 1050	950 - 1350	2300 - 3200



Carbon 4/CCR4 is for measuring helium gas temperatures during ramping and is not applicable to filling.



The above table is a guide only. Always compare the readings with those on the Magnet Acceptance Document.

5.2.2.2 Bottom fill procedure

For equipment required, see Filling the magnet (→ Equipment required / Page 136)

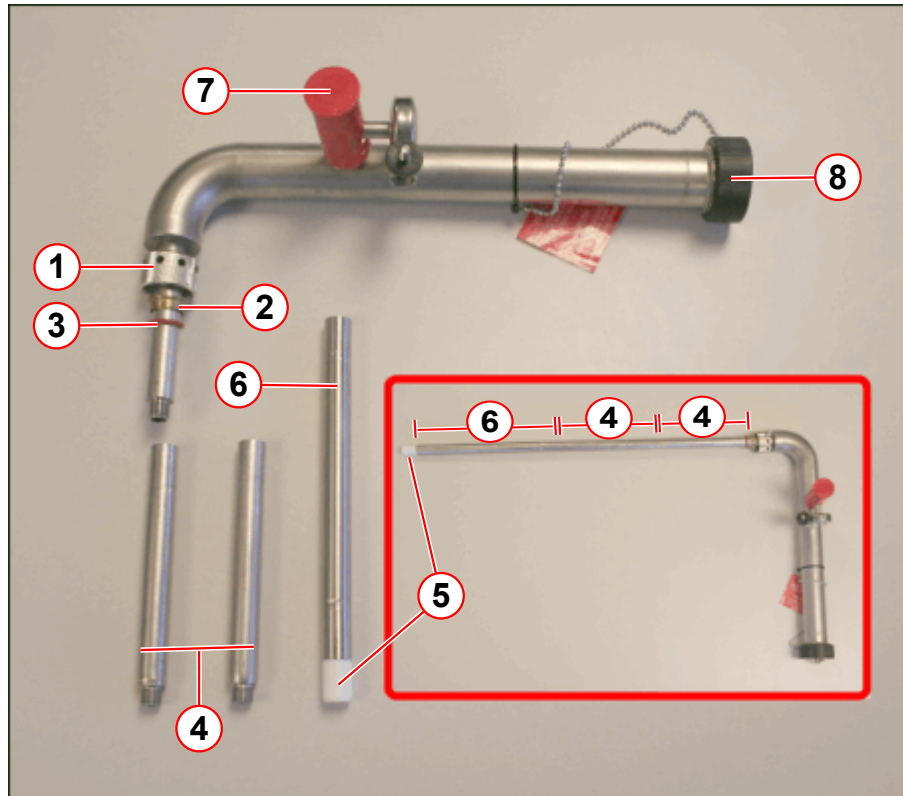


The bottom fill procedure must ONLY be performed when the magnet is at ZERO FIELD.

- Prepare the **dewar** and **dewar syphon** (→ Dewar preparation / Page 140), (→ Dewar syphon preparation / Page 142)
- Check the **ceiling height**. (→ Low ceiling height / Page 172)
 - Assemble the syphon (only if it can be fitted), or
 - Assemble the diffuser, bottom extension and support clamp.

- Assemble the **service syphon**.

Fig. 201: Magnet service syphon assembly - bottom filling

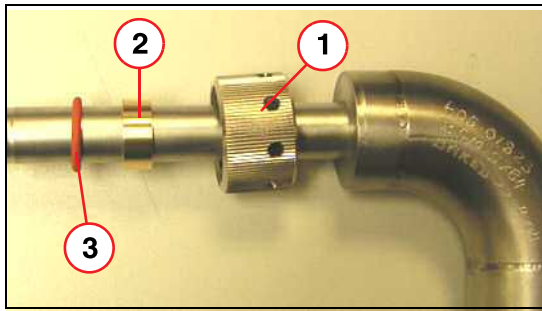


- (1) Ring nut
- (2) Brass spacer
- (3) O-ring seal
- (4) 140 mm syphon legs (2 off)
- (5) 22 mm PTFE seal (bottom fill)
- (6) 200 mm syphon leg (1 off)
- (7) Vacuum valve
- (8) Syphon end cap

Tab. 14 OR105 / OR122 Bottom filling syphon extension legs

Bottom Fill	PTFE seal	PTFE diffuser	*130 mm leg	*190 mm leg
OR105 Avanto	22 mm	N/A	2	1
OR122 Espree	22 mm	N/A	2	1
Note: *Effective extension leg length.				

Fig. 202: Assembling the Ring Nut Seal



- (1) Ring nut
- (2) Brass spacer
- (3) Silicon O-ring

- Fit the ring nut, brass spacer and the O-ring to the **service syphon leg**.

Refer to table above (→ Tab. 14 Page 181)

- Connect the **extension legs** and the 22 mm **PTFE seal**:
 - Screw the 2 x **140 mm legs** together, and connect to the **syphon**.
 - Fit the 1 x **200 mm leg**.
 - Fit the **22 mm PTFE seal** to the 200mm leg.



The minimum length required to reach the syphon cone is:

OR105 - 485 mm (+/- 5 mm)

OR122 - 547 mm (+/- 5 mm)

De-pressurize the helium vessel:-

- De-activate the **pressure heater**.



If the pressure service tool is not available, disable the pressure heater by removing the F3 from the ACC cabinet. This also switches off the compressor.

It may not be possible to measure, or monitor helium level during the transfer. This may result in helium wastage by overfilling the magnet.

- Open the **magnet bypass valve** slowly to depressurize the helium vessel.
 - » Do not open fully until most of the pressure has been released.
- Note the **gauge pressure** when the compressor switches off.
- **Close** the magnet **bypass valve**.
- Check the helium vessel has a **small positive pressure**.
- **Warm** and dry the **syphon blanking plug**, using a heat gun to ease removal.
- **Dry** the syphon **port**.
- **Remove** the syphon **end cap**.

- Remove the **syphon blanking plug**. (→ Fig. 164 Page 148), (→ Fig. 179 Page 159)



CAUTION

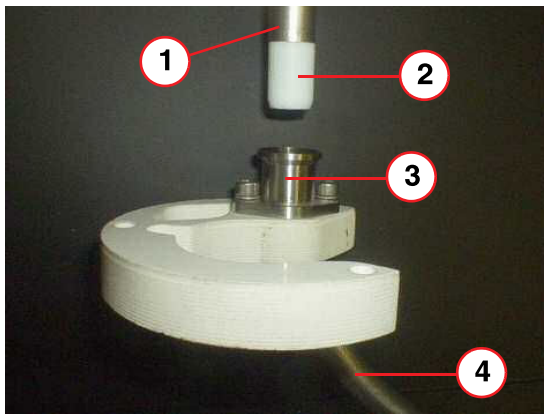
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

- » Stand clear when opening.
- » Wear protective gloves and safety glasses.

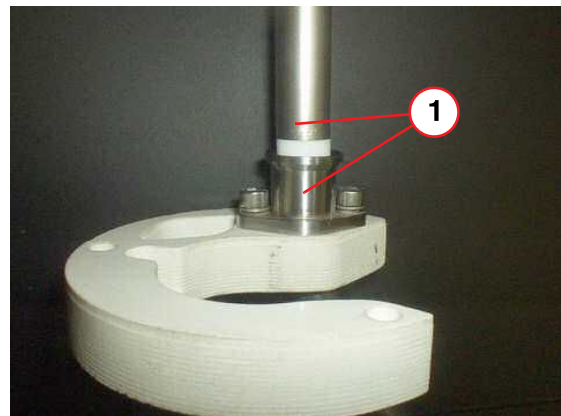
- **Mk2 only** - Fit the brass **syphon fill adaptor** and a new copper seal.
(→ Fig. 190 Page 167)
 - When fitting the adaptor, prevent **air ingress** by **temporarily** blocking the port with a plug of **paper towelling**.
- **Slowly** lower the **service syphon** fully into the neck.
- The **syphon leg** must locate fully into the **syphon cone** - make sure it is **fully engaged**.

Fig. 203: Bottom fill syphon cone (pictorial view)



- (1) Syphon leg
- (2) PTFE Seal (Bottom Fill)
- (3) Syphon cone
- (4) Syphon tube

Fig. 204: Bottom fill - syphon leg fully engaged (pictorial view)



- (1) Syphon leg - fully engaged



Helium efficiency will be reduced if the syphon leg is not fully engaged in the syphon cone.

- Tighten the **syphon port ring nut** (→ 1/ Fig. 202 Page 182) by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the **syphon end cap**.

5.2.2.3 Helium pressures at altitude

Helium transfer with magnet bypass valve closed

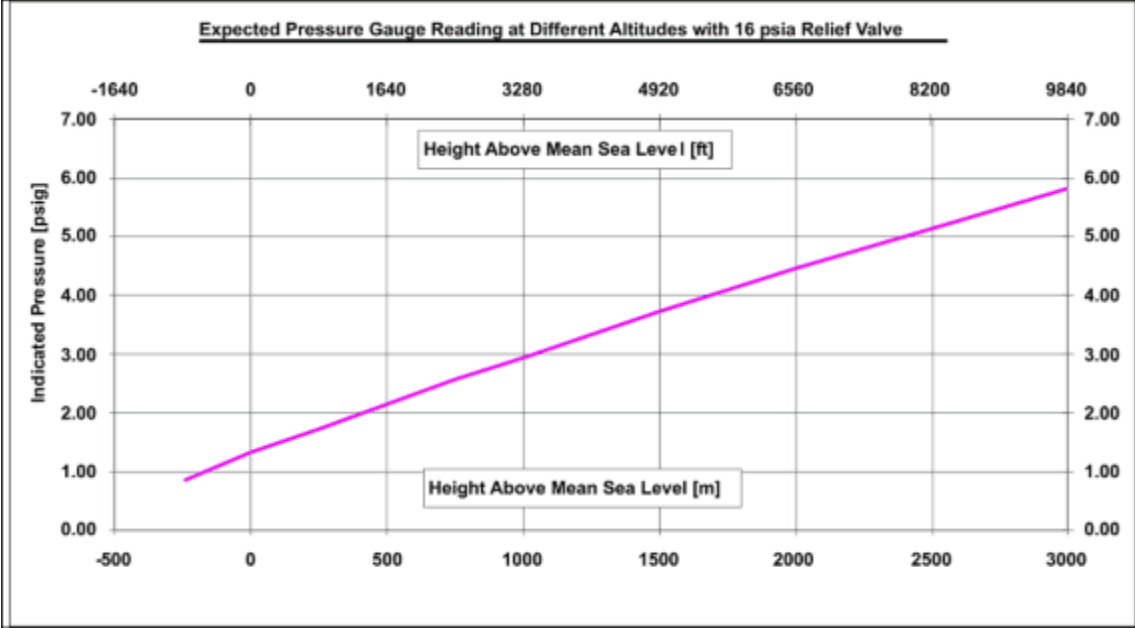
As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi. To calculate this:

- Determine the **magnet's altitude**
- Determine the **magnet gauge pressure** from the graph (→ Fig. 205 Page 184)

16 psi (A) valve

Fig. 205: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 15 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- Add 4 psi to this number to calculate the **required transfer pressure**
 - » **Example of 16 psi (A) vent valve assembly**
Altitude (1000m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)

i

Maximum dewar pressure is 10.5 psi (G).

5.2.2.4 Liquid helium transfer - bottom fill

i

The helium dewar must be positioned close to the magnet, with the dewar syphon fitted.

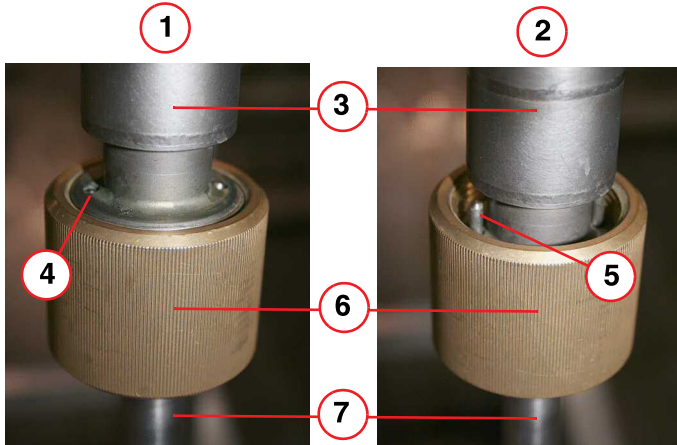


For bottom filling a magnet at zero, there is no need to pre-cool both syphons.

- Check the dewar gas supply pressure is set at **0.27bar (4 psi (G))**.
- Open the dewar syphon flow valve half way (2 turns) to **purge** the syphon of air.

Fig. 206: Dewar syphon flow valves - open and closed positions

Fig. 207: Helium filling

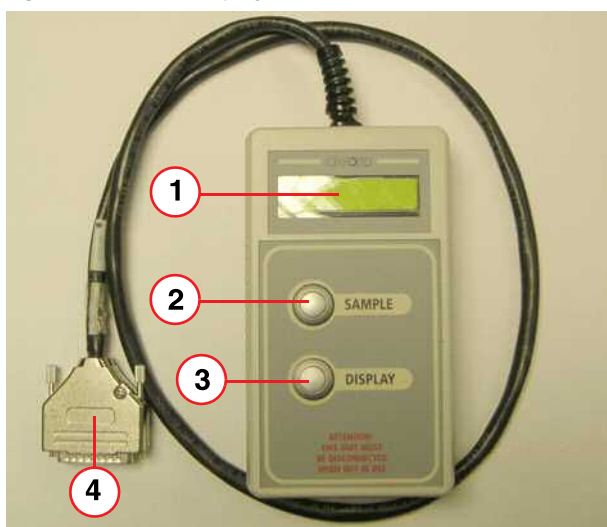


- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg



- Close the dewar syphon flow valve.
- Remove the service syphon end cap and **couple the two syphons together** and secure the **locknut**.
- Open the dewar syphon flow valve.
- Maximize the transfer by fully opening the dewar syphon flow valve.
 - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (→ 4/ Fig. 206 Page 185)
 - » From **fully closed** to **fully open**, is **4 turns**.

Fig. 208: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

- Monitor the **carbon resistors/CCRs** every **15 minutes** to measure the temperature drop.
 - » Refer to the **acceptance documentation** (Installation Protocol) for required values.
 - » If the values do not rise, **contact Magnet HSC**.
- When the required value is reached, monitor the **helium level** using the MSUP-Display Unit .



Approximately 600-700 liters of helium are required to achieve a 10% meter reading, from nitrogen temperature.



Opening the bypass valve will help to reduce the back pressure, enabling more efficient helium transfer.



Maintain a pressure of about 0.27 bar (4 psi (G)) at the helium dewar. Too much or too little pressure will result in inefficiency.

At **4 psi (G)**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**

5.2.2.5 Stopping the Helium Transfer

Fig. 209: Dewar manometer



- As soon as the level reads 95%, or the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
 - The **dewar syphon flow valve** must be closed **immediately** the dewar empties. (→ 6/Fig. 206 Page 185)
 - Set the dewar gas pressure regulator to **zero**.
 - Close the dewar **primary ball valve**. (→ 1/Fig. 157 Page 144)
 - Refer to (→ Changing supply dewars during transfer / Page 168) if the magnet is not full.
- Continue to **fill through the cone** until the desired **helium level** has been reached.



Do not fill above 95%.

Overfilling the magnet will cause the helium level to show S/C (short circuit).

If S/C is displayed, allow helium to boil off until a correct helium reading is confirmed. If this takes longer than 10 minutes contact Magnet HSC for advice.

The magnet may be ramped at a higher helium level as long as S/C is not displayed.

5.2.2.6 Removing the Service Syphon

- **Open** the **magnet bypass valve** (→ Fig. 160 Page 147), (→ Fig. 161 Page 147), (→ Fig. 175 Page 158), very slowly to **depressurize the helium vessel to 0.1 psi (G) (7 mbar)** above atmospheric pressure.
- **Close** the magnet **bypass valve**, before **disconnecting** the syphons.
- **Warm** the **syphon joint** to room temperature (using a heat gun), to remove ice.
- **Warm** the syphon **entry valve** to room temperature and **dry**.
- **Warm** the syphon **end cap** ready for fitment onto the syphon.
- **De-couple** the syphons.
 - ensure the magnet bypass valve is closed **BEFORE** disconnecting the syphons.

**CAUTION**

Risk of cold burns.

Cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting.
- » Wear protective gloves and safety glasses.

- **Immediately** close off the **end** of the service syphon with the **syphon end cap**.
- **Remove** the service syphon from the magnet.

**CAUTION**

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- For **Mk1** venting -
 - **Immediately** fit the **syphon blanking plug** using a new copper seal (→ Fig. 173 Page 156).
 - **Warm up** the **threads** on the plug and **carefully** tighten with the syphon C-spanner.
 - There might be **ice** on the threads - **do not use undue force**. Damaged threads will result in a leak.
 - **Warm** the **top plate** and **syphon plug** using a heat gun. Re-tighten the plug.
- For **Mk2** venting -
 - warm up and dry the **brass syphon fill adaptor**.
 - prevent **air ingress** by **temporarily** blocking the port using a plug of **paper towel-lin**.
 - remove the **brass syphon fill adaptor**.
 - fit the **syphon blanking plug**. (→ Fig. 191 Page 167)
- **Warm up** the magnet bypass **valve assembly** and **check** the bypass valve is **closed**
- **Re-activate** the pressure heater.
- Scroll to '**Pressure heater**' (**Off/On,%**) on the MSUP-Display Unit (#12).
 - Press the **SAMPLE** button to toggle on **100% duty cycle**.
 - When the pressure is at **15.3 psi (A)**, or above, press the **SAMPLE** button to toggle back to **20% duty cycle**.



Failure to do this may result in exceeding the 2 watt limit in the MSUP and disabling the pressure heater. An MSUP reset will be required to re-enable the heater.

- Check that the **pressure heater** activates with a 20% duty cycle. The yellow LED on the MSUP should also illuminate at the same duty cycle.
- Check that the **compressor** activates automatically as the pressure rises above atmospheric. Note the pressure at which it starts.
- **Warm the turret and venting valves** to room temperature with a heat gun.
 - **remove** any **surplus water** with paper towelling.
- **Warm the extension legs** using a heat gun.
- Remove the **bottom fill extension leg** and **PTFE seal**.
- **Move the dewar** from the imaging suite into a **well ventilated area**.
- **Remove the gas line** from the dewar. (→ 3/Fig. 157 Page 144)
- **Complete** depressurizing the dewar by opening the **primary ball valve**. (→ 5/Fig. 193 Page 170)
- When at the **gas pressure** is at zero, **remove the dewar syphon**.
- **Immediately** close the **syphon entry valve** on top of the dewar. (→ 4/Fig. 193 Page 170)
- **Open the low-pressure dewar relief valve**. (→ 1/Fig. 193 Page 170)
- **Close the primary dewar ball valve**. (→ 5/Fig. 193 Page 170)
- **Warm both syphons** with the heat gun until dry, and store safely.
- **Leak check** the turret and venting.



After a bottom fill, it is recommended that the first magnet energization is performed the next day. This will allow the magnet coils to thermally stabilize.

5.2.3 Filling the mobile system

5.2.3.1 General



CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmosphere.

- » **Always ensure the cold head bypass valve is CLOSED during helium filling.**

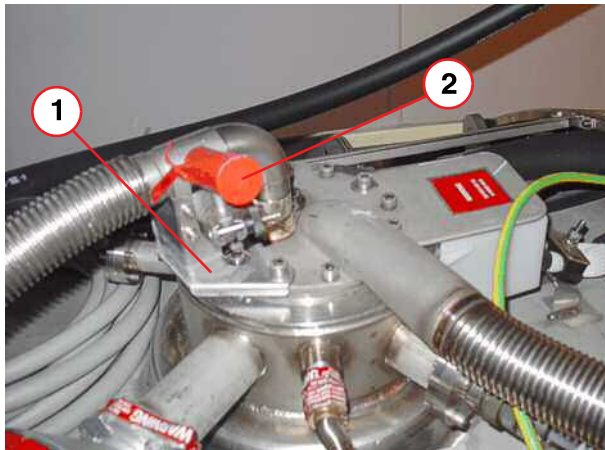
Mobile Systems

The helium fill process is very similar to that of static systems. Mobile systems differ from static systems in that they have a **permanently fitted high efficiency syphon (HES)**.

For convenience of operation, the coupling end of the syphon and services (i.e. pressure gauge) are located at the **rear** of the mobile trailer.

There must also be clear access for the **dewar** at the rear of the mobile trailer.

Fig. 210: Mobile syphon - turret transport bracket



- (1) Syphon transport bracket
(2) Syphon vacuum valve

- The mobile syphon also has an additional **turret transport bracket** (permanently fitted), to reduce vibration and movement.

This procedure can be performed on magnets at field and magnets at zero field.



Only Siemens trained and accredited CSEs are allowed to perform this work.

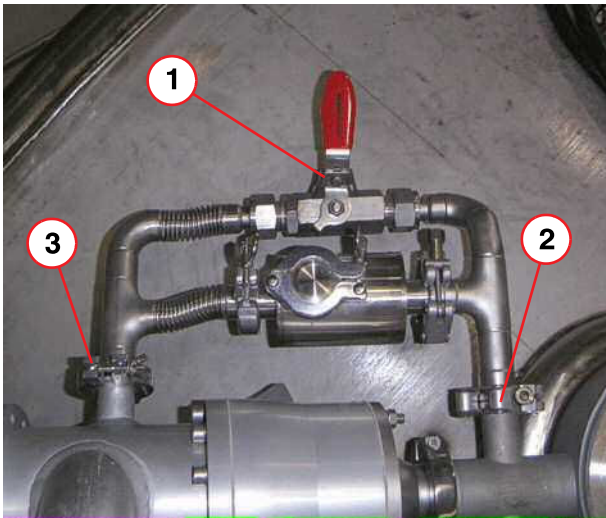


It is recommended that fills are performed during service visits and before ramping up. Do not fill above 95%. It may cause the helium level to show S/C on the MSUP-Display Unit which will stop the magnet being ramped.

Venting system

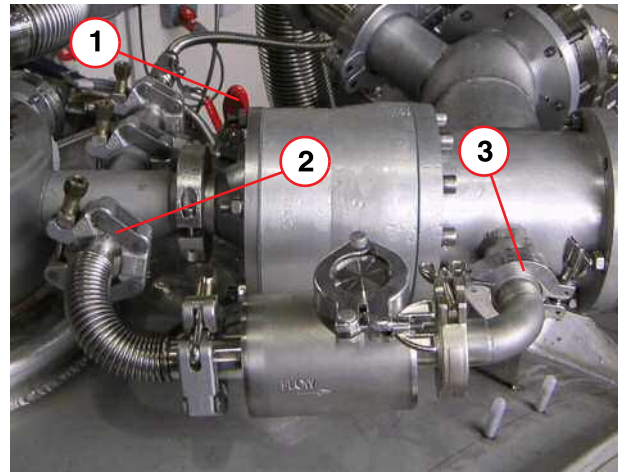
The document describes the procedure for mobile systems with **Mk1** and **Mk2** venting systems.

Fig. 211: OR105 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 212: OR122 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)



The venting bypass valve should remain CLOSED during the filling process.

With the syphon permanently fitted there is no need to enter the magnet room until the transfer is complete.

5.2.3.2 Safety



CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

- » Protective clothing, gloves and eye protection must be worn. Filling should be performed by trained personnel only.



WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the magnet trailer.



Ensure that the Quench Line has been correctly installed and safe before performing Installation and Startup procedures.

**WARNING**

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, helium gas may be vented to cool the syphon.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » Oxygen monitors must be activated at all times.

**CAUTION**

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the fittings without first depressurizing the helium vessel.

NOTICE

Mishandling of the dewar.

- » The dewar-relevant safety regulations of the manufacturer must be observed.



Monitor the filling procedure continuously by checking transfer pressure and the fill level.

Identify hazardous areas with warning signs.

5.2.3.3 Equipment required

- Dewar syphon
- Dewar neck adaptor, clamps, seals, and hose
- Pressure service tool (part no **03861221**)

Fig. 213: MSUP Display unit

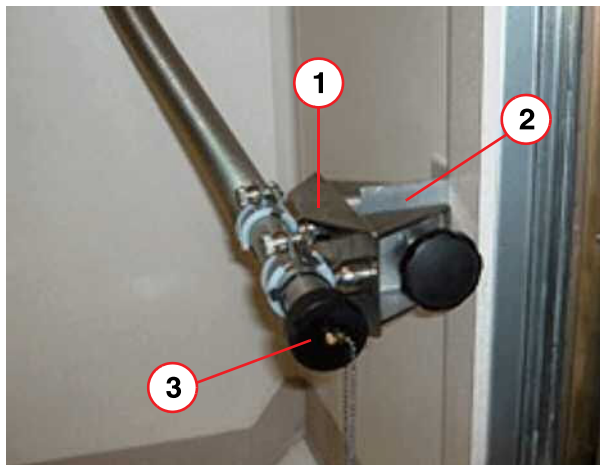


- The MSUP-Display Unit is supplied as part of the **mobile installation kit** (part no **10109296**).

- Heat gun
- Helium gas supply (at least 99.9% pure)
- Pressure regulator 200 bar/ 0 - 1.5 bar
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes
- O₂ Monitor - either room or personal
- Gas bottle wrench

5.2.3.4 System preparation

Fig. 214: Mobile syphon - transportation position (typical)



- (1) Syphon coupling clamp
- (2) Fixed wall bracket
- (3) Syphon end cap

The **High Efficiency Syphon** is fitted permanently into the magnet. Its main flexible tube is clamped into position around the trailer.

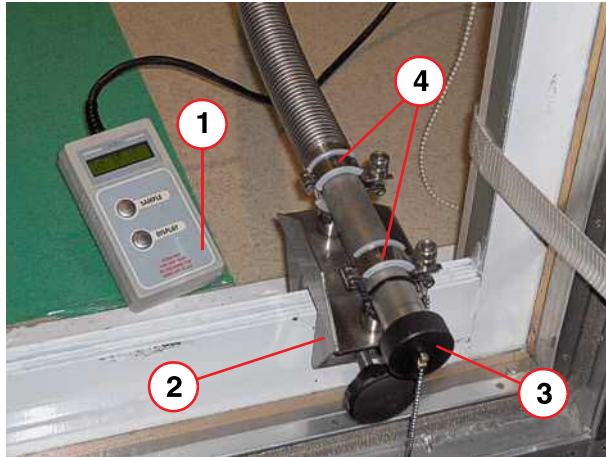
The coupling end of the syphon is fixed to a **syphon coupling clamp** (pos 1) positioned at the **rear** of the trailer.

- Remove the syphon coupling clamp from the fixed wall bracket.



The positioning of the clamp will vary, depending on the trailer design.

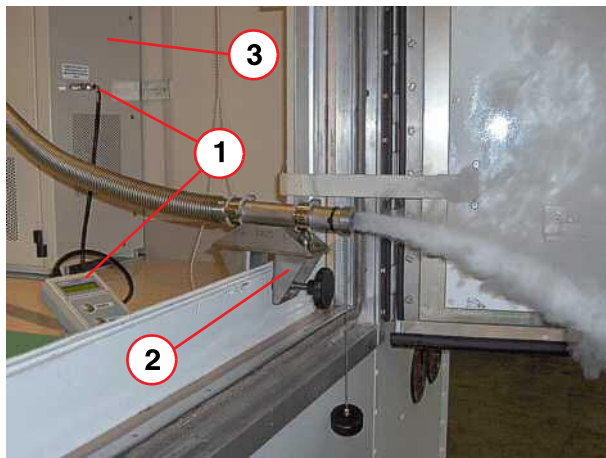
Fig. 215: Mobile syphon - rear door fitment (typical)



- (1) MSUP hand held display
- (2) Syphon clamped to rear door
- (3) Syphon end cap
- (4) Syphon bracket insulating bushes

- Secure the **syphon coupling clamp** to the lower edge of the rear trailer door.
 - Avoid twisting of the flexible syphon tube.
 - If required, loosen the **syphon bracket insulating bushes (pos 4)** to re-align the syphon.
- Ensure that there is open and clear access for **helium dewar(s)** at the rear of the trailer.

Fig. 216: Mobile syphon venting (cascading plume)



- (1) Hand held connection to MSUP
- (2) Clamp position for venting (variable)
- (3) Electronics cabinet (E-box)

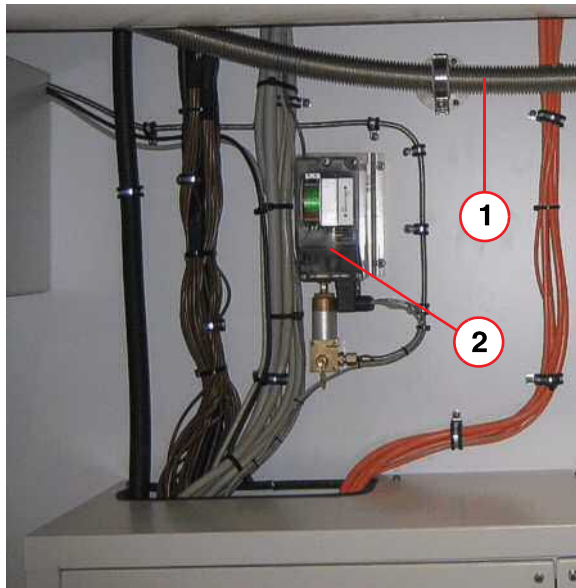
Fit the MSUP-Display Unit to enable continuous monitoring of the pressure and LHe levels during this procedure.

- Plug the **hand held remote unit** into the **MSUP connection (pos 1)**, located at the end of the E-box (pos 3).
- Ensure the hand held display is **clearly visible**.
- Ensure that there is **open and clear access** to the **syphon**.
- Switch **off** the **compressor**, (if it has not already tripped off at low pressure).
- Prepare the dewar (→ Dewar preparation / Page 140).



All helium dewars must be vented of any excess pressure away from the trailer.

Fig. 217: Positioning of pressure switch in trailer



- (1) Mobile syphon clamped to trailer
(2) Pressure switch location

The **pressure switch** is fitted above the Electronic cabinet (E-Box), and must be **clearly visible** from the end of the trailer.

Verifying the electronic low pressure switch

This procedure may be carried out in conjunction with any other process involving depressurization of the helium vessel, i.e. fitting the service syphon, cold head replacement.



DO NOT Switch off F3 (fuse) in the ACC - this will shut down the compressor and invalidate the Low Pressure Switch Test.

Pressure control of the helium vessel is managed by the **MSUP**, via the **Pressure Heater Control Unit**. The low pressure switch must be checked **whenever the magnet is depressurized**.

When pressure in the helium vessel drops to within 0.2 psi (G) above atmospheric pressure, the **electrical pressure switch** will open and turn **off** the compressor.



Atmospheric pressure may differ, depending on the altitude and local weather conditions.

As the helium vessel is pressurized, the switch closes and the compressor will re-start.



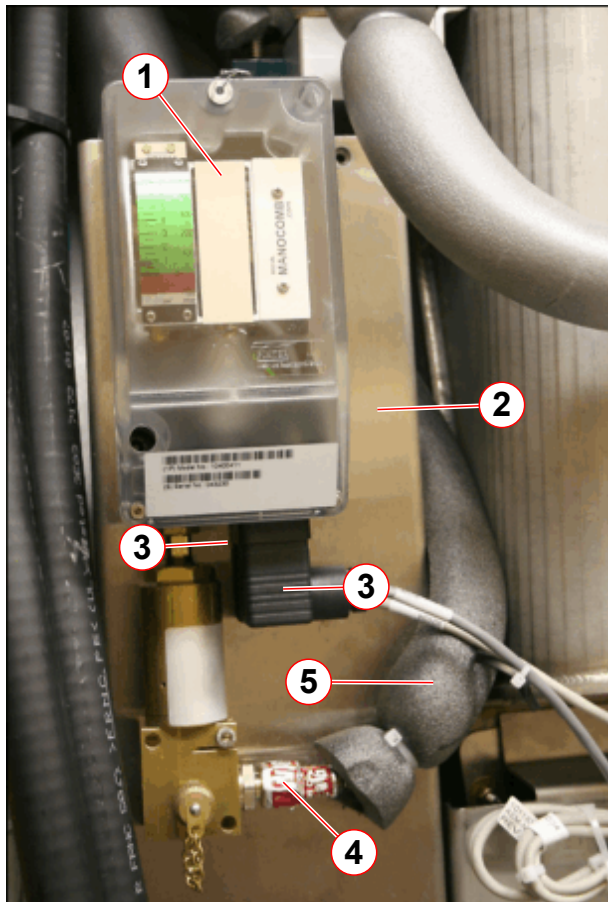
The pressure switch is factory calibrated to operate at 0.2 psi (G) above atmospheric pressure.



Perform this check after helium filling when the magnet syphon is *removed.
The filling port is closed and the refrigerator switched on.
Note: *Mobile systems have a fixed syphon.

Mk2 Manocomb Pressure switch

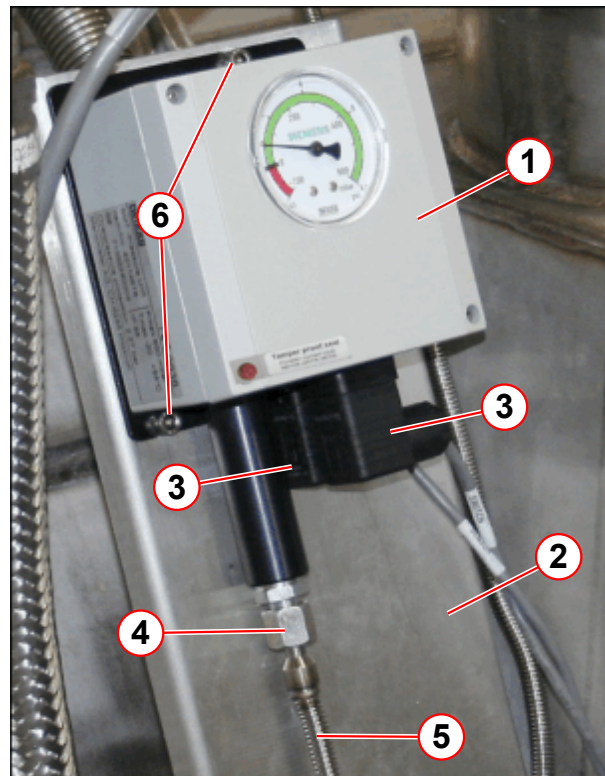
Fig. 218: Pressure switch assembly



- (1) Pressure switch assembly
- (2) Mounting plate
- (3) Magnet pressure sensor cables (front and rear)
- (4) Brass compression nut
- (5) Pressure line

WIKA Pressure switch

Fig. 219: WIKA Pressure switch assembly



- (1) Pressure switch assembly
- (2) Mounting plate
- (3) Magnet pressure sensor cables (front and rear)
- (4) Compression nut
- (5) Pressure line
- (6) M6 x 20mm cap head bolts (2 off)

- Check the **compressor** is switched **on** and running.
- Check **magnet pressure** is normal (15.3 psi (A)) (using the hand held display).

5.2.3.5 Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

It covers all dewars that are externally pressurized with helium gas.



Refer to the dewar manufacturer's documentation when using side fill and/or self-pressurizing dewars.



Read the dewar manufacturer's safety guidelines and check the dewar valves before commencing the LHe fill.

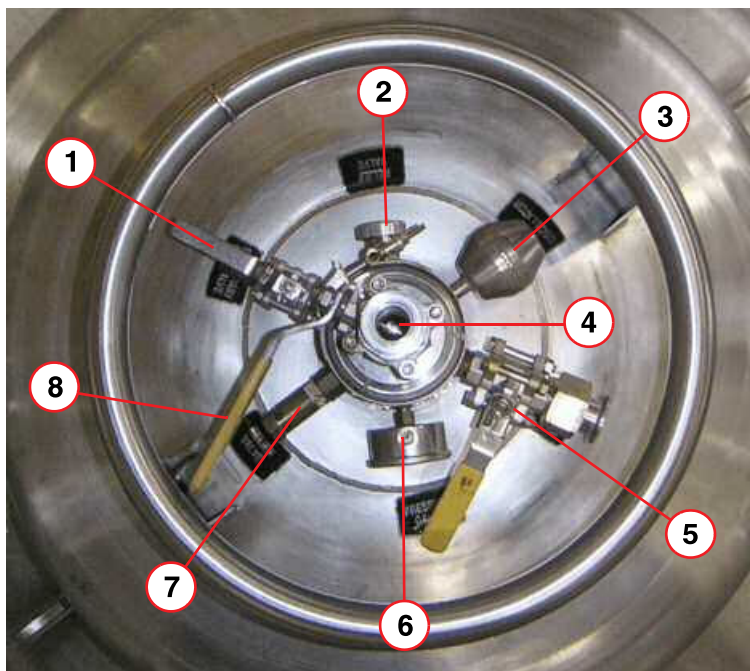
- If the dewar is **highly pressurized**, (greater than 0.3bar (4.4psi)), release the pressure by slowly opening the **primary ball valve** (→ 5/Fig. 220 Page 197) to minimize the flash losses of helium.
 - This must be done in a **well ventilated area outside of the trailer**.



Check for the cause of the high pressure, i.e. was the low pressure relief valve closed? Inform the supplier if the cause cannot be found.

- Fit the **dewar syphon adaptor** (→ 3/Fig. 223 Page 200) to the top of the dewar.

Fig. 220: Dewar Ports (typical)

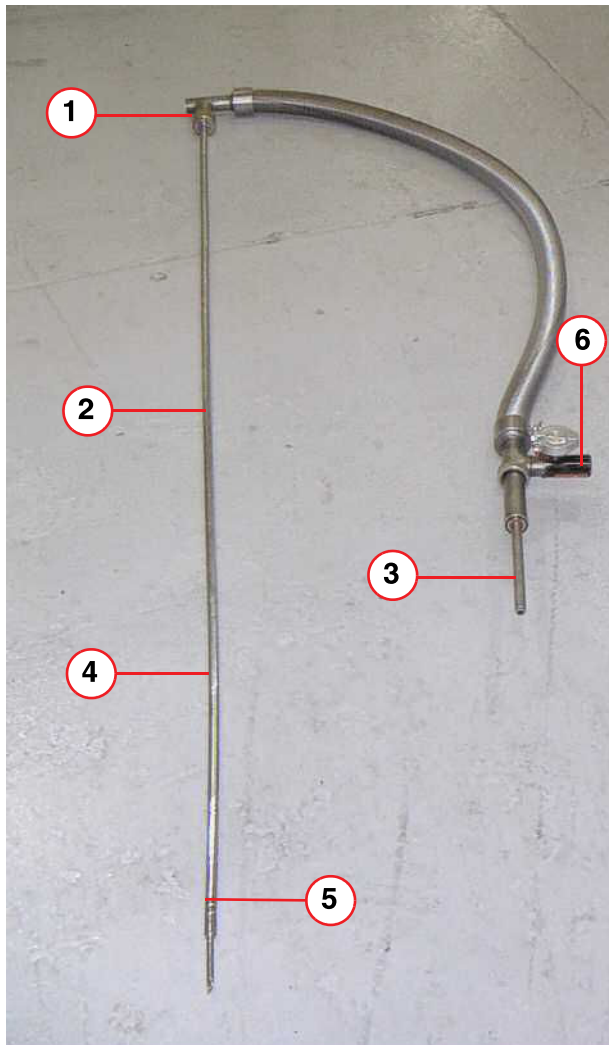


- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- **Close the primary ball valve** (→ 5/Fig. 220 Page 197) and **low pressure relief valve** (→ 1/Fig. 220 Page 197).

5.2.3.6 Dewar syphon preparation

Fig. 221: Dewar syphon



- Attach extensions as necessary, including the sludge inhibitor, to the dewar syphon leg.

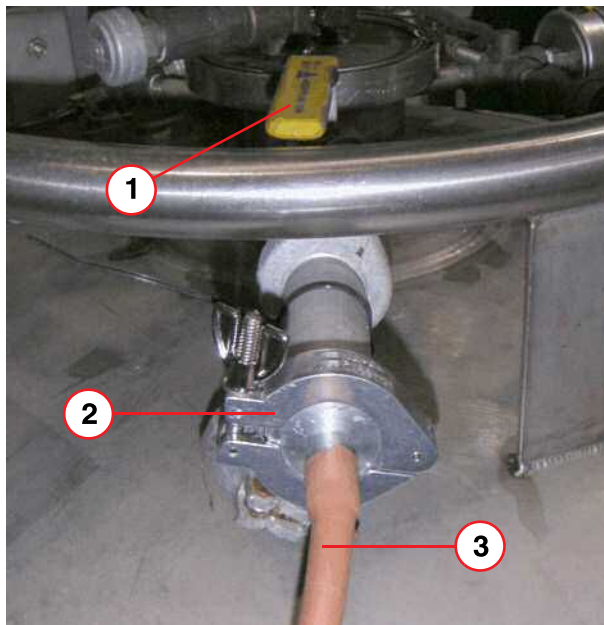
- (1) Syphon flow valve
- (2) Syphon leg
- (3) Bayonet connection (male)
- (4) Extension leg
- (5) Sludge inhibitor
- (6) Syphon vacuum valve



Do not open the syphon vacuum valve (6).

- Purge service and dewar syphons with helium gas to remove any debris.
- **Connect** the gas bottle to the dewar.

Fig. 222: Dewar gas connection

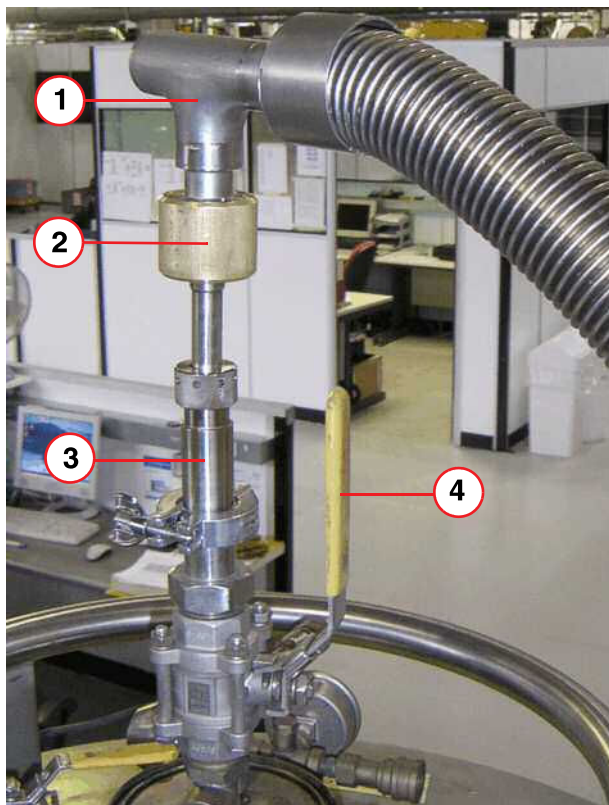


- (1) Primary ball valve
- (2) NW25 clamp and seal
- (3) Gas supply line

- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the end of the trailer, if the magnet is at field.
- Purge the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an NW25 clamp and seal (pos 2).

5.2.3.7 Pre-cool the dewar syphon

Fig. 223: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- Open the **syphon flow valve** (pos 2) two turns.
- Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.
(Do not fully tighten the ring nut at this stage).
- Open the **syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.

- Push the syphon fully down into the dewar, (very slowly to minimize pressure build-up and flash losses), until the sludge inhibitor touches the bottom.
- Fully tighten the ring nut and seal.
- Allow the syphon to cool until a dense white "cascading plume" appears. **Do not** allow high pressure to build in the dewar, only enough to cool the syphon.
- When the syphon is cold, close the **syphon flow valve** (→ 2/ Fig. 223 Page 200), and move the dewar so that the syphon can be connected without excessive bending.
- **Open the primary ball valve** (→ 5/ Fig. 220 Page 197) and pressurize the dewar to **0.27bar (4psi (G))**.

WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the trailer.

5.2.3.8 Pre-cool the mobile syphon

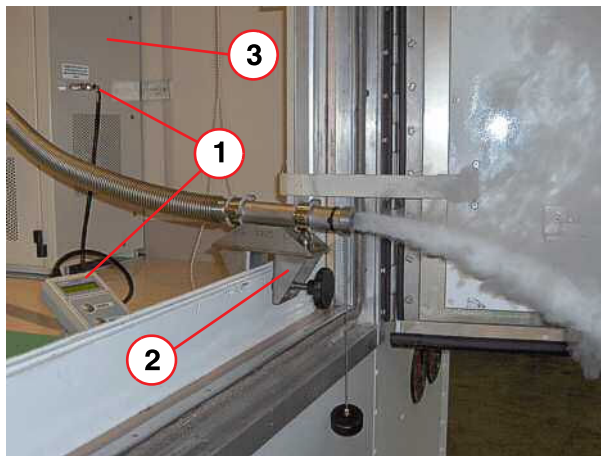


The helium dewar must be positioned close to the rear of the trailer with the dewar syphon fitted, ready for use.

Cool down the **mobile syphon** as follows:-

- Ensure the **syphon end cap** (→ 3/ Fig. 214 Page 193) is fitted and the **magnet bypass valve** is closed.
- **Activate the pressure heater.**

Fig. 224: Mobile syphon venting (cascading plume)



- (1) Hand held connection to MSUP
- (2) Clamp position for venting (variable)
- (3) Electronics cabinet (E-box)

- **Remove the mobile syphon end cap** and allow the syphon to vent until it is fully cold and a dense white “cascading plume” appears.

CAUTION

Risk of cold burns.

The “cascading plume” of cold gas is very cold.

- » Stand clear when removing the cap.
- » Wear protective gloves and safety glasses.

5.2.3.9 Helium pressures at altitude

Helium transfer with magnet bypass valve closed

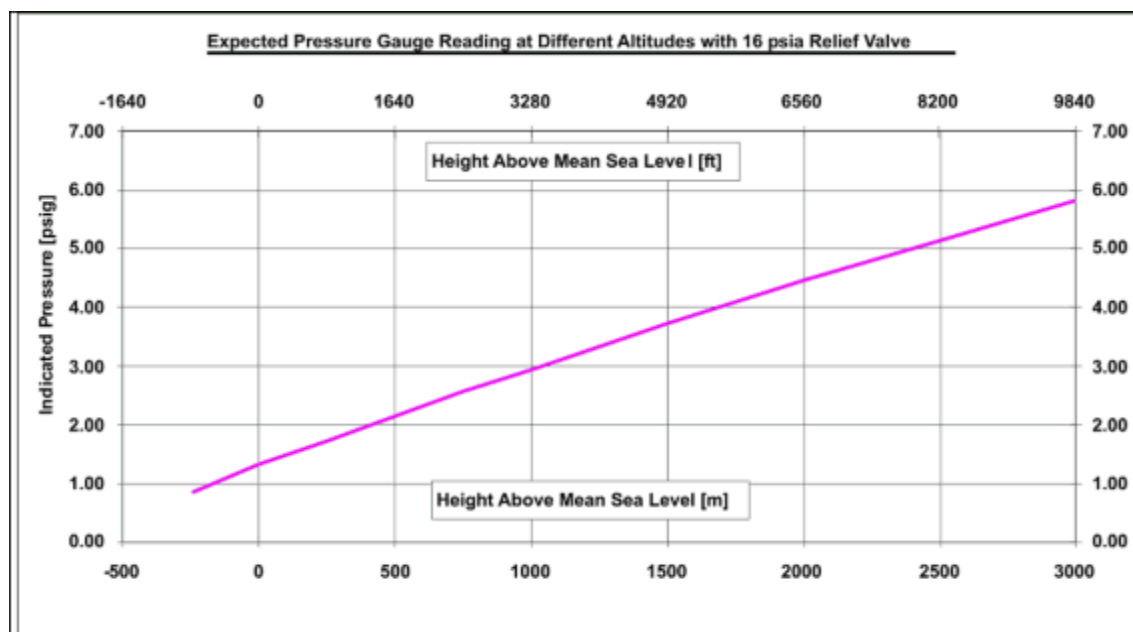
As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi. To calculate this:

- Determine the **magnet's altitude**
- Determine the **magnet gauge pressure** from the graph (→ Fig. 225 Page 202)

16 psi (A) valve

Fig. 225: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 16 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- Add 4 psi to this number to calculate the **required transfer pressure**
 - » **Example of 16 psi (A) vent valve assembly**
Altitude (1000m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)



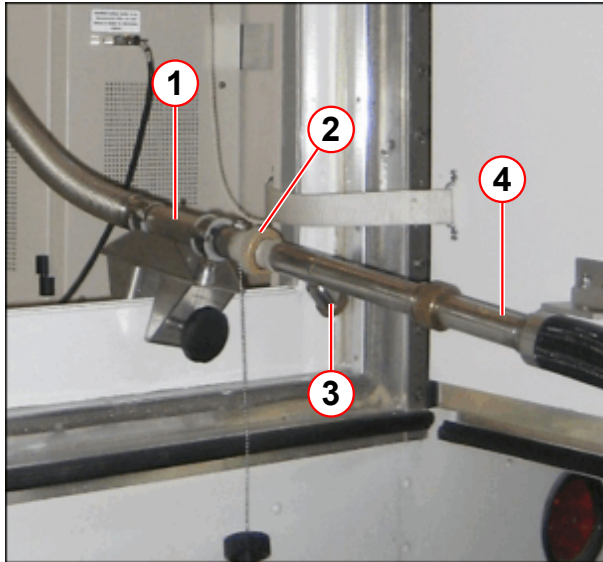
Maximum dewar pressure is 10.5 psi (G).

5.2.3.10 Liquid helium transfer - mobile

- Check the gas supply pressure is set at **0.27bar (4psi (G))**.

- As the **mobile syphon** nears the point of being fully cold, open the **dewar syphon flow valve** (→ 2/ Fig. 223 Page 200) (2 turns) and pre-cool the syphon again until a dense white “cascading plume” appears.

Fig. 226: Mobile - dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon vacuum valve
- (4) Dewar syphon

- When **both** syphons are giving a dense white “cascading plume” of helium gas, connect them together and **secure** the **lock nut** (pos 2).
- **OPEN** the magnet bypass valve. This will:
 - » Improve efficiency.
 - » Improve the filling speed.



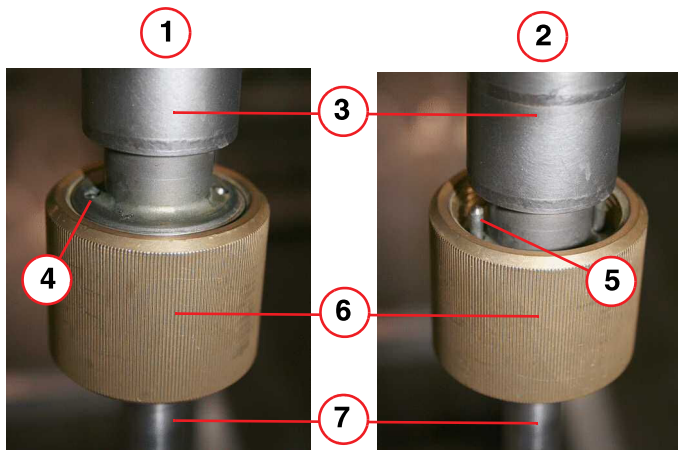
CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Fig. 227: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

- Fully open the **syphon flow valve** on the dewar syphon. (→ Fig. 227 Page 204)
 - » The flow valve is **fully open** when the **two guide pins** at the top of the brass nut are flush with the top of the stainless steel body.
 - » From fully closed to fully open is **4 turns**.



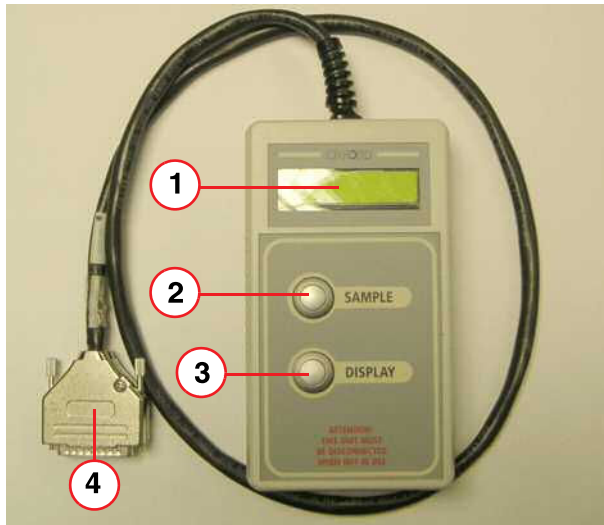
CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Fig. 228: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

- Take an initial helium level measurement from the MSUP-Display Unit.
- Repeat the level measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - If not, stop the transfer and investigate the reason for poor transfer efficiency.
- **NOTE: If F3 has been switched off, measurements can be taken, but only on battery for 2 minutes.**



Maintain a pressure of about 0.27bar (4psi (G)) at the helium dewar. Too much or too little pressure will result in inefficiency.

Fig. 229: Mobile - helium transfer



- At **4 psi (G)** the transfer rate is approximately **10** litres per minute.
- **DO NOT LEAVE THE TRANSFER UN-ATTENDED.**
- A 250 litre dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**



CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both syphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.
- » Frost on the syphon coupling is acceptable.

Stopping the Helium Transfer

Fig. 230: Dewar manometer



- As soon as the level reads 95%, or the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** (→ 6/Fig. 227 Page 204)
- Set the dewar gas pressure regulator to **zero**.
- Close the dewar **primary ball valve**. (→ 1/Fig. 222 Page 199)
- Refer to (→ Changing supply dewars during transfer / Page 168) if the magnet is not full.



Do not fill above 95%.

Overfilling the magnet will cause the helium level to show S/C (short circuit).

If S/C is displayed, allow helium to boil off until a correct helium reading is confirmed. If this takes longer than 10 minutes contact Magnet HSC for advice.

The magnet may be ramped at a higher helium level as long as S/C is not displayed.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- » Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon, or the dewar pressure falls.

5.2.3.11 Changing supply dewars during transfer

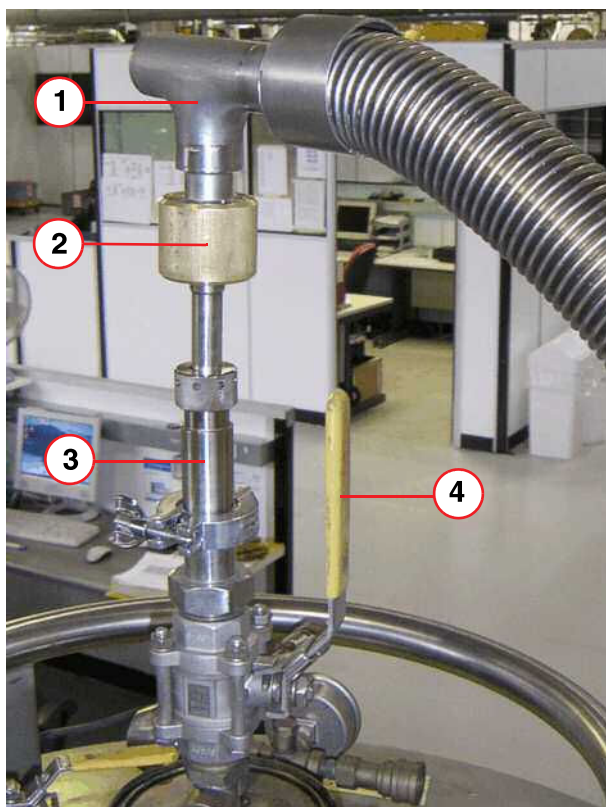


Whistling of the syphon (a high pitched sound), or a noticeable drop in dewar pressure are indications of an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, ready to receive the syphon.

Fig. 231: Dewar syphon fitment



- When the dewar empties, stop the transfer **immediately** by closing the dewar syphon flow valve (pos 2).

- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

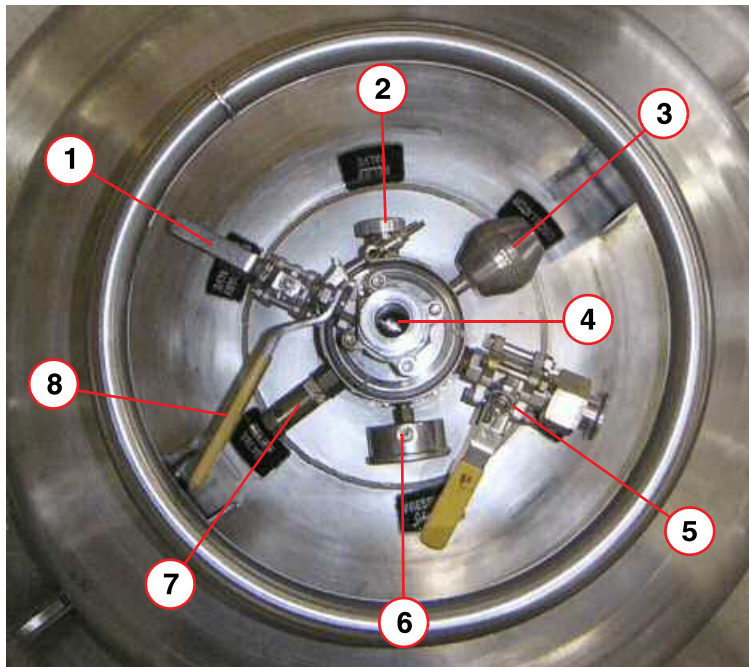
**CAUTION****Risk of quench**

Failure to close the valve immediately, when the magnet is at field, will result in a quench.

» Close the valve immediately.

- Turn **off** the dewar gas supply:
 - remove the **gas connection**
 - vent the dewar down to atmospheric pressure
 - **Ensure the area is well ventilated**
- Close the **primary ball valve**(→ 5/ Fig. 232 Page 208).

Fig. 232: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve

- Using a heat gun, **warm** the syphon connecting joint to room temperature.

- Disconnect the syphons.

**CAUTION**

Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting.
- » Wear protective gloves and safety glasses.

- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **syphon end cap**

**CAUTION**

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- » Ensure the syphons are capped and plugged immediately and securely.

- Move the empty dewar close to the full dewar.
- Vent both storage dewars down to **atmospheric pressure**.
 - » This must be done in a well ventilated area outside the trailer.
- **Open** the **low pressure relief valve** on the **empty** dewar (→ 1/Fig. 232 Page 208).
- Remove the dewar syphon and adaptor from the empty dewar and **immediately** Insert the syphon leg into the full dewar.
- **Close** the **syphon entry valve** (→ 4/Fig. 232 Page 208), of the **empty** dewar. and **open** the **low pressure relief valve** (→ 1/Fig. 232 Page 208).
- **Close** the **low pressure relief valve** of the **full** dewar.
- Purge the gas supply line.
- **Connect** to the full dewar and **open** the dewar **primary ball valve**.
- Set the **gas pressure** to 0.27bar (4 psi (G)).
- **Pre-cool** the **dewar syphon** (→ Pre-cool the dewar syphon / Page 143).
- Move the dewar next to the magnet system.
- Pre-cool the **service syphon**, (→ Pre-cool the service syphon / Page 162)
- When **both** syphons are giving a dense white "cascading plume" of helium gas, connect them together and secure the lock nut.
- **Continue** with the transfer.

5.2.3.12 Final work steps

After transfer, check:-

- The **magnet bypass valve** is **closed** and **locked**.
- The **pressure heater** is operating correctly.

- The **pressure heater LED** on the MSUP is illuminated.
- **No alarm signals** are evident on the MSUP-Display Unit, or the Alarm Box.
- The **compressor** is switched **on** and running.
- **Manually** operate the pressure heater to repressurize the magnet.
 - Clear average power by manually turning off pressure heater at **15.3 psi (A)**.
- The **magnet pressure** is stable at **15.3 psi (A)**.
- **Warm** the turret and venting system with heat gun.
- **Disconnect** the MSUP-Display Unit from the MSUP.
- Check the **filling port** and **venting system** for leaks. syphon to **Venting Leak Check**.

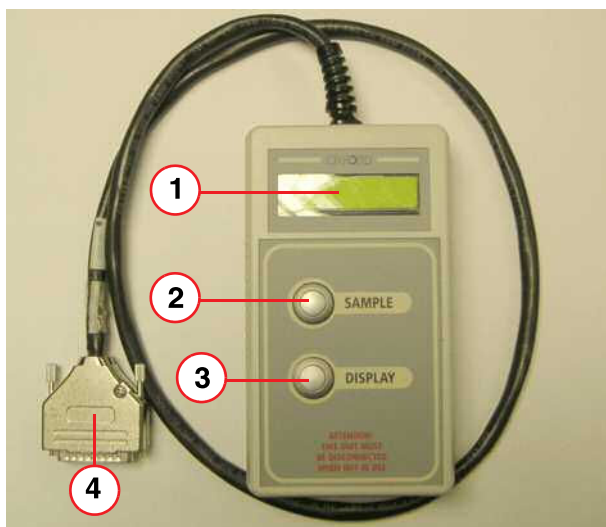
5.2.4 MSUP-Display Unit

The only operations that can be performed independently of the controlling host computer are through the **MSUP-display unit**.

Required tools

MSUP-display unit

Fig. 233: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Performing the procedure

- Connect the display unit to **X9** on the MSUP via the 25 way 'D' connector (4)
 - Ensure that power has been applied to the MSUP.
 - Switch on the hand held unit by simultaneously pressing the **Sample** button (2) and **Display** button (3).
 - Press **Display** (3).
 - The unit will **automatically** carry out a helium level test. The **helium fill level** is shown in the display (1).
- Scroll through the parameters by pressing the **DISPLAY** button, one parameter for each press.
- The parameters in order of presentation are:
1. He level (press sample to recheck level)
 2. Battery volts
 3. 50K link, bore (K, K)
 4. Carbon resistor 1, 2 (ohm, ohm)

5. Carbon resistor 3, 4 (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min timeout)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 60 min timeout)
15. Self-test



Disconnect the display unit from the MSUP after test is finished.

6.1 Verifying the pressure-controlled interlock



Verify the pressure-controlled interlock although the magnet is not refilled during start-up.

Refer to the procedure in the "filling the magnet" chapter (→ Verifying the electronic low pressure switch / Page 195)

6.2 Venting leak check

6.2.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1×10^{-4} mbar l/s or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as 1×10^{-4} mbar.

The magnet must be checked after any service operations have been performed, such as:

- Installation and Startup.
- After helium filling.
- After replacing a component on the high pressure side of the magnet. (cold head, venting).
- After a magnet quench. (Post quench).



CAUTION

Risk of icing

Small leaks may not show any change in pressure. Over time, they will contribute to air ingress and icing.

- » Leak check with the recommended equipment for small leaks greater than 1×10^{-4}

6.2.2 Safety



CAUTION

Pressurized helium vessel.

Risk of cold burns. Failure to observe the following safety information may result in injury.

- » Fittings on the high pressure side must not be disconnected without first depressurizing the helium vessel.

**WARNING**

Magnet at field.

Personal injury or damage to equipment may occur.

- » Do not use magnetic tools for repairing or tightening seals or screws.
- » Only non-magnetic equipment may be taken into the magnet room.

6.2.3 Tooling and Spare Parts

- Heat gun.
- MSUP-Display Unit.
- Paper towelling.
- Standard Service hand tools.
- Leak detection equipment:

Fig. 234: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Refer to the equipment manufacturer's operating instructions.



The function of the Edwards leak detector can be compromised by magnetic fields if placed near the magnet bore.

6.2.3.1 Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal or failed component.

- A number of seals are supplied in the spares kit. These must be replaced when one is used.

- Venting components will require ordering from CSML.

6.2.4 Venting Configurations

There are several configuration changes to the OR105 and OR122 magnet venting:

- Update to the 16 psi (A) magnet bypass valve assembly and its location.
- Reduction in size to the OVC vent and OVC burst disk.
- Removal of the OVC burst disk (DOP port).
- Pressure gauge updated to WIKA type.

Fig. 235: Mk1 OR105 Venting System



Fig. 236: OR122 Mk1 Venting System

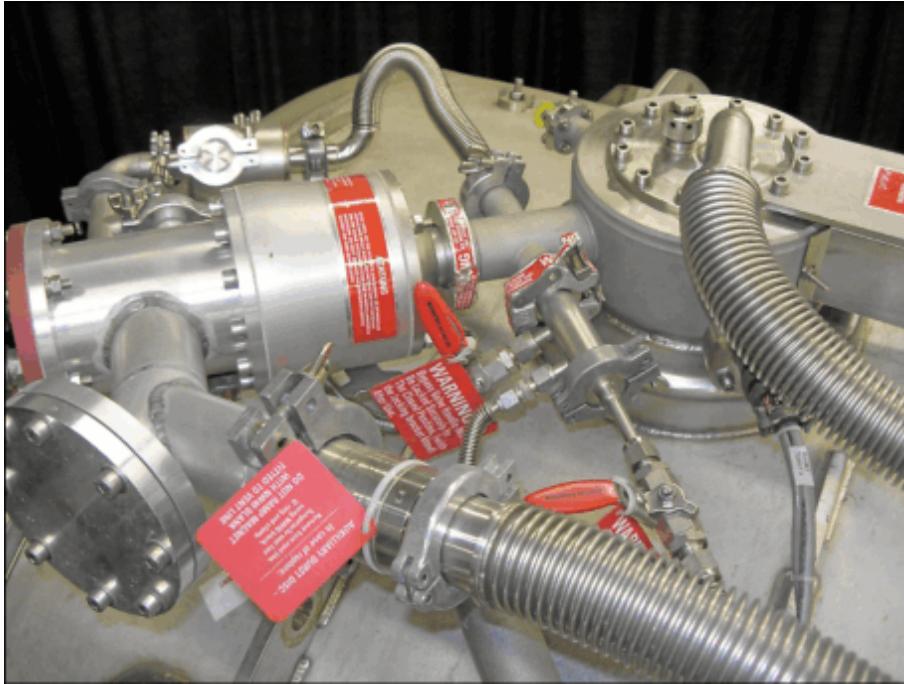


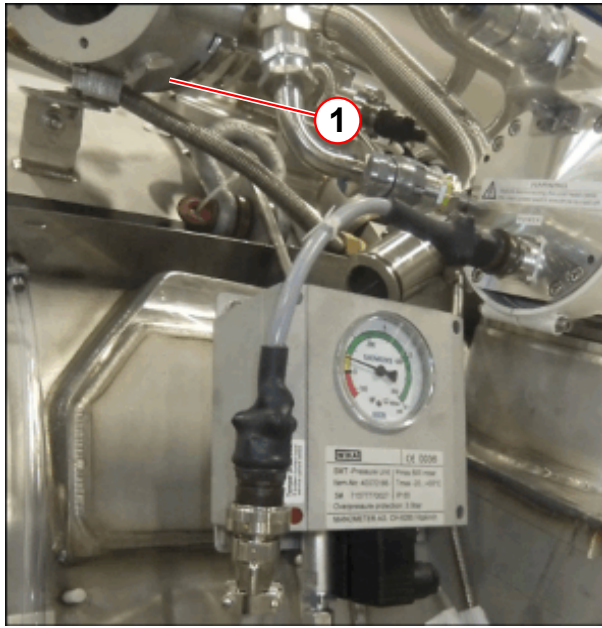
Fig. 237: Mk2 Venting System



Mk2 magnet bypass vent valve assembly new location.

Smaller OVC burst disk vent line and burst disk.

Fig. 238: Venting - No OVC vent port



(1) OVC vent port blank

Venting updated with the removal of the OVC burst disk and vent line to the quench valve.

WIKA pressure gauge fitted.

6.2.5 Leak Check Process at Installation and Service

6.2.5.1 Installation

- Check the system pressure on arrival. If the system has low or zero pressure then a gross leak has occurred during shipment.
- Zero pressure indicates a gross leak. Immediately check the burst disks.
- Check any joints (clamps and seals) that have been broken or damaged during transportation.
- If magnet pressure is still low, check for leaks through the magnet 17 psi (A) vent valve.

6.2.5.2 Service

- The turret, vent valve assembly and vent lines must be warmed up to room temperature after any service procedures have been performed, prior to leak check.
- Leak check all seals that have been subjected to low temperatures and the subsequent build up of frost. For example, magnets that have been helium filled, depressurized, or the magnet ramped.
 - Check joints broken and re-made during servicing. Replace damaged seals.
 - Check that all clamps and seals are tight. Do not over-tighten them, it may cause a leak, or make an existing leak worse.

6.2.5.3 Pre-requisites

- The helium vessel must be at normal operating pressure (15.5 psi (A)) prior to leak checking.

- Activate the pressure heater (if required) to raise pressure.
- If pressure heater activation continues for more than 30 min with NO increase in the magnet pressure:-
 - Stop heater activation and check for any obvious gross leaks such as the syphon ring nut or burst disks.
- All joints must be dry and at room temperature before starting a leak check.
 - » Warm all components up using a heat gun and dry. Moisture can cause false readings.

6.2.5.4 Leak Check Process

- Switch on the leak detector and allow it to warm up (refer to manufacturer's instructions).
- Check the equipment is working correctly (detecting helium). Check equipment before and after use.
 - Set the leak detector to its maximum sensitivity.
 - Check the response time of the detector.
- The background helium level in the magnet room must not be saturated with helium gas.
 - » **DO NOT** leak check immediately after a quench or helium filling.



The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s).

- Start at the lowest point.
- Hold the tip of the leak detector 5 mm above the surface of the component. **Keep the tip above water condensation.**
- Slowly move the detector around the seals or clamped joints.
- Hold the detector at a leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.
- Repeat the process to check that a leak is confirmed.
- Complete leak check of turret and venting.
- Fix all leaks (TSG) and repeat leak check of the component and seal.



Do not tighten any seals when the components are cold and covered in frost. Always warm up with a heat gun and dry with paper towelling.

6.2.5.5 Leak Check Requirements

The following points must be observed at all times:

- Before dismantling a joint, check that a replacement seal is available.
- Always **CLOSE** the magnet bypass valve as soon as the system is depressurized.
- Always plug open venting with paper towelling to prevent air ingress.

- **Do not** leave 2 joints open at the same time.
- **Do not** leave the system open for longer than is necessary to complete the repair.
- Check that the sealing surfaces are clean, dry, and undamaged.
- **Do not** replace metal seals with rubber or silicon seals.



Do not use any form of vacuum grease on O-rings and seals.
Vacuum grease dries over time and may cause a seal to leak.

For all leaks, first try tightening the clamp or the securing screws.

- **Do not** overtighten clamps or screws.
 - » If tightening does not solve the leak, disassemble the joint and check for any damage to the sealing surfaces.
 - » Replace the high load seal.
- **DO NOT** re-use metal seals. Assemble the joint using a new seal. Ensure that the seal is undamaged and clean, and is seated correctly.
- After repair, leak test the entire venting system.

6.2.6 High Pressure Turret and Venting Components

The venting and components on the high pressure side of the magnet turret require leak checking to ensure that the re-condensing performance is achieved. This includes all connections too:

- Cold Head
- Turret top plate, including syphon port and venting
- 16 psi (A) Vent valve assembly
- Pressure gauge assembly
- Vent line connections.



If no external leak is found but the system pressure is low, then it is possible that one of the internal valves is leaking. This can be caused by a cracked or broken burst disk, or a broken O-ring.



Before a leak check, the venting components must be warmed to room temperature, and dried with paper towelling.

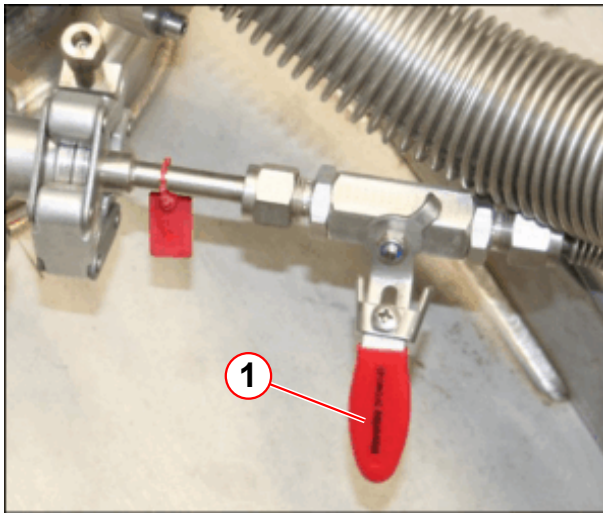
6.2.6.1 Cold Head

Leak check the cold head components:

- Check the hand-valve and cold head vent line seals at both ends.
- Cold head O-ring seals.
- Temperature sensor connector seals.

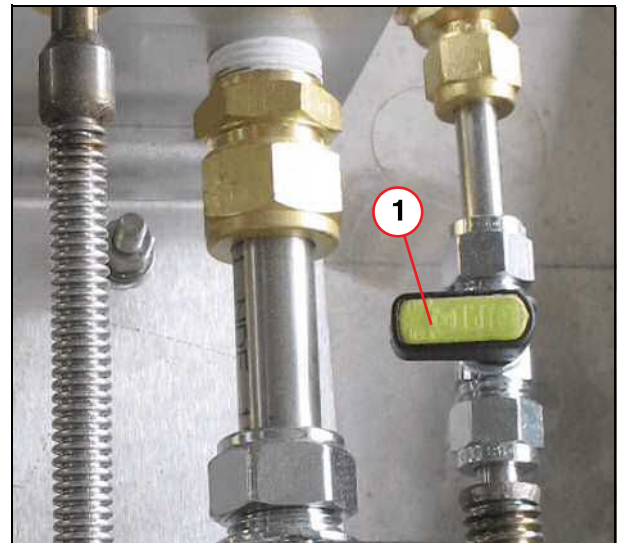
The cold head vent line is to remain connected with the hand-valve CLOSED during normal system operation.

Fig. 239: Mk1 Venting - Cold head vent line valve



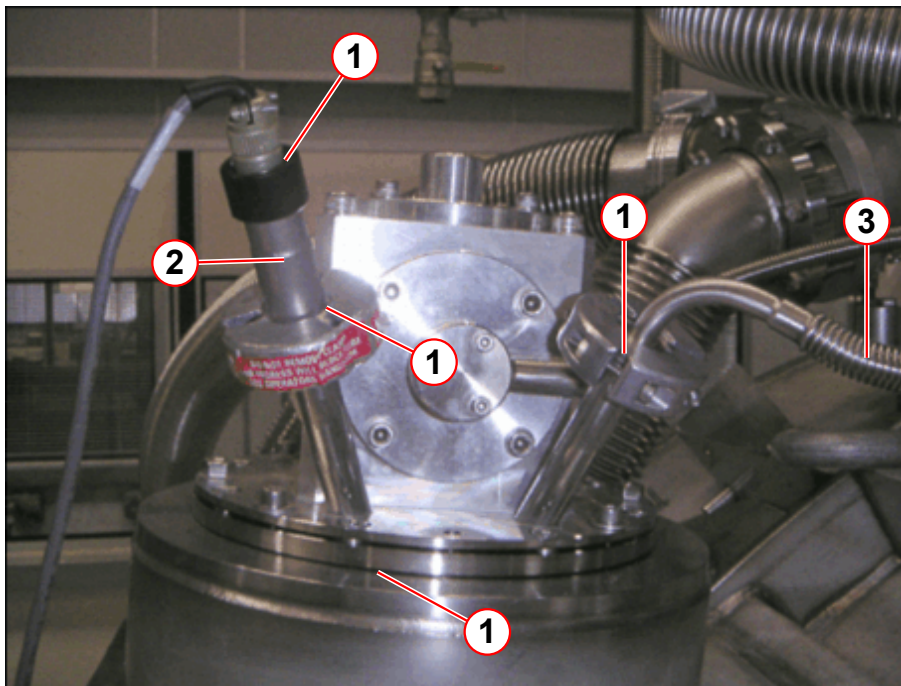
(1) Valve (shown closed)

Fig. 240: Cold head vent line valve is closed



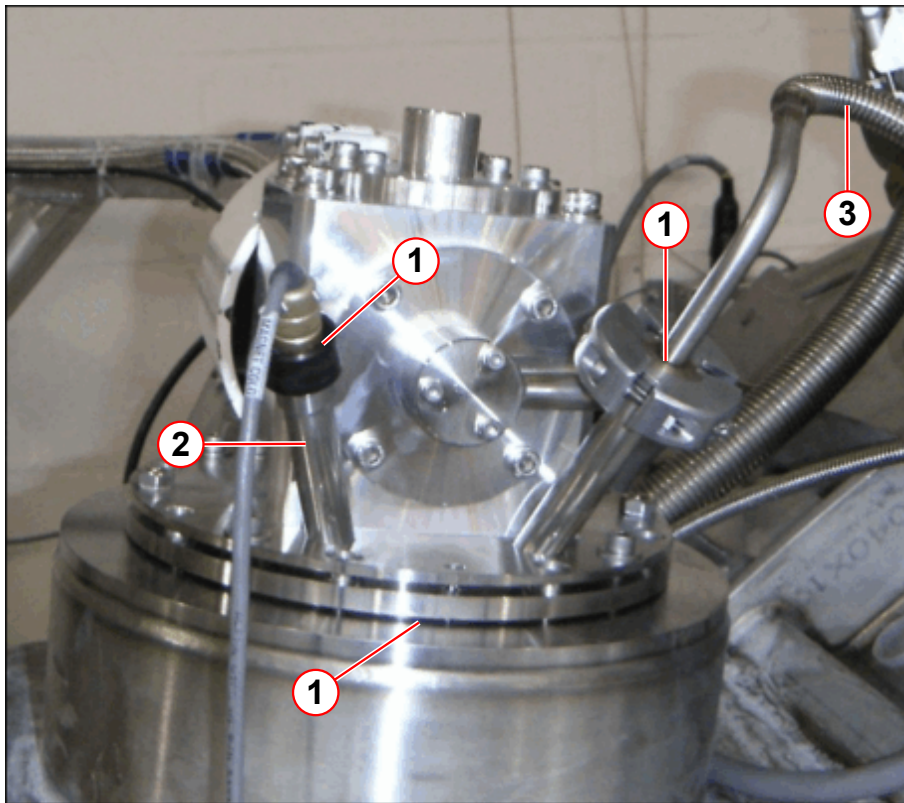
(1) Cold head vent line valve is closed

Fig. 241: Leak checking the cold head seals (early coldhead)



- (1) Cold head seals leak checkpoints
- (2) Temperature sensor feed-through
- (3) Cold head vent line

Fig. 242: Leak checking the cold head seals (later coldhead)



- (1) Cold head seals leak check points
- (2) Temperature sensor feedthrough
- (3) Cold head vent line



If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

6.2.6.2 Leak Checking the Temperature Sensors



WARNING

Risk of cold burns. High helium gas pressure.

Risk of personal injury may occur.

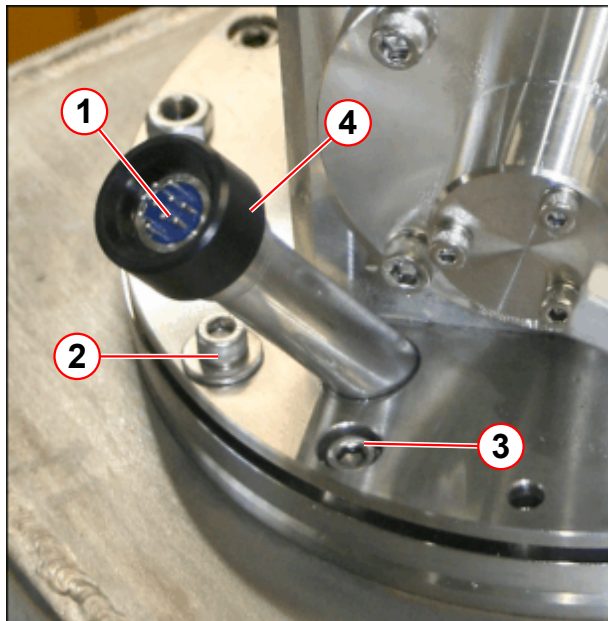
- » Do not unscrew or remove the black temperature sensor sealing cap.
- » Remove the temperature sensor cable with a straight pull away from the sealing cap.



Cold heads fitted from 2015 have a protective heat shrink and a warning label fitted around the temperature sensor connector.

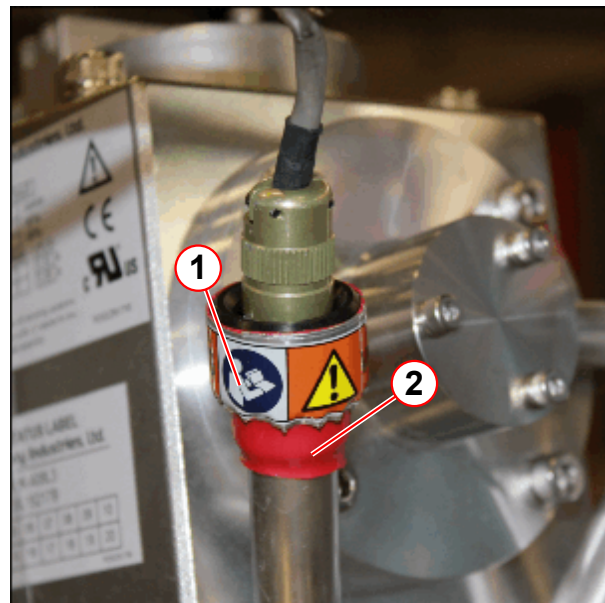
- **DO NOT** turn or try to remove the 'black' sealing cap (pos.4).
- Remove the temperature sensor cable with a straight pull away from the black sealing cap.
- Leak check the 10 pin seal connector,
- Immediately refit the cable if no leak is found.

Fig. 243: Cold head 10 pin seal



- (1) Leak check point
- (2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
- (3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
- (4) Black sealing cap

Fig. 244: Protective heat shrink & warning label



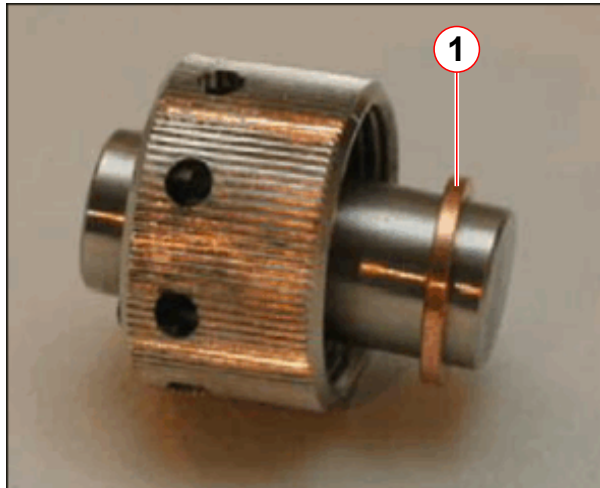
- (1) Warning label
- (2) Heat shrink

6.2.6.3 Syphon Port

There are several design revisions of the syphon port:

- Location remains on the turret top plate.
- Leak check and corrections are the same for each syphon port example.

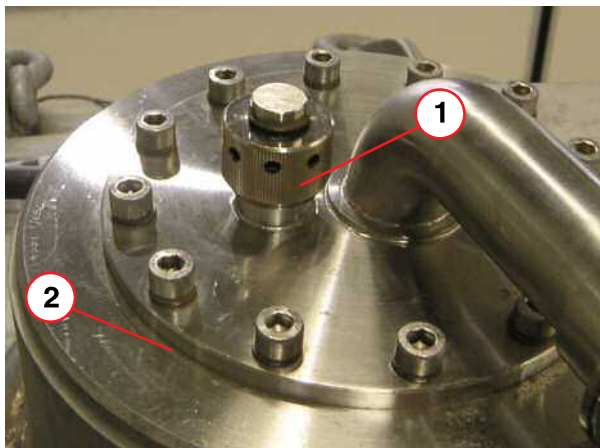
Fig. 245: Syphon blanking plug



(1) Copper seal

- After all helium fills warm up the syphon port. and fit a new, copper seal, (material number **03108094**).
- Tighten the ring nut firmly. Do not over- tighten the syphon plug.

Fig. 246: Mk1 Turret top plate



(1) Syphon entry
(2) Turret top plate seal

- Warm up and dry around the syphon port.
- Tighten the ring nut again. Do not over- tighten the syphon plug.
- Leak check. Repeat warm up of syphon plug and tightening of ring nut.
- If a leak persists, replace the copper seal.

Fig. 247: Syphon Brass Hex plug (example)

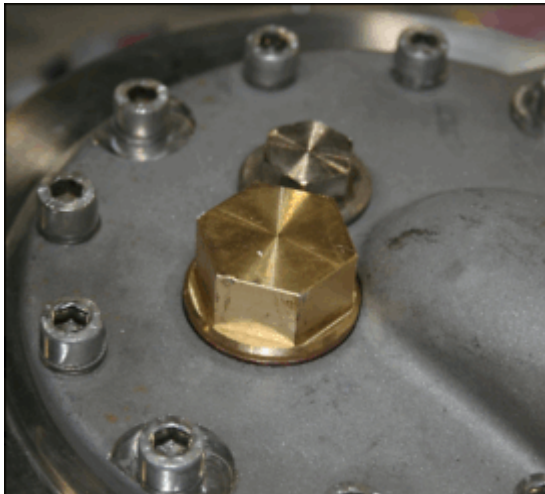


Fig. 248: Syphon plug (current 2017)

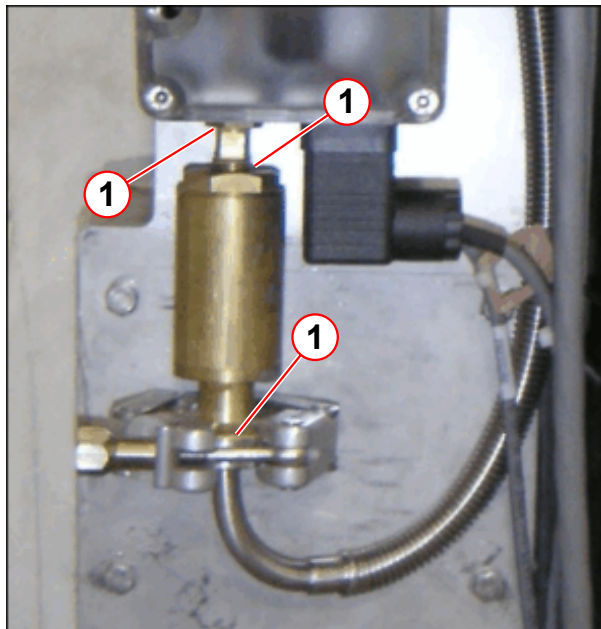


6.2.6.4 Pressure Gauge Assembly

Leak check the pressure switch transducer assembly. Note: there are different versions depending on system status level.

- Leak check around the fixings of the vent line to the pressure gauge.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws. **BEWARE OF OVER-TIGHTENING!**
- If a leak persists the seal must be replaced.

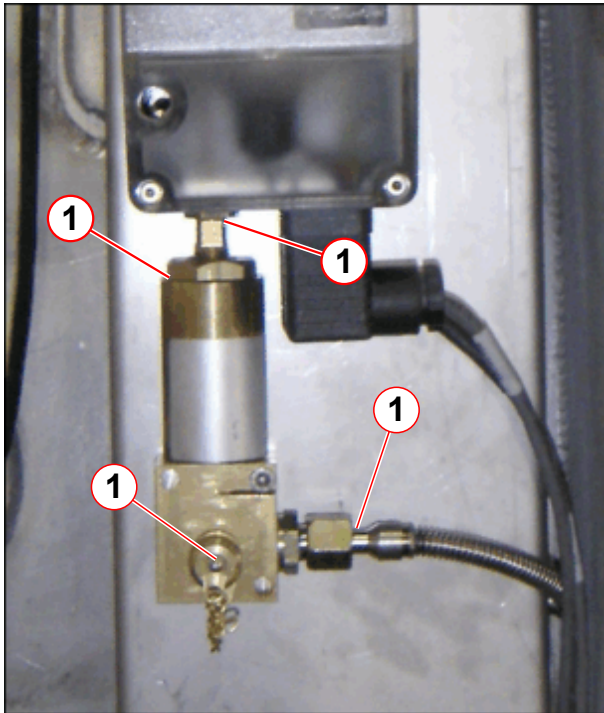
Fig. 249: Mk1 Pressure gauge - leak check



(1) Leak check points

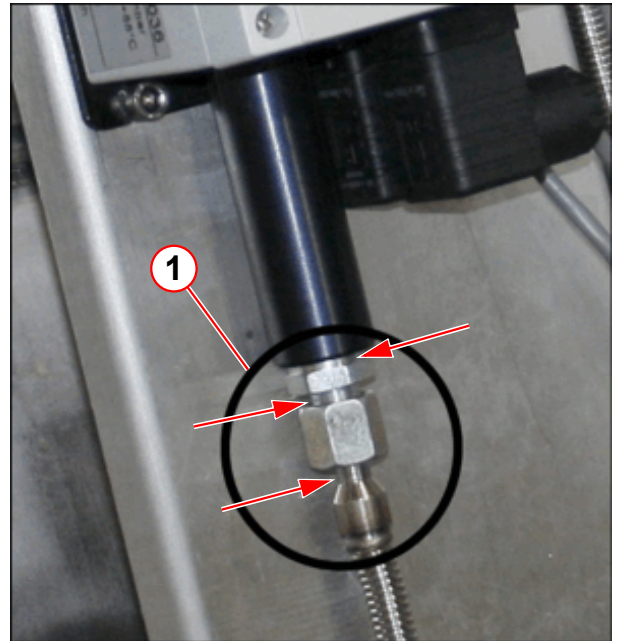
- Leak check around the high load seal and clamp at the pressure gauge.
- OR - leak check around the compression nut and high load seal.

Fig. 250: Pressure gauge - leak check



(1) Leak check points

Fig. 251: Pressure gauge - leak check



(1) Leak check points

6.2.6.5 FCL Lead Connector

The FCL lead connector passes through the turret in a ceramic interface. It is made leak tight by a high load seal. The FCL ceramic connector and seal should only be checked if no other leaks have been found.

- A broken ceramic seal will be seen as a gross leak.
- Report a damaged ceramic seal to the Magnet HSC. The FCL turret connector and internal wiring may need replacing.

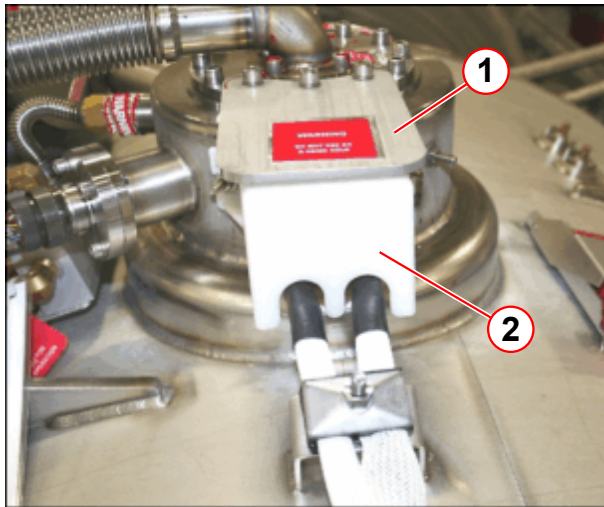
NOTICE

Air ingress

The ceramic seal is a fragile component

- » Take care when removing the FCL cover or working next to the ceramic seal.

Fig. 252: FCL cover and location



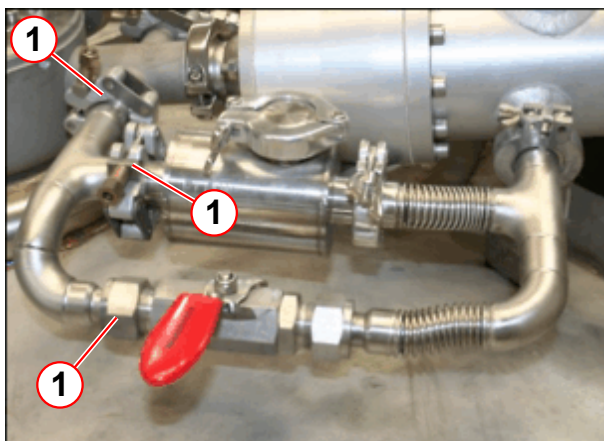
- (1) Protection plate
- (2) FCL cover

The FCL connector is located under the plastic protection cover.

- Only remove the cover if a leak is detected around this area.

6.2.6.6 16 psi (A) Vent Valve Assembly

Fig. 253: Mk1 Vent valve leak check

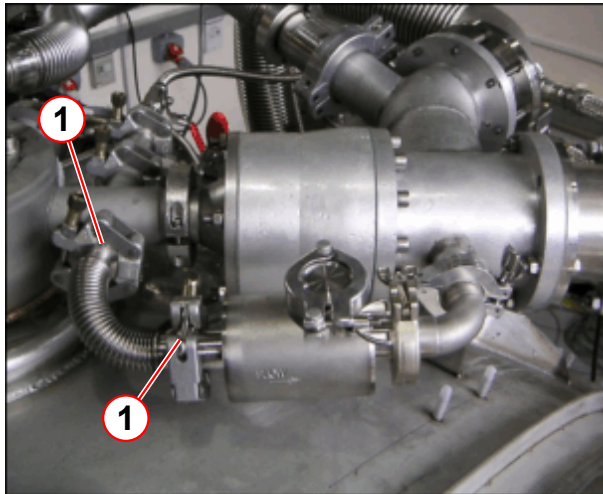


- (1) Leak check points

Mk1 valve and venting

- Check for a leak at the 16 psi (A) valve high pressure fittings.

Fig. 254: Vent valve leak check

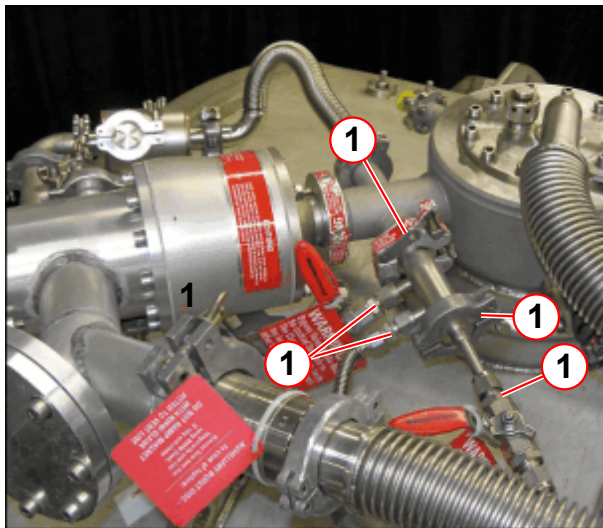


(1) Leak check points

Mk1 OR122 - valve and venting

- Check for a leak at the 16 psi (A) valve high pressure fittings (pos.1).

Fig. 255: Mk 1 Byass valve leak check

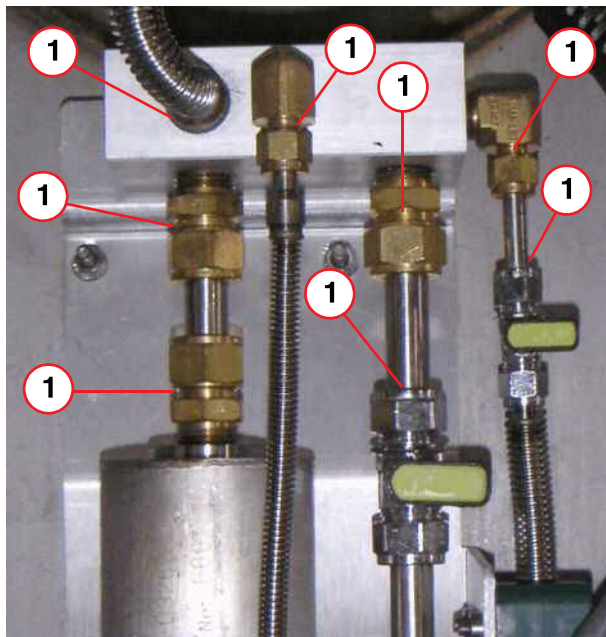


(1) Leak check points

Mk1 OR105 and OR122 - magnet by-pass valve venting.

- Check for a leak at the high pressure fittings (pos.1).

Fig. 256: Mk2 Venting 16 psi (A) Valve Assembly



(1) Leak test points

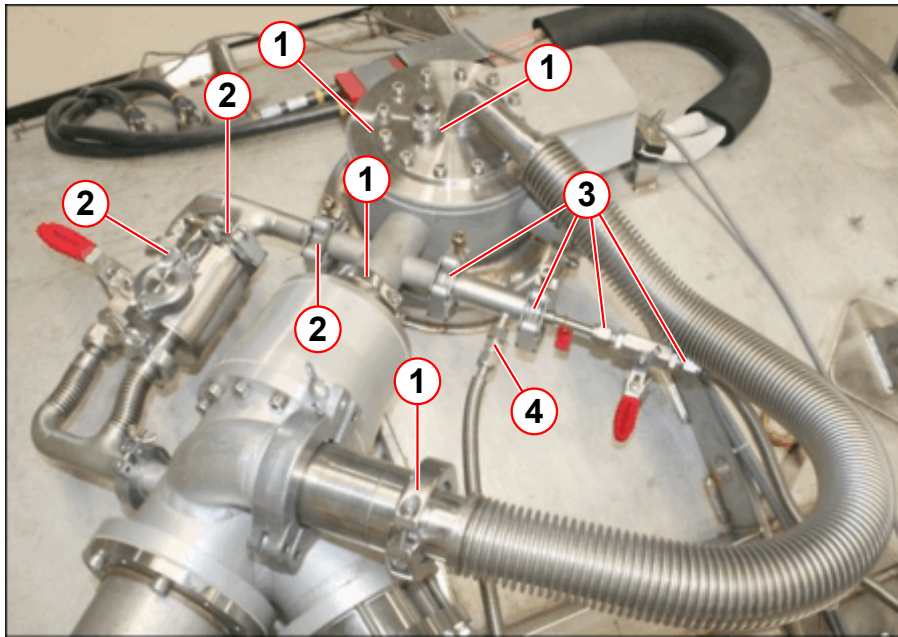
- Leak check the 16 psi (A) valve high pressure assembly.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws. **DO NOT OVERTIGHTEN!**

6.2.6.7 Turret and Venting

The figures below show the high pressure leak check locations for MK1 and MK2 turret configurations.

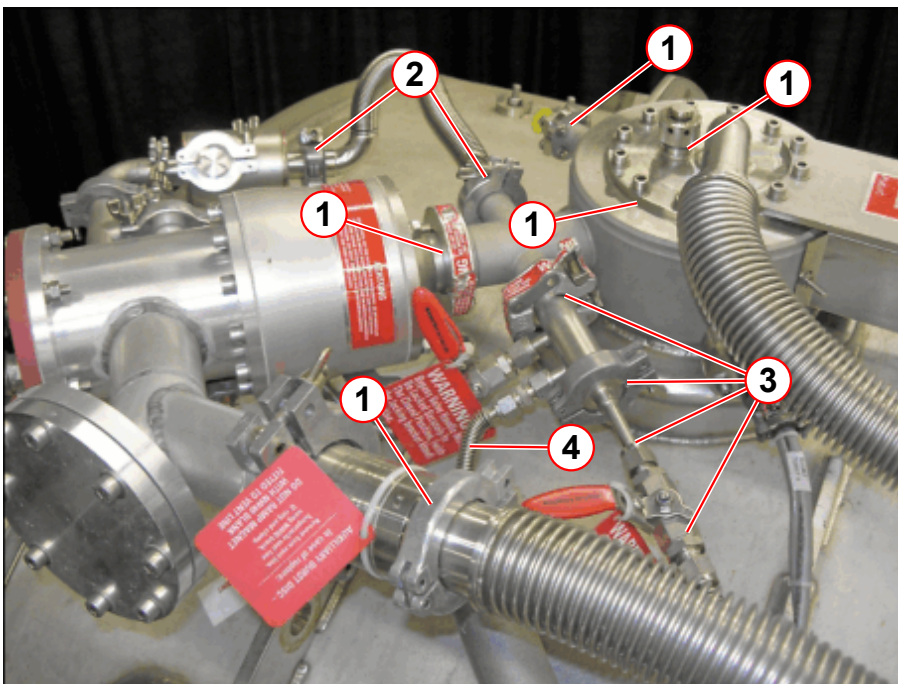
- Turret top plate, syphon port and O-rings (Pos. 1).
- Connections to the 16 psi (A) vent valve assembly (Pos. 2).
- Cold head line (Pos. 3).
- Pressure gauge line (Pos. 4).

Fig. 257: OR105 Mk1 - Turret and venting leak checks (high pressure)



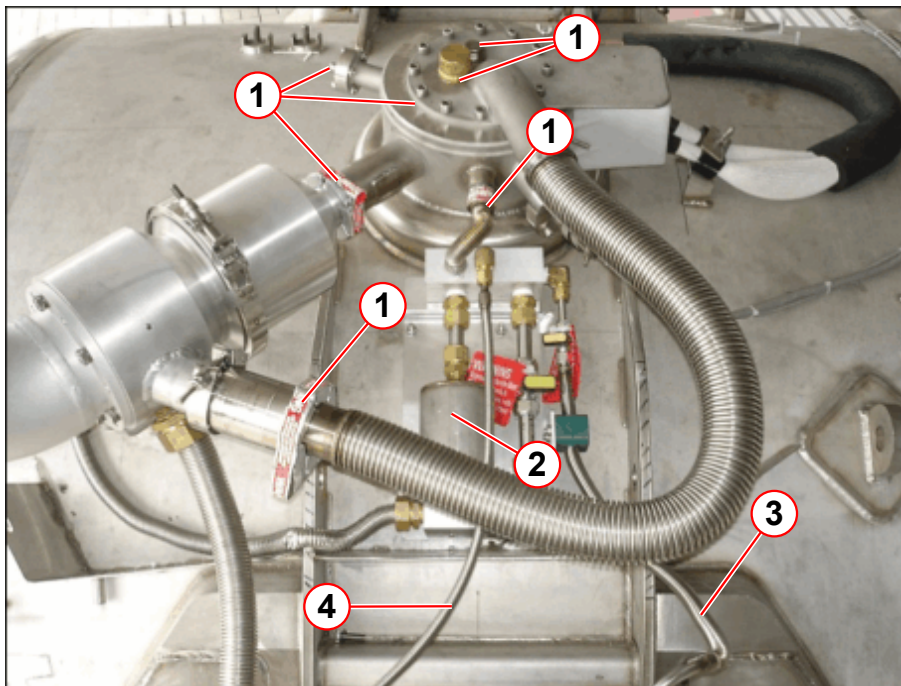
- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

Fig. 258: OR122 Mk1 - Turret and venting leak checks (high pressure)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

Fig. 259: OR105/122 Mk2 - Turret and venting leak checks (high pressure)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line

6.2.7 Internal Leak Checks

Leaks can occur internally through the magnet 16 psi (A) vent valve or burst disks:

- 1.8 bar auxiliary metal bursting disk.
- Quench valve and graphite burst disk.
- 16 psi (A) vent valve assembly.



CAUTION

Internal leak checks.

Risk of air ingress causing internal icing of turret.

- » Replace burst disk or valve immediately if found to be broken or leaking.



The magnet must be at **ZERO FIELD** before leak checking inside the quench valve or auxiliary vent.

**WARNING**

Risk of air ingress

If a burst disk is broken there is a high risk of air ingress and ice.

- » **IMMEDIATELY** fit the replacement burst disk from the spare parts kit.
- » **Order a replacement burst disk.**

6.2.7.1 Auxiliary Burst Disk

The auxiliary burst disk should not break during a magnet quench.

- If necessary warm up the burst disk and venting with a heat gun.
 - Activate the pressure heater.
 - Allow the system to pressurize.
- Leak check externally around the 1.8 bar high pressure seal (pos.3).
- If there is an internal leak or ruptured burst disk the magnet pressure will be zero.

To check for any leak path through the auxiliary metal bursting disk:

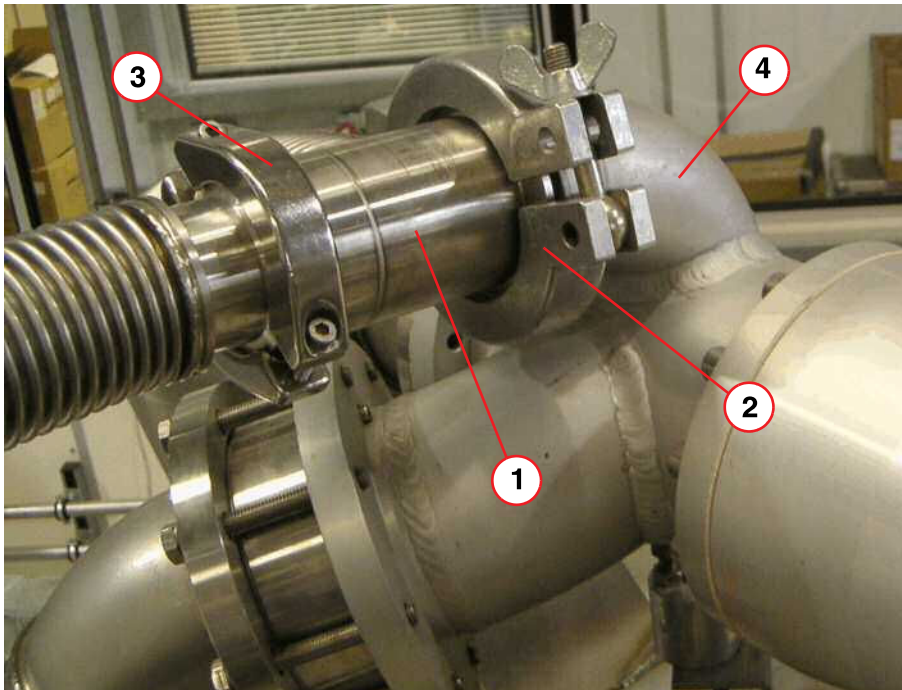
- Undo and remove the NW50 clamp on the low pressure side of the auxiliary burst disk (pos.2).



Do not touch the metal membrane; it is very fragile.

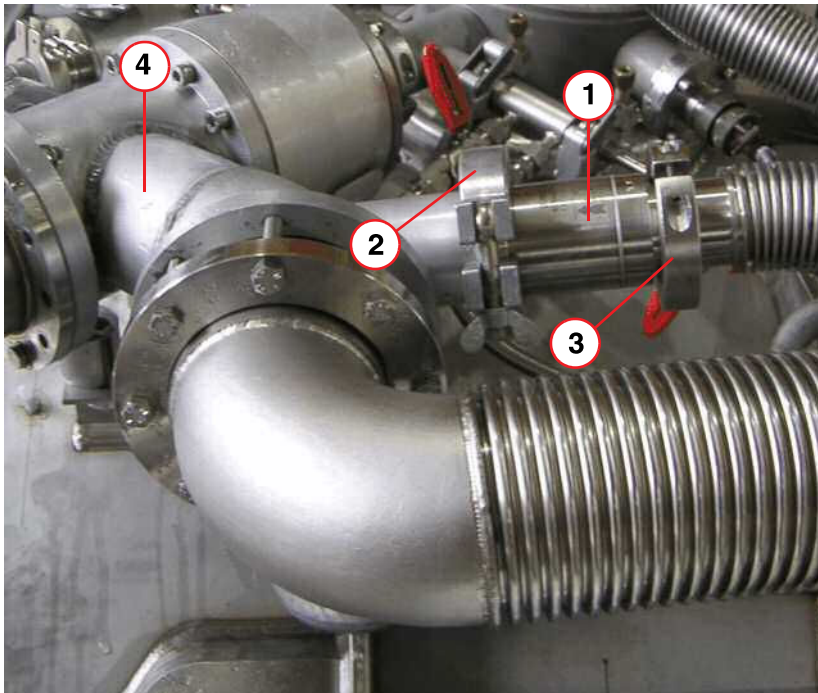
- Visually inspect the internal membrane of the burst disk for damage.
- Visually inspect the O-ring on the low pressure side.
- Replace the auxiliary metal burst disk if a leak is detected.

Fig. 260: OR105 Mk1 Auxiliary vent burst disc assy



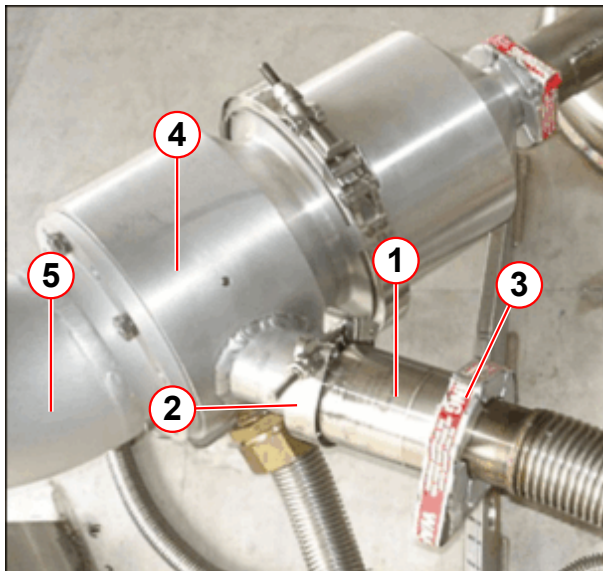
- (1) Aux vent burst disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

Fig. 261: OR122 Mk1 Auxilliary vent burst disc assy



- (1) Aux vent Burst Disc
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

Fig. 262: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

To check for any leak path through the Mk 2 auxiliary bursting disk:

- Undo and remove the 'V' clamp from the low pressure side of the auxiliary burst disk.
- Visually inspect the internal membrane of the burst disk for damage.
- Visually inspect the O-ring on the low pressure side.

6.2.7.2 Quench Valve Burst Disk Assembly



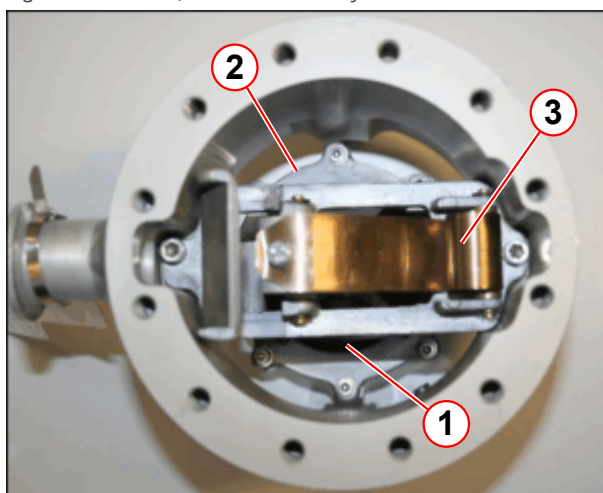
The magnet must be RUN TO ZERO before checking the quench valve.

If the magnet has a low gauge pressure and will not pressurize, it is possible that the quench valve has an internal leak.

The quench valve burst disk and spring plate must be leak checked at installation & Start Up and Post quench.

- Remove the quench valve elbow and manifold assembly from the turret to access the burst disk.

Fig. 263: Mk1 Quench valve assy

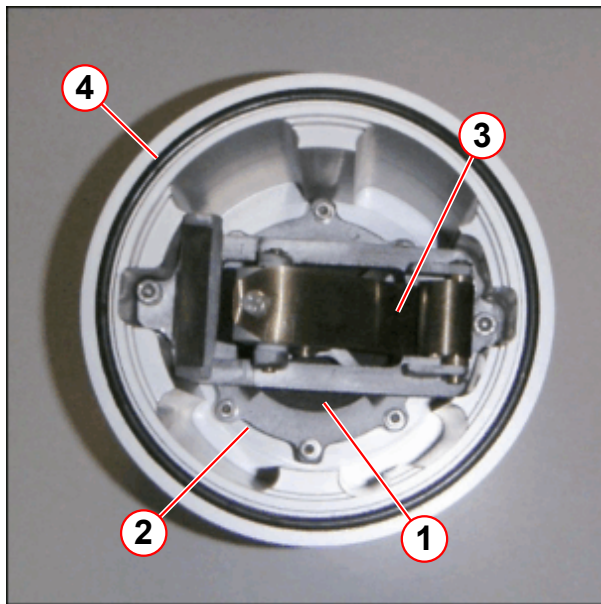


- (1) Graphite burst disk
- (2) O-ring sealing plate
- (3) Spring assy

Perform a leak check of:

- The graphite burst disk.
- Around the O-ring seal to the turret (pos.2).
- Check the condition of the spring and retaining screws.

Fig. 264: Mk2 Quench valve assy

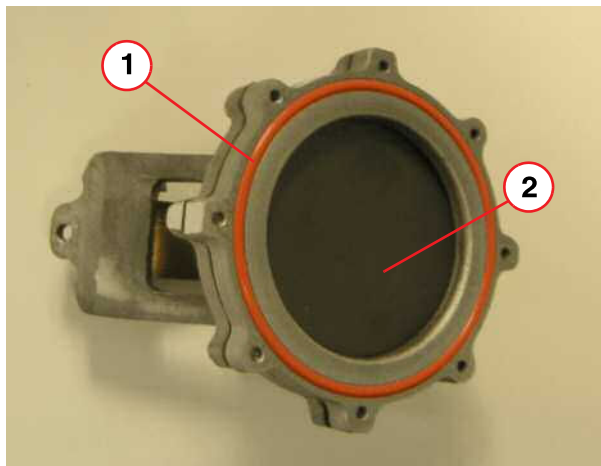


- (1) Graphite burst disk
- (2) O-ring sealing plate
- (3) Spring assy
- (4) O-ring (low pressure)

Perform a leak check of:

- The graphite burst disk.
- Around the O-ring seal to the turret (pos.2).
- Check the condition of the spring and retaining screws.

Fig. 265: Quench Valve Bursting Disk Assembly



- (1) O-ring
- (2) Burst disk

If a leak is seen around the burst disk or spring plate the magnet must be depressurized and the burst disk assembly removed.

- » Details of removing the quench valve is in the Replacement procedures.
- Inspect the burst disk.
 - » If the burst disk has ruptured or is cracked, it must be replaced.
- Inspect the burst disk O-ring.
 - » If the O-ring is split or damaged it must be replaced.

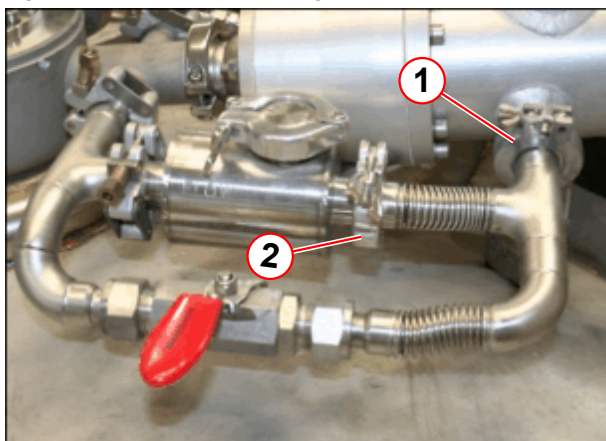


Do not use vacuum grease on the quench valve O-ring.

6.2.7.3 16 psi (A) Valve Assembly

If the magnet has a low gauge pressure and will not pressurize, it is possible that the vent valve assembly has a gross internal leak.

Fig. 266: Mk1 OR105 venting - internal leak checks

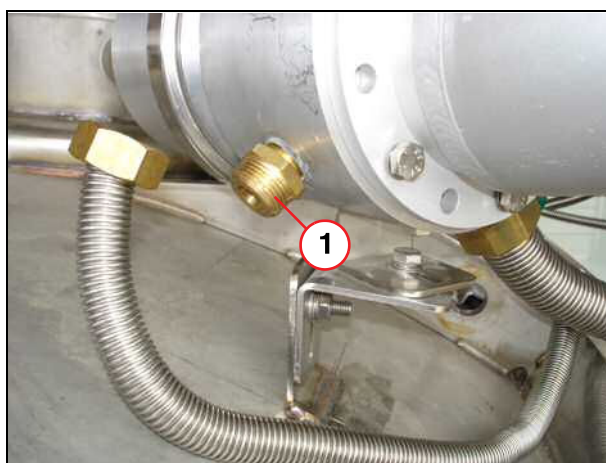


- (1) NW25 Clamp - quench valve port (low pressure)
 (2) NW25 Clamp - 16 psi (A) valve (low pressure)

Mk1 valve and venting

- Check for an internal leak through the 16 psi (A) valve.
- Disconnect the 16 psi (A) flexible vent line from the quench valve port.

Fig. 267: Mk2 16 psi (A) manifold line

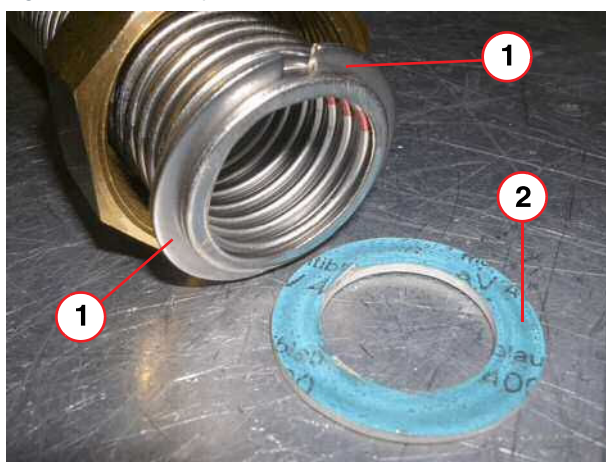


- (1) 16 psi (A) valve outlet

Mk2 valve and venting

- Disconnect the 16 psi (A) flexible vent line from the quench valve manifold.

Fig. 268: Mk2 16 psi (A) vent manifold connection

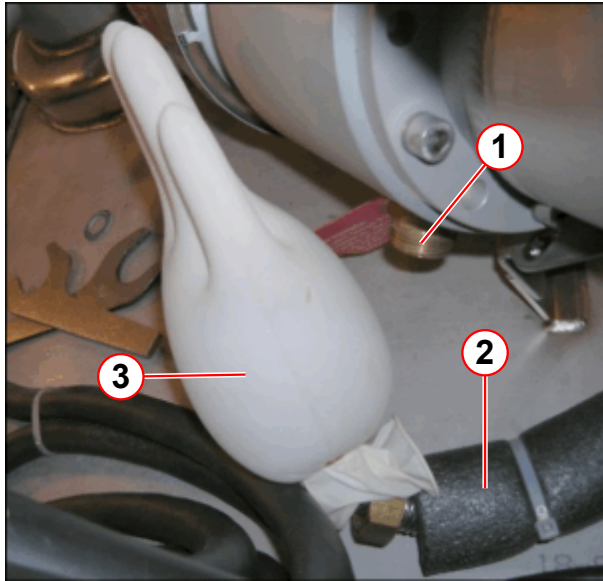


- (1) Split rings
 (2) Gasket

Take care not to lose the split rings during this process!

- Ensure the split rings do not fall out of position.
- Be careful to retain the gasket when disconnecting the vent line.
 - The gasket is not retained by the split rings and nut.

Fig. 269: 16 psi (A) valve - checking for gross internal leaks



- (1) 16 psi (A) quench valve outlet
- (2) 16 psi (A) vent line
- (3) Surgical glove

- Fit a surgical glove (or small airtight plastic bag) over the end of the open low pressure vent.
 - » Seal the glove with a cable tie.
 - » If the glove expands rapidly with the magnet pressure lower than 15.3 psi (A) (0.3 psi (G)), this indicates a gross leak.
 - » If the valve is not sealing, replace the bypass valve assembly.

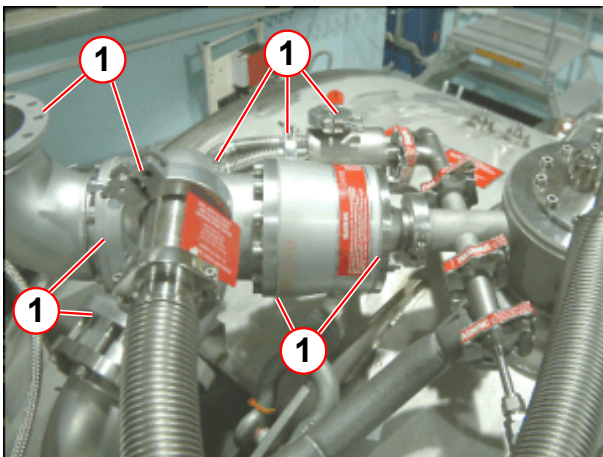
6.2.8 Low Pressure Turret and Venting Components



Low pressure venting leaks can only be checked visually during depressurization , helium filling or a magnet quench.

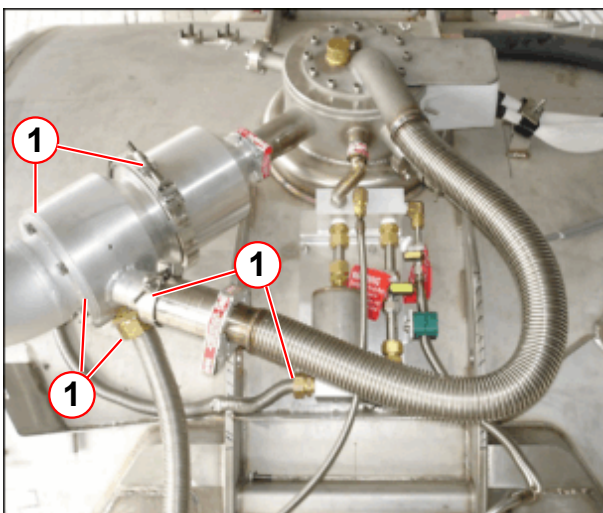
- Check visually for cold gas escaping around low pressure joints during:
 - depressurization, helium filling or a magnet quench.
- Check that all sealing nuts and flange screws on the venting are tight.
 - Warm venting and replace low pressure gasket seals and O-rings if found damaged.

Fig. 270: MK1 - Low pressure seals



(1) Leak check points

Fig. 271: Low pressure seals (example)



(1) Leak check points

6.2.9 Final Work Steps

- After the leak check, ensure that the leak detector is working correctly:
 - » Check the equipment function with a known helium gas source.
- Ensure that all parts of the venting system have been connected and fitted.
- Refit all system covers.
- Order replacement seals if spares have been used.

7.1 Testing the ERDU circuit



Before initially ramping up the magnet, the ERDU test has to be performed.

7.1.1 Required tools

- Magnet Display Unit (MDU)
- ERDU Test Load (ETL)



Perform the ERDU circuit test as described in the SeSo/Install & Maintenance/ Safety-Related Tests.

- Document the ERDU test in the safety-related test protocol.

7.1.2 Affix the tamper-proof seal

After the ERDU test is completed, the tamper-proof seal has to be affixed to the alarm box and ERDU buttons.

Fig. 272: Tamper-proof seal location



(1) Tamper-proof seal



The tamper-proof seal is delivered with the system.

7.2 Ramping the magnet - Avanto



Only Siemens CSEs trained in ramping magnets are allowed to perform the magnet ramping procedure.



Before ramping check that the quench line is correctly installed from the magnet to the outlet.



WARNING

Magnet safety checks

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Ramping is not allowed if the visually inspection of the quench line was not performed successfully.
- » Before initially ramping up the magnet, the 'Testing the ERDU circuit' has to be performed.
- » Procedure described in the safety related test can be used if ERDU test load is available.



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.
- » Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign has to be posted in a way that it is visible even when the door to the controlled area is opened.



CAUTION

Risk of asphyxiation and hypothermia, if helium gas is vented into the magnet room

Failure to observe the following safety warning may result in dizziness or loss of consciousness in personnel.

- » Quench valve must be connected to the Quench pipe.



Adhere to the general safety information included in this chapter when performing these steps.



Startup (1st) ramp helium level recommended at $\geq 65\%$.



Minimum helium level for ramping must be $> 40\%$.
Wait at least two hours after helium filling.



Check that the venting bypass valve is locked securely in the CLOSED position.



Check for leaks before ramping the magnet.

Always

- Always restrict access to the magnet room when ramping the magnet. Use flashing lights and signs.
- Always inspect all equipment and tools for defects prior to use.
- Always inspect the Anderson ramp lead connections at the filter plate.
- Clean the area of the magnet room, dispose off empty cartons or other packing materials
- Always inspect magnet room for the presence of non identified objects. Before ramping, the nature of those objects must be clarified.



Prepare a minimum of 4 pictures of the room **prior to first ramping**.

These pictures must be colored and should have 2 to 4 megapixel resolution. They must show the complete room, all 4 corners and walls. In case there are un-removable cupboards installed, their doors must be open to prove that they are empty.

Later those pictures are to be attached to the installation report.

Never

- Never use defective or damaged equipment.
- Never allow unauthorized people to stand too close to the magnet during ramping.
- Never block emergency exits.
- Never smoke or use open flames in the magnet room.
- Never ramp a magnet when you are unsure about magnetic parts inside the magnet room. Those parts could be hidden in e.g. supposed empty boxes or in closed cupboards or sliding of a desk.

Ramping a Mobile System



It is necessary to bump the magnet to account for the effect of the iron shielding.
Our recommendation for demand current on first ramping is to increase the magnet current by 2 A.

7.2.1 Tools required



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

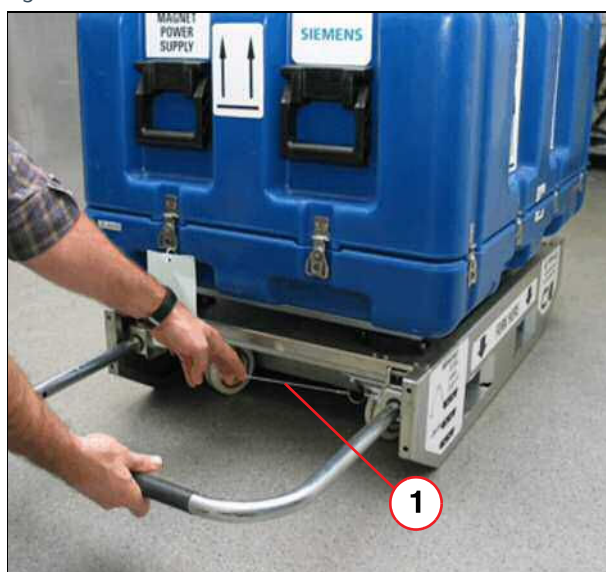
» Only non-magnetic equipment may be taken into the magnet room.

- Magnet Power Supply (3600 MPSU 3600); (order number: **8395613**). This includes:-
 - 3600 MPSU Power lead
 - CAN Open lead
 - RS232 lead
- Heat gun.
- Laptop or host computer.

7.2.2 Preparation

- Move the 3600 MPSU into the equipment room close to the RF filter plate.

Fig. 273: MPSU



(1) Wire locking strap

- Lower the trolley handle into the horizontal position. This will raise the wheels and allow the trolley to rest on the two skids.
- Pull the wire strap to unlock the handle, and slide the handle into the recess under the trolley.
- Remove the shipping case lid by unclipping the 12 spring clips.

Fig. 274: MPSU

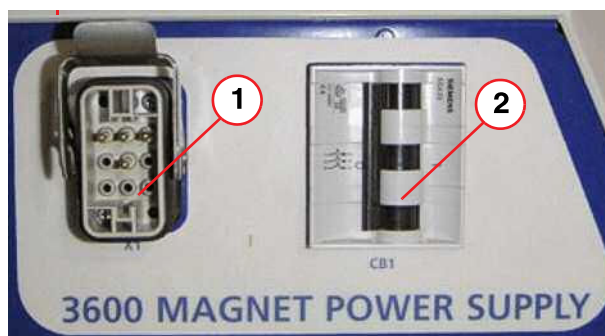


(1) MPSU LID

- Open the hinged lid of the 3600 MPSU by pulling the two silver knobs and turning them through a 1/4 turn.
- Raise the lid.

7.2.3 Installation

Fig. 275: 3600 MPSU Power connections



- (1) X1 (MPSU power)
- (2) CB1 (Mains circuit breaker)

- Ensure the mains circuit breaker **CB1** is in the **OFF** position on the 3600 MPSU.

Fig. 276: Adapter plate (view from Equipment Room)



- Unscrew the securing screws and remove the **adapter plate** from the **RF filter plate**.
- This will allow access to the magnet ramp leads in the **magnet** room.

- i

Do NOT connect the magnet ramp leads to the 3600 MPSU until AFTER the transport test has been completed. See (→ Transport Test / Page 263).
- i

The mandatory transport test was introduced from 3600 MPSU software revision level 17.

This test is only required the FIRST TIME time the 3600 MPSU is used on site.

Lead connections are detailed in the table below:-

Tab. 17 3600 MPSU lead connections

Lead	From	To
Magnet Ramp leads	MPSU X3 MPSU X4	Filter plate (grey) Filter plate (blue)
CAN OPEN lead	MPSU X7	ACC X125
RS232 lead	MPSU X6	RS232 lead X12 on ACC (open-end lead from Ser_Distribution Port 6 (Ser_MPSU) or P3)
MPSU Power lead	MPSU X1	ACC X130

Fig. 277: Avanto - 3600 Ramp Configuration

Avanto - 3600 Ramp Configuration

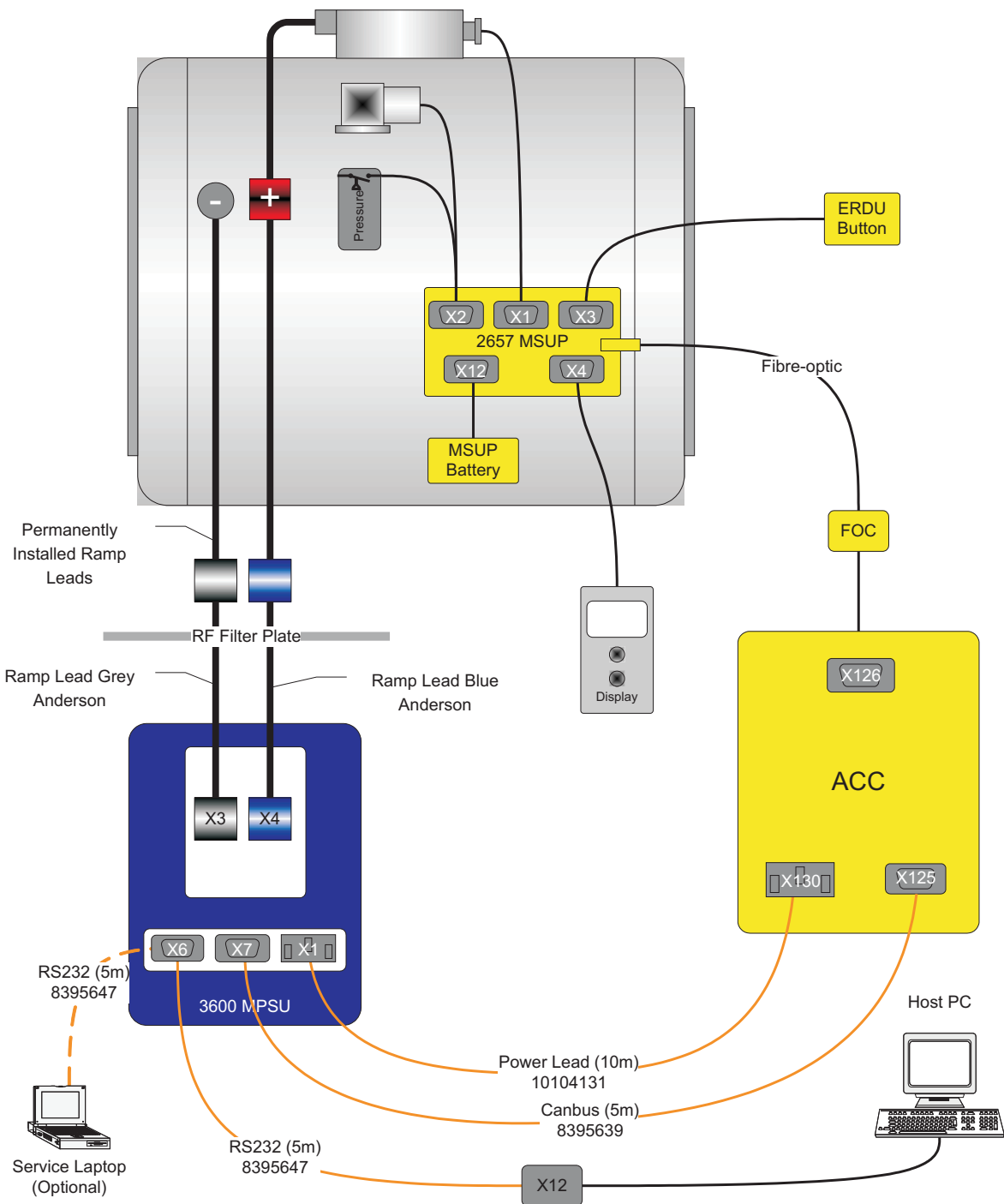
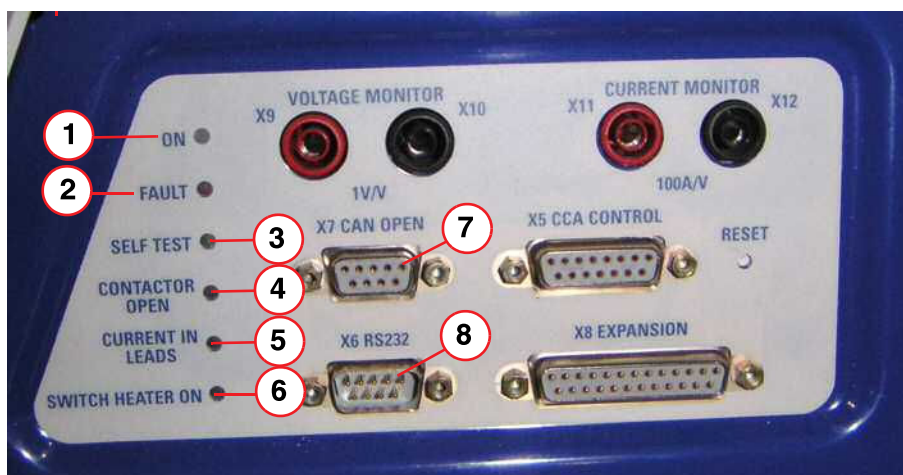


Fig. 278: 3600 MPSU Connector and indicator panel



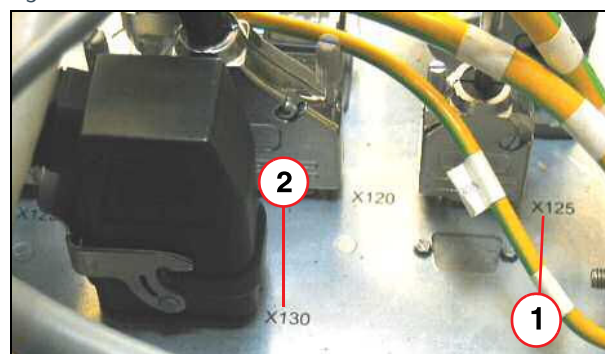
- (1) ON indicator
- (2) FAULT indicator
- (3) SELF TEST indicator
- (4) CONTACTOR OPEN indicator
- (5) CURRENT IN LEADS indicator
- (6) SWITCH HEATER ON indicator
- (7) X7 CAN OPEN connector
- (8) X6 RS232 connector

Fig. 279: 3600 MPSU leads



- (1) CAN OPEN lead
- (2) RS232 lead
- (3) MPSU power lead

Fig. 280: ACC



- (1) X125
- (2) X130

- Connect the **CAN OPEN**, and **RS232** leads..
- Connect 3600 MPSU **power lead** to X1 (→ 1/ Fig. 275 Page 243)
- Turn ON F28 in the ACC.

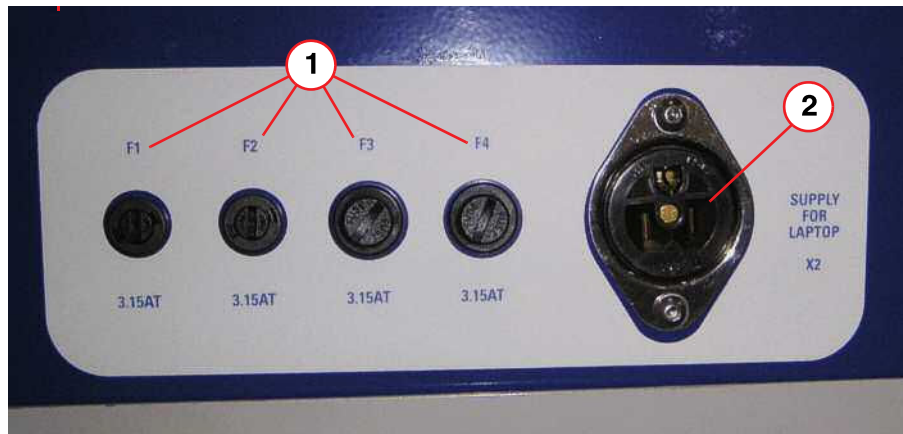
7.2.4 Set up of the 3600 MPSU

7.2.4.1 Connecting Host or Service-PC

To start ramping, the host or Service laptop has to be connected to **X6** of the 3600 MPSU (RS 232 interface) (→ 8/Fig. 278 Page 246). To establish communication, HyperTerminal software is required and must be configured. (Windows 2000 and above)

Power to a Service laptop can be supplied at **X2**.

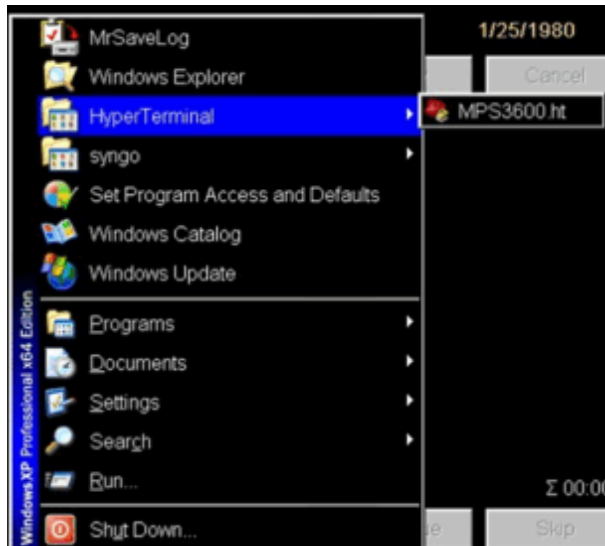
Fig. 281: 3600 MPSU Fuses and laptop power supply



- (1) Fuses F1 - F4
- (2) X2 Supply for Laptop computer

7.2.4.2 Connecting to the Host Computer

Fig. 282: Connecting to the Host Computer



To locate HyperTerminal on the host computer:

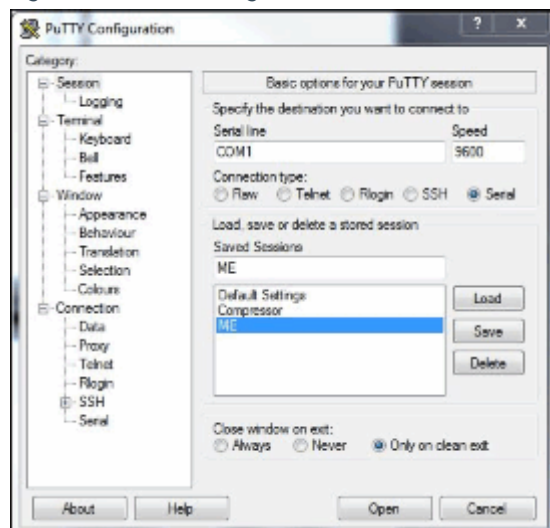
- Press Ctrl+Esc to activate the Windows Start menu.
- Click on the START button to display the menu.
- Navigate to the HyperTerminal program.

7.2.4.3 Setting up PuTTY

You can use software called PuTTY to connect to MSUP's (Magnet Supervisorys) and MPSU's (Magnet Power Supplies). This software has to be setup with the following settings in order to work.

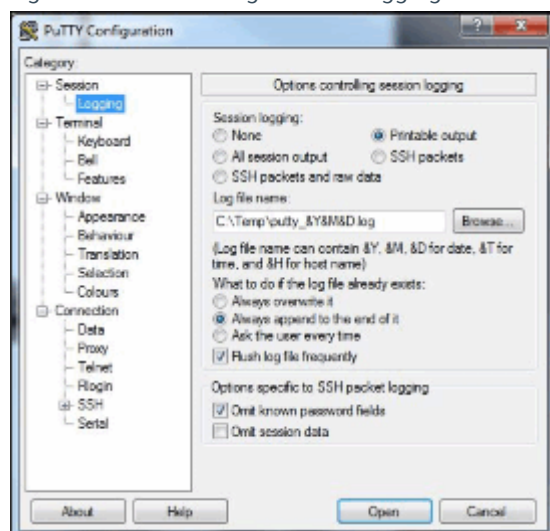
Ensure you have PuTTY version 0.60 or newer software installed on your PC.

Fig. 283: PuTTY Configuration - Session



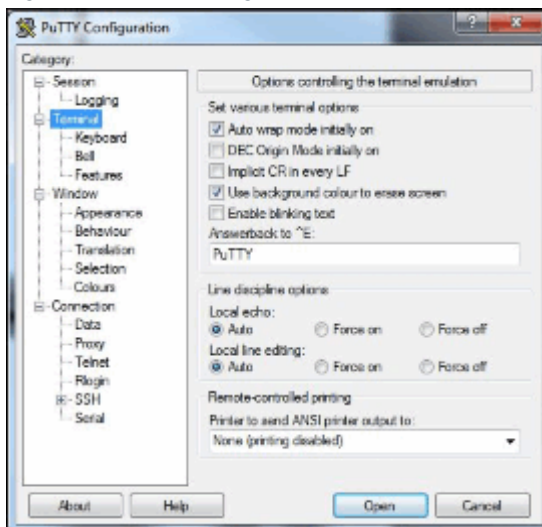
- Under Session tab.
- Set the 'Connection type:' to Serial.
- Set the 'Serial line' to the correct COM port being used. In this case COM1. Only state this if the com port that is being used. Change as required.
- Set the 'Speed' to 9600.
- Set the 'Close window on exit' to Only on clean exit.

Fig. 284: PuTTY Configuration - Logging



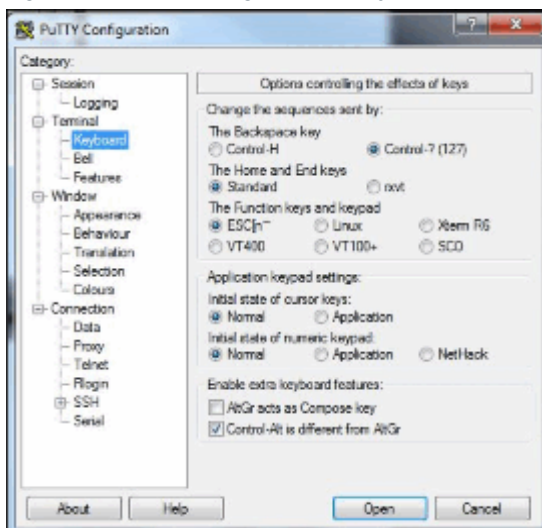
- Under Logging tab.
- Set the 'Session logging' to Printable output.
- Set the 'Log file name:' to C:\Temp\putty_&Y&M&D.log.
- Set the 'What to do if the log file already exists:' to Always append to the end of it and tick Flush log file frequently.
- Tick the 'Options specific to SSH packet logging' to Omit known password fields.

Fig. 285: PuTTY Configuration - Terminal



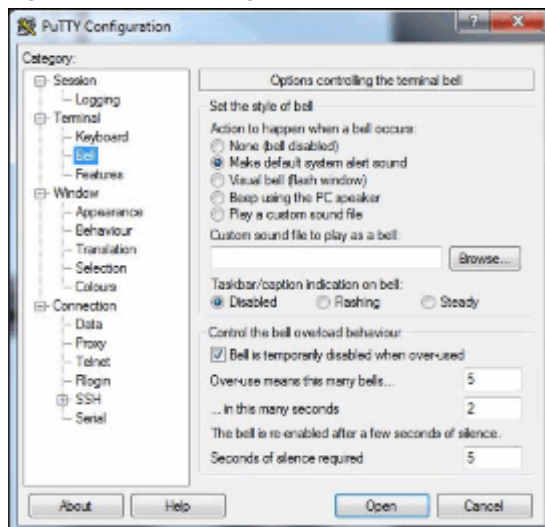
- Under Terminal tab.
- Under 'Set various terminal options' tick the following 'Auto wrap mode initially on' and 'Use background colour to erase screen'.
- Under the 'Answerback to ^E:' ensure it states 'PuTTY'.
- Under 'Local echo:' and 'Local line editing:' ensure 'Auto' is selected
- Under 'Printer to send ANSI printer output to:' select 'None (printing disabled)'.

Fig. 286: PuTTY Configuration - Keyboard



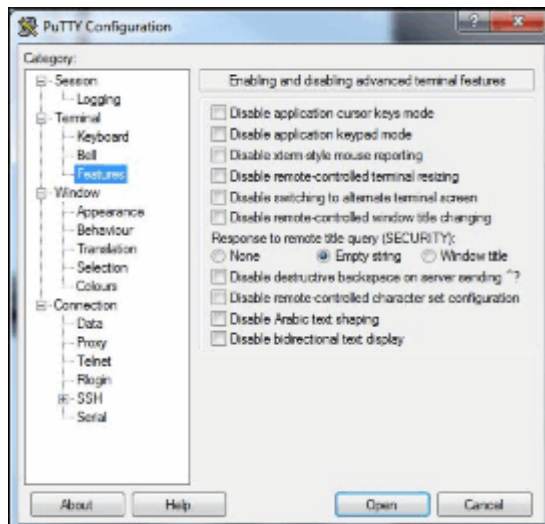
- Under the 'Keyboard' tab.
- Set 'The Backspace key' to 'Control-? (127)'.
- Set 'The Home and End keys' to 'Standard'.
- Set 'The Function keys and keyboard' to 'ESC{n~'.
- Set 'Initial state of cursor keys:' and 'Initial state of numeric keypad:' to 'Normal'.
- Tick 'Enable extra keyboard features' to 'Control-Alt is different is different from AltGr'.

Fig. 287: PuTTY Configuration - Bell



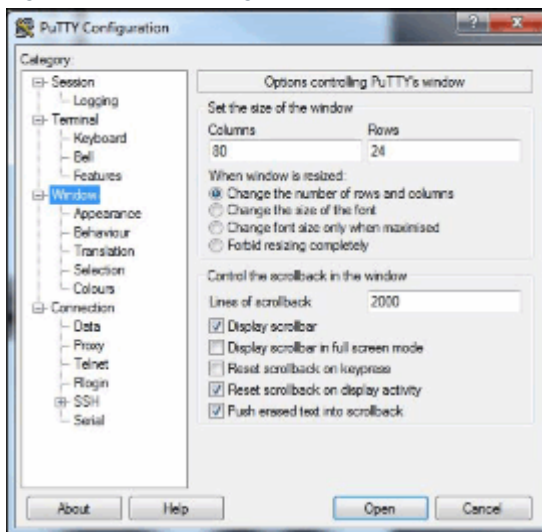
- Under the 'Bell' tab.
- Set 'Action to happen when a bell occurs' to 'Make default system alert sound'.
- Set 'Taskbar/caption indication on bell:' to 'Disabled',
- Tick 'Bell is temporarily disabled when over-used'.
- Set 'Over-use means this many bells...' to '5'.
- Set '...in this many seconds' to '2'.
- Set 'Seconds of silence required' to '5'.

Fig. 288: PuTTY Configuration - Features



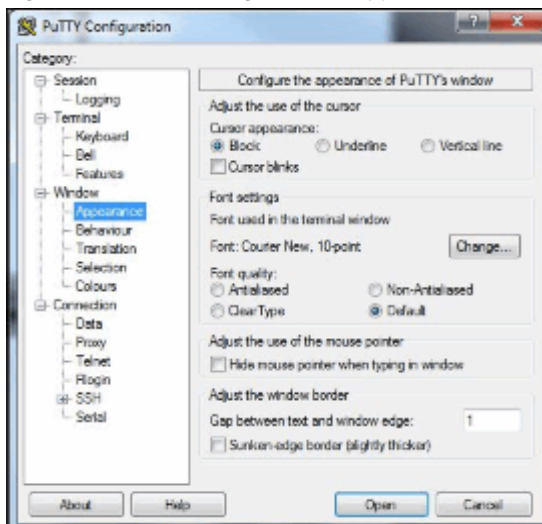
- Under the 'Features' Tab.
- Set 'Response to remote title query (SECURITY):' to 'Empty string'.
- Ensure nothing else is selected.

Fig. 289: PuTTY Configuration - Window



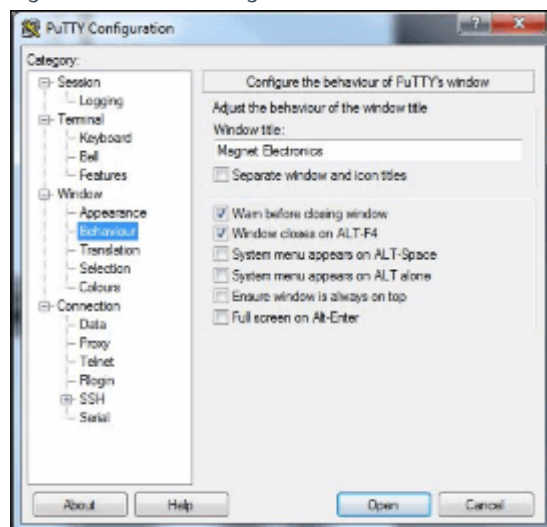
- Under the 'Window' Tab.
- Set the 'Columns' to '80' and the 'Rows' to '24'.
- Set 'When window is resized:' to 'Change the number of rows and columns'.
- Set the 'Lines of scrollbar' to '2000'.
- Ensure 'Display scrollbar', 'Reset scrollbar on display activity' and 'Push erased text into scrollbar' are all ticked.

Fig. 290: PuTTY Configuration - Appearance



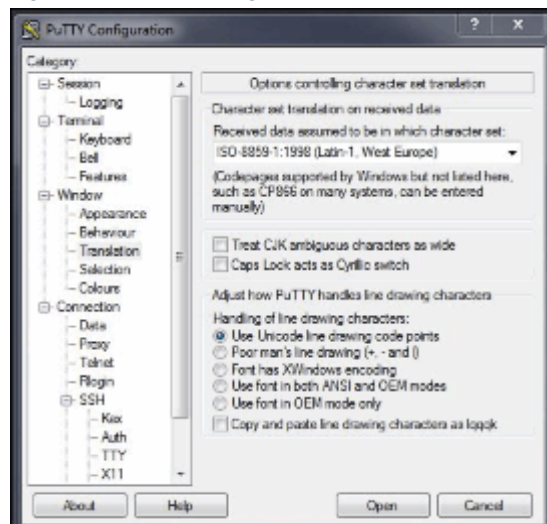
- Under 'Appearance' tab.
- Set 'Cursor appearance' to 'Block'.
- Set 'Font quality:' to 'Default'.
- Set 'Gap between text and window edge:' to '1'.

Fig. 291: PuTTY Configuration - Behaviour



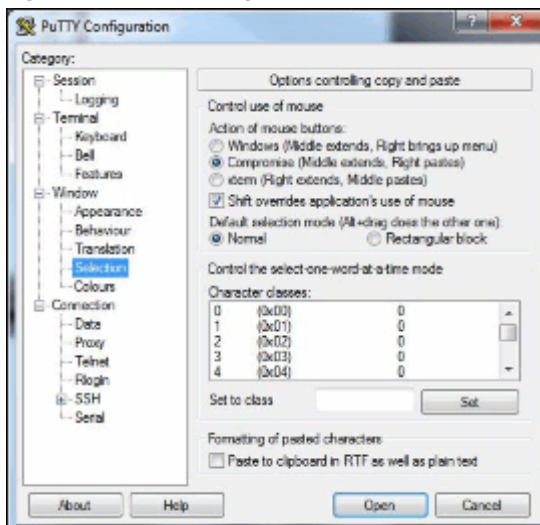
- Under 'Behavior' tab.
- Under 'Window title:' type 'Magnet Electronics'
- Ensure 'Warn before closing window', and 'Window closes on ALT-F4' are ticked.

Fig. 292: PuTTY Configuration - Translation



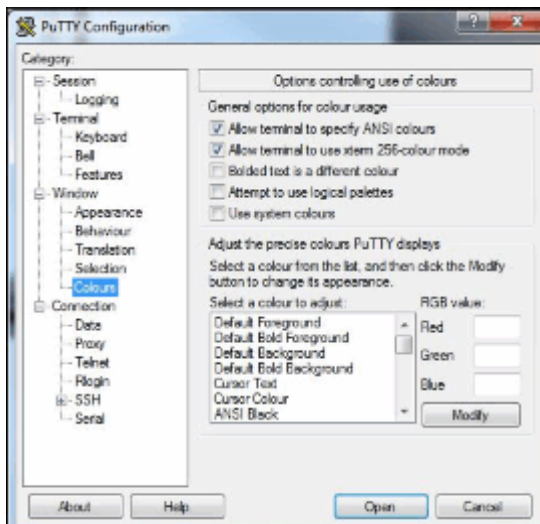
- Under 'Translation' tab.
- Set 'Received data assumed to be in which character set:' to 'ISO-8859-1:1998 (Latin-1, West Europe)'.
- Set 'Handling of line drawing characters:' to 'Use Unicode line drawing code points'.

Fig. 293: PuTTY Configuration - Selection



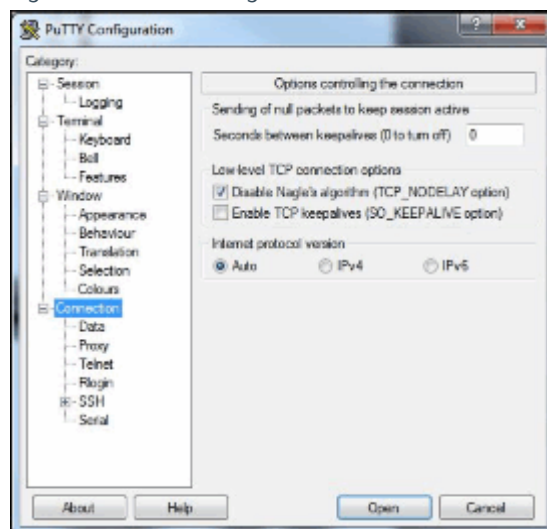
- Under 'Selection' tab.
- Set 'Action of mouse buttons:' to 'Compromise (Middle extends, Right pastes)'.
- Tick 'Shift overrides application's use of mouse'.
- Set 'Default selection mode (Alt + drag does the other one):' to 'Normal'.

Fig. 294: PuTTY Configuration - Colours



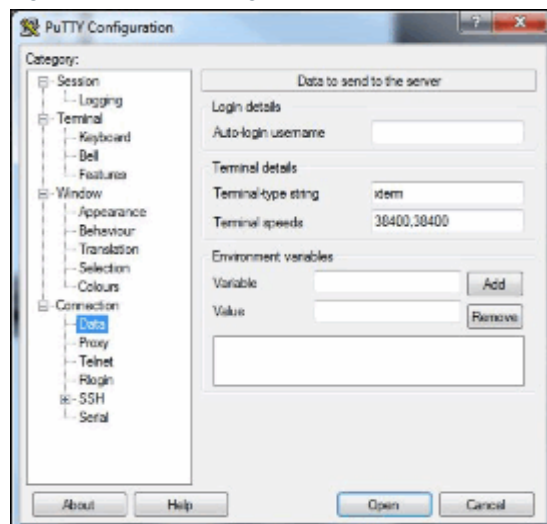
- Under 'Colours' tab.
- Tick 'Allow terminal to specify ANSI colours' and 'Allow terminal to use xterm 256-colour mode'.

Fig. 295: PuTTY Configuration - Connection



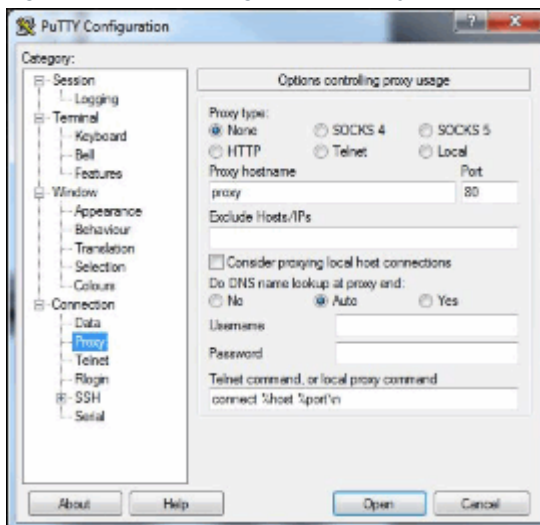
- Under 'Connection' tab.
- Set 'Low-level TCP connection options' to 'Disable Nagle's algorithm (TCP_NODELAY option)'.
- Set 'Internet protocol version' to 'Auto'.

Fig. 296: PuTTY Configuration - Data



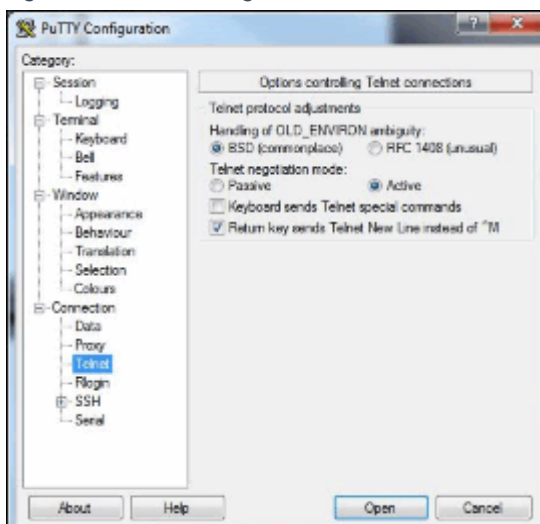
- Under 'Data' tab.
- Under the 'Terminal-type string' ensure it states 'xterm'.
- Under the 'Terminal speeds' ensure it states '38400.38400'.

Fig. 297: PuTTY Configuration - Proxy



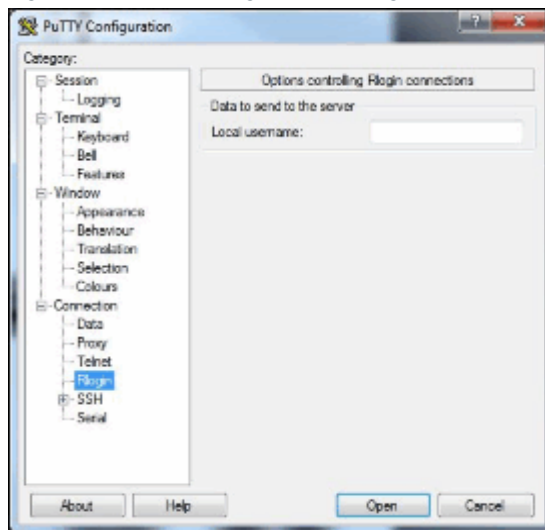
- Under 'Proxy' tab.
- Set 'Proxy type:' to 'None'.
- Under the 'Proxy hostname' ensure it states 'proxy'.
- Under the 'Port' ensure it states '80'.
- Set 'Do DNS name lookup at proxy end:' to 'Auto'.
- Under the 'Telnet command, or local proxy command' ensure it states 'connect %host %port\n'.

Fig. 298: PuTTY Configuration - Telnet



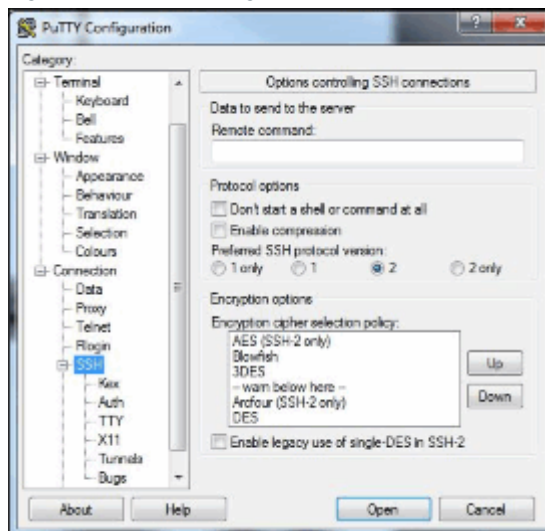
- Under 'Telnet' tab.
- Set 'Handling of OLD_ENVIRON ambiguity:' to 'BSD'.
- Set 'Telnet negotiation mode:' to 'Active'.
- Tick 'Return key sends Telnet New Line instead of ^M'.

Fig. 299: PuTTY Configuration - Rlogin



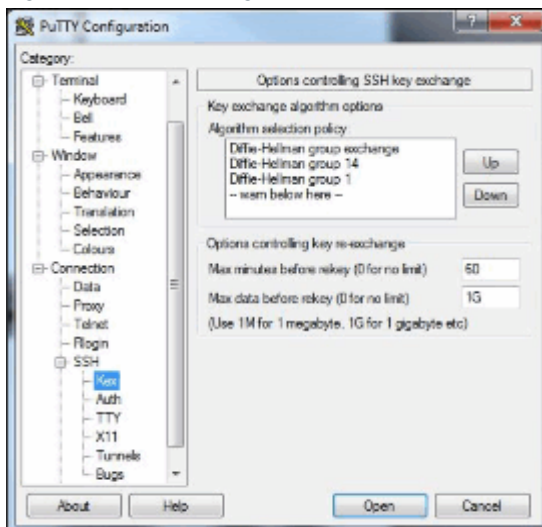
- Under 'Rlogin' tab.
- Nothing is selected.

Fig. 300: PuTTY Configuration - SSH



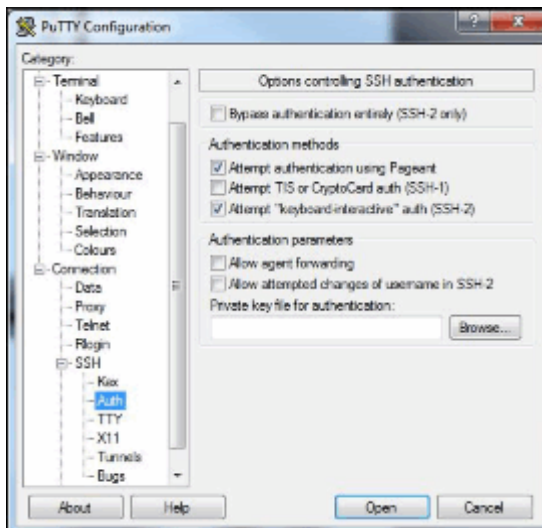
- Under 'SSH' tab.
- Set 'preferred SSH protocol version' to '2'.

Fig. 301: PuTTY Configuration - Kex



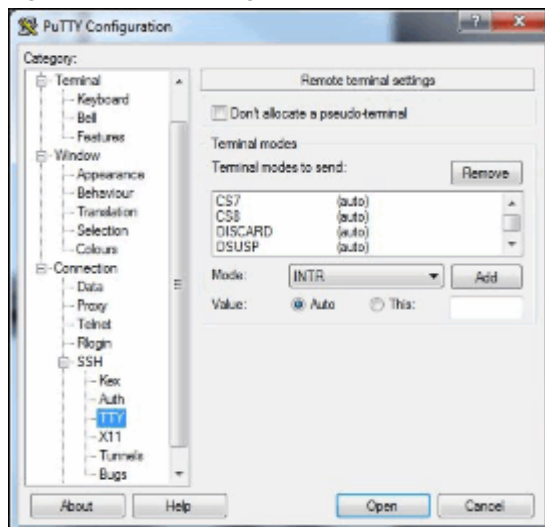
- Under 'Kex' tab.
- Under the 'Max minutes before rekey (0 for no limit)' ensure it states '60'.
- Under the 'Max data before rekey (0 for no limit)' ensure it states '1G'.

Fig. 302: PuTTY Configuration



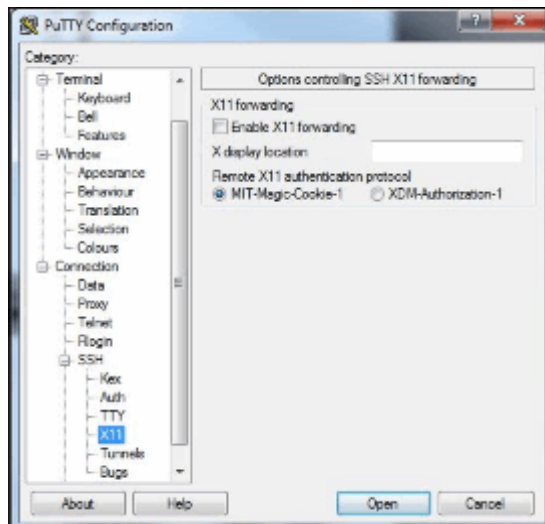
- Under 'Auth' tab.
- Tick 'Attempt authentication using Pageant' and 'Attempt "keyboard-interactive" auth (SSH-2)'.

Fig. 303: PuTTY Configuration - TTY



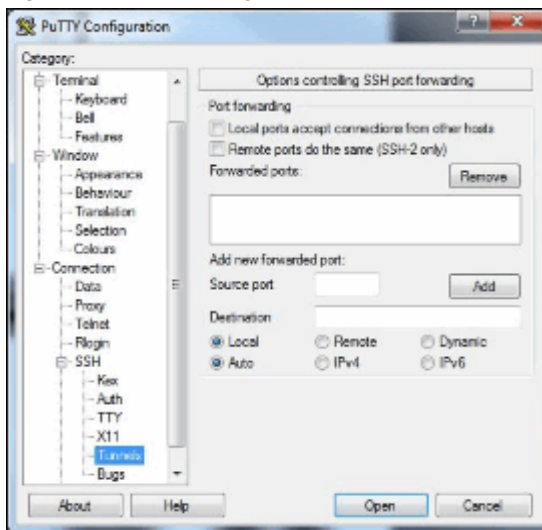
- Under 'TTY' tab.
- Set 'Value' to 'Auto'.

Fig. 304: PuTTY Configuration



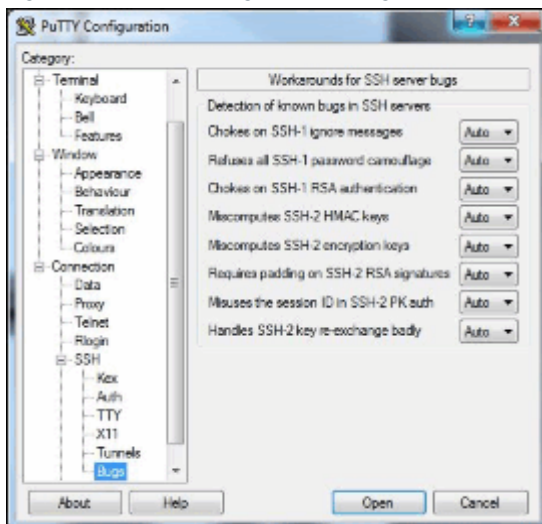
- Under 'X11' tab.
- Set 'Remote X11 authentication protocol' to 'MIT-Magic-Cookie-1'.

Fig. 305: PuTTY Configuration - Tunnels



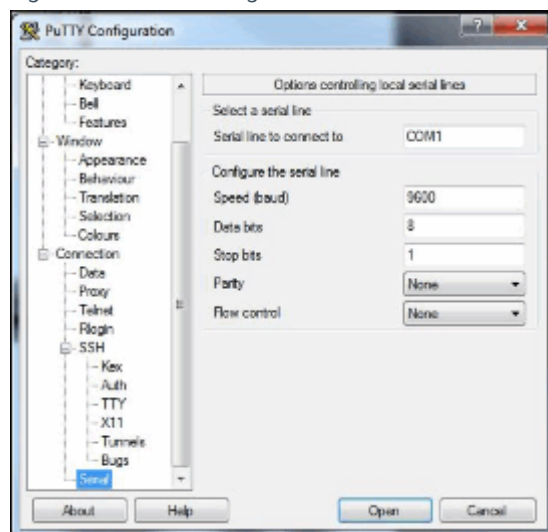
- Under 'Tunnels' tab.
- Set 'Destination' to 'Local' and 'Auto'.

Fig. 306: PuTTY Configuration - Bugs



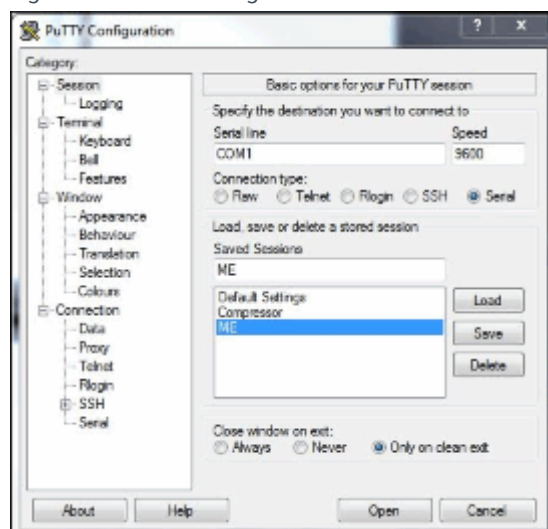
- Under 'Bugs' tab.
- Ensure all settings are set to 'Auto'.

Fig. 307: PuTTY Configuration - Serial



- Under 'Serial' tab.
- Under the 'Serial line to connect to' ensure it states 'COM1'. Only state this if the com port that is being used. Change as required.
- Under the 'Speed (baud)' ensure it states '9600'.
- Under the 'Data bits' ensure it states '8'.
- Under the 'Stop bits' ensure it states '1'.
- Under the 'Parity' ensure it states 'None'.
- Under the 'Flow control' ensure it states 'NONE'.

Fig. 308: PuTTY Configuration - Session



- Under 'Session' tab.
- Under 'Saved Sessions' type in 'ME' or any name which you want to save these settings under.
- Click the 'Save' button.
- Click the 'Open' button.
- PuTTY software will now open and run.

In future requirements to use PuTTY follow this short sequence.

- Open the PuTTY software.
- Under the Session tab select the session required. In the case above the session is called ME.
- Click the load button.
- Click the 'Open' button.
- PuTTY will now open and run.

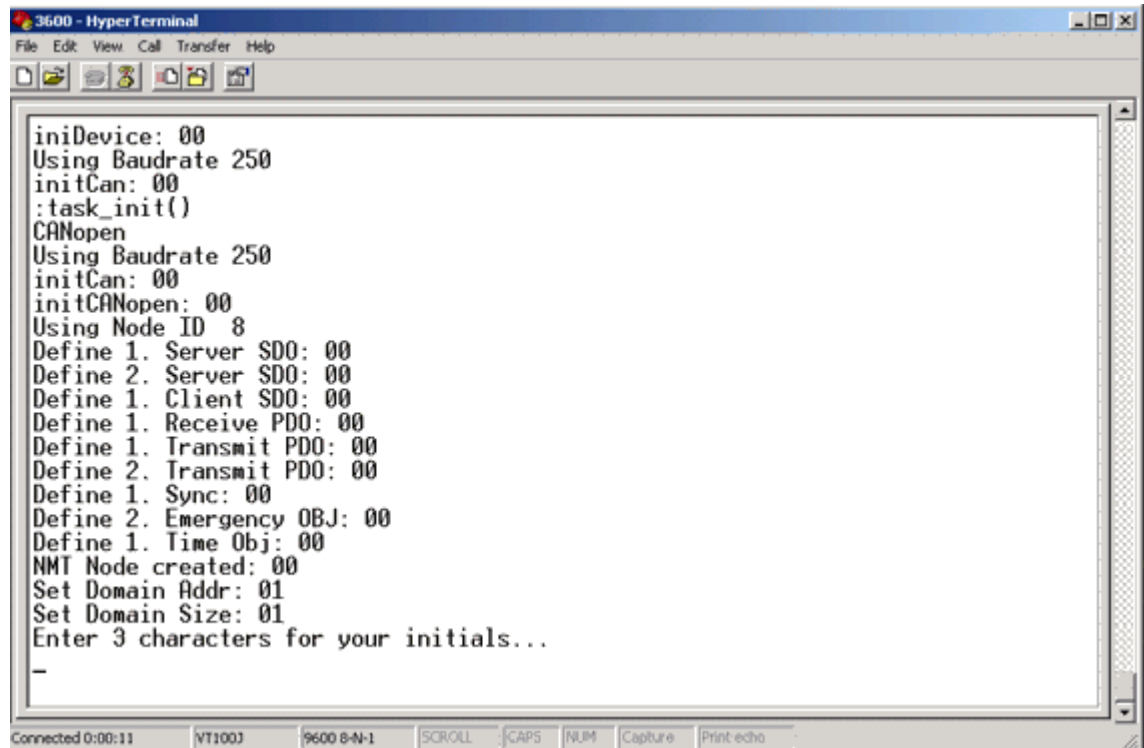
7.2.4.4 Initialising the 3600 MPSU



Ensure that there is a valid helium level reading before switching on the 3600 MPSU.

- Turn **ON** the 3600 MPSU using mains circuit breaker **CB1** on the 3600 MPSU.
(→ 2/Fig. 275 Page 243)
 - The green '**ON**' LED will light on the 3600 MPSU.(→ 1/Fig. 278 Page 246)
- The **boot-up** screen will be displayed.

Fig. 309: HyperTerminal Boot-up screen



If the screen does not appear, make sure that the correct Com port and settings have been selected. If they have, turn the 3600 MPSU off and then back on, using the mains circuit breaker CB1. (→ 1/Fig. 275 Page 243)



Reboot the scanner with the 3600 MPSU switched on.

A Reboot is necessary to update the loadware and to download the ramp logfile to the HOST.

The logfile is saved to C/MedCom/log directory.

Always reboot the scanner after the 3600 MPSU is switched off and on again. Rename the logfile, e.g. "MPSRampLog1.dat" after the ramp is finished.

If not renamed, the file will be overwritten with the next ramp logfile.

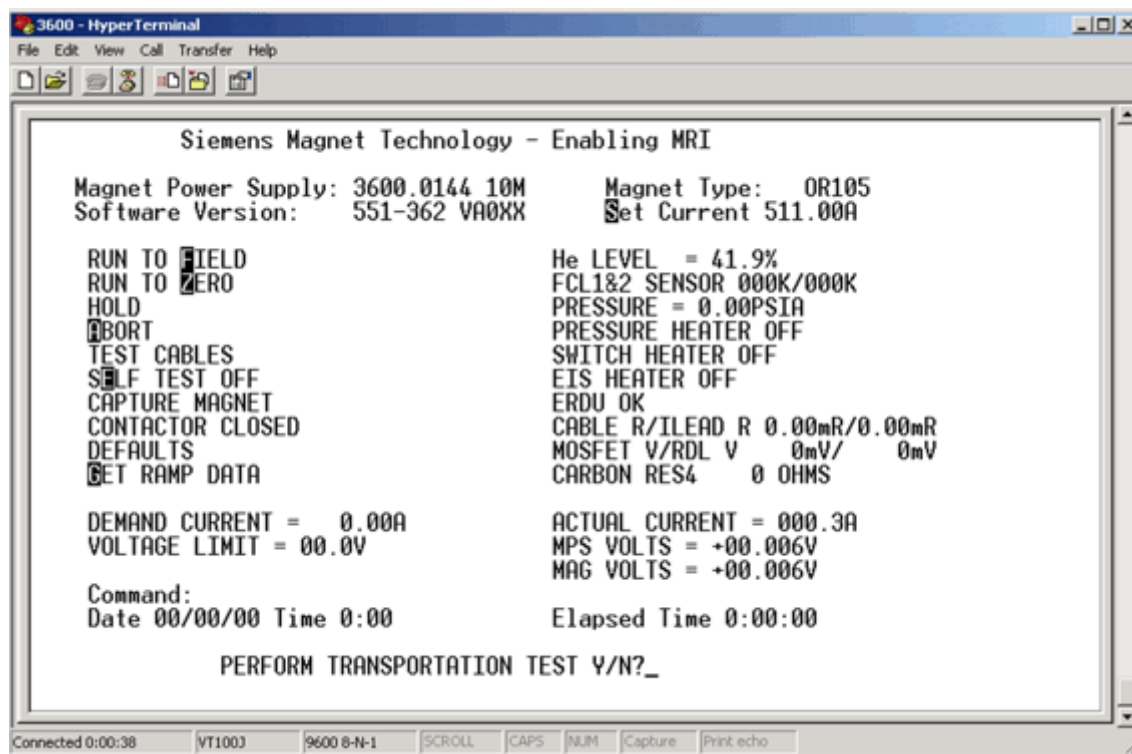
Repeat booting of the scanner if following warning message still persists: "MPS not operational"

- When prompted, enter your **initials** and **magnet serial number**.

The 3600 MPSU will now retrieve the **magnet type**, **last current set**, **He- level**, **date** and **time** from the MSUP.

- The following screen will prompt to '**Perform Transportation Test**':-

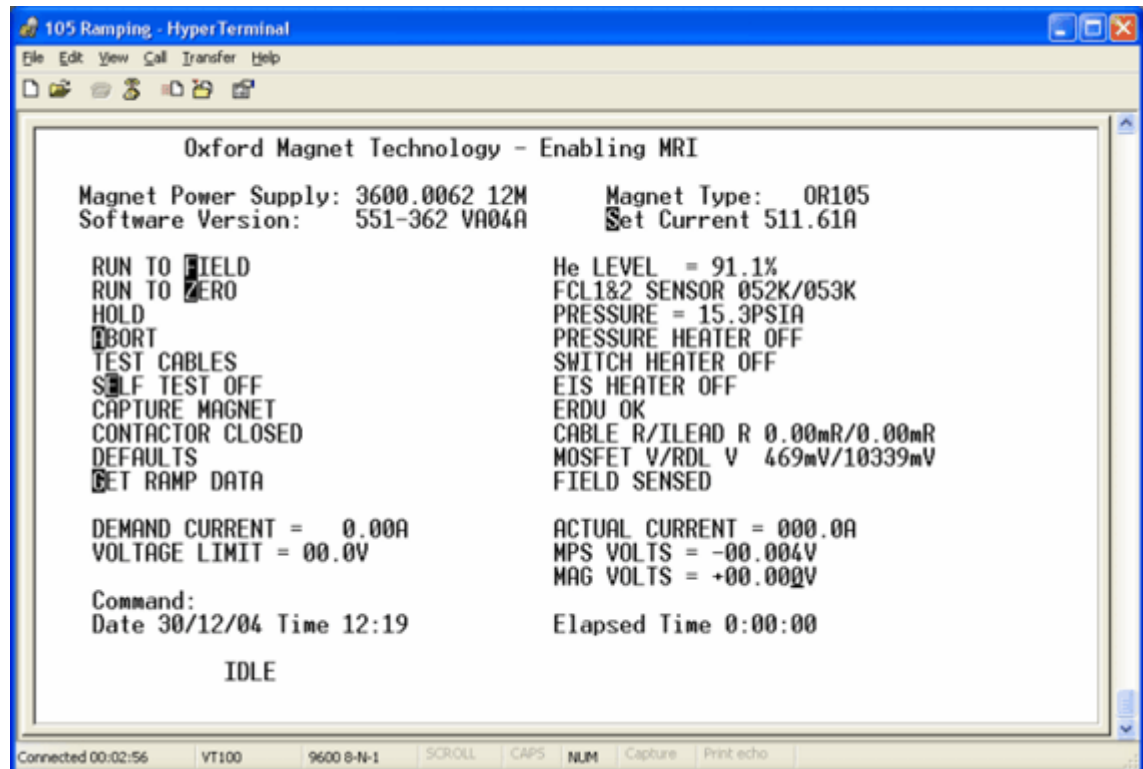
Fig. 310: Transport Test (example of Avanto screenshot)



- If the **Transport Test** is required, continue with (→ Transport Test / Page 263).
- If the test is **not** required:
 - » Enter **N<ENTER>**
 - » At the prompt '**ARE YOU SURE**', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 278 Page 246)

- When the self-test is complete, the **Operational screen** is displayed.

Fig. 311: Operational Screen (example of Avanto screenshot)



- Turn **OFF** the 3600 MPSU using mains circuit breaker **CB1** on the 3600 MPSU. (→ 2/Fig. 275 Page 243)
- Continue with (→ Ramp lead connection / Page 267).

Transport Test

During transport, it is possible that the 3600 MPSU may have suffered internal damage due to rough handling.



The mandatory transport test was introduced from 3600 MPSU software revision level 17.

For units prior to this revision, continue with (→ Ramp lead connection / Page 267).

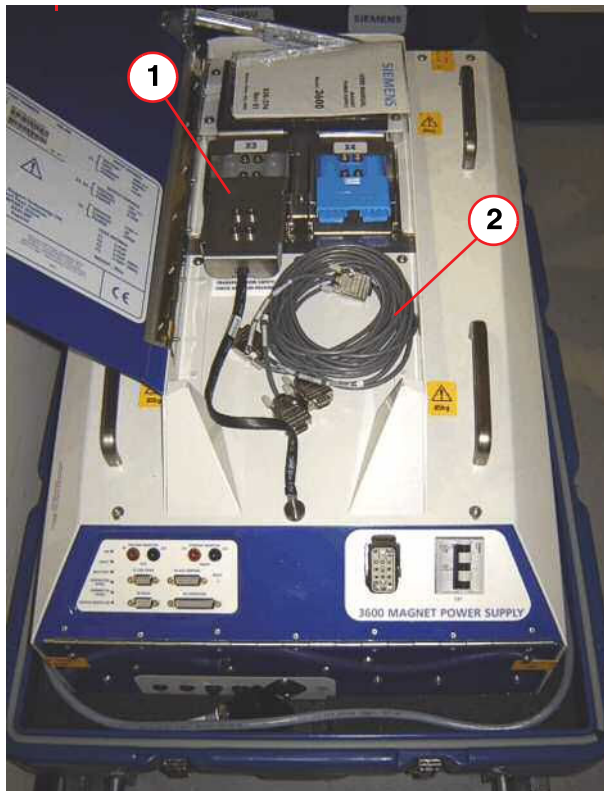
This test only needs to be completed the first time the 3600 MPSU is used on site.



This test must be performed **PRIOR** to connecting the ramp leads and using the 3600 MPSU on the magnet.

Transport Test Tool

Fig. 312: Transport test tool



- (1) Transport test plug (fitted)
- (2) RS232 cable

The **transport test tool** consists of 2 x 470 ohm resistors housed in a grey **Anderson connector**.

The center tap of the resistors is connected to a flying lead which is permanently bolted to the 3600 chassis.

The 3600 MPSU is supplied with the test plug already connected.

Transport Test Procedure

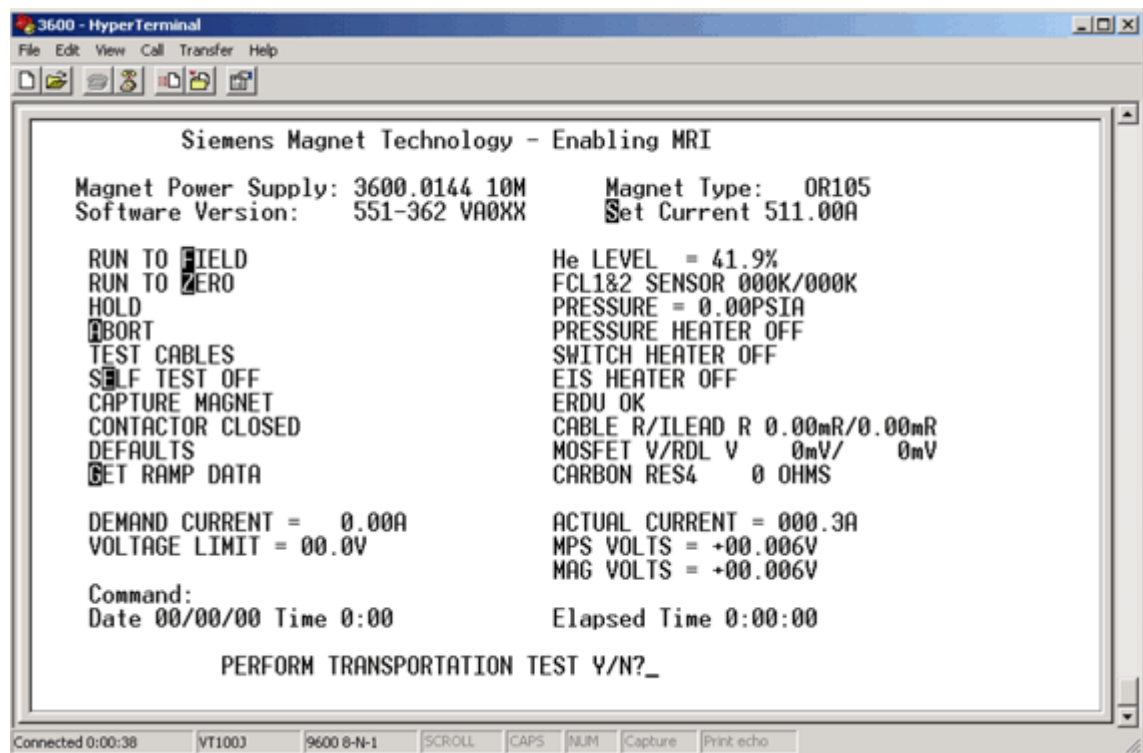
Check that the **transport test plug** is connected to **X3** as shown in (→ Fig. 312 Page 264) above.



This test must be passed before the test plug is removed and the ramp leads connected.

- Perform (→ Initialising the 3600 MPSU / Page 260).

Fig. 313: Transport Test (example of Avanto screenshot)



- The screen will then prompt to 'Perform Transportation Test'. Type **Y** <Enter>.



If this test has already been successfully completed, type 'N' at this point to cancel the test. At the prompt 'ARE YOU SURE', enter **Y**<ENTER>

The software will then continue to the next part of the ramping sequence.

Failure of the Transport test will be displayed briefly in the event of the test detecting an internal fault (→ Fig. 314 Page 266).

This will be followed by a request to check the connections and repeat the test (→ Fig. 315 Page 266).

Fig. 314: Transport Test failed (example Avanto screenshot)

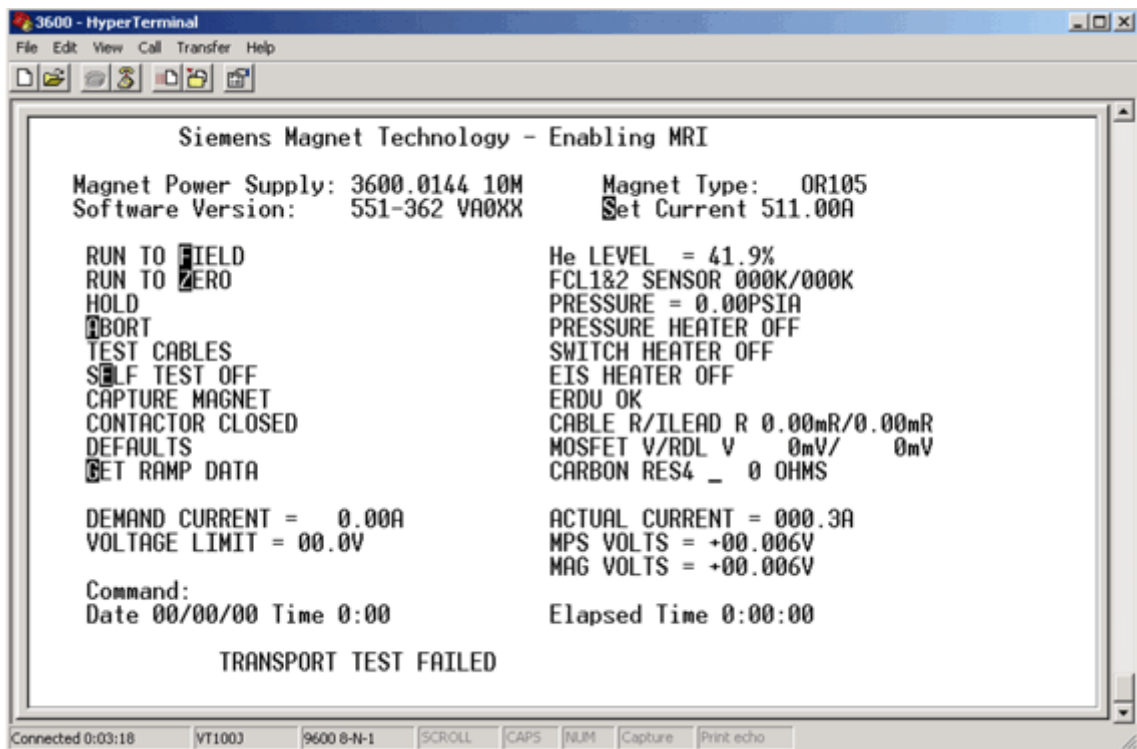
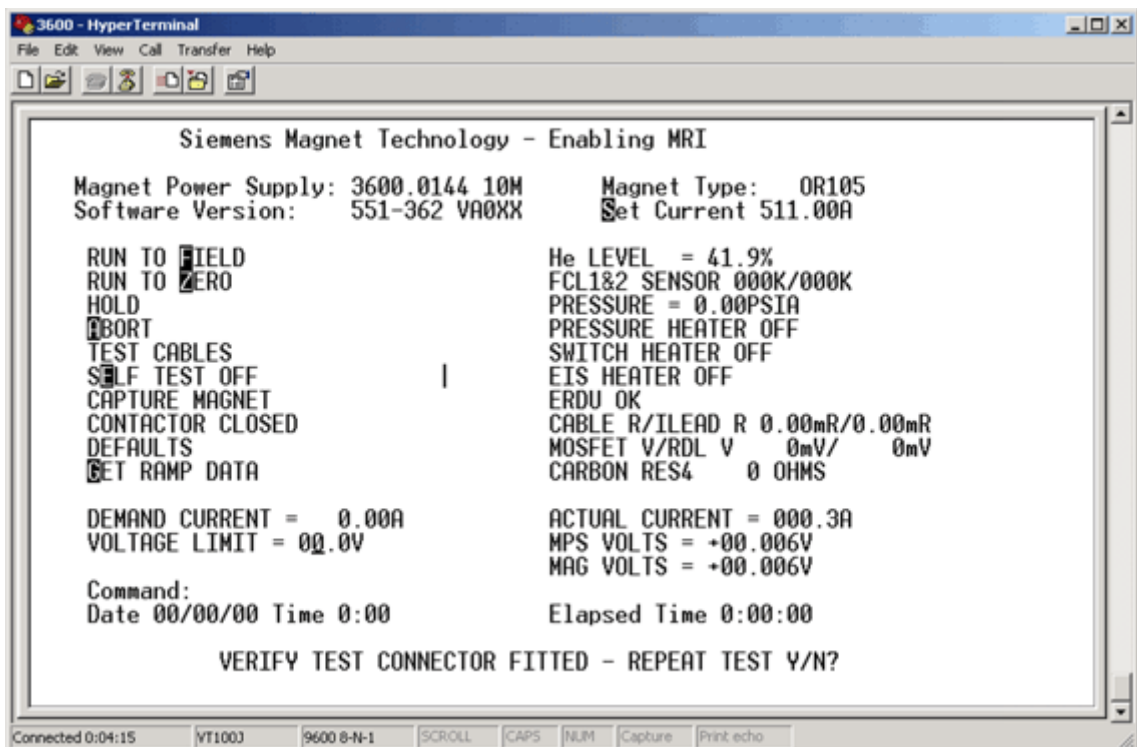


Fig. 315: Verify Test Connector (example Avanto screenshot)





After 3 successive Transport tests have FAILED, further operation of the 3600 MPSU is prevented.

Line power to the 3600 MPSU should be switched OFF then ON to enable further tests to be performed.



If all attempts fail, the 3600 MPSU should be returned for repair, and a replacement unit obtained.

If the test is **successful**, the power supply will perform an **internal self-test**.

- The **self-test LED** indicator will light during the test. (→ 3/ Fig. 278 Page 246)
- Once this test is completed turn off the 3600 via **CB1**. (→ 2/ Fig. 275 Page 243)

7.2.4.5 Ramp lead connection

After the Transport test has been successfully completed, (if applicable), the following connections must be made **prior to ramping the magnet**.

- Remove the **transport test plug**.

Fig. 316: MPSU - ramp cables fitted



- Connect the two **magnet extension ramp leads** to Anderson connectors **X3** and **X4** on the top of the 3600 MPSU.



CAUTION

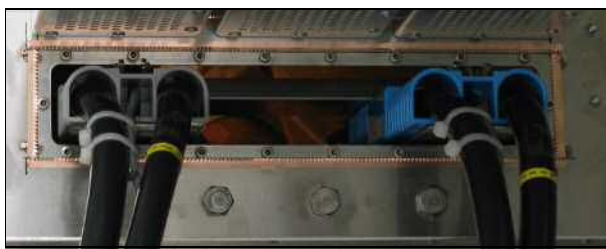
Incorrect locking of the ramp leads

If these instructions are not observed, the magnet and the 3600 MPSU can be damaged.

- » **Connect and secure the leads.**

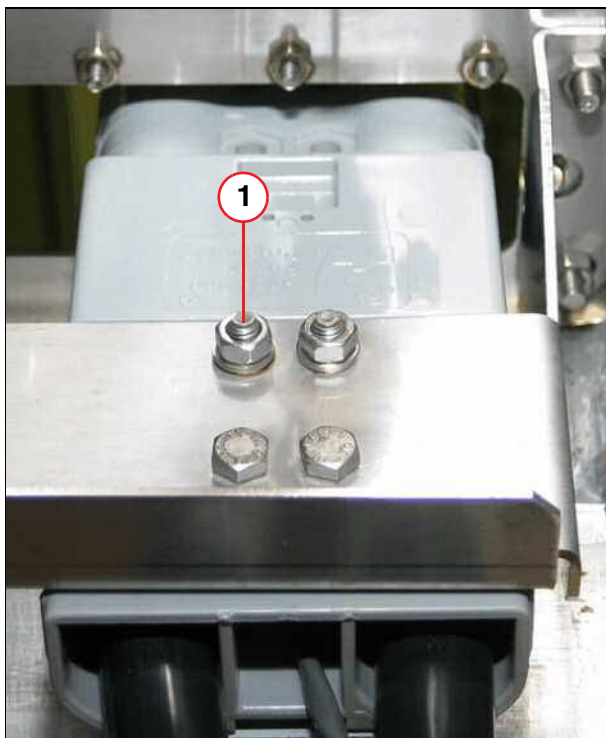
When shorting links **are not** fitted:-

Fig. 317: RF filter plate

**Equipment room**

- Connect the **extension leads** to the **magnet leads** at the **RF filter plate**.

Fig. 318: Power lead connection

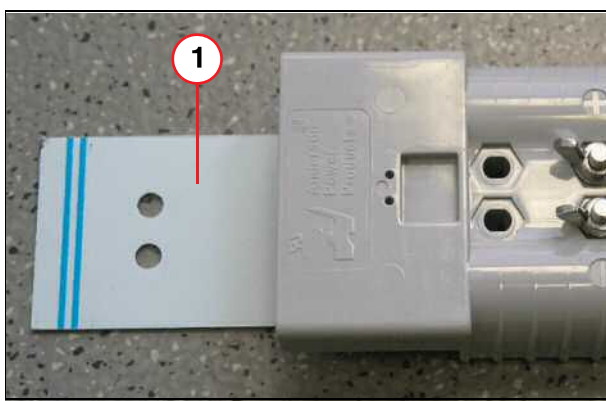


(1) Fastening bolts

Magnet room

- Prevent the **extension cables** from being accidentally **disconnected**.
- **Secure the leads** by tightening the **fastening bolts**.

Fig. 319: Connector



(1) Fastening plate

- From the equipment room, feed the 3600 MPSU leads and fastening plate through the RF filter plate.



Ramp cables are heavy - to prevent stress on the connection, use tie wraps to support the weight if locking bolts are missing.

- Turn **ON** the 3600 MPSU using mains circuit breaker **CB1** on the 3600 MPSU.
(→ 2/Fig. 275 Page 243)
 - The green '**ON**' LED will light on the 3600 MPSU.(→ 1/Fig. 278 Page 246)
- The **boot-up** screen will be displayed.

Fig. 320: HyperTerminal Boot-up screen

```

3600 - HyperTerminal
File Edit View Call Transfer Help

iniDevice: 00
Using Baudrate 250
initCan: 00
:task_init()
CANopen
Using Baudrate 250
initCan: 00
initCANopen: 00
Using Node ID 8
Define 1. Server SDO: 00
Define 2. Server SDO: 00
Define 1. Client SDO: 00
Define 1. Receive PDO: 00
Define 1. Transmit PDO: 00
Define 2. Transmit PDO: 00
Define 1. Sync: 00
Define 2. Emergency OBJ: 00
Define 1. Time Obj: 00
NMT Node created: 00
Set Domain Addr: 01
Set Domain Size: 01
Enter 3 characters for your initials...
-

Connected 0:00:11 VT1003 9600 8-N-1 SCROLL CAPS NUM Capture Print echo

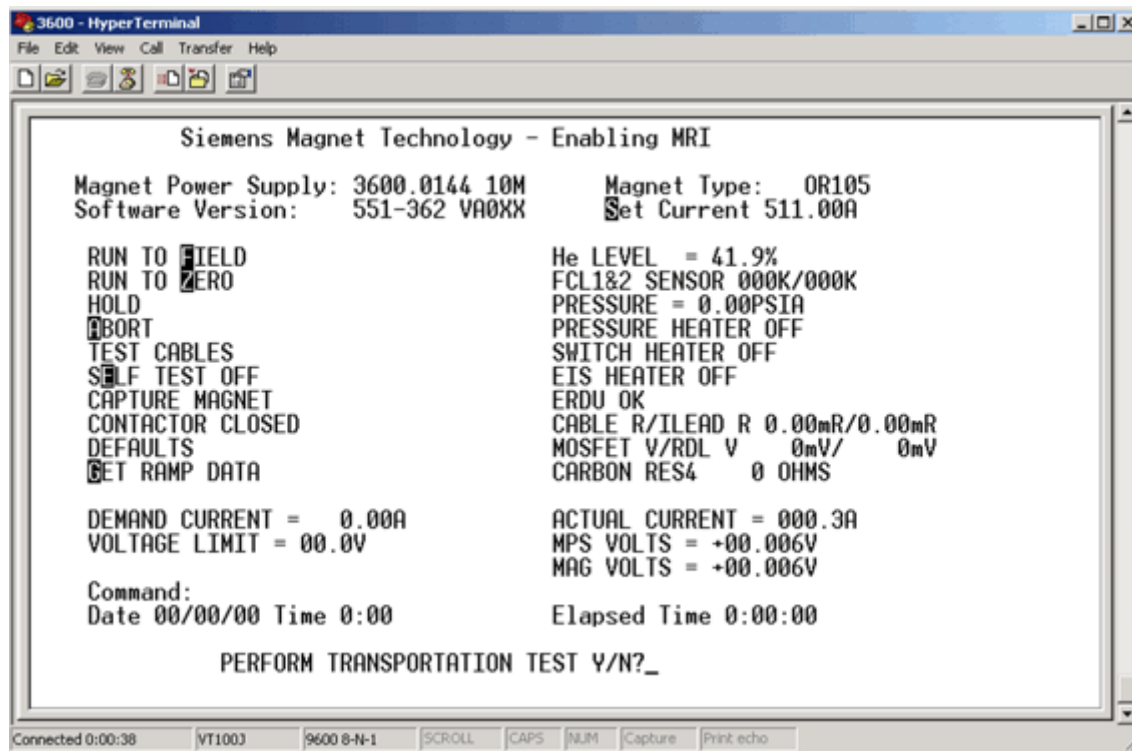
```

- When prompted, enter your **initials** and **magnet serial number**.

The 3600 MPSU will now retrieve the **magnet type**, **last current set**, **He- level**, **date** and **time** from the MSUP.

- The following screen will prompt to '**Perform Transportation Test**':-

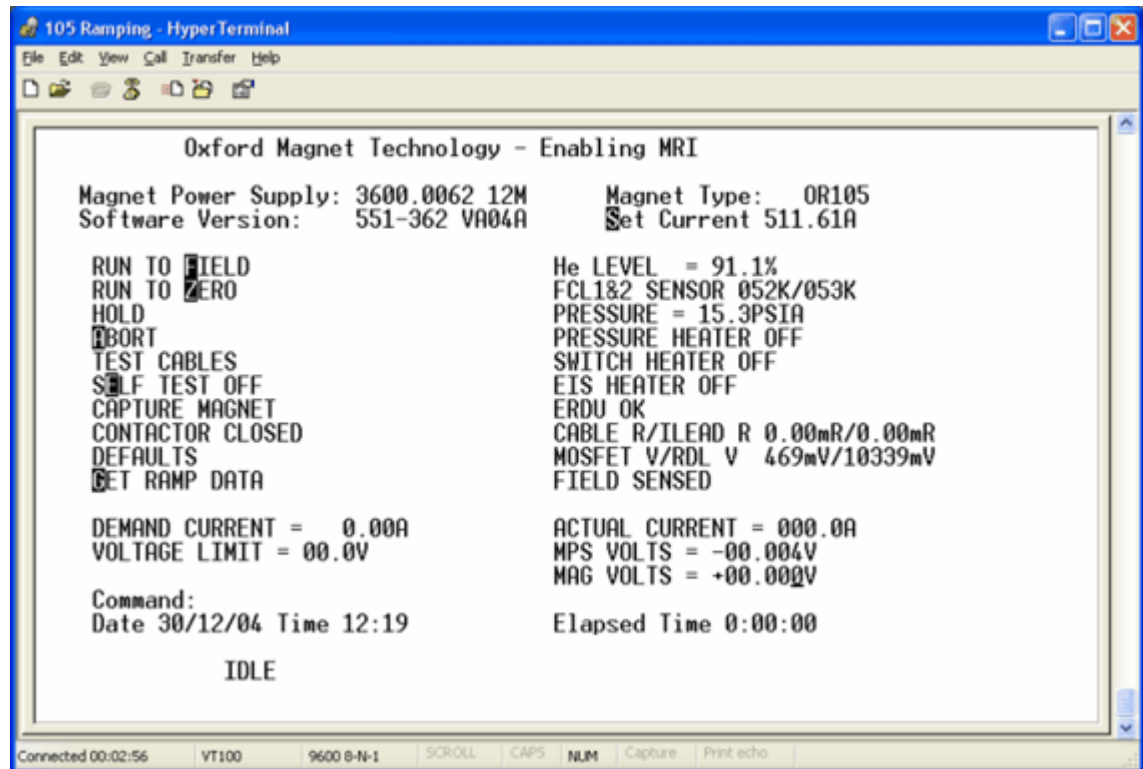
Fig. 321: Transport Test (example of Avanto screenshot)



- The test is **not** required:
 - » Enter **N<ENTER>**
 - » At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 278 Page 246)

- When the self-test is complete, the **Operational screen** is displayed.

Fig. 322: Operational Screen (example of Avanto screenshot)



7.2.5 Run To Field

7.2.5.1 Co-Siting of Magnets

Co-siting describes the ramping requirements where two MR systems with the same operating frequency are sited next to each other in adjacent rooms.



If two systems are installed next to each other in adjacent rooms:

- One magnet must be ramped to the upper frequency limit.
- The other magnet must be ramped to approx. 100 kHz less at the lower frequency limit.



Shimming is only optimized with both magnets ramped up.

Neither magnet is ramping, while the other is running applications. If one magnet is switched off, the other magnet has to be re-shimmed.

7.2.5.2 Run To Field Prerequisites:

- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).



Following a helium level fill, wait at least 2 hours before ramping.

- The helium level must be $\geq 65\%$ for the startup (1st) ramp up and $> 40\%$ for all future ramps.

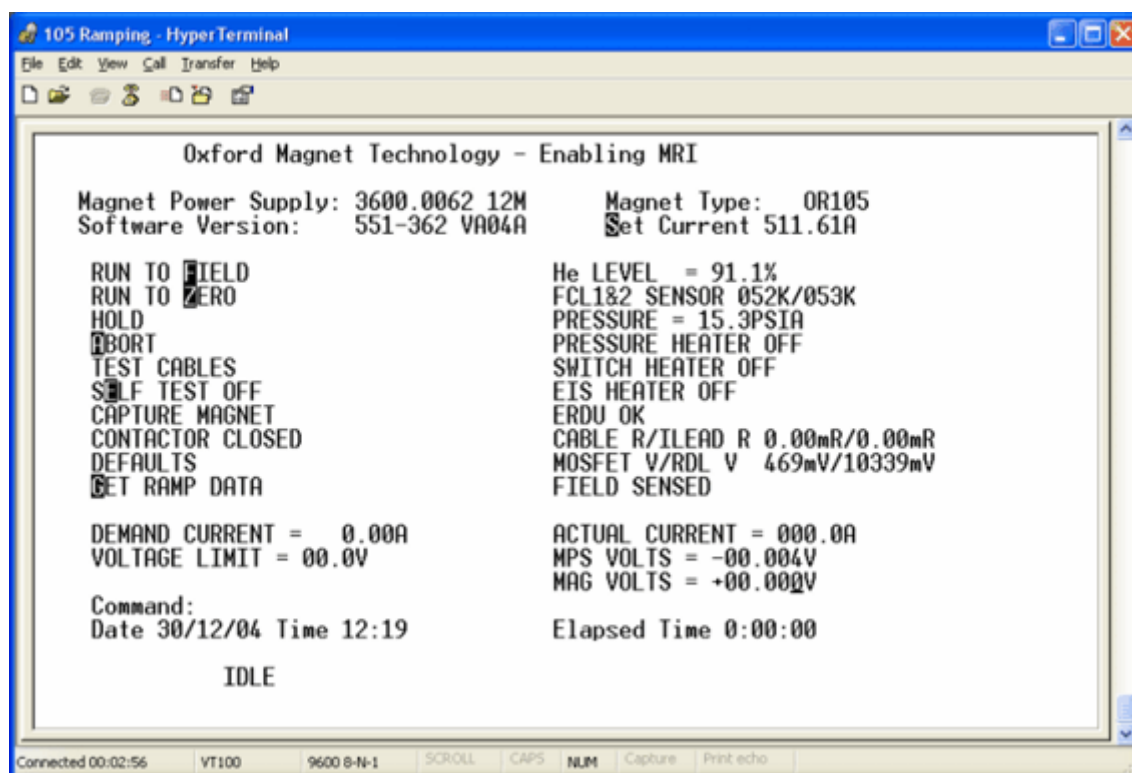
7.2.5.3 Setting the magnet current

- Enter the command: **S<space>514.05<ENTER>**
The numerical number may be varied as required.
- Enter the command for Run to field: **F <ENTER>**

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to field.

7.2.5.4 Run To Field

Fig. 323: Operational Screen (example of Avanto screenshot)



- Enter the command **F <ENTER>** for ramping up the magnet.

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to field.

7.2.6 Run To Zero

7.2.6.1 Run to zero prerequisites:

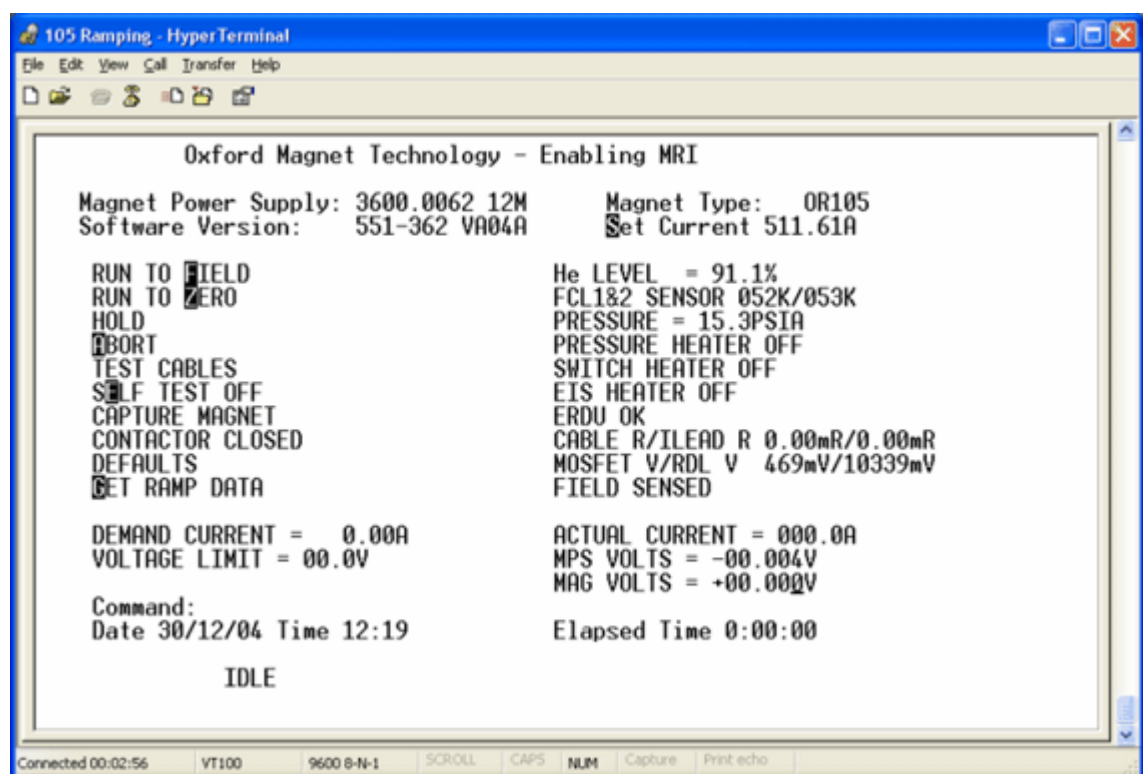
- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).



Following a helium level fill, wait at least 2 hours before ramping.

7.2.6.2 Run To Zero

Fig. 324: Operational Screen (example of Avanto screenshot)



- Enter the command **Z<ENTER>** for ramping down the magnet.

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to zero.

7.2.7 Ramping the magnet to the required field strength (bumping the magnet)

The magnet should be ramped to the upper limit of the frequency limit. Bumping the magnet involves making a small change to the magnetic field strength either up or down. The procedure is the same as ramping the magnet.

7.2.7.1 Magnet current setting

Check the **Magnet Acceptance Document/MSUP** to obtain the operating **current value**.



Avanto range of frequency:

Maximum frequency: 63,700 000 MHz + 0 kHz

Minimum frequency: 63,500 000 MHz

Use the mathematical formula below to calculate the current to determine the **centre frequency**.

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

$$\text{Magnetom Avanto: } I_{\text{final}} = (63,680 \text{ MHz} \times I_{\text{actual}}) / \text{Freq actual}$$

7.2.7.2 Bumping prerequisites:

- 3600 MPSU should be connected.
- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).



Following a helium level fill, wait at least 2 hours before ramping.

- Helium level must be **> 40 %**.

The last ramp current is stored in the magnet supervisory (MSUP). The ramp current will be displayed on-screen when the 3600 MPSU is switched on.

7.2.8 Work after ramping the magnet

7.2.8.1 Downloading the 3600 MPSU log file

The last ramp data is stored in a log file in non-volatile memory within the 3600 MPSU, even if the 3600 MPSU is switched off. Data is overwritten when the next ramp is started. It is recommended that after each ramp, the log file is downloaded and stored for future analysis.

The Magnet Peak Current is recorded. If the magnet should quench, the recording will tell you at what current level the magnet quenched.

To download the log file:

- **G** <Return>, this will ensure that you are in the error screen.
- From the HyperTerminal menu, select:-
 - Transfer

- Capture Text
- Browse
- Select the target directory and enter a filename e.g.C:\Medcom\Temp\MR25123_51099log.txt
- Start
- L <Return>, the data screen appears.
- A scrolling data screen will appear. Wait until the data scrolling has stopped.
The data screen contains seven columns. Each line displays the data stored at a specific period of time during magnet ramping.
- From the HyperTerminal menu, select:
 - Transfer
 - Capture Text
 - Stop
- Send the log file to immrqpinstallations.healthcare@siemens-healthineers.com

7.2.8.2 Switching off the 3600 MPSU



Verify that the yellow "CURRENT IN LEADS" LED is OFF when the 3600 MPSU is switched on.



If the yellow LED is flashing and the power LED is on, press reset on the front of the 3600 MPSU and run the magnet to zero or persist at field before proceeding further.



NEVER attempt to remove the magnet ramp leads when the "CURRENT IN LEADS" LED is flashing.

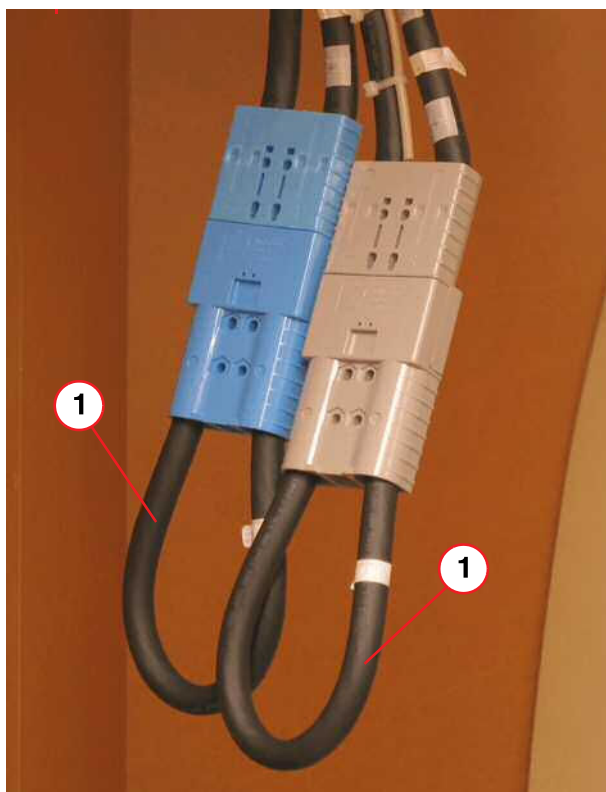
- Set the mains circuit breaker **CB1** to the **OFF** position. (→ 2/Fig. 275 Page 243)
- Set the circuit breaker **F28** in the **ACC** to the **OFF** position.

When shorting links are fitted:-



DO NOT disconnect **BOTH** extension leads at the same time.

Fig. 325: Magnet Shorting links.



(1) Shorting links

Shorting links (Newer systems do not have shorting links)

- Disconnect one extension lead, and fit the shorting link to the magnet lead.
- Fit the shorting link to the magnet lead.
- Repeat the process for the other shorting link.

- Pull the extension leads through the RF filter panel into the equipment room.
- Store the magnet leads on the lead tray in the magnet room.
- When shorting links are not fitted:-

When shorting links **are not** fitted:-

- Disconnect the extension leads from the magnet leads at the RF filter plate.
- Re-fit the the adapter plate using the securing screws.

Fig. 326: RF filter plate



Fig. 327: Adapter plate (view from Equipment Room)



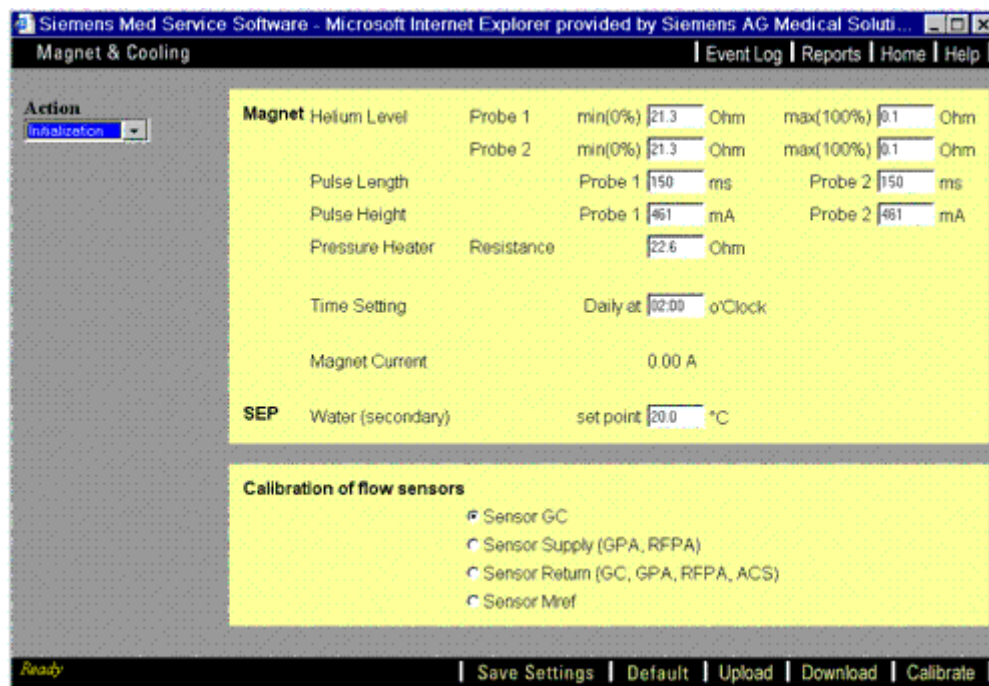
7.2.9 Final work steps

7.2.9.1 Save the magnet current to the HOST

- Access the service menu in the **SeSo at MRC**.

- Select menu item **"Magnet & Cooling"**.
- Select **"Initialization"** in the **"Action"** menu

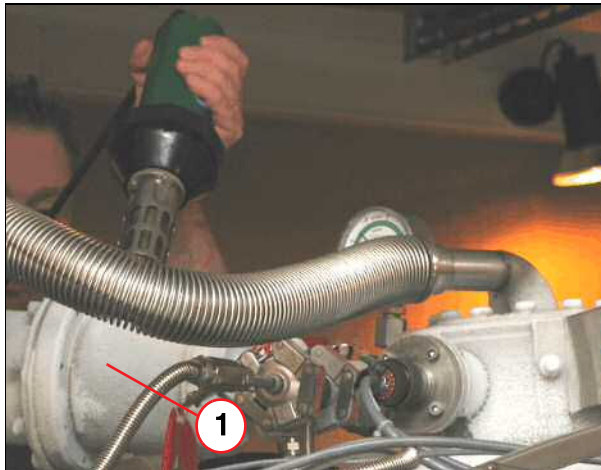
Fig. 328: Initialization



- Click **"Download"** to save the magnet-current value on the HOST.

7.2.9.2 Warm Turret & Venting

Fig. 329: Quench valve



(1) Quench valve

It is good practice to ensure that the **turret and venting** remains leak tight and the valves continue to function correctly.

- Warm up the **turret and venting** with a heat gun.

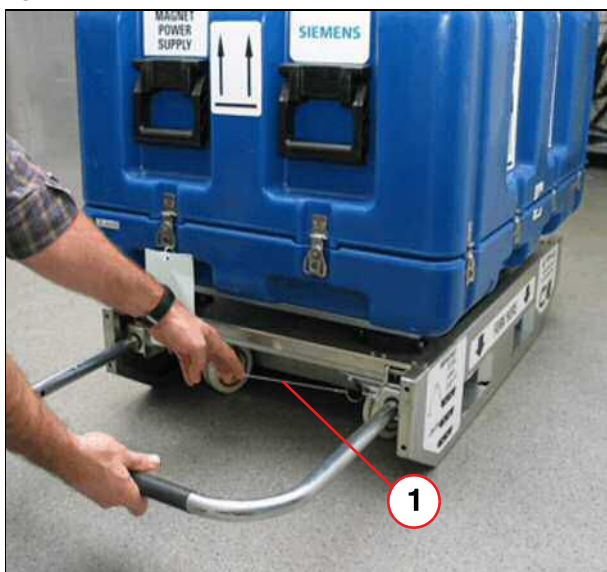
7.2.9.3 Pack the 3600 MPSU

Fig. 330: MPSU - ramp cables fitted



- Disconnect the two **3600 extension ramp leads** from the Anderson connectors **X3** and **X4**.
- Re-fit the **Transport Test Plug** into **X3**.
- Disconnect the remaining leads and pack inside the transport case.
- Close the lid and secure by pulling the two silver knobs and turning them through a 1/4 turn.

Fig. 331: MPSU



(1) Wire locking strap

- Replace the shipping case lid and secure using the 12 spring clips.
- Pull the wire strap to unlock the handle, and slide the handle out of the recess under the trolley.
- Release the wire strap to lock the handle in the extracted position.
- Raise the handle to lower the wheels and lift the trolley off of the two skids.
- The 3600 MPSU and trolley can now be safely moved.

7.3 Failure during ramping



CAUTION

If the supply to the MPSU fails while the magnet is ramping up or down, the magnet current will continue to flow.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Never remove the ramp cables from the magnet while current is in the magnet ramp cables.

Current is in the ramp cable whenever:

- » The yellow "CURRENT IN LEADS" LED is flashing.



Failure of the MPSU may prevent this LED from lighting up. So, in the event of a failure (e.g. mains failure), it is essential to measure the voltage across the terminals on the turret.

Any voltage on these terminals other than zero is an indication that there is current in the ramp cables.

- » Magnet was ramping up or down and there was a mains failure.
- » Magnet was ramping up or down and the MPSU detected a fault such as over temperature, which would shut down the MPSU.
- » Normal ramp up or down of the magnet.



CAUTION

If the magnet is suspected of ramping down through the ramp cables, owing to one of the reasons above.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Do not remove the ramp cables. The magnet must be run to zero or persist at field before removing the ramp cables.



If the cause of the shutdown was failure of the mains, nothing can be done until both the MPSU and the Magnet Supervisory (MSUP) are working again.

When the MPSU and the MSUP are operational, ramp the magnet to zero.

7.4 What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256)

7.4.1 OR105 TSG - Post Quench

7.4.1.1 General

If the magnet quenches, a large volume of helium gas is generated within the helium vessel. The quench valve opens, and stay open until the flow of gas subsides. The quench valve re-seals, preventing air (caused by cryo-pumping) from being drawn back into the magnet turret and helium vessel.

The MSUP automatically maintains the system pressure above atmospheric.



Magnet HSC must be notified of a quench (→ Siemens Healthineers Magnet WS CS HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

Fill out the **Quench Report** - (→ MR Product Information, <https://productinformation.healthcare.siemens.com/ProductInformation.aspx?uuid=7abf6bbb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (Quench Report)



The magnet helium level can be recovered if the magnet temperature is no warmer than nitrogen temperature.

Check the carbon resistor (CARBON RES 1;pins a-e) reading and compare to the calibration values on the Magnet Acceptance Document.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 1000 liters of liquid helium.

NOTICE

If the magnet temperature warms up above 77K, a full nitrogen pre-cool is required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

» A full nitrogen pre-cool will take 8 –10 days.

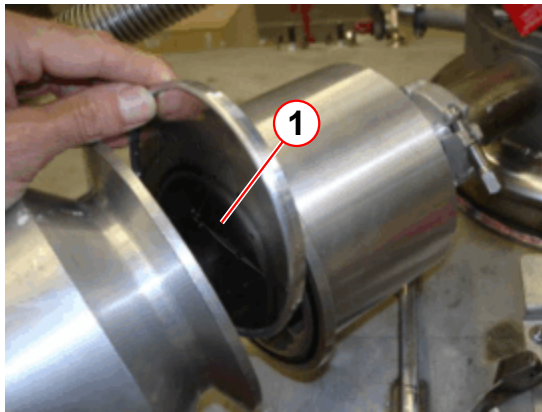
Mobile System

NOTICE**Risk of Air ingress**

The Mobile high altitude burst disk **WILL** rupture during a quench.

- » Post quench the high altitude 10 psi burst disk (10129467) and O-Ring (10113389) must be replaced.

Fig. 332: Removing the High Altitude disk



(1) Burst Disk

- Mobile systems are fitted with an additional High Altitude Burst Disk in front of the quench valve assembly.
 - » This disk will rupture if the magnet quenches.
 - » For replacement procedure: (→ Mobile Systems - High Altitude Burst Disk / Page 307)

Safety

Post quench procedures must be performed by CSE's that are trained to the MR2MAG-NET level. (or System level trained).



Switch off and isolate the magnet power supply from the mains before further action.

**WARNING****Risk of asphyxiation.**

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenics.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.

**CAUTION****RISK OF ICING!!**

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk and O-ring seals IMMEDIATELY.
- » Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.
- » If ice is suspected contact Magnet HSC immediately.

**CAUTION**

Cold helium gas.

Personal injury may occur.

- » Protective clothing (long sleeves), cryogenic gloves and eye protection must be worn.

7.4.1.2 Post Quench Steps

Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 1	■ Isolate the MPSU from the magnet. ■ Download the 3600 MPSU ramp log files.	■ Stop filling.	■ Check that no ERDU buttons are pressed. ■ Re-set ERDU button.
Step 2	From the MSUP-Display Unit. ■ Check the liquid helium level. ■ Check the carbon resistor temperatures. (→ Checking the Carbon Resistors (CCR) Values / Page 286)		
Step 3	■ Order 1000 liters of liquid helium.¹		
Step 4	■ Warm up and leak check the quench valve and burst disk. (→ Checking the Quench Valve Burst Disk / Page 293) ■ Warm up and leak check the auxiliary vent burst disk. (→ Auxiliary Burst Disk / Page 302) ■ MOBILE system - replace the high altitude burst disk. (→ Mobile Systems - High Altitude Burst Disk / Page 307)		

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 5	After 1 hour : ■ Download the Magnet Supervisory log files. (→ Log File Capture / Page 285) ■ Helium fill. Fill to recommended helium ramp level.		
Step 6	■ Leak check the quench valve and venting. (→ Final Checks / Page 312)		
Step 7	■ Fill out the Quench report.		
Step 8	■ Email the (→ Magnet HSC, mailto:magnet.wscshsc.team@siemens-healthineers.com) with: » 3600 Power Supply logs. » Magnet Supervisory logs and screen capture. » Magnet serial number.		
Step	■ Ramp the magnet.		
Step 10	■ Perform the shim check. » If the shim check exceeds the specifications, a re-shim is necessary.		

1. The quantity of the order depends on the helium level remaining in the system after a quench, and volume already on site.

7.4.1.3 Pre-requisites

Tools Required

The post quench tools required:

- Heat gun and paper toweling.
- Service tool kit.
- Pressure service tool (material number **03861221**).
- Laptop and RS232 cable.
- O₂ monitor.
- Personal protection equipment. To include, cryogenic gloves, safety shoes, glasses and a long sleeved protective clothing.
- Helium leak detector.

Fig. 333: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal.

- A number of seals are supplied in the spares kit. These must be replaced when used on the magnet.
- Venting components will require ordering from CSML.

Mk1 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 08396090).
- High load seal (material number 10100171).

Mk2 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 10121913)
- High load seal (material number 10100171).
- O-ring seal 44.5mm ID (material number 10100151).



Always replace the burst disks and seals used during replacement procedures. They are critical components for reducing the risk of air ingress. A replacement must always be available for immediate use.

Unattended Quench

Unattended quench indicators are:

- If the Magnet STOP (ERDU) has been activated:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the helium level is low:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the magnet pressure is very low, the
 - » cold head switches off.
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- Imaging is not possible if there is no magnetic field.

Press the AUDIO ALARM OFF button on the alarm box to silence any audible alarms.

Log File Capture

Magnet Supervisory

- The **quench data is lost**, if:
 - a) The system is re-booted within 1 hour after a quench
 - b) The system is turned off within 1 hour after a quench
 - c) The data is downloaded within 1 hour of after a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the Magnet Power Supply log file.



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data will be overwritten!

- Switch off and disconnect the MPSU from the magnet.

If the magnet quenches during helium fill:

- Stop the transfer.



Do not remove or disconnect the syphon as the magnet will be at high pressure.



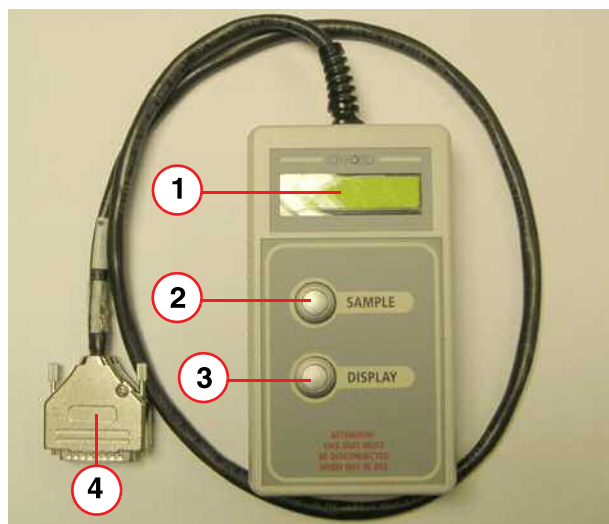
For all types of quench, allow up to 1 hour for the quench pressure to subside.

Use the MSUP Display unit:

- To check that the MSUP pressure control system is working. If a display unit is not available, use the SESO interface on the Host service screens.
- If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. The compressor switches off.

Checking the Carbon Resistors (CCR) Values

Fig. 334: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Using the MSUP-Display Unit:

- Connect the 25-pin 'D' connector to X9 on the MSUP.
- Activate the MDU by simultaneously pressing the SAMPLE button (2) and DISPLAY button (3).
- Press the Display button (3). The unit will automatically carry out a helium level test. The helium fill level is shown in the display (1).
- pressing the DISPLAY button, toggle through to Carbon resistor parameters.
- Check the carbon resistor values and compare the values with table with the expected values.



The magnet helium level can be recovered if the temperature of the carbon resistor (CARBON RES 1:Pins a-e) is still colder than nitrogen temperature.

The carbon resistor (CARBON RES 1) reading on the hand-held display must be in the range of 180 - 1500 ohms.

Confirm the readings, using the multimeter directly on the 41-pin connector.



Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.



Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.

- High readings = coldest temperature
- Low readings = warmest temperature
- Check temperatures with the figures on the Magnet Acceptance document.



If temperatures are below or equal to nitrogen temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

Tab. 18 Expected Carbon Resistor Values

41-pin plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e (Bottom)	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500
Carbon Four d - e (Top)	120 - 160	130 - 180	700 - 1500
All Ceramic Carbon Resistors (CCRs)	880 - 1050	1150 - 1350	2300 - 3380

Fig. 335: OR105/122 Magnet Acceptance Document (CR)

SIEMENS		Magnet Acceptance Documentation Installation Data		
Siemens Magnet Technology Quality Regulations				
Cryostat Number	21199812	SMT Reference	21201632 02.09	
System Description 1.494T Actively Shielded Superconducting Twin Turret Magnet System				
Measured Magnet Centre (Offset)				OR105
				-1 (S) mm
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet				
Operating Parameters				
Description	Specification value	Measured value		
Operating Current @ 1.494T	< 600 AMPS	509.95	AMPS	
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp	
Carbon Resistor Values				
POSITION	Room Temperature	Helium Temperature		
Top	162.4	1140		
One	134.4	903		
Two	132.3	994		
TDC	140.4	943		
Helium Level Probe Resistance (When using a 250 mA supply)				
		Probe 1	Probe 2	
@ 0 Resistance	Ohms	85.36	85.36	
0 % Calibration		774	774	
100 % Resistance	Ohms	0.22	0.22	
100 % Calibration		2	2	
Diode Table No		-1		
Pressure Heater Res	Ohms	19		
Final Temperature Sensor Recon Values				
50kK	50kK Lin	Coldhead	Recon Margin	
48	45	42	K 467 Millivolts	
Checked by	K Suraina		Signature	
Date	02/02/2009			

Fig. 336: OR105/122 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
MR Magnet Technology Quality Regulations			
Cryostat Number	21243234	SMT Reference	21245901 06.10
System Description		OR105	
1.494T Actively Shielded Superconducting Twin Turret Magnet System			
Measured Magnet Centre (Offset)		-2 (S) mm	
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet			
Operating Parameters			
Description	Specification value	Measured value	
Operating Current @ 1.494T	> 600 AMPS	510.05	AMPS
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp
Carbon Resistor Values			
POSITION	Room Temperature	Nitrogen(77K)	Helium Temperature
TDC d - e	878		2567
Top a - e	926		2938
Mid b - e	985		2715
Bottom c - e	877		2613
TDC Calibration Values:	850.4	1122.9	2731.7
Helium Level Probe Resistance (When using a 250 mA supply)			
		Probe 1	Probe 2
@ 0 Resistance	Ohms	85.36	85.36
0 % Calibration		774	774
100 % Resistance	Ohms	0.22	0.22
100 % Calibration		2	2
Diode Table No		-1	
Pressure Heater Res	Ohms	19	
Final Temperature Sensor Recon Values			
50kK	50kK Lin	Coldhead	Recon Margin
53	47	44 K	530 mW
Checked by	K Suraina	Signature	
Date	02/06/2010		
Page 1 of 1		25-AD EPS 000 01	

Check the Magnet Pressure

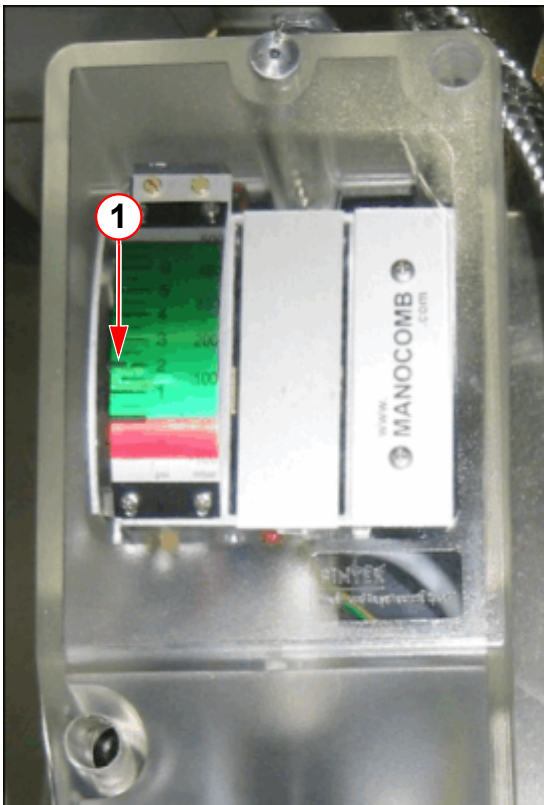
Check that the helium vessel has a positive pressure:

- Magnet pressure should read 15.3 psi (A) on the MSUP display unit.
- Check for positive pressure on the pressure gauge.
- If the pressure gauge is within the **RED zone**, the magnet is sub-atmospheric.
 - » **DO NOT** open the magnet bypass valve, or remove any high pressure fitting.
 - » If the magnet pressure does not stabilize via the pressure heater control circuit, manually induce pressure into the cryostat.



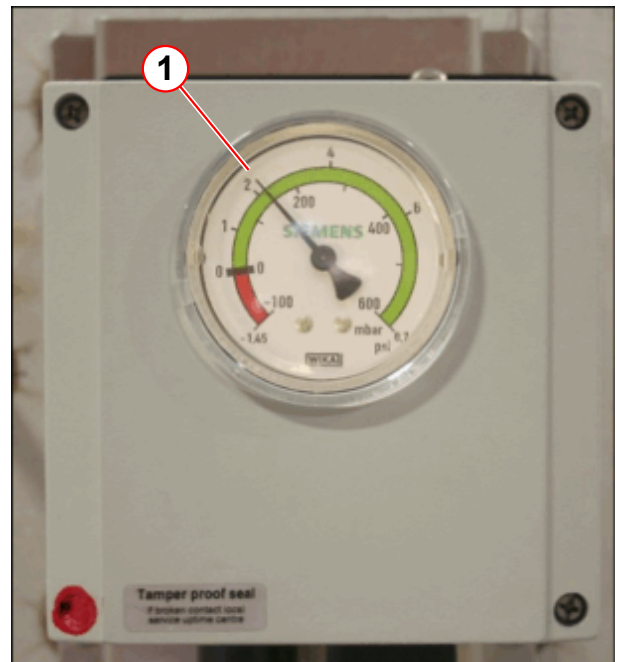
Wait until the magnet has gained a positive pressure before continuing with the burst disk and post quench leak checks.

Fig. 337: OR97 - Pressure gauge (expected pressure)



(1) Expected pressure 2.1 psi (G)

Fig. 338: OR97 - WIKA pressure gauge (expected pressure)



(1) Expected pressure 2.1 psi (G)

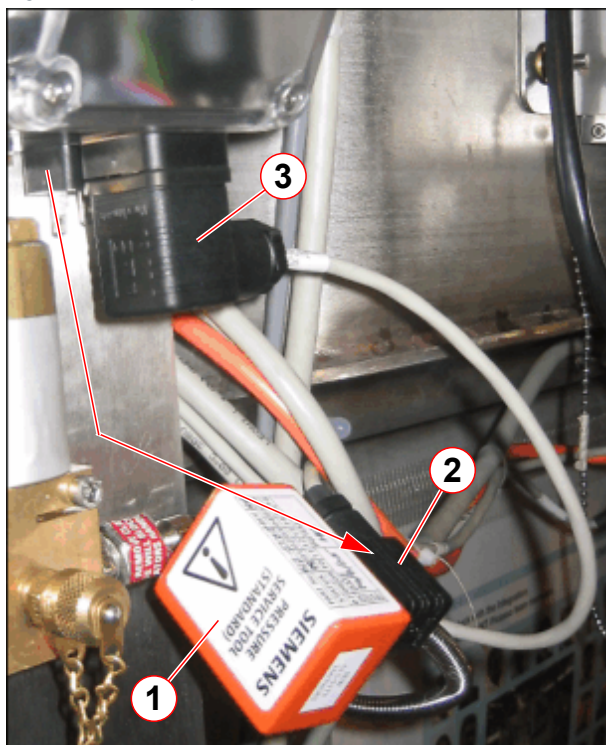
Deactivate the Pressure Heater

The Pressure Service Tool is used to deactivate the pressure heater control circuit. The electrical connectors are labeled pressure sensor (rear) and pressure switch (front).

- To fit the pressure service tool:
 - Disconnect the rear electrical connector on the pressure switch transducer.

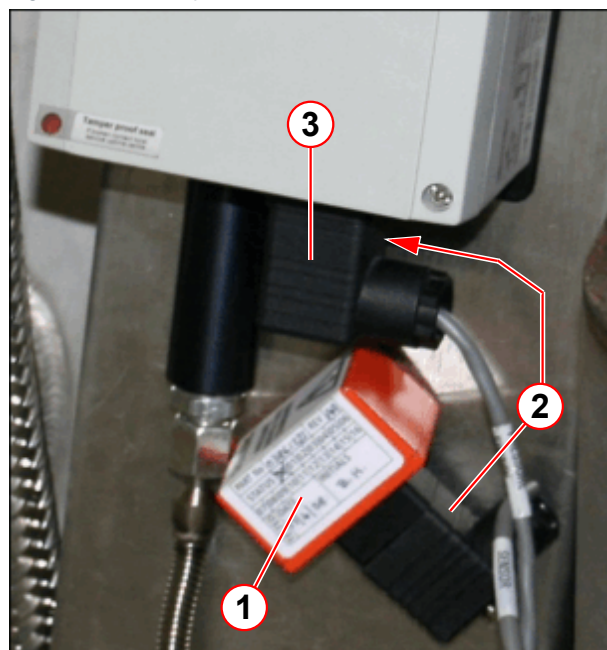
- Fit the pressure service tool to the connector lead.
- The Alarm Box will sound PRESSURE ERROR.
- Press the ACKNOWLEDGE button to cancel the alarm.
- The pressure reading will be set to ~18 psi (A), which will disable the pressure heater.
- To activate the pressure heater control circuit, connect the cable to the rear connector on the pressure switch transducer.
 - » Cancel any alarms.

Fig. 339: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

Fig. 340: Fit the pressure service tool

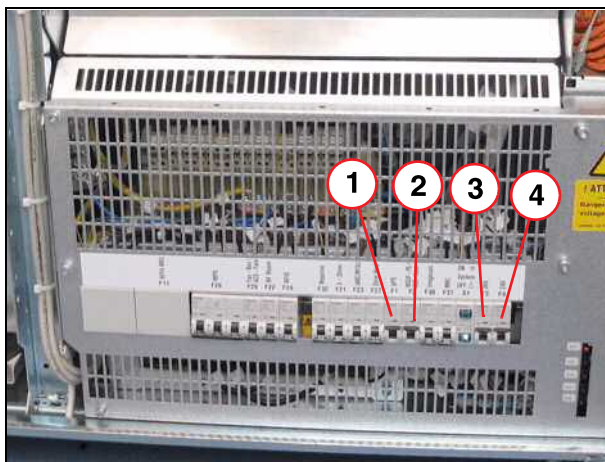


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 341: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service pressure tool is not available:

- Switch OFF the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » Pressure heater is powered down.
- The ERDU circuit is still functional.
- Helium level and magnet temperatures can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, only 2 minutes of power is available.

Depressurize the Helium Vessel



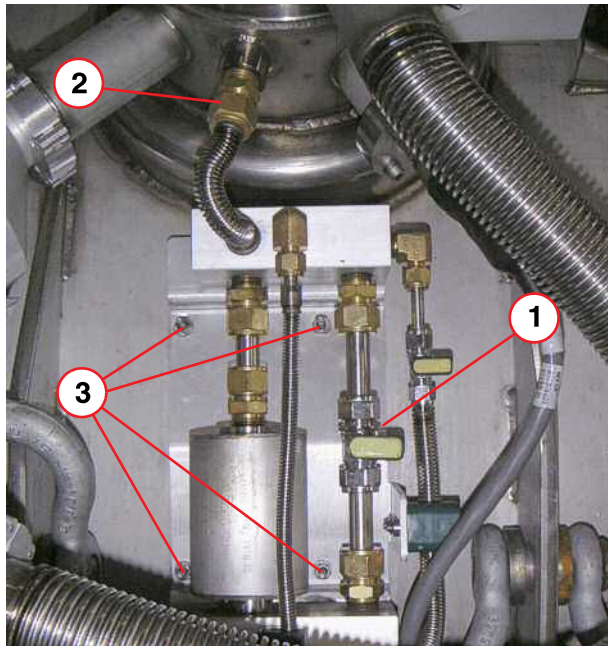
CAUTION

Risk of ice formation.

The magnet bypass valve is only **OPENED** for depressurization of the helium vessel. The bypass valve is **CLOSED** during normal operation.

- » To avoid air ingress, **CLOSE** the magnet bypass valve immediately after the helium vessel is depressurized.

Fig. 342: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

If depressurization of the helium vessel is required:

- Fully open the magnet bypass valve to depressurize the helium vessel.
- Wait until zero psi is displayed on the pressure gauge.
- The pressure switch will activate.
 - » the cold head will turn off.
- Close the magnet bypass valve to prevent air ingress.
- Warm up the vent valve assembly.
 - » Take care not to damage any components when using a heat gun.

7.4.1.4 Checking the Quench Valve Burst Disk

Post quench if there is no magnet pressure a burst disk has ruptured. Check the graphite (quench valve) burst disk.

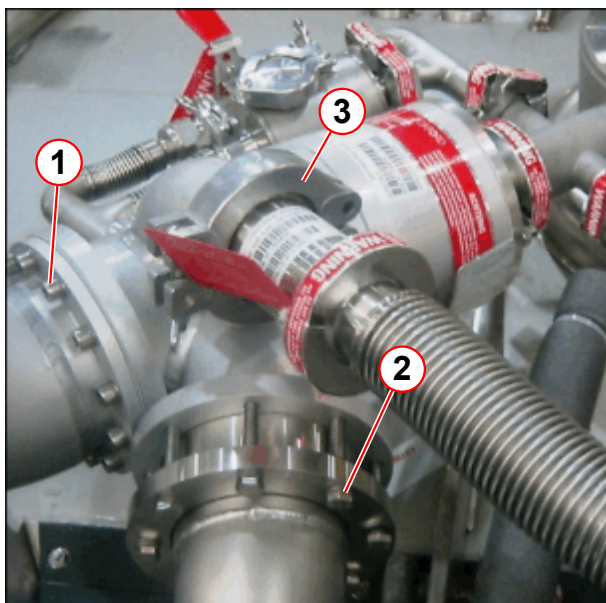
- Disable the pressure heater by fitting the pressure service tool.
- Warm up the quench valve and venting to room temperature, using a heat gun.
- The elbow assembly and manifold must be removed for inspection and leak check of the graphite burst disk and O-ring.



Check venting configuration fitted and follow applicable procedure for Mk1 or MK2 versions.

Mk1 Venting Configuration

Fig. 343: Mk1 - Venting removal

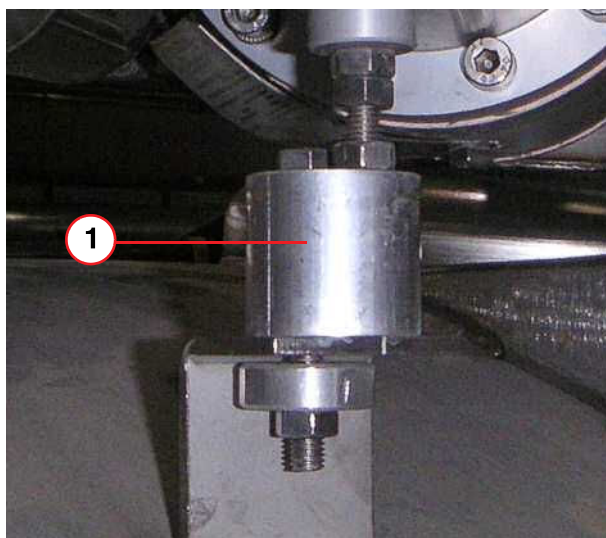


- (1) Elbow connection
- (2) OVC burst disk
- (3) Auxiliary burst disk

To access the quench valve and burst disk:

- Remove the 12 x M8 Cap hd screws and retaining the elbow assembly to the manifold venting (pos.1).
- Remove 8 x M8 Hex hd screws retaining the OVC burst disk to the manifold (pos.2).
- Disconnect the NW50 clamp (pos.3) and remove the auxiliary vent line.

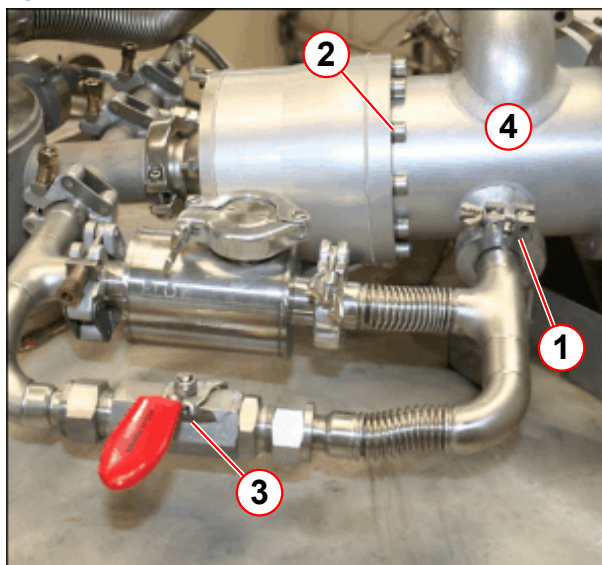
Fig. 344: Mk1 Support Bracket



- (1) Support Bracket

- Loosen the support bracket (1) screw underneath the quench valve interface assembly.

Fig. 345: Mk1 - Manifold removal

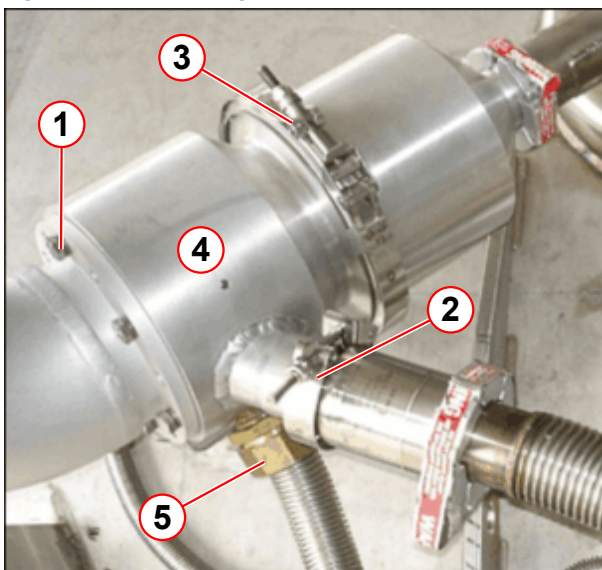


- (1) NW25 Bypass valve connection
- (2) Manifold screws (M8 x 12)
- (3) Bypass valve
- (4) Manifold

- Remove the NW25 (low pressure) clamp to release the 16 psi (A) valve assembly (pos.1).
- Remove the M8 Cap hd screws (x12) securing the manifold to the quench valve (pos.2).
 - » Place the manifold to one side, allowing access to the quench valve.
- Check that the magnet bypass valve is closed.

Mk2 Venting Configuration

Fig. 346: Mk2 Venting removal



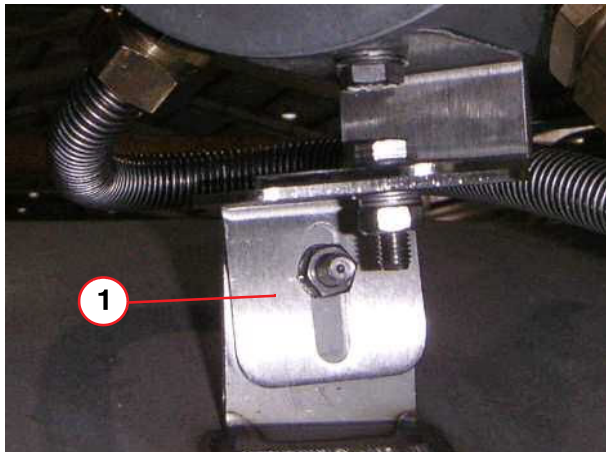
- (1) Elbow connection (M8 x 6)
- (2) V-clamp - Auxiliary vent
- (3) V-clamp manifold to quench valve
- (4) Manifold assembly
- (5) OVC vent connection

To access the quench valve and burst disk:

- Remove the 6 x M8 screws retaining the elbow assembly to the manifold (pos.1).
- Remove the V-clamp retaining the auxiliary vent to the manifold (pos. 2).
- Remove the V-clamp retaining the manifold to the quench valve (pos. 3).

NOTE: it should not be necessary to remove the OVC vent line connection (pos.5).

Fig. 347: Mk2 Support bracket



(1) Support bracket

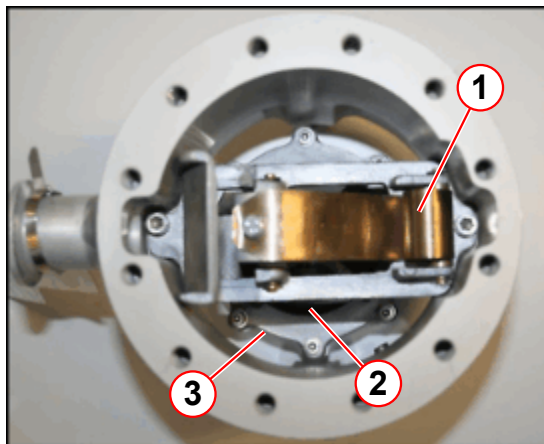
- Loosen the support bracket (1) screw underneath the quench valve interface assembly.
- Move the manifold assembly away from the quench valve.

The quench valve and graphite burst disk can now be accessed to check for leaks.



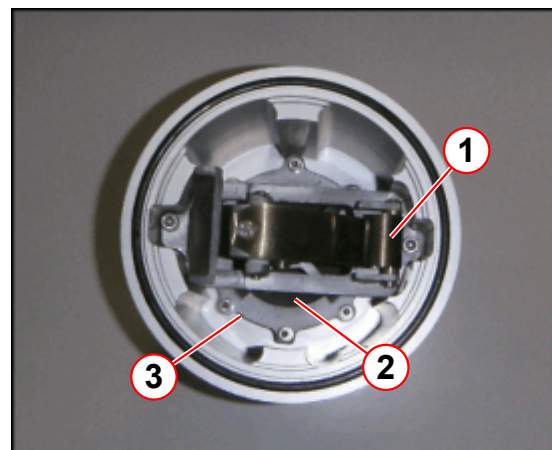
The spring plate and burst disk assembly is the same for MK1 and MK2 quench valves. The next steps show the MK2.

Fig. 348: Mk1 Quench valve assembly



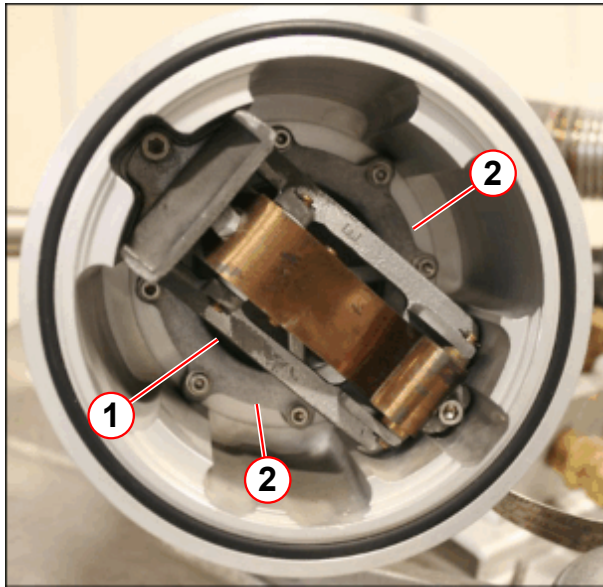
- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

Fig. 349: MK2 - Quench valve assembly



- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

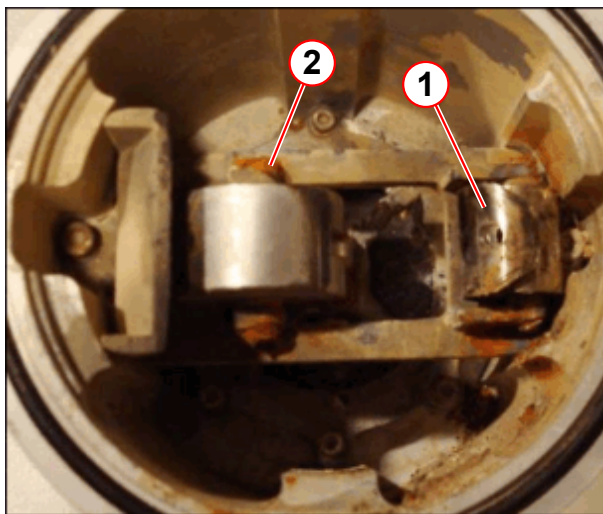
Fig. 350: Leak check



- (1) Graphite burst disk
- (2) Burst disk O-ring seal

- Check the graphite burst disk for breakage or cracks.
- If the graphite burst disk is broken or cracked, it must be replaced **immediately**.

Fig. 351: Water corrosion (example)



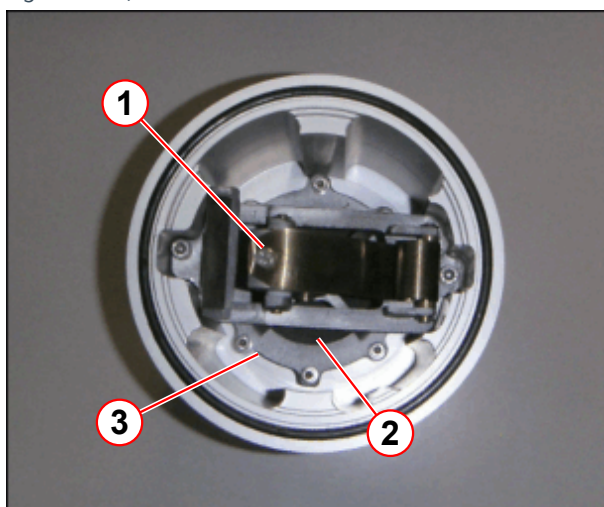
- (1) Spring broken
- (2) Corroded screws

Check for ice and water corrosion or breakage of:

- The spring plate assembly.
- All screws.
- Corroded quench valve components must be replaced.

- With no damage to the burst disk or spring plate visible, continue to leak check around the quench valve O-ring and retaining plate.
- Check that the magnet pressure is a minimum of 0.5 psi (G) for leak checking.
- Leak check around the burst disk and spring plate assembly.

Fig. 352: Quench Valve leak check

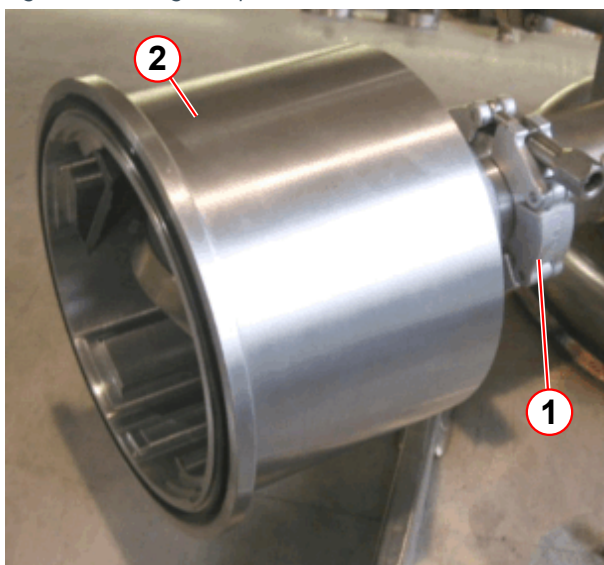


- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

- If there is no leak found the quench valve O-ring seal behind the spring plate is OK.
- There is no requirement to remove the burst disk spring plate.
- Continue to check the auxiliary burst disk and refit the quench valve.
- If a leak is detected the burst disk spring plate assembly will need removing to replace the O-ring.

Replacing the Burst Disk and O-ring Seal

Fig. 353: Fitting the quench valve



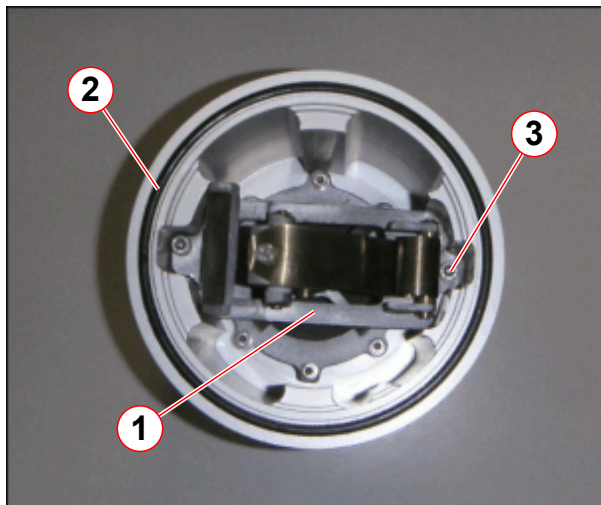
- (1) NW40 seal and clamp
- (2) Quench valve assembly

- Open the magnet bypass valve to depressurize (if necessary).
- De-ice the turret flange.
- Mark the orientation of the quench valve
- Remove the NW40 (high load) turret clamp and seal.
- Remove the quench valve from the turret flange.
- To minimize air ingress, immediately place a ball of paper toweling into the open turret port.
- Close the magnet bypass valve.



The spring exerts significant force on these screws.

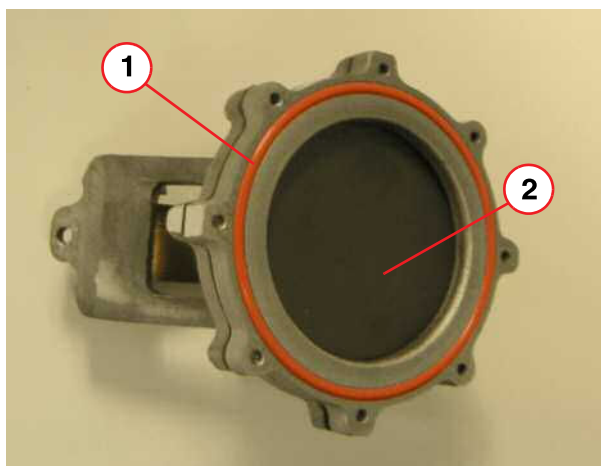
Fig. 354: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Unscrew and remove the (2 off) M5 cap head screws.
- Lift out the burst disk assembly from the quench valve body.

Fig. 355: Quench Valve Bursting Disk Assembly



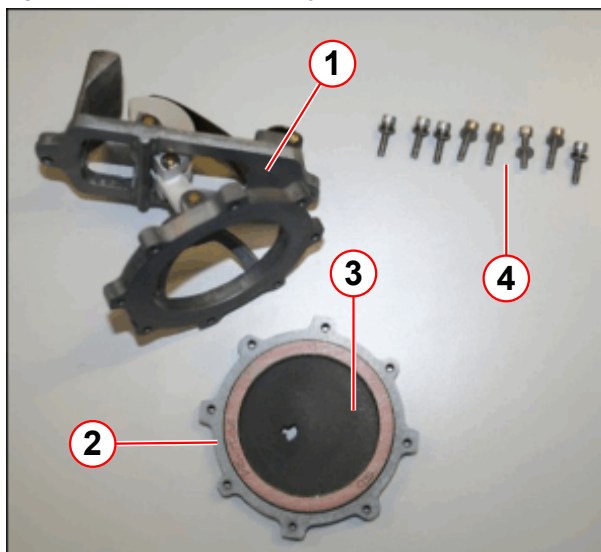
- (1) O-ring
- (2) Burst disk

- Inspect the burst disk 'O-ring and replace if damaged.
- Clean the O-ring groove and O-ring before re-fitting.
- Check that the O-ring is seated evenly around the groove.



The O-ring must be replaced each time the quench valve is removed. DO NOT apply grease to the O-ring.

Fig. 356: Burst disk assembly

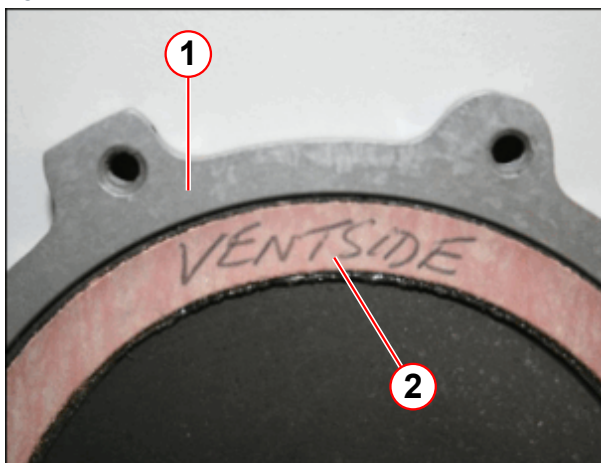


- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

To replace the burst disk:

- Unscrew and remove the 8 x M4 cap screws.
- Remove the broken burst disk and discard.
- Check that the new burst disk and flange faces are clean and undamaged.

Fig. 357: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

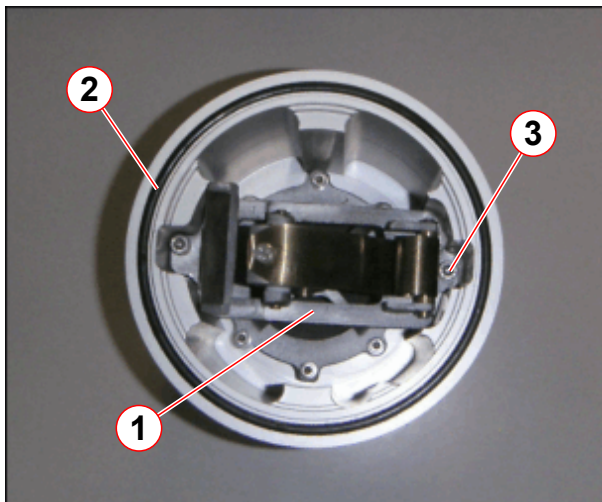
- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 358: Thread locking compound



- Fit the burst disk assembly. (Do not apply grease).
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound..
- Diametrically torque the screws evenly to 0.8 Nm (0.6 ft. lbs).

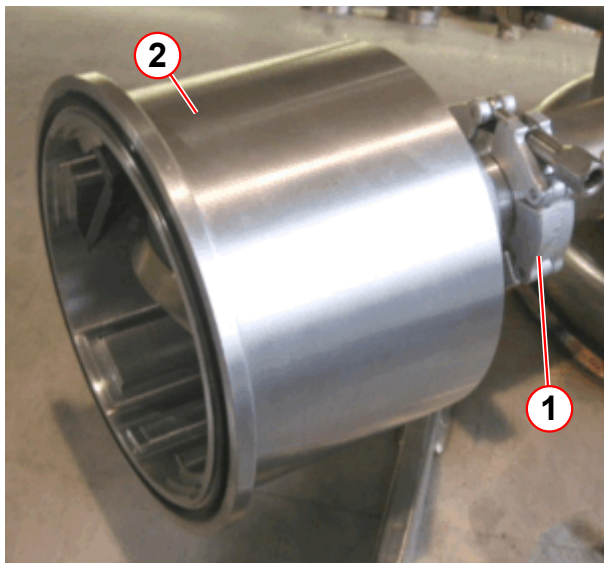
Fig. 359: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Re-fit the burst disk assembly into the quench valve body.
- Torque the 2 x M5 cap hd screws to 5 Nm. (3.8 ft.lbs).
- Check the quench valve manifold O-ring is in good condition (pos.2).
- Replace if damaged.

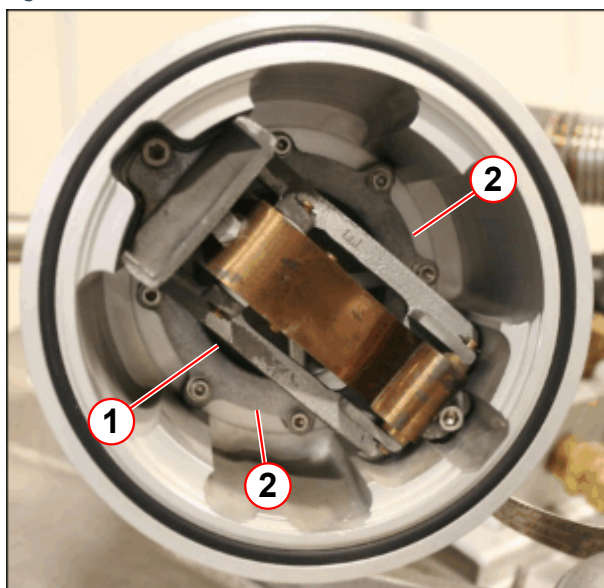
Fig. 360: Fitting the quench valve



- (1) NW40 seal and clamp
(2) Quench valve assembly

- **Remove the paper toweling** if used (or NW40 blank) from the turret port. Ensure the surfaces are clean and dry.
- Fit the quench valve assembly to the turret flange.
 - » Re-fit using a new NW40 high load seal (material number 10100171).
 - » Check the correct orientation.
 - » Warm with heat gun when tightening the NW40 clamp.

Fig. 361: Leak check



- (1) Graphite burst disk
(2) Burst disk O-ring seal

- Check that the magnet bypass valve is closed and pressurize the system.
- Leak check the burst disk and O-ring seal.
- Leak check the NW40 clamp and seal at the turret port.
- If no leaks are detected:
 - » Fit the quench valve elbow interface with the V-clamp to the quench valve.
 - » Fit the support bracket screw to the elbow interface assembly.
 - » Check the auxiliary burst disk.

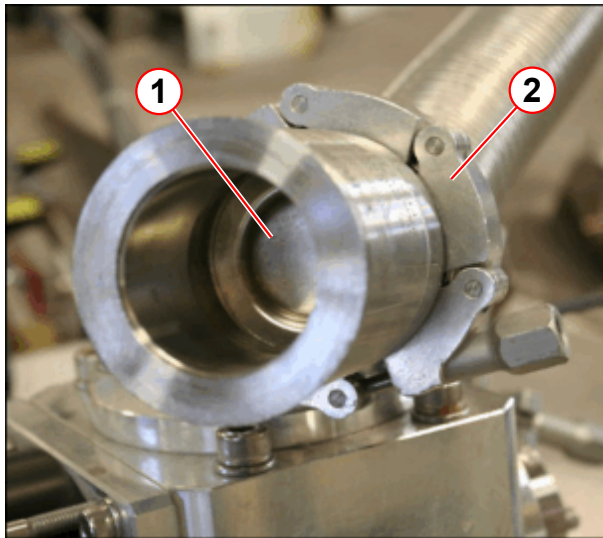
7.4.1.5 Auxiliary Burst Disk

Post quench the auxiliary burst disk vent line is removed when disconnecting the venting to leak check the quench valve.



Do not touch the metal membrane, it is very fragile.

Fig. 362: Auxiliary burst disk check



- (1) Burst disk
(2) High load seal

- Visually inspect the internal metal membrane of the burst disk for damage (pos.1).
- Leak check the high load seal (pos. 2).



WARNING

Risk of air ingress

If the auxiliary burst disk is broken, IMMEDIATELY fit the blanking gasket seal.

- » The blanking gasket seal is supplied in the magnet spares kit.
- » Order a replacement auxiliary burst disk and fit before ramping the magnet.



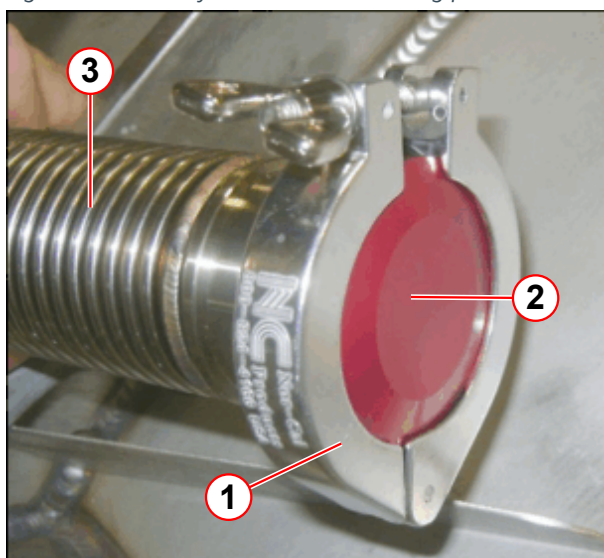
WARNING

Safety device removed

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » Replace the auxiliary vent burst disk before ramping.

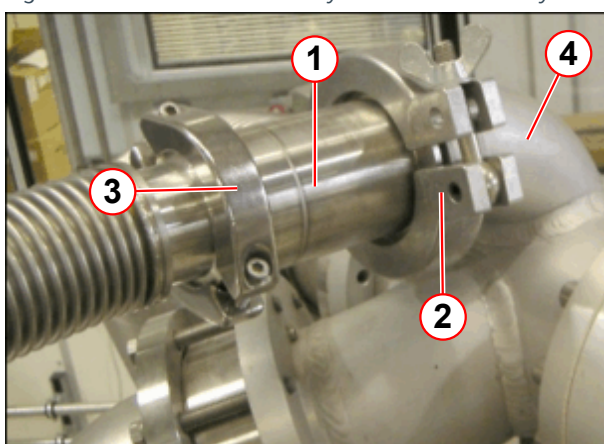
Fig. 363: Auxiliary vent - NW40 blanking plate



- (1) NW40 clamp
- (2) NW40 red blanking plate
- (3) Auxiliary vent line

- If the membrane has ruptured, disconnect the high load clamp and remove the burst disk from the vent line.
- Fit the red anodized blank immediately from the spares kit.
- Order a replacement auxiliary burst disk. Fit as soon as available.
- With the blank fitted allow magnet to pressurize.
- Leak check red blanking plate and seal on the end of the auxiliary vent line.

Fig. 364: OR105 Mk1 Auxiliary burst disk assembly



- (1) Aux burst disk assy
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

For the Mk1 venting:

- Connect the auxiliary burst disk to the manifold assembly with a NW50 clamp and seal.

Fig. 365: Auxiliary vent interface - O-ring seal



- (1) Quench valve manifold
- (2) O-ring seal
- (3) Auxiliary bursting disk
- (4) NW40 clamp (high pressure)

- Check that the O-ring seal (Pos.2) is not damaged and that the surface is clean.
- If damaged, replace the O-ring.
- Connect the auxiliary burst disk to the manifold assembly with the V-clamp.

7.4.1.6 OVC Burst Disk

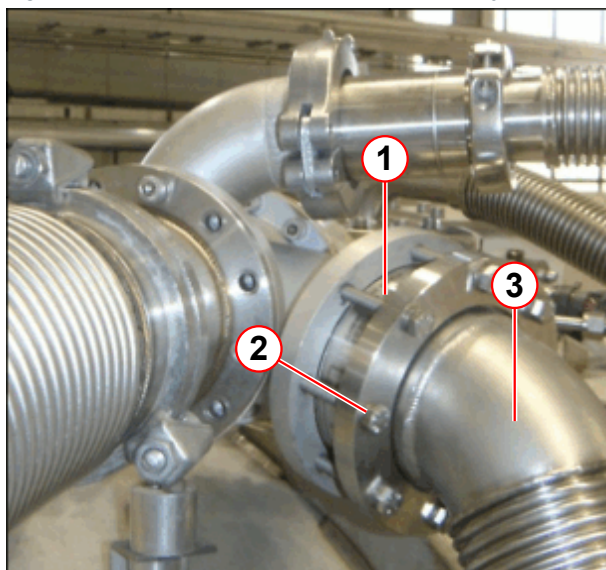


The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 19 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk1	Before System MR25575	Before System MR30125
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 366: OR105 Mk1 OVC burst disk assembly



- (1) OVC burst disk
- (2) Screws (8 off)
- (3) OVC vent tube

Mk 1 OVC burst disk

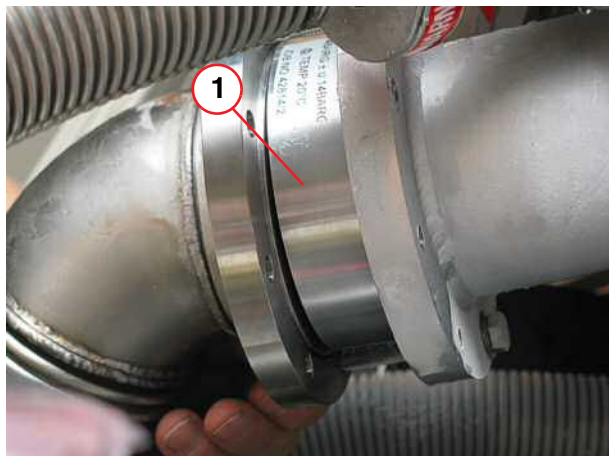
A quench will not cause the OVC metal burst disk to rupture.

- During the post quench checks the burst disk can inspected when the venting removed to inspect the quench valve.
- Remove the (8 off) **screws** holding the OVC burst disk in place.



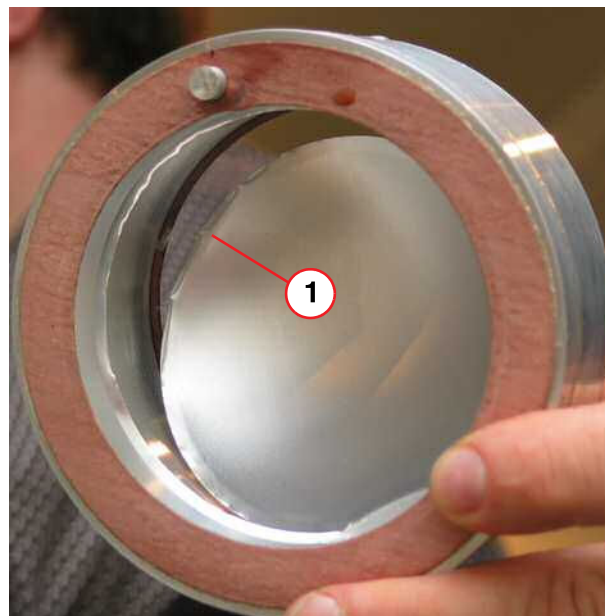
The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 367: OVC burst disk



- (1) OVC burst disk

Fig. 368: OVC burst disk

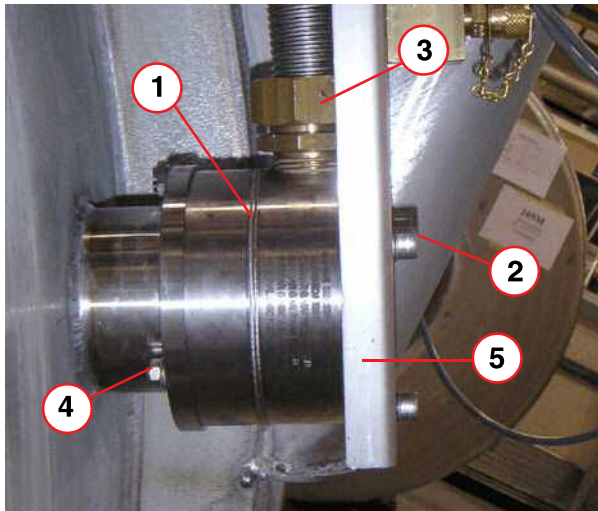


- (1) Broken OVC burst disk



The burst disk has a metal peg in one face to ensure correct orientation of the disk.

Fig. 369: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

Mk 2 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. The OVC burst disk requires checking:

- During routine safety checks and ice inspection of the venting.

7.4.1.7 Mobile Systems - High Altitude Burst Disk



Mobile systems are fitted with a high altitude burst disk, in addition to the quench valve burst disk.

If the magnet quenches, the high altitude disk will rupture and must be replaced.

Spare Parts

Post quench the high altitude burst disk will need to be replaced and the O-ring seal checked.

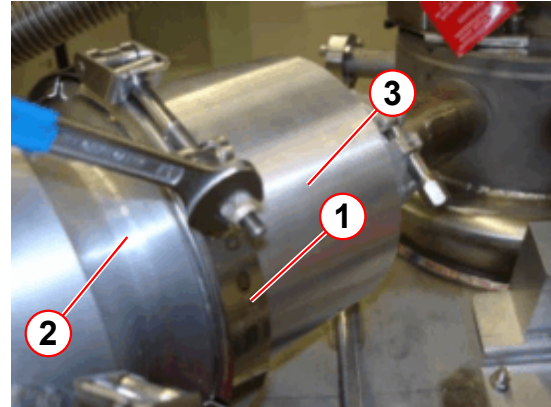
- 10 psi burst disk (Composite) - (material number **10129467**).
- O-ring seal - (material number **10113389**).

High Altitude Bursting Disk Removal

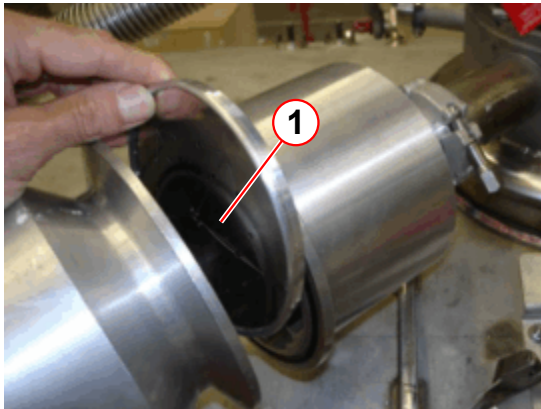
- Undo the M6 screw and nut and remove the retaining clamp on the manifold support bracket.
- Undo the M6 nut and release the V-clamp securing the manifold assembly to the quench valve.

Fig. 370: Manifold Support Bracket

(1) Support Clamp

Fig. 371: Manifold V Clamp

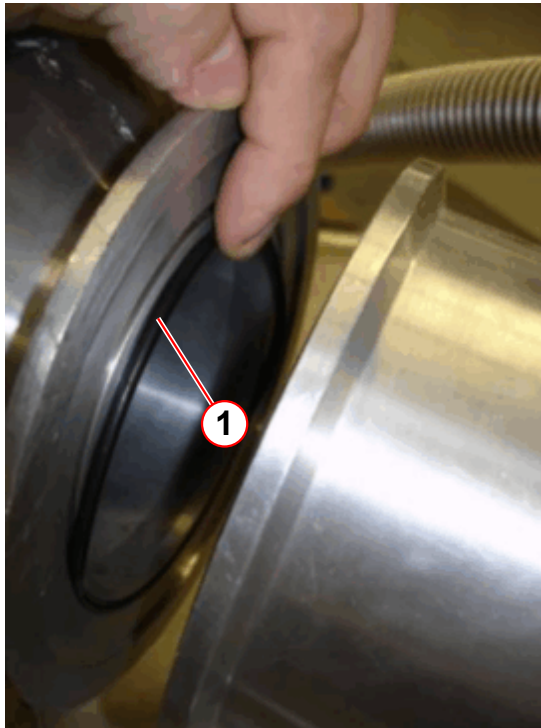
(1) V Clamp
(2) Manifold
(3) Quench Valve

Fig. 372: Removing the High Altitude disk

(1) Burst Disk

- Carefully pull the manifold away from the quench valve.
- The burst disk is held in position by the V-clamp seal. There are no screws.
- Remove the ruptured high altitude burst disk.

Fig. 373: O Ring Fitment



(1) O Ring and Groove

- Remove the O-ring in the manifold flange.
- Clean the flange mating surfaces.
- Complete the quench valve burst disk checks.



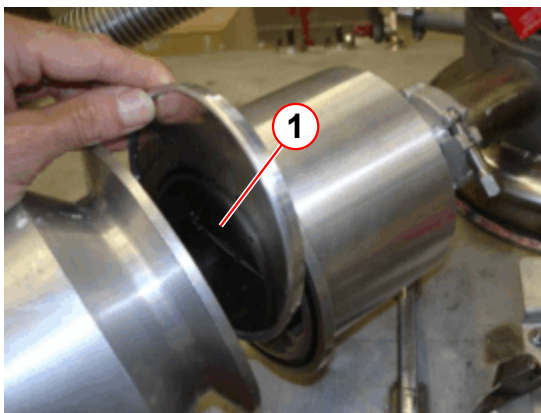
As part of the Post Quench checks at this stage the standard quench valve and graphite burst disk checks will be completed.

Fitting the High Altitude Bursting Disk



Take care not to touch the center of the disk. It can be damaged easily.

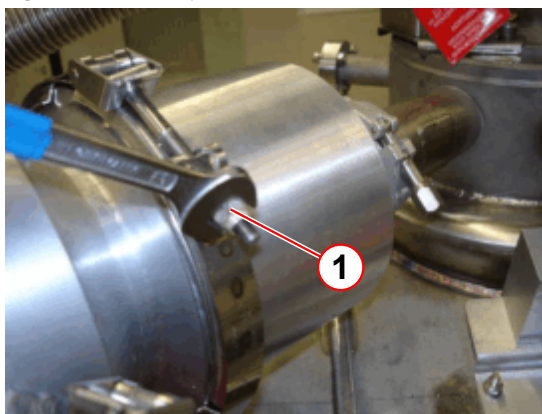
Fig. 374: Burst Disc Fitment



(1) Burst Disc Words "VENT SIDE"

- Check that the manifold O-ring is clean and not damaged.
 - » Replace if necessary.
- Check that the 2 flange surfaces are clean.
- On the disk, check that the wording 'Vent Side' is facing away from the quench valve.
- Fit the new high altitude burst disk between the manifold and quench valve flanges.

Fig. 375: V Clamp Fitment



(1) Self Locking Nut

- Hold the burst disk in position and join the manifold flange and burst disk to the quench valve.
- Check that the burst disk is still centered and refit the V-clamp.
- Tighten the nut until there is a sharp increase in resistance indicating that the self locking nut is fully tight.

Fig. 376: Manifold Support Bracket



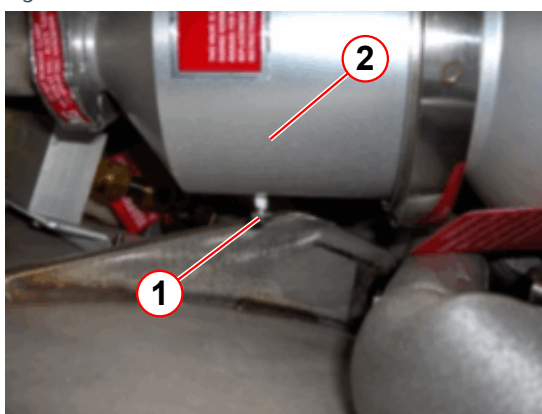
(1) M8 Nuts & Bolts

- Tighten the screw and M6 nut on the manifold support bracket.



At high altitudes the venting is only effective with the drain screw fitted in the quench valve.

Fig. 377: Drain Screw



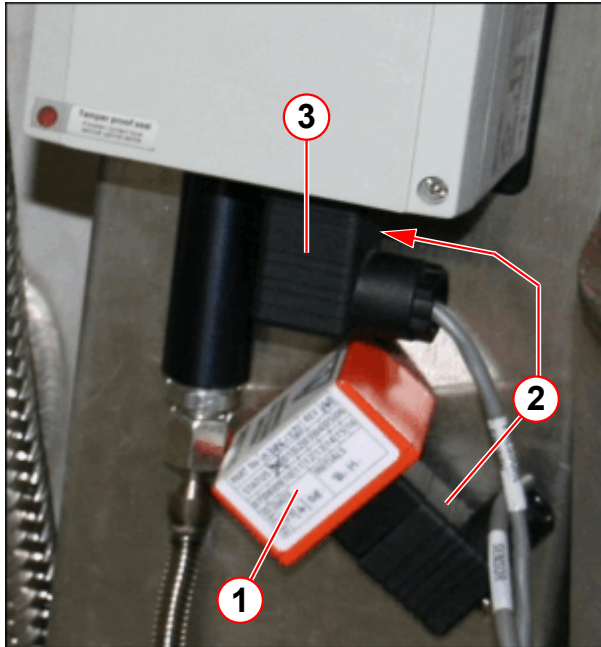
(1) Drain Screw
(2) Quench Valve

- Check that the drain screw is fitted in the quench valve body.

7.4.1.8 Final Work Steps

Reactivate the Pressure Heater

Fig. 378: Remove the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

Re-configure the system to enable the pressure heater depending on method used:

1. Switch ON the MSUP at circuit breaker F3 (fuse) in the ACC.
2. Remove the pressure service tool from the pressure sensor cable. Re-fit the connector to the pressure display unit.

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Set the pressure heater mode to **On 100%**.
- Select SAVE.
- Monitor the magnet pressure.
 - Select Update to refresh data until pressure reaches ≥ 15.3 psi (A).



Data is only accurate from the last refresh/update of the page, or from when the page was opened last.

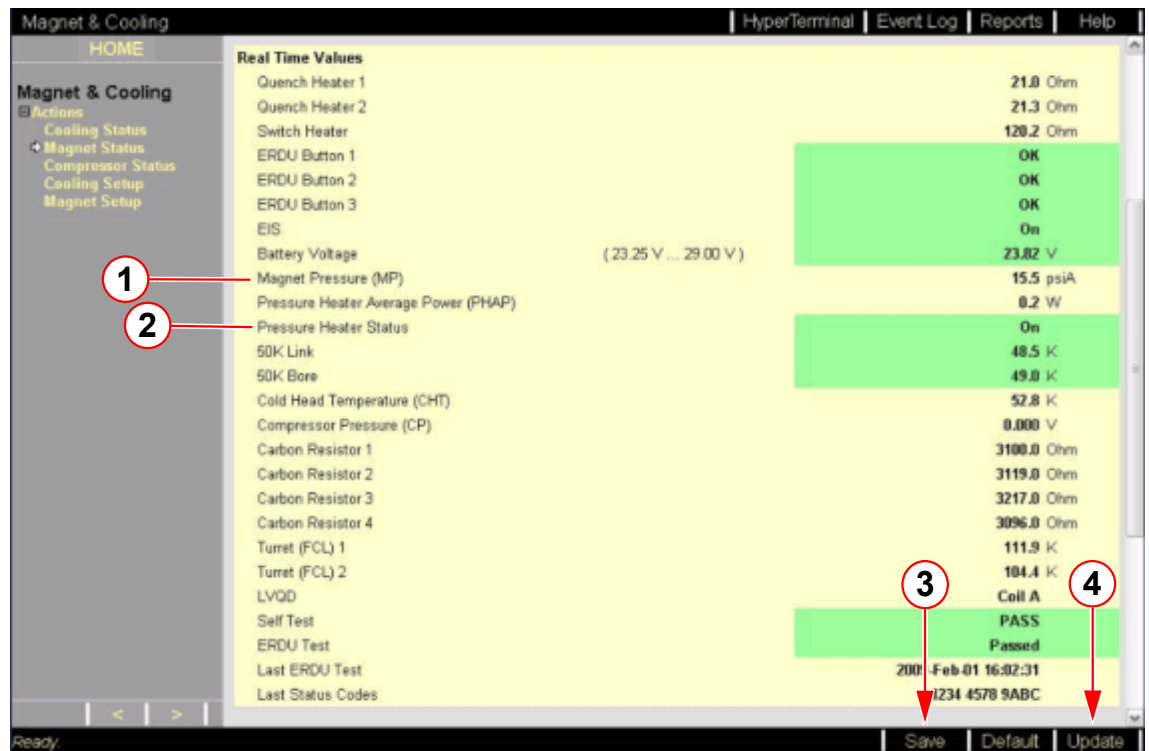


During pressurization, the cold head will switch on automatically.
If the cold head was turned off at MRC, select 'pressure heater status'.

- At MSUP Control - Set the pressure heater mode to Automatic mode.

- Select SAVE.

Fig. 379: Check magnet pressure level

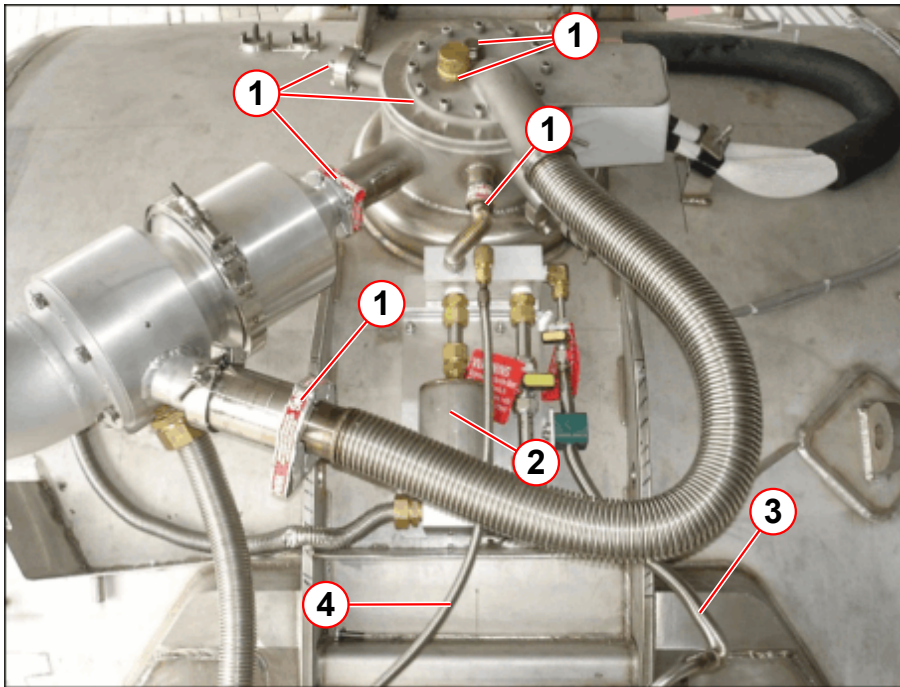


- (1) Magnet pressure (MP)
- (2) Pressure heater status
- (3) Save button
- (4) Update button

Final Checks

- Helium fill magnet to the required level for ramping the magnet.
- Order replacement components if the spare parts have been used (burst disk, seals).
- Perform a leak check of the turret venting and components.
 - » All vent pipes and flanges on the high pressure side of the helium vessel.

Fig. 380: OR105/122 - Turret and venting leak checks (Example Mk2)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line



Allow a minimum of 2 hours after filling before ramping the magnet.

Quench Report

- Fill out the Quench report - MR - Quality Database and Quench Report.

TSG - Checking the Carbon Resistors

The Ceramic Carbon Resistors (CCR) values can be checked from the MSUP or manually by using a multimeter.

- The measurements are taken across pins **a-e**, **b-e**, **c-e**, **d-e** on the 41-pin turret connector.



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » **The magnet must be RAMPED DOWN before attempting this procedure.**

Tooling requirements

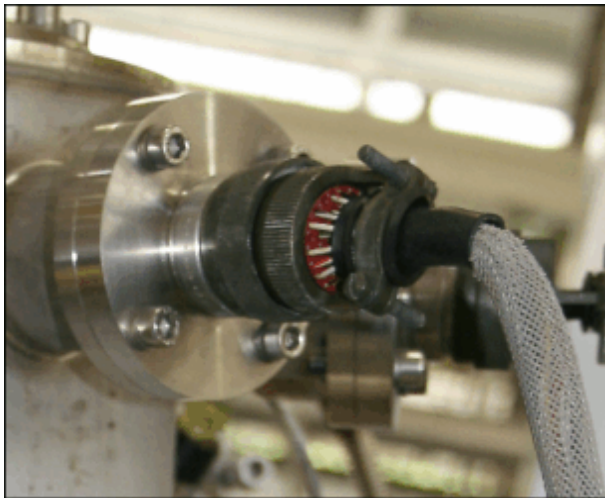
- Multimeter and pin leads.
- Laptop/PC with Terminal Emulation software, (PuTTY).
- RS232 cable.

Procedure

Using a multimeter:

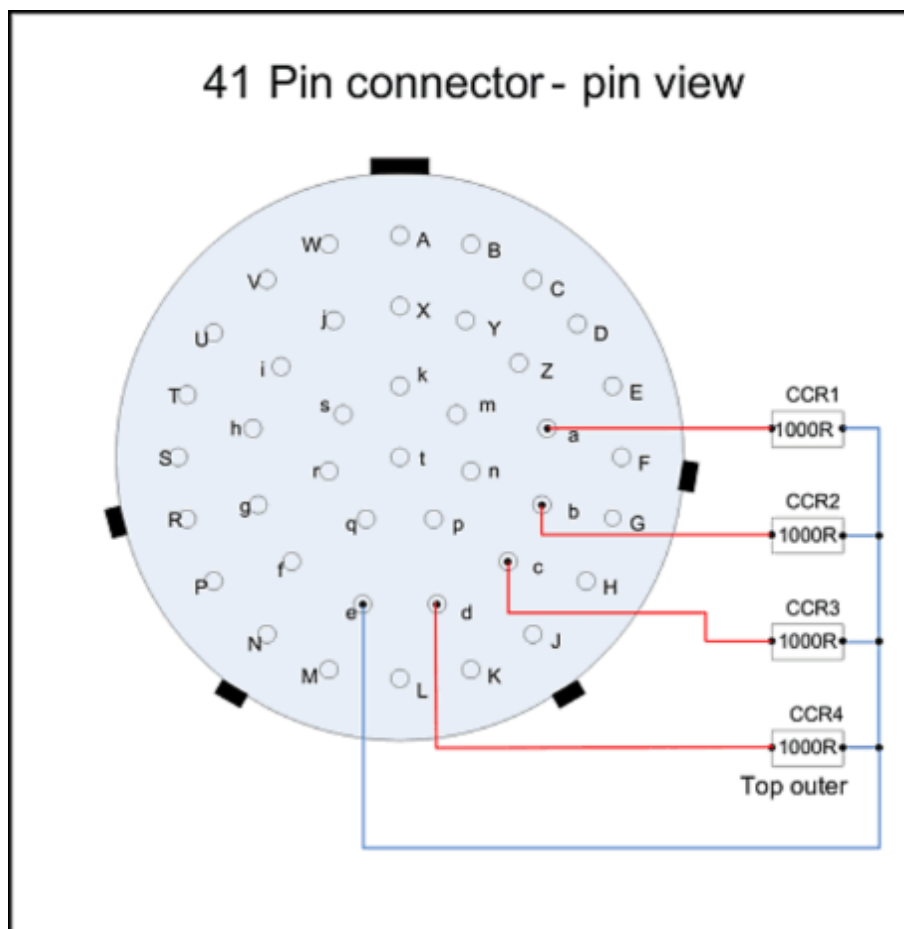
- Connect the pin leads.
- Turn the mode selector to resistor mode Ω .

Fig. 381: Turret - 41pin connector



- Disconnect the 41-pin connector.
- Measure across pins **a-e**, **b-e**, **c-e**, **d-e**
- Compare the values with the Magnet Acceptance Document.
- Re-connect the 41-pin connector.

Fig. 382: 41pin turret connector - ceramic carbon resistor pins (room temperature values)



8.1 Testing the ERDU circuit



Before initially ramping up the magnet, the ERDU test has to be performed.

8.1.1 Required tools

- Magnet Display Unit (MDU) (only for Avanto Dot Upgrade)
- ERDU Test Load (ETL) (only for Avanto Dot Upgrade)



Perform the ERDU circuit test as described in the SeSo/Install & Maintenance/ Safety-Related Tests.

- Document the ERDU test in the safety-related test protocol.

8.1.2 Affix the tamper-proof seal

After the ERDU test is completed, the tamper-proof seal has to be affixed to the alarm box and ERDU buttons.

Fig. 383: Alarm Box - Avanto Dot Ex-factory



(1) Tamper-proof seal

Fig. 384: Alarm box - Avanto Dot Upgrade



(1) Tamper-proof seal



The tamper-proof seal is delivered with the system.

8.2 Ramping the Magnet - Avanto Dot



Only Siemens CSEs trained in ramping magnets are allowed to perform the magnet ramping procedure.



Before ramping check that the quench line is correctly installed from the magnet to the outlet.



WARNING

Magnet safety checks

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Ramping is not allowed if the visually inspection of the quench line was not performed successfully.
- » Before initially ramping up the magnet, the 'Testing the ERDU circuit' has to be performed.
- » Procedure described in the safety related test can be used if ERDU test load is available.



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.
- » Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign has to be posted in a way that it is visible even when the door to the controlled area is opened.



CAUTION

Risk of asphyxiation and hypothermia, if helium gas is vented into the magnet room

Failure to observe the following safety warning may result in dizziness or loss of consciousness in personnel.

- » Quench valve must be connected to the Quench pipe.



Adhere to the general safety information included in this chapter when performing these steps.



Initial installation ramp helium level recommended to be >60 %.



Minimum helium level for ramping must be > 40%.
Wait at least two hours after helium filling.



Check that the venting bypass valve is locked securely in the CLOSED position.



Check for leaks before ramping the magnet (refer to Turret Leak Check).

Always

- Always restrict access to the magnet room when ramping the magnet. Use flashing lights and signs.
- Always inspect all equipment and tools for defects prior to use.
- Always inspect the Anderson ramp lead connections at the filter plate.
- Clean the area of the magnet room, dispose off empty cartons or other packing materials
- Always inspect magnet room for the presence of non identified objects. Before ramping, the nature of those objects must be clarified.



Prepare a minimum of 4 pictures of the room **prior to first ramping**.

These pictures must be colored and should have 2 to 4 megapixel resolution. They must show the complete room, all 4 corners and walls. In case there are un-removable cupboards installed, their doors must be open to prove that they are empty.

Later those pictures are to be attached to the installation report.

Never

- Never use defective or damaged equipment.
- Never allow unauthorized people to stand too close to the magnet during ramping.
- Never block emergency exits.
- Never smoke or use open flames in the magnet room.
- Never ramp a magnet when you are unsure about magnetic parts inside the magnet room. Those parts could be hidden in e.g. supposed empty boxes or in closed cupboards or sliding of a desk.

Ramping a Mobile System



It is necessary to bump the magnet to account for the effect of the iron shielding.
Our recommendation for demand current on first ramping is to increase the magnet current by 2 A.

8.2.1 Tools required



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

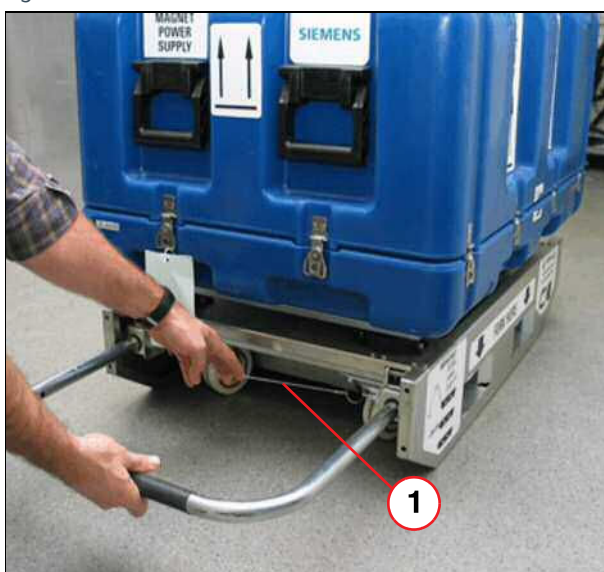
» Only non-magnetic equipment may be taken into the magnet room.

- Magnet Power Supply (3600 MPSU 3600); (order number: **8395613**). This includes:-
 - 3600 MPSU Power lead
 - CAN Open lead
 - RS232 lead
- Heat gun.
- Laptop or host computer.

8.2.2 Preparation

- Move the 3600 MPSU into the equipment room close to the RF filter plate.

Fig. 385: MPSU



(1) Wire locking strap

- Lower the trolley handle into the horizontal position. This will raise the wheels and allow the trolley to rest on the two skids.
- Pull the wire strap to unlock the handle, and slide the handle into the recess under the trolley.
- Remove the shipping case lid by unclipping the 12 spring clips.

Fig. 386: MPSU



(1) MPSU LID

- Open the hinged lid of the 3600 MPSU by pulling the two silver knobs and turning them through a 1/4 turn.
- Raise the lid.

8.2.3 Installation

Fig. 387: 3600 MPSU Power connections



- (1) X1 (MPSU power)
- (2) CB1 (Mains circuit breaker)

- Ensure the mains circuit breaker **CB1** is in the **OFF** position on the 3600 MPSU.

Fig. 388: Adapter plate (view from Equipment Room)



- Unscrew the securing screws and remove the **adapter plate** from the **RF filter plate**.
- This will allow access to the magnet ramp leads in the **magnet** room.

i

Do NOT connect the magnet ramp leads to the 3600 MPSU until AFTER the transport test has been completed. See (→ Transport Test / Page 328).

i

The mandatory transport test was introduced from 3600 MPSU software revision level 17.
This test is only required the FIRST TIME time the 3600 MPSU is used on site.

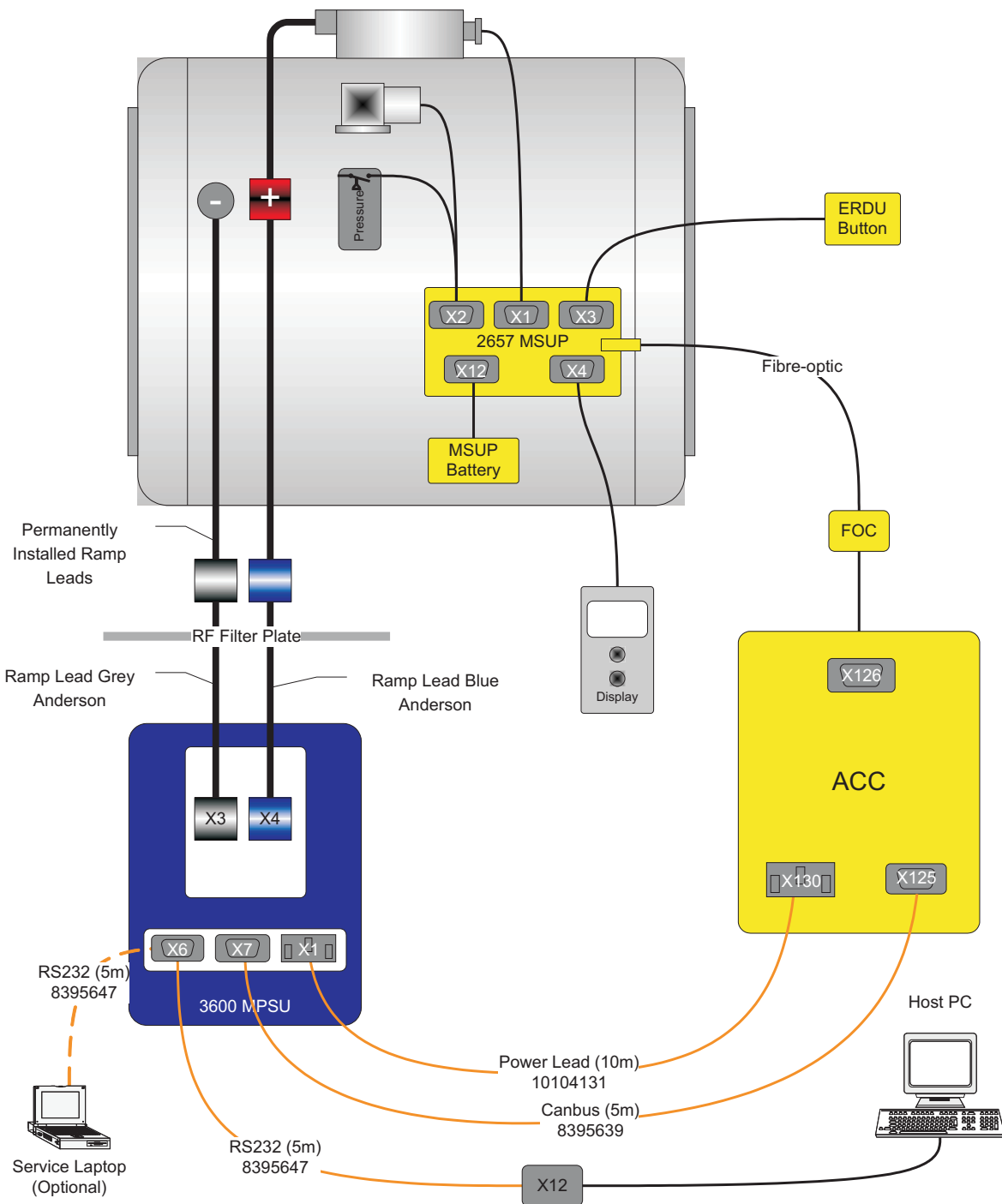
Lead connections are detailed in the table below:-

Tab. 20 3600 MPSU lead connections

Lead	From	To
Magnet Ramp leads	MPSU X3 MPSU X4	Filter plate (grey) Filter plate (blue)
CAN OPEN lead	MPSU X7	ACC X125
RS232 lead	MPSU X6	RS232 lead X12 on ACC (open-end lead from Ser_Distribution Port 6 (Ser_MPSU) or P3)
MPSU Power lead	MPSU X1	ACC X130

Fig. 389: Avanto - 3600 Ramp Configuration

Avanto - 3600 Ramp Configuration



The configuration with the MSUP Display unit is valid for Avanto, Avanto Dot Upgrade.

Fig. 390: 3600 MPSU Connector and indicator panel



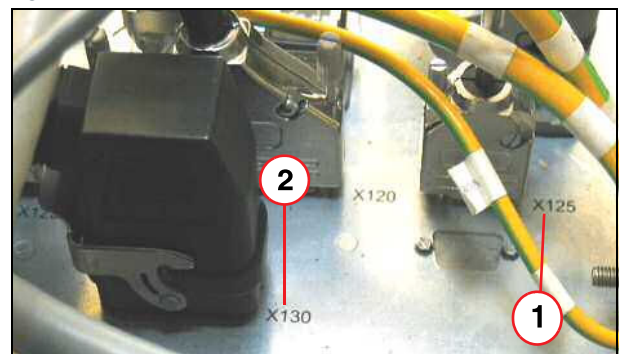
- (1) ON indicator
- (2) FAULT indicator
- (3) SELF TEST indicator
- (4) CONTACTOR OPEN indicator
- (5) CURRENT IN LEADS indicator
- (6) SWITCH HEATER ON indicator
- (7) X7 CAN OPEN connector
- (8) X6 RS232 connector

Fig. 391: 3600 MPSU leads



- (1) CAN OPEN lead
- (2) RS232 lead
- (3) MPSU power lead

Fig. 392: ACC



- (1) X125
- (2) X130

- Connect the **CAN OPEN**, and **RS232** leads. (refer to (→ Tab. 20 Page 322)).
- Connect 3600 MPSU **power lead** to **X1** (→ 1/ Fig. 387 Page 321)
- Turn ON **F28** in the **ACC**.

8.2.4 Set up of the 3600 MPSU

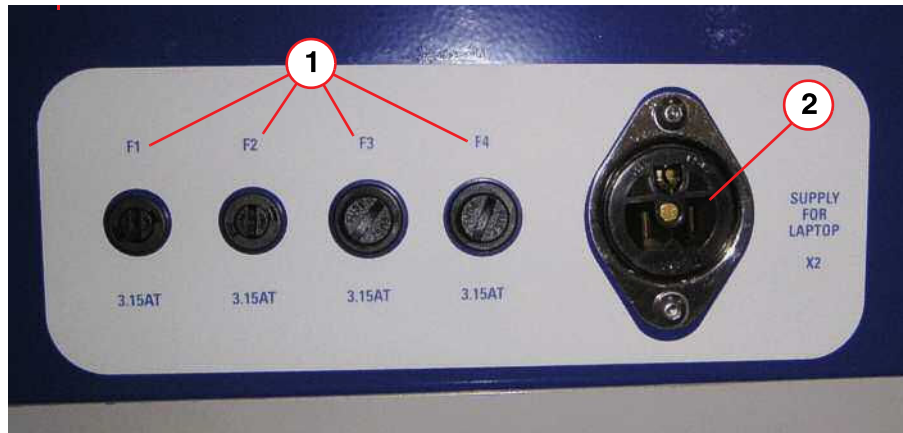
8.2.4.1 Connecting Host or Service-PC

To start ramping, the host or Service laptop is connected to **X6** of the 3600 MPSU (RS 232 interface) (→ 8/ Fig. 390 Page 324).

To establish communication, **Terminal emulation software** is required.

Power to a Service laptop can be supplied at **X2**.

Fig. 393: 3600 MPSU Fuses and laptop power supply

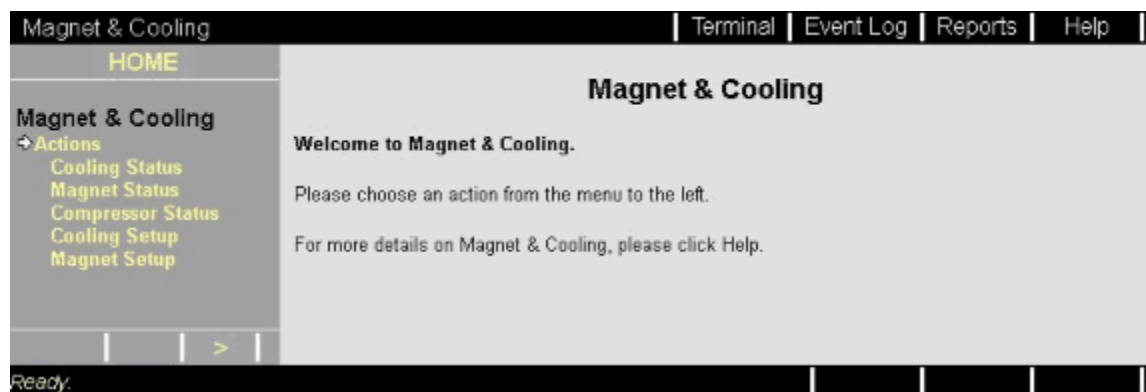


- (1) Fuses F1 - F4
- (2) X2 Supply for Laptop computer

8.2.4.2 Configuring the Host

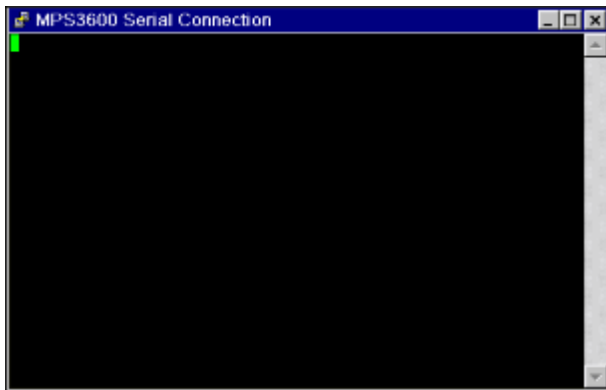
- Open the **Magnet & Cooling** service platform.
- Select **Terminal** from the **toolbar**.

Fig. 394: Open the Terminal via Magnet&Cooling



- **Terminal** option: the program will **open**. No configuration is required.

Fig. 395: MPS3600 Serial Connection



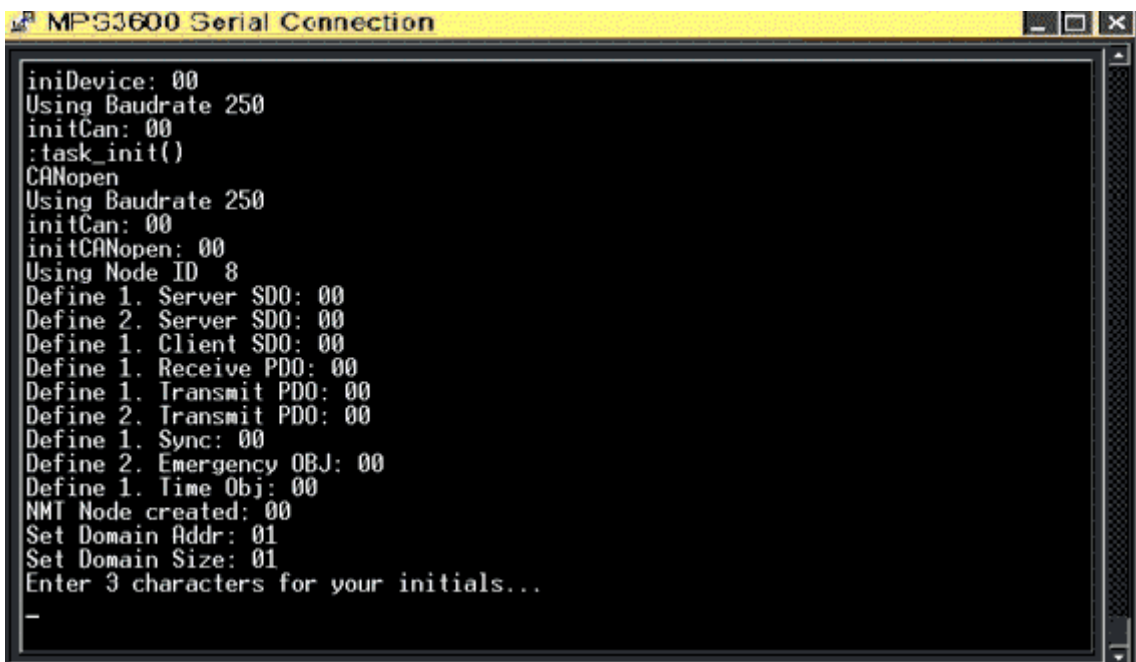
8.2.4.3 Initializing the 3600 MPSU



Ensure that there is a valid helium level reading before switching on the 3600 MPSU.

- Turn **ON** the 3600 MPSU using mains circuit breaker **CB1** on the 3600 MPSU. (→ 2/ Fig. 387 Page 321)
 - The green '**ON**' LED will light on the 3600 MPSU. (→ 1/ Fig. 390 Page 324)
- The **boot-up** screen will be displayed.

Fig. 396: Boot-up screen



If the screen does not appear, make sure that the correct Com port and settings have been selected. If they have, turn the 3600 MPSU off and then back on, using the mains circuit breaker CB1. (→ 1/ Fig. 387 Page 321)



Reboot the scanner with the 3600 MPSU switched on.

A Reboot is necessary to update the loadware and to download the ramp logfile to the HOST.

The logfile is saved to C/MedCom/log directory.

Always reboot the scanner after the 3600 MPSU is switched off and on again. Rename the logfile, e.g. "MPSRampLog1.dat" after the ramp is finished.

If not renamed, the file will be overwritten with the next ramp logfile.

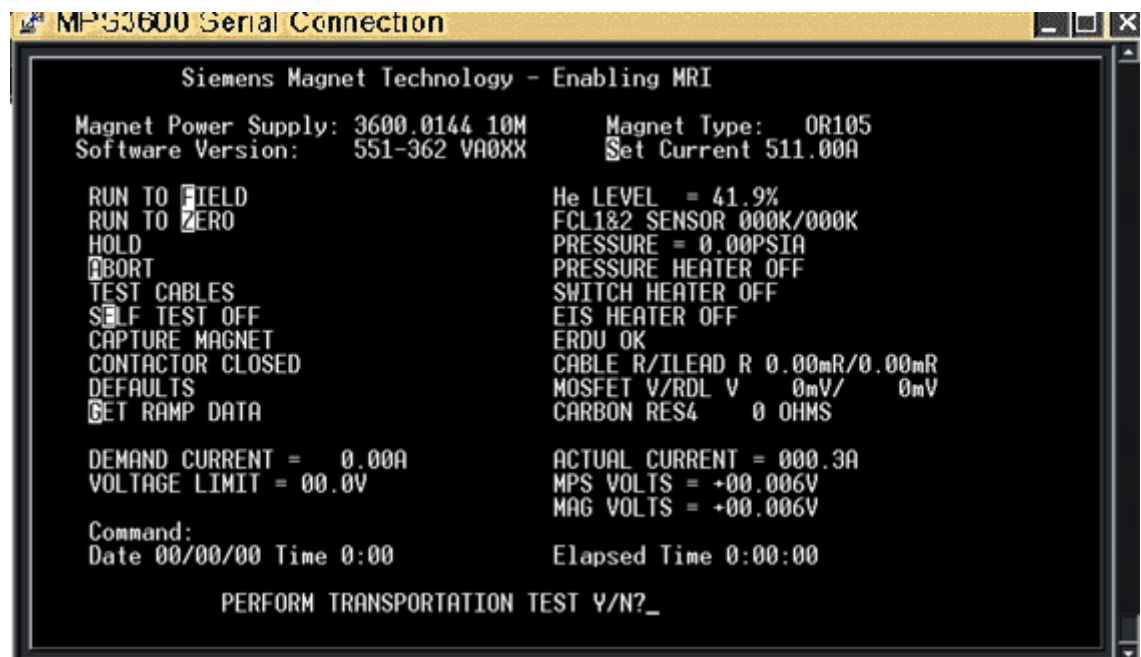
Repeat booting of the scanner if following warning message still persists: "MPS not operational"

- When prompted, enter your **initials**.

The 3600 MPSU will now retrieve the **magnet type, magnet serial number, last current set, He- level, date and time** from the MSUP.

- The following screen will prompt to '**Perform Transportation Test**':-

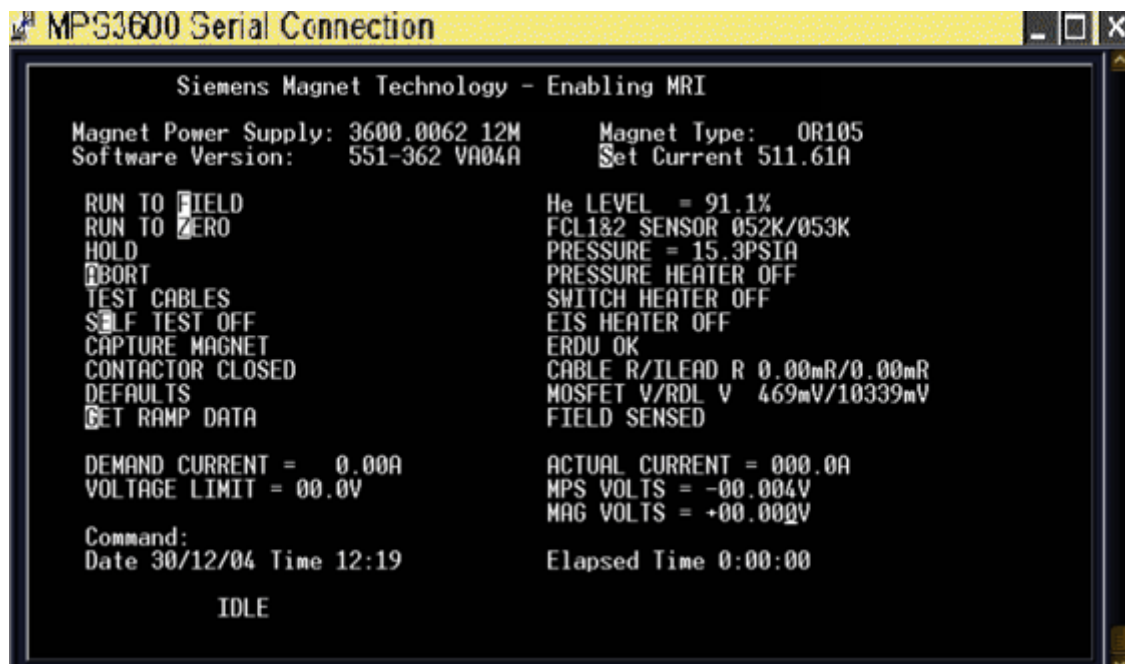
Fig. 397: Transport Test (example of Avanto screenshot)



- If the **Transport Test** is required, continue with (→ Transport Test / Page 328).
- If the test is **not** required:
 - » Enter **N<ENTER>**
 - » At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 390 Page 324)

- When the self-test is complete, the **Operational screen** is displayed.

Fig. 398: Operational Screen (example of Avanto screenshot)



- Turn off the MPSU 3600 via CB1.
- Continue with (→ Ramp lead connection / Page 331).

8.2.5 Transport Test

During transport, it is possible that the 3600 MPSU may have suffered internal damage due to rough handling.



The mandatory transport test was introduced from 3600 MPSU software revision level 17.

For units prior to this revision, continue with (→ Ramp lead connection / Page 331)

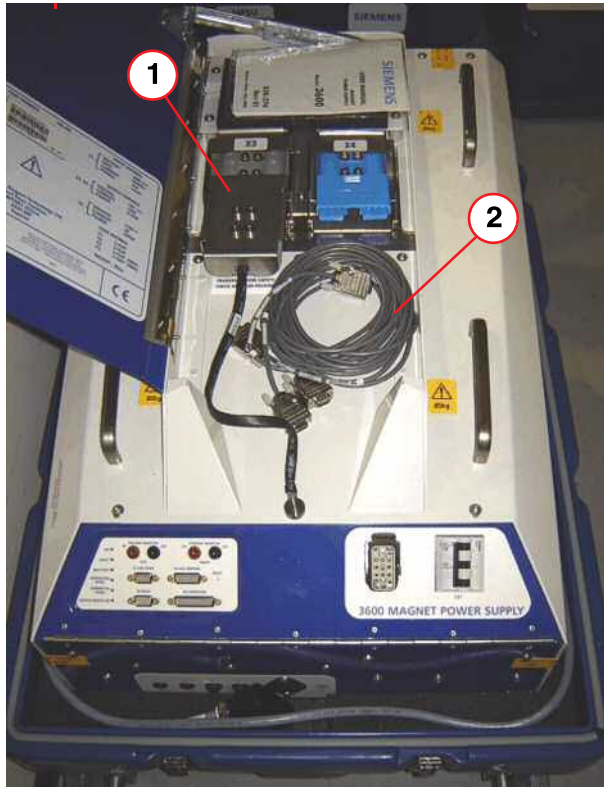
This test only needs to be completed the first time the 3600 MPSU is used on site.



This test must be performed PRIOR to connecting the ramp leads and using the 3600 MPSU on the magnet.

8.2.5.1 Transport Test Tool

Fig. 399: Transport test tool



- (1) Transport test plug (fitted)
- (2) RS232 cable

The **transport test tool** consists of 2 x 470 ohm resistors housed in a grey **Anderson connector**.

The center tap of the resistors is connected to a flying lead which is permanently bolted to the 3600 chassis.

The 3600 MPSU is supplied with the test plug already connected.

8.2.5.2 Transport Test Procedure

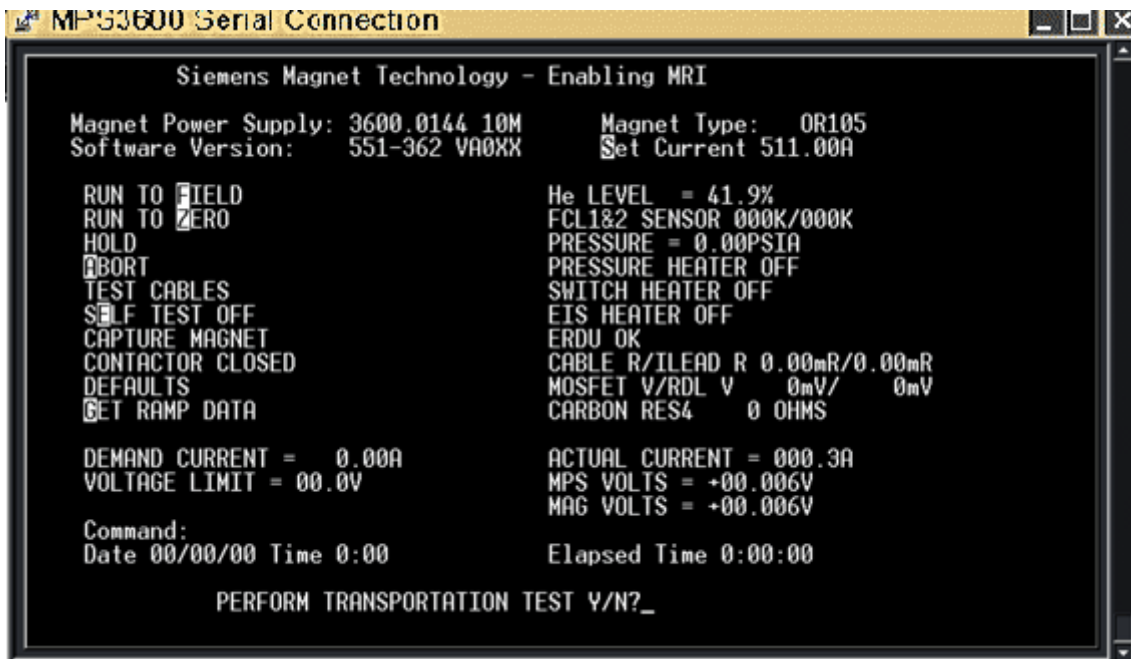
Check that the **transport test plug** is connected to **X3** as shown in (→ Fig. 399 Page 329) above.



This test must be passed before the test plug is removed and the ramp leads connected.

- Perform (→ Initializing the 3600 MPSU / Page 326).

Fig. 400: Transport Test (example of Avanto screenshot)



- The screen will then prompt to 'Perform Transportation Test'. Type Y <Enter>.

Failure of the Transport test will be displayed briefly in the event of the test detecting an internal fault (→ Fig. 400 Page 330).

This will be followed by a request to check the connections and repeat the test (→ Fig. 402 Page 331).

Fig. 401: Transport Test failed (example Avanto screenshot)

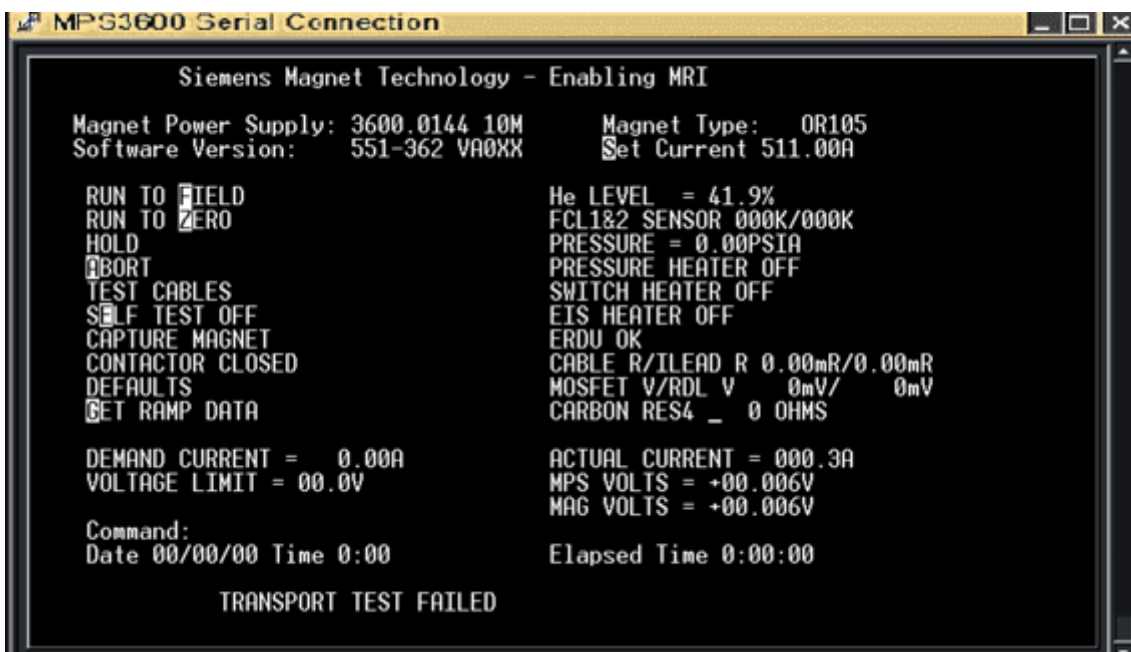
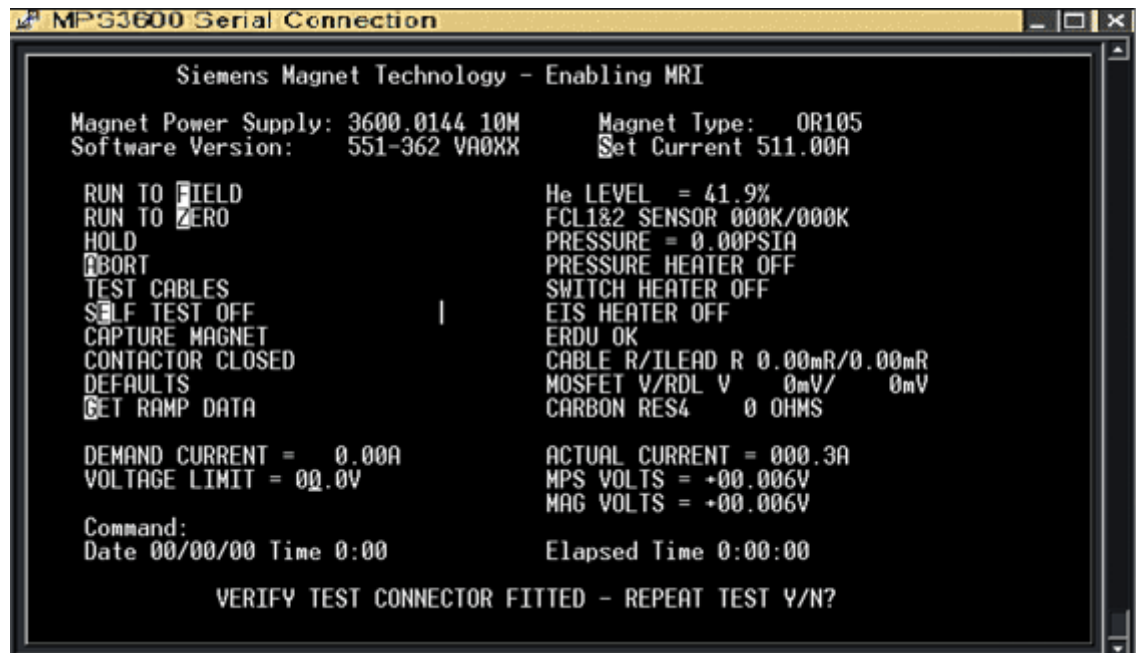


Fig. 402: Verify Test Connector (example Avanto screenshot)



After 3 successive Transport tests have FAILED, further operation of the 3600 MPSU is prevented.

Line power to the 3600 MPSU should be switched OFF then ON to enable further tests to be performed.



If all attempts fail, the 3600 MPSU should be returned for repair, and a replacement unit obtained.

If the test is **successful**, the power supply will perform an **internal self-test**.

- When the self test is completed turn off the 3600 MPSU via CB1.

8.2.6 Ramp lead connection

After the Transport test has been successfully completed, (if applicable), the following connections must be made **prior to ramping the magnet**.

- Remove the **transport test plug**.

Fig. 403: MPSU - ramp cables fitted



- Connect the two **magnet extension ramp leads** to Anderson connectors **X3** and **X4** on the top of the 3600 MPSU.

**CAUTION**

Incorrect locking of the ramp leads

If these instructions are not observed, the magnet and the 3600 MPSU can be damaged.

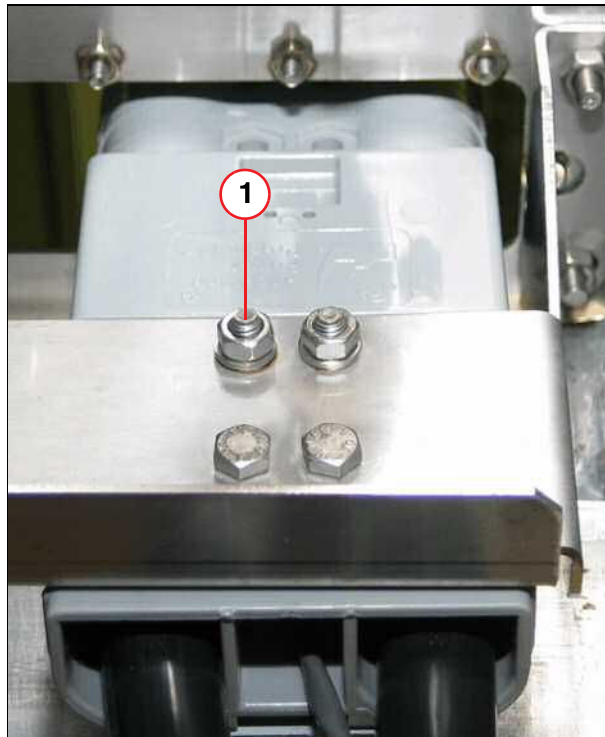
- » Connect and secure the leads.

Fig. 404: RF filter plate

**Equipment room**

- Connect the **extension leads** to the **magnet leads** at the **RF filter plate**.

Fig. 405: Power lead connection

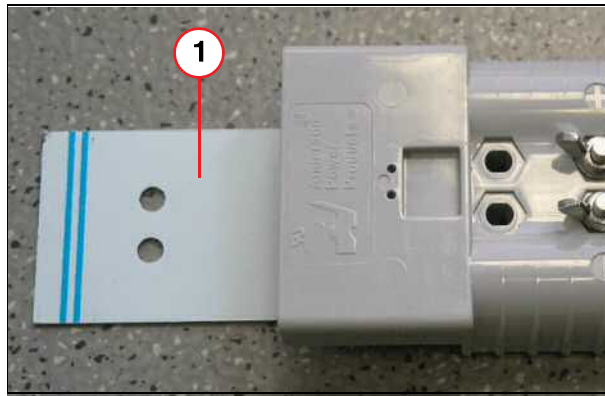


(1) Fastening bolts

Magnet room

- Prevent the **extension cables** from being accidentally **disconnected**.
- **Secure the leads** by tightening the **fastening bolts**.

Fig. 406: Connector



(1) Fastening plate

- From the equipment room, feed the 3600 MPSU leads and fastening plate through the RF filter plate.



Ramp cables are heavy - to prevent stress on the connection, use tie wraps to support the weight if locking bolts are missing.

8.2.7 Run To Field

8.2.7.1 Co-Siting of Magnets

Co-siting describes the ramping requirements where two MR systems with the same operating frequency are sited next to each other in adjacent rooms.



If two systems are installed next to each other in adjacent rooms:

- One magnet must be ramped to the upper frequency limit.
- The other magnet must be ramped to approx. 100 kHz less at the lower frequency limit.



Shimming is only optimized with both magnets ramped up.

Neither magnet is ramping, while the other is running applications. If one magnet is switched off, the other magnet has to be re-shimmed.

8.2.7.2 Run To Field Prerequisites:

- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).

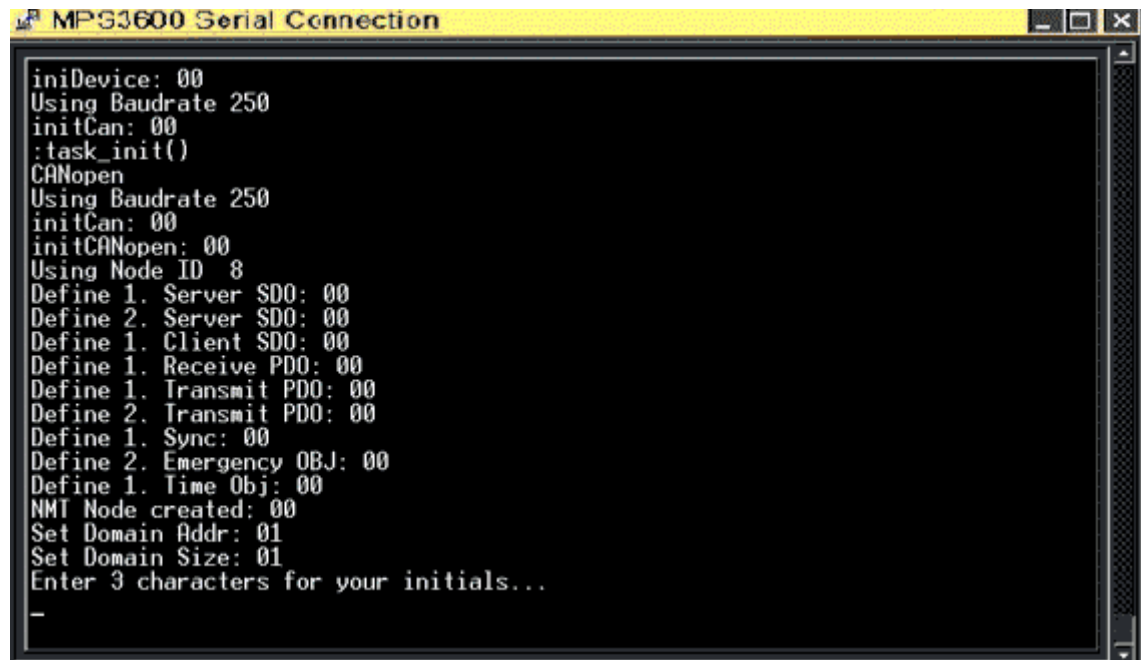


Following a helium level fill, wait at least 2 hours before ramping.

- Helium level must be **> 40%**.
- Turn on the 3600 MPSU via CB1.

- The **boot-up** screen will be displayed.

Fig. 407: Boot-up screen

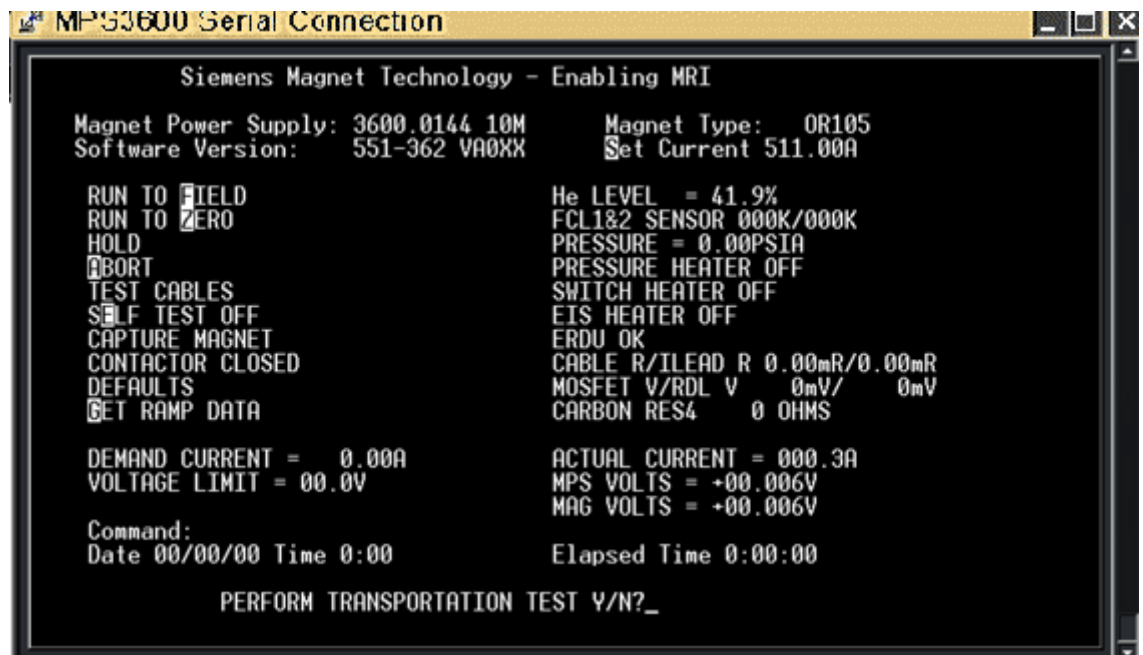


- When prompted, enter your **initials**.

The 3600 MPSU will now retrieve the **magnet type**, **magnet serial number**, **last current set**, **He- level**, **date** and **time** from the MSUP.

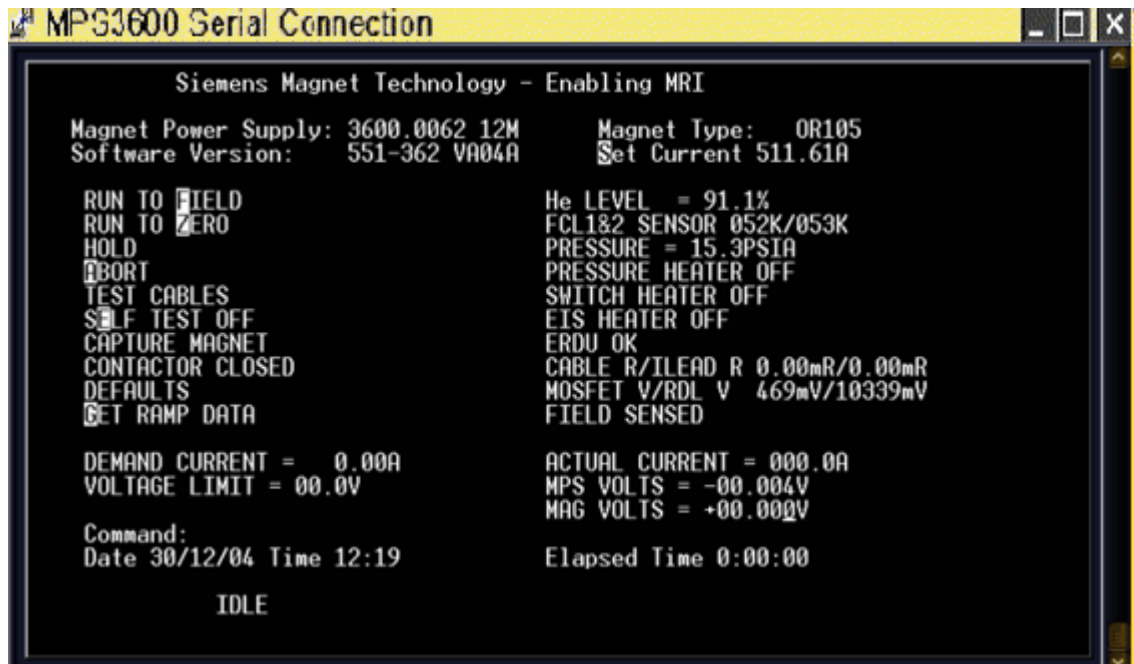
- The following screen will prompt to '**Perform Transportation Test**':-

Fig. 408: Transport Test (example of Avanto screenshot)



- The **Transport Test** is not required at this point.
 - » Enter **N<ENTER>**
 - At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 390 Page 324)
- When the self-test is complete, the **Operational screen** is displayed.

Fig. 409: Operational Screen (example of Avanto screenshot)



8.2.7.3 Setting the magnet current

- Enter the command: **S<space>514.05<ENTER>**
 - The numerical number may be varied as required.
- Enter the command for Run to field: **F <ENTER>**

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to field.

8.2.8 Run To Zero

8.2.8.1 Run to zero prerequisites:

- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).

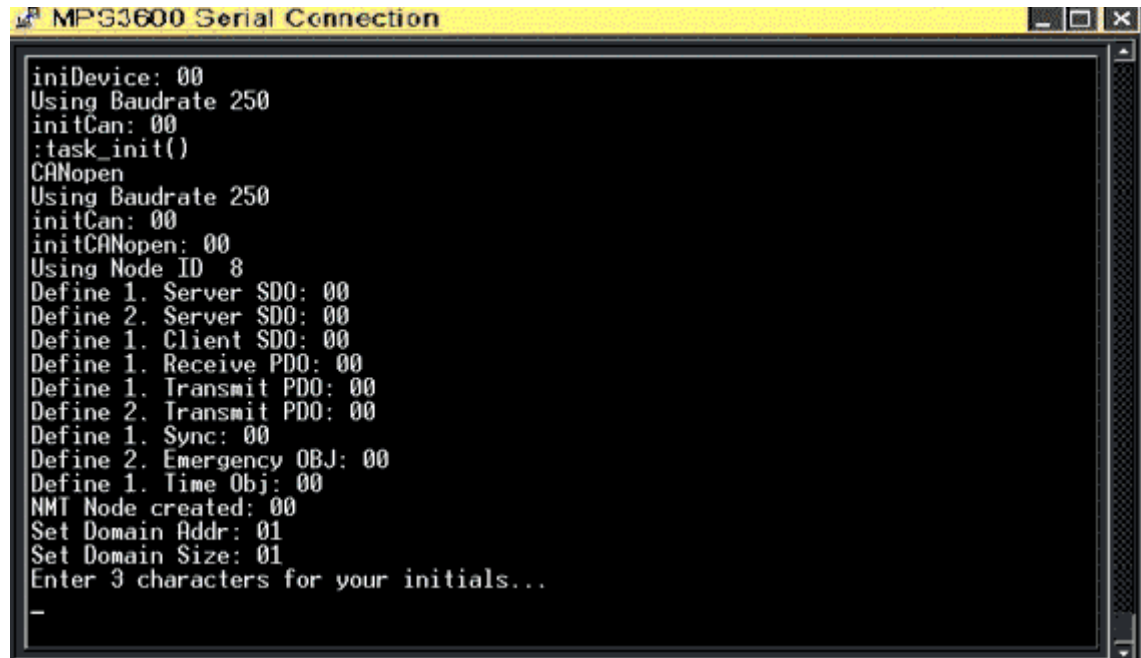


Following a helium level fill, wait at least 2 hours before ramping.

8.2.8.2 Run To Zero

- Turn on the 3600 MPSU via CB1.
- The **boot-up** screen will be displayed.

Fig. 410: Boot-up screen

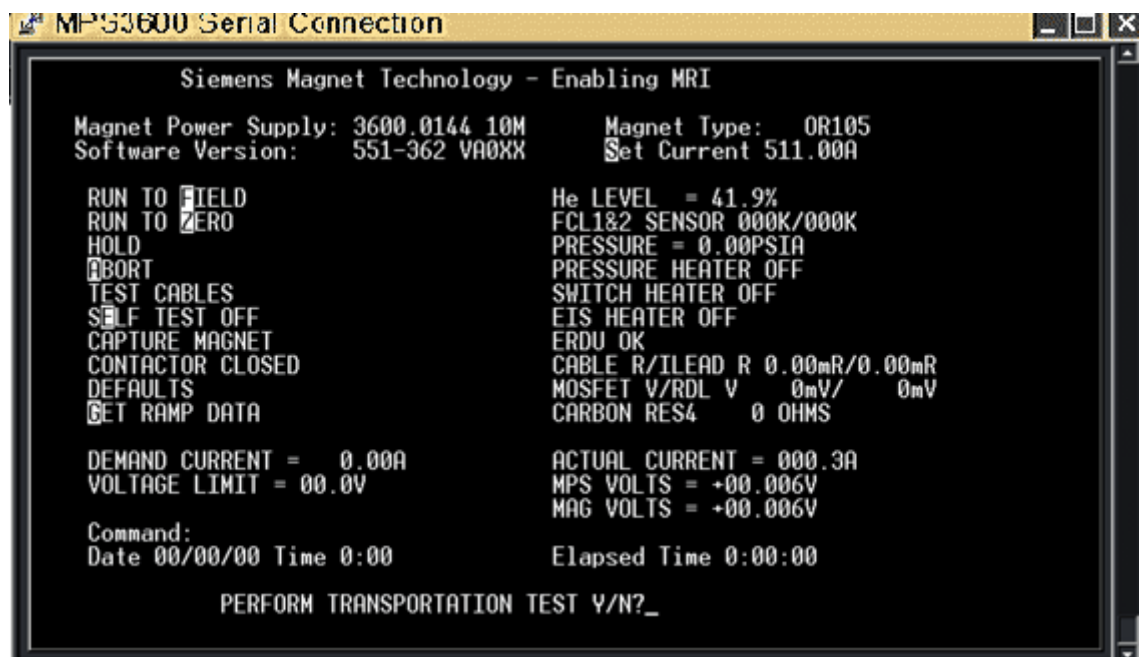


- When prompted, enter your **initials**.

The 3600 MPSU will now retrieve the **magnet type**, **magnet serial number**, **last current set**, **He- level**, **date** and **time** from the MSUP.

- The following screen will prompt to '**Perform Transportation Test**':-

Fig. 411: Transport Test (example of Avanto screenshot)



- The **Transport Test** is not required at this point.
 - » Enter **N<ENTER>**
 - At the prompt 'ARE YOU SURE', enter **Y<ENTER>**
- The 3600 MPSU will now perform an **internal self-test**.
 - » The **self-test** LED indicator will light during the test. (→ 3/ Fig. 390 Page 324)
- When the self-test is complete, the **Operational screen** is displayed.

Fig. 412: Operational Screen (example of Avanto screenshot)

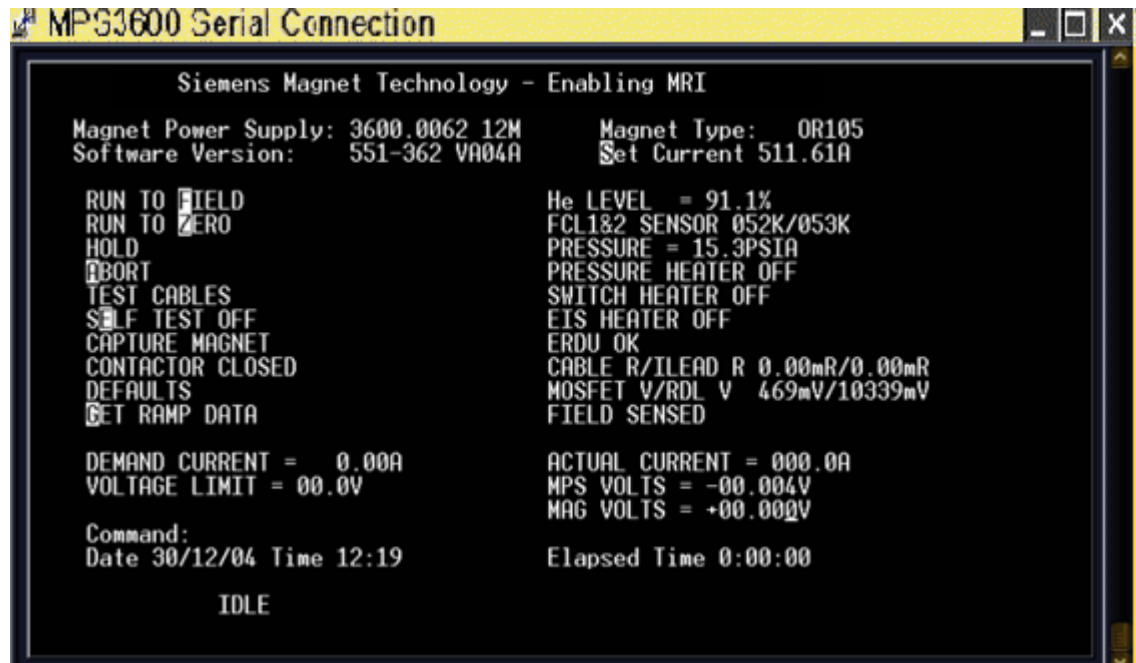
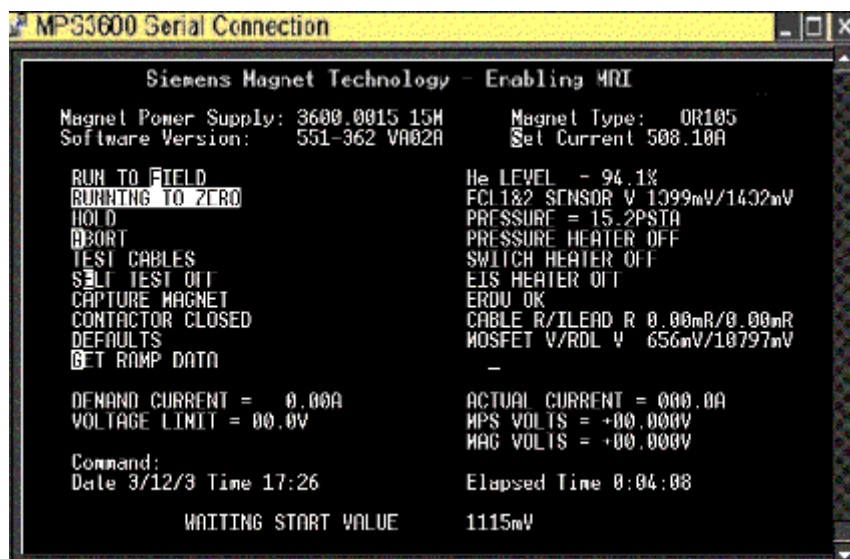


Fig. 413: Ramp to zero



- Enter the command **Z<ENTER>** for ramping down the magnet.

Checks will be carried out and the 3600 MPSU will then perform an automatic ramp to zero.

8.2.9 Ramping the magnet to the required field strength (bumping the magnet)

The magnet should be ramped to the upper limit of the frequency limit. Bumping the magnet involves making a small change to the magnetic field strength either up or down. The procedure is the same as ramping the magnet.

8.2.9.1 Magnet current setting

Refer to the **Magnet Acceptance Document/MSUP** to obtain the operating **current value**.



Avanto range of frequency:

Maximum frequency: 63,700 000 MHz + 0 kHz

Minimum frequency: 63,500 000 MHz

Use the mathematical formula below to calculate the current to determine the **centre frequency**.

Between ramps, the magnet system's frequency will slowly reduce over a period of time. To increase the period between ramps, aim to achieve for the **upper sector** of the frequency window.

$$\text{Magnetom Avanto: } I_{\text{final}} = (63,680 \text{ MHz} \times I_{\text{actual}}) / \text{Freq actual}$$

8.2.9.2 Bumping prerequisites:

- 3600 MPSU should be connected.
- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including lead checks).



Following a helium level fill, wait at least 2 hours before ramping.

- Helium level must be > **40%**.

The last ramp current is stored in the magnet supervisory (MSUP). The ramp current will be displayed on-screen when the 3600 MPSU is switched on.

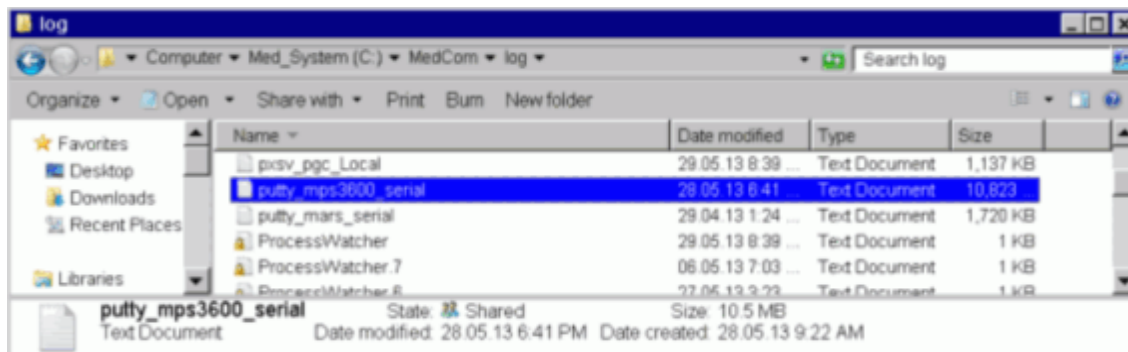
8.2.10 Work after ramping the magnet

8.2.10.1 Downloading the MPSU log file



The MPSU log file is saved under MedCom/log.

Fig. 414: MPS_logfile



- Navigate to the location C:\MedCom\log\putty_MPS3600_serial.log, zip this file, and send the log file to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens-healthineers.com>).

8.2.10.2 Switching off the 3600 MPSU



Verify that the yellow “CURRENT IN LEADS” LED is OFF when the 3600 MPSU is switched on.



If the yellow LED is flashing and the power LED is on, press reset on the front of the 3600 MPSU and run the magnet to zero or persist at field before proceeding further.



NEVER attempt to remove the magnet ramp leads when the “CURRENT IN LEADS” LED is flashing.

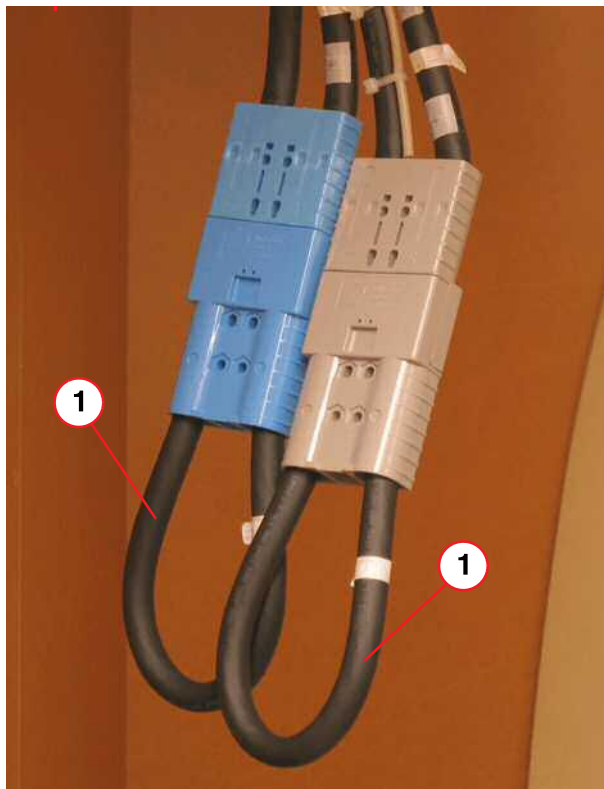
- Set the mains circuit breaker **CB1** to the **OFF** position. (→ 2/ Fig. 387 Page 321)
- Set the circuit breaker **F28** in the **ACC** to the **OFF** position.

When shorting links are fitted:-



DO NOT disconnect **BOTH** extension leads at the same time.

Fig. 415: Magnet Shorting links.



(1) Shorting links

Shorting links

- Disconnect one extension lead, and fit the shorting link to the magnet lead.
- Fit the shorting link to the magnet lead.
- Repeat the process for the other shorting link.

- Pull the extension leads through the RF filter panel into the equipment room.
- Store the magnet leads on the lead tray in the magnet room.
- When shorting links are not fitted:-

When shorting links **are not** fitted:-

- Disconnect the extension leads from the magnet leads at the RF filter plate.
- Re-fit the the adapter plate using the securing screws.

Fig. 416: RF filter plate



Fig. 417: Adapter plate (view from Equipment Room)



8.2.11 Final work steps

8.2.11.1 Save the magnet current to the HOST

- Access the service menu in the **SeSo at MRC**.

- Select menu item **"Magnet & Cooling"**.
- Select **"Magnet Setup"** in the **"Action"** menu

Fig. 418: Magnet Setup - Dot Upgrade

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup
 - Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.1	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.1	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Warning	Min.	55	%
Pressure Heater Resistance		20.5	Ohm
Shield Temperature Warning 50K Link	Max.	74.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	0.0	Ohm
	Nitrogen Temperature	0.0	Ohm
	Helium Temperature	0.0	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1054	
	Revision Number	0	
Magnet	Serial Number	21176154	
	Revision Number	0	
Magnet Current		466.64	A

MSUP Data

Downloads MSUP data from MSUP to host Download 1

Uploads MSUP data from host to MSUP Upload

MSUP Logfiles

Downloads all MSUP logfiles to the host Download 2

☐ Save values as default

Ready: | Save | Default | Update

- (1) Download of MSUP data
- (2) Download of MSUP files

- Click **"Download"** to save the magnet-current value on the HOST.

8.2.11.2 Warm Turret & Venting

Fig. 419: Warm the turret and venting with a heat gun



Ensure the **turret and venting** remains **leak tight** and the valves continue to function correctly.

- Warm up the **turret and venting** with a heat gun.

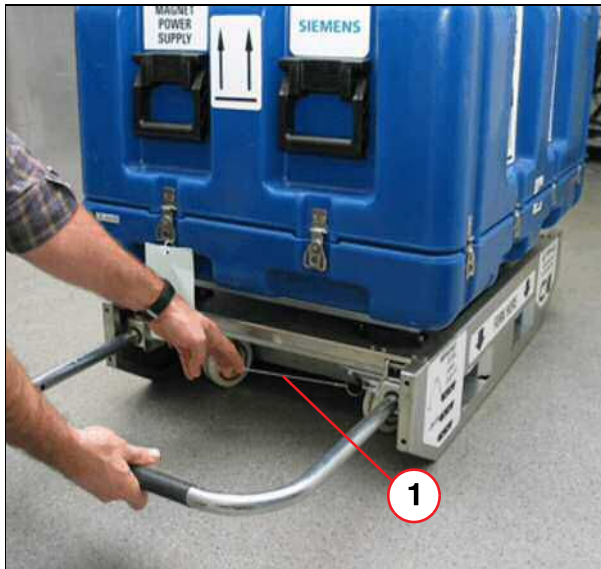
8.2.11.3 Pack the 3600 MPSU

Fig. 420: MPSU - ramp cables fitted



- Disconnect the two **3600 extension ramp leads** from the Anderson connectors **X3** and **X4**.
- Re-fit the **Transport Test Plug** into **X3**.
- Disconnect the remaining leads and pack inside the transport case.
- Close the lid and secure by pulling the two silver knobs and turning them through a 1/4 turn.

Fig. 421: MPSU



(1) Wire locking strap

- Replace the shipping case lid and secure using the 12 spring clips.
- Pull the wire strap to unlock the handle, and slide the handle out of the recess under the trolley.
- Release the wire strap to lock the handle in the extracted position.
- Raise the handle to lower the wheels and lift the trolley off of the two skids.
- The 3600 MPSU and trolley can now be safely moved.

8.3 Failure during ramping



CAUTION

If the supply to the MPSU fails while the magnet is ramping up or down, the magnet current will continue to flow.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Never remove the ramp cables from the magnet while current is in the magnet ramp cables.

Current is in the ramp cable whenever:

- » The yellow "CURRENT IN LEADS" LED is flashing.



Failure of the MPSU may prevent this LED from lighting up. So, in the event of a failure (e.g. mains failure), it is essential to measure the voltage across the terminals on the turret.

Any voltage on these terminals other than zero is an indication that there is current in the ramp cables.

- » Magnet was ramping up or down and there was a mains failure.
- » Magnet was ramping up or down and the MPSU detected a fault such as over temperature, which would shut down the MPSU.
- » Normal ramp up or down of the magnet.



CAUTION

If the magnet is suspected of ramping down through the ramp cables, owing to one of the reasons above.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Do not remove the ramp cables. The magnet must be run to zero or persist at field before removing the ramp cables.



If the cause of the shutdown was failure of the mains, nothing can be done until both the MPSU and the Magnet Supervisory (MSUP) are working again.

When the MPSU and the MSUP are operational, ramp the magnet to zero.

8.4 What to do after a Quench (Avanto Dot ex-factory, Espree with MSUP K2306)

8.4.1 General

What to do after a Quench (Avanto Dot Ex-factory, Espree with MSUP K2306)

If the magnet quenches, a large volume of helium gas is generated within the helium vessel. The quench valve opens, and stay open until the flow of gas subsides. The quench valve re-seals, preventing air (caused by cryo-pumping) from being drawn back into the magnet turret and helium vessel.

The MSUP should automatically maintain the system pressure above atmospheric.



Magnet HSC must be notified of a quench (→ Siemens Healthineers Magnet WS CS HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

Fill out the **Quench Report** - (→ MR Product Information, <https://productinformation.healthcare.siemens.com/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (Quench Report)

NOTICE

The magnet helium level can be recovered if the magnet temperature is between LN2 and He temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms. Compare to the values on the Magnet Acceptance Document, or with HSC.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 2000 liters of liquid helium.

» Contact Magnet HSC IMMEDIATELY before proceeding.



CAUTION

If the magnet is allowed to warm up above 77K, a full nitrogen pre-cool will be required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

A full nitrogen pre-cool takes about 10 days.

» Contact Magnet HSC IMMEDIATELY before proceeding.

8.4.1.1 Safety



Post quench procedures must be performed by CSE's that are trained to the MR2MAG-NET level. (or System level trained).



Switch off and isolate the magnet power supply from the mains before further action.

**WARNING**

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenics.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.

**CAUTION**

RISK OF ICING!!

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk and O-rings IMMEDIATELY.
- » Do not leave turret ports open to atmosphere.

**CAUTION**

Cold helium gas.

Personal injury may occur.

- » Protective clothing (long sleeves), cryogenic gloves and eye protection must be worn.

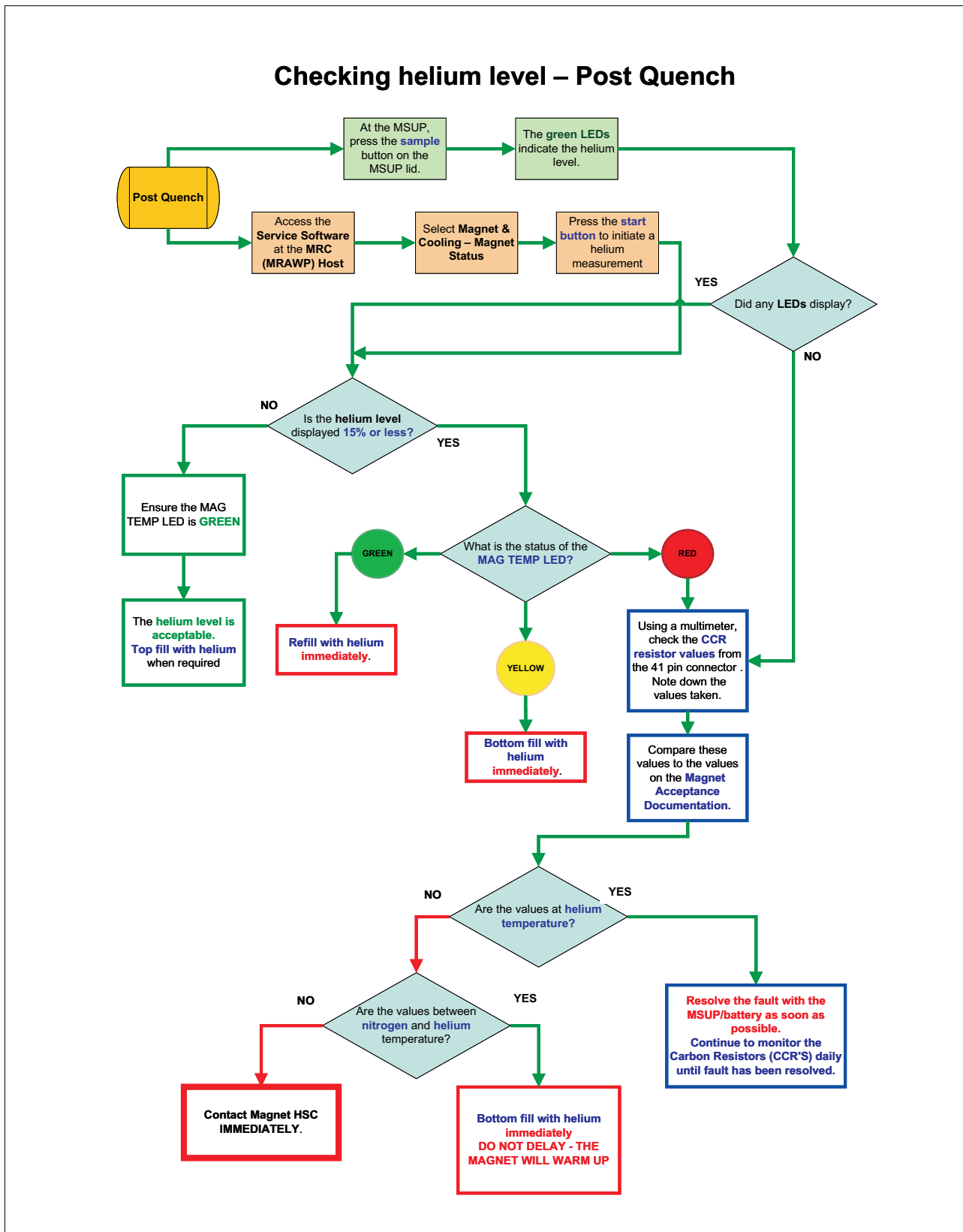
8.4.2 Post Quench Steps

Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 1	- Isolate the MPSU from the magnet. - Download the 3600 MPSU ramp log files.	Stop filling.	- Check that no ERDU buttons are pressed. - Re-set ERDU button.
Step 2	- Check the liquid helium level. - Check the ceramic carbon resistor temperatures at the MRC Host. (→ Checking the Carbon Resistors (CCR) Values / Page 352)		
Step 3	Order 1000 liters of liquid helium.¹		
Step 4	- Warm up and leak check the quench valve. (→ Checking the Quench Valve Burst Disk / Page 357) - Warm up and leak check the auxiliary vent burst disk.(→ Auxiliary Burst Disk / Page 364)		
Step 5	After 1 hour : - Download the Magnet Supervisory log files. (→ Log File Capture / Page 351) - Helium fill. Fill to recommended helium ramp level.		
Step 6	Leak check the quench valve and venting. (→ Final Checks / Page 368)		
Step 7	Fill out the Quench report.		
Step 8	- Email the (→ Magnet HSC, mailto:magnet.wscshsc.team@siemens-healthineers.com) with: » 3600 Power Supply logs. » Magnet Supervisory logs and screen capture. » Magnet serial number.		
Step 9	Ramp the magnet.		
Step 10	- Perform the shim check. » If the shim check exceeds the specifications, a re-shim is necessary.		

1. The quantity of the order depends on the helium level remaining in the system after a quench, and volume already on site.

Fig. 422: Check helium level - Post Quench



8.4.3 Pre-requisites

8.4.3.1 Tools Required

The post quench tools required:

- Heat gun and paper toweling.
- Service tool kit.
- Laptop and RS232 cable.
- O₂ monitor.
- Personal protection equipment. To include cryogenic gloves, safety shoes, glasses and a long sleeved protective clothing.
- Helium leak detector.

Fig. 423: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

8.4.3.2 Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal.

- A number of seals are supplied in the spares kit. These must be replaced when used on the magnet.
- Venting components will require ordering from CSML.

Mk2 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 10121913)
- High load seal (material number 10100171).
- O-ring seal 44.5mm ID (material number 10100151).



Always replace the burst disks and seals used during replacement procedures. They are critical components for reducing the risk of air ingress. A replacement must always be available for immediate use.

8.4.3.3 Unattended Quench

Unattended quench indicators are:

- If the Magnet STOP (ERDU) has been activated:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the helium level is low:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
- If the magnet pressure is very low, the
 - » cold head switches off.
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.

- Imaging is not possible if there is no magnetic field.

Press the AUDIO ALARM OFF button on the alarm box to silence any audible alarms.

8.4.3.4 Log File Capture

Magnet Supervisory

- The **quench data is lost**, if:
 - a) The system is re-booted within 1 hour after a quench
 - b) The system is turned off within 1 hour after a quench
 - c) The data is downloaded within 1 hour of after a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the Magnet Power Supply log file.



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data will be overwritten!

- Switch off and disconnect the MPSU from the magnet.

If the magnet quenches during helium fill:

- Stop the transfer.



Do not remove or disconnect the syphon as the magnet will be at high pressure.



For all types of quench, allow up to 1 hour for the quench pressure to subside.

Use the MSUP Display unit:

- To check that the MSUP pressure control system is working. If a display unit is not available, use the SESO interface on the Host service screens.
- If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. The compressor switches off.

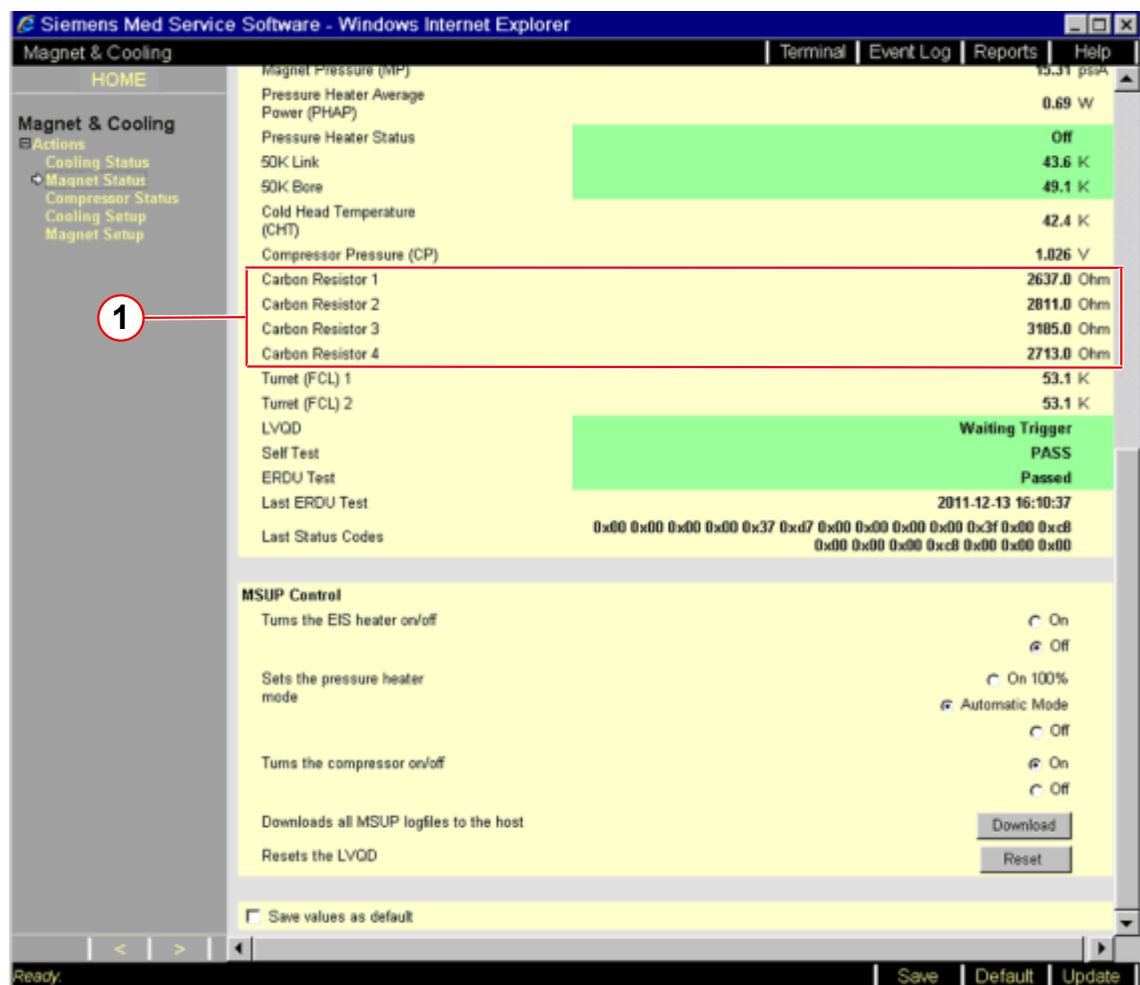
8.4.3.5 Checking the Carbon Resistors (CCR) Values



Compare the temperatures with the values on the Magnet Acceptance Document.

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.

Fig. 424: Checking the Ceramic Carbon Resistors



(1) Resistor values

Tab. 21 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380

If the host (or laptop), is not available, check CCR values directly from the pins (a-e) on the 41 pin connector on the turret.



The magnet helium level can be recovered if the temperature of the CCR is still colder than nitrogen temperature.

The CCR (CARBON RES 1) reading must be in the range of between 1200 and 3400 ohms.



If temperatures are between N2 and helium temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

Fig. 425: OR105 Magnet Acceptance Document

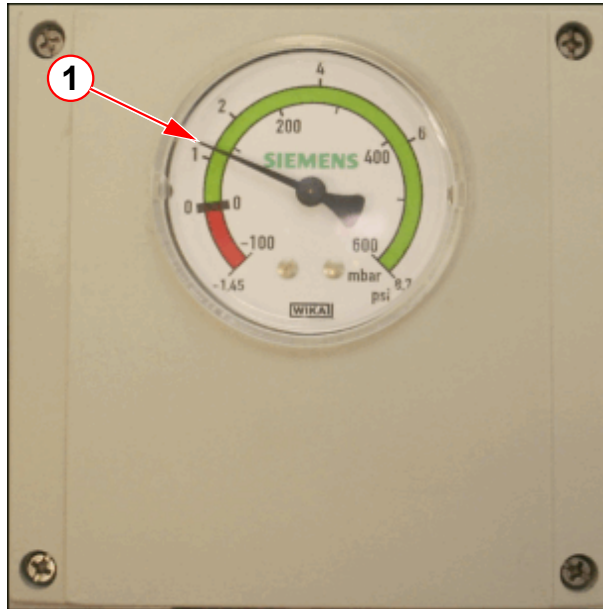
SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).			
 This system requires 0 Z2 rings and 2.92 kg of iron to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms	Calibration values	
Liquid nitrogen temperature (77K)	1309 Ohms	1009 Ohms	
Liquid helium temperature (4.2K)	2994 Ohms	1309 Ohms	
		2994 Ohms	
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	Signed <u>ANOTHER</u>		
V Turner			
	Date 23/12/2011		
	Page 1 of 1		
	25-AD EPS 000 01		

8.4.3.6 Check the Magnet Pressure

Check that the helium vessel has a positive pressure:

- Magnet pressure should read 15.3 psi (A) on the MSUP display unit.
- Check for positive pressure on the pressure gauge.

Fig. 426: WIKA pressure switch - expected pressure



(1) Expected pressure 1.3 psi (G)

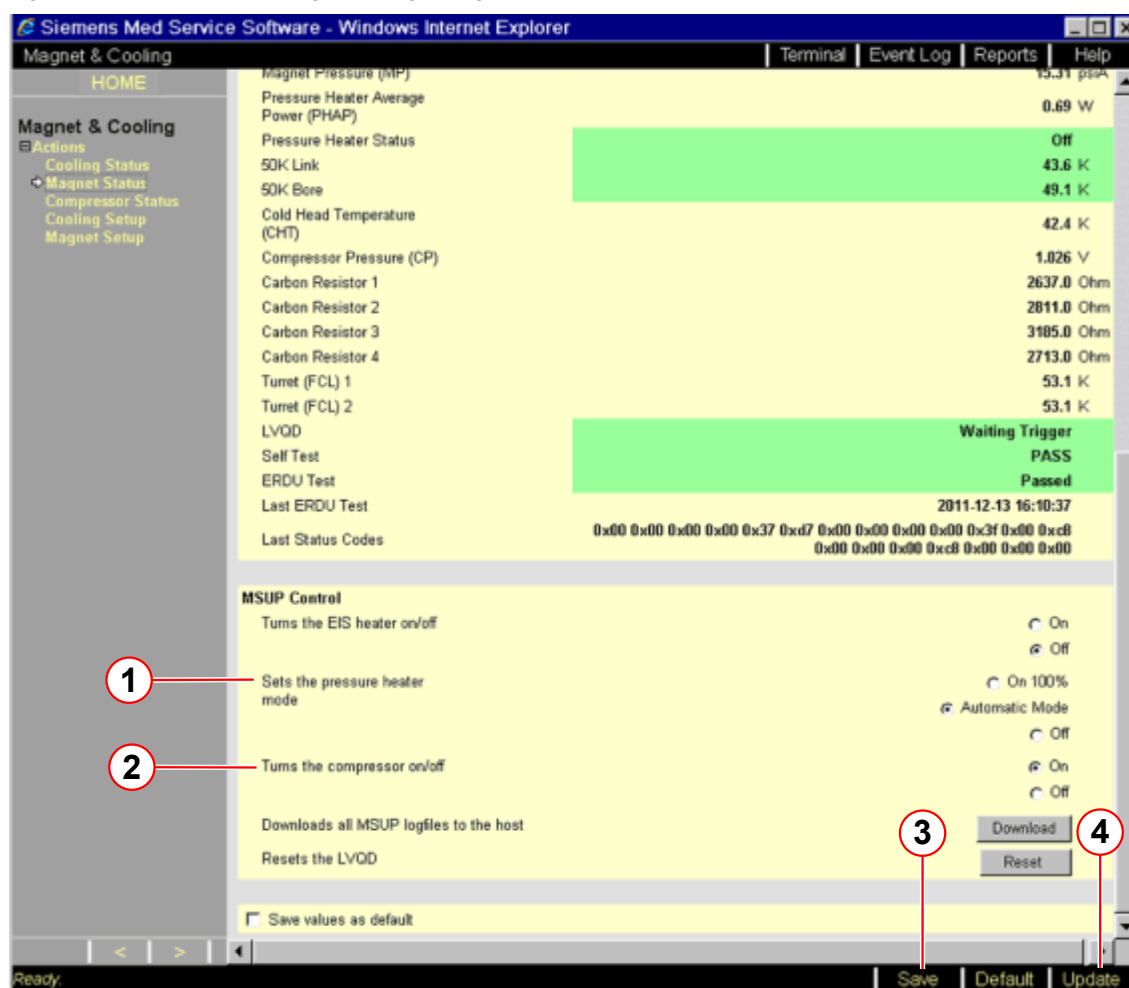
- If the pressure gauge is within the **RED zone**, the magnet is sub-atmospheric.

» **DO NOT** open the magnet bypass valve, or disturb any high pressure fitting.

8.4.3.7 Deactivate the Pressure Heater

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Sets the pressure heater mode select the OFF button.
- Select SAVE.

Fig. 427: Service screen Mag&Cooling - Magnet Status - MSUP Control



- (1) Sets the pressure heater mode
- (2) Sets the compressor on/off
- (3) Save
- (4) Update



The pressure heater will reactivate within 60 min.

8.4.3.8 Depressurize the Helium Vessel



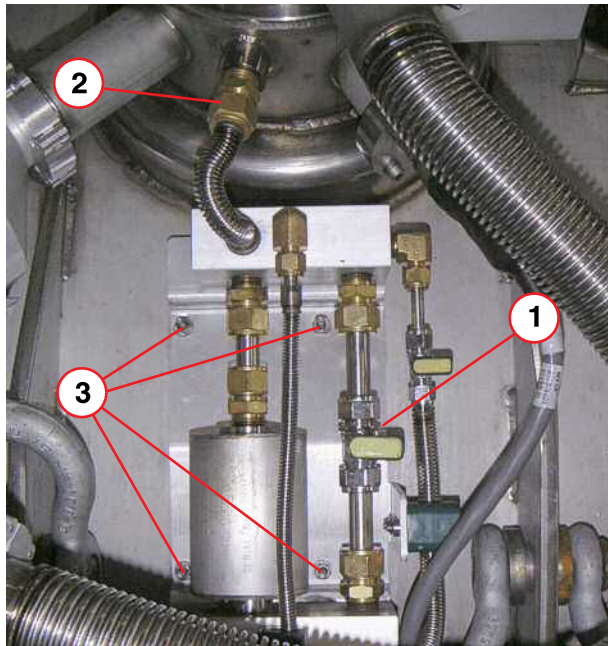
CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressurization of the helium vessel. The bypass valve is CLOSED during normal operation.

- » To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.

Fig. 428: Mk2 Vent valve assembly

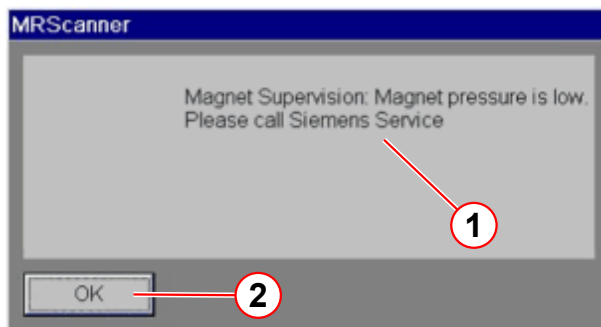


- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

If depressurization of the helium vessel is required:

- Fully open the magnet bypass valve to depressurize the helium vessel.
- Wait until zero psi is displayed on the pressure gauge.
- The pressure switch will activate.
 - » the cold head will turn off.
- Close the magnet bypass valve to prevent air ingress.
- Warm up the vent valve assembly.
 - » Take care not to damage any components when using a heat gun.

Fig. 429: MSUP - Magnet pressure is low



- (1) Pop-up message
- (2) Click OK to clear

On the MR Scanner:

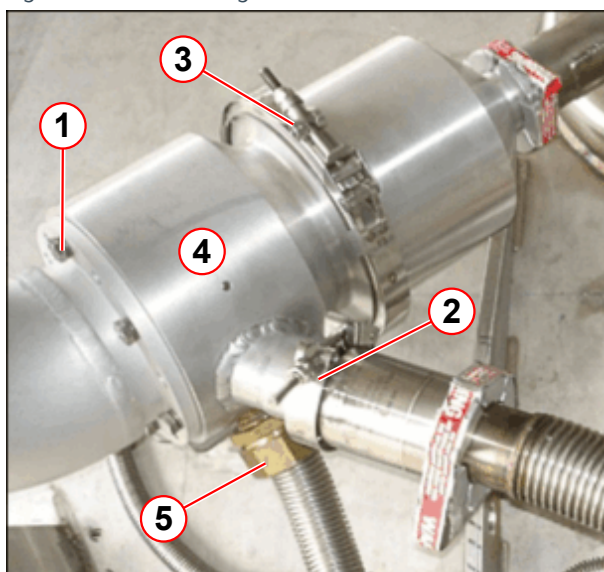
- A pop-up alarm message is displayed.
 - » Magnet pressure is low.
 - » Click OK to clear the alarm.
- The alarm box will sound when pressure falls to alarm limits.
 - » The yellow WARNING LED lights on the alarm box.
 - » The alarm buzzer is activated
 - » Press AUDIO ALARM OFF to silence the alarm.

8.4.4 Checking the Quench Valve Burst Disk

Post quench if there is no magnet pressure a burst disk has ruptured. Check the graphite (quench valve) burst disk.

- Disable the pressure heater by fitting the pressure service tool.
- Warm up the quench valve and venting to room temperature, using a heat gun.
- The elbow assembly and manifold must be removed for inspection and leak check of the graphite burst disk and O-ring.

Fig. 430: Mk2 Venting removal



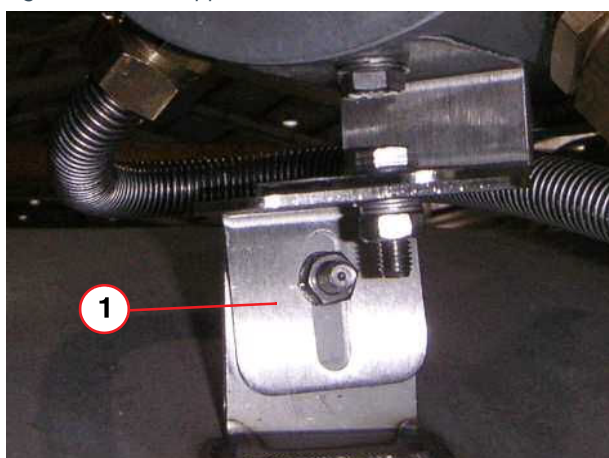
- (1) Elbow connection (M8 x 6)
- (2) V-clamp - Auxiliary vent
- (3) V-clamp manifold to quench valve
- (4) Manifold assembly
- (5) OVC vent connection

To access the quench valve and burst disk:

- Remove the 6 x M8 screws retaining the elbow assembly to the manifold (pos.1).
- Remove the V-clamp retaining the auxiliary vent to the manifold (pos. 2).
- Remove the V-clamp retaining the manifold to the quench valve (pos. 3).

NOTE: it should not be necessary to remove the OVC vent line connection (pos.5).

Fig. 431: Mk2 Support bracket

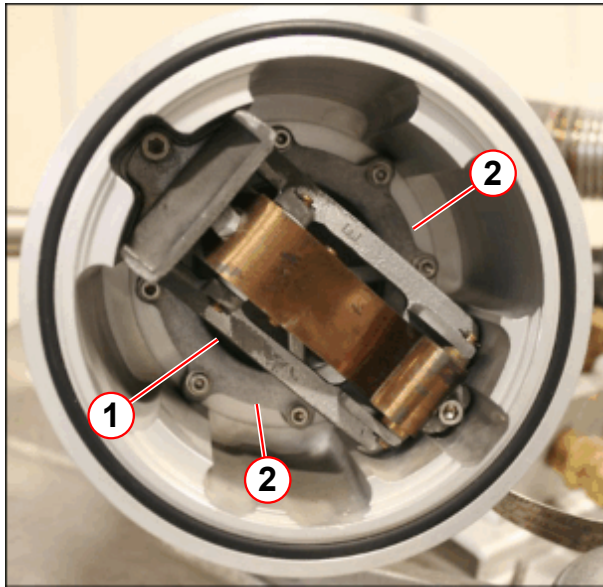


- (1) Support bracket

- Loosen the support bracket (1) screw underneath the quench valve interface assembly.
- Move the manifold assembly away from the quench valve.

The quench valve and graphite burst disk can now be accessed to check for leaks.

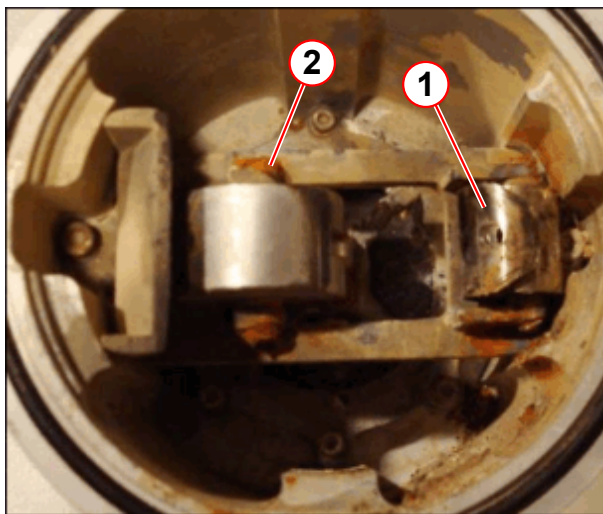
Fig. 432: Leak check



- (1) Graphite burst disk
- (2) Burst disk O-ring seal

- Check the graphite burst disk for breakage or cracks.
- If the graphite burst disk is broken or cracked, it must be replaced **immediately**.

Fig. 433: Water corrosion (example)



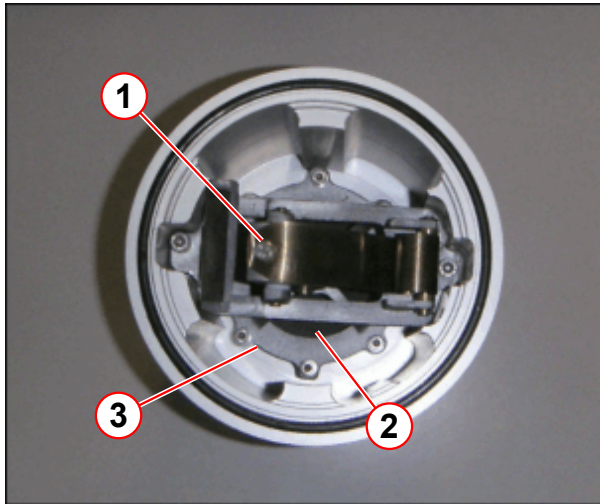
- (1) Spring broken
- (2) Corroded screws

Check for ice and water corrosion or breakage of:

- The spring plate assembly.
- All screws.
- Corroded quench valve components must be replaced.

- With no damage to the burst disk or spring plate visible, continue to leak check around the quench valve O-ring and retaining plate.
- Check that the magnet pressure is a minimum of 0.5 psi (G) for leak checking.
- Leak check around the burst disk and spring plate assembly.

Fig. 434: Quench Valve leak check

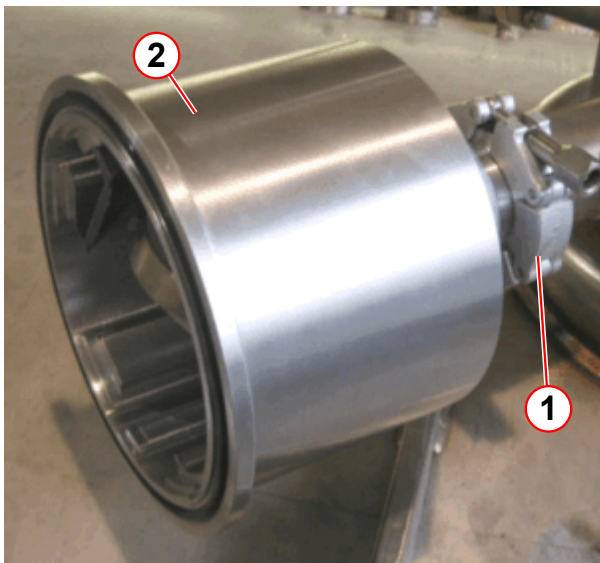


- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

- If there is no leak found the quench valve O-ring seal behind the spring plate is OK.
- There is no requirement to remove the burst disk spring plate.
- Continue to check the auxiliary burst disk and refit the quench valve.
- If a leak is detected the burst disk spring plate assembly will need removing to replace the O-ring.

8.4.4.1 Replacing the Burst Disk and O-ring Seal

Fig. 435: Fitting the quench valve



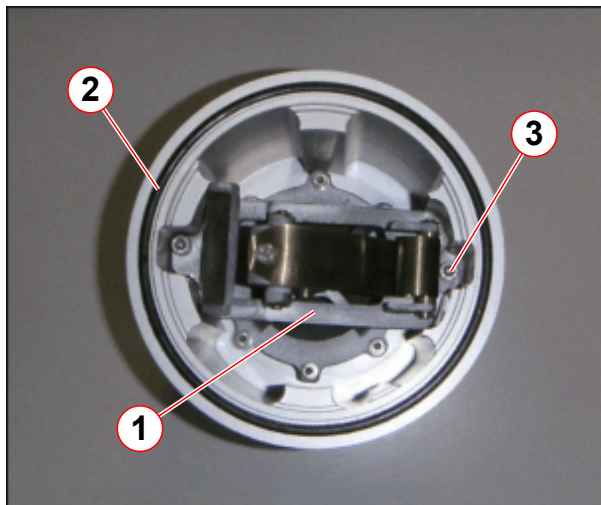
- (1) NW40 seal and clamp
- (2) Quench valve assembly

- Open the magnet bypass valve to depressurize (if necessary).
- De-ice the turret flange.
- Mark the orientation of the quench valve
- Remove the NW40 (high load) turret clamp and seal.
- Remove the quench valve from the turret flange.
- To minimize air ingress, immediately place a ball of paper toweling into the open turret port.
- Close the magnet bypass valve.



The spring exerts significant force on these screws.

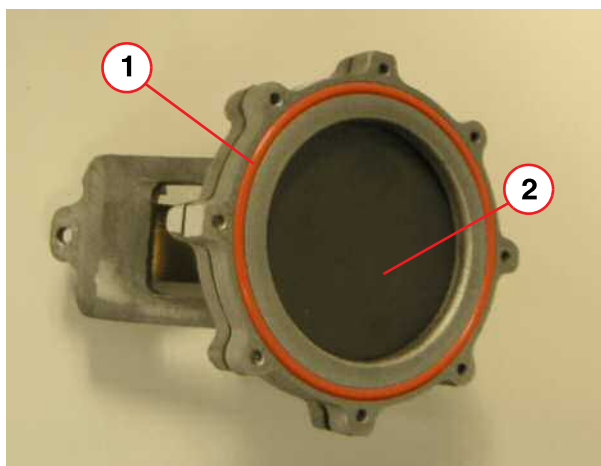
Fig. 436: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Unscrew and remove the (2 off) M5 cap head screws.
- Lift out the burst disk assembly from the quench valve body.

Fig. 437: Quench Valve Bursting Disk Assembly



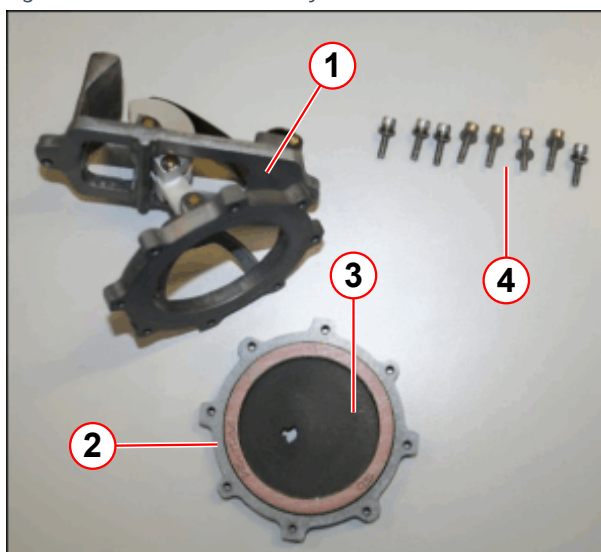
- (1) O-ring
- (2) Burst disk

- Inspect the burst disk 'O-ring and replace if damaged.
- Clean the O-ring groove and O-ring before re-fitting.
- Check that the O-ring is seated evenly around the groove.



The O-ring must be replaced each time the quench valve is removed. DO NOT apply grease to the O-ring.

Fig. 438: Burst disk assembly

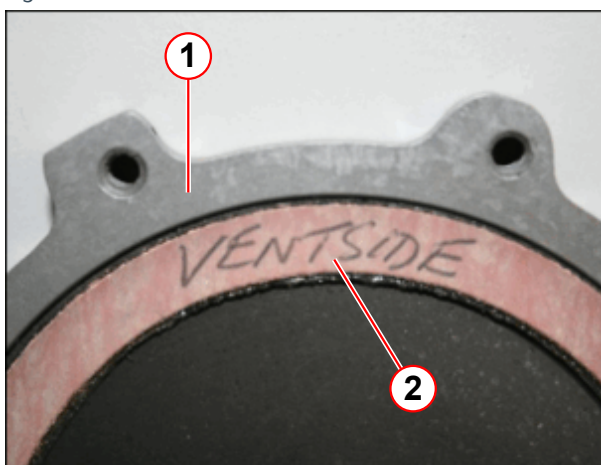


- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

To replace the burst disk:

- Unscrew and remove the 8 x M4 cap screws.
- Remove the broken burst disk and discard.
- Check that the new burst disk and flange faces are clean and undamaged.

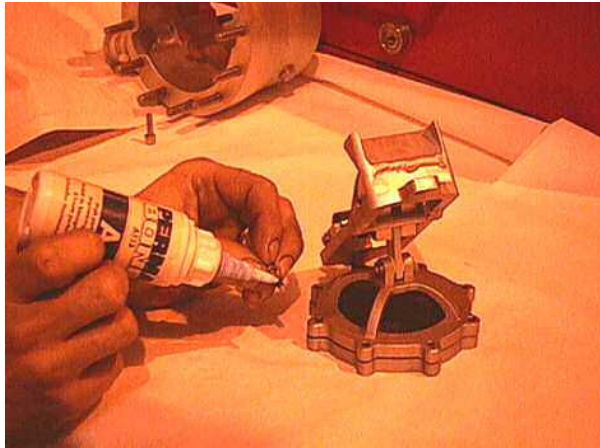
Fig. 439: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

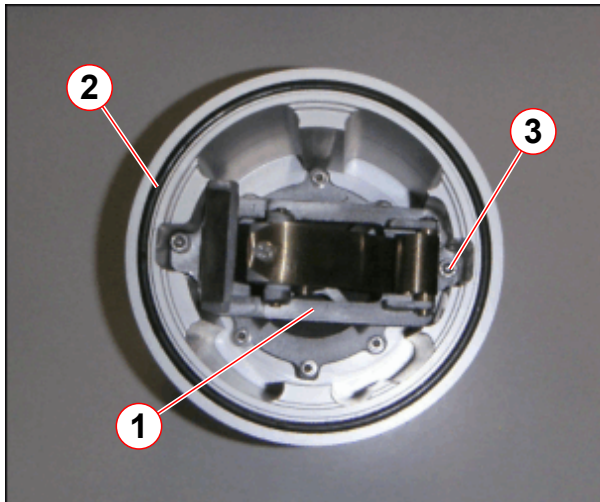
- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 440: Thread locking compound



- Fit the burst disk assembly. (Do not apply grease).
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound..
- Diametrically torque the screws evenly to 0.8 Nm (0.6 ft. lbs).

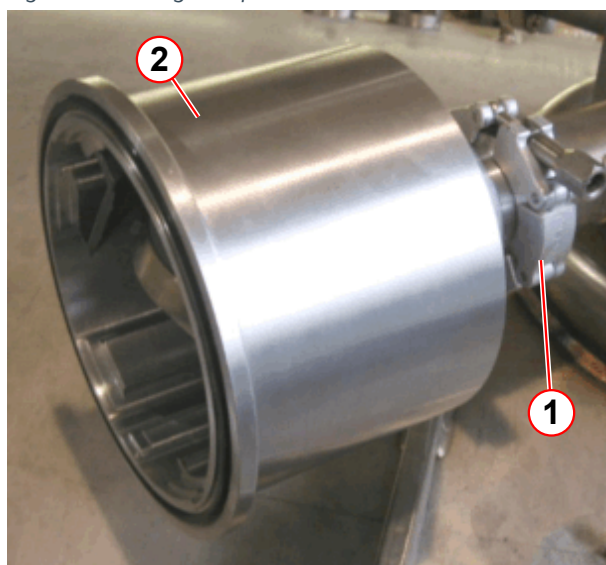
Fig. 441: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Re-fit the burst disk assembly into the quench valve body.
- Torque the 2 x M5 cap hd screws to 5 Nm. (3.8 ft.lbs).
- Check the quench valve manifold O-ring is in good condition (pos.2).
- Replace if damaged.

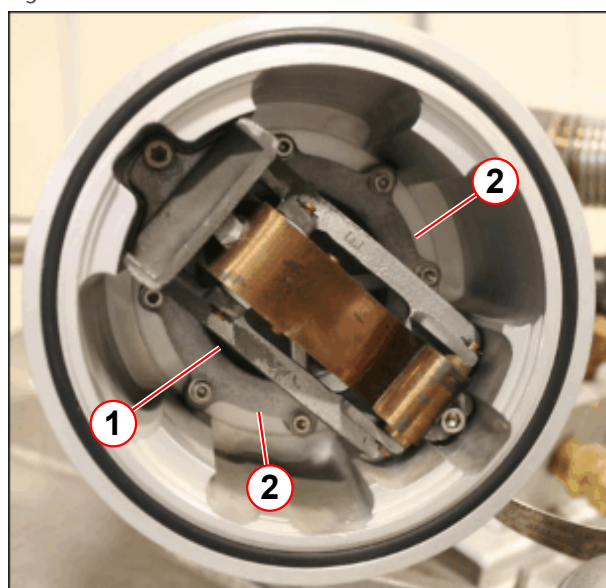
Fig. 442: Fitting the quench valve



- (1) NW40 seal and clamp
(2) Quench valve assembly

- **Remove the paper toweling** if used (or NW40 blank) from the turret port. Ensure the surfaces are clean and dry.
- Fit the quench valve assembly to the turret flange.
 - » Re-fit using a new NW40 high load seal (material number 10100171).
 - » Check the correct orientation.
 - » Warm with heat gun when tightening the NW40 clamp.

Fig. 443: Leak check



- (1) Graphite burst disk
(2) Burst disk O-ring seal

- Check that the magnet bypass valve is closed and pressurize the system.
- Leak check the burst disk and O-ring seal.
- Leak check the NW40 clamp and seal at the turret port.
- If no leaks are detected:
 - » Fit the quench valve elbow interface with the V-clamp to the quench valve.
 - » Fit the support bracket screw to the elbow interface assembly.
 - » Check the auxiliary burst disk.

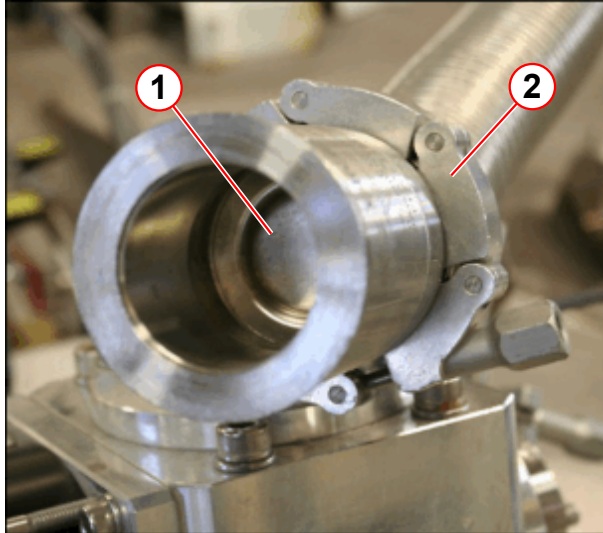
8.4.5 Auxiliary Burst Disk

Post quench the auxiliary burst disk vent line is removed when disconnecting the venting to leak check the quench valve.



Do not touch the metal membrane, it is very fragile.

Fig. 444: Auxiliary burst disk check



- (1) Burst disk
- (2) High load seal

- Visually inspect the internal metal membrane of the burst disk for damage (pos.1).
- Leak check the high load seal (pos. 2).



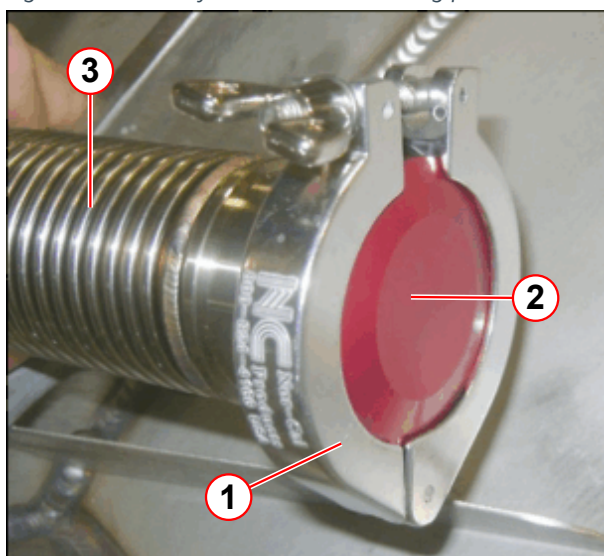
WARNING

Safety device removed

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » Replace the auxiliary vent burst disk before ramping.

Fig. 445: Auxiliary vent - NW40 blanking plate



- (1) NW40 clamp
- (2) NW40 red blanking plate
- (3) Auxiliary vent line

- If the membrane has ruptured, disconnect the high load clamp and remove the burst disk from the vent line.
- Fit the red anodized blank immediately from the spares kit.
- Order a replacement auxiliary burst disk. Fit as soon as available.
- With the blank fitted allow magnet to pressurize.
- Leak check red blanking plate and seal on the end of the auxiliary vent line.

Fig. 446: Auxiliary vent interface - O-ring seal



- (1) Quench valve manifold
- (2) O-ring seal
- (3) Auxiliary bursting disk
- (4) NW40 clamp (high pressure)

- Check that the O-ring seal (Pos. 2) is not damaged and that the surface is clean.
- If damaged, replace the O-ring.
- Connect the auxiliary burst disk to the manifold assembly with the V-clamp.

8.4.6 OVC Burst Disk

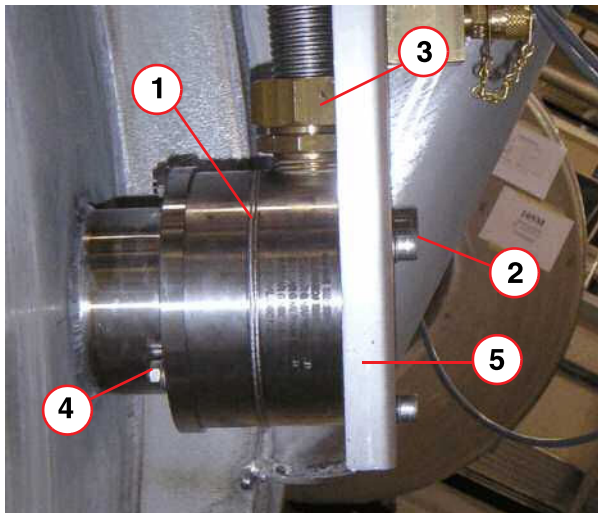


The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 22 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 447: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. There is no requirement to check the condition of the OVC burst disk 'Post quench'.

8.4.7 Final Work Steps

8.4.7.1 Reactivate the Pressure Heater

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Set the pressure heater mode to **On 100%**.
- Select SAVE.
- Monitor the magnet pressure.
 - Select Update to refresh data until pressure reaches ≥ 15.3 psi (A).



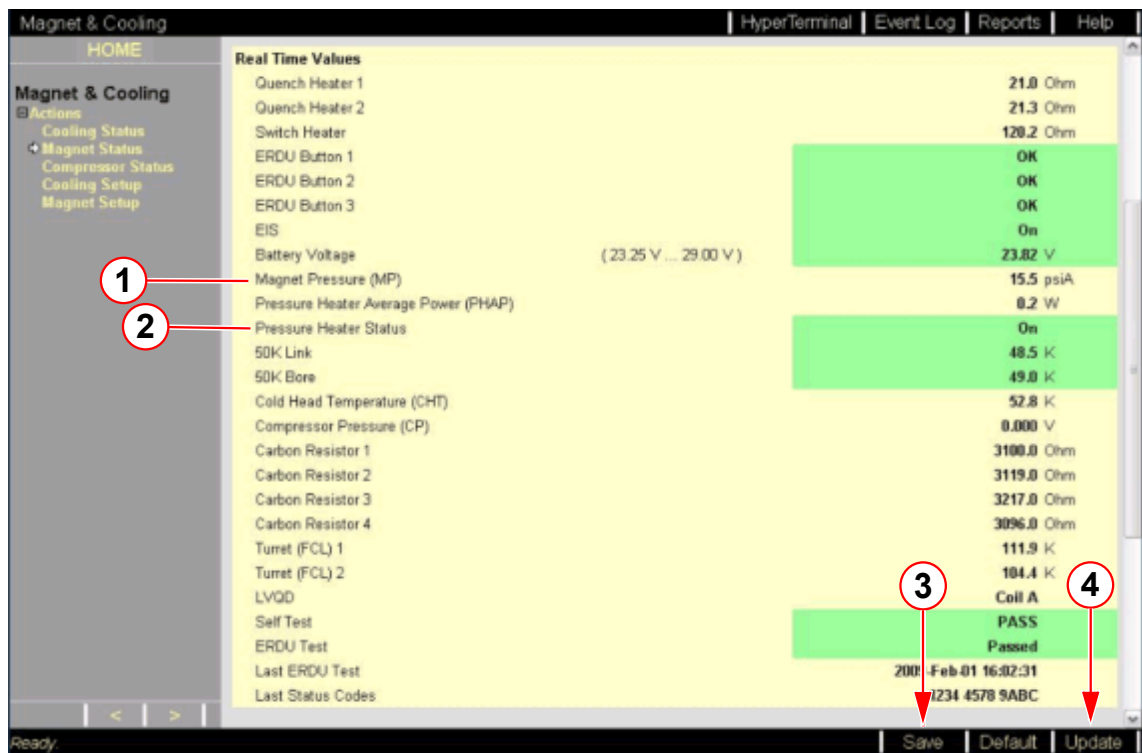
Data is only accurate from the last refresh/update of the page, or from when the page was opened last.



During pressurization, the cold head will switch on automatically.
If the cold head was turned off at MRC, select 'pressure heater status'.

- At MSUP Control - Set the pressure heater mode to Automatic mode.
- Select SAVE.

Fig. 448: Check magnet pressure level

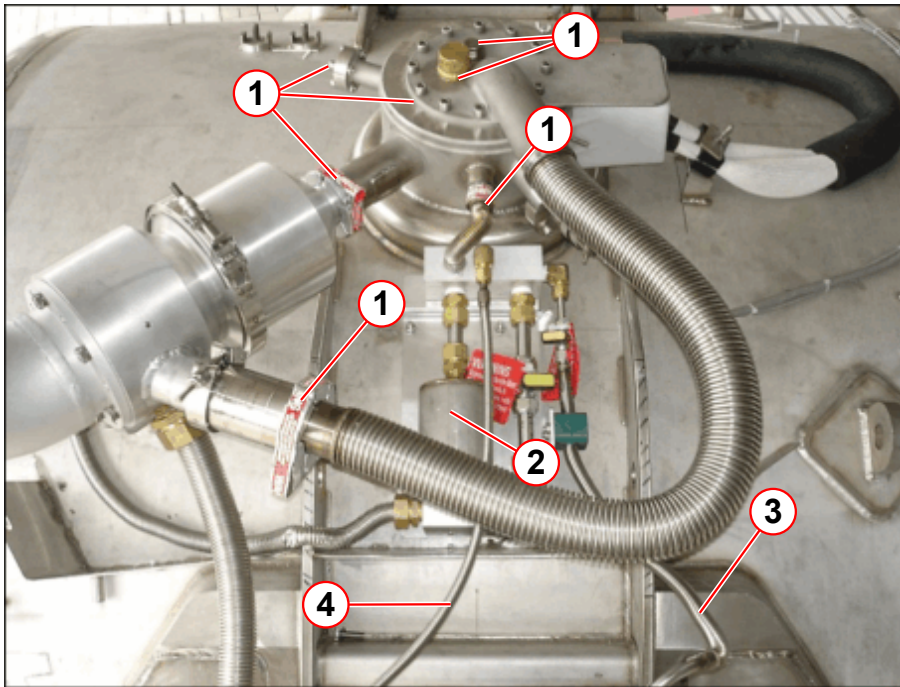


- (1) Magnet pressure (MP)
- (2) Pressure heater status
- (3) Save button
- (4) Update button

8.4.7.2 Final Checks

- Helium fill magnet to the required level for ramping the magnet.
- Order replacement components if the spare parts have been used (burst disk, seals).
- Perform a leak check of the turret venting and components.
 - » All vent pipes and flanges on the high pressure side of the helium vessel.

Fig. 449: OR105/122 - Turret and venting leak checks (Example Mk2)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line



Allow a minimum of 2 hours after filling before ramping the magnet.

8.4.7.3 Quench Report

- Fill out the Quench report - MR - Quality Database and Quench Report.

8.4.7.4 TSG - Checking the Carbon Resistors

The Ceramic Carbon Resistors (CCR) values can be checked from the MSUP or manually by using a multimeter.

- The measurements are taken across pins **a-e**, **b-e**, **c-e**, **d-e** on the 41-pin turret connector.



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » **The magnet must be RAMPED DOWN before attempting this procedure.**

Tooling requirements

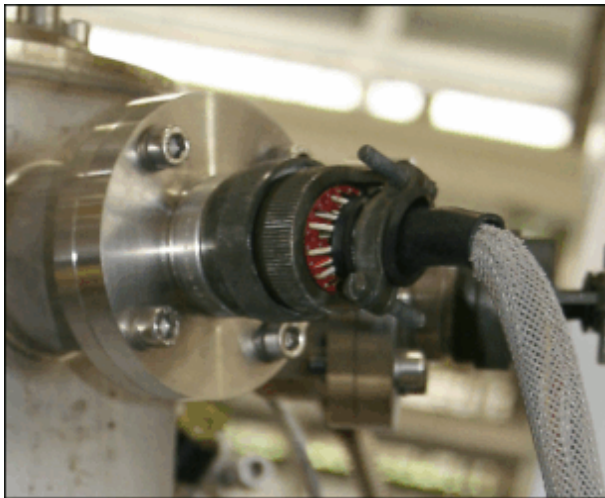
- Multimeter and pin leads.
- Laptop/PC with Terminal Emulation software, (PuTTY).
- RS232 cable.

Procedure

Using a multimeter:

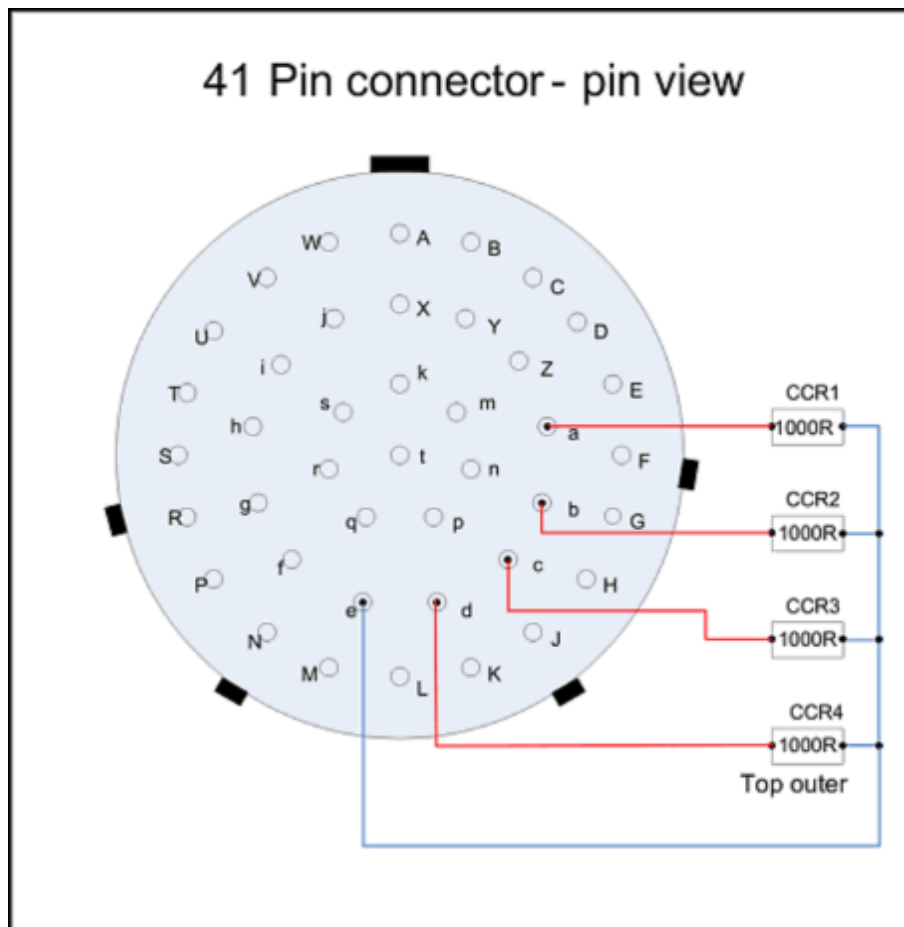
- Connect the pin leads.
- Turn the mode selector to resistor mode Ω .

Fig. 450: Turret - 41pin connector



- Disconnect the 41-pin connector.
- Measure across pins **a-e**, **b-e**, **c-e**, **d-e**
- Compare the values with the Magnet Acceptance Document.
- Re-connect the 41-pin connector.

Fig. 451: 41pin turret connector - ceramic carbon resistor pins (room temperature values)



8.5 What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256)

8.5.1 General

If the magnet quenches, a large volume of helium gas is generated within the helium vessel. The quench valve opens, and stay open until the flow of gas subsides. The quench valve re-seals, preventing air (caused by cryo-pumping) from being drawn back into the magnet turret and helium vessel.

The MSUP automatically maintains the system pressure above atmospheric.



Magnet HSC must be notified of a quench (→ Siemens Healthineers Magnet WS CS HSC, <mailto:magnet.wscshsc.team@siemens-healthineers.com>)

Fill out the **Quench Report** - (→ MR Product Information, <https://productinformation.healthcare.siemens.com/ProductInformation.aspx?uuid=7abf6bb6-5daf-440c-9920-fab2d4640084>) - Links - MR - Quality Database (Quench Report)



The magnet helium level can be recovered if the magnet temperature is no warmer than nitrogen temperature.

Check the carbon resistor (CARBON RES 1; pins a-e) reading and compare to the calibration values on the Magnet Acceptance Document.

The magnet must be re-filled immediately with helium.

To re-fill the magnet from zero helium, or warm carbon resistor temperatures, perform a BOTTOM FILL procedure.

If proceeding with a helium fill, immediately place an order for 1000 liters of liquid helium.

NOTICE

If the magnet temperature warms up above 77K, a full nitrogen pre-cool is required.

Helium filling will not cool the magnet sufficiently to collect liquid helium.

» A full nitrogen pre-cool will take 8 –10 days.

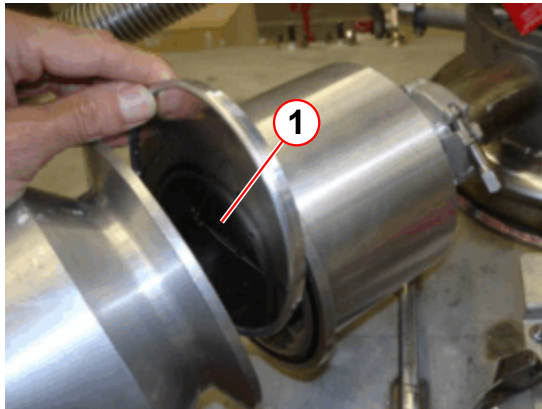
8.5.1.1 Mobile System

NOTICE**Risk of Air ingress**

The Mobile high altitude burst disk WILL rupture during a quench.

- » Post quench the high altitude 10 psi burst disk (10129467) and O-Ring (10113389) must be replaced.

Fig. 452: Removing the High Altitude disk



(1) Burst Disk

- Mobile systems are fitted with an additional High Altitude Burst Disk in front of the quench valve assembly.
 - » This disk will rupture if the magnet quenches.
 - » For replacement procedure: (→ Mobile Systems - High Altitude Burst Disk / Page 399)

8.5.1.2 Safety



Post quench procedures must be performed by CSE's that are trained to the MR2MAG-NET level. (or System level trained).



Switch off and isolate the magnet power supply from the mains before further action.

**WARNING****Risk of asphyxiation.**

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogenes.

- » Do not vent helium gas into the magnet room during helium transfer. As an exception, a small quantity of helium gas can be vented to cool the syphon.
- » Ensure the room is ventilated.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » O₂ monitors must be used in the imaging suite.

**CAUTION****RISK OF ICING!!**

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk and O-ring seals IMMEDIATELY.
- » Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.
- » If ice is suspected contact Magnet HSC immediately.

**CAUTION**

Cold helium gas.

Personal injury may occur.

- » Protective clothing (long sleeves), cryogenic gloves and eye protection must be worn.

8.5.2 Post Quench Steps

Follow the work steps in the table below for the different types of quench.

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 1	■ Isolate the MPSU from the magnet. ■ Download the 3600 MPSU ramp log files.	■ Stop filling.	■ Check that no ERDU buttons are pressed. ■ Re-set ERDU button.
Step 2	From the MSUP-Display Unit. ■ Check the liquid helium level. ■ Check the carbon resistor temperatures. (→ Checking the Carbon Resistors (CCR) Values / Page 378)		
Step 3	■ Order 1000 liters of liquid helium.¹		
Step 4	■ Warm up and leak check the quench valve and burst disk. (→ Checking the Quench Valve Burst Disk / Page 385) ■ Warm up and leak check the auxiliary vent burst disk. (→ Auxiliary Burst Disk / Page 394) ■ MOBILE system - replace the high altitude burst disk. (→ Mobile Systems - High Altitude Burst Disk / Page 399)		

	Type of quench		
	Quench during ramping	Quench during He filling	Persistent mode quench or ERDU button pressed
Step 5	After 1 hour : ■ Download the Magnet Supervisory log files. (→ Log File Capture / Page 377) ■ Helium fill. Fill to recommended helium ramp level.		
Step 6	■ Leak check the quench valve and venting. (→ Final Checks / Page 404)		
Step 7	■ Fill out the Quench report.		
Step 8	■ Email the (→ Magnet HSC, mailto:magnet.wscshsc.team@siemens-healthineers.com) with: » 3600 Power Supply logs. » Magnet Supervisory logs and screen capture. » Magnet serial number.		
Step	■ Ramp the magnet.		
Step 10	■ Perform the shim check. » If the shim check exceeds the specifications, a re-shim is necessary.		

1. The quantity of the order depends on the helium level remaining in the system after a quench, and volume already on site.

8.5.3 Pre-requisites

8.5.3.1 Tools Required

The post quench tools required:

- Heat gun and paper toweling.
- Service tool kit.
- Pressure service tool (material number **03861221**).
- Laptop and RS232 cable.
- O₂ monitor.
- Personal protection equipment. To include, cryogenic gloves, safety shoes, glasses and a long sleeved protective clothing.
- Helium leak detector.

Fig. 453: Edwards Leak detector (example)



- Use the Edwards hand-held leak detector (material number **10614850**). This is the recommended equipment for leaks greater than 1×10^{-4}

NOTE: The hand-held device is calibrated in ml/sec:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

8.5.3.2 Spare Parts

Helium leaks from the magnet pressure vessel are generally repaired by replacing a damaged seal.

- A number of seals are supplied in the spares kit. These must be replaced when used on the magnet.
- Venting components will require ordering from CSML.

Mk1 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 08396090).
- High load seal (material number 10100171).

Mk2 Spares required:

- Graphite Burst disk 26 psi (material number 10129362).
- O-ring seal burst disk - (material number 10100140).
- Auxiliary burst disk (material number 10121913)
- High load seal (material number 10100171).
- O-ring seal 44.5mm ID (material number 10100151).



Always replace the burst disks and seals used during replacement procedures. They are critical components for reducing the risk of air ingress. A replacement must always be available for immediate use.

8.5.3.3 Unattended Quench

Unattended quench indicators are:

- If the Magnet STOP (ERDU) has been activated:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
 - If the helium level is low:
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
 - If the magnet pressure is very low, the
 - » cold head switches off.
 - » the yellow WARNING LED lights on the alarm box.
 - » the alarm buzzer is activated.
 - Imaging is not possible if there is no magnetic field.
- Press the AUDIO ALARM OFF button on the alarm box to silence any audible alarms.

8.5.3.4 Log File Capture

Magnet Supervisory

- The **quench data is lost**, if:
 - a) The system is re-booted within 1 hour after a quench
 - b) The system is turned off within 1 hour after a quench
 - c) The data is downloaded within 1 hour of after a quench
- Download the MSUP log files.

If the magnet quenches during ramping:-

- Download the Magnet Power Supply log file.



The MPSU log files **MUST** be downloaded **BEFORE** the next ramp is performed, otherwise previous data will be overwritten!

- Switch off and disconnect the MPSU from the magnet.

If the magnet quenches during helium fill:

- Stop the transfer.



Do not remove or disconnect the syphon as the magnet will be at high pressure.



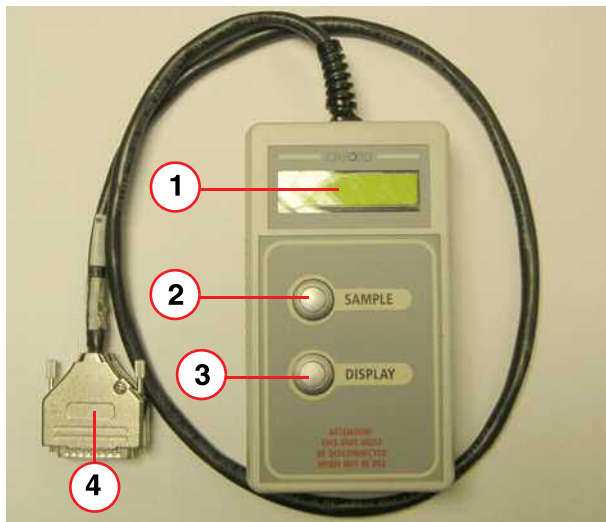
For all types of quench, allow up to 1 hour for the quench pressure to subside.

Use the MSUP Display unit:

- To check that the MSUP pressure control system is working. If a display unit is not available, use the SESO interface on the Host service screens.
- If the pressure control system is NOT working, the pressure inside the cryostat will fall to atmospheric levels. The compressor switches off.

8.5.3.5 Checking the Carbon Resistors (CCR) Values

Fig. 454: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Using the MSUP-Display Unit:

- Connect the 25-pin 'D' connector to X9 on the MSUP.
- Activate the MDU by simultaneously pressing the SAMPLE button (2) and DISPLAY button (3).
- Press the Display button (3). The unit will automatically carry out a helium level test. The helium fill level is shown in the display (1).
- pressing the DISPLAY button, toggle through to Carbon resistor parameters.
- Check the carbon resistor values and compare the values with table with the expected values.



The magnet helium level can be recovered if the temperature of the carbon resistor (CARBON RES 1:Pins a-e) is still colder than nitrogen temperature.

The carbon resistor (CARBON RES 1) reading on the hand-held display must be in the range of 180 - 1500 ohms.

Confirm the readings, using the multimeter directly on the 41-pin connector.



Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.



Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.

- High readings = coldest temperature
- Low readings = warmest temperature
- Check temperatures with the figures on the Magnet Acceptance document.



If temperatures are below or equal to nitrogen temperature (77K), a BOTTOM FILL must be performed IMMEDIATELY.

Tab. 23 Expected Carbon Resistor Values

41-pin plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e (Bottom)	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500
Carbon Four d - e (Top)	120 - 160	130 - 180	700 - 1500
All Ceramic Carbon Resistors (CCRs)	880 - 1050	1150 - 1350	2300 - 3380

Fig. 455: OR105/122 Magnet Acceptance Document (CR)

SIEMENS		Magnet Acceptance Documentation Installation Data		
Siemens Magnet Technology Quality Regulations				
Cryostat Number	21199812	SMT Reference	21201632 02.09	
System Description 1.494T Actively Shielded Superconducting Twin Turret Magnet System OR105				
Measured Magnet Centre (Offset)		-1 (S) mm		
If magnetic Offset is outside the spec (+_5mm) the magnet will be marked (+_5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet				
Operating Parameters				
Description	Specification value	Measured value		
Operating Current @ 1.494T	< 600 AMPS	509.95	AMPS	
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp	
Carbon Resistor Values				
POSITION	Room Temperature	Helium Temperature		
Top	162.4	1140		
One	134.4	903		
Two	132.3	994		
TDC	140.4	943		
Helium Level Probe Resistance (When using a 250 mA supply)				
		Probe 1	Probe 2	
@ 0 Resistance	Ohms	85.36	85.36	
0 % Calibration		774	774	
100 % Resistance	Ohms	0.22	0.22	
100 % Calibration		2	2	
Diode Table No		-1		
Pressure Heater Res	Ohms	19		
Final Temperature Sensor Recon Values				
50kK	50kK Lin	Coldhead	Recon Margin	
48	45	42	K 467 Millivolts	
Checked by	K Suraina		Signature	
Date	02/02/2009			

Fig. 456: OR105/122 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
MR Magnet Technology Quality Regulations			
Cryostat Number	21243234	SMT Reference	21245901 06.10
System Description		OR105	
1.494T Actively Shielded Superconducting Twin Turret Magnet System			
Measured Magnet Centre (Offset)		-2 (S) mm	
If magnetic Offset is outside the spec (+.5mm) the magnet will be marked (+.5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet			
Operating Parameters			
Description	Specification value	Measured value	
Operating Current @ 1.494T	> 600 AMPS	510.05	AMPS
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp
Carbon Resistor Values			
POSITION	Room Temperature	Nitrogen(77K)	Helium Temperature
TDC d - e	878		2567
Top a - e	926		2938
Mid b - e	985		2715
Bottom c - e	877		2613
TDC Calibration Values:	850.4	1122.9	2731.7
Helium Level Probe Resistance (When using a 250 mA supply)			
		Probe 1	Probe 2
@ 0 Resistance	Ohms	85.36	85.36
0 % Calibration		774	774
100 % Resistance	Ohms	0.22	0.22
100 % Calibration		2	2
Diode Table No		-1	
Pressure Heater Res	Ohms	19	
Final Temperature Sensor Recon Values			
50kK	50kK Lin	Coldhead	Recon Margin
53	47	44 K	530 mW
Checked by	K Suraina	Signature	
Date	02/06/2010		
Page 1 of 1		25-AD EPS 000 01	

8.5.3.6 Check the Magnet Pressure

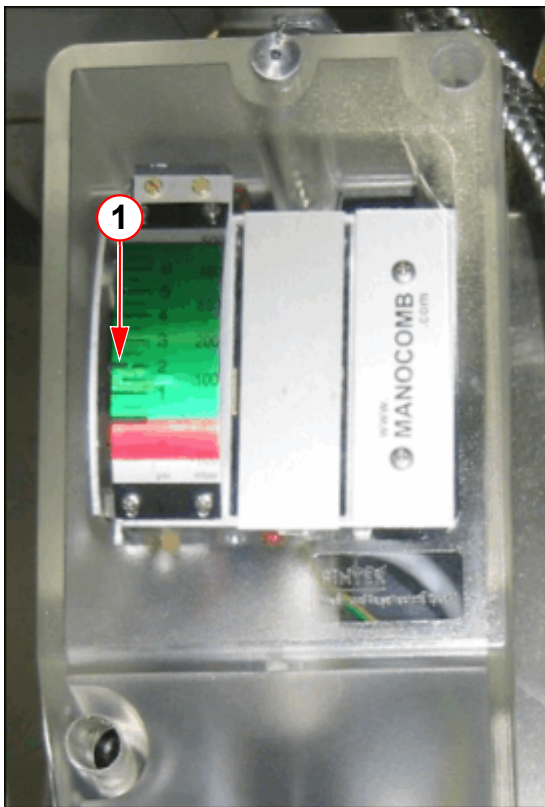
Check that the helium vessel has a positive pressure:

- Magnet pressure should read 15.3 psi (A) on the MSUP display unit.
- Check for positive pressure on the pressure gauge.
- If the pressure gauge is within the **RED zone**, the magnet is sub-atmospheric.
 - » **DO NOT** open the magnet bypass valve, or remove any high pressure fitting.
 - » If the magnet pressure does not stabilize via the pressure heater control circuit, manually induce pressure into the cryostat.



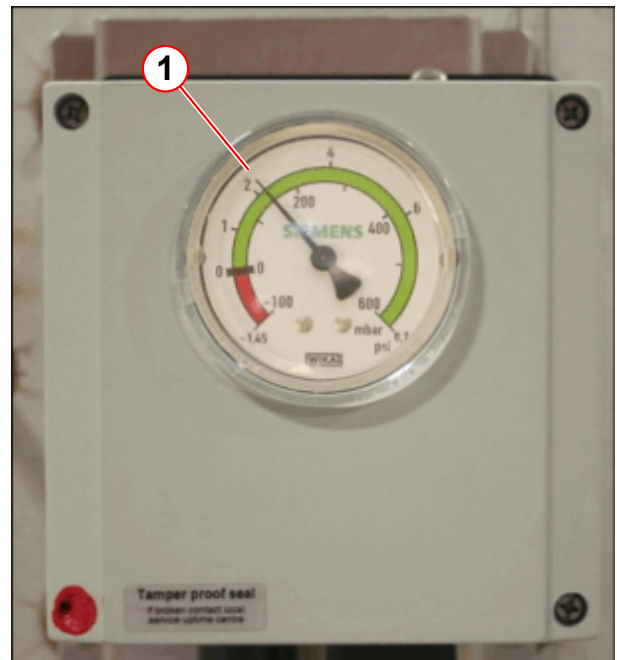
Wait until the magnet has gained a positive pressure before continuing with the burst disk and post quench leak checks.

Fig. 457: OR97 - Pressure gauge (expected pressure)



(1) Expected pressure 2.1 psi (G)

Fig. 458: OR97 - WIKA pressure gauge (expected pressure)



(1) Expected pressure 2.1 psi (G)

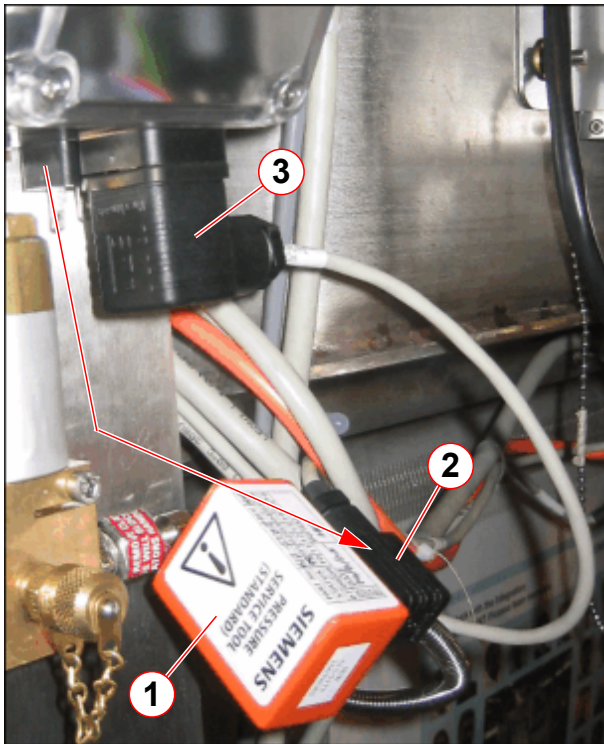
8.5.3.7 Deactivate the Pressure Heater

The Pressure Service Tool is used to deactivate the pressure heater control circuit. The electrical connectors are labeled pressure sensor (rear) and pressure switch (front).

- To fit the pressure service tool:
 - Disconnect the rear electrical connector on the pressure switch transducer.
 - Fit the pressure service tool to the connector lead.

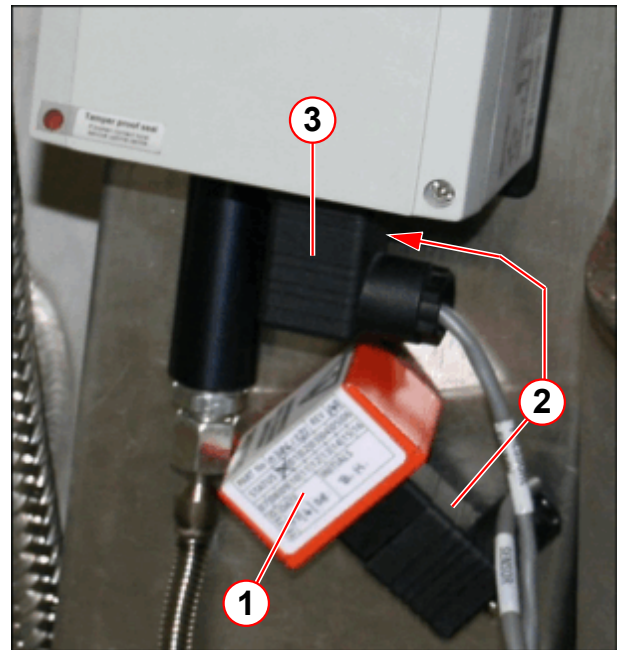
- The Alarm Box will sound PRESSURE ERROR.
- Press the ACKNOWLEDGE button to cancel the alarm.
- The pressure reading will be set to ~18 psi (A), which will disable the pressure heater.
- To activate the pressure heater control circuit, connect the cable to the rear connector on the pressure switch transducer.
 - » Cancel any alarms.

Fig. 459: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

Fig. 460: Fit the pressure service tool

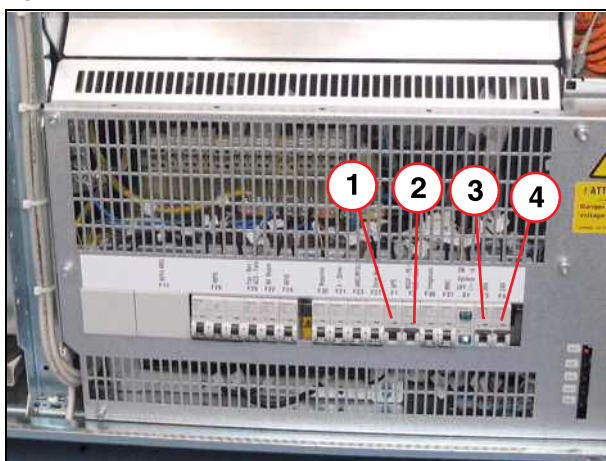


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 461: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service pressure tool is not available:

- Switch OFF the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » Pressure heater is powered down.
- The ERDU circuit is still functional.
- Helium level and magnet temperatures can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, only 2 minutes of power is available.

8.5.3.8 Depressurize the Helium Vessel



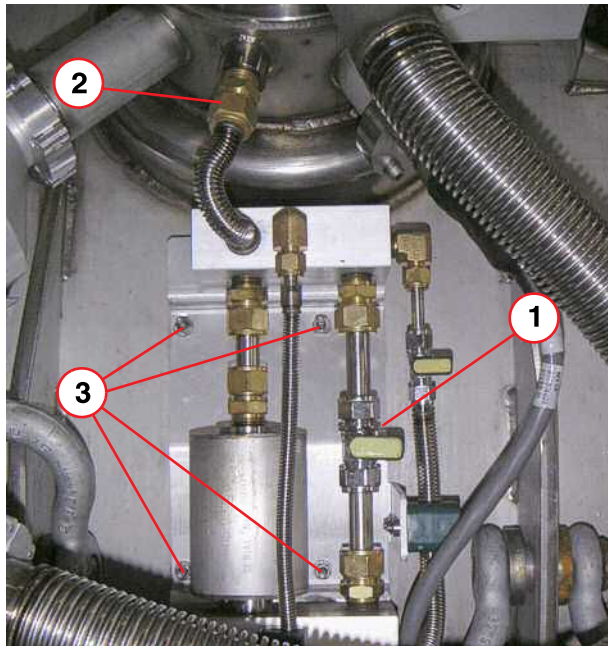
CAUTION

Risk of ice formation.

The magnet bypass valve is only **OPENED** for depressurization of the helium vessel. The bypass valve is **CLOSED** during normal operation.

- » To avoid air ingress, **CLOSE** the magnet bypass valve immediately after the helium vessel is depressurized.

Fig. 462: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

If depressurization of the helium vessel is required:

- Fully open the magnet bypass valve to depressurize the helium vessel.
- Wait until zero psi is displayed on the pressure gauge.
- The pressure switch will activate.
 - » the cold head will turn off.
- Close the magnet bypass valve to prevent air ingress.
- Warm up the vent valve assembly.
 - » Take care not to damage any components when using a heat gun.

8.5.4 Checking the Quench Valve Burst Disk

Post quench if there is no magnet pressure a burst disk has ruptured. Check the graphite (quench valve) burst disk.

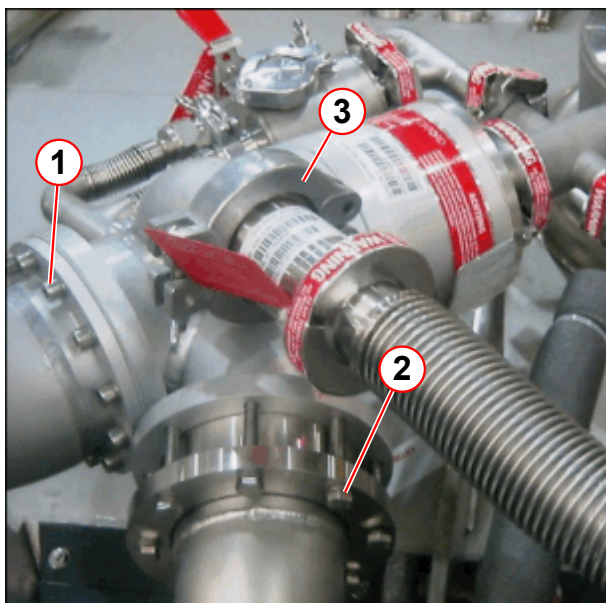
- Disable the pressure heater by fitting the pressure service tool.
- Warm up the quench valve and venting to room temperature, using a heat gun.
- The elbow assembly and manifold must be removed for inspection and leak check of the graphite burst disk and O-ring.



Check venting configuration fitted and follow applicable procedure for Mk1 or MK2 versions.

Mk1 Venting Configuration

Fig. 463: MK1 - Venting removal

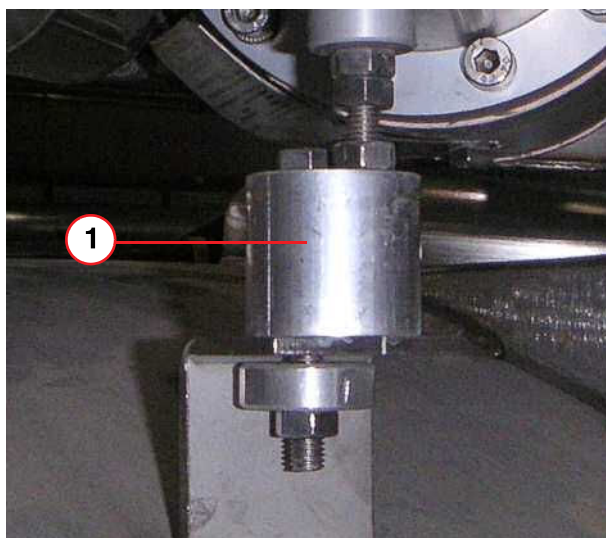


- (1) Elbow connection
- (2) OVC burst disk
- (3) Auxiliary burst disk

To access the quench valve and burst disk:

- Remove the 12 x M8 Cap hd screws and retaining the elbow assembly to the manifold venting (pos.1).
- Remove 8 x M8 Hex hd screws retaining the OVC burst disk to the manifold (pos.2).
- Disconnect the NW50 clamp (pos.3) and remove the auxiliary vent line.

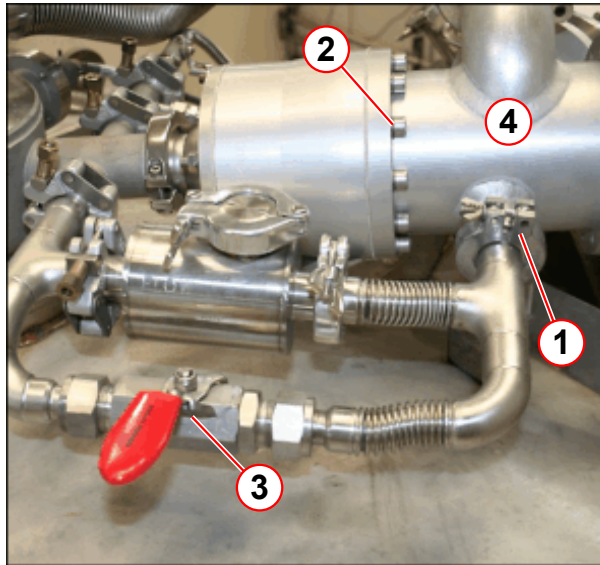
Fig. 464: Mk1 Support Bracket



- (1) Support Bracket

- Loosen the support bracket (1) screw underneath the quench valve interface assembly.

Fig. 465: Mk1 - Manifold removal

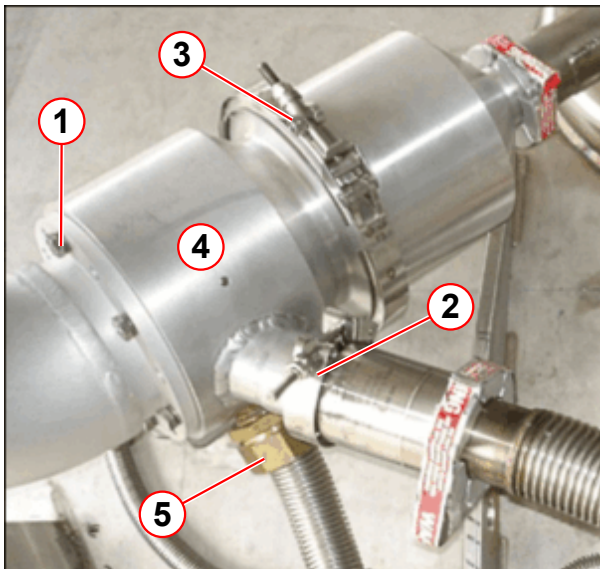


- (1) NW25 Bypass valve connection
- (2) Manifold screws (M8 x 12)
- (3) Bypass valve
- (4) Manifold

- Remove the NW25 (low pressure) clamp to release the 16 psi (A) valve assembly (pos.1).
- Remove the M8 Cap hd screws (x12) securing the manifold to the quench valve (pos.2).
 - » Place the manifold to one side, allowing access to the quench valve.
- Check that the magnet bypass valve is closed.

Mk2 Venting Configuration

Fig. 466: Mk2 Venting removal



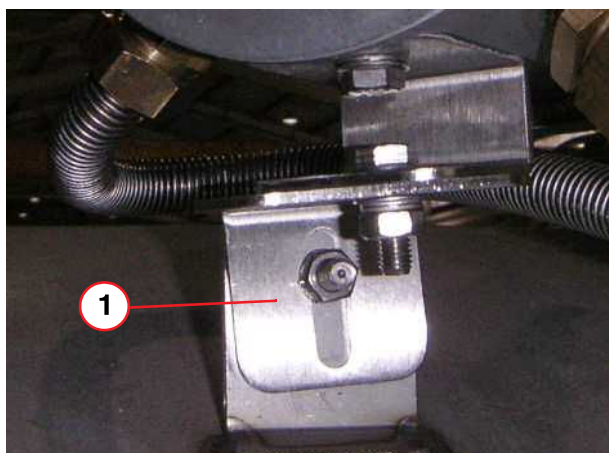
- (1) Elbow connection (M8 x 6)
- (2) V-clamp - Auxiliary vent
- (3) V-clamp manifold to quench valve
- (4) Manifold assembly
- (5) OVC vent connection

To access the quench valve and burst disk:

- Remove the 6 x M8 screws retaining the elbow assembly to the manifold (pos.1).
- Remove the V-clamp retaining the auxiliary vent to the manifold (pos. 2).
- Remove the V-clamp retaining the manifold to the quench valve (pos. 3).

NOTE: it should not be necessary to remove the OVC vent line connection (pos.5).

Fig. 467: Mk2 Support bracket



(1) Support bracket

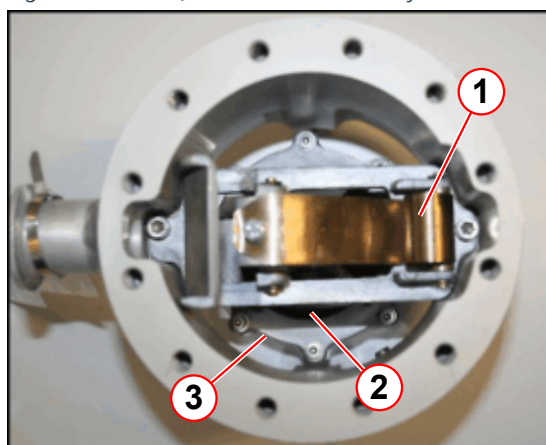
- Loosen the support bracket (1) screw underneath the quench valve interface assembly.
- Move the manifold assembly away from the quench valve.

The quench valve and graphite burst disk can now be accessed to check for leaks.



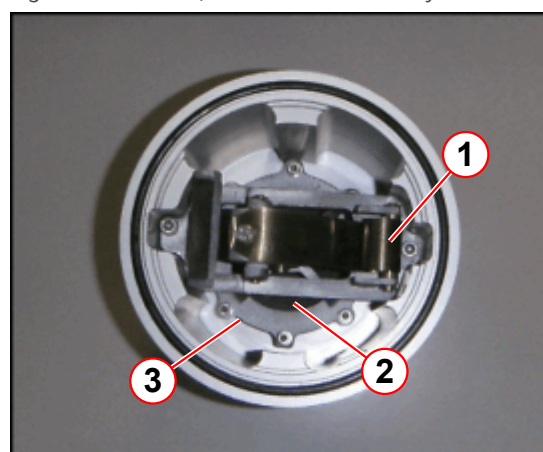
The spring plate and burst disk assembly is the same for MK1 and MK2 quench valves. The next steps show the MK2.

Fig. 468: Mk1 Quench valve assembly



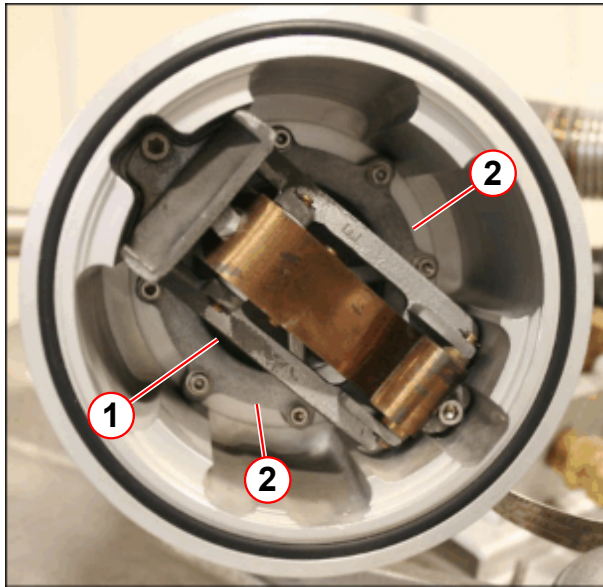
- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

Fig. 469: MK2 - Quench valve assembly



- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

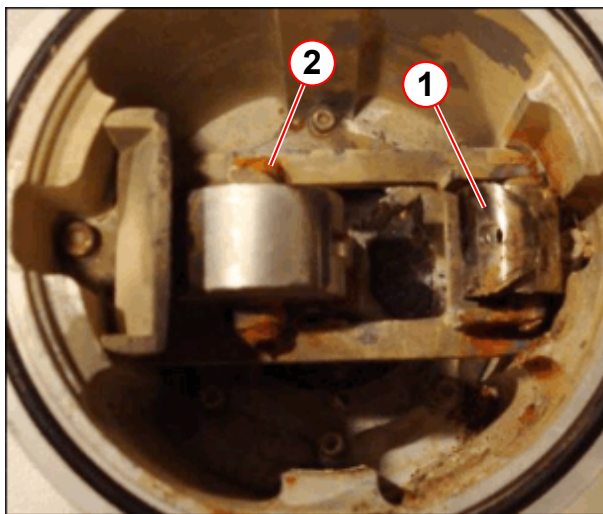
Fig. 470: Leak check



- (1) Graphite burst disk
- (2) Burst disk O-ring seal

- Check the graphite burst disk for breakage or cracks.
- If the graphite burst disk is broken or cracked, it must be replaced **immediately**.

Fig. 471: Water corrosion (example)



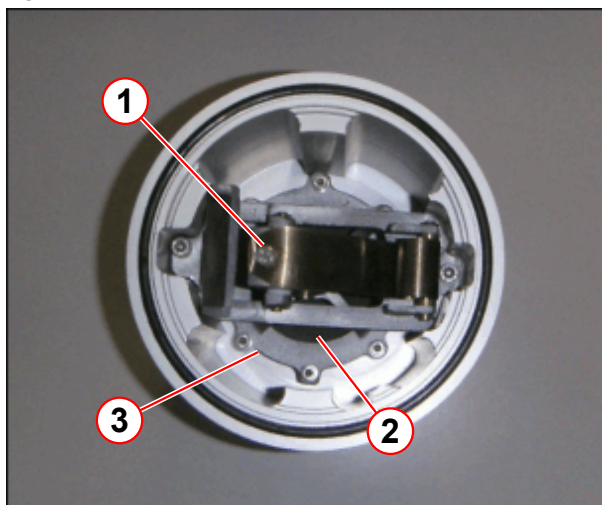
- (1) Spring broken
- (2) Corroded screws

Check for ice and water corrosion or breakage of:

- The spring plate assembly.
- All screws.
- Corroded quench valve components must be replaced.

- With no damage to the burst disk or spring plate visible, continue to leak check around the quench valve O-ring and retaining plate.
- Check that the magnet pressure is a minimum of 0.5 psi (G) for leak checking.
- Leak check around the burst disk and spring plate assembly.

Fig. 472: Quench Valve leak check

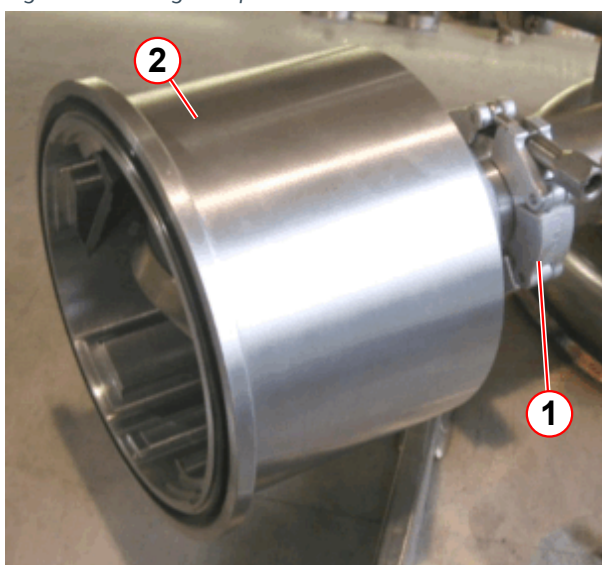


- (1) Spring plate assembly
- (2) Graphite burst disk
- (3) O-ring seal

- If there is no leak found the quench valve O-ring seal behind the spring plate is OK.
- There is no requirement to remove the burst disk spring plate.
- Continue to check the auxiliary burst disk and refit the quench valve.
- If a leak is detected the burst disk spring plate assembly will need removing to replace the O-ring.

8.5.4.1 Replacing the Burst Disk and O-ring Seal

Fig. 473: Fitting the quench valve



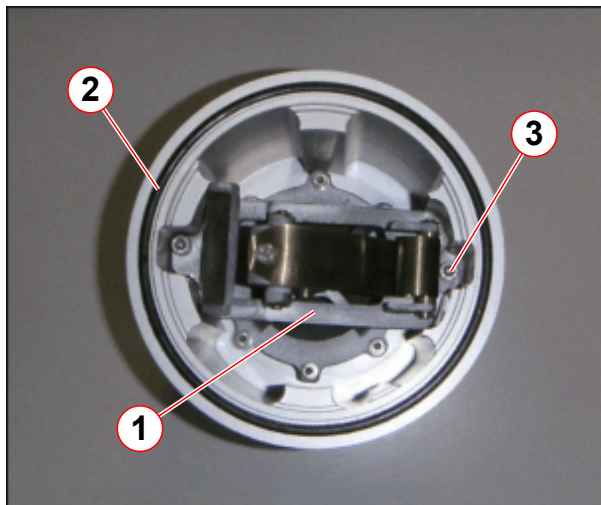
- (1) NW40 seal and clamp
- (2) Quench valve assembly

- Open the magnet bypass valve to depressurize (if necessary).
- De-ice the turret flange.
- Mark the orientation of the quench valve
- Remove the NW40 (high load) turret clamp and seal.
- Remove the quench valve from the turret flange.
- To minimize air ingress, immediately place a ball of paper toweling into the open turret port.
- Close the magnet bypass valve.



The spring exerts significant force on these screws.

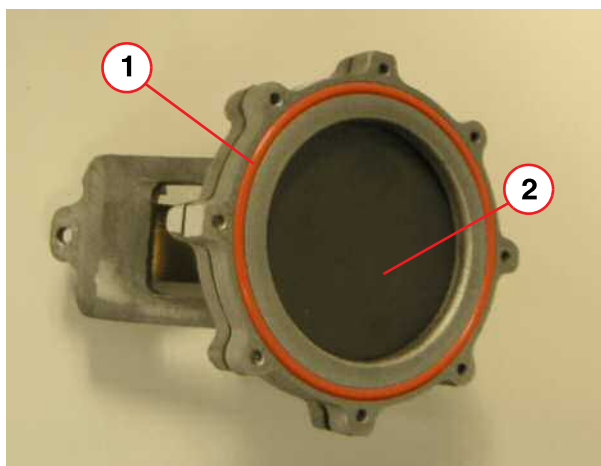
Fig. 474: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Unscrew and remove the (2 off) M5 cap head screws.
- Lift out the burst disk assembly from the quench valve body.

Fig. 475: Quench Valve Bursting Disk Assembly



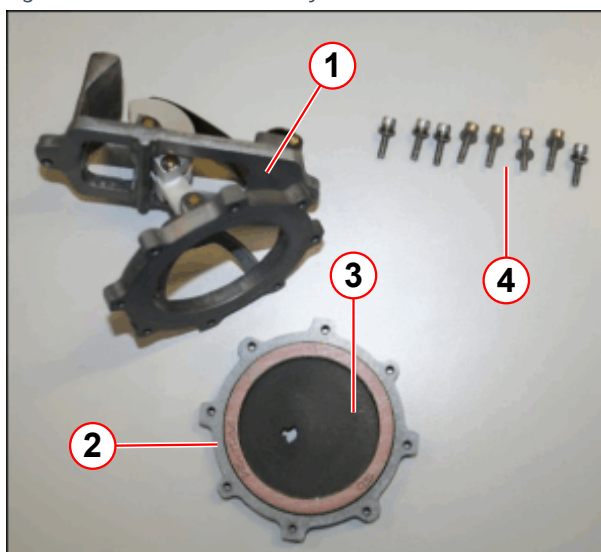
- (1) O-ring
- (2) Burst disk

- Inspect the burst disk 'O-ring and replace if damaged.
- Clean the O-ring groove and O-ring before re-fitting.
- Check that the O-ring is seated evenly around the groove.



The O-ring must be replaced each time the quench valve is removed. DO NOT apply grease to the O-ring.

Fig. 476: Burst disk assembly

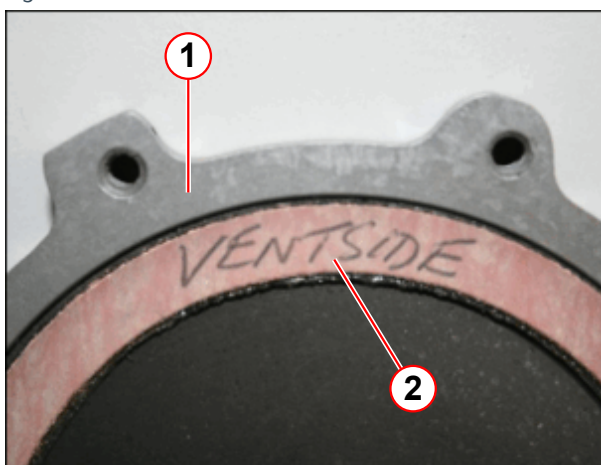


- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

To replace the burst disk:

- Unscrew and remove the 8 x M4 cap screws.
- Remove the broken burst disk and discard.
- Check that the new burst disk and flange faces are clean and undamaged.

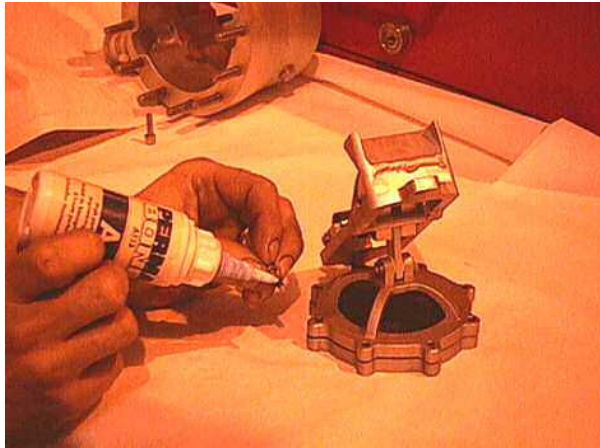
Fig. 477: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

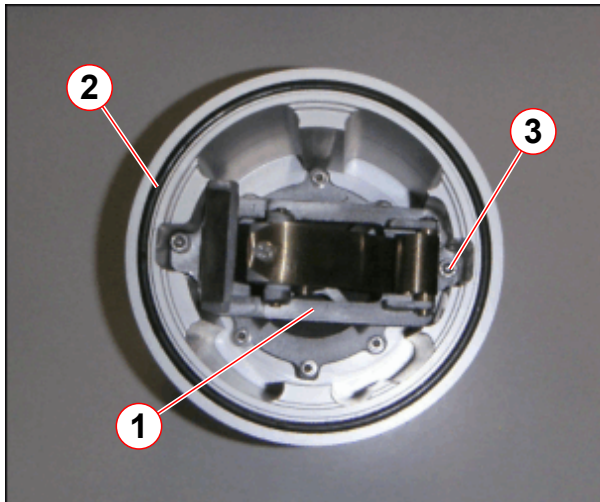
- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 478: Thread locking compound



- Fit the burst disk assembly. (Do not apply grease).
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound..
- Diametrically torque the screws evenly to 0.8 Nm (0.6 ft. lbs).

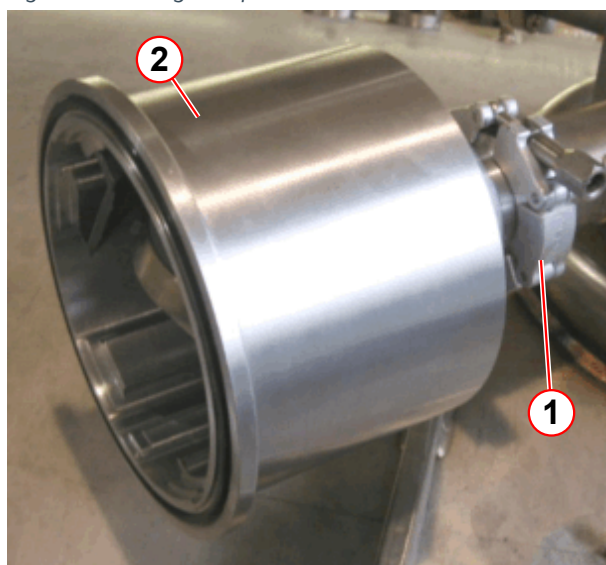
Fig. 479: Quench Valve



- (1) Spring plate assembly
- (2) O-ring
- (3) M5 cap screws (2 off)

- Re-fit the burst disk assembly into the quench valve body.
- Torque the 2 x M5 cap hd screws to 5 Nm. (3.8 ft.lbs).
- Check the quench valve manifold O-ring is in good condition (pos.2).
- Replace if damaged.

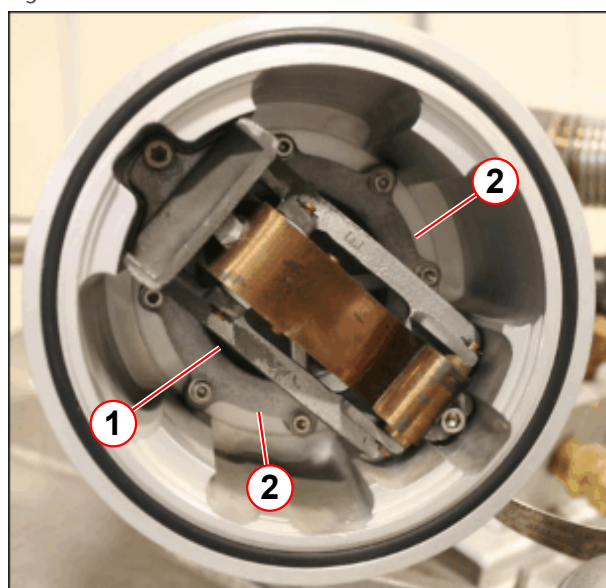
Fig. 480: Fitting the quench valve



- (1) NW40 seal and clamp
(2) Quench valve assembly

- **Remove the paper toweling** if used (or NW40 blank) from the turret port. Ensure the surfaces are clean and dry.
- Fit the quench valve assembly to the turret flange.
 - » Re-fit using a new NW40 high load seal (material number 10100171).
 - » Check the correct orientation.
 - » Warm with heat gun when tightening the NW40 clamp.

Fig. 481: Leak check



- (1) Graphite burst disk
(2) Burst disk O-ring seal

- Check that the magnet bypass valve is closed and pressurize the system.
- Leak check the burst disk and O-ring seal.
- Leak check the NW40 clamp and seal at the turret port.
- If no leaks are detected:
 - » Fit the quench valve elbow interface with the V-clamp to the quench valve.
 - » Fit the support bracket screw to the elbow interface assembly.
 - » Check the auxiliary burst disk.

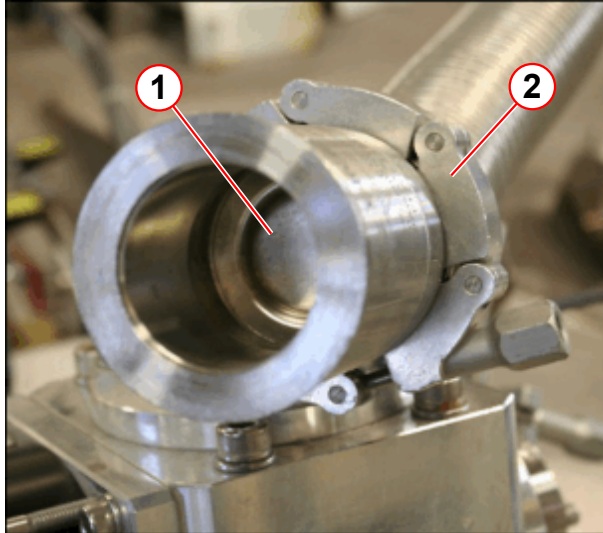
8.5.5 Auxiliary Burst Disk

Post quench the auxiliary burst disk vent line is removed when disconnecting the venting to leak check the quench valve.



Do not touch the metal membrane, it is very fragile.

Fig. 482: Auxiliary burst disk check



- (1) Burst disk
- (2) High load seal

- Visually inspect the internal metal membrane of the burst disk for damage (pos.1).
- Leak check the high load seal (pos. 2).



WARNING

Risk of air ingress

If the auxiliary burst disk is broken, IMMEDIATELY fit the blanking gasket seal.

- » The blanking gasket seal is supplied in the magnet spares kit.
- » Order a replacement auxiliary burst disk and fit before ramping the magnet.



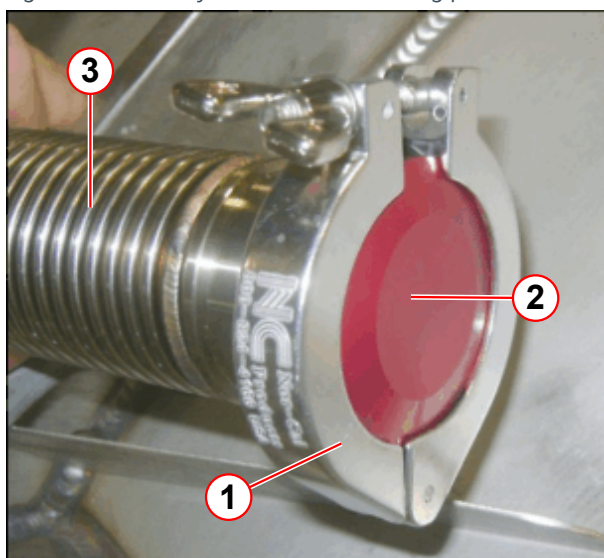
WARNING

Safety device removed

The magnet must not be ramped with the auxiliary burst disk removed or broken.

- » Replace the auxiliary vent burst disk before ramping.

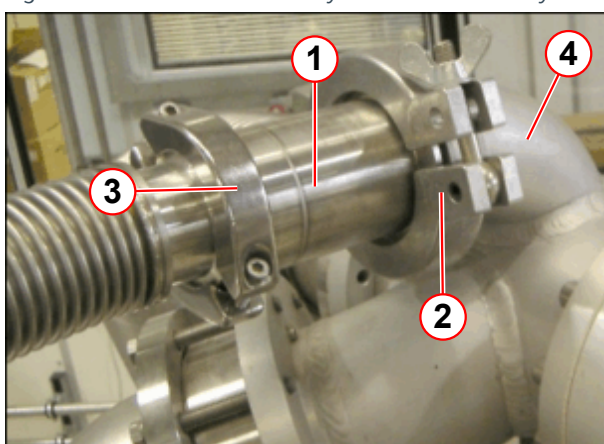
Fig. 483: Auxiliary vent - NW40 blanking plate



- (1) NW40 clamp
- (2) NW40 red blanking plate
- (3) Auxiliary vent line

- If the membrane has ruptured, disconnect the high load clamp and remove the burst disk from the vent line.
- Fit the red anodized blank immediately from the spares kit.
- Order a replacement auxiliary burst disk. Fit as soon as available.
- With the blank fitted allow magnet to pressurize.
- Leak check red blanking plate and seal on the end of the auxiliary vent line.

Fig. 484: OR105 Mk1 Auxiliary burst disk assembly



- (1) Aux burst disk assy
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

For the Mk1 venting:

- Connect the auxiliary burst disk to the manifold assembly with a NW50 clamp and seal.

Fig. 485: Auxiliary vent interface - O-ring seal



- (1) Quench valve manifold
- (2) O-ring seal
- (3) Auxiliary bursting disk
- (4) NW40 clamp (high pressure)

- Check that the O-ring seal (Pos.2) is not damaged and that the surface is clean.
- If damaged, replace the O-ring.
- Connect the auxiliary burst disk to the manifold assembly with the V-clamp.

8.5.6 OVC Burst Disk

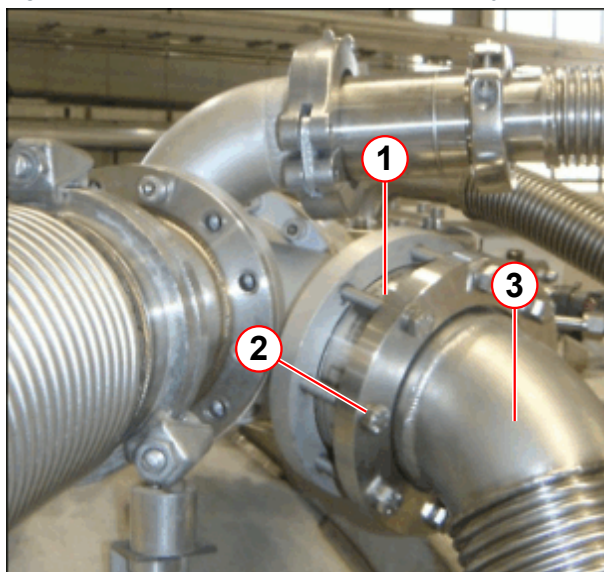


The OVC (DOP) port burst disk and vent line has been removed from the Mk2 magnet design. See Table for system serial number change.

Tab. 24 Venting configurations

Venting	OR105 Avanto serial #	OR122 Espree serial #
Mk1	Before System MR25575	Before System MR30125
Mk2	From System MR25575	From System MR30125
Mk 2 (No DOP)	From System MR57232	From System MR31427

Fig. 486: OR105 Mk1 OVC burst disk assembly



- (1) OVC burst disk
- (2) Screws (8 off)
- (3) OVC vent tube

Mk 1 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture.

- During the post quench checks the burst disk can inspected when the venting removed to inspect the quench valve.
- Remove the (8 off) **screws** holding the OVC burst disk in place.



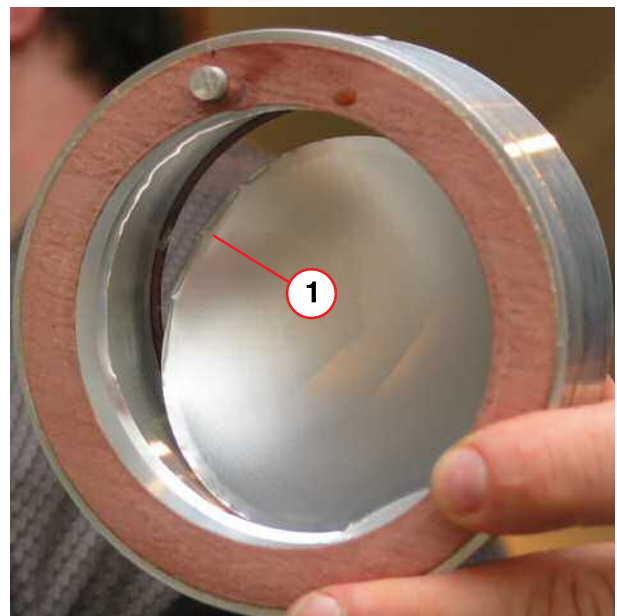
The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 487: OVC burst disk



- (1) OVC burst disk

Fig. 488: OVC burst disk

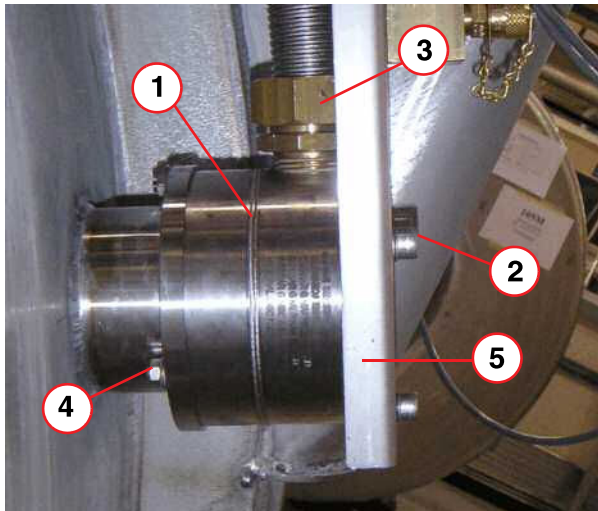


- (1) Broken OVC burst disk



The burst disk has a metal peg in one face to ensure correct orientation of the disk.

Fig. 489: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

Mk 2 OVC burst disk

A quench will not cause the OVC metal burst disk to rupture. The OVC burst disk requires checking:

- During routine safety checks and ice inspection of the venting.

8.5.7 Mobile Systems - High Altitude Burst Disk



Mobile systems are fitted with a high altitude burst disk, in addition to the quench valve burst disk.

If the magnet quenches, the high altitude disk will rupture and must be replaced.

Spare Parts

Post quench the high altitude burst disk will need to be replaced and the O-ring seal checked.

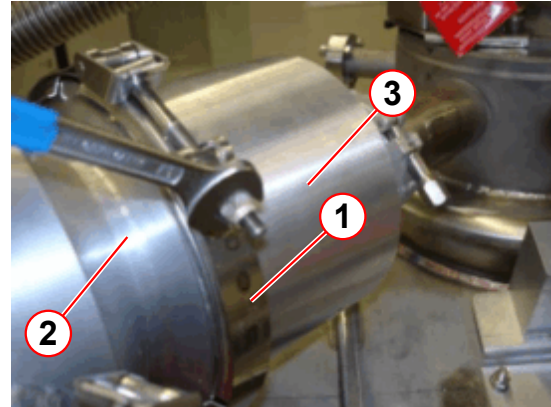
- 10 psi burst disk (Composite) - (material number **10129467**).
- O-ring seal - (material number **10113389**).

8.5.7.1 High Altitude Bursting Disk Removal

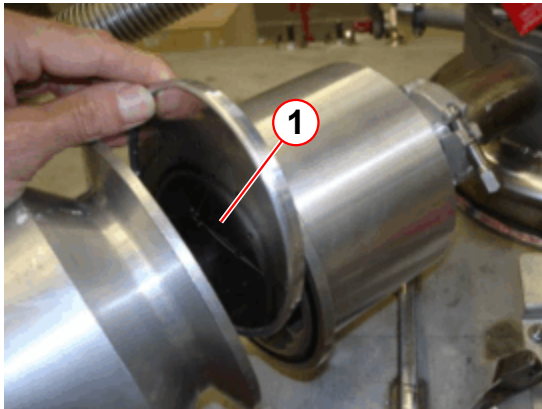
- Undo the M6 screw and nut and remove the retaining clamp on the manifold support bracket.
- Undo the M6 nut and release the V-clamp securing the manifold assembly to the quench valve.

Fig. 490: Manifold Support Bracket

(1) Support Clamp

Fig. 491: Manifold V Clamp

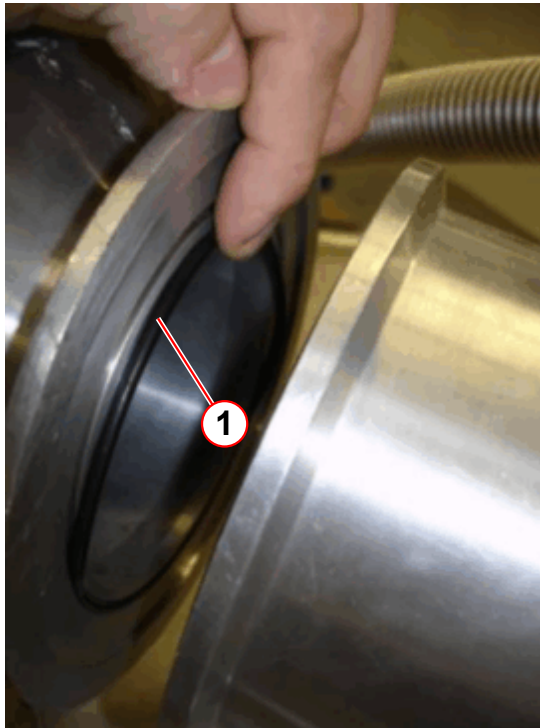
(1) V Clamp
(2) Manifold
(3) Quench Valve

Fig. 492: Removing the High Altitude disk

(1) Burst Disk

- Carefully pull the manifold away from the quench valve.
- The burst disk is held in position by the V-clamp seal. There are no screws.
- Remove the ruptured high altitude burst disk.

Fig. 493: O Ring Fitment



(1) O Ring and Groove

- Remove the O-ring in the manifold flange.
- Clean the flange mating surfaces.
- Complete the quench valve burst disk checks.



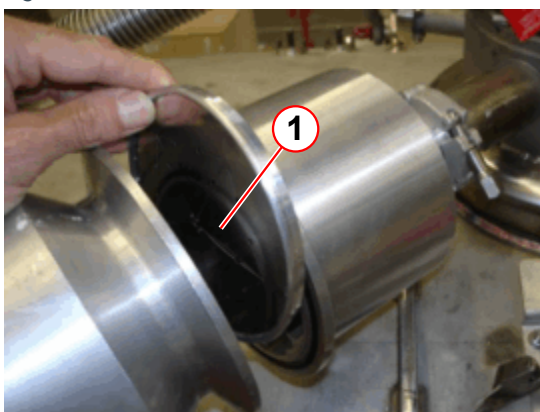
As part of the Post Quench checks at this stage the standard quench valve and graphite burst disk checks will be completed.

8.5.7.2 Fitting the High Altitude Bursting Disk



Take care not to touch the center of the disk. It can be damaged easily.

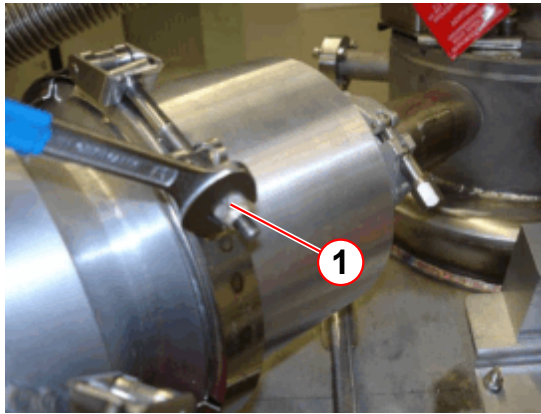
Fig. 494: Burst Disc Fitment



(1) Burst Disc Words "VENT SIDE"

- Check that the manifold O-ring is clean and not damaged.
 - » Replace if necessary.
- Check that the 2 flange surfaces are clean.
- On the disk, check that the wording 'Vent Side' is facing away from the quench valve.
- Fit the new high altitude burst disk between the manifold and quench valve flanges.

Fig. 495: V Clamp Fitment



(1) Self Locking Nut

- Hold the burst disk in position and join the manifold flange and burst disk to the quench valve.
- Check that the burst disk is still centered and refit the V-clamp.
- Tighten the nut until there is a sharp increase in resistance indicating that the self locking nut is fully tight.

Fig. 496: Manifold Support Bracket



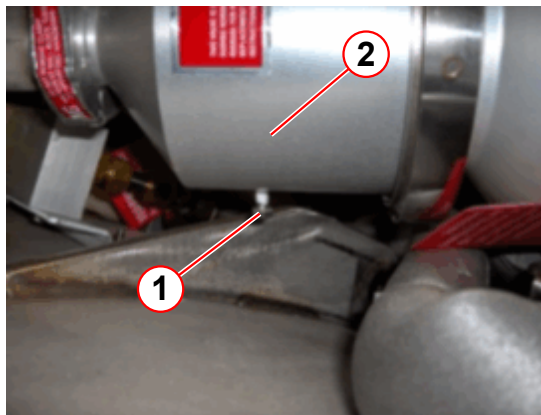
(1) M8 Nuts & Bolts

- Tighten the screw and M6 nut on the manifold support bracket.



At high altitudes the venting is only effective with the drain screw fitted in the quench valve.

Fig. 497: Drain Screw



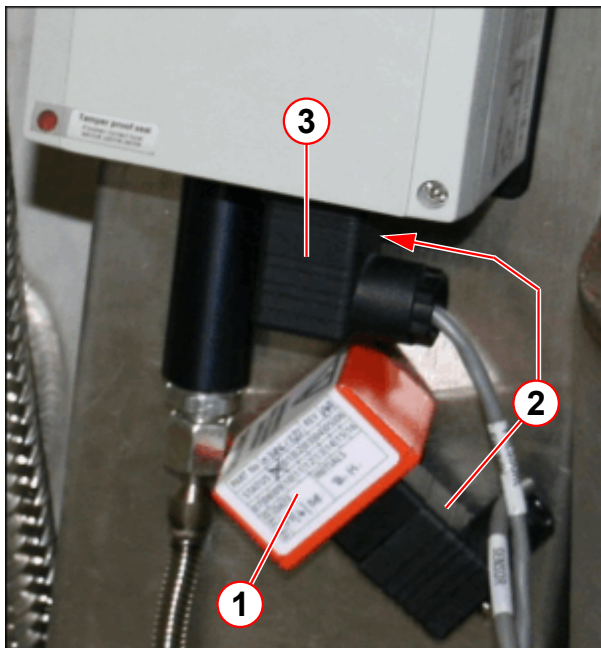
- (1) Drain Screw
- (2) Quench Valve

- Check that the drain screw is fitted in the quench valve body.

8.5.8 Final Work Steps

8.5.8.1 Reactivate the Pressure Heater

Fig. 498: Remove the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

Re-configure the system to enable the pressure heater depending on method used:

1. Switch ON the MSUP at circuit breaker F3 (fuse) in the ACC.
2. Remove the pressure service tool from the pressure sensor cable. Re-fit the connector to the pressure display unit.

- Access the Service Software at the MRC (MRAWP) Host.
- Select the menu item Magnet & Cooling - Magnet Status.
- At MSUP Control - Set the pressure heater mode to **On 100%**.
- Select SAVE.

- Monitor the magnet pressure.
 - Select Update to refresh data until pressure reaches ≥ 15.3 psi (A).



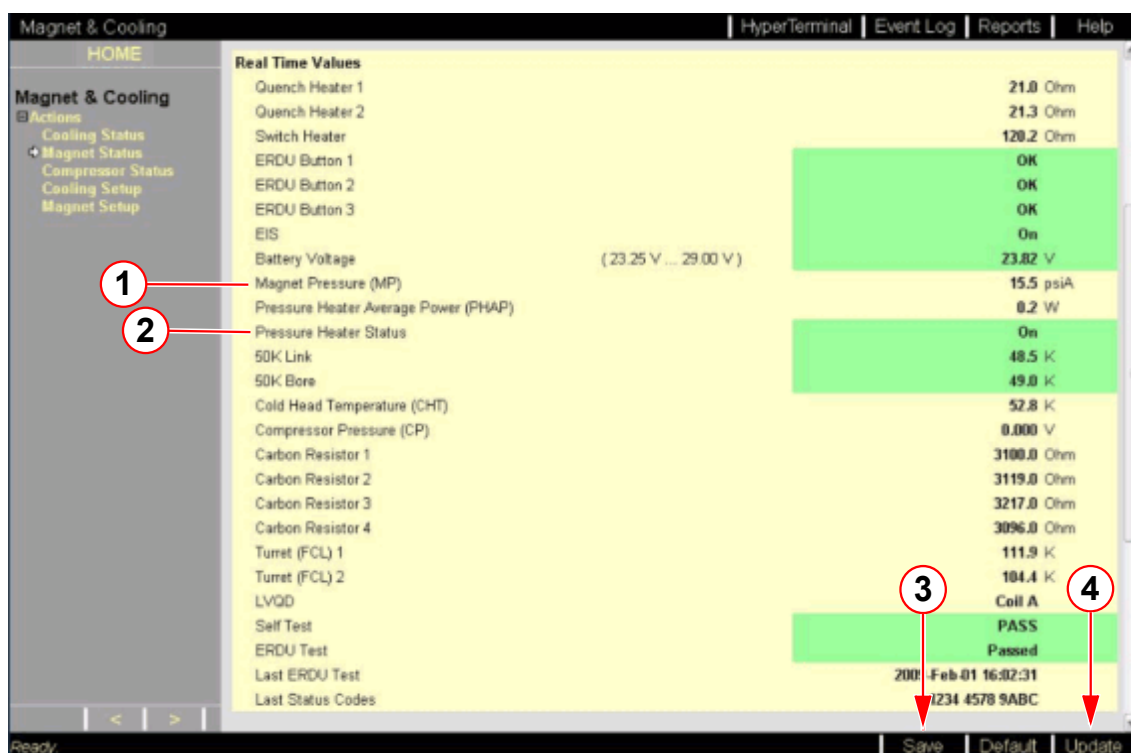
Data is only accurate from the last refresh/update of the page, or from when the page was opened last.



During pressurization, the cold head will switch on automatically.
If the cold head was turned off at MRC, select 'pressure heater status'.

- At MSUP Control - Set the pressure heater mode to Automatic mode.
- Select SAVE.

Fig. 499: Check magnet pressure level

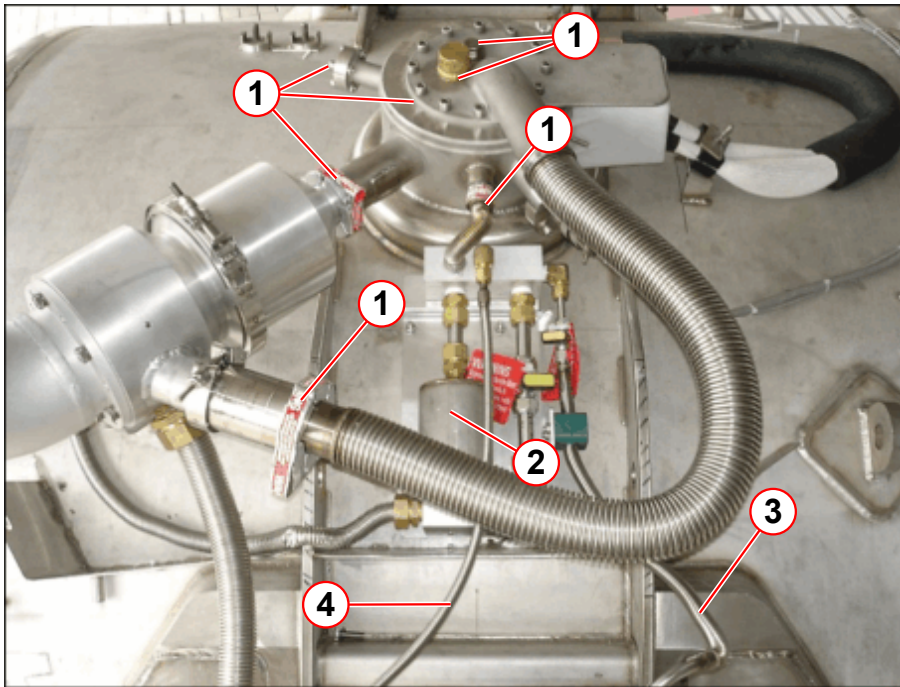


- (1) Magnet pressure (MP)
- (2) Pressure heater status
- (3) Save button
- (4) Update button

8.5.8.2 Final Checks

- Helium fill magnet to the required level for ramping the magnet.
- Order replacement components if the spare parts have been used (burst disk, seals).
- Perform a leak check of the turret venting and components.
 - » All vent pipes and flanges on the high pressure side of the helium vessel.

Fig. 500: OR105/122 - Turret and venting leak checks (Example Mk2)



- (1) Turret top plate, syphon port and O-rings
- (2) 16 psi (A) vent valve
- (3) Cold head line
- (4) Pressure gauge line



Allow a minimum of 2 hours after filling before ramping the magnet.

8.5.8.3 Quench Report

- Fill out the Quench report - MR - Quality Database and Quench Report.

8.5.8.4 TSG - Checking the Carbon Resistors

The Ceramic Carbon Resistors (CCR) values can be checked from the MSUP or manually by using a multimeter.

- The measurements are taken across pins **a-e**, **b-e**, **c-e**, **d-e** on the 41-pin turret connector.



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » The magnet must be **RAMPED DOWN** before attempting this procedure.

Tooling requirements

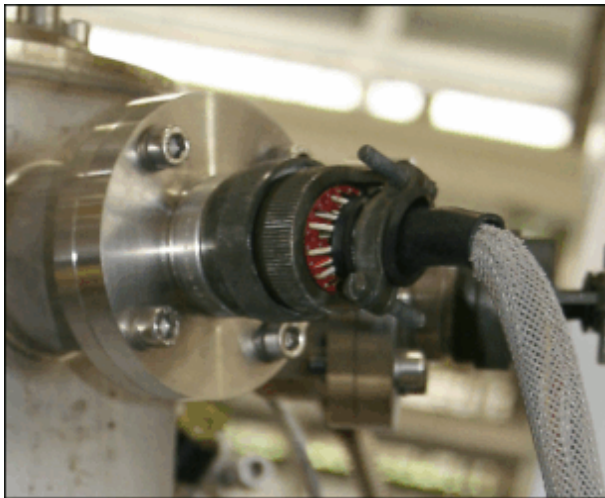
- Multimeter and pin leads.
- Laptop/PC with Terminal Emulation software, (PuTTY).
- RS232 cable.

Procedure

Using a multimeter:

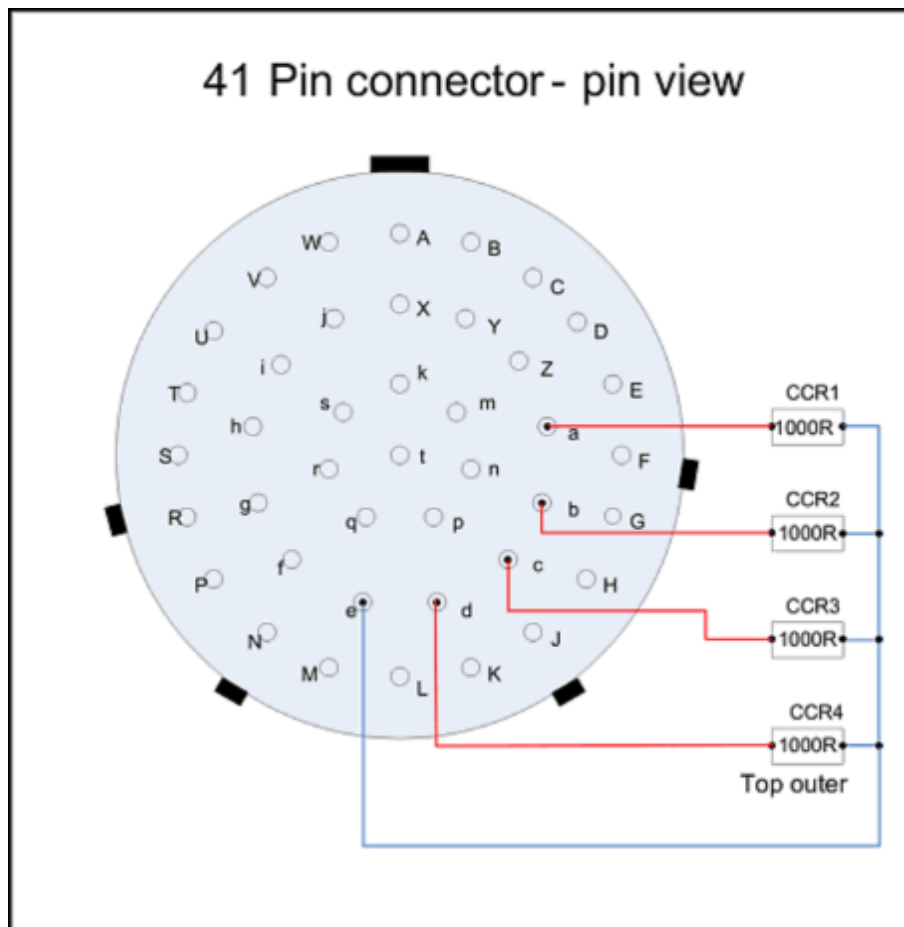
- Connect the pin leads.
- Turn the mode selector to resistor mode Ω .

Fig. 501: Turret - 41pin connector



- Disconnect the 41-pin connector.
- Measure across pins **a-e**, **b-e**, **c-e**, **d-e**
- Compare the values with the Magnet Acceptance Document.
- Re-connect the 41-pin connector.

Fig. 502: 41pin turret connector - ceramic carbon resistor pins (room temperature values)



9.1 Check the mechanical adjustment of the patient table



For the mechanical adjustment of the patient table wear protective gloves.

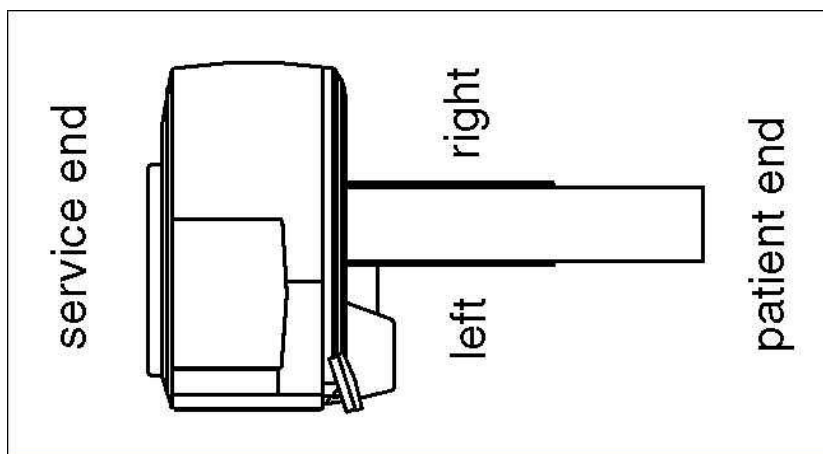


The mechanical adjustment of the patient table may be influenced by the magnetic field. For this reason the final mechanical patient table adjustment has to be performed with the magnet at field.



The directions of the adjustments e.g. left, right, relates to the front view (patient end) of the magnet.

Fig. 503: View directions from patient end to the magnet



9.1.1 Check the free horizontal reference moving of the patient table



To prevent problems when referencing the patient table along the vertical axis, ensure that the table can find the reference point within the 3-8 mm clearance between the support frame and magnet cover without any obstruction.



Perform this test prior to starting up the patient table by referencing the highest, center, and lowest positions. Also, ensure that the tabletop does not collide with the covers.

- Visually examine the covers along the seam to the table. Ensure that they are LEVEL.
- Monitor vertical movement(s) at the narrowest position between the table and covers. There should be a clearance of **3-8 mm**, remain at this position.
- Power off the table, manually (F22) move the tabletop to the outer mechanical end stop.
- Power on the table and monitor the subsequent automatic tabletop movement for referencing: During the reference phase, the tabletop moves toward the magnet or

cover. This should result in the table reference position at the front of the support frame.

- **Check:** The front of the table should be flush with the front of the support frame.



The table should NOT touch the cover. Shortly before reaching the cover, the table should move away from the magnet.

- Reboot



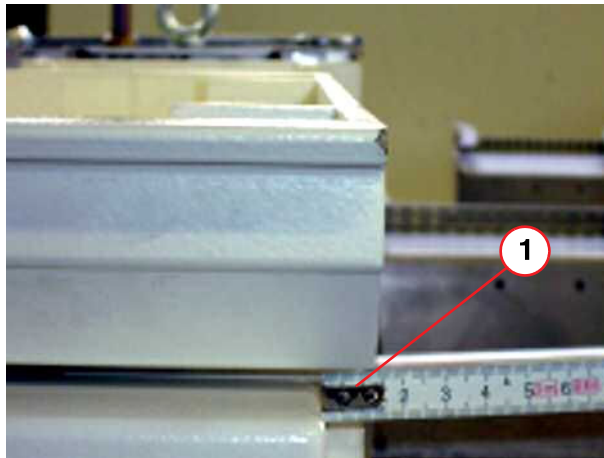
If the clearance between the table and the magnet cover is not between 3-8 mm, the patient table needs to be adjusted to the magnet.

9.1.1.1 Checking the reference switch S10 of the patient table



After each readjustment, circuit breaker F22 (ACC RF-Room) has to be switched off and then on again to ensure that the new reference position is transferred.

Fig. 504: Adjustment value for threaded pin (example)

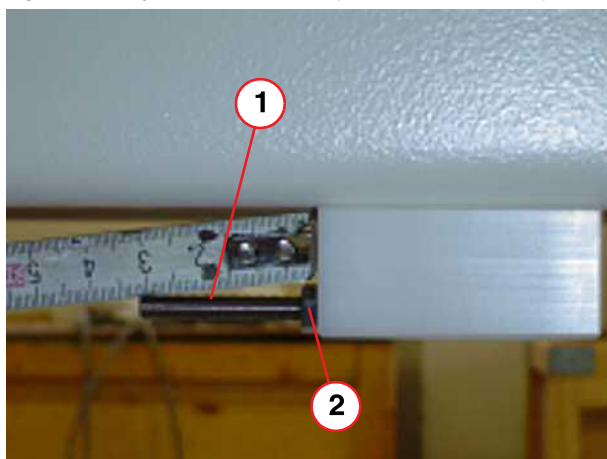


(1) Table excess length

Checking and setting the reference switch S10:

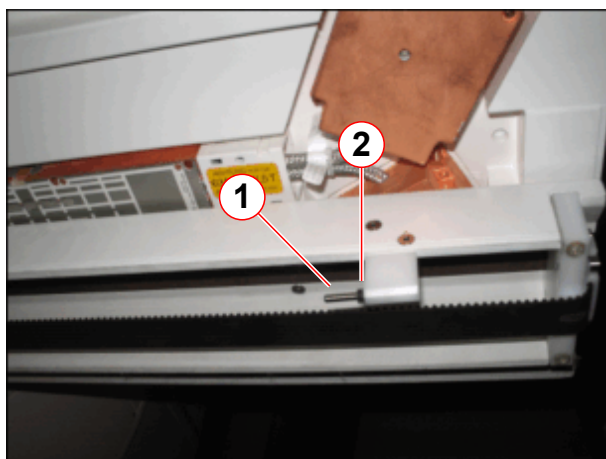
- Move the table into the home position by pressing the home position key.
- Measure the projection of the patient table.
- Move patient table all the way through the magnet.

Fig. 505: Adjustment value required for threaded pin



- (1) Threaded pin
(2) Counter nut

Fig. 506: Reference point adjustment (partial drive unit)



- (1) Threaded pin
(2) Locknut

- Turn the threaded pin (1) by the adjustment measured and tighten the counter nut (2).
- Move the patient table forward (home position).
- Switch off fuse F22 (RF room) in the ACC.
- Switch on fuse F22 (RF room) in the ACC (reference search).
- Check whether the patient table is aligned in vertical direction with the support arm. Also ensure that the patient table does not collide with the panels when moving it down (distance 3-8 mm).

Readjust, if necessary.

9.1.2 Checking the distance between patient table and cover

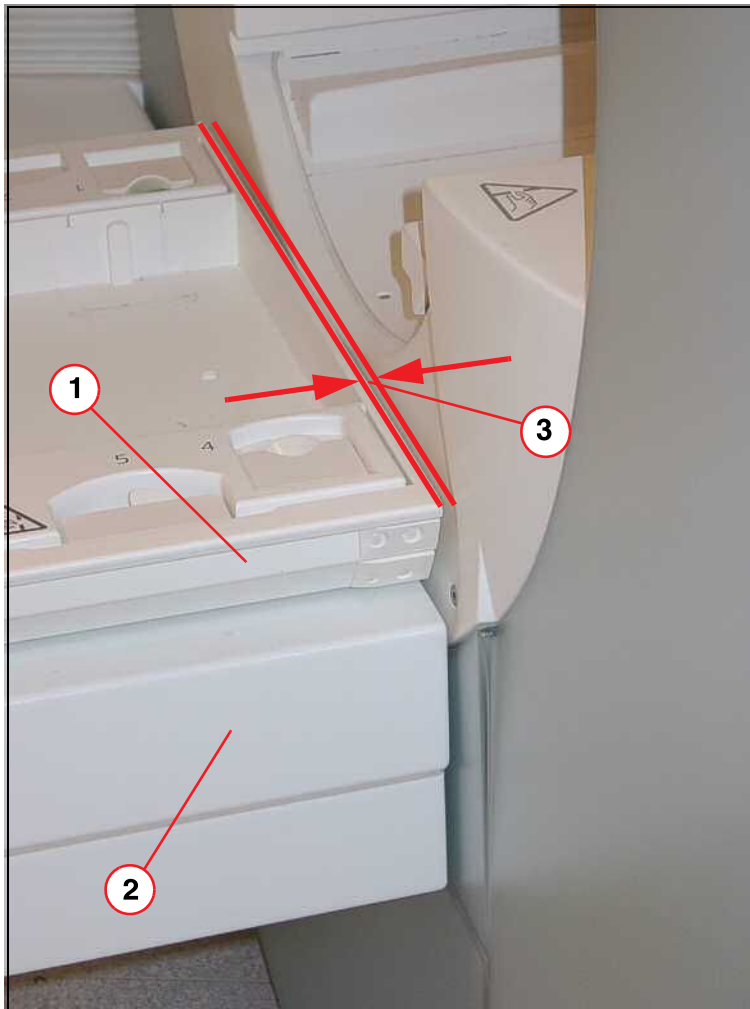


The translucent front cover may not be exactly vertical. Therefore, the translucent front cover and table may not be in parallel. In this case, do not try to compensate for the angle between the table and the cover using the X-radial adjustment procedure because this will also change the travel path of the tabletop inside the magnet bore.



The following pictures provide general information on how to perform the distance checks. The cover design differs slightly among Tim systems.

Fig. 507: Clearance, vertical gap



- (1) Tabletop
- (2) Support frame
- (3) Clearance < 8 mm between table and magnet cover

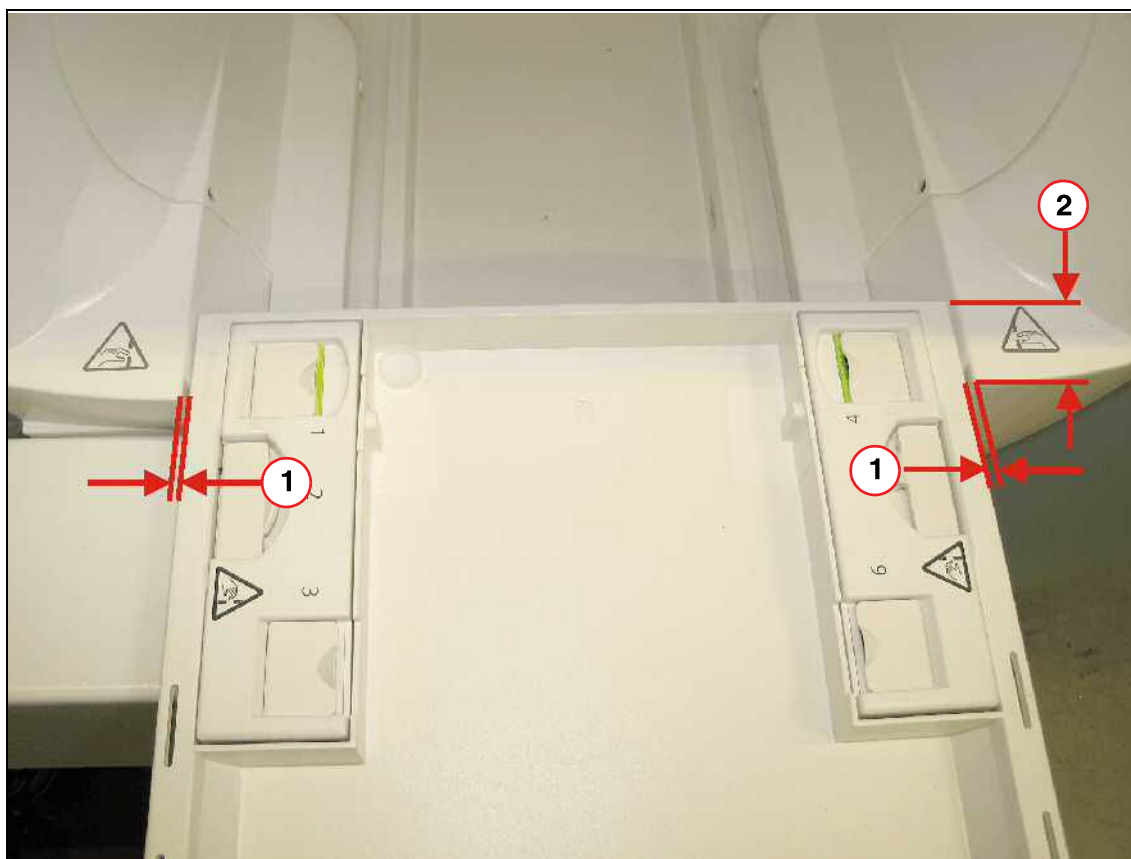
1. Move the patient table up to the highest position and then lower it about 80 mm.
2. Measure the distance of the vertical gap (→ 3/ Fig. 507 Page 411) between the tabletop and the magnet cover.
 - » The distance between the patient tabletop and the magnet cover (→ 3/ Fig. 507 Page 411) has to be within the range of **3-8 mm**.
 - » If the distance is not within the range, adjustment of the patient table is required.

The patient table should run in parallel to the Body Coil / covers with a sufficient gap across the entire patient board to avoid scratching.



There is always a clearance between the exchangeable tabletop and the table carrier, as well as between the table carrier and the slide way. For this reason, the exchangeable tabletop has to be pushed as described.

Fig. 508: Case 1: Checking the distance between patient table and cover (patient end)



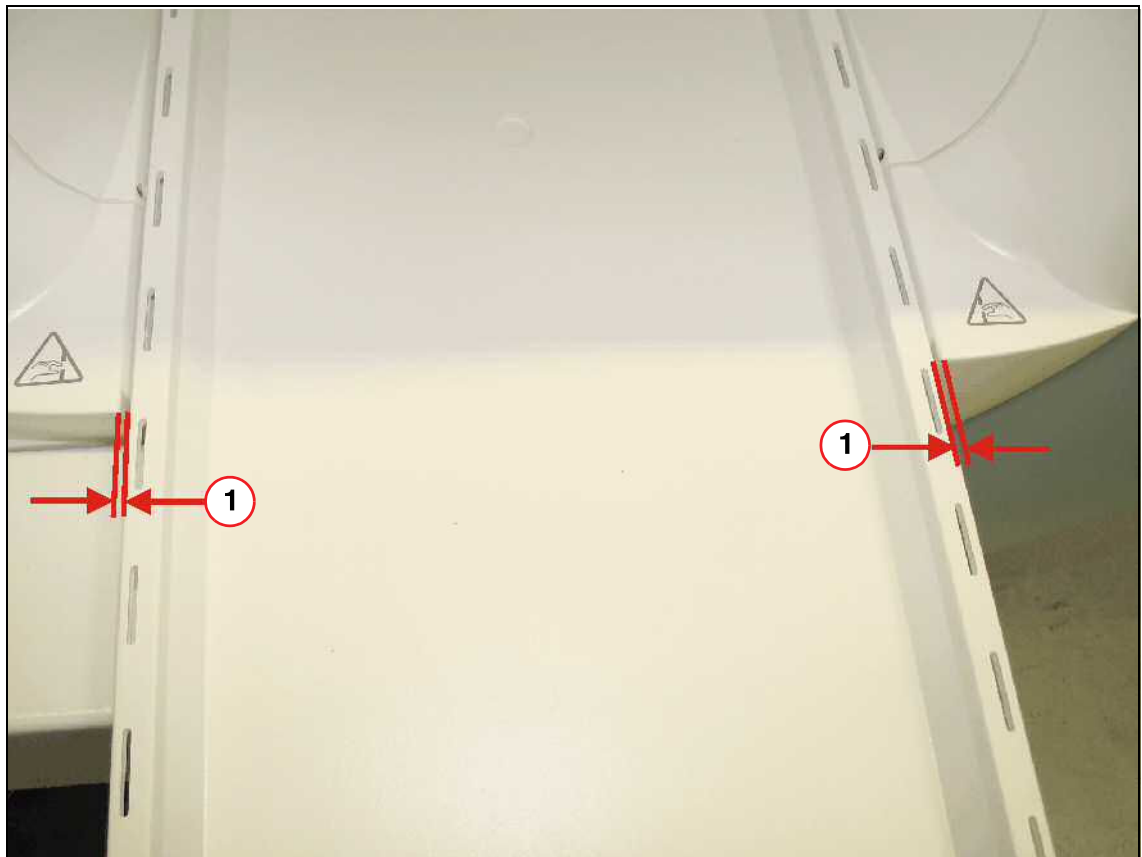
(1) Distance to be measured

(2) Approx. 50 mm

Case 1:

1. Drive the patient table approx. 50 mm (Symphony, A Tim System: approx. 400 mm) into the magnet.
2. Push the tabletop (exchangeable tabletop) to the **left** side and check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop to the **right** side and check the distance of < 8 mm on the **left** side, at the front of the magnet.
 - » The distance in both tests has to be within the range of < 8 mm.
 - » If the distance is not within the range of < 8 mm, adjustment of the patient table is required.

Fig. 509: Case 2: Checking the distance between patient table and cover / body coil (patient end)

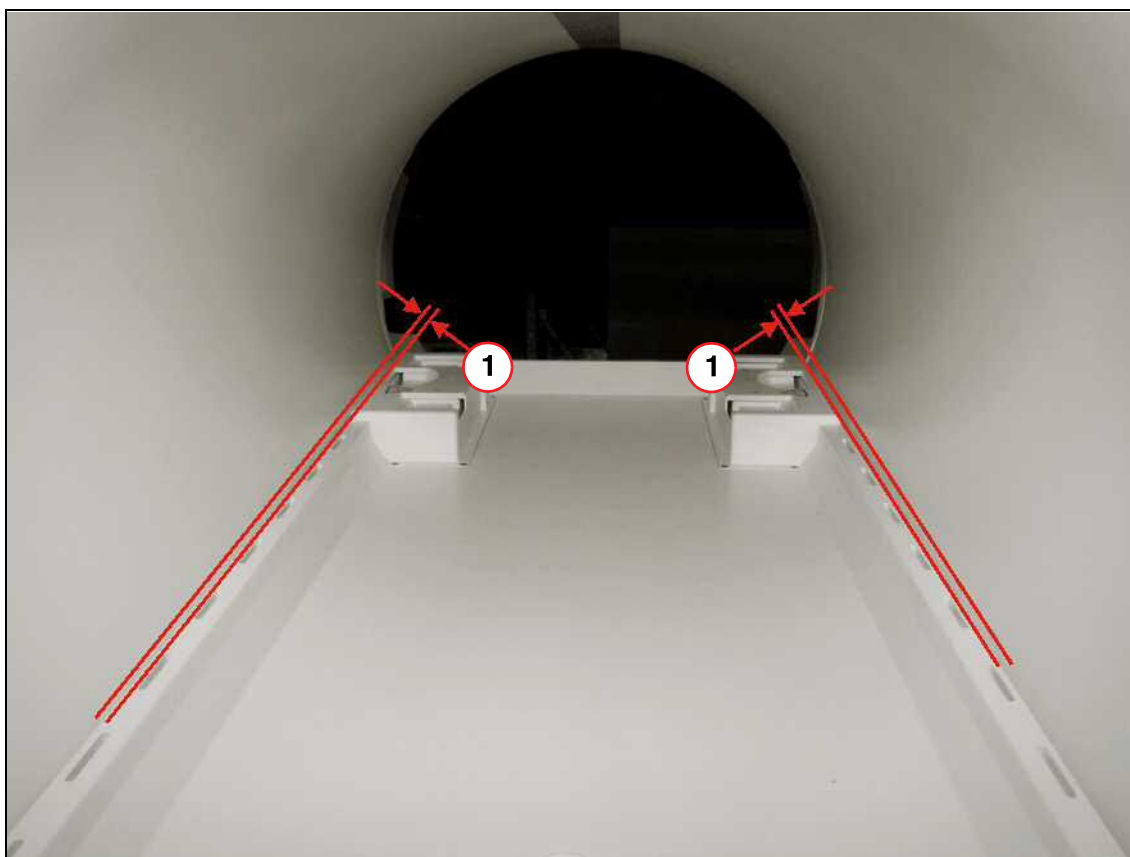


(1) Distance to be measured

Case 2:

1. Drive the patient table approx. 1200 mm into the magnet.
2. Push the tabletop (exchangeable tabletop) to the **left** side and check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop to the **right** side and check the distance of < 8 mm on the **left** side, at the front of the magnet.
 - » The distance in both tests has to be within the range of < 8 mm.
 - » If the distance is not within the range of < 8 mm, adjustment of the patient table is required.

Fig. 510: Case 3: Checking the distance between patient table and Body Coil / cover (service end)



(1) Distance to be measured

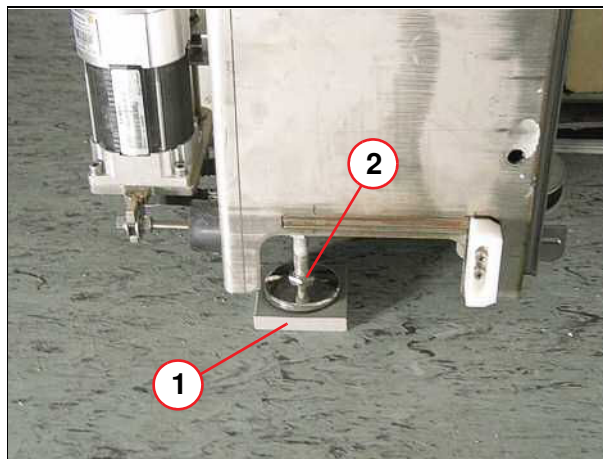
Case 3:

1. Drive the complete patient table into the magnet.
2. Check the distance of < 8 mm between the patient table and cover / body coil on both sides along the entire length of the tabletop.
 - » The distance has to be within the range of < 8 mm.
 - » If the distance is not within the range of < 8 mm, adjustment of the patient table is required.

9.1.3 Remove the covers to adjust the patient table.

- Remove the service access cover
- Removing the covers of the lifting column

Fig. 511: Support Patient Table

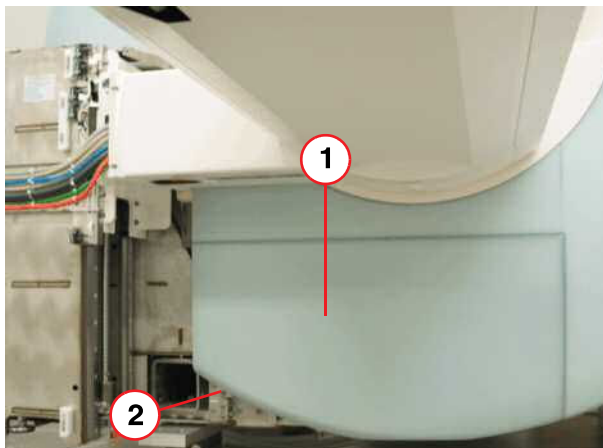


- (1) Sylomer plate (black)
- (2) Support leg

Loosening the support of the patient table

- Loosen (cw) the counter nut and turn up the support leg (2) (ccw).

Fig. 512: Front funnel, bottom



- (1) Front funnel, bottom
- (2) Screw

- Removing the front funnel, bottom (1).

9.1.4 Setting the patient table

The following chapters describe how to set the patient table in the various axes.



CAUTION

Consideration

Orange market screws and washers are extremely magnetic.

- » Do not remove/replace while magnet is ramped.
- » Do not exchange with non magnetic parts.

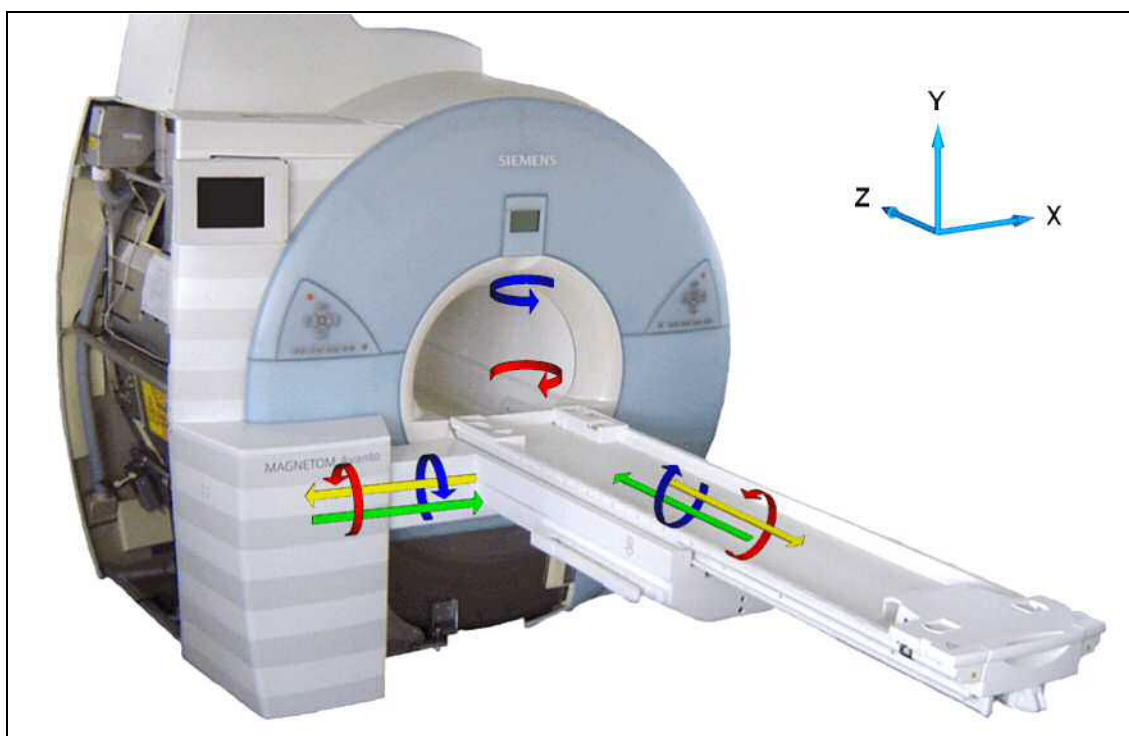


The final adjustment is noticeable only after all screws and counter nuts have been tightened again.



The travel path of the patient table on the slide rails has to be checked under load as well. For example: using one person or a weight.

Fig. 513: Patient table adjustment overview (example Avanto)



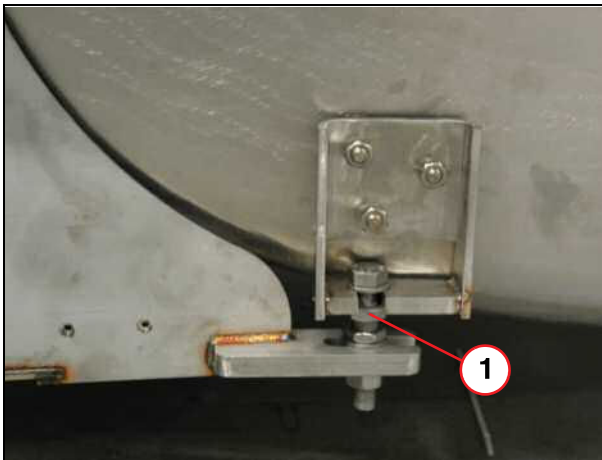
9.1.5 Adjusting the Z-axis

- Position the patient table at the most narrow location.

Version 01:

- Loosen (cw) the counter nut (1).

Fig. 514: Setting screws, front

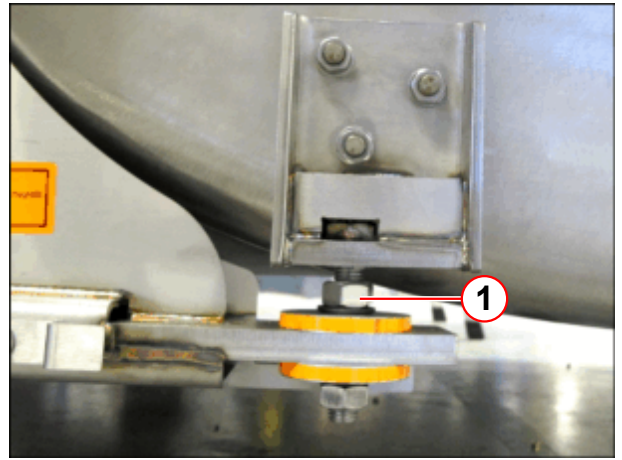


(1) Counter nut

Version 02:

- Loosening (ccw) the counter nuts(1).

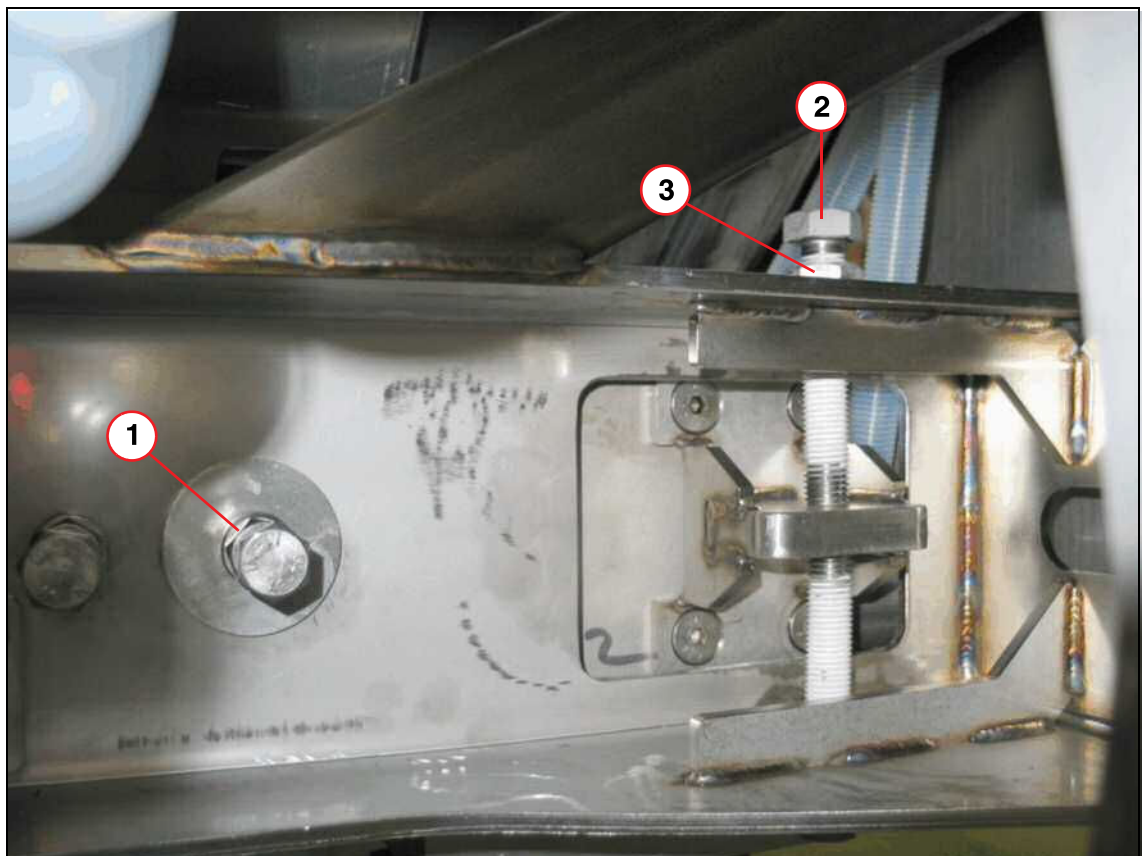
Fig. 515: Setting nuts, patient end (version 02)



(1) Counter nut

- Loosen (ccw) the nut (1,3) and the screw (2).

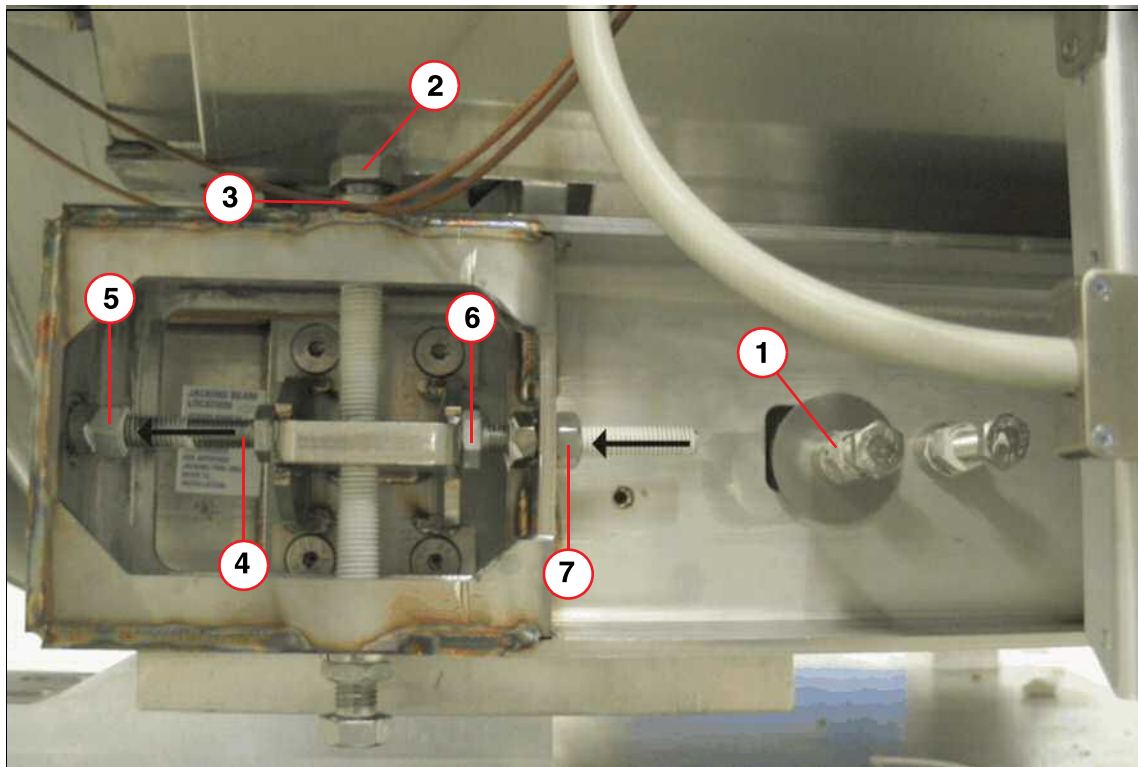
Fig. 516: Patient table, setting screws, front (patient end)



- (1) Setting nut
- (2) Screw
- (3) Counter nut

- Loosen the nuts (1,3,5,7) and the screws (2,4,6).

Fig. 517: Patient table, setting screws, back (service end, Avanto)

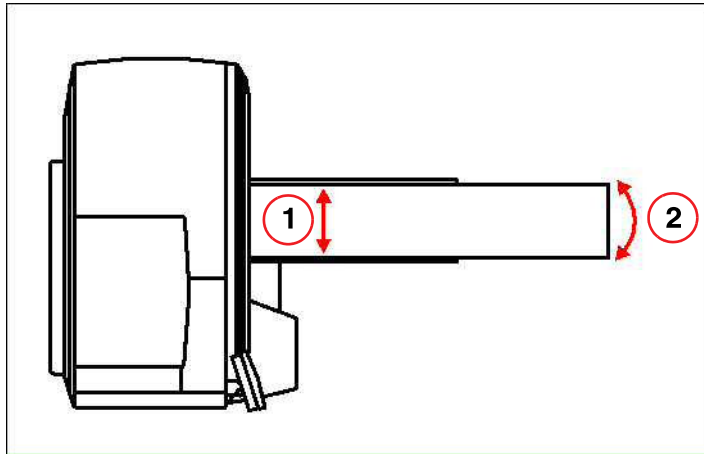


- (1) Setting nut
- (2) Screw
- (3) Counter nut
- (4) Screw (Z-axis)
- (5) Counter nut
- (6) Screw (Z-axis)
- (7) Counter nut

- Set the screws (→ 4,6/Fig. 517 Page 418) until the distance between the patient table and the magnet covers is between **3-8 mm** .
 - To enlarge the distance between patient table and magnet cover:
 - » Turn the screws (→ 4,6/Fig. 517 Page 418) **in** the direction of the arrow. (cw).
 - To shorten the distance between patient table and magnet cover:
 - » Turn the screws (→ 4,6/Fig. 517 Page 418) **counterclockwise** to the direction of the arrow. (ccw).
- Tightening the loosened screws and counter nuts. (→ 1-3/Fig. 516 Page 417)and (→ 1-7/Fig. 517 Page 418).
- Tighten the patient table support again.
- Check whether the distance is **3-8 mm**.

9.1.6 Setting the X-axis and Y-axis radial

Fig. 518: Top view, X-axis and Y-axis radial



- (1) X- axis
- (2) Y-axis radial

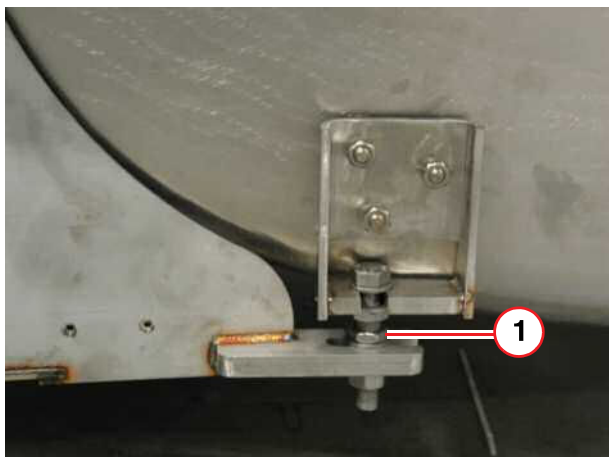
- Remove the covers as previously described under "removing covers."

- Loosen nut (1) (ccw)

Avanto:

Version 01:

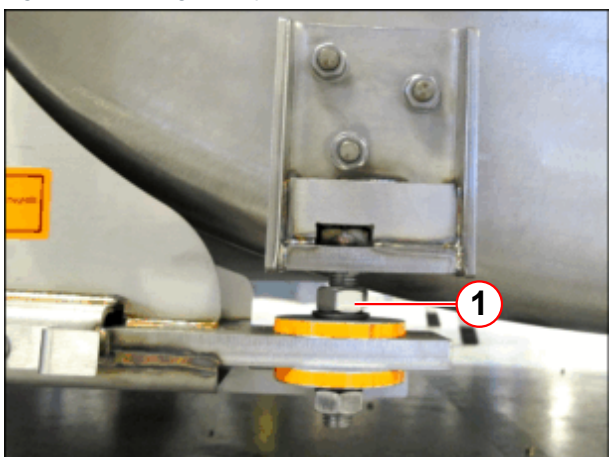
Fig. 519: Setting screw, patient end



(1) Nut

Version 02:

Fig. 521: Setting nuts, patient end (version 02)

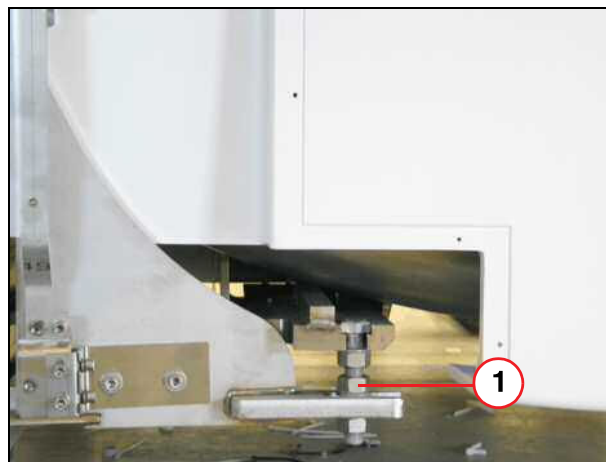


(1) Counter nut

Espree:

Version 01:

Fig. 520: Setting screw, patient end



(1) Nut

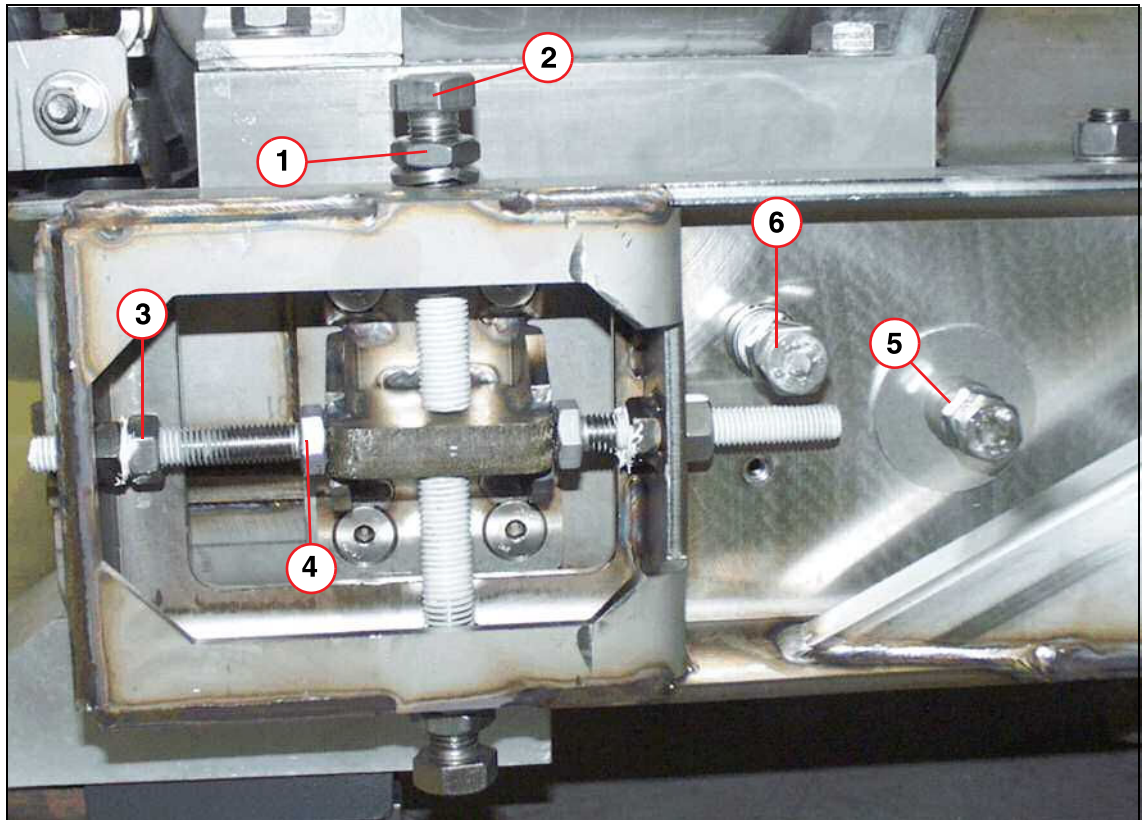
- Loosen nuts (1,3,5) and screws (2,4).

Espreo: (→ Fig. 522 Page 421)

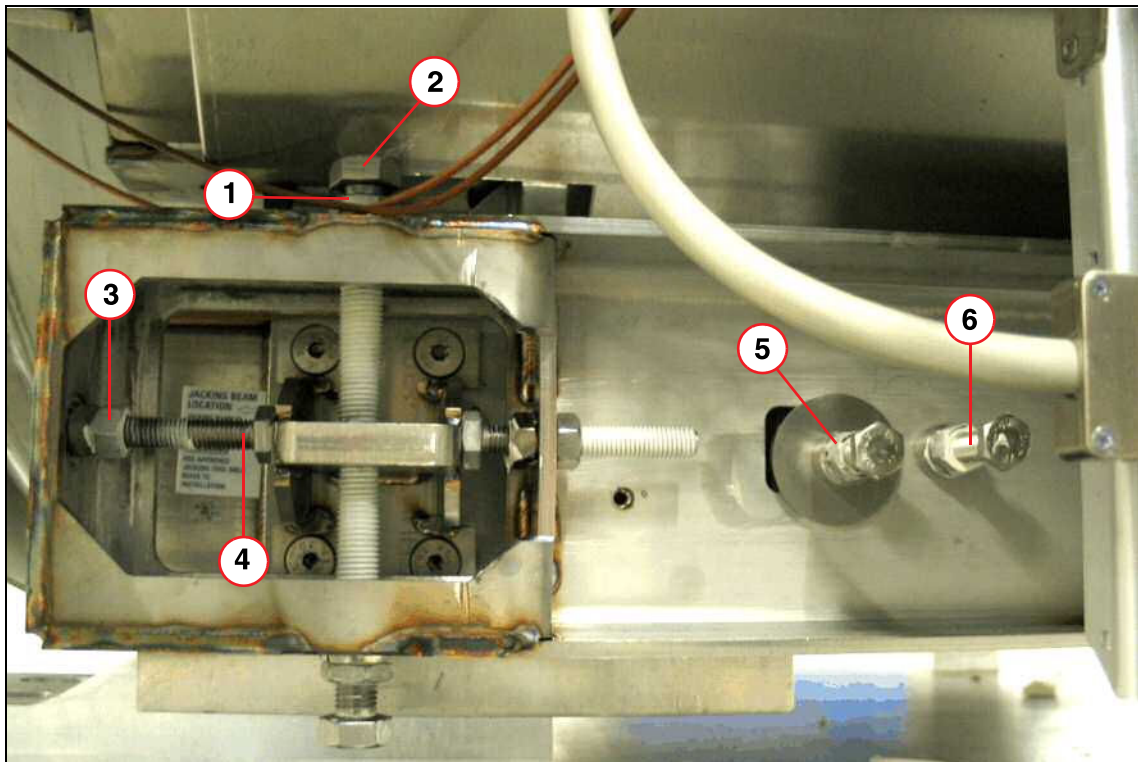
Avanto: (→ Fig. 523 Page 422)

Espreo:

Fig. 522: Patient table, setting screws, back (service end, Espreo)



- (1) Counter nut
- (2) Screw
- (3) Counter nut
- (4) Screw
- (5) Setting nut (X-axis / Y-axis radial) - pull
- (6) Counter nut and screw (X-axis / Y-axis radial) - push

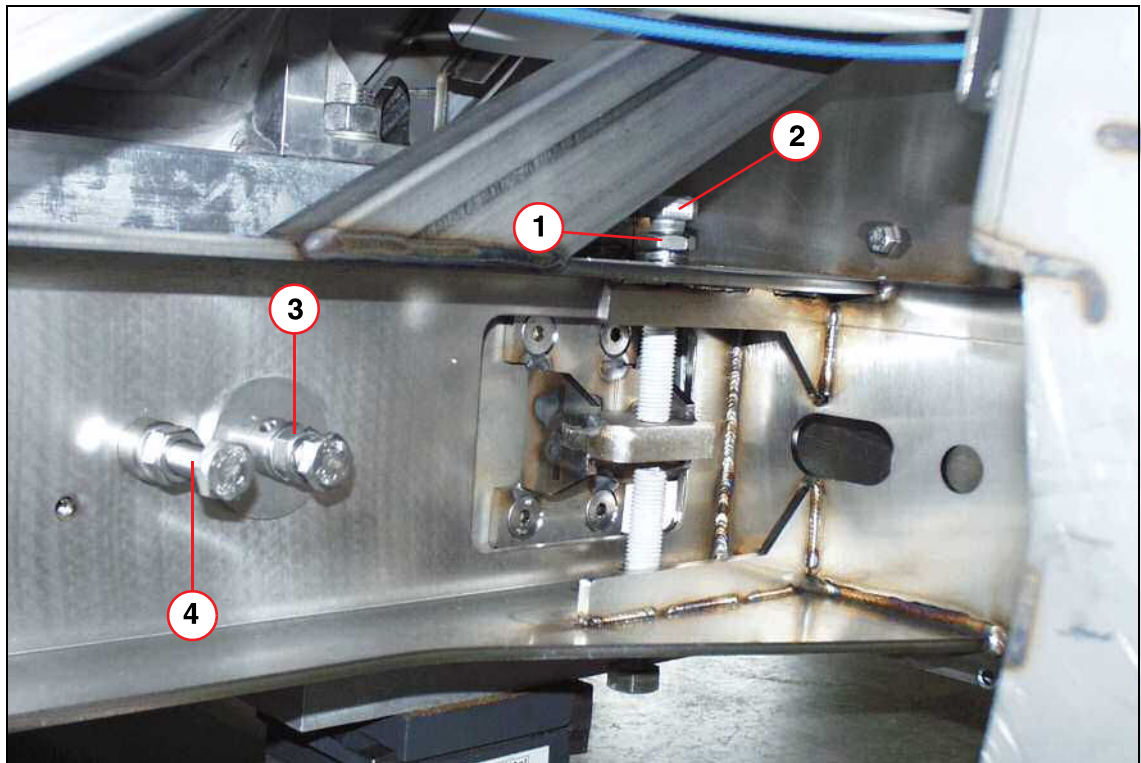
Avanto:*Fig. 523: Patient table, setting screws, back (service end, Avanto)*

- (1) Counter nut
- (2) Screw
- (3) Counter nut
- (4) Screw
- (5) Setting nut (X-axis / Y-axis radial) - pull
- (6) Counter nut and screw (X-axis / Y-axis radial) - push

- Loosen (ccw) the nut (1,3) and the screw (2).

Avanto/Espree:

Fig. 524: Patient table, setting screws, front (patient end)

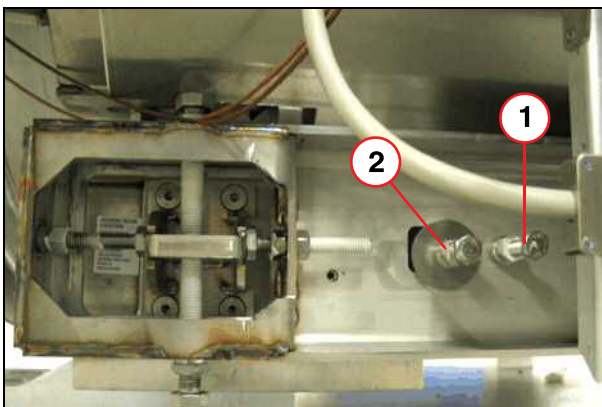


- (1) Counter nut
- (2) Screw
- (3) Setting nut (X-axis / Y-axis radial) - pull
- (4) Counter nut and screw (X-axis / Y-axis radial)- push

Setting:

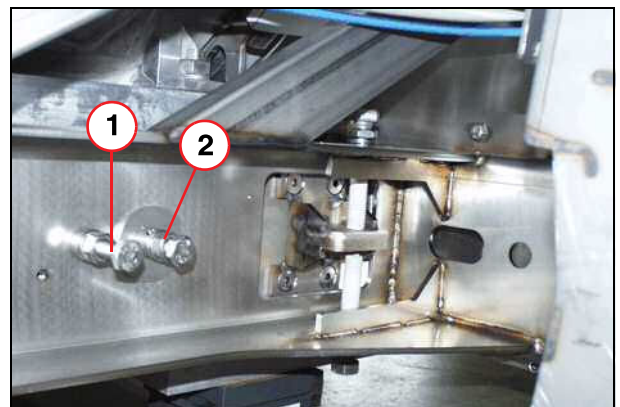
Avanto:

Fig. 525: Setting screws, back (service end, Avanto)



- (1) Screw
- (2) Setting nut

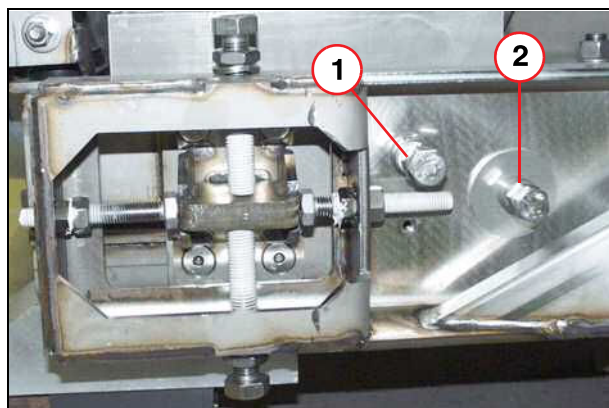
Fig. 526: Setting screws, front (patient end)



- (1) Screw
- (2) Setting nut

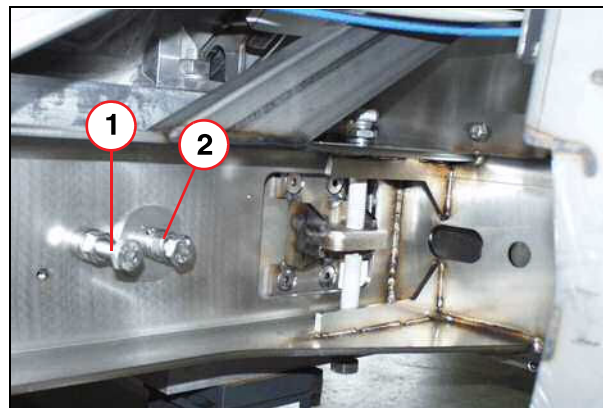
Espre:

Fig. 527: Setting screws, back (service end, Espre)



- (1) Screw
(2) Setting nut

Fig. 528: Setting screws, front (patient end)



- (1) Screw
(2) Setting nut

Use screws(1) and nuts (2) to set the distance between the box section and the attachment frame of the table. This will allow you to set the shift in the X axis and Y axis in the radial direction.

X-axis:

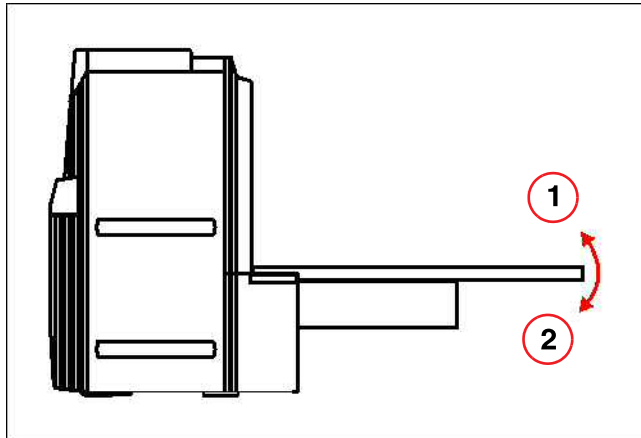
- Move the patient table in **parallel** to the **right** (→ 1/Fig. 518 Page 419):
 - Loosen screws (1) **uniformly** (e.g.: by two turns) and tighten the setting nuts (2).
- Move the patient table in **parallel** to the **left**:
 - Loosen setting nuts (2) **uniformly** and tighten screws (1).

Y-axis radial:

- Turn the patient table to the **right**(→ 2/Fig. 518 Page 419):
 - **Service end:** Loosen setting nut (2) and tighten screw (1), if setting is not sufficient,
 - **Patient end:** Loosen screw (1) and tighten setting nut (2),
- Turn the patient table to the **left**:
 - **Service end:** Loosen screw (1) and tighten setting nut (2), if setting is not sufficient,
 - **Patient end:** Loosen setting nut (2) and tighten screw (1),
- Tighten all setting screws and setting nuts. Use counter nuts to secure the screws.
- Another parallel correction may be necessary after the radial correction.

9.1.7 Setting the X-axis radial

Fig. 529: Side view, X-axis radial



- (1) top
- (2) bottom

- Loosen (ccw) counter nut (1).

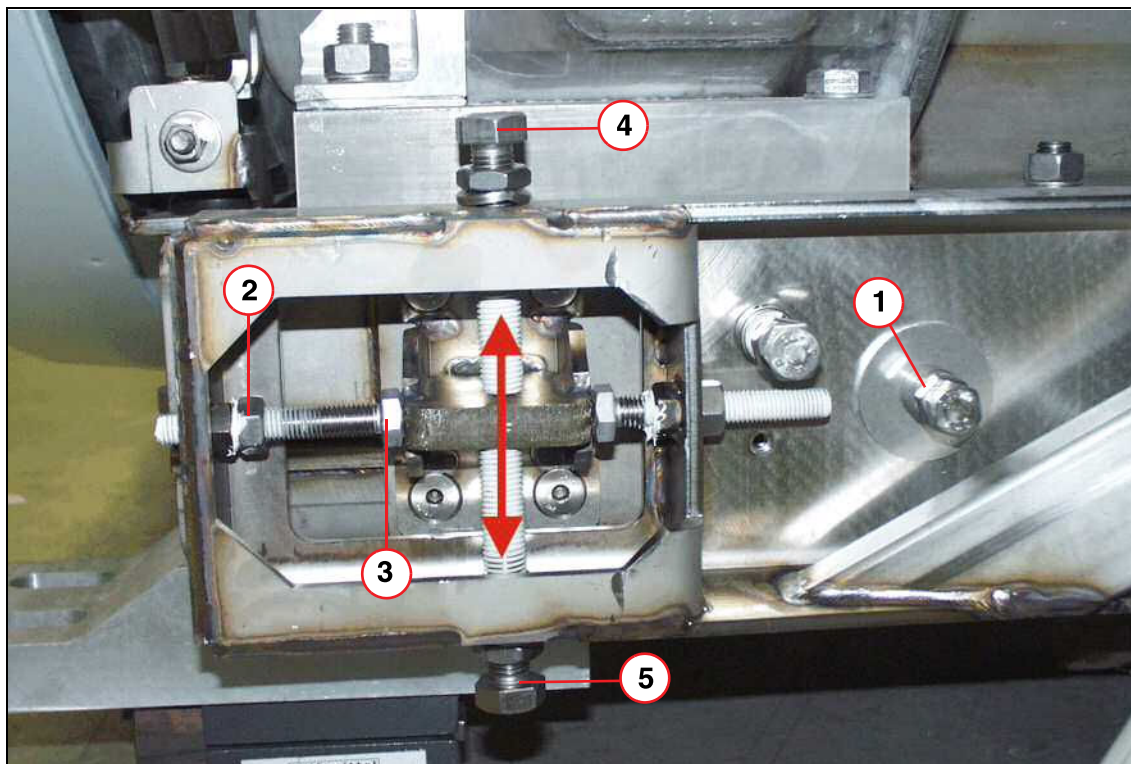
Fig. 530: Patient table, setting screws, front (patient end)



- (1) Setting nut

- Loosen nuts (1,2) and screw (3) as well as all counter nuts of screws (4,5).

Fig. 531: Patient table, setting screws, back (service end, Espree)



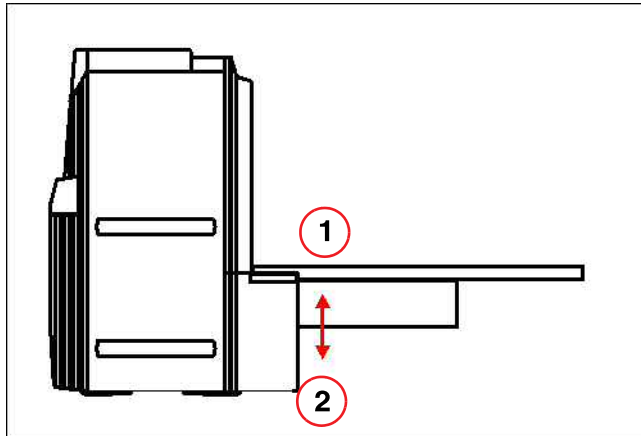
- (1) Setting nut
- (2) Counter nut
- (3) Screw
- (4) Counter nut and screw (X-axis radial)
- (5) Counter nut and screw (X-axis radial)

X-axis radial:

- Turn the patient table to the **top**(in the direction of) (→ 1/ Fig. 529 Page 425)
 - Loosen screw (4) and tighten screw (5).
- Turn the patient table to the **bottom**(in the direction of) (→ 2/ Fig. 529 Page 425)
 - Loosen screw (5) and tighten screw (4).
- Tightening all screws and counter nuts.

9.1.8 Setting the Y-axis

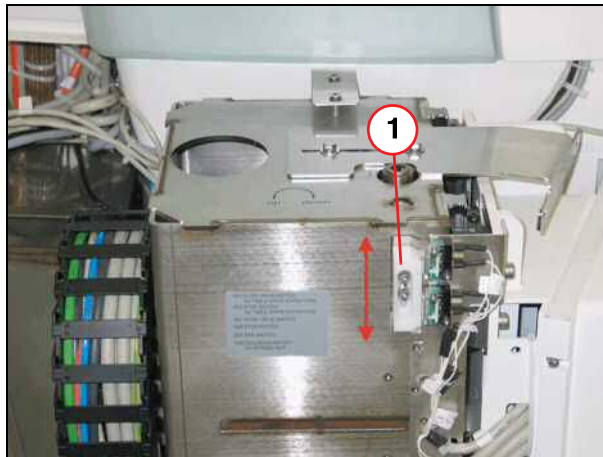
Fig. 532: Side view, Y-axis



- (1) Upper
- (2) Lower

- Loosen the two screws at the end stop (1).

Fig. 533: End stop for micro-switch S20 and S21



- (1) Switch cam

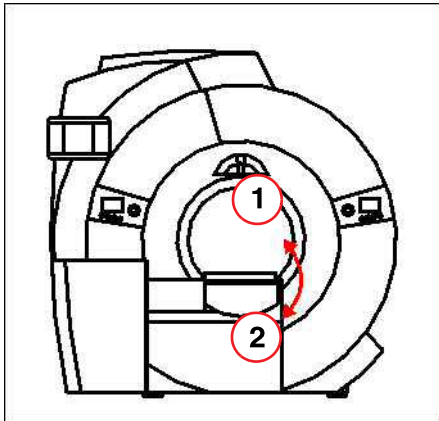
By changing the end stop (up-down) for the microswitch, the upper end position of the patient table is changed (moving-in height of patient table at both sides of the slide rails)

Y-axis:

- Moving-in height of patient table **higher**:
 - Loosen the screws and move the switch cam upward.
- Moving-in height of patient table **lower**:
 - Loosen the screws and move the switch cam downward.
- Tightening the screws.

9.1.9 Setting the Z-axis radial

Fig. 534: Front view, Z-axis radial

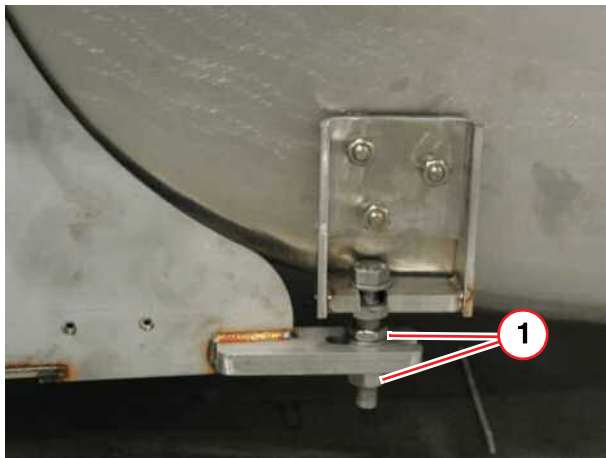


- (1) enlarge
- (2) shorten

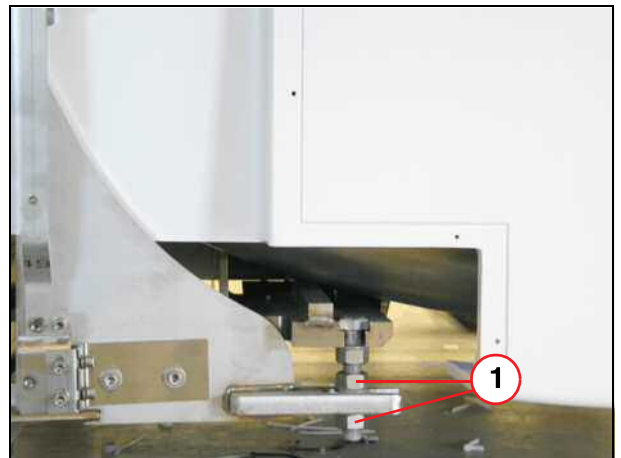
By moving the nuts (1), the patient table is changed in relationship to the slide rails:

Z-axis radial:

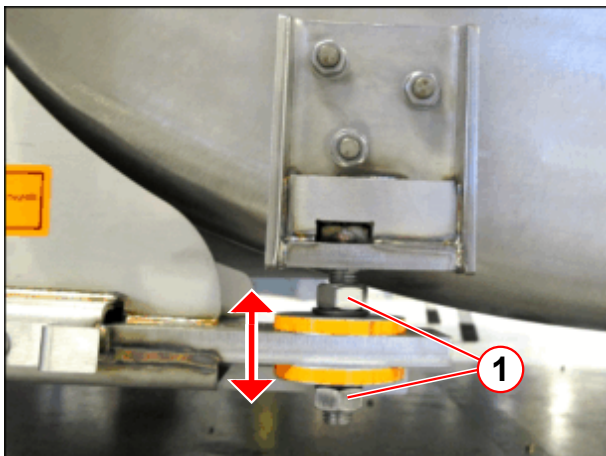
- Distance to the right slide rail **increase**
 - Turn nut upward (1)
- Distance to the right slide rail **shorten**
 - Turn nuts downward (1)

Avanto:**Version 01:***Fig. 535: Setting screw, patient end*

(1) Counter nut (Z-axis radial)

Espre:**Version 01:***Fig. 536: Setting screw, patient end*

(1) Counter nut (Z-axis radial)

Version 02:*Fig. 537: Setting nuts, patient end (version 02)*

(1) Counter nut (Z-axis radial)

- Tightening all screws and counter nuts.

9.1.10 Checking the travel path

- Checking the travel path of the patient tabletop on the slide rails while moving it in or out. Check to see that the tabletop moves quietly and in parallel in both directions. It should be free of warpings.
- Reinstall the cover panels.

9.2 Shimming the Magnet



Information regarding Shim is not published as hardcopy documentation. For more information, refer to the Online Help.

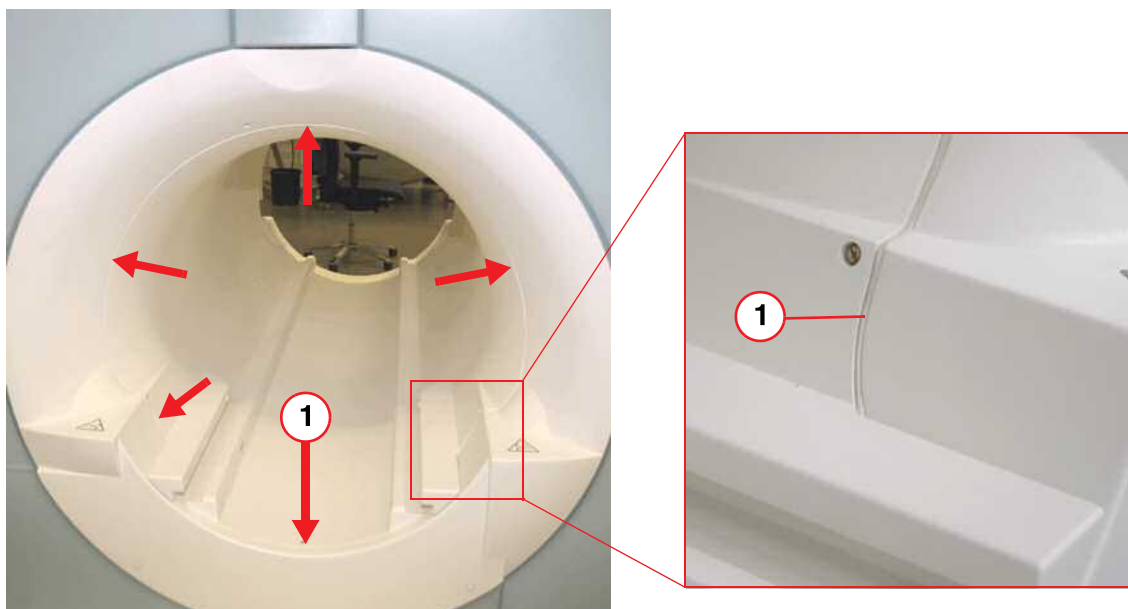
- Start the Shim workflow from the SeSo platform.
Popup windows will inform you of required actions.
Refer to the Online Help for detailed information.



Check the rubber seals between the front/back funnel and the body coil.

The rubber seal must be aligned (no overlapping) with the funnel and the body coil, refer to the following picture.

Fig. 538: Rubber seal at the front and back funnel (example shows front funnel)



(1) Rubber seal (between the funnel and body coil)

9.3 Tim Whole Body Suite (optional)



If the system is provided with the option “Tim Whole Body Suite” perform the following steps, described in (→ Whole Body / MR00-000.814.31), Technical Documents Volume 2, Options register.

9.4 Spike Check



Perform a QA Expert Spike Check with the Head Coil (without phantom) in center position and ± 600 mm off-center position, before assembling the cover. (Loose cables, screws, cover frame could cause spikes)

10.1 General



Valid for Germany and certain other EU countries.

If the monitor of an MR system is used for medical reporting, a special acceptance test is required in accordance with DIN V6868-57. The monitor must be labeled as such by the installation personnel.

According to the documentation listed below:

1. Acceptance testing for image display devices (→ Acceptance testing for image display devices / MR00-000.820.01).
2. Acceptance test protocol for image display devices. (→ Acceptance test protocol for image display devices / MR00-000.820.02)



The monitor only needs to be checked (and adjusted as necessary) if at least one of the parts listed below is replaced.

- Monitor
- Graphics board
- PC

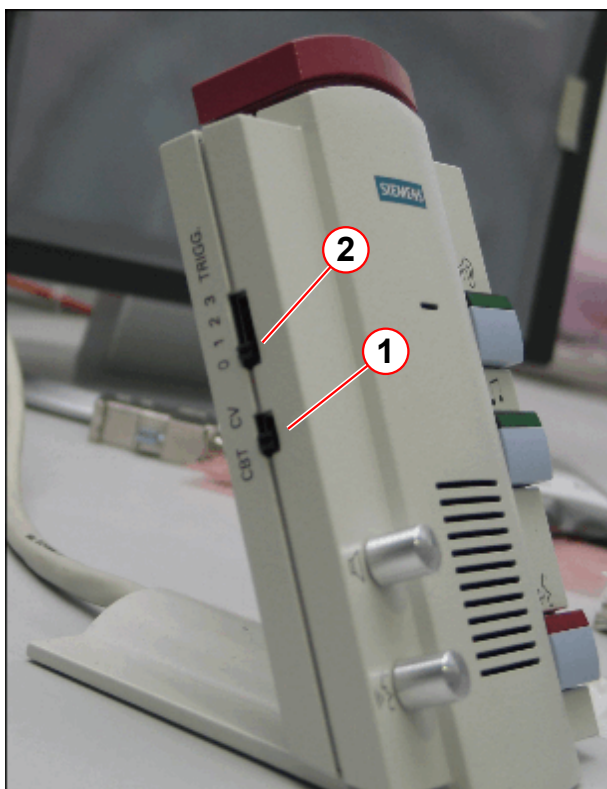
Refer to (→ Host / Imager / M6-020.841.12)"Replacement of parts"

11.1 Avanto Dot / Verio Dot Intercom

11.1.1 Settings

- Switch on and boot the system.
- Switch the operating unit (p.n. 4757261) to position "CBT". This switch is located on the left side of the operating unit.

Fig. 539: Intercom

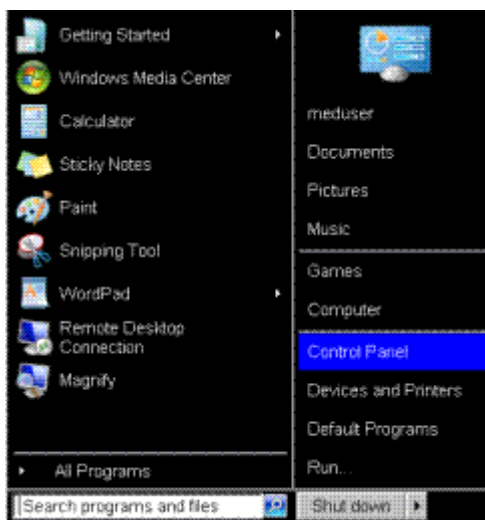


- (1) CBT
(2) TRIGG

- Close the patient application (Patient/Close Patient) if a patient is active.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Advanced User and enter the password: meduser1. Click <OK> to exit.

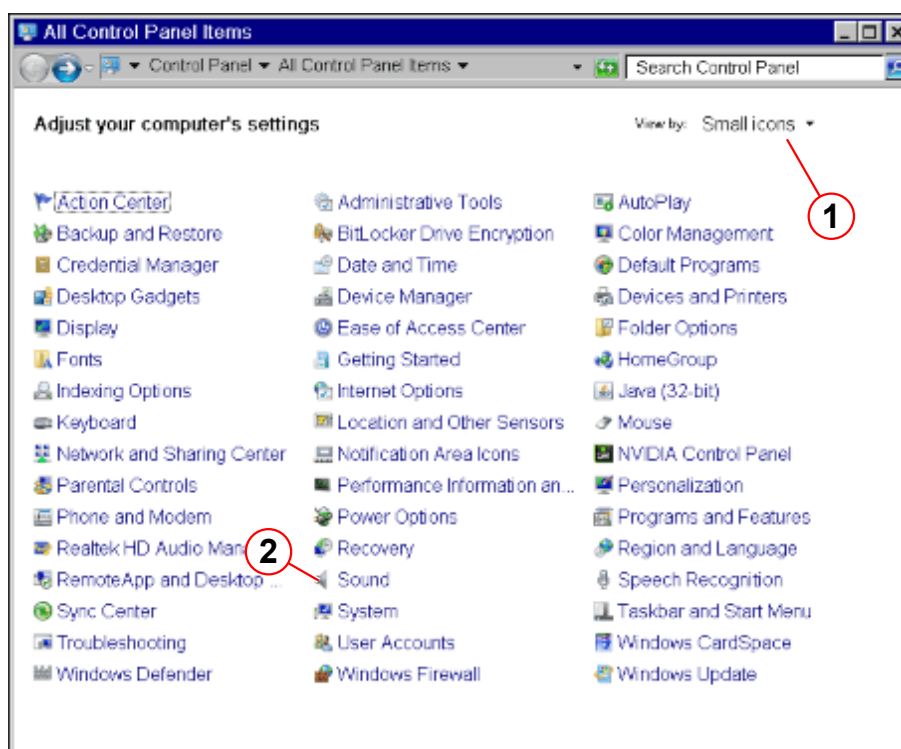
- Select Control Panel

Fig. 540: Control Panel



- Select Small Icons (→ 1/ Fig. 541 Page 435)

Fig. 541: Control Panel



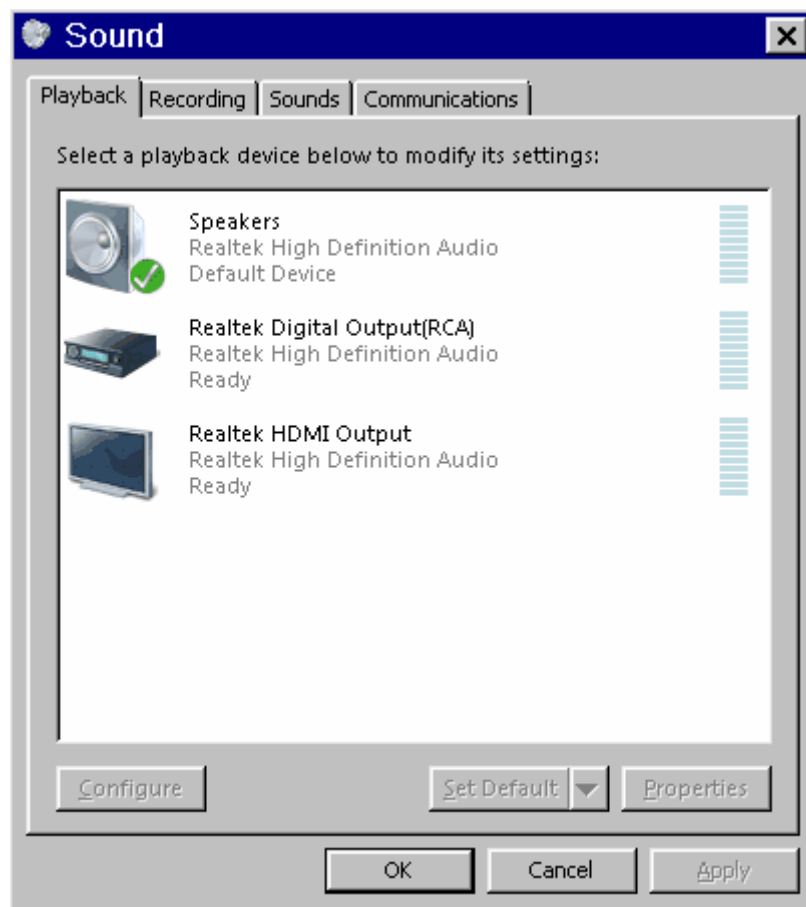
- (1) Small Icons
- (2) Sound



The screenshots can be different. They are dependent on the Host type.

- Click <Sound>” (→ 2/ Fig. 541 Page 435)

Fig. 542: Sound



- Click <Speakers> <Levels>

Fig. 543: Speakers Properties

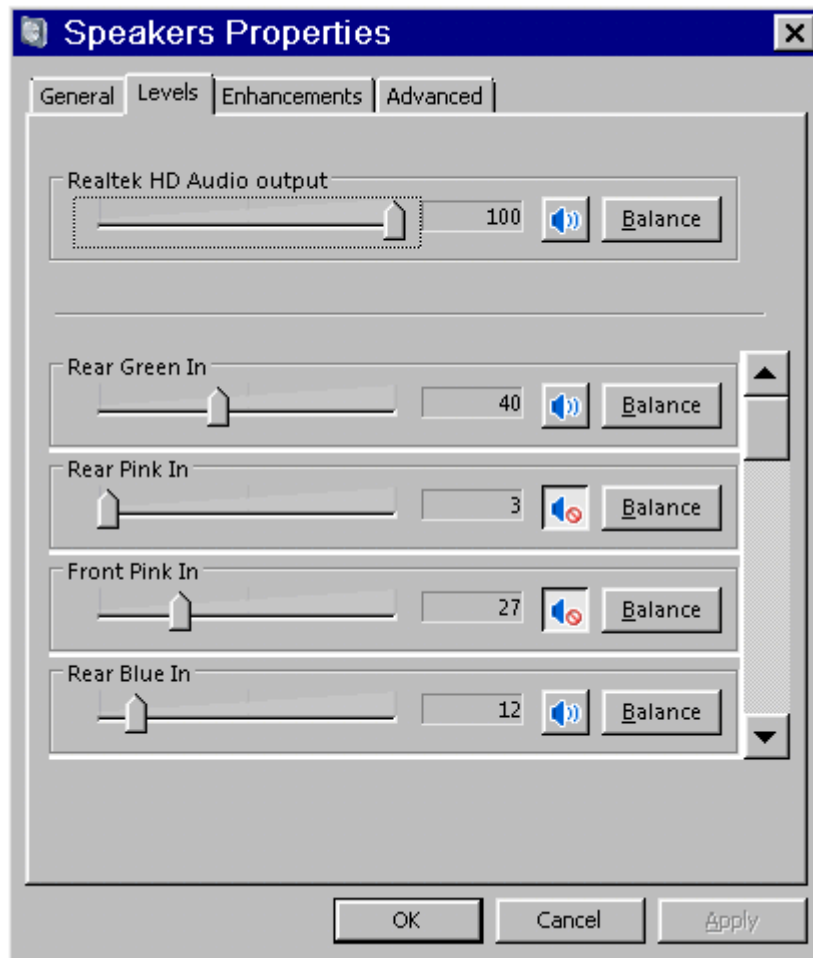
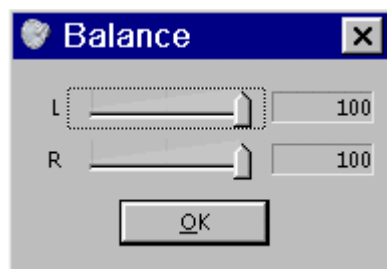
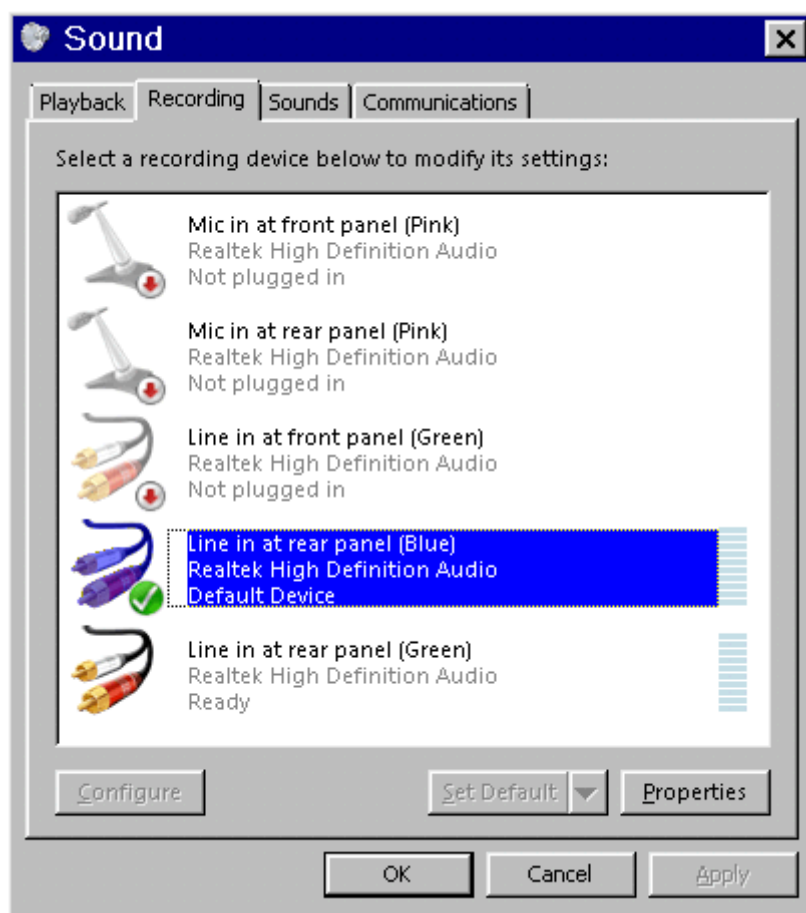


Fig. 544: Balance



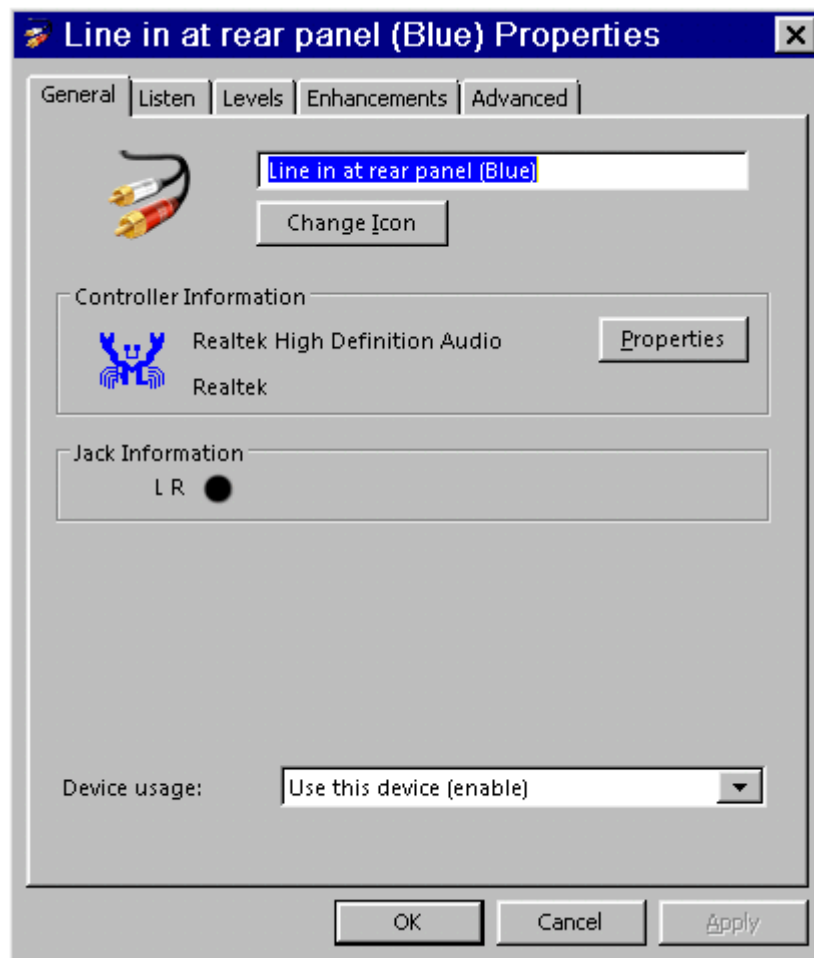
- Select <ok>
- Select <Sound>
 - Select <Recording>

Fig. 545: Sound



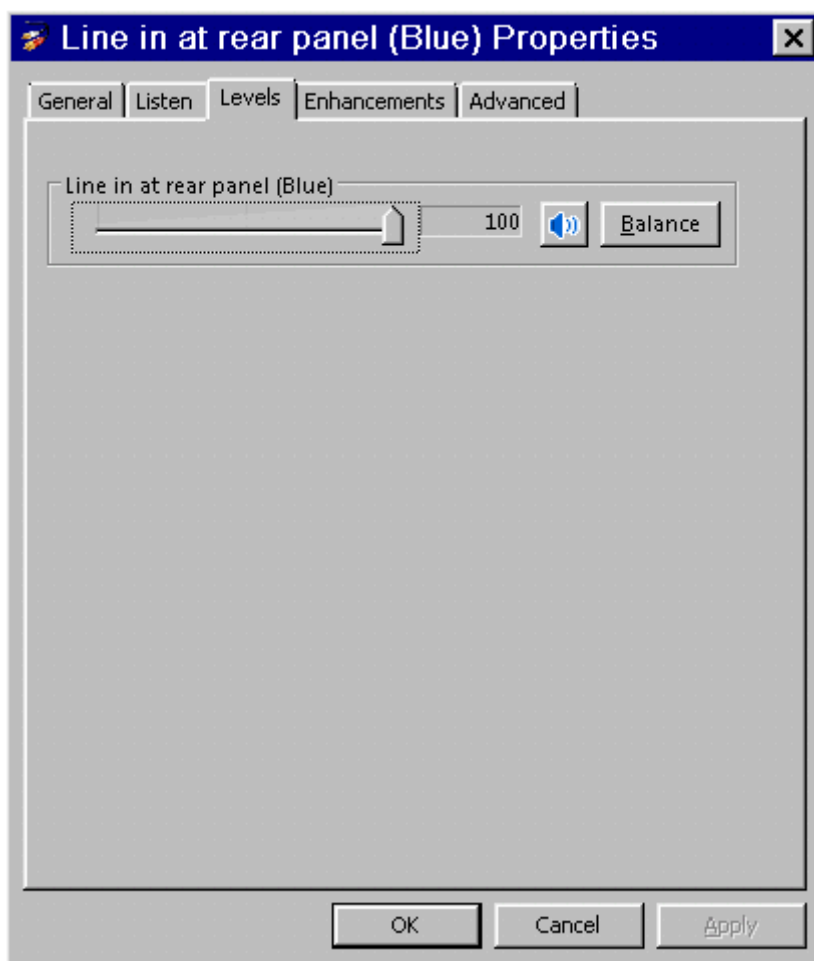
- Select <Line in at rear panel (Blue)>

Fig. 546: Line in at rear panel Properties



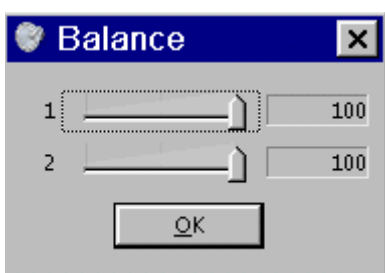
- Select <Levels>

Fig. 547: Line in at rear panel Properties



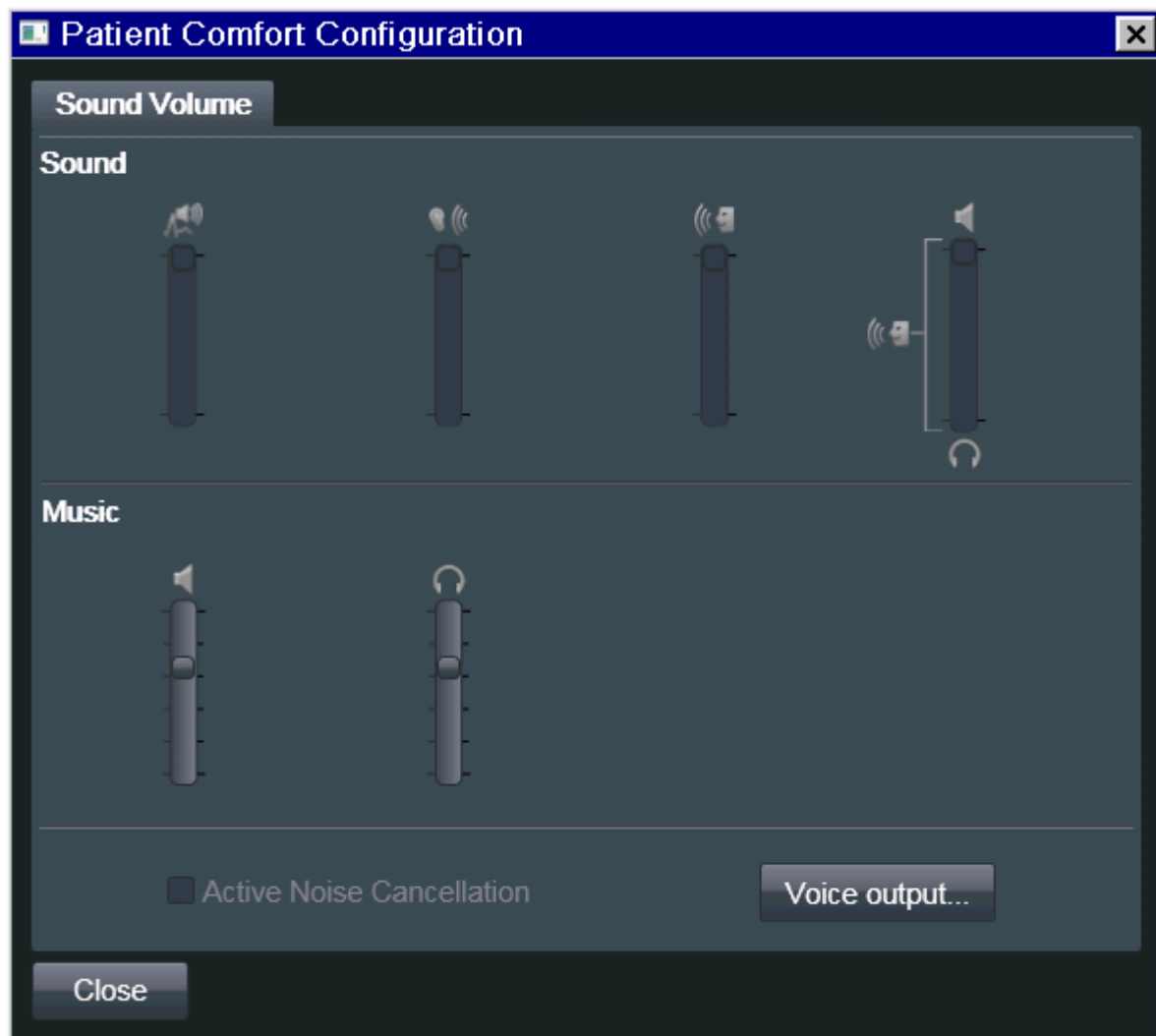
- Select <Balance>

Fig. 548: Balance



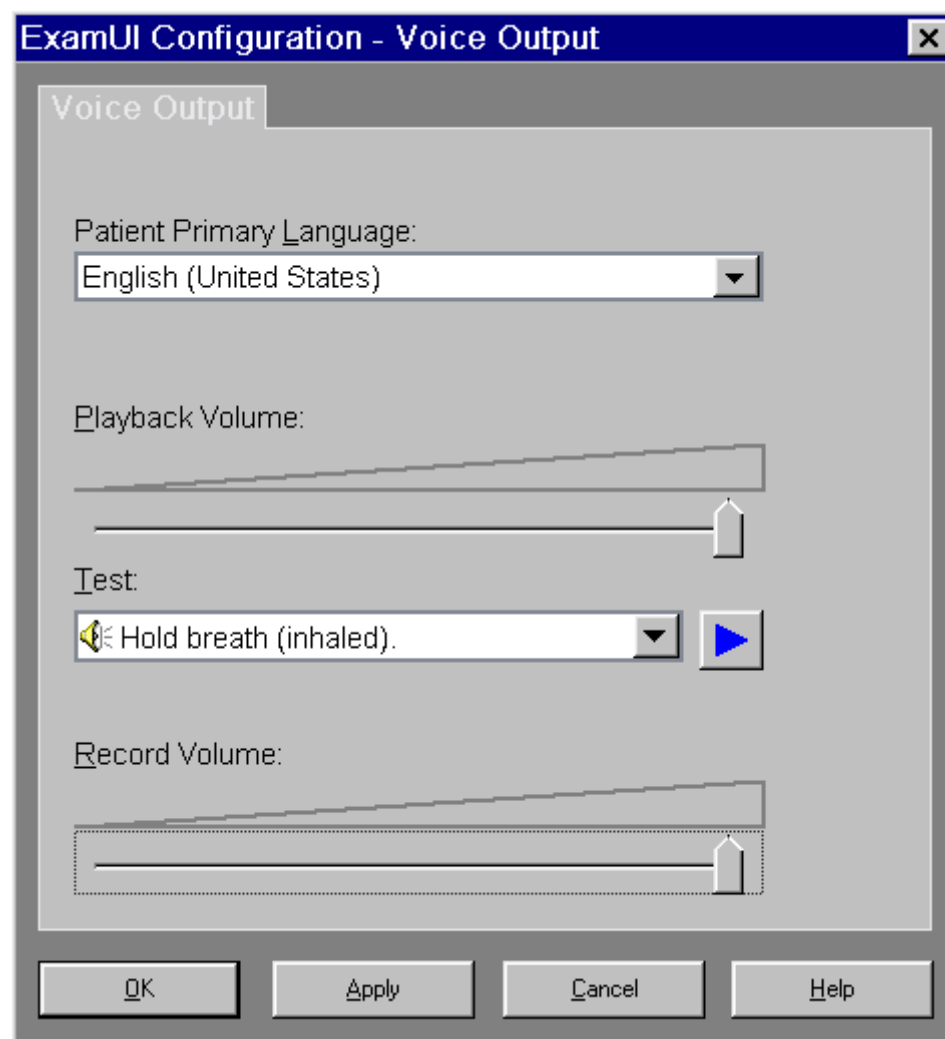
- Select <Options> / <Configuration> / <Patient Comfort>

Fig. 549: Patient Comfort Configuration



- Select <Voice output>

Fig. 550: ExamUI Configuration



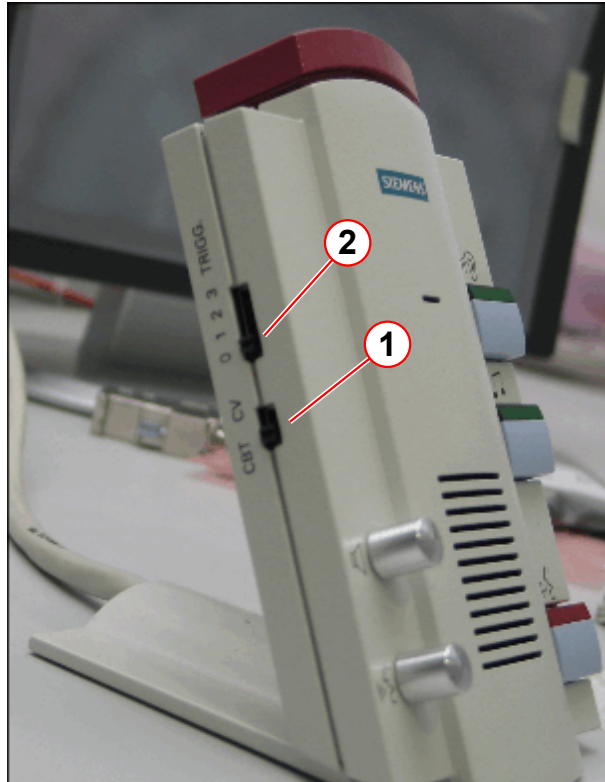
- Set the Record Volume to maximum.

11.2 Avanto / Espree / Verio Intercom

11.2.1 Start-up

- Switch on and boot the system.
- Switch the operating unit (p.n. 4757261) to position "CBT". This switch is located on the left side of the operating unit.

Fig. 551: Intercom



- (1) CBT
(2) TRIGG

- Close the patient application (Patient/ Close Patient) in case patient is active.
- Press <Ctrl> and <ESC> and select Programs/Accessories/Advanced User and enter the password: meduser1. Click <OK> to exit.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Multimedia/ Volume Control. Set all six regulators in the volume control window to maximum (upper position), as shown in (→ Fig. 552 Page 444). Deactivate the four controls (CD Audio Bal., Synt., AuxB, ILS) on the right side of the window by clicking mute. "Mute" does not have to be activated for the two controls on the left side (Vol. Contr., Wave Output). In case the volume control menu has a different design than the figure below, set the "Volume Control" and wave output to maximum, and set all other controls to "Mute". Select Options/ Properties in the same window and click <Recording>. Press <OK> and set the regulators for the "Recording Control" as shown in (→ Fig. 553 Page 444). Set the recording control volume to approx. 25% and the volume for the other four regulators to approx. 50%. Click <Select> for the microphone. In case the recording control menu has a different design than the figure below, set the volume to approx. 25% and all others to 50%. To select the microphone, click <Select> with all others deselected.

ted, as shown in the example below. Refer to online help, if one of the controls for the device is missing. Close the window via Options/Exit.

Fig. 552: Volume Control

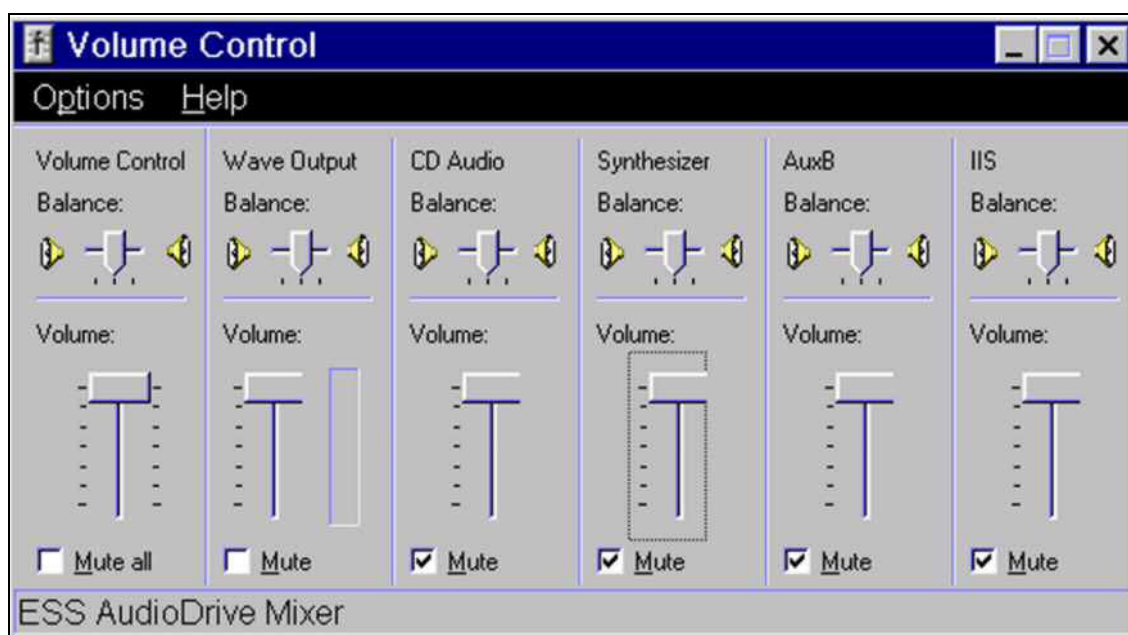
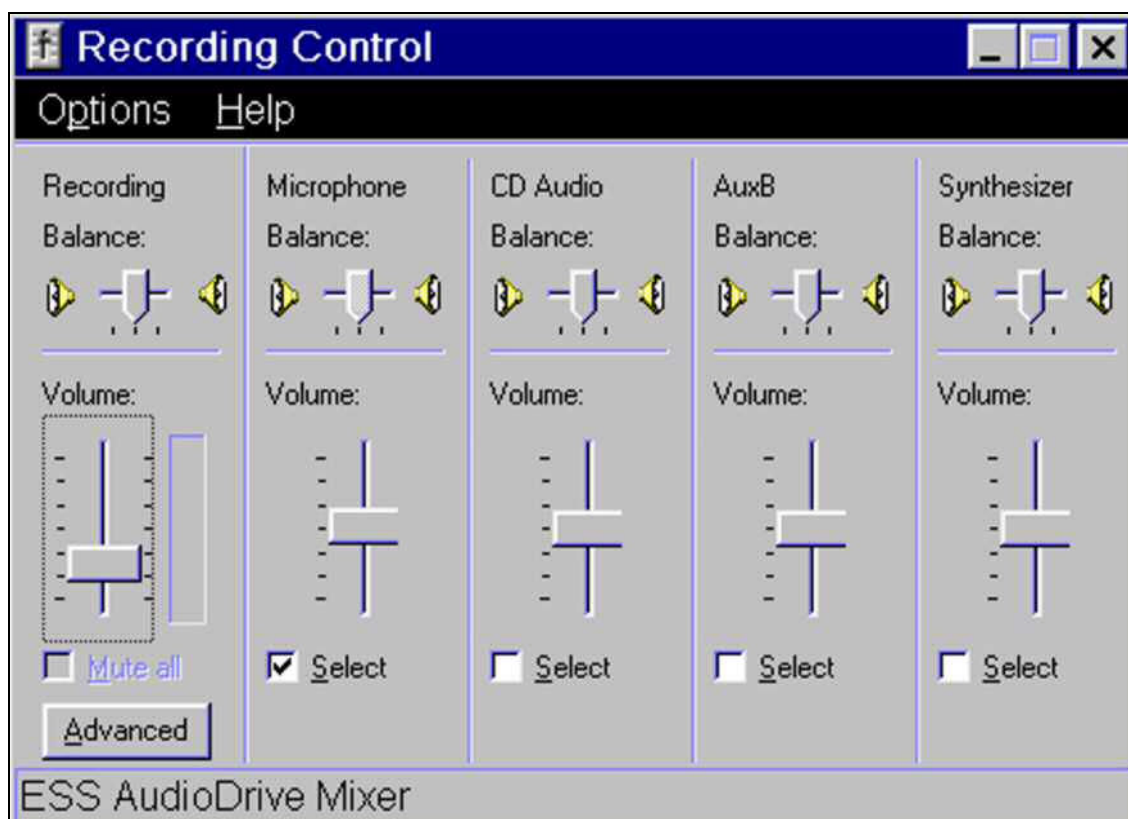


Fig. 553: Recording Control



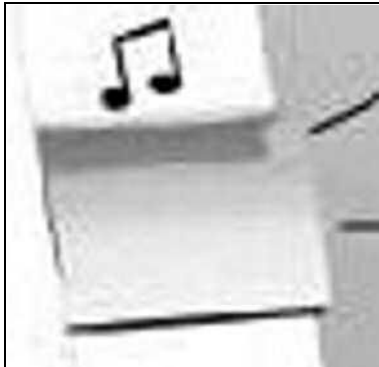
- Setting the computer voice (CV) volume: Select Options/ Configuration/ ExamUI and set the volume at approx. 70% of maximum. To save the settings and exit the windows, select <Apply> and <OK>.

- Setting the music volume, i.e. CD player: Turn both potentiometers on the left patient table control in clockwise direction to maximum. Switch on the music source, i.e. CD player, and adjust the output volume control on the music source to a volume just below the level when the music starts to sound distorted (open the door to the RF room to listen). Keep this setting for the music source and adjust the two potentiometers on the left patient table control panel to a comfortable volume level (depending on the customer) for the speaker and the head phones.

11.2.2 Final Check

- Perform a final check for the voice output (new option as of VA21A). Close the patient application (Patient/ Close Patient) in case Patient is active. Select Queue/ Voice Output/ Organize... and click <New...>. Enter a name, e.g. "test", and select <OK> and click the ● button. Press the lower right-hand button on the intercom's operating unit and keep it pressed during this procedure. Click the ● button in the record voice window. Speak the command into the operating unit (keep a distance of approx. 40 cm). After finishing the text, click the ■ button in the record voice window and release the lower right-hand button on the operating unit. The voice recording is now finished. To check the recording, switch the operating unit from CBT to CV and click the >▶ button. The recorded test command should be complete. A snapping sound at the beginning of the recording means that the procedure has been performed incorrectly. Compare the volume of your test recording with that of a default recording; both recordings should have just about the same sound level. If there is a difference in levels between the two recordings, readjust the volume of the recording control. Refer to the online help for more information regarding this new option. Close the voice output windows.
- The potentiometers for the operating unit (p.n. 4757261) affect the volume of the computer voice (CV) as well as the voice commands for the operator. The potentiometers on the operating unit do not require a basic volume adjustment, since these are set individually by the user. To select a starting value, turn the two potentiometers on the operating unit to approx. 50% of the volume. The user decides on an individual basis whether to use music or computer voice. The computer voice should not be used at the same time as music, because it may be difficult for the patient to understand the computer voice commands.
- If the customer prefers music: Switch the operating unit to "CBT" and press the play music button located in the center of the right side (music active = light switches on) as shown in (→ Fig. 554 Page 446).
- If the customer prefers the computer voice (CV): Switch the operating unit to "CV" and switch off the play music button in the center of the right side (music deactivated = switch light off). Notice: For computer voice recordings, the operating unit has to be switched back to "CBT". After the recording is complete, switch back to "CV".

Fig. 554: Music switch



11.2.3 Customer Information

Inform the customer about this new option by explaining how to record automatic voice output and what to keep in mind when using this new option.

12.1 General

12.1.1 Functional description of the VLM VESDA device

The SIEMENS_VLM incorporates a highly sensitive, laser-based smoke detection unit that can detect overheated connections in a very early phase. A small fan continuously draws air through a hose system that connects to the relevant connections which can overheat in a fault scenario.

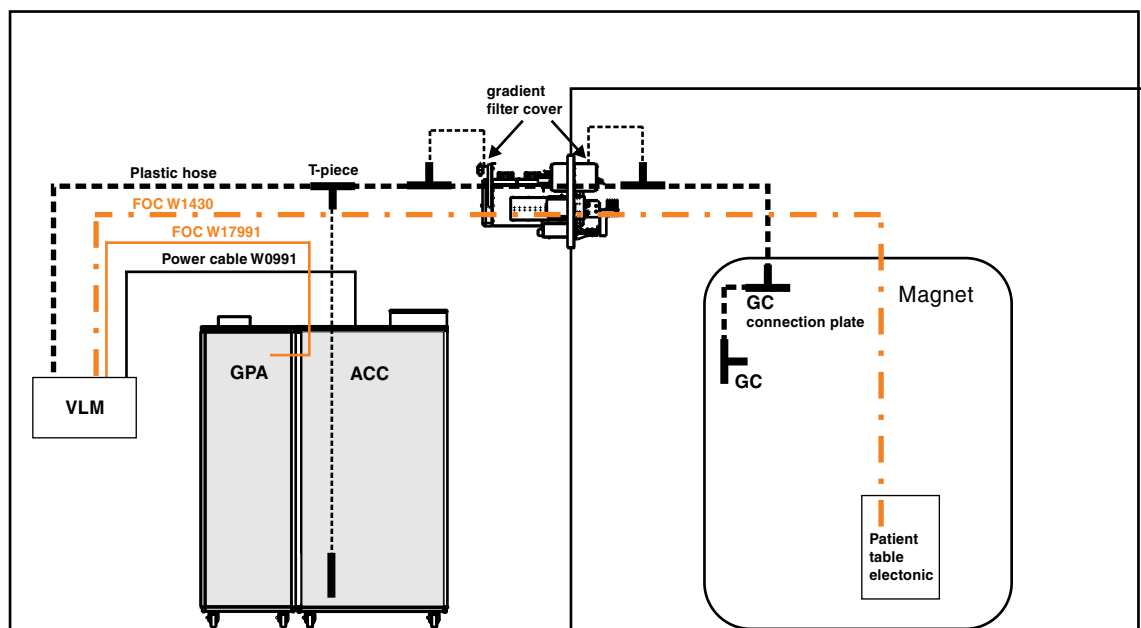
The aspirated air samples are fed through a filter into the laser detection chamber inside the unit.

Inside the calibrated detection chamber, the air sample is exposed to a laser light source. If the sample contains small particles (ones that might be emitted by overheated gradient connections, for example), then light is scattered within the detection chamber and is immediately recognized by the highly sensitive optical receiver system.



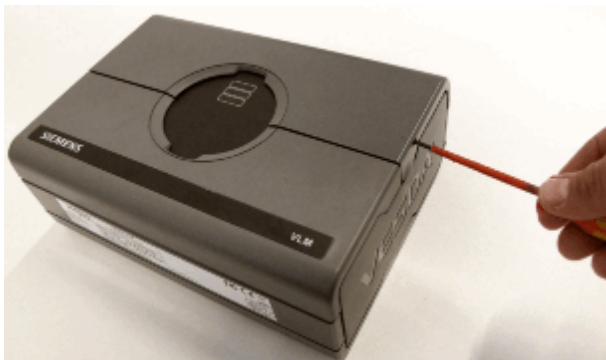
The VLM device is not designed to work as or to replace a fire warning system, but rather as an additional safety precaution. It switches the MR system into "stand-by mode" if one of the monitored connections overheats.

Fig. 555: Gradient Connection Supervision (Tim systems, Biograph mMR)



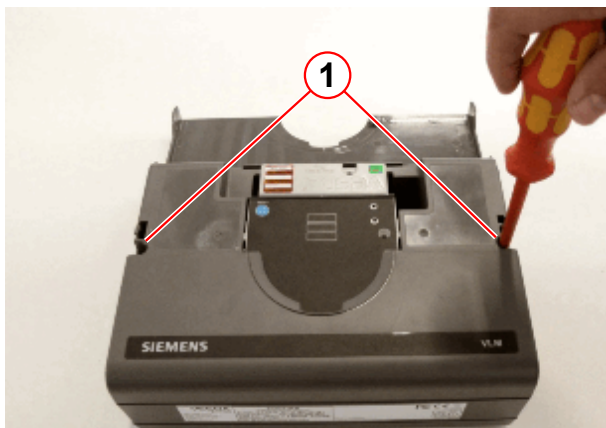
12.1.2 VLM opening

Fig. 556: Opening the upper cover



- Open the upper cover of the device.

Fig. 557: Opening the lower housing

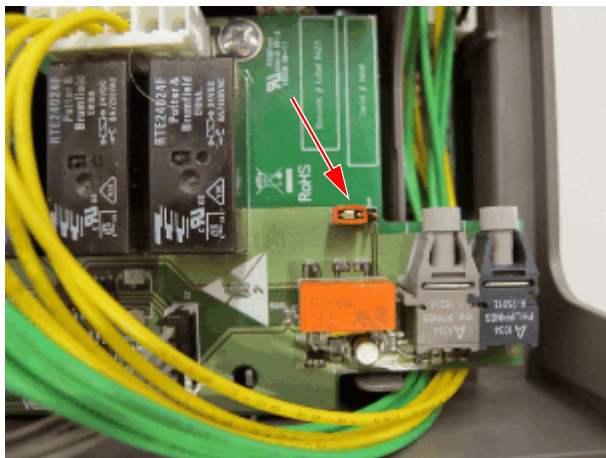


- Remove the 2 Phillips screws (1) and swing out the housing.

(1) 2x Phillips screws

12.1.3 VLM Jumper Settings

Fig. 558: Jumper on 2-3 default setting - correct for Tim systems and Biograph mMR



- There is one jumper left from the fiber optic connectors. The default jumper setting is **2-3**, which is appropriate for all **Tim systems**.

12.2 Display

Fig. 559: VESDA device



- (1) OK (green)
- (2) FAULT (orange)
- (3) FIRE (red)
- (4) RESET

Legend	ON	FLASHING	Consequence
OK	Detector is powered and in normal operation		
FAULT	Indicates a major fault	Indicates a minor fault	» Normalize VLM device » Call Service
FIRE			» GPA-Modulator will be disabled » MR scanner will be switched off 1 minute later

12.3 Detector Control Button

There is only one button on front of the VLM device, which is the RESET button.

Icon	Button	Function
	RESET	■ Press and hold down this button to test the function of the LEDs on the unit. ■ Release this button to clear latched faults and alarms. » Alarm and fault lights will switch off, and if the system is still in alarm or fault mode, the lights will reappear after a certain delay. » The reset button does not operate during the delay fire notification interval. ■ Press and hold down this button for 10 sec. until the 3 LEDs start flashing to activate the set-mode for 100% air flow. » For details refer to (→ Start-Up Procedure of the VLM Device / Page 452).

12.4 Start-Up Procedure of the VLM Device



The VLM Device was powered on with switching on the System.

- Close the RF-cabin door.
- The VLM device is powered now and needs a 90 sec warm-up time.
- Open the front cover of the device and push the RESET button for 10 sec. until all lamps are flashing.
- Release the RESET button and wait until the lamps go off.
- When the lamps start flashing again, press the RESET button once more to confirm the actual setting.
 - » The current air flow through the device (which depends on the individual system configuration and hose length) is now set to 100%.
 - » Deviations from this nominal Air flow $100 \% \pm 20 \%$ will result in an internal error.
 - » A removed, broken, or blocked hose will result in a FAULT now.



In case the normalization procedure described above does not work properly and the VLM device goes into failure mode, there could be a differential pressure between the RF-cabin and the equipment room, which exceeds the limits of the VLM device. In this case a pressure equalization tube must be fed from the exhaust of the VLM device back into the RF-cabin, according to (→ Pressure Equalization (optional) / Page 459).



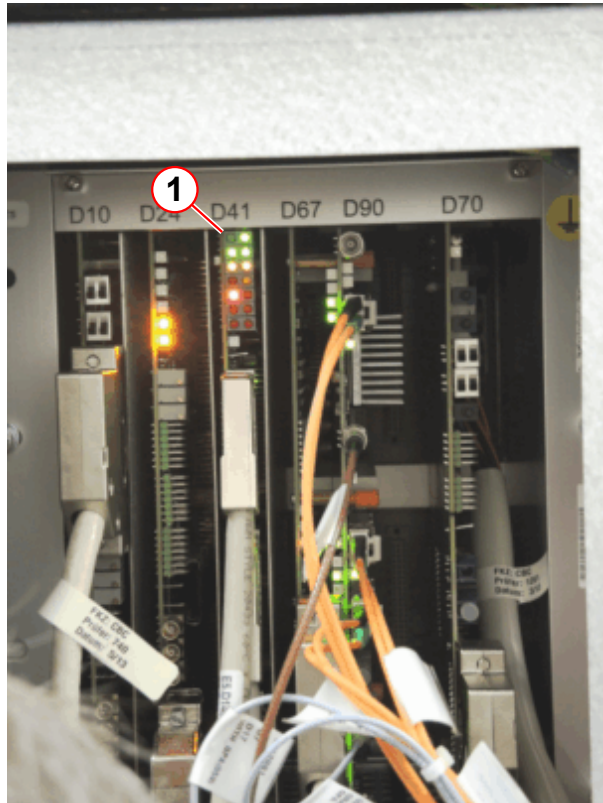
A final test with a smoke wick has to be performed. This test may activate the on-site smoke detectors.

As a precautionary measure, inform the local fire safety engineer/fire department about the test in order to avoid false alarms.

12.5 Final Check

Modulator disable test

Fig. 560: Modulator LED in GPA cabinet off



(1) Modulator LED (V120)

- Remove the FOC (W1430) from U2
 - » The GPA modulator should be disabled (scanner signal at MRC is crossed through and modulator LED in GPA cabinet is off; refer to (→ Fig. 560 Page 453))

- » GPA error should be reported in the event log:

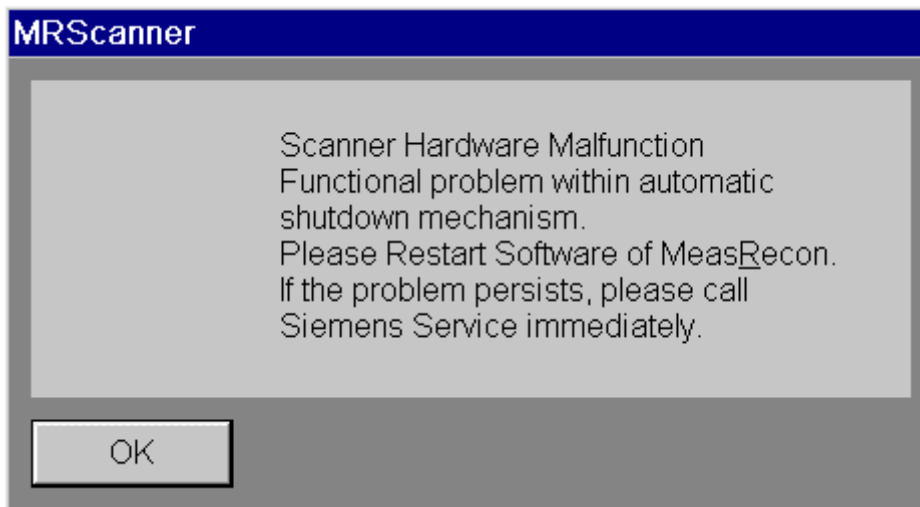
Date/Time	ID	Source	Seq.	Message Text
22-May-15 17:24:25	2016	MRI_GDV	0	Gradient Power Amplifier error: Hardware error. Please reboot

- Reattach the FOC W1430 to U2.
- Reassemble the VLM device.
- Perform "Reboot Scanner."

Air flow failure simulation

- Remove the hose from the VLM device.
 - » ~30 sec. later a fault should be displayed on front of the VLM device due to high air flow and a pop-up should be displayed at the console (→ Fig. 561 Page 454).

Fig. 561: Pop-up "FAULT" condition



- Refit the hose.
 - » The green OK lamp should go on.

Differential pressure test

- Open the RF-cabin door and wait for 5 minutes.
 - » The green OK lamp on the VLM device should remain on.



In case the VLM device goes into failure mode, there is a differential pressure between the RF-cabin and the equipment room, which exceeds the limits of the VLM device. In this case a pressure equalization tube must be fed from the exhaust of the VLM device back into the RF-cabin, according to (→ Pressure Equalization (optional) / Page 459).



In order to prove the proper function of the detector, a smoke test must be performed.



WARNING

Accumulation of oxygen during helium filling or magnet ramping.

The use of glowing smoke wicks during Helium filling or magnet ramping could result in an open fire in the area of the magnet service turret.

- » Do NOT perform a smoke test during helium filling or magnet ramping!



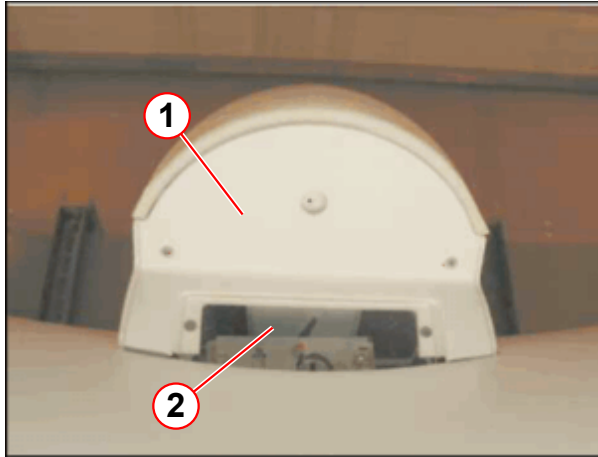
The smoke wicks, located in a box for the test below, are part of the Update-Kit. Each smoke wick will provide up to 30 minutes of continuous smoke, or 60 individual 30 second tests.

Smoke test

■ System with Camera option

- Remove the cover of the camera (→ 2/Fig. 562 Page 455)

Fig. 562: Patient Video



(1) Camera

(2) Cover

■ System without Camera option



CAUTION

Hazardous voltages may be present at the connections of the gradient cables.
(→ 2/Fig. 563 Page 456)

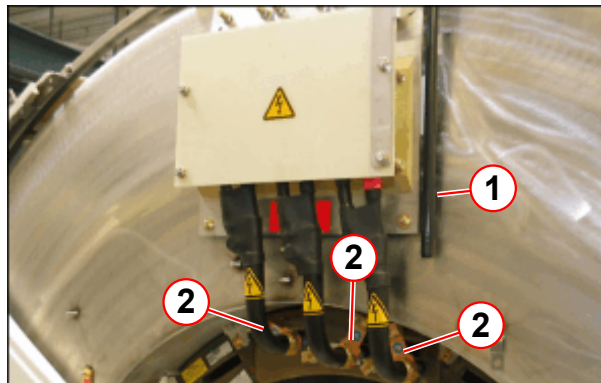
Observe the safety requirements. Lack of compliance may lead to death or serious bodily injury.

- » Switch off the gradient amplifier via circuit breaker F2/GSSU and the F1/100A contactor in the GPA. Wait at least 20 minutes before you start working on GC.

- Remove the left rear ring segment
- Remove the right rear ring segment

- Remove the top rear cover.

Fig. 563: Gradient supervision



- (1) Pipe of the lower sample plate
- (2) Gradient cable connections

- Take one smoke wick out of the box (→ Fig. 564 Page 456).
- Hold wick at the end with black plastic handle mounted (→ Fig. 565 Page 456).
- Light the tip of the wick and allow it to burn for 10-15 seconds (→ Fig. 565 Page 456).

Fig. 564: Smoke wicks in a box

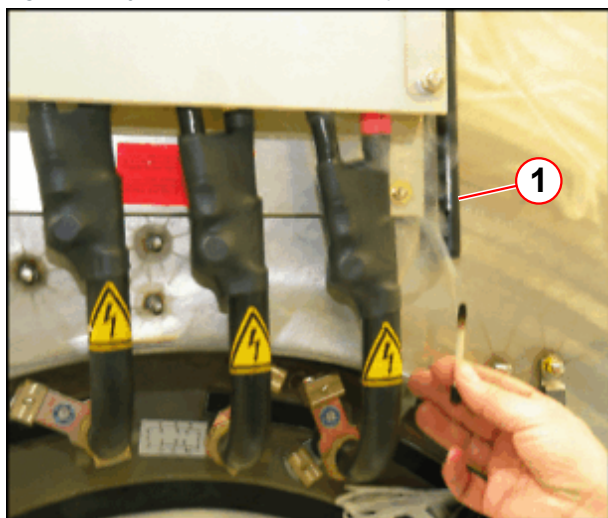


Fig. 565: Lighted smoke wick



- Extinguish the flame and hold the smoke wick behind the rear cover in the area of the lower sample plate for 5 seconds (→ Fig. 566 Page 457) or (→ Fig. 567 Page 457).
- Extinguish the wick by simply placing it in the empty box and close lid carefully. This will extinguish the wick after approximately 15 seconds.
 - » With a delay of about 20-40 seconds (depending on the hose length) the "FIRE" lamp on the VLM device should light up (→ Fig. 568 Page 457).

Fig. 566: System without camera option



(1) Pipe of the lower sample plate

Fig. 567: Smoke test at the rear side with Camera option

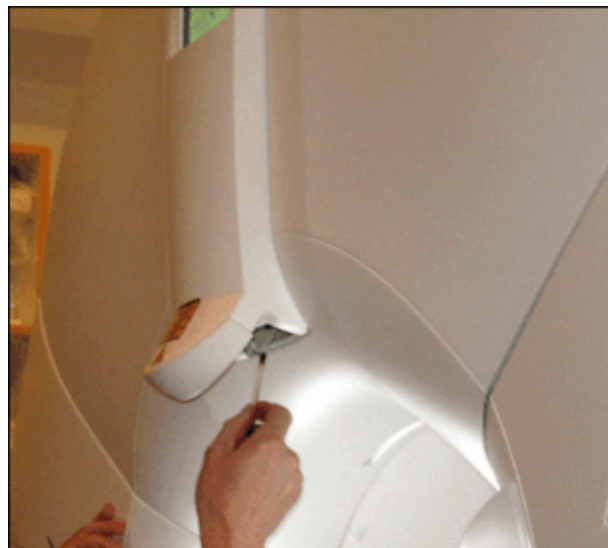
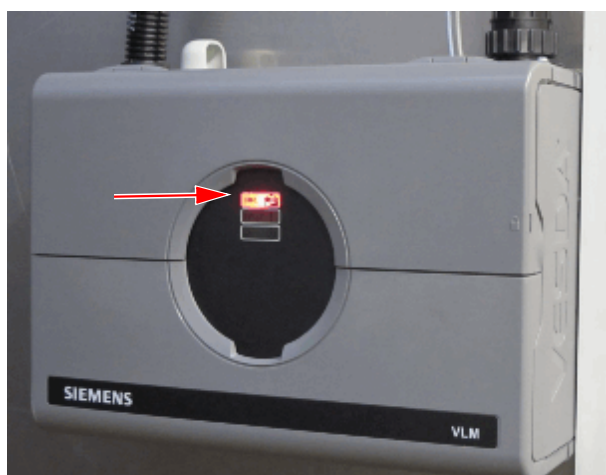
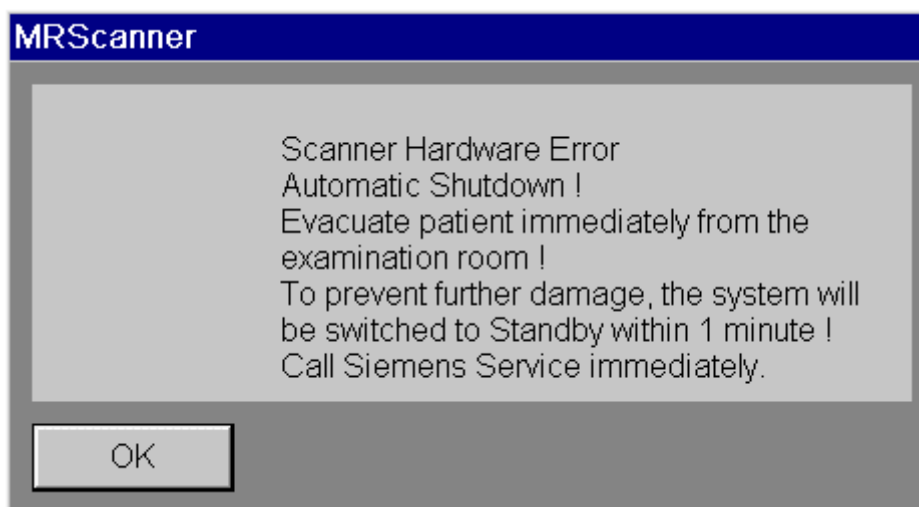


Fig. 568: Fire lamp ON



- » Subsequently the modulator in the GPA cabinet should be disabled and the message (→ Fig. 569 Page 458) should appear on the screen of the host.
- » 1 minute later the scanner should be switched off (to stand-by).

Fig. 569: Pop-up "Automatic Shutdown"



Once the FIRE lamp has been activated, it must be reset with the RESET button (→ 4/ Fig. 559 Page 450). It may take some minutes until the background level of smoke particles has returned to a normal level and a rest is possible. After a successful reset, the system can be switched on again.

- To reuse a smoke wick, simply take it out of the box and follow the steps as described above.
- The smoke wicks have a long shelf time and should be stored in a cool and dry place.



Use only as directed, do not eat and please keep away from children.

- Reassembly the covers.

12.6 Pressure Equalization (optional)



This procedure is only valid if the differential pressure test fails.

Fig. 570: VLM device exhaust cap removal

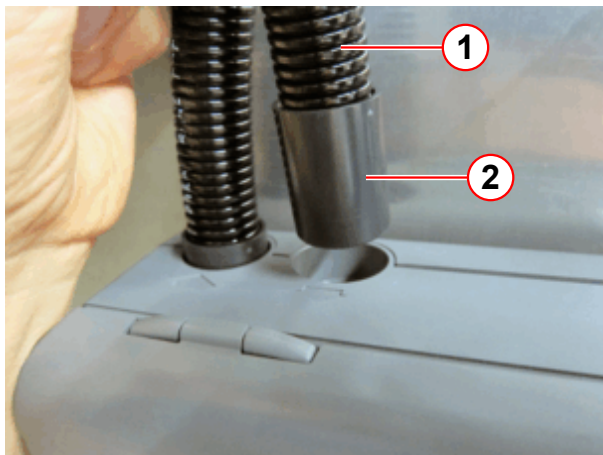


(1) Exhaust cap

Removal of the exhaust cap at the VLM device

- Remove the exhaust cap (1) on top of the VLM device

Fig. 571: VLM exhaust hose connection

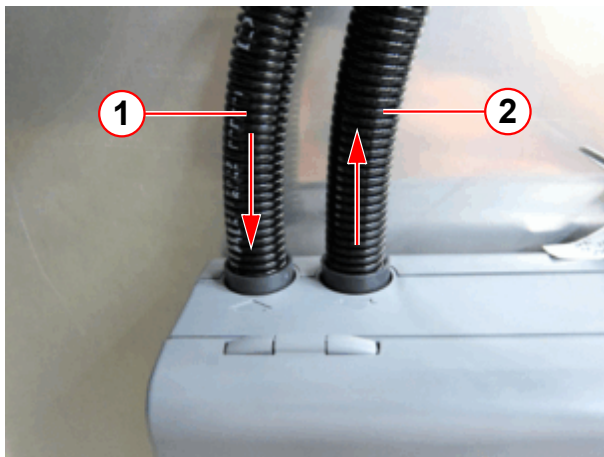


(1) Exhaust hose
(2) Bushing

Mounting of a pressure equalization hose

- Push-fit the bushing (2) onto one end of the remaining 22 mm hose roll.
- Push the bushing with the hose into the exhaust on top of the VLM device (→ Fig. 571 Page 459).

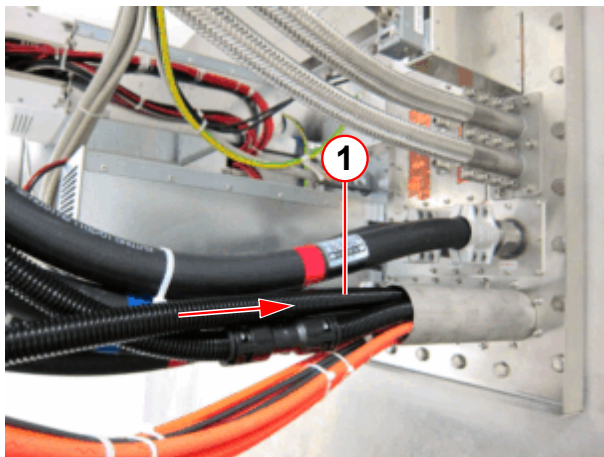
Fig. 572: VLM device with pressure equalization hose



- (1) Inlet
- (2) Exhaust

The air flow is now via hose (1) into the VLM device inlet and into the outlet hose (2).

Fig. 573: Hose fed from the equipment room

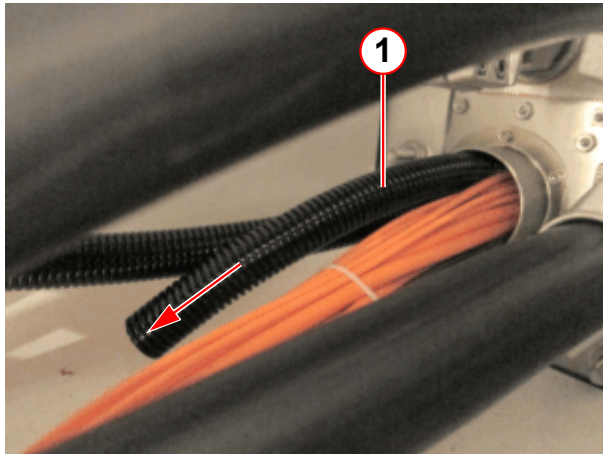


- (1) Pressure equalization hose

Feeding the pressure equalization hose through the filter plate

- Lay the pressure equalization hose (1) through the wave duct into the magnet room.

Fig. 574: Hose fed into the examination room



(1) Pressure equalization hose

Cutting the pressure equalization hose

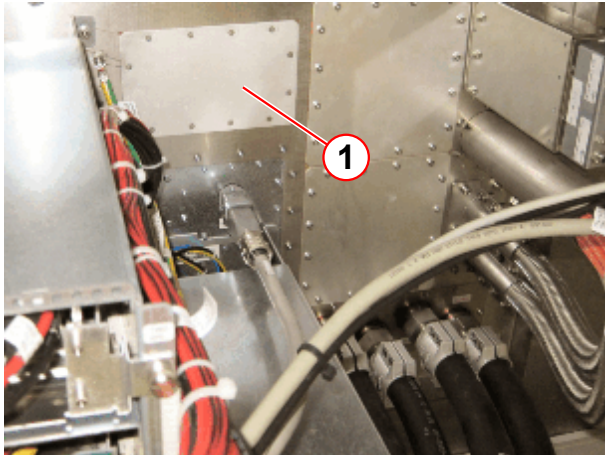
- Cut the pressure equalization hose (1) inside the magnet room at ~20-30 cm behind the wave duct.

12.7 Mounting of an additional Wave Duct (optional)



This procedure is only valid for systems where the original wave duct has not enough space for keeping the hose (-s). In this case, additional hardware is required and must be ordered from CS ML. Part-no.: 10684487

Fig. 575: RF filter plate (equipment room)



(1) Blind plate

Fig. 576: RF feed-through (optional)



Mounting the additional wave duct

- Remove the blind plate (1) from the filter plate.
- Replace it with the new wave duct (which can also be used with the MNO-option)
 - » Now the additional wave duct can be used for the hoses.

13.1 Prerequisites for Tune-Up/QA

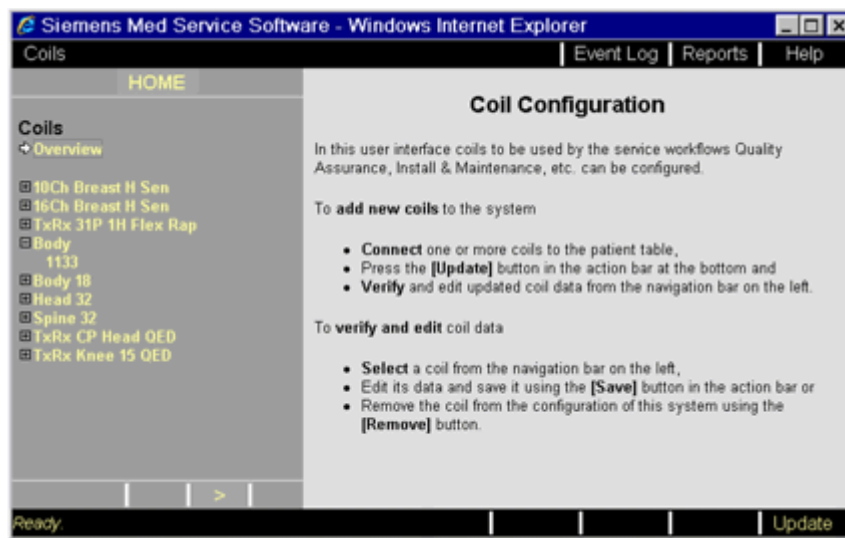
13.1.1 Configuration checks (Num4_VD13)



The status of the Coil QA will be set to "ToDo" after the coil configuration. This is necessary because of the new coil asset mechanism.

- Select **Coils** in Local Service

Fig. 577: Coil Configuration



- Check the correct configuration of the Body coil and the surface coils.
- Add missing coil configuration.

13.2 Tune-Up



If the system is not to be used with the option “Tim Whole Body Suite”, or there is only limited patient table travel, the PAMatrix Coil may only be selected in the configuration as F_PAMatrix (feet first).

This only indicates the coil in the service software QA and installation protocol.

The H_PAMatrix may not be used for quality assurance without the “Tim Whole Body Suite”.

The reason is that the patient table travel distance is too short to measure all of the coil elements.

This would result in an incorrect value in the installation protocol.



Tune-Up information is not published as hardcopy documentation. For more information, refer to the Online Help.

Performing the Tune-Up of the system (Options/ Service/ Local Service/ Tune-Up). Most Tune-Up steps are performed completely automatically. Follow the instructions shown on-screen. In addition, pop-up windows will inform you of the actions required. Refer to the Online Help for detailed information.

13.3 Testing pick-up coils/Body Coil



This test is part of the Quality Assurance (QA) with software version N4 VB15A and N4 VD13A.



This test is valid for all systems with software version N4 VB13.
Except for the Espree Straight Bore system and the Tim system with 8 Receiver Channels configuration.

Auxiliary tools:

- Loader with 24 cm spherical phantom

Preliminary steps:

- Register a patient (any)
- Move the table to HOME position
- Coil and phantom selection:
 - Remove all local transmit coils
 - Remove all local receiver coils
 - Position the loader together with the 24 cm spherical phantom (centered in the loader) in the upper part of the table and move it under the light localizer.
- Switch the light localizer on and off.
- Move the table into the CENTER position.

13.3.1 Performing the measurement

- Start the measurement with one of the protocols (e.g. a localizer).
 - » Measurement can be stopped immediately after it begins.

13.3.2 Evaluation

- Open the file "RFSWDHistoryListNew.log"
(it is located in c:\MedCom\Log Directory)
- Scroll back from the last entry in the file to the "604 DiCoTest Tra Power[0]" column
- Once there, ensure that the entries for the following parameters reach a minimum of 200 V or maximum 2500 V
 - 2500 > DiCoTest PickUp Probe Ampl. [0][0] > 200 V
 - 2500 > DiCoTest PickUp Probe Ampl. [0][1] > 200 V

This concludes the test.

- » If the conditions are not met, check the connections from the output BC to the input receiver.

13.4 QA (Quality Assurance)



Information regarding QA is not published as hardcopy documentation. For more information, refer to the Online Help.



The service software supports only the surface coil that has been configured in the software configuration menu (SeSo/Configuration/Measurement/Avail.Coil).

Perform the QA (Quality Assurance) of the system and of all available surface coils (Options/ Service/ Local Service/ QA). Most QA steps are performed completely automatically. Follow the instructions shown on-screen. Pop-up windows will inform you of the actions required. Refer to the Online Help for detailed information.

13.4.1 TestTools



Before handing the system over to the customer, please ensure that all tests of the TestTools have been performed and are within specifications. Please refer to the chapter "TestTools" in "Software Configuration".

13.4.2 Safety-related tests

Perform the "Safety-related tests".



Information regarding Safety-Related Tests is not published as a hardcopy. Refer to the SeSo/Install & Maintenance/Safety-Related Tests

14.1 Final work steps

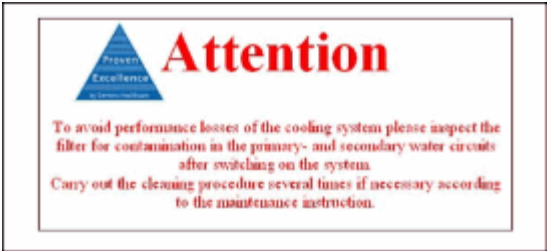
14.1.1 Cleaning the water filter



To avoid performance losses of the cooling system please inspect the filter for contamination in the primary and secondary water circuits of the SEP.

This working step is important for refurbished systems.

Fig. 578: Refurbished system label



- Clean the water filter of the primary and secondary water circuit.

14.1.2 Check the water pressure

Check the water pressure via SeSo “Magnet & Cooling”.

Fig. 579: Cooling Status



- (1) Water pressure supply
(2) Water pressure return

14.1.2.1 System with SEP

■ Open the Status platform

Read out the water pressure with the running water pump:

- » ACS water pressure supply
 - » ACS water pressure return
 - » Calculate the mathematical difference between the water pressure supply and water pressure return
-
- » The difference (water pressure supply minus water pressure return) must be at 50 Hz: 3.0 ± 0.3 bar
 - » The difference (water pressure supply minus water pressure return) must be at 60 Hz: 3.8 ± 0.3 bar

If water pressure return < 1.3 bar or difference pressure out of specification refill water.

Warning level at 0.8 bar.

Refer to the Installation Instructions in (→ Magnet Cooling / M6-020.812.01).

14.1.2.2 System with Chiller / IFP

Bleeding the Chiller water pump

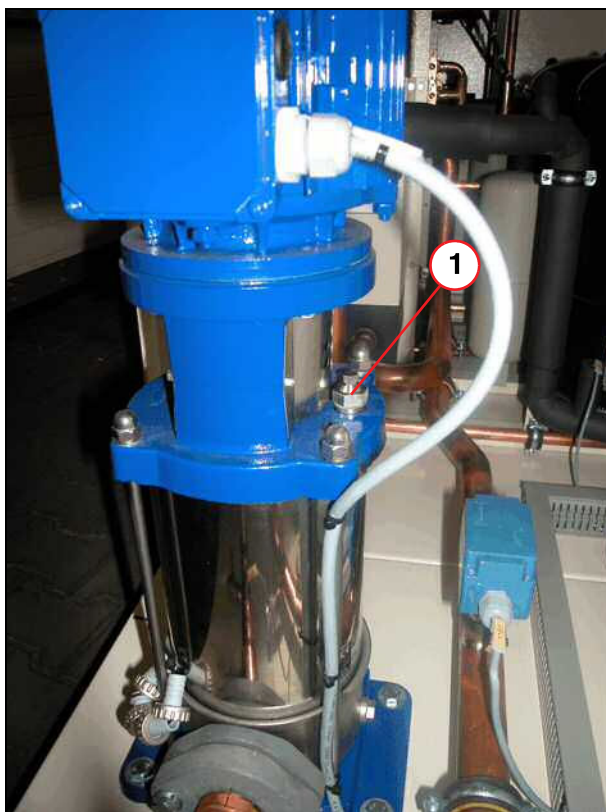


It is possible that air bubbles in the water cooling circuit collect in the water pump of the Chiller.

The bubbles prevent sufficient cooling of the water pump.

As a remedy, bleed the pump after operating it for a few days.

Fig. 580: Water pump



(1) Bleeding valve

- Open the vent valve of the pump
- Leave it open until air escapes.

Fig. 581: Water pump



(1) Bleeding valve

14.1.2.3 System with Chiller / IFP

Check the water pressure via SeSo

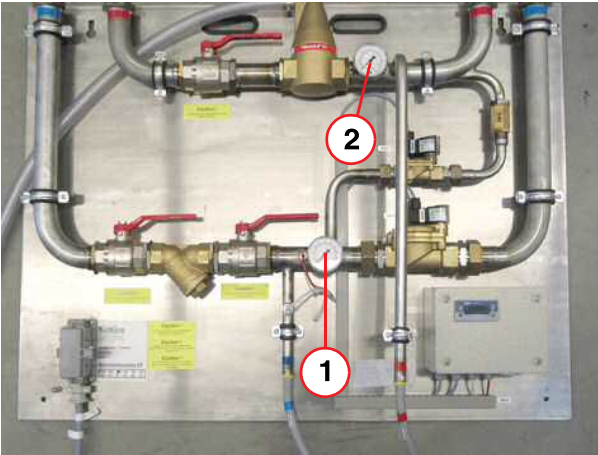
■ Open the Status platform

Check the water pressure, the chiller is on:

- » ACS water pressure, supply
 - » ACS water pressure, return
 - » Calculate the mathematical difference between the water pressure supply and water pressure return
-
- » Mathematical difference (water pressure supply minus water pressure return) must range between 4.0 bar and 4.5 bar for 50 and 60 Hz.

Check the water pressure at the IFP

Fig. 582: IFP with pressure manometer



- (1) Water pressure supply
- (2) Water pressure return

» Pressure difference at the IFP (water pressure supply - water pressure return):
pressure difference at IFP > pressure difference at ACS

Filling pressure:

Refill the chiller if the water pressure return < 0.5 bar or if the pressure difference is out of specification.

Tab. 25 Filling pressure

System configuration	Fill pressure at Chiller manometer Chiller is off
MR system & Chiller are on the same level and pipe routing is higher than MR system or Chiller is above the MR system.	0.1 - 0.5 bar (overpressure) + 0.1 bar/m (difference of level)
Example: Chiller or pipe routing 10m above MR system.	0.5 bar (overpressure) + 10 m * 0.1 bar/m (difference of level) = 1.5 bar (fill pressure)
MR system & Chiller & pipe routing are on same level.	0.1 - 0.5 bar (overpressure)

System configuration	Fill pressure at Chiller manometer Chiller is off
MR system is higher than the Chiller.	0.1 - 0.5 bar (overpressure) + 0.1 bar/m (difference of level)
Example: Chiller is 5 m below MR system.	0.5 bar (overpressure) + 5m * 0.1 bar/m (difference of level) = 1.0 bar (fill pressure)

14.1.3 Checking the water circuit to the gradient coil



To be sure that cooling water is flowing through the gradient coil, perform the following test:

- Open the Cooling Status platform

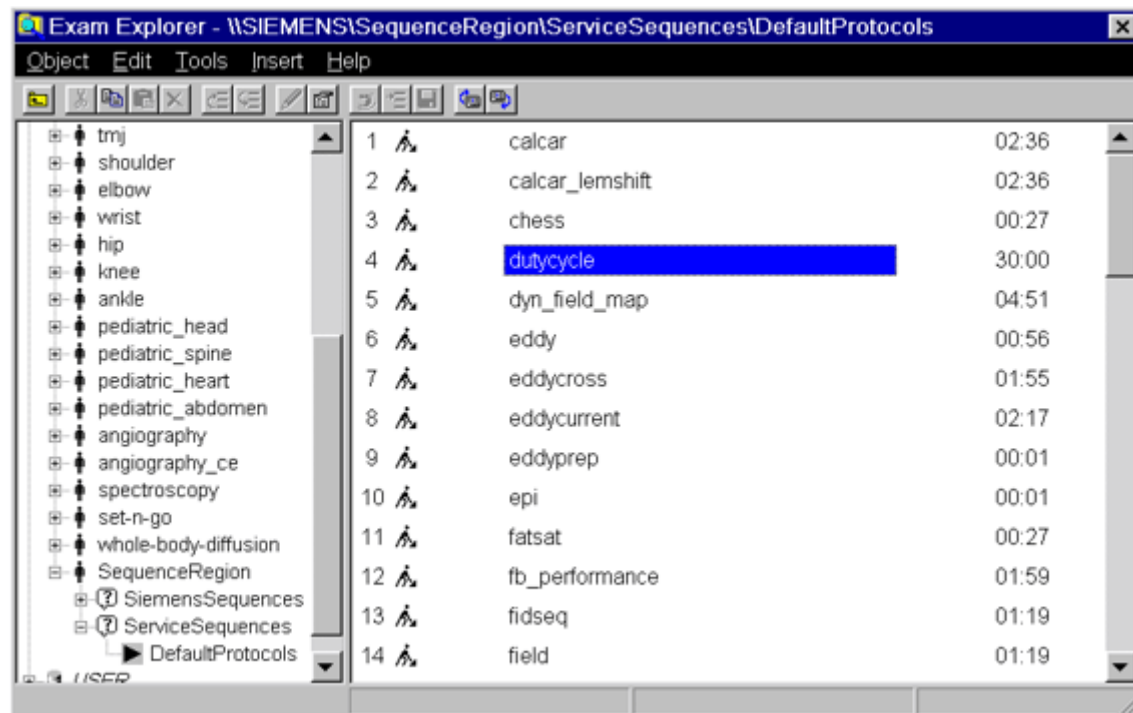
Fig. 583: Water Temperature GC

Siemens Med Service Software - Microsoft Internet Explorer provided by Siemens			
Magnet & Cooling Event Log Reports Home Help			
Action			
Cooling Status			
Cabinet			
	Air Temperature Supply		26.1°C
	Air Temperature Return		28.5°C
ACS			
	Water Temperature GC		22.4°C
	Water Temperature Cabinet		OK
	Water Pressure Supply		3.9bar
	Water Pressure Return		0.7bar
	Water Flow GC		OK
	Water Flow Cabinet		OK
	Water Flow Total		OK
	Water Flow Mref		OK

- Note the "Water Temperature GC"

- Run the duty cycle sequence for about 5 minutes

Fig. 584: Duty cycle



The temperature of the "Water Temperature GC" must increase while the duty cycle sequence is running.

Increase in temperature about 5 °C within 5 minutes.

If there is no increase in temperature, check the water circuit to the gradient coil.

14.1.4 Installation&Maintenance Protocol

Complete the software-based "Installation&Maintenance Protocol".

14.1.5 Quench Line Acceptance Protocol

Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer if present.

14.1.6 Dynamical Backup

Perform "Backup of dynamic data." For more information, refer to CB Doc-> Installation Software.



Keep the following parts for later use onsite:

- CD's / DVD's with "Recovery Image"
- CD's / DVD's with "Backup Package"
- Blue ring binder

14.1.7 Sign for bruising hazards



The bruising hazard sign must be posted in back of the patient table if the distance between patient table and back wall is less than 500 mm.

Fig. 585: Service end area



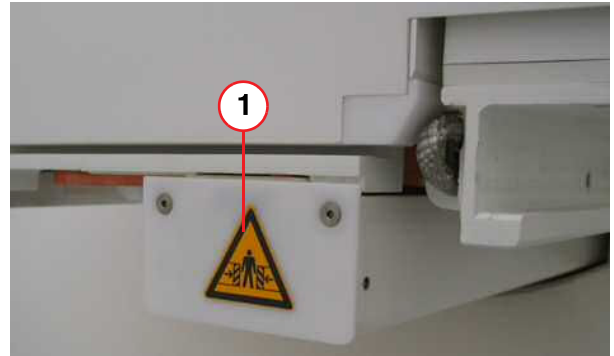
- (1) Patient table
- (2) Wall of RF room
- (3) Distance < 500mm

- Affix the label

Fig. 586: Bruising hazard sign



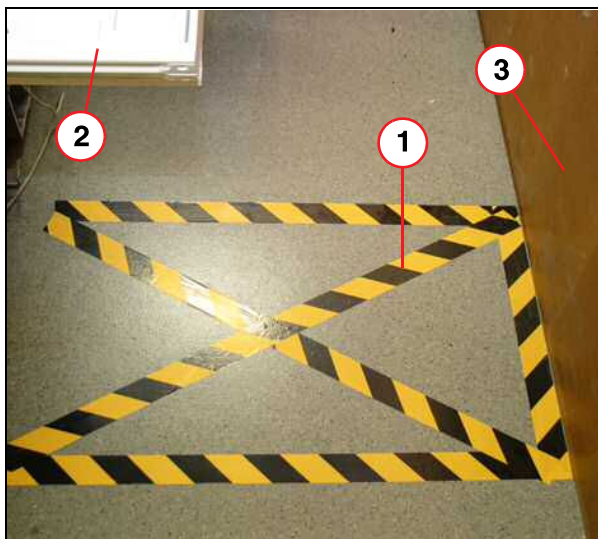
Fig. 587: Bruising hazard



(1) Label

- Protect the controlled area with yellow and black tape.

Fig. 588: Service end area



- (1) Marking on the floor
- (2) Patient table
- (3) Wall of the RF room

14.1.8 Hearing protection sign

The hearing protection sign is packed together with the warning signs.



The hearing protection sign must be posted at the main entrance door of the examination room. The sign should be positioned at eye level height as shown in (→ 1/ Fig. 590 Page 477).

Fig. 589: Hearing protection sign



Fig. 590: Warning Signs



(1) Hearing protection sign

14.1.9 Cleaning up



This completes the start-up of your MR system. If no other options are being installed in your system, you should now complete the following final steps. If you are installing other options, complete these final steps after installation of the options is finished. Please consider that your customer has purchased an excellent, high-quality MR system with a modern design. The visual impression of the system is equally important to your customer's satisfaction.

- Assemble the surface coil cart according to the installation manual attached.
- Close all magnet covers.
- Close all electronics cabinet doors.
- Touch up paint chips with matching color paint.
- Clean fingerprints off the monitor screens.
- Clean fingerprints off the table and PMU displays.
- Clean fingerprints off the system with a mild cleaner.
- Clean the patient table and the surface coils with a mild cleaner.
- Clean the area of the system, dispose of empty cartons or other packing materials.
- Make sure that the MR system rooms are cleaned by the cleaning crew.

14.1.10 Documentation for Review

Send back the following mandatory documents
to Siemens Healthcare GmbH MR QT department.

immrqpininstallations.healthcare@siemens-healthineers.com

- Safety-related Test Report
- Start-up Report
- Quench Line Acceptance Protocol incl. pictures of the quench line (provided by the project manager)
if not done by other participants like project manager or mechanical vendor.
- Pictures of the magnet room, shortly before ramping.

Version 02

- Chapter 02
Check whether the circuit breaker F1 (ACS) is switched to the "ON" position.
- Chapter 03
TestTools added
- Chapter 05
Changes in "Work after ramping"
- Chapter 06
Editorial changes
- Chapter 08
Remote Diagnostics: Note
- Chapter 11
Check the proper horizontal reference moving of the p-tab added.

Version 03

- Chapter 04
(→ Setting the warning level (only for systems with IFP) / Page 64)
- Chapter 07
(→ Check the mechanical adjustment of the patient table / Page 408) added.
(→ Spike Check / Page 432) added.
- Chapter 11
(→ Bleeding the Chiller water pump / Page 469)
(→ Check the water pressure at the IFP / Page 472)

Version 04

- Chapter 02
Checking the fuses F3 and F4
- Chapter 05
New venting system Mk2 added.
- Chapter 06
New venting system Mk2 added.
- Chapter 07
(→ Checking the distance between patient table and cover / Page 410) added.

Version 05

- Chapter 08
TFT color monitor updated.

Version 06

- Chapter 03
Check the MeasSettings added.

- Chapter 05
Filling of static system updated.
- Chapter 06
Turret Leak Check updated
- Chapter 07
(→ Check the mechanical adjustment of the patient table / Page 408) updated.

Version 07

- Chapter 06
(→ Verifying the pressure-controlled interlock / Page 212) added.
- Chapter 07
(→ Ramping the magnet - Avanto / Page 240) updated, transport test added.
(→ Tim Whole Body Suite (optional) / Page 431) added

Version 08

- Chapter 08
TFT color monitor updated.

Version 09

- Chapter 01
RF-cable 2 m BNC (→ Service Tools / Page 10)
- Chapter 02
Checking the water hoses.
(→ T-class (optional) / Page 36) added.
Compressor F70 added.
- Chapter 04
Checking the water hoses.
- Chapter 05
Filling the magnet, reused from SMT document.
Filling the mobile system, reused from SMT document.
Bottom filling updated
- Chapter 06
Turret leak check, reused from SMT document
- Chapter 07
Ramping the magnet, reused from SMT document
Failure during ramping, reused from SMT document
Post Quench Procedure, reused from SMT document.
- Chapter 11
Testing the water circuit to the GC

Version 10

- Chapter 01
 - Link to the safety check of the service platform added.
- Chapter 02
 - Static pressure tolerance corrected.
- Chapter 05
 - Service platform installation added.
 - Updated syphon leg lengths.
- Chapter 07
 - (→ Checking the reference switch S10 of the patient table / Page 409) added.
- Chapter 08
 - Transferring monitor information to the REP.
- Chapter 11
 - Hearing protection sign updated.

Version 11

- Chapter 2
 - Update of warning signs
- Chapter 4
 - Update of setting the helium level warning.
- Chapter 5
 - Updated syphon leg lengths.
- Chapter 6
 - Update of leak check
- Chapter 7
 - Update of safety instructions.
 - Update of post quench procedure.
- Chapter 11
 - Update of the Installation & Maintenance Protocol chapter.

Version 12

- Chapter 2
 - (→ Checking the gradient connections / Page 15) was added.
- Chapter 5
 - Siphon port de-icing information was added.
 - Helium pressure at altitude added.
 - Service platform safety updated.
- Chapter 7
 - Note of quench line acceptance protocol was added.

Version 13

- Chapter 2
 - Warning sign updated.
- Chapter 04
 - Update of section "Setting the correct the water pressure"
 - Update of the pressure value.
- Chapter 5
 - New section: Bottom filling the magnet
- Chapter 7
 - Update of links.
 - Note of ERDU test updated
- Chapter 11
 - Update of section "Check the water pressure"
 - Update of the pressure value.

Version 14

- Chapter 02
 - (→ Fig. 16 Page 20) has been updated.

Version 15

- Chapter 5
 - Note: Cold head bypass valve closed.
- Chapter 6
 - New leak detector
- Chapter 11
 - Company name changed.

Version 16

- Chapter 10
 - Gradient Supervision added.

Version 17

- Chapter 02: Switch on
 - New Alarm box added.
- Chapter 04: Magnet & Cooling
 - Avanto Dot update
- Chapter 05: Filling the magnet
 - Filling procedure for Avanto Dot Ex-factory (MSUP K2306) added.
- Chapter 08: Ramping the magnet Avanto Dot
 - Ramping procedure for Avanto Dot added.
- Chapter 11: Intercom
 - Avanto Dot / Verio Dot Intercom added.

Version 18

- Chapter 4: Configuring the "Magnet&Cooling"
 - Msup K2306 for Espree added
 - Chapter re-ordered
- Chapter 13: Tune-up/QA
 - Prerequisites for Tune-up/QA
- Chapter 14: Final work
 - Cleaning the water filter added.
 - Quench Line Acceptance Protocol added.

Version 19

- Chapter 09: Work after ramping
 - Patient table adjustment updated.
(→ Fig. 537 Page 429)

Version 20

- Chapter 02:
 - Gradient cable tolerances 35 ± 1 Nm changed to 35 ± 2 Nm .
- Chapter 06
 - Cold head temperatur sensor updated. Charm MR_00450528
- Chapter 07:
 - Ramping the magnet- Avanto, updated. Charm MR_00453494
(→ What to do after a Quench (Avanto, Avanto Dot Upgrade, Espree with MSUP K2256) / Page 372)
- Chapter 08:
 - Ramping the magnet- Avanto Dot, updated. Charm MR_00453494
(→ What to do after a Quench (Avanto Dot ex-factory, Espree with MSUP K2306) / Page 346)
- Chapter 14:
 - Installation & Maintenance Protocol
Deleted: Send back the sketch of the magnet room. Charm MR_00454236 (→ Installation&Maintenance Protocol / Page 474)

Version 21

- Chapter 02:
 - Gradient cable tolerances 35 ± 2 Nm changed to 40 ± 4 Nm (Charm MR_00459191).
- Chapter 05:
 - Warning added regarding visual inspection of quench tube.
 - Quench line acceptance protocol: sign-off by the project manager deleted.

- Chapter 06
 - Instructions for protective heat shrink and warning label graphics added (→ Fig. 244 Page 222).
 - Leak check reworked
- Chapter 07:
 - Ramping the magnet - Avanto, updated. Transport test procedure: Charm MR_00455599.
 - Warning for visual inspection of quench tube added. (→ Ramping the magnet - Avanto / Page 240)
- Chapter 08:
 - Ramping the magnet - Avanto Dot, updated. Transport test procedure: Charm MR_00455599.
 - Warning for visual inspection of quench tube added. (→ Ramping the Magnet - Avanto Dot / Page 318)
- Chapter 12:
 - Gradient coil temperature monitoring test reworked (Charm MR_00458861)
- Chapter 14:
 - Bruising hazard sign added.
 - Changed Siemens AG into Siemens Healthcare GmbH.
 - Quench line acceptance protocol: sign-off by the project manager deleted.

Version 22

- Chapter 14
 - Minor editorial changes.
 - New Section, Documentation for Review (→ Documentation for Review / Page 478) added.

Version 23

- Chapter 05
 - Minor editorial changes: magnet bypass must be open during filling. (ID 211623)
- Chapter 06
 - Chapter "Turret Leak Check" updated
- Chapter 07
 - Information regarding shorting link revised. (ID 209125)
 - New address, graphic updated (ID 209675).
 - Update of "What to do after a quench" (ID 212345)
- Chapter 08
 - Information regarding shorting link revised. (ID 209125)
 - New address, graphic updated (ID 209675).
 - Update of "What to do after a quench" (ID 212345)

- Chapter 14
 - Email address updated.

There are no Hazard IDs in this document.

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